Graal Technical Reference Manual

build of 55c00ec

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0 Front matter and conventions

This chapter fixes the scope, structure, citation rules and notation of the Graal Technical Reference Manual. Every later chapter uses the terms defined here and does not redefine them.

0.1 Scope

The manual describes **Graal itself**: its clients from the 1998 relay builds through 6.x, its wire protocol, file formats, game servers, listserver, RC and NC, NPC scripting, Graal3D, the Torque-era 3D clients, and the tools that define formats.

These are out of scope:

- Any reimplementation's own design, including Seihai's boards, systems and gap lists. These notes are marked `excluded` in `manual/sources.tsv` (→ App. F).
- How the notes were produced: lab procedure, the order of discovery, and tooling for the notes themselves (the navigator, the release builders).
- Trainers and hacks, except where they document a client integrity check as behaviour (→ §4.10).

An implementer should be able to build a compatible client or server from the manual alone. Where the notes do not reach that far, the gap is listed in App. E rather than filled.

0.2 Structure and numbering

Unit	Form	Example	File
Part	one number	§2 Wire protocol	`manual/02-protocol/00-part.md`
Chapter	part.chapter	§2.2 Ciphers and compression	`manual/02-protocol/02-ciphers.md`
Section	chapter.section	§2.2.2 Generation 3 insertion	heading inside the chapter
Single-file part	number, then sections	§5.1 GServer13 (1999)	`manual/05-servers.md`
Appendix	letter	App. A Message index	`manual/app-a-messages.md`
Appendix entry	letter + # + number	`App. E#12`	an `### E#12` heading in App. E

Every heading that can be cited begins with its number. Part 0 is this chapter.

Cross-references. `→ §2.2.2` points to a section; `→ App. E#12` points to an appendix entry. A cross-reference replaces a repeated fact: every fact has exactly one canonical section, and other sections point to it. The checker (`tools/manual/check_manual.py`) fails on a reference that does not resolve to a heading.

Version differences go in columns or rows of the same table, never in parallel chapters. A rule that changes across builds is one table with one row per build, or per range of builds where the notes measured the whole range.

0.3 Evidence tags and citations

Every factual statement carries an **evidence tag** and the **path of the note** it comes from, in this form:

```
[tag] (path/to/note.md#anchor)
```

- The tag says what kind of evidence backs the claim. The full legend is App. G; build tags such as `[2.05]` mean "read from that client build".
- The path is relative to the repository root. The `#anchor` is optional and names the heading in the note.
- Several tags may share one path: `[srv][2.05] (protocol/codec-and-framing.md)`. Several sources are separated by `;` inside the parentheses, or written as several tag-and-path pairs.
- In a table, the tags and paths go in a **Source** column. Each body row carries its own source.

- A sentence that only arranges tagged facts, such as a cross-reference or a table caption, carries no tag.

`[inf]` marks an inference that the source note itself marked as one. The manual never upgrades an `[inf]` to a measured tag, and never adds an inference of its own without `[inf]`.

The manual adds **no new measurements**. A fact missing from the notes goes to App. E, or to a lab note in the notes first, before it can appear here.

0.4 Table cells

A table cell holds exactly one of:

Cell	Meaning
a value	measured or read, with the row's source
`unmeasured`	no build in the notes was measured for this cell; no neighbouring build's value is copied in
`disputed → App. E#n`	two notes disagree; App. E quotes both and says what would settle it
`n/a`	the field or rule does not exist in that build or generation
–	used only where a source table itself leaves a cell empty and gives no reason; treated as `unmeasured`

A value is never interpolated from builds on either side.

0.5 Numbers, byte order and bits

Item	Convention
Hexadecimal	`0x` prefix, lower or upper case as in the source note: `0x20`, `0x04A80B38`
Decimal	no prefix; message ids are given in decimal
Byte strings	hex pairs separated by spaces: `FF FF FF`
Bit numbering	bit 0 is the least significant bit; a mask is written in hex (`0x7F`)
Bit streams	read least-significant-bit first unless the section says otherwise (→ §3.1)
Multi-byte integers	each layout names its byte order; there is no global default
`u8`, `u16be`, `u16le`, `u32le`	raw unsigned integers of 8, 16 or 32 bits, big- or little-endian
Signed values	a signed field is named `s8`, `s16le` and so on; a G-type sent as signed is its unsigned bit pattern [srv] (protocol/codec-and-framing.md#graal-integer-types)
Text	bytes as sent, with no character set implied; `\n` is the byte `0x0A`

0.6 Field types

Graal's own integer types add `0x20` to each 7-bit group so that a message stays printable and never contains `\n` [srv][2.05] (protocol/codec-and-framing.md#graal-integer-types). Their encodings are defined once in §2.1.1. Field layouts in this manual use these short names:

Name	Short	Size	Defined in
GCHAR	`c`	1 byte	→ §2.1.1
GSHORT	`s`	2 bytes	→ §2.1.1
GINT, GINT3	`i3`	3 bytes	→ §2.1.1
GINT4	`i4`	4 bytes	→ §2.1.1
GINT5	`i5`	5 bytes	→ §2.1.1
GSTRING	`str`	c length, then that many bytes	→ §2.1.1
rest of message	`rest`	all bytes up to the terminator	→ §2.1.2
raw integer	`u8`, `u16le`, …	as in §0.5	→ §0.5
half-tile value	`x×2`	the value is in half-tiles; the reader divides by 2	→ §0.8

These short names follow the notation of the protocol notes [doc] (protocol/packet-layouts.md). A layout is written left to right in wire order, for example `c x, c y, str text`.

0.7 Messages and their ids

Term	Meaning
message	one logical unit: an id, a payload and, in most generations, a \n terminator (→ §2.1.2)
frame	one transport unit that carries one or more messages (→ §2.1.4)
PLI	a message from client to server; names PLI_*
PLO	a message from server to client; names PL0_*
UDP id	a message on the peer-to-peer UDP channel (→ §2.9)
logical id	the decimal id used throughout this manual
wire byte	the byte on the wire; for a GCHAR id it is the logical id + 0x20 [srv] (protocol/packet-ids.md)

Message and prop **names** are the conventional ones the notes use, restated rather than copied from any server source [doc] (protocol/packet-ids.md). Every id appears in App. A (messages) or App. B (props) with a pointer to its canonical section.

0.8 Units

Unit	Definition
pixel (px)	one pixel of the 2D game view
tile	one board cell; 16 px on a side, since a walk step of 8 px is half a tile [cap] (protocol/v5v6-lab-play.md)
half-tile	8 px; the unit of most coordinate fields in 2D messages [srv][2.05] (protocol/packet-layouts.md)
board	one level's tile grid; its size per format is in §3.1
tick	one step of the client's game loop; its length per build is in §4.1
seconds	used only where the source measured wall time; ticks are not converted to seconds unless the source gives the tick length for that build

0.9 Versions, builds and generations

Five numbering schemes describe "which Graal" a fact applies to. They change on different calendars and are never mixed.

Scheme	What it names	Written as	Canonical table
build	one client executable, by its release name and revision	`2.05 rev 2`, `1.39`, `4.20r8`; in tags `[2.05]`	→ §1.1
archive name	the archive a build was found in	`5thgraalalpha`, `graal121`	→ §1.1
version token	the login version string a build sends	`GNW03081`, `G3D14097`	→ §1.2
client type	the login type value that selects a server code path	type 0, 4, 5, 6	→ §2.3
encryption generation	the transport scheme, in the GServer-v2 numbering 1–6	gen 3	→ §2.2.1
level-format generation	the level file family and revision	Z3-V1.03, GR-V1.02, GLEVNW01, GRMAP001	→ §3.1
save generation	the save-game magic	Graalsav5, Graalsv11	→ §3.7

The encryption generation numbering is that of GServer-v2, as the notes use it [srv] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering); the build catalogue uses the same numbering [bin] (client-internals/build-catalogue.md). PE FileVersion is recorded but is not used as a build name, because it is sometimes stale (2.05 still says 2.0.4.0) [bin] (client-internals/build-catalogue.md#how-each-column-was-read).

Eras. For brevity the manual groups builds into eras: the 1998 relay and Zelda Online, 1.x, NewMain/NewWorld, 2.x, 3.x, 4.x, 5.x, 6.x (including the 6.11 server-defined protocol), Graal3D (2000) and the Torque-era 3D clients. An era name is a heading aid only. A fact is always attached to the builds it was measured on, never to a whole era by default.

0.10 Servers, hosts and redaction

Item	Convention
Public hosts	never named; written `<listserver-host>`, `<login-host>`, `<public-host>`, `<lan-host>`, or described ("the 1999 list host")
Lab hosts	"the loopback address"; no dotted literal
Ports	lab ports only (14802, 14803, 14804, 14900, 14921–14923); a historical default port that is part of a format or protocol is given as the notes give it
Accounts	the clients' built-in guest login only; no other account names, passwords or licence secrets
Binary internals	no code addresses, no disassembler names, no decompiled text, no raw string dumps
Outside projects	GServer-v2, gs2lib and similar sources are summarised and cited through the notes, never quoted

These rules come from the repository's agent rules [doc] (AGENTS.md#no-real-servers-no-binary-addresses) and are enforced by the checker (→ §0.11).

0.11 Checking and building

Task	Command
check the manual	`python tools/manual/check_manual.py`
check before the final audit	`python tools/manual/check_manual.py --final` (also fails on any `pending` row in `sources.tsv`)
run the checker's tests	`python -m pytest tools/manual -q`
build the PDF	`python tools/manual/build_manual.py` (pandoc, then Chromium prints A4; default output `manual/pdf/graal-trm.pdf`)

Each manual file begins with a header comment `<!-- manual: kind=K status=S -->`:

Field	Values	Effect in the checker
kind	`reference`	every body line, list item and table row needs a tag and a path, except cross-references and captions (a line ending in `:`)
kind	`front`	definitions and rules; tags and paths are checked where present
kind	`index`	headings and cross-references only
status	`stub`	only headings, cross-references and > status notes are allowed
status	`draft`, `complete`	the kind rules apply; a `> Status:` line may say what is still missing

`manual/sources.tsv` lists every note in the repository with one of three states: `pending` (not yet represented), `cited` (with the sections that cite it) or `excluded` (with a reason). App. F is its readable form.

1 Client generations and builds

This part names every client build the notes cover. Every other part keys its version columns off the build names in §1.1 and the schemes defined in → §0.9.

1.1 Build catalogue

1.1.1 Delphi and Java clients, 1998 to 2001

Most 1998 to 2001 executables come from one shared archive of 73 executables in 55 folders [file] (client-internals/drive-clients-inventory.md#versions-present); [file] (version-profiles/drive-clients-exes.tsv). *FileVersion* is read from the PE version resource, not from program logic [bin] (client-internals/version/VERSION.md#primary-method-pe-version-resource). One row per executable, sorted by link date. *Compiled* is the resource-directory timestamp in local time; *Gen* is the encryption generation (→ §1.4); *Levels and saves* lists the level, map and save magics the loader compares against (→ §1.5). Default hosts and list names are in → §6.

Archive	Release	Compiled	File Version	MD5	Size	Token	Gen	Levels and saves	Source
zeldaonline2	Zelda Online (Java applet)	1997–98	–	–	–	none	1998 relay (→ §2.3.1)	`Z3-V1.02/1.03`	[bin][file] (client-internals/build-catalogue.md#catalogue)
1stgraal-1	first Delphi beta	1998-09-06 23:54	–	07452f8941fdfcdce878b035156e701c	654336	none	1	Z3-V1.03/1.04 only; Graalsav1	[bin][file] (client-internals/build-catalogue.md#catalogue)
2ndgraal	beta	1998-09-16 01:33	–	f74b35fc7975a4873087f7fd2aa1d79b	777216	none	1	+ GR-V1.00–1.01; Graalsav1	[bin][file] (client-internals/build-catalogue.md#catalogue)
3rdgraal	beta	1998-09-30 01:47	–	fc78fbc96bdbc6ca22ae56d3ed8635ab	665600	none	1	GR-V1.00–1.01; Graalsav1	[bin][file] (client-internals/build-catalogue.md#catalogue)
4thgraal	beta	1998-10-05 01:33	–	e040af2e19912419ab80aa6dd202b4ea	666112	none	1	GR-V1.00–1.01; Graalsav1	[bin][file] (client-internals/build-catalogue.md#catalogue)
4thgraal alpha	beta (alpha)	1998-10-12 22:54	–	8cae23a58e1b1b10f3149bdbe53d0431	711680	none	1	GR-V1.00–1.01; Graalsav2	[bin][file] (client-internals/build-catalogue.md#catalogue)
5thgraal	beta	1998-10-19 02:27	–	0e7c3573f0a916f8b2376060e768fe96	723968	none	1	GR-V1.00–1.02; Graalsav2	[bin][file] (client-internals/build-catalogue.md#catalogue)
5thgraal alpha	beta (alpha)	1998-10-29 02:21	–	e54397e7d4f59d30b77d5cf0475335a1	739840	none	1	GR-V1.00–1.02; Graalsav2	[bin][file] (client-internals/build-catalogue.md#catalogue)

Archive	Release	Compiled	File Version	MD5	Size	Token	Gen	Levels and saves	Source
6thgraal	beta5	1998-11-09 02:08	–	53c3caf3d930534db3f44f346f5d651a	750080	none	1	GR-V1.00–1.02; Graalsav3	[bin][file] (client-internals/build-catalogue.md#catalogue)
7thgraal	beta7	1998-12-04 00:28	–	deb40c31268e499fc4436b8d0687f05d	766976	none	1	GR-V1.00–1.03; Graalsav4	[bin][file] (client-internals/build-catalogue.md#catalogue)
8thgraal	beta8	1998-12-08 01:13	–	a49aa1b773e41741f211867fd5ea55d7	775680	none	1	GR-V1.00–1.03; Graalsav4	[bin][file] (client-internals/build-catalogue.md#catalogue)
9thgraal alpha	beta9 (alpha)	1998-12-16 02:36	–	e226e1b83fd437b0aede968fb395b911	788992	none	1	GR-V1.00–1.03; Graalsav5	[bin][file] (client-internals/build-catalogue.md#catalogue)
9thgraal	beta9	1998-12-18 01:53	–	f0185ff8525a13f2f4e5997f649bc250	793088	none	1	GR-V1.00–1.03; Graalsav5	[bin][file] (client-internals/build-catalogue.md#catalogue)
10thgraal	V1.0	1998-12-22 03:24	–	f1f34b4479a8ad7614bd8f7ce56112c6	795648	none	1	GR-V1.00–1.03; Graalsav5	[bin][file] (client-internals/build-catalogue.md#catalogue)
11thgraal	V1.0	1998-12-23 01:00	–	a255212621907affc94a4483c6dfc8e3	795648	none	1	GR-V1.00–1.03; Graalsav5	[bin][file] (client-internals/build-catalogue.md#catalogue)
12thgraal	V1.0	1999-01-05 01:04	–	788ae59e9419806024d14d3e5b611971	794624	none	1	GR-V1.00–1.03; Graalsav5	[bin][file] (client-internals/build-catalogue.md#catalogue)
13thgraal	V1.1 beta1	1999-01-12 01:15	–	a211a0ecf3b58daa2f46759444e18bd3	797184	none	1	GR-V1.00–1.03; Graalsav6	[bin][file] (client-internals/build-catalogue.md#catalogue)
14thgraal	V1.1	1999-01-15 02:20	–	d0999d44fcd9c226f350fcf204618885	858112	none	1	GR-V1.00–1.03; Graalsav7	[bin][file] (client-internals/build-catalogue.md#catalogue)
15thgraal	V1.1	1999-01-25 00:12	–	9a7d66dd5c8c506ad4d3e63ee4a9a1f1	878592	none	1	GR-V1.00–1.03; Graalsav7	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal1.1.9	V1.1.9 Gold Edition	1999-02-11 23:50	–	0a6eef88acb4c0fce6e84eee3e76c04a	882176	none	1	GR-V1.00–1.03; Graalsav7/8	[bin][file] (client-internals/build-catalogue.md#catalogue)

Archive	Release	Compiled	File Version	MD5	Size	Token	Gen	Levels and saves	Source
graal1.1	V1.1 (readme)	1999-02-14 23:03	–	1a9580655cce686c602c01daa251c453	882176	none	1	GR-V1.00–1.03; Graalsav7/8	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal121	V1.2.1	1999-05-10 18:29	–	8da2e8284dd8b315a28e2d4fccae79ac	9738224	GSERV007	2	GR-V1.00–1.03; reads Graalsav9/Graalsv10/Graalsv11, writes Graalsv11	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal128b2	v1.28 beta2	1999-11-23 01:09	–	8e3c1a7233864da994a0db26f1811212	11965444	GSERV013	2	GR-V1.00–1.03; reads Graalsav9/Graalsv10/Graalsv11, writes Graalsv11	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal13b	v1.3 test	1999-11-29 02:03	–	b62aaa4d606a519e7f4689b33983429b	13117744	GSERV014	2	GR-V1.00–1.03; reads Graalsav9/Graalsv10/Graalsv11, writes Graalsv11	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal131	V1.31	1999-12-07 02:56	–	bff87a95a34399b2d35b243f281169a7	13184000	GSERV016	2	GR-V1.00–1.03; reads Graalsav9/Graalsv10/Graalsv11, writes Graalsv11	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal132	V1.32	2000-01-12 01:13	–	42b2596af4b75bcf82db03e0fc274f7d	13281128	GSERV018	2	GR-V1.00–1.03; reads Graalsav9/Graalsv10/Graalsv11, writes Graalsv11	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal3d	Graal3D	2000-02-05 01:57	–	305c985a020cec6ab8d81ac95d447343	605184	none	own (→ §9)	none	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal133	V1.33	2000-02-21 00:36	–	a3667ce55a5c67cf025882014b9540b0	13501444	GNW20020	2	GR-V1.00–1.03; carries Graalsav9/Graalsv10/Graalsv11 (string scan)	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal134	V1.34	2000-03-23 00:38	–	9e403616e73a6079ac37d65d475c81cd	13496632	GNW22030	2	GR-V1.00–1.03; carries Graalsav9/Graalsv10/Graalsv11 (string scan)	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal135	V1.35	2000-04-13 23:17	–	fe9d99c4bf6e6bc9b1b127823f39646f	13895568	GNW13040	2	GR-V1.00–1.03; carries Graalsav9/Graalsv10/Graalsv11 (string scan)	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal136	V1.36	2000-05-12 17:34	–	018f51662c7c8862cafb9bfbd14dea41	13992296	GNW10050	2	GR-V1.00–1.03; carries Graalsav9/Graalsv10/Graalsv11 (string scan)	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal137	V1.37	2000-05-23 02:40	1.3.7.0	5e966a9d1a079af6d682b2618674edb1	14013444	GNW21050	2	GR-V1.00–1.03; carries Graalsav9/Graalsv10/Graalsv11 (string scan)	[bin][file] (client-internals/build-catalogue.md#catalogue)

Archive	Release	Compiled	File Version	MD5	Size	Token	Gen	Levels and saves	Source
graal138	V1.38	2000-06-23 22:11	1.3.8.0	4401f7d02966af709c2df662cebe34a1	14264432	GNW22060	2	GR-V1.00–1.03; carries Graalsav9/Graalsv10/Graalsv11 (string scan)	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal139	V1.39	2000-08-07 02:49	1.3.8.0	62513ab884f1802a0bcd8114abe349b9	14489600	GNW07080	2	GR-V1.00–1.03; carries Graalsav9/Graalsv10/Graalsv11 (string scan)	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal_gr1411	V1.41 rev1 (-patched copy)	– (resource stamp missing)	1.4.1.1	a97740e711ffe9b698ec03712d4c06af	15068116	GNW13110	3	GR-V1.00–1.03, GLEVNW01; carries Graalsav9/Graalsv10/Graalsv11 (string scan)	[bin][file] (client-internals/build-catalogue.md#catalogue)
graalnm24	NewMain client (1.42 notes)	2001-01-10 19:55	1.4.1.1	229f05de7a43b98e223d8e1b58bffc63	17832296	GNW08011	3	GR-V1.00–1.03, GLEVNW01	[bin][file] (client-internals/build-catalogue.md#catalogue)
graalnm26	NewMain client (1.42 notes)	2001-01-12 21:19	1.4.1.1	22a08ed154f67948e01b969b68cdb05e	17843320	GNW08011	3	GR-V1.00–1.03, GLEVNW01	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.0	v2.00	2001-02-01 14:54	2.0.0.0	c93e659e2cda36379aaea4706db20b38	18124800	GNW24011	3	GR-V1.00–1.03 (no `.nw`)	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.0-rev1	v2.00 rev1	2001-02-02 19:19	2.0.0.1	30bd230e578d3a347866a87060efec20	18129992	GNW24011	3	as above	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.0-rev2	v2.00 rev2	2001-02-05 02:32	2.0.0.2	fc4bb7e3b7ca9cd8ea6693a033635534	18150400	GNW24011	3	as above	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.0-rev3	v2.00 rev3	2001-02-07 01:05	2.0.0.3	1ce5c1662b65246ae1c6559af5ff71f0	18150400	GNW24011	3	as above	[bin][file] (client-internals/build-catalogue.md#catalogue)
GraalX (`.rar`, separate upload)	2.01-era side build	2001-03-07 21:40	2.0.0.3	d3ee52ef9d22a0d99b154b47c5f56f60	18216696	GNW23021	3	GR-V1.00–1.03, **GLEVNW01**	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.01	v2.01	2001-03-08 14:59	2.0.1.0	08340a8868f1d11e0285cc70e51200fe	18155552	GNW23021	3	as above	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.01-rev1	v2.01 rev1	2001-03-15 21:10	2.0.1.0	010a32afce66729b838258809a97c0d9	18155552	GNW23021	3	as above	[bin][file] (client-internals/build-catalogue.md#catalogue)

Archive	Release	Compiled	File Version	MD5	Size	Token	Gen	Levels and saves	Source
Graal-v2.02	v2.02	2001-04-15 20:48	2.0.2.0	60018a19ef671c506743826728c49718	18083844	GNW15041	3	as above	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.02-rev1	v2.02 rev1	2001-04-15 23:16	2.0.2.0	e06ff79aded885a223b933617d456577	18088896	GNW15041	3	as above	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.02-rev2	v2.02 rev2	2001-04-20 18:48	2.0.2.2	993a1829ddbb6acaa2b46063f5efe636c	18078872	GNW20041	3	as above	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.03	v2.03	2001-05-07 15:31	2.0.3.0	36972c74baad09ceb33dad94b0eb1e33	18360032	GNW06051	3	as above	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.04	v2.04	2001-06-25 07:42	2.0.4.0	a40608308d4f890cb5e4efdd45b38ba4	18723844	GNW24061	3	as above	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.05	v2.05	2001-08-03 19:12	2.0.4.0 (stale)	9abce477945a1307bb2be9ed47d2f170	18913288	GNW03081	3	GR-V1.00–1.03, **GLEVNW01, GRMAP001**	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.05-rev1	v2.05 rev1	2001-08-04 03:26	2.0.5.0	2780f2f370396c580e1648696698c871	18959936	GNW03081	3	as above	[bin][file] (client-internals/build-catalogue.md#catalogue)
graalnwb1	NewWorld beta 1	2001-08-06 04:13	2.0.5.0	88d1f84c96da1dcc3942b139fbd741c9	18206672	G2N15041	3	as 2.05	[bin][file] (client-internals/build-catalogue.md#catalogue)
Graal-v2.05-rev2	v2.05 rev2	2001-08-16 17:24	2.0.5.2	f3fdf54d943fa91c89acba5bb695db99	18918840	GNW03081	3	as 2.05	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal21test2	v2.1 test 2	2001-09-21 17:54	2.1.0.0	d15d877822ad4145578107857c90fb3c	17484800	GNW03081	3	as 2.05	[bin][file] (client-internals/build-catalogue.md#catalogue)
graal21test3	v2.1 test 3	2001-09-27 21:11	2.1.0.0	2116a7ba03648dcc004035534ec94de8	17536000	GNW03081	3	as 2.05	[bin][file] (client-internals/build-catalogue.md#catalogue)
graalnwb2	NewWorld beta 2	2001-10-03 20:46	2.1.0.0	af2d21066659bed6af6c77e52b2cc0e1	16138824	G2N15041	3	as 2.05	[bin][file] (client-internals/build-catalogue.md#catalogue)

Notes on the catalogue:

- The "Nth graal" archive names are not chronological: 4thgraalalpha and 5thgraalalpha were linked after their namesakes and 9thgraalalpha before 9thgraal; graal1.1 has the same size as graal1.1.9 but was linked three days later [bin] (client-internals/build-catalogue.md#patterns-worth-knowing).
- FileVersion exists only from 1.37 on and is sometimes stale (2.05 says 2.0.4.0; 1.39 says 1.3.8.0) [bin] (client-internals/build-catalogue.md#how-each-column-was-read).
- The only 1.41 copy (graal_gr1411) is patched by a third party: its resource timestamp is zeroed and its list host and port may not be the original values; the token and code-level facts look untouched [bin][inf] (client-internals/build-catalogue.md#patterns-worth-knowing).
- GraalX is 2.01 with the GLEVNW01 reader and writer kept in: 6350 functions identical to 2.01, 7 differing, and the same command table and packet and prop ids [bin] (client-internals/build-catalogue.md#graalx).
- graal_v2.05 rev 2 is the same file as `Graal_v2.05_rev2.exe`, the main decompile target of the notes [bin] (client-internals/build-catalogue.md#catalogue).

1.1.2 Compilers

Builds	Compiler	Evidence	Source
all Delphi builds	Borland Delphi, linker 2.25, DelphiX for DirectDraw, RTL copyright "1983,97 Borland"	PE header and RTL strings	[bin] (client-internals/build-catalogue.md#patterns-worth-knowing)
2nd Graal; 1.28 beta 2 and later	Delphi 4	Delphi 3+ class-metadata layout (self pointer 76, name 44 bytes before the VMT) and `TMultiReadExclusiveWriteSynchronizer` linked	[bin] (client-internals/build-catalogue.md#patterns-worth-knowing)
1st Graal; 3rd Graal to 1.1.9; 1.21	disputed → App. E#1	Delphi 2 class-metadata layout (64 and 32 bytes) but a 1997 RTL string	[bin][inf] (client-internals/build-catalogue.md#patterns-worth-knowing)

1.1.3 Companion executables in the 1998 to 2001 archives

Archive	File	Compiled	MD5	Token	Port	Source
graal1.1.9	Gserver.exe	1999-01-22	7882dba6ebb2ab1026de260c3043fa47	none	14802	[bin] (client-internals/build-catalogue.md#other-executables-in-the-archives)
graal121	GServer.exe	1999-05-09	bfcdaae0217d953124a9b21ff7858967	GSERV007	14804	[bin] (client-internals/build-catalogue.md#other-executables-in-the-archives)
graal121	remotecontrol.exe	1999-05-10	5932d6baa08289a65515342dfc4b9f5a	GSERV007	14804	[bin] (client-internals/build-catalogue.md#other-executables-in-the-archives)
graal128b2	GServerv.exe	1999-11-22	8d6bff85bda6e6c9558503afa6b7259d	accepts GSERV007–013	14804	[bin] (client-internals/build-catalogue.md#other-executables-in-the-archives)
graal128b2	RemoteControl1.28.exe	1999-10-20	d7150d1385ea11b812d10d30212f9757	GSERV011	14804	[bin] (client-internals/build-catalogue.md#other-executables-in-the-archives)
graal13b	GServer13.exe	1999-11-29	d1d0470e53a60b8f000b9ea5da9d37b6	accepts GSERV007–014	14804	[bin] (client-internals/build-catalogue.md#other-executables-in-the-archives)
graal13b	RemoteControl13.exe	1999-11-29	1f2c19ab385c8b8af6236197f98ad735	GSERV011	14804	[bin] (client-internals/build-catalogue.md#other-executables-in-the-archives)
graal3d	GServer.exe	2000-02-05	aab1a34a1f196d4f4fddf12112ccbf52	none	14901	[bin] (client-internals/build-catalogue.md#other-executables-in-the-archives)

Archive	File	Compiled	MD5	Token	Port	Source
graalnm24/26	RemoteControl-NewMain.exe	2001-01-08	bc2b8a2e80acc09a6bb418970cff5a64	GSERV021	14804	[bin] (client-internals/build-catalogue.md#other-executables-in-the-archives)
graalnwb1	RemoteControl-NW.exe	2001-05-22	6fe24bdd4e135900fc40d7fb70bdcdd2	–	14804	[bin] (client-internals/build-catalogue.md#other-executables-in-the-archives)
graalnwb2	RemoteControl-NW.exe	2001-08-27	ce454437af8880f61ac2173b0bd11047	–	14804	[bin] (client-internals/build-catalogue.md#other-executables-in-the-archives)

The archives also hold gif2mng, GraalShop, GraalLevelGen and GraalLevelGenNW, NPCControl, LevelFiller with levelviewer, TerrainGenerator, PNGView, YoshiChess and Greaditor [file] (client-internals/build-catalogue.md#other-executables-in-the-archives). Their formats and behaviour are in → §11.

1.1.4 Later builds

Builds after 2001 are known from their tokens, lab runs and static reads; there is no per-build hash catalogue for them in the notes. Tokens are in → §1.2.

Build	Token	Source
2.1.0.0 to 2.1.0.3	GNW31101	[doc] (client-internals/version-tokens.md)
2.1.4.0 (first with server-list tabs)	GNW30042	[doc] (client-internals/version-tokens.md)
2.1.7.0, 2.1.7.1 (2.171, `Graal_2171.exe`)	GNW22122	[bin][cap] (client-internals/version-tokens.md)
2.1.9.0 (bz2 and new encryption)	GNW15053	[doc] (client-internals/version-tokens.md)
2.2.1.1 (extra compression byte)	GNW01113	[doc] (client-internals/version-tokens.md)
2.22 (`Graal_222.exe`)	GNW03014	[bin][cap] (client-internals/version-tokens.md)
2.3.0.0	GNW14015	[doc] (client-internals/version-tokens.md)
2.31 (`Graal_231.exe`; editor separate, editor FileVersion 2.3.1.1)	GNW28015	[bin][cap] (client-internals/version-tokens.md)
Linux 3.04 (Torque rewrite, 2.22 protocol)	G3D03014	[bin][cap] (client-internals/version-tokens.md)
Mac OS X 3.153 (2.31 protocol)	G3D28015	[bin][cap] (client-internals/version-tokens.md)
Windows 4.04r2 (`GraalEngine.dll`, March 2006)	G3D17026	[bin][cap] (client-internals/version-tokens.md)
Mac OS X 4.20r6	G3D27076	[bin][cap] (client-internals/version-tokens.md)
Windows 4.20r8 (`GraalEngine.dll`)	G3D20126	[bin][cap] (client-internals/version-tokens.md)
Windows 5.00r7 (engine DLL, Jul 2007)	G3D22067	[bin][cap] (client-internals/version-tokens.md)
Windows 5.12	G3D14097	[bin][cap] (client-internals/version-tokens.md)
Windows 6.00 beta1	G3D03048	[bin][cap] (client-internals/version-tokens.md)
Windows 6.00 beta2, 5.24r3, 5.24r4	G3D15010	[bin][cap] (client-internals/version-tokens.md)
Windows 5.30r3, 5.31r4, 6.00r2, 6.00r4	G3D26090	[bin][cap] (client-internals/version-tokens.md)
Linux 6.01r1	G3D0108A	[bin][cap] (client-internals/version-tokens.md)

Build	Token	Source
Windows 6.03r7	G3D0311C	[bin][cap] (client-internals/version-tokens.md)
Linux 6.03r7	G3D0511C	[bin][cap] (client-internals/version-tokens.md)

From 2.3 the editor and the online client are separate executables [doc] (client-internals/version-tokens.md). Graal3D (2000) is a separate client with its own server and no Graal level loaders [bin] (client-internals/build-catalogue.md#catalogue); it is described in → §9.

1.2 Version tokens

The version token is the eight-character string a client sends in its login (→ §2.3).

1.2.1 Token schemes

Scheme	Form	Builds	Rule	Source
none	—	1998 betas to 1.1.9	no token is sent	[bin] (client-internals/build-catalogue.md#catalogue)
protocol revision	GSERV + 3 digits	1.21 to 1.32, RC to Jan 2001	the number is a protocol revision: 007 (1.21), 011 (RC 1.28), 013 (1.28b2), 014 (1.3 test), 016 (1.31), 018 (1.32), 021 (RC-NewMain)	[bin] (client-internals/build-catalogue.md#patterns-worth-knowing)
build date	GNW + DDMMY	1.33 to 2.31	DD day, MM month, Y last digit of the year; the date is on or a few days before the link date	[bin] (client-internals/build-catalogue.md#patterns-worth-knowing) ; [inf] (client-internals/decomp-findings-2026-10-24-c/02-login-token-and-version.md#reading-the-digits-inferred)
NewWorld	G2N + DDMMY	NewWorld betas 1 and 2	same date form	[bin] (client-internals/build-catalogue.md#catalogue)
3D era	G3D + date digits	3.x to 6.x	the 3.x tokens carry the same date stamps as their 2.x protocol twins (G3D03014 and GNW03014); later suffixes include hex-like letters (G3D0311C, G3D0108A)	[bin] (client-internals/version-id-map.md#protocol-token--version); [cap] (client-internals/version-tokens.md)

A revision that does not change the protocol keeps the previous token: 2.0 rev1 to rev3, 2.01 rev1, 2.02 rev1, 2.05 rev1 and rev2, and the 2.1 test builds, which still send the 2.05 token GNW03081 [bin] (client-internals/build-catalogue.md#patterns-worth-knowing). Among those revisions only 2.02 rev2 changes the token, and it also adds the `PLI_LANGUAGE` send after login [bin] (client-internals/decomp-findings-2026-10-24-c/02-login-token-and-version.md#what-changed-in-the-login-function-between-revisions); [bin] (client-internals/decomp-findings-2026-10-24-c/version_tokens.tsv); [bin] (version-profiles/version-tokens-by-build.tsv). From 1.38 the token is built one character at a time in code rather than stored as a string literal [bin] (client-internals/build-catalogue.md#how-each-column-was-read).

A server accepts a range of tokens rather than one: GServer 1.28 accepts GSERV007 to GSERV013 and GServer13 accepts GSERV007 to GSERV014 [bin] (client-internals/build-catalogue.md#patterns-worth-knowing). A lab GServer-v2 admits a client when its token is in the allowed-versions list, which for 2.00 is a configuration entry, not a code change [lab] (client-internals/version-tokens.md). Server-side token handling is in → §5.4.

1.2.2 Token table

Token	Build	Source
GSERV007	1.21	[bin] (client-internals/build-catalogue.md); [srv] (client-internals/version-id-map.md)

Token	Build	Source
GSERV 009	1.25	[srv] (client-internals/version-id-map.md#full-gs2lib-client-token-table-2026-10)
GSERV 011	1.27; RemoteControl 1.28 and 1.3	[srv] (client-internals/version-id-map.md); [bin] (client-internals/build-catalogue.md)
GSERV 013	1.28 beta 2	[bin] (client-internals/build-catalogue.md)
GSERV 014	1.3 test (1.30)	[bin] (client-internals/build-catalogue.md)
GSERV 016	1.31	[bin] (client-internals/build-catalogue.md)
GSERV 018	1.32	[bin] (client-internals/build-catalogue.md)
GSERV 021	RemoteControl-NewMain	[bin] (client-internals/build-catalogue.md)
GNW20 020	1.33	[bin] (client-internals/build-catalogue.md)
GNW22 030	1.34	[bin] (client-internals/build-catalogue.md)
GNW13 040	1.35	[bin] (client-internals/build-catalogue.md)
GNW10 050	1.36	[bin] (client-internals/build-catalogue.md)
GNW21 050	1.37	[bin] (client-internals/build-catalogue.md)
GNW22 060	1.38	[bin] (client-internals/build-catalogue.md); [bin] (client-internals/version-tokens.md)
GNW07 080	1.39	[bin] (client-internals/build-catalogue.md); [bin] (client-internals/version-tokens.md)
GNW12 080	1.391	[srv] (client-internals/version-id-map.md)
GNW13 110	1.41 rev1	[bin] (client-internals/build-catalogue.md); [file] (client-internals/version-id-map.md)
GNW08 011	NewMain client (1.42 notes)	[bin] (client-internals/build-catalogue.md)
GNW24 011	2.00, rev1 to rev3	[bin] (client-internals/version-tokens.md)
GNW23 021	2.01, 2.01 rev1, GraalX	[bin] (client-internals/version-tokens.md); [bin] (client-internals/build-catalogue.md)
GNW15 041	2.02, 2.02 rev1	[bin] (client-internals/version-tokens.md)
GNW20 041	2.02 rev2	[bin] (client-internals/version-tokens.md)
GNW06 051	2.03	[bin] (client-internals/version-tokens.md)
GNW24 061	2.04	[bin] (client-internals/version-tokens.md)
GNW03 081	2.05, rev1, rev2; 2.1 test 2 and 3	[bin] (client-internals/version-tokens.md)
G2N15 041	NewWorld betas 1 and 2	[bin] (client-internals/build-catalogue.md)
GNW31 101	2.1 (2.1.0.0 to 2.1.0.3)	[srv] (client-internals/version-id-map.md); [doc] (client-internals/version-tokens.md)
GNW01 012	2.12	[srv] (client-internals/version-id-map.md)
GNW23 012	2.13	[srv] (client-internals/version-id-map.md)
GNW30 042	2.14 (2.1.4.0)	[srv] (client-internals/version-id-map.md); [doc] (client-internals/version-tokens.md)

Token	Build	Source
`GNW19052`	2.15	[srv] (client-internals/version-id-map.md)
`GNW20052`	2.151	[srv] (client-internals/version-id-map.md)
`GNW12102`	2.16	[srv] (client-internals/version-id-map.md)
`GNW22122`	2.17 (2.1.7.0, 2.1.7.1)	[srv] (client-internals/version-id-map.md); [cap] (client-internals/version-tokens.md)
`GNW21033`	2.18	[srv] (client-internals/version-id-map.md)
`GNW15053`	2.19 (2.1.9.0)	[srv] (client-internals/version-id-map.md); [doc] (client-internals/version-tokens.md)
`GNW28063`	2.2	[srv] (client-internals/version-id-map.md)
`GNW01113`	2.21 (2.2.1.1)	[srv] (client-internals/version-id-map.md); [doc] (client-internals/version-tokens.md)
`GNW03014`	2.22 (2.2.2.0) game login	[srv] (client-internals/version-id-map.md); [bin] (client-internals/version-tokens.md)
`GNW14015`	2.3 (2.3.0.0)	[srv] (client-internals/version-id-map.md)
`GNW28015`	2.31 (2.3.1.0) game login	[srv] (client-internals/version-id-map.md); [cap] (client-internals/version-tokens.md)
`GNW30123`	listserver hello of 2.22 and 2.31, sent with `newmain`; not a game build (→ §6)	[bin][lab] (client-internals/version-tokens.md)
`G3D16053`	3.0	[srv] (client-internals/version-id-map.md)
`G3D27063`	3.01	[srv] (client-internals/version-id-map.md)
`G3D03014`	Linux 3.04 rev 1 (3.041)	[bin] (client-internals/version-id-map.md)
`G3D28015`	Mac OS X 3.153	[bin] (client-internals/version-id-map.md)
`G3D30123`	3.x listserver login	[bin] (client-internals/version-id-map.md)
`G3D17026`	Windows 4.04r2	[bin] (client-internals/version-id-map.md)
`G3D27076`	Mac OS X 4.20r6	[bin] (client-internals/version-id-map.md)
`G3D20126`	Windows 4.20r8	[bin] (client-internals/version-id-map.md)
`G3D22067`	Windows 5.00r7	[bin] (client-internals/version-tokens.md)
`G3D14097`	Windows 5.12	[cap] (client-internals/version-tokens.md)
`G3D03048`	Windows 6.00 beta1	[cap] (client-internals/version-tokens.md)
`G3D15010`	Windows 6.00 beta2, 5.24r3, 5.24r4	[cap] (client-internals/version-tokens.md)
`G3D26090`	Windows 5.30r3, 5.31r4, 6.00r2, 6.00r4	[cap] (client-internals/version-tokens.md)
`G3D0108A`	Linux 6.01r1	[cap] (client-internals/version-tokens.md)
`G3D0311C`	Windows 6.03r7	[cap] (client-internals/version-tokens.md)
`G3D0511C`	Linux 6.03r7	[cap] (client-internals/version-tokens.md)

The older GServer-v2 version table writes the 1.33 to 1.38 tokens without the leading `G` (`NW20020` and so on) [srv] (client-internals/version-id-map.md#full-gs2lib-client-token-table-2026-10). The gs2lib table lists `G3D28095` and later for 4.x and up without per-build rows [srv] (client-internals/version-id-map.md#full-gs2lib-client-token-table-2026-10).

1.3 Version ids

Server code identifies a client by an internal version id; ids sort, so 2.05 rev 2 sorts below 2.1 [srv] (client-internals/version-id-map.md#protocol-token--version).

Version id	Build	Token	Source
`CLVER_1_39`	1.39	`GNW07080`	[srv] (client-internals/version-id-map.md#protocol-token--version)
`CLVER_2_052`	2.05 rev 2; sorts below CLVER_2_1	`GNW03081`	[srv] (client-internals/version-id-map.md#protocol-token--version)
`CLVER_2_17`	2.17	`GNW22122`	[srv] (client-internals/version-id-map.md#protocol-token--version)
`CLVER_2_22`	2.22	`GNW03014`	[srv] (client-internals/version-id-map.md#protocol-token--version)
`CLVER_2_31`	2.31	`GNW28015`	[srv] (client-internals/version-id-map.md#protocol-token--version)

The notes give version ids only for the builds above; ids for other builds are unmeasured [srv] (client-internals/version-id-map.md#protocol-token--version).

1.4 Encryption generation by build

Gen	Builds	How it was determined	Source
1	1st Graal to 1.1.9	no zlib linked, no decompression-error option	[bin] (client-internals/build-catalogue.md#patterns-worth-knowing)
2	1.21 to 1.39	zlib 1.0.4 linked; the 1.2 notes say "packet compression"; `nodecompressionerror` from 1.28b2	[bin][doc] (client-internals/build-catalogue.md#patterns-worth-knowing)
3	1.41 rev1, NewMain 1.42, 2.00 to 2.05, 2.1 tests, NewWorld betas, 2.17	the connection initialiser seeds the insertion LCG and draws a 7-bit key; 1.39 lacks it	[bin] (client-internals/build-catalogue.md#patterns-worth-knowing); [bin] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering)
4	2.19 to 2.21 and 3.0 to 3.01 (client type 4)	server source	[srv] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering)
5	2.22 and later (client type 5); RemoteControl as type 6	server source; 2.22 and 2.31 send type 5	[srv][bin] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering)
6	v6	not implemented in the server source the notes summarise	[srv] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering)
own	Graal3D	own protocol	[bin] (client-internals/build-catalogue.md#catalogue)

The schemes themselves are defined in → §2.2. Generations of the 3.x to 6.x clients beyond these rows are in → §2.3.

1.5 Level-format generation by build

Builds	Level and map magics read	Save magic	Source
Zelda Online (Java)	Z3-V1.02, Z3-V1.03	n/a	[bin] (client-internals/build-catalogue.md#catalogue)
1st Graal	Z3-V1.03, Z3-V1.04	Graalsav1	[bin] (client-internals/build-catalogue.md#catalogue)
2nd Graal to 4th Graal	+ GR-V1.00, GR-V1.01	Graalsav1	[bin] (client-internals/build-catalogue.md#catalogue)
4thgraalalpha	GR-V1.00 to GR-V1.01	Graalsav2	[bin] (client-internals/build-catalogue.md#catalogue)
5th Graal, 5thgraalalpha	GR-V1.00 to GR-V1.02	Graalsav2	[bin] (client-internals/build-catalogue.md#catalogue)
6th Graal	GR-V1.00 to GR-V1.02	Graalsav3	[bin] (client-internals/build-catalogue.md#catalogue)
7th, 8th Graal	GR-V1.00 to GR-V1.03	Graalsav4	[bin] (client-internals/build-catalogue.md#catalogue)
9th Graal to 12th Graal (1.0)	GR-V1.00 to GR-V1.03	Graalsav5	[bin] (client-internals/build-catalogue.md#catalogue)
13th Graal	GR-V1.00 to GR-V1.03	Graalsav6	[bin] (client-internals/build-catalogue.md#catalogue)
14th, 15th Graal	GR-V1.00 to GR-V1.03	Graalsav7	[bin] (client-internals/build-catalogue.md#catalogue)
1.1.9, 1.1	GR-V1.00 to GR-V1.03	Graalsav7/8	[bin] (client-internals/build-catalogue.md#catalogue)
1.21 to 1.39	GR-V1.00 to GR-V1.03	reads Graalsav9, Graalsv10, Graalsv11; writes Graalsv11	[bin] (client-internals/build-catalogue.md#catalogue)
1.41 rev1, NewMain 1.42	GR-V1.00 to GR-V1.03, GLEVNW01	as 1.21 (1.41 rev1); unmeasured (NewMain)	[bin] (client-internals/build-catalogue.md#catalogue)
2.00 to 2.04	GR-V1.00 to GR-V1.03 (no GLEVNW01, no GRMAP001)	unmeasured	[bin] (client-internals/build-catalogue.md#catalogue)
GraalX	GR-V1.00 to GR-V1.03, GLEVNW01	unmeasured	[bin] (client-internals/build-catalogue.md#catalogue)
2.05 and later 2001 builds	GR-V1.00 to GR-V1.03, GLEVNW01, GRMAP001	unmeasured	[bin] (client-internals/build-catalogue.md#catalogue)

GR-V1.03 is the last `.graal` revision any catalogued build reads [bin] (client-internals/build-catalogue.md#catalogue). `.nw` support is not monotonic: 1.41 rev1 and NewMain 1.42 read GLEVNW01, 2.00 to 2.04 contain neither GLEVNW01 nor GRMAP001, and both return in 2.05 [bin] (client-internals/build-catalogue.md#patterns-worth-knowing). That 2.0 branched from a pre-NewMain tree is an inference from strings [inf] (client-internals/build-catalogue.md#patterns-worth-knowing). The 2.05 string table holds exactly Z3-V1.03, Z3-V1.04, GR-V1.00 to GR-V1.03 and GLEVNW01 as level magics [2.05] (formats/level-binary.md). The save writer was traced only in 1.21 and 1.32; for 1.33 to 1.41 rev1 the save magics are inferred from the same string arrangement [bin] [inf] (client-internals/build-catalogue.md#patterns-worth-knowing). The formats themselves are in → §3.1, → §3.2 and → §3.7.

1.6 Era timeline

The manual groups builds into thirteen eras. Inside an era the clients share their engine and wire format [bin] (version-profiles/matrix/README.md#eras). An era letter is a heading aid; facts stay attached to builds (→ §0.9).

Era	Builds	Dates	Engine	Source
Z	Zelda Online 1 and 2 (Java applets), GServ Standalone	May to Jul 1998	Java	[bin][doc] (version-profiles/matrix/README.md#eras)

Era	Builds	Dates	Engine	Source
A	1st Graal to 1.1.9 Gold (20 builds)	Sep 1998 to Feb 1999	Delphi 2-level compiler (2nd Graal: Delphi 4), DelphiX	[bin][doc] (version-profiles/matrix/README.md#eras)
B	1.21, 1.28 beta 2, 1.3 test, 1.31, 1.32	May 1999 to Jan 2000	Delphi 4 from 1.28b2, DelphiX	[bin][doc] (version-profiles/matrix/README.md#eras)
3D0	Graal3D	Feb 2000	Delphi, DelphiX Direct3D retained mode	[bin][doc] (version-profiles/matrix/README.md#eras)
C	1.33 to 1.39	Feb to Aug 2000	Delphi 4, DelphiX	[bin][doc] (version-profiles/matrix/README.md#eras)
D	1.41 rev1	Nov 2000	as C	[bin][doc] (version-profiles/matrix/README.md#eras)
E	NewMain 1.42, 2.00 to 2.05 rev2, 2.1 tests, NewWorld betas	Jan to Oct 2001	as C, plus DirectDraw 7	[bin][doc] (version-profiles/matrix/README.md#eras)
F	2.1 to 2.17 (only 2.17.1 survives as a binary)	Oct 2001 to 2002	as E, BASS audio from 2.17 at the latest	[bin][doc] (version-profiles/matrix/README.md#eras)
G	2.22, 2.3 (no binary), 2.31	2004 to 2005	as F	[bin][doc] (version-profiles/matrix/README.md#eras)
H	3.04 Linux, 3.153 Mac	2004 to 2005	C++ on the Torque Game Engine	[bin][doc] (version-profiles/matrix/README.md#eras)
I	4.04r2, 4.20r6, 4.20r8	2006 to 2007	Delphi launcher + C++ `GraalEngine.dll` (+ `Graal3DEngine.dll`)	[bin][doc] (version-profiles/matrix/README.md#eras)
J	5.00r7, 5.12; then 5.24 to 5.31	2007 to 2011	5.00 and 5.12 as I; from 5.24 one MSVC executable	[bin][doc] (version-profiles/matrix/README.md#eras)
K	6.00 betas to 6.03r7, 6.11 Mac	2010 to 2014	one native C++ program per platform, no Torque layer	[bin][doc] (version-profiles/matrix/README.md#eras)

Change	Between	Source
largest single code step: about 800 more functions, partly new DirectDraw 7 and grid library code	1.41 rev1 and NewMain 1.42	[bin] (client-internals/cross-version-timeline.md#function-maps-onto-205-rev2)
NewWorld beta 1 cut from the 2.05 code base; NewWorld beta 2 cut from the 2.1 test code base	2001	[bin] (client-internals/cross-version-timeline.md#classes)
2.1 test builds are a later branch of the 2.05 code: more functions than 2.05 but fewer unique to 2.05 rev2	Sep 2001	[bin][inf] (client-internals/cross-version-timeline.md#function-maps-onto-205-rev2)

Era-by-era wire, scripting and id-set differences are tabulated in → App. C.

1.7 Precursors and side builds

These builds sit around the Delphi client line. The uploads hold 9 distinct items under 14 names [file] (client-internals/precursors-and-fan-builds.md#what-the-uploads-are).

Date	Build	What it is	Source
1998-05	Zelda Online 1	Java 1.0 applet, single player, no socket code; `Z3-V1.02` and `Z3-V1.03` levels, start `level1.zelda`	[bin] (client-internals/precursors-and-fan-builds.md#zelda-online-1-may-1998)

Date	Build	What it is	Source
1998-07	Zelda Online 2 "local version"	Java 1.1 application with one-player and multiplayer modes, the GServ Standalone lobby on 14801 and the author's commented source; 51 `Z3-V1.03` levels	[src] (client-internals/precursors-and-fan-builds.md#zelda-online-2-local-version-july-1998)
1998-07	ZeldaEditor	Delphi 3 level editor; writes only `Z3-V1.03`	[bin] (client-internals/precursors-and-fan-builds.md#zeldaeditor-july-1998)
1999-09	Graal Game Boy Color demo	homebrew by the Graal author, built with GBDK 2.x; header title `GRAAL`, MBC1, 256 KB ROM of which banks 0 to 5 are used	[bin] (client-internals/precursors-and-fan-builds.md#graal-game-boy-color-demo-1999); [bin] (client-internals/gameboy-demo.md#header-and-bank-map)
2000-07	Java Graal	Zelda Online 2's game and client with Graal art, reading `GR-V1` levels; relay on 14802 (→ §2.3.1.3)	[bin][src] (client-internals/precursors-and-fan-builds.md#java-graal-july-2000)
2001-03	GraalX	a 2.01 pre-release that keeps the `.nw` reader and writer (→ §1.1.1)	[bin] (client-internals/precursors-and-fan-builds.md#graalx)
2014-04	39ster client 1.00	an offline fan remake on an SDL engine with GameMonkey script; reads `.nw` and `GR-V1.00` to `1.03`; NPC scripts are not run	[bin] (client-internals/precursors-and-fan-builds.md#39sters-graal-client-100-2014)

Rule	Source
1st Graal grew out of ZeldaEditor: it carries the editor's sign, link, object-name and sign-code dialogs under the same class names; the baddy dialog and the main form were renamed	[bin] (client-internals/precursors-and-fan-builds.md#zeldaeditor-july-1998)
The Graal sign alphabet (→ §3.6) is already in Zelda Online 1	[bin] (client-internals/precursors-and-fan-builds.md#zelda-online-1-may-1998)
Zelda Online 2's options hold a 10-character save code of 60 bits (position, level index, rupees, darts, power, item and story bits, a 6-bit count of set bits), XORed with a fixed mask and transposed before writing	[src] (client-internals/precursors-and-fan-builds.md#password-saves)
ZeldaEditor's `objects.dat` stamp list: a name line, one width byte and one height byte, then width × height 32-bit little-endian tile ids; tile 2047 is transparent; 105 stamps in 1998, and Graal 1.21 ships the same format	[bin] (client-internals/precursors-and-fan-builds.md#objectsdat-stamps)
The Java games read `arrays.dat` in the same header form as the Delphi clients (→ §3.5.3), in a fixed order of 14 lists	[src] (client-internals/precursors-and-fan-builds.md#shared-java-game-data)
Java Graal reads `GR-V1` NPC lines but shows only the image, dropping the script, and skips the chest section	[src] (client-internals/precursors-and-fan-builds.md#level-format-as-the-java-code-reads-it)
Zelda Online, the Java client of 1997–98, keeps text-line relay traffic on port 14801 (→ §2.3.1)	[bin] (client-internals/zelda-online-1998.md#server-protocol-text-lines-port-14801)

1.7.1 Behaviour of the Java precursors

Zelda Online 2 and Java Graal share one engine, and every number below is the same in both [src] (client-internals/precursor-runs-2026-10-30.md#behaviour-constants-zelda-online-2-and-java-graal).

What	Value	Source
main loop	50 ms per tick (20 Hz)	[src] (client-internals/precursor-runs-2026-10-30.md#behaviour-constants-zelda-online-2-and-java-graal)
walk	half a tile per tick while a key is held; a quarter step sideways around a block corner	[src] (client-internals/precursor-runs-2026-10-30.md#behaviour-constants-zelda-online-2-and-java-graal)

What	Value	Source
sword	5 ticks, hits tested on ticks 1 to 3	[src] (client-internals/precursor-runs-2026-10-30.md#behaviour-constants-zelda-online-2-and-java-graal)
hurt	40 ticks, half a heart	[src] (client-internals/precursor-runs-2026-10-30.md#behaviour-constants-zelda-online-2-and-java-graal)
pushing a wall	push pose and ledge check from tick 18 (0.9 s); a push stone moves at tick 28 (1.4 s); the counter loops from 38 back to 18	[src][live] (client-internals/precursor-runs-2026-10-30.md#walking-into-a-wall)
bush drop	one roll of 0 to 99: 25 % a green rupee, 5 % a heart, 2 % five rupees, else nothing; a lifted bush rolls the same	[src] (client-internals/precursor-runs-2026-10-30.md#bushes-stones-and-pots)
regrowth	none during a visit; every board change is cleared when the level loads	[src] (client-internals/precursor-runs-2026-10-30.md#bushes-stones-and-pots)
carrying	the player is drawn with carry sprites and the object above the head, not hidden under it	[src] (client-internals/precursor-runs-2026-10-30.md#bushes-stones-and-pots)
chat	bold serif black text with a one-pixel white offset, above the head, with no bubble; the nickname is drawn the same way under the feet	[src][live] (client-internals/precursor-runs-2026-10-30.md#chat)

1.7.2 The Game Boy Color demo

Rule	Source
No title screen and no HUD: the ROM goes from the Nintendo logo to room 0; a sign there reads as a welcome from the author	[live] (client-internals/gameboy-demo.md#seen-in-an-emulator); [bin] (client-internals/precursors-and-fan-builds.md#graal-game-boy-color-demo-1999)
Seven rooms of 32 × 32 tiles in banks 4 and 5; their links use the PC level link record with a room number for the file name	[bin] (client-internals/gameboy-demo.md#room-format)
Select opens a carousel over a table of 31 character types, and the pick changes the player at once	[bin][live] (client-internals/precursor-runs-2026-10-30.md#game-boy-color-demo-in-an-emulator)
The text font runs space, A–Z, a–z, 0–9, then punctuation, from byte 32; it is not the PC sign alphabet	[bin] (client-internals/precursors-and-fan-builds.md#graal-game-boy-color-demo-1999)
Start mutes the game; the music format is open	[live] (client-internals/precursor-runs-2026-10-30.md#open)

2 Wire protocol

> Part index. Each chapter carries its own status line.

- → §2.1 Framing and encodings (`01-framing.md`)
- → §2.2 Ciphers and compression (`02-ciphers.md`)
- → §2.3 Login and handshake by generation (`03-login.md`)
- → §2.4 PLI message reference (client to server) (`04-pli.md`)
- → §2.5 PLO message reference (server to client) (`05-plo.md`)
- → §2.6 Props (`06-props.md`)
- → §2.7 File transfer (`07-file-transfer.md`)
- → §2.8 Chat and private messages (`08-chat-pm.md`)
- → §2.9 UDP peer-to-peer (`09-udp.md`)
- → §2.10 The 6.11 server-defined protocol (`10-server-defined.md`)

2.1 Framing and encodings

Notation is fixed in → §0.5 to → §0.7. Ciphers that act on messages or frames are in → §2.2; the 1998 text relay, which has no binary framing, is in → §2.3.1.

2.1.1 Integer and string encodings

Graal's own types add `0x20` to every 7-bit group, so an encoded value is printable and never contains `\n` [srv][2.05] (protocol/codec-and-framing.md#graal-integer-types).

Type	Short	Bytes	Encoding, first byte first	Range	Source
GCHAR	`c`	1	`v + 0x20`	0–223	[srv][2.05] (protocol/codec-and-framing.md#graal-integer-types)
GSHORT	`s`	2	`(v >> 7) + 0x20`, `(v & 0x7F) + 0x20`	0–28767	[srv] (protocol/codec-and-framing.md#graal-integer-types)
GINT (GINT3)	`i3`	3	three 7-bit groups, high first, each + `0x20`	0–3682399	[srv] (protocol/codec-and-framing.md#graal-integer-types)
GINT4	`i4`	4	four 7-bit groups, high first, each + `0x20`	unmeasured	[srv] (protocol/codec-and-framing.md#graal-integer-types)
GINT5	`i5`	5	first group 4 bits (`v >> 28`), then four 7-bit groups, each + `0x20`	32-bit	[srv] (protocol/codec-and-framing.md#graal-integer-types)
GSTRING	`str`	1 + n	GCHAR length, then n bytes	n ≤ 223	[srv][2.05] (protocol/codec-and-framing.md#graal-integer-types)

Rule	Builds	Source
On overflow the server clamps rather than wraps (a GSHORT high byte is at most 223)	server	[srv] (protocol/codec-and-framing.md#graal-integer-types)
A signed value is sent as its unsigned bit pattern	server	[srv] (protocol/codec-and-framing.md#graal-integer-types)
In the login message, an account or password of 224 bytes or more is sent as length byte `0xFF` followed by its first 223 bytes	4.20r6, 4.20r8, 5.00r7, 6.01r1, 6.03r7	[bin] (protocol/v4x-clients.md#login-packet); [bin] (protocol/v5x-clients.md#login); [bin] (protocol/v6x-clients.md#login-message-601-and-603)
Numbers in 1998 relay payloads use the same +32 single-character encoding as GCHAR	1st Graal to 1.1.9	[bin] (protocol/relay-1998.md#d-payloads)

2.1.2 Messages and terminators

Rule	Builds	Source
A message is `GCHAR(id)`, the payload, then `\n`	1.21 to 6.x	[bin] (protocol/gserv-1999.md#framing); [srv][2.05] (protocol/codec-and-framing.md#messages); [bin] (protocol/v6x-clients.md#cipher-and-framing-classic-mode)
Several messages are concatenated in one frame	1.21 to 6.x	[srv][2.05] (protocol/codec-and-framing.md#messages); [bin] (protocol/gserv-1999.md#framing)
The wire byte of the id is the logical id + `0x20`; an id above 223 cannot be sent as one GCHAR	all binary generations	[srv] (protocol/packet-ids.md)
The sender refuses a payload that already contains `\n`; binary payloads use the raw-data escape (→ §2.1.3)	1.39, 2.05	[2.05] (protocol/codec-and-framing.md#messages); [bin] (client-internals/decomp-findings-2026-10-24/05-205-props-misc.md); [bin] (client-internals/decomp-findings-2026-10-24/07-graal-139.md#framing-and-encryption)

Rule	Builds	Source
An outgoing message containing \n is refused with "Illegal characters in the outgoing data"	6.01r1, 6.03r7	[bin] (protocol/v6x-clients.md#cipher-and-framing-classic-mode)
The receiver splits an inflated frame on \n	1.21 to 2.05	[bin] (protocol/gserv-1999.md#framing); [2.05] (protocol/codec-and-framing.md#205-send-pipeline-update-2026-10-21-205)
An id the receive dispatcher has no case for is ignored silently; 2.05 has 59 cases	2.05	[2.05] (protocol/codec-and-framing.md#205-send-pipeline-update-2026-10-21-205)
Every 2.05 handler checks a minimum length first and drops a short message silently	2.05	[2.05] (protocol/packet-layouts.md#how-205-reads-a-message-205)

2.1.3 The raw-data escape

Rule	Builds	Source
PL0_RAWDATA (100, wire 0x84) carries an i3 length; the next *length* bytes are read as one message, newline or not	server; 1.21 to 6.x clients	[srv] (protocol/codec-and-framing.md#raw-data-escape); [bin] (protocol/gserv-1999.md#framing); [bin] (protocol/v4x-clients.md#cipher-and-framing)
The server sends 100 in front of PL0_BOARDPACKET (101) and PL0_FILE (102)	server	[srv] (protocol/codec-and-framing.md#raw-data-escape)
2.05 has a case for 100 in its main dispatch but none for 101 or 102, consistent with 101 and 102 only arriving behind 100	2.05	[2.05][inf] (protocol/codec-and-framing.md#raw-data-escape)
In the other direction, before a payload sent as id 93 the client sends id 50 with an i3 length, capped at 0x37FFFF	4.20r8, 5.00r7	[bin] (protocol/v4x-clients.md#cipher-and-framing); [bin] (protocol/v5x-clients.md#login)

The board and file layouts behind 100 are in → §2.5 and → §2.7.

2.1.4 Frames by generation

Encryption generations are defined in → §2.2.1; builds per generation are in → §1.4.

Generation	Builds	Client → server frame	Server → client frame	Source
1	1st Graal to 1.1.9	text lines ending in \n, no length (→ §2.3.1)	same	[bin] (protocol/relay-1998.md#summary)
2	1.21 to 1.39	u16be length, then a zlib stream of that length; every flush is compressed, with no size threshold; zlib 1.0.4	same	[bin] (protocol/gserv-1999.md#framing)
3	1.41 rev1 to 2.17	u16be length + zlib stream of the whole out buffer; gen-3 insertion applied per message before compression (→ §2.2.2)	u16be length + zlib, no extra byte, no cipher	[2.05] (protocol/codec-and-framing.md#frame); [bin] (protocol/client-versions.md#game-server-login); [srv][cap] (protocol/codec-and-framing.md#frame)
4	client type 4 (2.19 to 2.21, 3.0, 3.01)	u16be length + bz2 body, then partial XOR (→ §2.2.3)	same	[srv] (protocol/codec-and-framing.md#frame)
5	2.22 and later; RemoteControl type 6	login frame: u16be length + zlib, no type byte, no XOR (fixed mode 3); later frames: u16be (len + 1), a type byte, the body (→ §2.2.4)	same layout, type byte present from the first frame	[srv] (protocol/codec-and-framing.md#frame); [bin] (protocol/client-versions.md#gen-5-send-on-the-client-222-231-remotecontrol-bin); [2.22][cap] (protocol/codec-and-framing.md#gen-45-partial-xor)

Generation	Builds	Client → server frame	Server → client frame	Source
5, long frames	6.01r1, 6.03r7	a frame of `0xFFFF` bytes or more is sent as length `0xFFFF` followed by a 4-byte big-endian length	unmeasured	[bin] (protocol/v6x-clients.md#cipher-and-framing-classic-mode)
own	Graal3D	→ §9	→ §9	[bin] (client-internals/build-catalogue.md#catalogue)

2.1.4.1 Flush and buffer limits

Rule	Builds	Source
1.39 flushes when the queue would pass 4095 bytes and writes the frame to the socket in 256-byte slices; it drops the link when more than 256 KiB of compressed output is pending	1.39	[bin] (client-internals/decomp-findings-2026-10-24/07-graal-139.md#framing-and-encryption)
Messages collect in an out buffer, compressed and flushed at `0x1000` bytes or at the end of the tick	2.00 to 2.05, 2.17	[2.05] (protocol/codec-and-framing.md#205-send-pipeline-update-2026-10-21-205); [bin] (version-profiles/gen3-insertion-200-205.tsv); [bin] (protocol/client-versions.md#game-server-login)
The link is dropped when more than `0x40000` bytes are queued unsent	2.00 to 2.05, 2.17, 2.22, 2.31	[2.05] (protocol/codec-and-framing.md#205-send-pipeline-update-2026-10-21-205); [bin] (version-profiles/gen3-insertion-200-205.tsv); [bin] (protocol/client-versions.md#gen-5-send-on-the-client-222-231-remotecontrol-bin)
The same limit is `0x800000` bytes	Remote Control (gen 5)	[bin] (protocol/client-versions.md#gen-5-send-on-the-client-222-231-remotecontrol-bin)
Buffers flush at `0x1000`	6.01r1, 6.03r7	[bin] (protocol/v6x-clients.md#cipher-and-framing-classic-mode)
The client links zlib 1.1.2 (Delphi ZLib unit)	2.05	[2.05] (protocol/codec-and-framing.md#frame)
zlib is a compiled-in unit in 2.00 to 2.01 rev1 and from 2.03 on; 2.02, 2.02 rev1 and 2.02 rev2 import an external `zlib.dll` and need it beside the program; the wire format is the same	2.00 to 2.1 test	[bin] (client-internals/decomp-findings-2026-10-24-c/03-framing-and-encryption.md#where-zlib-comes-from-the-one-real-difference)
The NewWorld betas and NewMain 24 and 26 use the same transport and login shape as 2.00 to 2.05	NewWorld, NewMain	[bin] (client-internals/decomp-findings-2026-10-24-c/03-framing-and-encryption.md#encryption-generations)
The 1998 relay read buffer is 1 KB	1st Graal to 1.1.9	[bin] (protocol/relay-1998.md#summary)
A UDP peer message is not framed this way (→ §2.9)	1.28 beta 2 and later	[2.05] (protocol/codec-and-framing.md#205-send-pipeline-update-2026-10-21-205)

2.1.4.2 zlib stream header

Rule	Builds	Source
Real clients compress at the best zlib level: the stream starts `78 DA`	2.05, 2.17.1, 2.22, 2.31	[cap] (protocol/login-frames-by-version-2026-10-25.md#one-byte-level-difference-that-applies-to-every-seihai-frame)
A listserver accepts gen-5 frames that start `78 9C` (default level); whether a classic frame with that header is accepted is unmeasured	public listserver	[cap] (protocol/login-frames-by-version-2026-10-25.md#one-byte-level-difference-that-applies-to-every-seihai-frame)

2.2 Ciphers and compression

Frame layouts are in → §2.1.4. Which build uses which generation is in → §1.4.

2.2.1 Encryption generations

The numbering is GServer-v2's (→ §0.9).

Gen	Selected by	Users	Scheme	Source
1	web client	web client; the 1998 relay builds have the same "no compression" property	none, uncompressed	[srv] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering); [bin] (protocol/relay-1998.md)
2	client type 0 whose first 8 login bytes are a known token	1.21 to 1.39	zlib, no cipher	[srv] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering); [bin] (protocol/gserv-1999.md#gen-1-to-gen-2-in-one-table)
3	client type 0 whose first 8 bytes are not a known token (the key byte comes first)	1.41 rev1 to 2.17; RC, NC and NPC-server connections in the server's numbering	zlib frame; one junk byte inserted into each client-to-server message (→ §2.2.2)	[srv][2.05] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering); [bin] (protocol/client-versions.md#game-server-login)
4	client type 4	2.19 to 2.21, 3.0, 3.01	bz2, then partial XOR (→ §2.2.3)	[srv] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering)
5	client type 5; RC2 (type 6)	2.22, 2.31, 3.04, 3.153, 4.x, 5.x, 6.01, 6.03, Delphi RemoteControl	compression byte, then partial XOR (→ §2.2.3, → §2.2.4)	[srv] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering); [bin] (protocol/v3x-clients.md#game-server-login-and-cipher); [bin] (protocol/v4x-clients.md#cipher-and-framing); [cap] (protocol/v5v6-windows-unpacked.md); [bin] (protocol/v6x-clients.md#cipher-and-framing-classic-mode)
6	v6	unmeasured	not implemented in the server source the notes summarise (→ §2.2.5)	[srv] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering)

Rule	Builds	Source
The NC socket logs in as type 3 and stays zlib with no cipher	Delphi RemoteControl	[bin] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering)
Nothing in generation 2 is encrypted; the DES code in those clients is used only for local files	1.21 to 1.32	[bin] (protocol/gserv-1999.md#gen-1-to-gen-2-in-one-table)
All 2.x clients share one LCG, `it = it * 0x08088405 + key`, seed `0x04A80B38`; the multiplier is Borland's `Random` constant, but the key is added where Borland adds 1	2.05, 2.17, 2.22, 2.31	[bin] (protocol/client-versions.md)
The shared multiplier with the RTL `Random` is a toolchain coincidence, not shared state	2.05	[2.05] (protocol/codec-and-framing.md#corrections-to-earlier-notes)

Rule	Builds	Source
The same seed is also used for web-level cache file encryption from 1.3 on; that use is not the connection cipher (→ §3.8)	1.3 test and later	[bin] (client-internals/build-catalogue.md#how-each-column-was-read)

2.2.2 Generation 3 insertion

Client to server only. Server-to-client traffic for a gen-3 client is plain zlib [srv][cap] (protocol/codec-and-framing.md#gen-3-205).

State. `it` is a u32 seeded `0x04A80B38` when the connection object is created. The key `k` is `Random & 0x7F`, sent once as a GCHAR in the login message (→ §2.3.3) [2.05] (protocol/codec-and-framing.md#gen-3-205); [bin] (version-profiles/gen3-insertion-200-205.tsv).

Per message, starting with the second message of the session (the login message is sent clean; the messages that follow it in the same first frame, such as `PLI_LANGUAGE`, already carry the byte) [2.05][cap] (protocol/codec-and-framing.md#gen-3-205):

Step	Operation	Source
1	`it = it * 0x08088405 + k` (mod 2^32)	[2.05] (protocol/codec-and-framing.md#gen-3-205)
2	`len` = message length including the trailing `\n`; `pos0 = (it & 0xFFFF) mod len` (0-based)	[2.05] (protocol/codec-and-framing.md#gen-3-205); [bin] (version-profiles/gen3-insertion-200-205.tsv)
3	insert the byte `((it >> 16) & 0x7F) + 0x20` at 0-based index `pos0`, before compression	[2.05] (protocol/codec-and-framing.md#gen-3-205)

Property	Source
`pos0` can be 0, which puts the junk byte before the id byte; a decoder must not take offset 0 as the id until the byte is removed	[2.05][cap] (protocol/codec-and-framing.md#gen-3-205)
The inserted byte is never `\n` and never lands after the terminating `\n`	[2.05][cap] (protocol/codec-and-framing.md#gen-3-205)
Insertion is per message, after the receiver's decompression, not over the whole frame	[srv] (protocol/codec-and-framing.md#gen-3-205)
The server mirrors the step and deletes the byte at 0-based `(it & 0xFFFF) mod len`, where `len` is the split-off line without `\n` but with the junk byte, which equals the client's length	[srv][2.05] (protocol/codec-and-framing.md#gen-3-205)
The server's own gen-3 encoder always inserts `)` instead of an iterator-derived byte; harmless, because the receiver discards it	[srv] (protocol/codec-and-framing.md#gen-3-205)
Three lab 2.05 sessions: every message after the login had the predicted byte at the predicted index, several at index 0, no mismatches	[cap] (protocol/codec-and-framing.md#gen-3-205)

Builds	Insertion	Source
1.39	absent	[bin] (client-internals/build-catalogue.md#patterns-worth-knowing)
1.41 rev1, NewMain 1.42	present (seed and 7-bit key in the connection initialiser)	[bin] (client-internals/build-catalogue.md#patterns-worth-knowing)
2.00	the formula above, read byte for byte	[bin] (version-profiles/gen3-insertion-200-205.tsv)
2.01 to 2.04	same call structure and endpoints as 2.00 and 2.05	[bin] (version-profiles/gen3-insertion-200-205.tsv)
2.05, 2.05 rev2	the formula above	[2.05] (protocol/codec-and-framing.md#gen-3-205)
2.17	the same gen-3 send code	[bin] (protocol/client-versions.md#game-server-login)

2.2.3 Generation 4 and 5 partial XOR

Same LCG and key as generation 3, with one iterator per direction, both seeded `0x04A80B38` [srv] [2.22] (protocol/codec-and-framing.md#gen-45-partial-xor); [bin] (protocol/client-versions.md#gen-5-send-on-the-client-222-231-remotecontrol-bin).

Step	Operation	Source
1	on send, compress the body first (→ §2.2.4)	[srv][2.22] (protocol/codec-and-framing.md#gen-45-partial-xor)
2	`groups = min(ceil(len / 4), limit)`, with `limit` 12 for an uncompressed body (`0x02`) and 4 for zlib or bz2 (`0x04`, `0x06`)	[srv][2.22][cap] (protocol/codec-and-framing.md#gen-45-partial-xor)
3	for each group: advance `it` once, XOR the 4 body bytes with the little-endian bytes of `it`	[srv][2.22] (protocol/codec-and-framing.md#gen-45-partial-xor)
4	a short last group still advances `it` and is XORed byte by byte with the low bytes of `it`	[srv][2.22][cap] (protocol/codec-and-framing.md#gen-45-partial-xor) ; [cap] (protocol/lab-checks-2026-10-24.md#gen-5-xor-confirmation)
5	on receive, undo the XOR first, then decompress	[srv] (protocol/codec-and-framing.md#gen-45-partial-xor)

Evidence for the step rule	Source
27 of 27 server-to-client frames from four public 2.22 game servers and both listserver frames, including zlib and bz2 bodies with a partial last group	[cap] (protocol/codec-and-framing.md#gen-45-partial-xor)
610 lab frames, both directions	[cap] (protocol/codec-and-framing.md#gen-45-partial-xor)
a real 2.22 client sent 580 uncompressed and 17 zlib frames; all 17 compressed bodies had a length that was not a multiple of 4 and decode only under this rule	[cap] (protocol/codec-and-framing.md#gen-45-partial-xor)
ten 5.x and 6.x Windows captures decode with this rule	[cap] (protocol/v5v6-windows-unpacked.md)
an earlier rule, `floor(len / 4)` groups for compressed bodies, is wrong and leaves the iterator one step behind	[cap] (protocol/codec-and-framing.md#gen-45-partial-xor)

Builds	Step count for zlib and bz2 bodies	Source
2.22, 2.31	`ceil(len / 4)`, capped at 4	[2.22][cap] (protocol/codec-and-framing.md#gen-45-partial-xor)
3.04, 3.153	`ceil(len / 4)`, capped at 4	[bin] (protocol/v3x-clients.md#game-server-login-and-cipher)
4.04r2, 4.20r6, 4.20r8	cap 4; rounding unmeasured	[bin] (protocol/v4x-clients.md#cipher-and-framing)
5.12	disputed → App. E#2	[bin][wire] (protocol/v5v6-lab-play.md#512s-cipher-step-count); [cap] (protocol/v5v6-windows-unpacked.md)
5.24r3 and later 5.x, 6.00 to 6.03r7	`ceil(len / 4)`, capped at 4	[cap] (protocol/v5v6-windows-unpacked.md); [bin][wire] (protocol/v5v6-lab-play.md#512s-cipher-step-count)

2.2.4 Compression choice

The type byte of a generation-5 frame (→ §2.1.4):

Code	Body	XORed	Source
1, 2	raw	only 2	[bin] (protocol/client-versions.md#gen-5-send-on-the-client-222-231-remotecontrol-bin)
3, 4	zlib	only 4; 3 is the fixed "zlib, no cipher" mode of the login frame and of the whole NC session	[bin] (protocol/client-versions.md#gen-5-send-on-the-client-222-231-remotecontrol-bin)
5, 6	bz2	only 6	[bin] (protocol/client-versions.md#gen-5-send-on-the-client-222-231-remotecontrol-bin)

Which code a client picks for each flush of `n` bytes:

Builds	`n < 0x30`	`0x30 ≤ n < 0x1000`	`n ≥ 0x1000`	Source
2.22, 2.31, RemoteControl	`0x04` if zlib is actually shorter, otherwise `0x02`	`0x04`	`0x06`	[bin] (protocol/client-versions.md#gen-5-send-on-the-client-222-231-remotecontrol-bin); [2.22][cap] (protocol/codec-and-framing.md#gen-45-partial-xor)
3.04, 3.153	`0x02` or `0x04`	`0x04`	`0x06`	[bin] (protocol/v3x-clients.md#game-server-login-and-cipher)
4.x	`0x02`	`0x04`	`0x06`	[bin] (protocol/v4x-clients.md#cipher-and-framing)
5.12	`0x04` even for short bodies	`0x04`	unmeasured	[cap] (protocol/v5v6-windows-unpacked.md)
5.24r3 and later	`0x02`, like 2.22	unmeasured	unmeasured	[cap] (protocol/v5v6-windows-unpacked.md)
6.01r1, 6.03r7	`0x02`	`0x04`	`0x06`	[bin] (protocol/v6x-clients.md#cipher-and-framing-classic-mode)

Rule	Source
The u16be length counts the type byte	[2.22][cap] (protocol/codec-and-framing.md#gen-45-partial-xor)
The login frame carries no type byte and no XOR, because the key travels inside it; the socket switches to dynamic mode after it is flushed	[2.22][cap] (protocol/codec-and-framing.md#gen-45-partial-xor); [bin] (protocol/client-versions.md#game-server-login)
No client-originated bz2 frame has been captured	[cap] (protocol/codec-and-framing.md#gen-45-partial-xor)
Generations 2 and 3 compress every flush with zlib	[bin] (protocol/gserv-1999.md#framing); [2.05] (protocol/codec-and-framing.md#frame)

2.2.5 Generation 6 and script-defined transports

Rule	Builds	Source
Generation 6 is named in the server source but not implemented there	server	[srv] (protocol/codec-and-framing.md#encryption-generations-gs2lib-numbering)
The classic transport of 6.01 and 6.03 is generation 5, unchanged	6.01r1, 6.03r7	[bin] (protocol/v6x-clients.md#cipher-and-framing-classic-mode)
A server-sent connector script can replace the framing, the cipher (RC4, AES or TLS) and both id tables; generation 5 stays the default until the script switches protocol (→ §2.10)	6.11 Mac	[bin] (protocol/login-session-known.md); [bin][cap][doc] (version-profiles/matrix/README.md#thinks-wire-and-scripting)

2.3 Login and handshake by generation

Tokens are listed in → §1.2.2, frames in → §2.1.4 and ciphers in → §2.2. The listserver exchange that comes before a game login is in → §6; RC and NC logins are in → §7.

2.3.1 1998 relay (generation 1)

2.3.1.1 Connections

Rule	Builds	Source
The client connects over TCP to a lobby and relay server that knows nothing about the game; it groups connections into named games, gives each an 8-character id, and forwards `D` lines between partners	1st Graal to 1.1.9	[bin] (protocol/relay-1998.md#summary)
A client opens two connections to the same port: one in the fixed game name and one in a room named after the current level; both pair with the other player	1.1.9 (captured); 1st, 3rd, 4th alpha, 5th Graal (live)	[cap][live] (protocol/relay-1998.md#game-names)
Every message is a text line ending in `\n`; the first character is the command	1st Graal to 1.1.9	[bin] (protocol/relay-1998.md#summary)
There is no login token and no peer-to-peer mode	1st Graal to 1.1.9	[bin] (protocol/relay-1998.md#summary)
Levels are not sent over the relay; web levels are fetched over plain HTTP/1.0 by a separate small client	1st Graal to 1.1.9	[bin] (protocol/relay-1998.md#transport-by-build)
Default server: the loopback address on 14801; from 3rd Graal the connect dialog is pre-filled with a relay on 14802; 4th and 5th alpha have no pre-filled relay host	1st Graal to 1.1.9	[bin] (protocol/relay-1998.md#summary)

Builds	Socket layer	Source
1st, 2nd Graal	ICS `TWSocket`, event driven	[bin] (protocol/relay-1998.md#transport-by-build)
3rd Graal to 1.1.9	plain WinSock 1.1: blocking `connect`, then `FIONBIO`, then polled `recv`; host tried with `inet_addr`, then `gethostbyname`	[bin] (protocol/relay-1998.md#transport-by-build)
15th Graal on	as above, plus a hostname-to-address cache in front of the resolver	[bin] (protocol/relay-1998.md#transport-by-build)

2.3.1.2 Lobby lines

Direction	Line	Meaning	Source
C→S	`N<game>#<name>`	join `<game>` as a player; the clients always send the name `def` and send the nickname later inside `D`	[bin] (protocol/relay-1998.md#lobby-lines)
C→S	`P<id>`	ask to play with player `<id>`, or accept a request	[bin] (protocol/relay-1998.md#lobby-lines)
C→S	`C`	leave the current party	[bin] (protocol/relay-1998.md#lobby-lines)

Direction	Line	Meaning	Source
C→S	-	leave the available list; sent on a deliberate disconnect	[bin] (protocol/relay-1998.md#lobby-lines)
C→S	D<data>	relay <data> to the party (→ §2.3.1.4)	[bin] (protocol/relay-1998.md#lobby-lines)
S→C	I<id8>	your id: 0000 plus a 4-digit counter	[bin] (protocol/relay-1998.md#lobby-lines)
S→C	+ <id8>	player in the same game; if it is the only known peer the client answers P<id8> at once	[bin] (protocol/relay-1998.md#lobby-lines)
S→C	- <id8>	player left	[bin] (protocol/relay-1998.md#lobby-lines)
S→C	P<id8>	partnership confirmed; the client then pushes its whole state to the peer	[bin] (protocol/relay-1998.md#lobby-lines)
S→C	C<id8>	the partner left the party	[bin] (protocol/relay-1998.md#lobby-lines)
S→C	D<data>	relayed data	[bin] (protocol/relay-1998.md#lobby-lines)
S→C	NR, NL	sent right after N by the 1.1.9 server: NR = relay levels, NL = local levels (the client then starts in onlinestartlocal.graal); a server with local levels sends NR then NL	[bin][cap] (protocol/relay-1998.md#lobby-lines)
C→S	O, S, V<id>	observer, super user, observe a player; accepted by the 1.1.9 server	[bin] (protocol/relay-1998.md#lobby-lines)
C→S	O<game># <name>	in Knorr's GServ Standalone (1997): join as an observer, who sees the players of that game already in a party	[src] (client-internals/precursors-and-fan-builds.md#network-gserv-standalone-and-the-java-clients)
C→S	S<name>	GServ Standalone: super user, who sees every player of every game as name (game)	[src] (client-internals/precursors-and-fan-builds.md#network-gserv-standalone-and-the-java-clients)
C→S	+	GServ Standalone: available again after -	[src] (client-internals/precursors-and-fan-builds.md#network-gserv-standalone-and-the-java-clients)
C→S	V<id8>	GServ Standalone: watch a player and receive their party's D traffic	[src] (client-internals/precursors-and-fan-builds.md#network-gserv-standalone-and-the-java-clients)
C→S	H<score>	high score; GServ Standalone ignores it	[src] (client-internals/precursors-and-fan-builds.md#network-gserv-standalone-and-the-java-clients)
S→C	+ <id8> <name>	GServ Standalone sends the name directly after the id; D is relayed unchanged, without the sender	[src] (client-internals/precursors-and-fan-builds.md#network-gserv-standalone-and-the-java-clients)

2.3.1.3 Game names

Builds	Name sent with N	Source
1st, 2nd Graal	Zelda2	[bin] (protocol/relay-1998.md#game-names)
3rd, 4th Graal	Graal	[bin] (protocol/relay-1998.md#game-names)
4thgraalalpha	Graal2 (room prefix G2)	[bin] (protocol/relay-1998.md#game-names)
5th Graal, 5thgraalalpha	Graal3	[bin] (protocol/relay-1998.md#game-names)
6th Graal	Graal4	[bin] (protocol/relay-1998.md#game-names)
7th to 9th Graal	Graal5	[bin] (protocol/relay-1998.md#game-names)
10th to 12th Graal	Graal6	[bin] (protocol/relay-1998.md#game-names)
13th Graal	Graal7	[bin] (protocol/relay-1998.md#game-names)
14th, 15th Graal, 1.1, 1.1.9	Graal8	[bin] (protocol/relay-1998.md#game-names)

In 1st Graal, `Zelda2` is used by one branch only; the other, chosen when the network client is created, builds the name from the level file's base name as `Z` + base + `2` (`level1.zelda` gives `Zlevel12`); a live 1st Graal joined `Zhouse12` and `Zelda2` (table below) [bin] (client-internals/decomp-findings-2026-10-24/06-first-graal.md#game-room-name-new-detail); [bin] (client-internals/decomp-findings-2026-10-24/08-contradictions.md#contradicts-or-corrects).

The constant is the fallback. On a level the client sends `G<n>` plus the level's location (a web level's URL without `http://`), so each level is its own room and players meet only on the same level and a compatible build [bin][live] (protocol/relay-1998.md#game-names).

Build	Level room	Game-wide room	Source
Zelda Online 2 (Java, 1998)	`Z` + level base name + `1.6` (`Zlevel11.6`)	none	[src][live] (client-internals/precursors-and-fan-builds.md#zelda-online-2-client); [live] (client-internals/precursor-runs-2026-10-30.md#what-the-wire-showed)
1st Graal	`Z` + level base name + `2` (`Zhouse12`)	`Zelda2`	[live] (client-internals/relay-room-bridge.md#rooms-each-build-joins)
3rd Graal	`G` + level file (`Ghouse1.graal`)	`Graal`	[live] (client-internals/relay-room-bridge.md#rooms-each-build-joins)
4th Graal alpha	`G2` + level file (`G2level13.graal`)	`Graal2`	[live] (client-internals/relay-room-bridge.md#rooms-each-build-joins)
5th Graal	`G3` + level file	`Graal3`	[live] (client-internals/relay-room-bridge.md#rooms-each-build-joins)
Java Graal (July 2000)	`G2` + level file	none; a `Graal2` mode exists but is never selected	[live] (client-internals/relay-room-bridge.md#rooms-each-build-joins); [src] (client-internals/precursors-and-fan-builds.md#java-graal-client)

Rule	Source
Every Delphi beta opens two relay connections, one per level room and one game-wide; the Java clients open one	[live] (client-internals/relay-room-bridge.md#rooms-each-build-joins)
Java Graal dials its own codebase host on 14802 and speaks 4th Graal alpha's dialect: the two met and chatted on a stock GServ	[live] (client-internals/relay-room-bridge.md#result)
Java Graal and 5th Graal met through a proxy that renamed the `G2` level room to `G3`; the `D` lines needed no change, since each side ignores the types it does not know	[live] (client-internals/relay-room-bridge.md#result); [inf] (client-internals/relay-room-bridge.md#what-this-means)
A renaming bridge must keep the level part; mapping a level room onto a game-wide room would put players on different levels together	[inf] (client-internals/relay-room-bridge.md#what-this-means)
When two clients each answer the other's `+` with `P`, GServ links the pair twice and delivers every later `D` line twice; Zelda Online 2 and Java Graal both did this	[live] (client-internals/precursor-runs-2026-10-30.md#what-the-wire-showed)
`D` types: Zelda Online 2 uses `P C N S M H B T D E O W A`; Java Graal drops `S` (colours) and adds `L` (level name), so Java players show default colours	[src] (client-internals/precursors-and-fan-builds.md#java-graal-client)
The level-leader rule (earliest login time, ties to the lower id) comes from Zelda Online 2	[src] (client-internals/precursors-and-fan-builds.md#zelda-online-2-client)

2.3.1.4 `D` payloads

```
D <sid:2> <type:1> <payload>
```

Field	Encoding	Source
`sid`	sender id as two base-128 characters, `(c0 - 32) * 128 + (c1 - 32)`; the receiver writes it back as `0000` plus 4 zero-padded digits; the server does not stamp the sender	[bin] (protocol/relay-1998.md#d-payloads)

Field	Encoding	Source
coordinates	half tiles, `(c - 32) / 2`	[bin] (protocol/relay-1998.md#d-payloads)
knock-back and fine offsets	`(c - 96) / 16`	[bin] (protocol/relay-1998.md#d-payloads)

Builds	Types sent	Source
1st Graal	`A B C E F G H L M N O P R S T W` (can also receive `D`)	[bin] (protocol/relay-1998.md#d-payloads)
2nd to 4th Graal	adds `Q`	[bin] (protocol/relay-1998.md#d-payloads)
4thgraalalpha to 5thgraalalpha	adds `I U X Y Z`	[bin] (protocol/relay-1998.md#d-payloads)
6th Graal	adds sending `D`	[bin] (protocol/relay-1998.md#d-payloads)
7th Graal	adds `0` to `4`	[bin] (protocol/relay-1998.md#d-payloads)
8th Graal, 9thgraalalpha	adds 5	[bin] (protocol/relay-1998.md#d-payloads)
9th to 14th Graal	adds 6, 7	[bin] (protocol/relay-1998.md#d-payloads)
15th Graal, 1.1, 1.1.9	adds 8	[bin] (protocol/relay-1998.md#d-payloads)

Type	Payload	Meaning	Source
`A`	x, y, dir/flags, kind, one more byte	arrow fired; in a 1.1.9 capture, level NPCs firing every 0.5 s produced steady `A` lines from both clients, because each client runs the level NPCs itself	[bin][cap] (protocol/relay-1998.md#d-payloads)
`B`	x, y, w·8+h, then w·h tiles as two base-128 characters each	board modify	[bin] (protocol/relay-1998.md#d-payloads)
`C`	text	chat line	[bin] (protocol/relay-1998.md#d-payloads)
`D`	one character per object id	listed baddies or NPCs are dead	[bin] (protocol/relay-1998.md#d-payloads)
`E`	x, y, item index	item laid down	[bin] (protocol/relay-1998.md#d-payloads)
`F`	x, y, a, b	bomb laid	[bin] (protocol/relay-1998.md#d-payloads)
`G`	x, y, item index	item picked up	[bin] (protocol/relay-1998.md#d-payloads)
`H`	object id, dx, dy, damage (dx, dy in the /16 encoding)	baddy hurt	[bin] (protocol/relay-1998.md#d-payloads)

Type	Payload	Meaning	Source
`I`	none	player left the level or went to a link; cleared by the next `P`	[bin] (protocol/relay-1998.md#d-payloads)
`L`	level name	current level	[bin] (protocol/relay-1998.md#d-payloads)
`M`	repeated 6-byte records: id, x, y, sprite, dir·4+mode, state	baddy movement batch	[bin] (protocol/relay-1998.md#d-payloads)
`N`	text	nickname	[bin] (protocol/relay-1998.md#d-payloads)
`O`	carried object id (space = none)	carry object	[bin] (protocol/relay-1998.md#d-payloads)
`P`	x, y, sprite (animation frame·4 + direction, 0 up, 1 left, 2 down, 3 right)	position and animation, sent every 100 ms while moving	[bin][cap] (protocol/relay-1998.md#d-payloads)
`Q`	NPC id	`playertouchsme` for an NPC on the receiver's side	[bin] (protocol/relay-1998.md#d-payloads)
`R`	x, y	object at x, y removed	[inf] (protocol/relay-1998.md#d-payloads)
`S`	count, then that many colour indices (4 in 1st Graal, 5 from 4th alpha)	sprite colours	[bin] (protocol/relay-1998.md#d-payloads)
`T`	10 characters, base 128, little end first	login time; the peer with the earliest time is level leader and runs baddies and NPCs	[bin] (protocol/relay-1998.md#d-payloads)
`U`	x, y, kind, id	baddy spawned	[bin] (protocol/relay-1998.md#d-payloads)
`W`	sprite	throws the carried object	[bin] (protocol/relay-1998.md#d-payloads)
`X`	target id	hit or kill aimed at one player; only the named player applies it	[inf] (protocol/relay-1998.md#d-payloads)
`Y`	sword power, shield power, power	sword and shield change	[bin] (protocol/relay-1998.md#d-payloads)
`Z`	none	state flag	[inf] (protocol/relay-1998.md#d-payloads)

Type	Payload	Meaning	Source
`0`	horse image, or empty	horse on or off	[bin] (protocol/relay-1998.md#d-payloads)
1, 4	x, y	horse and bush actions	[inf] (protocol/relay-1998.md#d-payloads)
2	dx, dy	fireball	[bin] (protocol/relay-1998.md#d-payloads)
3	len, image, script	carrying an NPC; the image can be an `http://` path	[bin] (protocol/relay-1998.md#d-payloads)
5, 6, 7	none	pose flags (sitting, sleeping)	[inf] (protocol/relay-1998.md#d-payloads)
8	registration-key string	duplicate key check; a peer with the same key disconnects the client with a message	[bin] (protocol/relay-1998.md#d-payloads)

Several `D` types live on in 1.21 as numbered messages with the same meaning and similar payloads: board modify, bomb, item, horse and throw [bin] (protocol/relay-1998.md#what-changed-in-121).

2.3.2 1999 GSERV login (generation 2)

The first message after `connect` is an id-0 line [bin] (protocol/gserv-1999.md#login):

Offset	Field	Type	Value	Source
0	type	`c`	0 = player; the server treats 1 as RemoteControl	[bin] (protocol/gserv-1999.md#login)
1	token	8 bytes	`GSERVnnn` (→ §1.2.2)	[bin] (protocol/gserv-1999.md#login)
9	account	`str`	the connect dialog defaults to the built-in guest login	[bin] (protocol/gserv-1999.md#login)
—	password	`str`		[bin] (protocol/gserv-1999.md#login)
—	terminator	`\n`		[bin] (protocol/gserv-1999.md#login)

GServer	Accepts	Minimum for a player connection	Source
1.21 (May 1999)	`GSERV007`	7	[bin] (protocol/gserv-1999.md#what-the-gservers-accept)
1.28 (Nov 1999)	`GSERV007` to `GSERV013`	13	[bin] (protocol/gserv-1999.md#what-the-gservers-accept)
GServer13 (Nov 1999)	`GSERV007` to `GSERV014`	14	[bin] (protocol/gserv-1999.md#what-the-gservers-accept)

Rule	Source
The server matches the token against a fixed list, stores its number (7 to 14) on the connection, then checks the minimum for the connection type; old tokens stay on the list so that RemoteControl (`GSERV011`) can log in	[bin] (protocol/gserv-1999.md#what-the-gservers-accept)

Rule	Source
Refusals and login errors (version too old, account disabled, account in use, no such account) are sent as id 16 (DISCMESSAGE)	[bin] (protocol/gserv-1999.md#what-the-gservers-accept)
A token newer than the server's list is refused like an old one: GServer13 answers `GSERV016` and `GSERV018` with id 16	[cap] (protocol/gserv-1999.md#what-the-gservers-accept)
The 1.33 to 1.39 clients use the same generation with `GNW` tokens and default to the built-in guest login	[bin] (protocol/gserv-1999.md); [bin] [cap][doc] (version-profiles/matrix/README.md#thinks-wire-and-scripting)
1.39 sends id 0 with the token `GNW07080`, then `str` account and `str` password; it dials port 14900 with no listserver step; the layout of 1.33 to 1.38 is unmeasured	[bin] (client-internals/decomp-findings-2026-10-24/07-graal-139.md#login)

Message ids and props handled per build in this generation are in → App. C.

2.3.3 2.x login (generation 3, type 0)

Offset	Field	Type	Value	Source
0	client type	c	0 (wire `0x20`)	[2.05][srv] (protocol/login-and-walk.md#login-line-first-message-sent-clean)
1	key	c	`rand & 0x7F`, the gen-3 key (→ §2.2.2)	[2.05] (protocol/login-and-walk.md#login-line-first-message-sent-clean); [bin] (version-profiles/gen3-insertion-200-205.tsv)
2	token	8 bytes	e.g. `GNW03081` (2.05), `GNW22122` (2.17)	[2.05] (protocol/login-and-walk.md#login-line-first-message-sent-clean); [bin] (protocol/client-versions.md#game-server-login)
10	account	`str`		[2.05] (protocol/login-and-walk.md#login-line-first-message-sent-clean)
—	password	`str`		[2.05] (protocol/login-and-walk.md#login-line-first-message-sent-clean)
—	terminator	`\n`	no identity string follows the password	[2.05][bin] (protocol/login-and-walk.md#how-the-server-parses-it-srv)

Rule	Builds	Source
The login message is sent without the gen-3 insertion; the messages after it carry it	2.00 to 2.17	[2.05][cap] (protocol/codec-and-framing.md#gen-3-205); [bin] (version-profiles/gen3-insertion-200-205.tsv)
The token is built one character at a time in code, so no string literal exists	1.38 to 2.31	[2.05] (protocol/login-and-walk.md#login-line-first-message-sent-clean); [bin] (protocol/client-versions.md#game-server-login)
`PLI_LANGUAGE` (37) follows the login, with a language name (`English` observed)	2.02 rev2 and later	[2.05][cap] (protocol/login-and-walk.md#login-line-first-message-sent-clean); [bin][cap][doc] (version-profiles/matrix/README.md#thinks-wire-and-scripting)
`PLI_TAMPERCHECK` (39) follows 37 (→ §2.4, → §4.10)	2.17 and later Delphi clients	[bin] (protocol/client-versions.md#game-server-login)
The account name is kept obfuscated in memory and decoded only to build the login message; the map is `dec(b) = (~b - 10) & 0xFF`	2.17, 2.22, 2.31	[bin] (protocol/client-versions.md#game-server-login)
The password is obfuscated the same way	2.31	[bin] (protocol/client-versions.md#game-server-login)
2.00 sends the same type 0 shape with the token `GNW24011`; the account and password are neither hidden nor hashed	2.00	[bin] (client-internals/decomp-findings-2026-10-24-b/04-v2.00-client.md#login-string-gserver)
A lab GServer-v2 admits 2.00 once `GNW24011` is in its allowed-versions list	2.00	[lab] (client-internals/version-tokens.md) ; [lab] (client-internals/decomp-findings-2026-10-24-b/04-v2.00-client.md#token-allow-list)

2.3.4 2.22, 2.31 and 3.x login (generation 5, type 5)

The fields are those of → §2.3.3 with client type 5. The message travels in the fixed mode-3 login frame (→ §2.2.4) [bin] (protocol/client-versions.md#game-server-login); [bin] (protocol/v3x-clients.md#game-server-login-and-cipher).

Builds	Type	Token	After the login	Credentials in memory	Source
2.22	5	`GNW03014`	37, then 39	obfuscated (→ §2.3.3)	[bin] (protocol/client-versions.md#game-server-login)
2.31	5	`GNW28015`	37, then 39	obfuscated	[bin] (protocol/client-versions.md#game-server-login)
3.04 Linux	5	`G3D03014`	37 only; no 39 (confirmed on the wire)	plain	[bin] (protocol/v3x-clients.md#game-server-login-and-cipher)
3.153 Mac	5	`G3D28015`	37 only; no 39	plain	[bin] (protocol/v3x-clients.md#game-server-login-and-cipher)

2.3.5 4.x to 6.03 login (generation 5, type 5, machine ids)

Order	Field	Type	Source
1	client type 5	c	[bin] (protocol/v4x-clients.md#login-packet); [bin] (protocol/v5x-clients.md#login)
2	key	`c, rand & 0x7F`; wire bytes above `0x7F` occur because the key goes up to `0x7F`	[bin] (protocol/v5x-clients.md#login); [cap] (protocol/v5v6-windows-unpacked.md#login-message)
3	token	8 bytes	[bin] (protocol/v4x-clients.md#login-packet)
4	account	`str`; 224 bytes or more → `0xFF` + first 223 bytes	[bin] (protocol/v4x-clients.md#login-packet)
5	password	`str`; same long-string rule	[bin] (protocol/v4x-clients.md#login-packet)
6	machine list	comma-separated text, below	[bin] (protocol/v4x-clients.md#login-packet)

Builds	Field 1	Field 2	Field 3	Field 4	Field 5	Source
Windows 4.20r8, 5.00r7	`win`	MD5 hex of the Windows `DigitalProductId`	MD5 hex of the network adapter MAC (fallback: node part of `UuidCreateSequential`, or `UuidCreate`)	MD5 hex of the `C:\` volume serial	none	[bin] (protocol/v4x-clients.md#login-packet); [bin] (protocol/v5x-clients.md#login)
Mac 4.20r6	`mac`	hash of the primary Ethernet MAC	hash of 4 bytes of `stat("/etc")`	empty	none	[bin] (protocol/v4x-clients.md#login-packet)
Windows 5.12, 5.24r3 to 6.03r7	`win`	MD5 hex of the product id (5.24 and later also read `DigitalProductId4`, so the hash differs from 5.12)	MD5 hex of the adapter MAC	MD5 hex of the `C:\` serial	quoted Windows version "`<major>.<minor> <build> <service pack>`"	[cap][bin] (protocol/v5v6-windows-unpacked.md#login-message)
Linux 6.01r1, 6.03r7	`linux`	MD5 hex of the `eth0` MAC	MD5 hex of the inode number of `/etc`	empty	kernel release and version string	[bin] (protocol/v6x-clients.md#login-message-601-and-603)

Rule	Builds	Source
The machine fields let a server ban per machine; 2.x and 3.x send no machine id	4.x and later	[bin][inf] (protocol/v5x-clients.md#login); [bin] (protocol/v4x-clients.md#login-packet)
`PLI_LANGUAGE` (37) follows, then the socket switches to dynamic mode	4.x to 6.03	[bin] (protocol/v5x-clients.md#login); [bin] (protocol/v6x-clients.md#login-message-601-and-603)
After 37 the client requests `basepackage.gupd` (23, twice), then `arial.ttf`, `bigmap.txt` or `minimap.txt` by build, then 34 for a GUI image (→ §2.7)	5.12 to 6.03r7 Windows	[cap] (protocol/v5v6-windows-unpacked.md#login-message)
Each build logs in by itself as the built-in guest once it reaches the login screen; the guest password is fixed per build family and is decrypted at run time on Linux 6.x	5.12 to 6.03r7	[cap][inf] (protocol/v5v6-windows-unpacked.md#login-message); [bin][inf] (protocol/v6x-clients.md#login-message-601-and-603)
The first connection goes to a game server named `Login`, not to a separate listserver; the client reaches a game only after S→C 178 (`PLO_SERVERWARP`) and logs in again to that host (→ §6)	4.x to 6.03	[bin] (protocol/v4x-clients.md#login-server-and-licensegraal); [bin][cap] (protocol/v5x-clients.md#login); [bin] (protocol/v6x-clients.md#login-server)
6.11 asks for a connector script before the game socket; the script builds the login itself, so its token and layout are unmeasured (→ §2.10)	6.11 Mac	[bin] (protocol/login-session-known.md)

2.3.6 Server parse and reply sequence

2.3.6.1 How a GServer-v2 parses the login

Step	Rule	Source
1	read the type GCHAR; `type = 1 << value`	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)
2	for type 0, read 8 bytes as a token; if they are not a known token (the gen-3 key comes first), rewind one byte, switch to generation 3, read the key, then the real token	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)
3	read the account and password `str`; an optional identity string read exists for later clients	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)
4	map the token to a version id (→ §1.3); `GNW03081` sorts below 2.1, so 2.05 gets every pre-2.1 code path (no file modification times, `PLPROP_GANI` sent as the bow slot, the 1.41 level sequence)	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)

Type value	Connection	Source
0	client (type 0: generation 2 or 3)	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)
1	RC	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)
2	NPC-server	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)
3	NC	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)
4	client 2.19 to 2.21	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)
5	client 2.22 and later	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)
6	RC2	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)
8	web	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)

Other login tokens the server knows: NC `NCL11012` and `NCL21075`, RC `GSERV023` to `GSERV025`, NPC-server `GRNS0000` and `LNX00001` [srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv).

2.3.6.2 Reply sequences

Server	Messages after a good player login, in order	Source
GServer13	9 (own props), 28 (flags, only if the account has any), 14 (warp), 100 then a separate frame with 101 and the board, 6 (level name), 1 (links), 5 (signs), 39 (level modification time), 3 (NPC props with scripts), 10	[cap] (protocol/gserv-1999.md#what-the-gservers-accept)
GServer 1.28	as GServer13 without 39	[cap] (protocol/gserv-1999.md#what-the-gservers-accept)
GServer 1.21	9, 14, 100, 101 with the board, 6, 1, 10, all in one zlib frame; no 39	[cap] (protocol/gserv-1999.md#what-the-gservers-accept)
any of the above, another player already on the level	8 (other player's props) right after 9	[cap] (protocol/gserv-1999.md#what-the-gservers-accept)
GServer-v2 as a `Login` stand-in, to 5.x and 6.x clients	190, the level load (14, 6, 100, 101, chests, 156, 3), staff and status lists (47, 180), start and HTML messages (179, 41), 182 (the client answers with its window titles on 44); 178 is not in this burst	[wire] (protocol/login-session-known.md#the-lab-burst-these-clients-accepted)

The bytes a public 5.x or 6.x `Login` server sends are unobserved (→ App. E#3).

2.4 PLI message reference (client to server)

Field notation is in → §0.6, the message envelope in → §2.1.2, and the login message (the first message of a connection) in → §2.3. Server-to-client messages are in → §2.5, prop streams in → §2.6, and the staff tools' own ids in → §7. Every id, with its wire byte, is indexed in → App. A.

> Status: draft. Layouts are given where a note measured or read them; ids with no layout in the notes say `unmeasured`.

2.4.1 Conventions

Rule	Builds	Source
A PLI is `GCHAR(id)`, the payload, then `\n`, sent inside the frame of the build's generation	1.21 to 6.x	[srv][2.05] (protocol/packet-layouts.md); [bin] (protocol/v6x-clients.md#client--server-ids-601-603)
Ids are logical; the wire byte is the id + `0x20`	all binary generations	[srv] (protocol/packet-ids.md)
Coordinates marked x×2 are half-tiles; the receiver divides by 2	2.05 and the server	[srv][2.05] (protocol/packet-layouts.md)
Id 0 is overloaded: the login line starts with GCHAR 0 because that byte is the client type, and later `PLI_LEVELWARP` messages also use id 0; the server tells them apart by connection state	2.x	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
In 6.x the login message is sent under id 5 (`PLI_BOMBDEL`) inside the mode-3 frame	6.01r1, 6.03r7	[bin] (protocol/v6x-clients.md#client--server-ids-601-603)
6.x sends every PLI through one send method taking `(id, body)`; it writes `id + 32`, the body and `\n`	6.01r1, 6.03r7	[bin] (protocol/v6x-clients.md#client--server-ids-601-603)
Before ids 93 and 163, 6.x first sends id 50 (`PLI_RAWDATA`) with a 3-byte length (three 7-bit groups, capped at `0x37FFFF`)	6.01r1, 6.03r7	[bin] (protocol/v6x-clients.md#client--server-ids-601-603)
Before a long id 93 payload, 4.20r8 and 5.00r7 send id 50 with an `i3` length	4.20r8, 5.00r7	[bin] (protocol/v4x-clients.md#cipher-and-framing); [bin][inf] (protocol/v5x-clients.md#client--server-ids-500r7)
A server re-emits most level-object PLIs to the rest of the level with `s player id` added in front; the PLO forms are in §2.5	server	[srv] (protocol/packet-layouts.md#combat-and-objects-both-directions)

Whether a level-object PLI also goes, or only goes, over the UDP peer channel is in → §2.9.

2.4.2 Ids sent by build

The *First sent* column of the tables below names the earliest build in this table that sends the id; a later gap is noted only where the later list is complete. A list marked *floor* holds only the ids the binary passes to its send routine as constants; sends that pass the id in a register are not counted [bin] (client-internals/version-profiles/v121-v132.toml).

Build	Ids sent	Complete?	Source
1.21	0, 10–13, 15–26	floor; disputed → App. E#6	[bin] (client-internals/version-profiles/v121-v132.toml)
1.28 beta 2	0, 15–21, 23, 25, 28, 29	floor (fewer constants than 1.21 for the reason above)	[bin] (client-internals/version-profiles/v121-v132.toml)

Build	Ids sent	Complete?	Source
1.3 test	0, 14–21, 23, 25, 28–30	floor	[bin] (client-internals/version-profiles/v121-v132.toml)
1.31, 1.32	0, 10–21, 23–30	floor; 1.32 disputed → App. E#6	[bin] (client-internals/version-profiles/v121-v132.toml)
1.33 to 1.37	0–31	yes for constant ids; a few register sends uncounted	[bin] (client-internals/version-profiles/v133-v139.toml)
1.38	0–24, 26–33	as above	[bin] (client-internals/version-profiles/v133-v139.toml)
1.39	0–24, 26–33, 80, 81	as above	[bin] (client-internals/version-profiles/v133-v139.toml)
1.41 rev1	0–24, 26–30, 32, 33, 80, 81	unmeasured	[bin] (client-internals/version-profiles/141-217.toml)
1.42 (NewMain), NewMain 24 and 26, 2.00 (rev0–rev3)	0–24, 26–30, 32–34, 80, 81	unmeasured	[bin] (client-internals/version-profiles/141-217.toml); [bin] (version-profiles/pli-sent-ids-by-build-20x.tsv) ; [bin] (client-internals/decomp-findings-2026-10-24-c/pli_sent_ids_by_build.tsv)
2.01 (+rev1), 2.02, 2.02 rev1	0–24, 26–30, 32–36, 80, 81	unmeasured	[bin] (version-profiles/pli-sent-ids-by-build-20x.tsv) ; [bin] (client-internals/decomp-findings-2026-10-24-c/pli_sent_ids_by_build.tsv)
2.02 rev2	0–24, 26–30, 32–37, 80, 81	unmeasured	[bin] (version-profiles/pli-sent-ids-by-build-20x.tsv) ; [bin] (client-internals/decomp-findings-2026-10-24-c/pli_sent_ids_by_build.tsv)
2.03, 2.04, 2.05 (rev0–rev2), 2.1 test 2 and test 3	0–24, 26–30, 32–38, 80, 81	unmeasured	[bin] (version-profiles/pli-sent-ids-by-build-20x.tsv) ; [bin] (client-internals/decomp-findings-2026-10-24-c/pli_sent_ids_by_build.tsv)
NewWorld beta 1, beta 2	the 2.1 test set without 7, 8, 11, 15, 16, 17, 20; disputed → App. E#11	unmeasured	[bin] (client-internals/version-profiles/141-217.toml); [bin] (client-internals/decomp-findings-2026-10-24-c/08-test-builds-newworld-newmain.md#newworld-beta1--beta2)
2.14 to 2.16	the 2.1 set plus 41	inferred; no binaries of 2.1 to 2.16 in the archive	[inf] (client-internals/version-profiles/141-217.toml)
2.17 (the 2.17.1 binary)	0–24, 26–30, 32–41, 43–45, 80, 81	unmeasured	[bin] (client-internals/version-profiles/141-217.toml)
2.22	0–24, 26–30, 32–35, 37–41, 43, 44, 80	floor; 36, 42 and 45 not matched to a call site	[bin][cap] (version-profiles/2.22-2.31.md#wire-protocol); [bin] (version-profiles/2.22-2.31.toml)
2.31	the 2.22 set without 21, plus 46	floor	[bin][cap] (version-profiles/2.22-2.31.md#wire-protocol); [bin] (version-profiles/2.22-2.31.toml)
3.04 (Linux)	0–24, 26–30, 32–38, 40, 41, 43, 80, 81	yes, every game-code send site; listserver sends excluded	[bin] (version-profiles/3.04-3.153.md#client--server-ids); [bin] (version-profiles/3.04-3.153.toml)
3.153 (Mac OS X)	the 3.04 set plus 46, 50, 93, 96–98, 152, 154–158	no: 2, 3, 15, 24, 29 and 80 assumed from 3.04	[bin][inf] (version-profiles/3.04-3.153.md#client--server-ids)

Build	Ids sent	Complete?	Source
4.04r2	1, 2, 4–10, 12–19, 21–24, 26, 28, 29, 32, 33, 35–37, 41, 44, 50, 79–81, 92, 96, 130, 152, 154, 157, 158, 161	floor	[inf][cap] (version-profiles/4.04-4.20.toml)
4.20r6 (Mac)	the 4.04r2 set plus 34, 46, 162	floor	[inf][cap] (version-profiles/4.04-4.20.toml)
4.20r8	1, 2, 4–10, 12–19, 21–24, 26, 28, 29, 32, 33, 35–37, 41, 44, 50, 79–81, 92, 96, 130, 152, 154, 157, 158, 161, 162	floor; some sends pass the id in a register	[bin] (protocol/v4x-clients.md#client--server-ids); [bin] (version-profiles/4.04-4.20.toml)
5.00r7	1, 2, 4–10, 12–19, 21–24, 26, 28, 29, 32–37, 41, 44, 47, 50, 79–81, 92, 93, 96–98, 130, 152, 154–159, 161	floor; whether 11 is in the list is disputed → App. E#4	[bin] (protocol/v5x-clients.md#client--server-ids-500r7); [bin][cap] (version-profiles/5.00-6.03.toml)
5.12	1, 2, 11, 23, 24, 34, 37, 38, 44, 47, 157 (seen in capture)	floor	[cap] (version-profiles/5.00-6.03.toml)
5.24r4	the 5.12 set plus 158 (seen in capture)	floor	[cap] (version-profiles/5.00-6.03.toml)
5.30r3	1, 11, 37, 38 (seen in capture)	floor	[cap] (version-profiles/5.00-6.03.toml)
5.31r4	2, 37 (seen in capture)	floor	[cap] (version-profiles/5.00-6.03.toml)
6.00r2, 6.015	0–20, 22–24, 26–30, 32–38, 41, 44–48, 50, 79, 92, 93, 96–98, 130, 152, 154–159, 161, 162	inferred from the 6.01 and 6.03 reads	[inf] (version-profiles/5.00-6.03.toml)
6.01r1	the 6.03r7 set without 160	unmeasured	[bin] (protocol/v6x-clients.md#client--server-ids-601-603)
6.034	as 6.03r7	inferred	[inf] (version-profiles/5.00-6.03.toml)
6.03r7	0–20, 22–24, 26–30, 32–38, 41, 44–48, 50, 79, 92, 93, 96–98, 130, 152, 154–162	yes	[bin] (protocol/v6x-clients.md#client--server-ids-601-603); [bin] (version-profiles/5.00-6.03.toml)

Rule	Builds	Source
No sender was found for 25 (`PLI_UNKNOWN25`) or 31 (`PLI_PACKETCOUNT`)	2.05	[2.05] (protocol/packet-ids.md#client--server-pli_)
6.x sends none of 21 (`PUTNPC`), 25, 31, 39 (`TAMPERCHECK`) or 40 (`SHOOT`) as a constant	6.01r1, 6.03r7	[bin] (protocol/v6x-clients.md#client--server-ids-601-603)
3.04 and 3.153 send no 39 or 44, unlike their Windows twin, and add 36 and 81	3.04, 3.153	[bin] (version-profiles/3.04-3.153.md#client--server-ids)
6 (`PLI_TOALL`) is a real sender in every build of the family, with the chat text cut to 200 characters before sending	2.00 to 2.1 test	[bin] (client-internals/decomp-findings-2026-10-24-c/05-pli-client-to-server.md); [bin] (client-internals/decomp-findings-2026-10-24-c/10-open-questions-and-corrections.md#corrections-to-batch-b--earlier-notes)
No id is removed anywhere from 2.00 to 2.1 test 3, and none of 25, 31 or any id from 39 up has a constant send, so the tamper check (39) is not part of this family	2.00 to 2.1 test	[bin] (client-internals/decomp-findings-2026-10-24-c/05-pli-client-to-server.md); [bin] (client-internals/decomp-findings-2026-10-24-c/id_changes_between_builds.tsv)
36 and 38 look like TCP copies of UDP 25 and 26: one routine sends the UDP message, then repeats it over TCP when a flag says the server needs it; what sets the flag is unknown	2.01 to 2.1 test	[inf] (client-internals/decomp-findings-2026-10-24-c/05-pli-client-to-server.md)

Rule	Builds	Source
2.02 rev1 has no PLI_LANGUAGE (37) send site and sends none in the lab; rev2 is the first build that sends 37	2.02 rev1, rev2	[bin][cap] (client-internals/version-profiles/141-217.toml)
Ids 4.x adds over 3.x: 44 (window list for PLO 182), 50 (large-packet length), 157 and 161 (file requests carrying an i5 time), 158 (weapon by name), 162	4.20 r8	[bin] (protocol/v4x-clients.md#client--server-ids)

The summary lists for 1.21 and 1.32 in the 1999 protocol note disagree with the per-build profile by one id each → App. E#6.

2.4.3 Level navigation

Id	Name	Layout	First sent	Server action	Source
0	PLI_LEVELWARP	c x×2, c y×2, rest = level name	1.21	warps the player and sends the level (→ §2.5)	[srv][2.05] (protocol/packet-layouts.md#level-navigation-client--server)
30	PLI_LEVELWARPMOD	i5 modtime, c x×2, c y×2, rest = level name	1.3 test	as 0, but the server can answer with only the changes since modtime	[srv][2.05] (protocol/packet-layouts.md#level-navigation-client--server)
35	PLI_ADJACENTLEVEL	i5 modtime, rest = level name; modtime is all zeros when the level is not cached	2.01	sends a gmap neighbour for preloading	[srv][2.05] (protocol/packet-layouts.md#level-navigation-client--server)
41	PLI_SERVERWARP	rest = server name	2.17 (2.14 inferred)	the game server asks the listserver and answers with PLO_SERVERWARP (178)	[cap] (protocol/packet-ids.md#client--server-pli_); [cap] (version-profiles/lab-serverwarp178.tsv)
46	PLI_ENTERLEVEL	c 1	2.31; not 3.04	none recorded	[cap] (protocol/packet-ids.md#client--server-pli_); [bin] (protocol/v6x-clients.md#client--server-ids-601-603)

Rule	Builds	Source
2.05 sends 30 instead of 0 when it has the level cached, in memory or in its web-level cache	2.05	[2.05] (protocol/packet-layouts.md#level-navigation-client--server)
A warp to the level the player is already in only moves the player and sends nothing, in most states	2.05	[2.05] (protocol/packet-layouts.md#level-navigation-client--server)
A real warp also sends UDP id 24 to gmap neighbours (→ §2.9)	2.05	[2.05] (protocol/packet-layouts.md#level-navigation-client--server)
2.17 has one send site for 30 instead of two	2.17	[bin] (client-internals/version-profiles/141-217.toml)
46 is sent on every level entry (login, gmap edge crossings, doors, edge links, server warps), just before the first props message of the new level, in the same frame	2.31	[cap] (protocol/packet-ids.md#client--server-pli_); [cap] (version-profiles/lab-enterlevel-231.tsv); [cap] (client-internals/decomp-findings-2026-10-24-b/07-lab-findings-46-186-200.md#pli_enterlevel-46-real-231-client-observed); [cap] (client-internals/decomp-findings-2026-10-24-b/08-lab-findings-levelchange-24-178.md#a-pli_enterlevel-46-on-doors-edge-links-and-server-warps); [cap] (version-profiles/lab-levelchange-46-222-231.tsv)

Rule	Builds	Source
46 comes from one send site at the start of the routine that switches the current level, whenever connected; its `GCHAR 1` is a constant in the code; 2.17.1 and 2.22 have no send site	2.31; 2.17.1, 2.22	[bin] (client-internals/decomp-findings-2026-10-25-g/02-id46-level-switch.md)
At login, 46 is the first message of the client's second frame, before 35 and 44	2.31	[cap] (version-profiles/lab-enterlevel-231.tsv)
On a gmap cell crossing the props after 46 carry the new cell in props 43 or 44 and a local coordinate that starts again at the cell edge	2.31	[cap] (version-profiles/lab-enterlevel-231.tsv)
2.22 sends no 46; 2.31 does not re-send id 39 on level entry	2.22, 2.31	[cap] (protocol/packet-ids.md#client--server-pli_)

The client's handling of the `PLO_SERVERWARP` reply is in → §2.5; the listserver side is in → §6.

2.4.4 Props

Id	Name	Layout	First sent	Server action	Source
2	`PLI_PLAYERPROPS`	`rest` = PLPROP stream (→ §2.6)	1.33	stores the props and forwards them to the level as `PLO_OTHERPLPROPS`	[srv][2.05] (protocol/packet-layouts.md#other-client--server-senders-205); [srv] (protocol/chat-pm-rc.md#chat-bubble)
3	`PLI_NPCPROPS`	`rest` = NPCPROP stream (→ §2.6)	1.33	passed through	[srv][2.05] (protocol/packet-layouts.md#other-client--server-senders-205)
15	`PLI_BADDYPROPS`	`rest` = `c id` + BDPROP stream (→ §2.6)	1.21; not NewWorld betas	re-emitted as `PLO_BADDYPROPS` (2)	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)

Rule	Builds	Source
There is no movement message: a walk step is a `PLI_PLAYERPROPS` carrying X (15), Y (16) and SPRITE (17), so a walk frame starts with `"` (`0x22`)	2.05, 2.17	[srv][2.05] (protocol/login-and-walk.md#walk-frames-are-pli_playerprops)
Example bodies: `1-` is SPRITE 13; `/[` is X 59 (29.5 tiles); `0f` is Y 70 (35.0 tiles); `*$walk/I` is GANI `walk` then X 41	capture of 2026-10-04; the `walk` frame from 2.17	[cap][inf] (protocol/login-and-walk.md#walk-frames-are-pli_playerprops)
`sprite % 4` is the direction: 0 up, 1 left, 2 down, 3 right	2.22, server	[srv][doc] (protocol/login-and-walk.md#walk-frames-are-pli_playerprops)
2.05 sends prop groups {0, 11, 13, 18, 31, 33} at session start, {42, 15, 16} on attach or a position change, and a per-tick diff of changed props	2.05	[2.05] (protocol/login-and-walk.md#walk-frames-are-pli_playerprops)
Local chat is not a message id: it is the CURCHAT prop (12) inside `PLI_PLAYERPROPS`, and server commands typed in chat are parsed from it	server	[srv] (protocol/chat-pm-rc.md#chat-bubble)
The login props go out as one `PLI_PLAYERPROPS` right after the first server message, with the head (11) as an index below 100	1.41 rev1	[cap] (client-internals/version-profiles/141-217.toml)
Player props stay on TCP while the UDP channel is active	2.05	[cap] (protocol/packet-layouts.md#combat-and-objects-both-directions)

2.4.5 Combat and level objects

Layouts read from the 2.05 senders and the server source. The server copy (PLO) and its receiver rules are in → §2.5.

Id	Name	Layout	First sent	Server action	Source
1	PLI_BOARDMODIFY	`c x, c y, c w, c h` (whole tiles), then w × h `s tile`, row-major; 2.05 sends exactly one run	1.33	rebroadcast as PLO_BOARDMODIFY (7)	[srv][2.05] (protocol/packet-layouts.md#board-modify-correction)
4	PLI_BOMBADD	`c x×2, c y×2, c power` (+8 when the bomb is the sender's), `c fuse`	1.33	re-emitted as PLO_BOMBADD (11)	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
5	PLI_BOMBDEL	`c x×2, c y×2`	1.33	re-emitted as PLO_BOMBDEL (12) with no player id	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
7	PLI_HORSEADD	`c x×2, c y×2, c (dir + 4 × bushes)`, `rest` = image file name	1.33; not NewWorld betas	re-emitted as PLO_HORSEADD (17)	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205)
8	PLI_HORSEDEL	`c x×2, c y×2`	1.33; not NewWorld betas	re-emitted as PLO_HORSEDEL (18)	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205)
9	PLI_ARROWADD	`c x×2, c y×2, c flags, c sprite`, then the projectile type sent as `v + 33` (0 arrow, 1 fireball, 2 fireblast, 3 nuke); flags = `dir & 3`, +4 first removes a matching shot, +8 owned	1.33	re-emitted as PLO_ARROWADD (19)	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
10	PLI_FIRESPY	`c (a × 8 + b)`	1.21	re-emitted as PLO_FIRESPY (20)	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
11	PLI_THROWCARRIED	empty	1.21; not NewWorld betas	re-emitted as PLO_THROWCARRIED (21)	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
12	PLI_ITEMADD	`c x×2, c y×2, c item`	1.21	re-emitted as PLO_ITEMADD (22)	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
13	PLI_ITEMDEL	`c x×2, c y×2`	1.21	re-emitted as PLO_ITEMDEL (23) to the other players only	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions); [cap] (protocol/packet-layouts.md#board-change-on-the-wire-cap)
16	PLI_BADDYHURT	`c id, c (dx + 4) × 16, c (dy + 4) × 16, c power`	1.21; not NewWorld betas	re-emitted as PLO_BADDYHURT (27); applied only by the level leader	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
17	PLI_BADDYADD	`c x×2, c y×2, c type, c power`, `rest` = image	1.21; not NewWorld betas	answers with PLO_BADDYPROPS (2)	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
20	PLI_OPENCHEST	`c x, c y` (whole tiles)	1.21; not NewWorld betas	replies PLO_LEVELCHEST (4) with opened = 1	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)

Id	Name	Layout	First sent	Server action	Source
26	PLI_HURTPLAYER	`s victim, c (dx + 4) × 16, c (dy + 4) × 16, c power`, optional `i3 npc`	1.21	sends PLO_HURTPLAYER (40) with the attacker's id	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
27	PLI_EXPLOSION	`c radius, c x×2, c y×2, c power` (+8 when the explosion is the sender's)	1.31	re-emitted as PLO_EXPLOSION (36)	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
32	PLI_ITEMTAKE	`c x×2, c y×2, c item`	1.38	unmeasured	[2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
36	PLI_HITOBJECTS	`c power × 2, c x×2, c y×2`, optional `i3 npc`	2.01; not matched in 2.22 or 2.31	re-emitted as PLO_HITOBJECTS (46)	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
38	PLI_TRIGGERACTION	`i3 npc` (0 = none), `c x×2, c y×2`, `rest` = `action` or `action,params`	2.03	re-emitted as PLO_TRIGGERACTION (48)	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions); [srv] (protocol/packet-ids.md#client--server-pli_)
40	PLI_SHOOT	unmeasured	2.17; not 6.03r7	unmeasured	[bin] (client-internals/version-profiles/141-217.toml); [bin] (protocol/v6x-clients.md#client--server-ids-601-603)
48	PLI_SHOOT2	unmeasured	6.01r1	unmeasured	[bin] (protocol/v6x-clients.md#client--server-ids-601-603)

Rule	Builds	Source
GServer-v2 reads the bomb's third byte as `(player << 2) | power`; 2.05 writes `power + 8`, and a receiver takes the power as `v & 7` and bit 3 as "mine"	2.05 vs server	[2.05] (protocol/packet-layouts.md#board-change-on-the-wire-cap)
Captured bomb: x 78, y 28, power byte 9 (power 1 plus the owner flag), fuse 55	2.05	[cap] (protocol/packet-layouts.md#board-change-on-the-wire-cap)
The fuse is probably in 0.05 s steps (55 ≈ 2.75 s)	2.05	[inf] (protocol/packet-layouts.md#board-change-on-the-wire-cap)
When a bomb destroyed bushes, the other player (not the bomb's owner) sent one PLI_BOARDMODIFY per destroyed 2 × 2 patch, eight in one frame; why that player reported is not settled	2.05	[cap][inf] (protocol/packet-layouts.md#board-change-on-the-wire-cap)
Captured board runs: a cut bush 2 × 2 = tiles 1834, 1835, 1850, 1851; cut grass = 677, 678, 693, 694	2.05	[cap] (protocol/packet-layouts.md#board-change-on-the-wire-cap)
Item drops after a bush cut came from the server (`bushitems` on); no client sent PLI_ITEMADD	2.05 with GServer-v2	[cap] (protocol/packet-layouts.md#board-change-on-the-wire-cap)
The bush and sign pickup path also sends PLI_BOARDMODIFY	2.05	[2.05] (protocol/packet-layouts.md#board-modify-correction)
2.05 sends only weapon type 0 in id 33 (→ §2.4.6)	2.05	[2.05] (protocol/packet-layouts.md#weapons)

The server's respawn of board changes is behaviour and is in → §4.5.

2.4.6 Flags, NPCs and weapons

Id	Name	Layout	First sent	Server action	Source
18	PLI_FLAGSET	rest = flag (name or name=value)	1.21	stores the flag	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205)
19	PLI_FLAGDEL	rest = flag name	1.21	removes the flag	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205)
21	PLI_PUTNPC	str image, str script file, c x×2, c y×2	1.21; not 2.31 or 6.03r7	unmeasured	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205); [bin] (version-profiles/2.22-2.31.md#wire-protocol)
22	PLI_NPCDEL	i3 npc id; only for NPCs with id > 0	1.21	unmeasured	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205)
29	PLI_NPCWEAPONDEL	rest = weapon name	1.28 beta 2	removes the weapon	[srv][2.05] (protocol/packet-layouts.md#weapons)
33	PLI_WEAPONADD	c type; type 0 → c item index; type 1 → i3 npc id (copy an NPC's script as a weapon)	1.38	adds the weapon and answers with PLO_NPCWEAPONADD (33)	[srv][2.05] (protocol/packet-layouts.md#weapons)

Rule	Builds	Source
Flags whose name starts with local. are never sent	2.05	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205)
When the deleted NPC is one of the player's weapons, 2.05 sends PLI_NPCWEAPONDEL (29) with the weapon name instead of PLI_NPCDEL	2.05	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205)
2.05 only sends weapon type 0	2.05	[2.05] (protocol/packet-layouts.md#weapons)

2.4.7 Files, scripts and server values

The download replies and their framing are in → §2.7.

Id	Name	Layout	First sent	Server action	Source
23	PLI_WANTFILE	rest = file name	1.21	sends the file (→ §2.7)	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205)
34	PLI_UPDATEFILE	i5 modtime, rest = file name	1.42 (New Main)	sends the file if it is newer than modtime	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205)
47	PLI_VERIFYWANTSEND	i5 crc32, str file name	5.00r7	sends the file if the client's copy is stale	[srv] (protocol/packet-ids.md#client--server-pli_); [bin][inf] (protocol/v5x-clients.md#client--server-ids-500r7)
50	PLI_RAWDATA	i3 length of the next message (→ §2.4.1)	3.153	reads the next *length* bytes as one message	[bin] (protocol/v6x-clients.md#client--server-ids-601-603); [bin] (protocol/v4x-clients.md#cipher-and-framing)
130	PLI_REQUESTUPDATEBOARD	c level-name length, level name, i5 modtime, s x, s y, s width, s height	4.04r2	unmeasured	[srv] (protocol/packet-ids.md#client--server-pli_); [bin] (protocol/v6x-clients.md#client--server-ids-601-603)
152	PLI_REQUESTTEXT	unmeasured (reads a server value)	3.153	answers with the value	[srv] (protocol/packet-ids.md#client--server-pli_); [bin][inf] (protocol/v5x-clients.md#client--server-ids-500r7)

Id	Name	Layout	First sent	Server action	Source
154	PLI_SENDTEXT	unmeasured (writes a server value)	3.153	stores the value	[srv] (protocol/packet-ids.md#client--server-pli_); [bin][inf] (protocol/v5x-clients.md#client--server-ids-500r7)
157	PLI_UPDATEGANI	unmeasured beyond an `i5` time (a gani request)	3.153	sends the gani script	[srv] (protocol/packet-ids.md#client--server-pli_); [bin] (protocol/v4x-clients.md#client--server-ids)
158	PLI_UPDATESCRIPT	`rest` = script name	3.153	sends the weapon or class script	[srv] (protocol/packet-ids.md#client--server-pli_); [bin][inf] (protocol/v5x-clients.md#client--server-ids-500r7)
159	PLI_UPDATEPACKAGEREQUESTFILE	an order value (meaning unmeasured), package file name, modtime	5.00r7	unmeasured	[srv] (protocol/packet-ids.md#client--server-pli_)
161	PLI_UPDATECLASS	`i5 modtime`, `rest` = class name	4.04r2	sends the class	[srv] (protocol/packet-ids.md#client--server-pli_); [bin] (protocol/v4x-clients.md#client--server-ids)

Rule	Builds	Source
5.00r7's update-package downloader uses 34 and 47 (with and without a modification time)	5.00r7	[bin][inf] (protocol/v5x-clients.md#client--server-ids-500r7)
158 is sent after a cache miss in the client's script cache; 4.x uses it for the getweapon path	4.20r8, 5.00r7	[bin] (protocol/v4x-clients.md#client--server-ids); [bin][inf] (protocol/v5x-clients.md#client--server-ids-500r7)
154 is also used by the options and secure-URL code	5.00r7	[bin][inf] (protocol/v5x-clients.md#client--server-ids-500r7)

2.4.8 Chat, private messages and profiles

How chat is shown, and the server-to-client forms, are in → §2.8.

Id	Name	Layout	First sent	Server action	Source
6	PLI_TOALL	`str message`; 2.05 sends the chat-bar text after a `toall:` prefix, trimmed, at most 200 bytes	1.33	sends PLO_TOALL (13) with the sender's id	[2.05] (protocol/chat-pm-rc.md#toall)
28	PLI_PRIVATEMESSAGE	`s count`, `count` × `s player id`, `rest` = message (gtokenized)	1.28 beta 2	sends PLO_PRIVATEMESSAGE (37) to each listed player	[srv][2.05] (protocol/chat-pm-rc.md#private-messages)
80	PLI_PROFILEGET	`rest` = account	1.39	answers with PLO_PROFILE (75), relayed from the listserver	[srv][2.05] (protocol/chat-pm-rc.md#profiles)
81	PLI_PROFILESET	ten fields, each `c length` + text (→ §2.8)	1.39; not 6.03r7	forwards to the listserver	[2.05] (protocol/chat-pm-rc.md#profiles)

Rule	Builds	Source
The chat-bar text is sent as 6 only when, lower-cased and trimmed, it starts with `toall:`; all other text goes out as the CURCHAT prop (→ §2.4.4)	2.05	[2.05] (protocol/chat-pm-rc.md#chat-bubble)
5.00r7 sends 6 for chat-bar `toall` commands	5.00r7	[bin][inf] (protocol/v5x-clients.md#client--server-ids-500r7)
Field 0 of PLI_PROFILESET is the profile owner's account, which can be another online player's; a server must not take it as proof of who is editing	2.05	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)

2.4.9 Client checks and housekeeping

Id	Name	Layout	First sent	Server action	Source
14	PLI_CLAIMPKER	`s player id`	1.3 test	unmeasured	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205)
24	PLI_SHOWIMGPLAYER	`rest`, passed through	1.21	relays it unchanged as PLO_SHOWIMGPLAYER (32) with `s sender id` in front	[2.05] (protocol/packet-layouts.md#other-client--server-senders-205); [cap] (version-profiles/lab-showimg24-relay.tsv)
25	PLI_UNKNOWN25	unmeasured	1.21; last seen 1.37	unmeasured	[bin] (client-internals/version-profiles/v121-v132.toml); [bin] (client-internals/version-profiles/v133-v139.toml)
31	PLI_PACKETCOUNT	unmeasured	1.33; last seen 1.39	unmeasured	[bin] (client-internals/version-profiles/v133-v139.toml)
37	PLI_LANGUAGE	`rest` = language name (English seen)	2.02 rev2	stored	[cap] (version-profiles/2.22-2.31.md#wire-protocol); [bin] (protocol/v6x-clients.md#client--server-ids-601-603)
39	PLI_TAMPERCHECK	10 raw bytes, a constant per build (not two i5)	2.17, 2.22, 2.31; not 3.04, 3.153 or 6.x	GServer-v2 reads and discards it	[cap][bin] (protocol/packet-ids.md#client--server-pli_); [cap][srv] (version-profiles/2.22-2.31.md#wire-protocol)
43	PLI_MUTEPLAYER	`s player id, c on` (1 mute, 0 unmute); one message per click in the player list	2.17; not 6.03r7	unmeasured	[cap] (protocol/packet-ids.md#client--server-pli_) ; [cap] (protocol/lab-checks-2026-10-24.md#buddy-ignore-mute-222)
44	PLI_PROCESSLIST	the client's window list, comma-separated window titles	2.17; not 3.04	the reply to PLO_LISTPROCESSES (182)	[bin][cap] (protocol/packet-ids.md#client--server-pli_); [cap] (client-internals/version-profiles/141-217.toml)
45	PLI_PROCESSRUNNING	`c flag`: 1 if the server's string occurs in the client's own process list	2.17	the reply to PLO_HASPROCESSRUNNING (183)	[bin] (protocol/packet-ids.md#client--server-pli_); [bin] (protocol/v6x-clients.md#client--server-ids-601-603)
162	PLI_RC_UNKNOWN162	disputed → App. E#5	4.20r6	disputed → App. E#5	[bin] (protocol/v4x-clients.md#client--server-ids); [bin][inf] (protocol/v6x-clients.md#client--server-ids-601-603); [srv] (protocol/packet-ids.md#client--server-pli_)
252	PLI_SET_ENC_KEY	sets the AES or RC4 key and IV; above 223, so not sendable as one GCHAR (→ §2.10)	newer gs2lib only	unmeasured	[srv] (protocol/packet-ids.md#client--server-pli_)
253	PLI_BUNDLE	several messages in one (→ §2.10)	newer gs2lib only	unmeasured	[srv] (protocol/packet-ids.md#client--server-pli_)

Rule	Builds	Source
24 is also sent on its own about every 5.1 s with one byte (value 9)	2.31; 2.22 only while another player is in the level; 2.17 every 5 s after login	[cap] (protocol/packet-ids.md#client--server-pli_); [cap] (client-internals/version-profiles/141-217.toml); [cap] (client-internals/decomp-findings-2026-10-24-b/08-lab-findings-levelchange-24-178.md#b-pli_showimgplayer-24)

Rule	Builds	Source
The meaning of the 24 payload: receivers showed no visible reaction to the values 0, 1, 5, 9, 14, 58, 95, to the bytes `xx.png`, or to an empty payload, and the server neither disconnected nor logged an invalid packet; the client reads show it is the weapon-image list of → §2.4.9.1	2.31 sender, 2.x receiver	[cap] (version-profiles/lab-showimg24-relay.tsv); [bin] (client-internals/decomp-findings-2026-10-25-g/01-id24-image-list.md#what-the-text-is)
39 is sent right after `PLI_LANGUAGE` at login	2.17, 2.22, 2.31	[cap] (protocol/packet-ids.md#client--server-pli_)
39 bytes: 2.17 `2a 7b ad 19 5e 23 12 43 d8 2c`, 2.22 `25 0e 4d 0f 96 2b 03 ed e6 31`, 2.31 `29 16 55 b7 60 2e 87 c9 eb 82`; identical across logins, accounts and login keys	2.17, 2.22, 2.31	[cap] (version-profiles/tampercheck-values.tsv)
39 re-sends: 2.17 once at login; 2.22 at login, on every gmap edge crossing and after every client-initiated level-link request, but not after a server-side warp; 2.31 at login and then about every 3.2 s	2.17, 2.22, 2.31	[cap] (version-profiles/2.22-2.31.md#wire-protocol); [cap] (version-profiles/tampercheck-values.tsv) ; [cap] (protocol/lab-checks-2026-10-24.md#tampercheck-client-pli-39); [bin] [cap] (version-profiles/tampercheck-words.tsv)
How the 39 bytes are computed is a client integrity check (→ §4.10)	2.17, 2.22, 2.31	[bin] (version-profiles/2.22-2.31.md#wire-protocol)
Because GServer-v2 discards 39, whether a different value would be rejected is unmeasured	server	[srv] (version-profiles/2.22-2.31.md#wire-protocol)
44 is sent just before the login props	2.17	[cap] (client-internals/version-profiles/141-217.toml)

2.4.9.1 The id 24 weapon-image list

Id 24 carries the images that the sender's weapon scripts show (`showimg` and the `changeimg*` commands), as changes against the list sent last; the relay is `PLO_SHOWIMGPLAYER` (32):

Rule	Builds	Source
a record starts with 0x2a + the image index (0 to 199); tag bytes 0x20 + tag follow with their data; a byte of 0x2a or more starts the next record; an unknown tag skips one data byte; a tag with no data ends the text	2.05, 2.17.1, 2.22, 2.31	[bin] (client-internals/decomp-findings-2026-10-25-g/01-id24-image-list.md#what-the-text-is); [2.05] (client-internals/id24-205.md#what-205-sends)
tag 0 name (`GCHAR` length + bytes; empty removes the image), 1 x, 2 y (half tiles, or screen units at layer 10 and over; units unconfirmed), 3 layer − 1, 4 visible part (0x20 whole image, or 0x21 + left and top as two 7-bit bytes each + width and height, clamped to 0..16000 and 0..220), 5 red, green, blue and alpha over 200, 6 zoom over 10, 7 draw order − 0x52, 8 mode	2.17.1 to 2.31 (receiver read in 2.31)	[bin] (client-internals/decomp-findings-2026-10-25-g/image-text-format.tsv)
2.05 sends tags 0 to 6 only: layer only when not 1, part only when its size is above 0, colours only when one is not 1.0, zoom only when not 1.0	2.05	[2.05] (client-internals/id24-205.md#what-205-sends)
a leading 0x29 (`GCHAR` 9) marks a full list; after it the receiver removes every image of that player the text did not name; with no images the full list is the marker alone, the one byte every real client was seen to send	2.05 to 2.31	[bin] (client-internals/decomp-findings-2026-10-25-g/01-id24-image-list.md#what-the-text-is); [2.05] (client-internals/id24-205.md#what-205-sends)
a full list goes out when a countdown wraps: every 61st call in 2.05, every 102 ticks (101 calls, the call after a send skipped) in 2.17.1 to 2.31, which fits the 5.1 s seen at 50 ms a tick; the first tick after login is a wrap; an empty text is not sent	2.05; 2.17.1 to 2.31	[2.05] (client-internals/id24-205.md#what-205-sends); [bin][inf] (client-internals/decomp-findings-2026-10-25-g/01-id24-image-list.md#sender)

Rule	Builds	Source
a UDP copy goes out when it can (id 15 in 2.05, → §2.9); the TCP copy goes out when UDP is off for the client or does not reach some player in the level; 2.31 always sends the TCP copy; a wrap that finds the test false is lost	2.05, 2.17.1, 2.22; 2.31	[2.05] (client-internals/id24-205.md#what-205-sends); [bin] (client-internals/decomp-findings-2026-10-25-g/01-id24-image-list.md#the-tcp-test)
a new image gets layer 1, colours 200 each, zoom 10 and mode 0; fields other than the name apply only to an image that exists	2.31	[bin] (client-internals/decomp-findings-2026-10-25-g/01-id24-image-list.md#receiver)
the sender's own value formatting, the "UDP reaches this player" flag of the 2.22 test, and how the images are drawn are unmeasured	2.17.1 to 2.31	[doc] (client-internals/decomp-findings-2026-10-25-g/04-open-points.md)
an earlier note's four-field form (index, x, y, name) for 2.05 and earlier is wrong for 2.05; builds before 2.05 were not read	2.05	[2.05] (client-internals/id24-205.md#correction)

2.4.10 Staff messages sent by game clients

The RC and NC ids (51 to 119, 150, 151, 155, 156, 160) are documented with the staff tools in → §7. Some game clients send a few of them:

Id	Name	Builds	Use in a game client	Source
79	PLI_RC_CHAT	4.x, 5.00r7, 6.x	unmeasured	[bin] (protocol/v4x-clients.md#client--server-ids); [bin] (protocol/v6x-clients.md#client--server-ids-601-603)
92, 93	PLI_RC_FILEBROWSER_DOWN, _UP	92: 4.x, 5.00r7, 6.x; 93: 3.153, 5.00r7, 6.x	93 carries a file-name string and one data chunk, behind id 50	[bin] (protocol/v6x-clients.md#client--server-ids-601-603); [bin] (version-profiles/3.04-3.153.md#client--server-ids)
96, 97, 98	PLI_RC_FILEBROWSER_MOVE, _DELETE, _RENAME	3.153, 5.00r7, 6.x; 96 also 4.x	file browser	[bin] (protocol/v6x-clients.md#client--server-ids-601-603); [bin] (version-profiles/3.04-3.153.md#client--server-ids)
155, 156	PLI_RC_LARGEFILESTART, _END	3.153, 5.00r7, 6.x	the upload queue: 155, then 93 chunks of at most 32,000 bytes, then 156; the queue pauses while about 64 KB is waiting to go out	[bin] (protocol/v6x-clients.md#client--server-ids-601-603)
160	PLI_RC_FOLDERDELETE	6.03r7 only	folder delete	[bin] (protocol/v6x-clients.md#client--server-ids-601-603)

The 6.11 client sends no fixed ids: a script-loaded table maps each outgoing message to a wire id, so its named send slots are in → §2.10.

2.4.11 Login-time order

The login message itself and the server's reply are in → §2.3. What the client sends next:

Build	Messages after the login message, in order	Source
1.41 rev1	one PLI_PLAYERPROPS with the login props, right after the first server message; no file requests	[cap] (client-internals/version-profiles/141-217.toml)
2.02 rev1	file requests and the login props; no 37	[cap] (client-internals/version-profiles/141-217.toml)
2.17	37, then 39 once; 44 with the window titles just before the login props	[cap] (client-internals/version-profiles/141-217.toml); [cap] (version-profiles/tampercheck-values.tsv)

Build	Messages after the login message, in order	Source
2.22	37 `English`, 39, then 34 for the default images (sword, shield, body, bomb), 23, and 2; 44 in answer to the server's 182	[cap] (version-profiles/2.22-2.31.md#wire-protocol)
2.31	37, 39, then in the second frame 46 before the props, 35 and 44	[cap] (version-profiles/2.22-2.31.md#wire-protocol); [cap] (version-profiles/lab-enterlevel-231.tsv)
6.01r1, 6.03r7	37 right after login; no 39	[bin] (protocol/v6x-clients.md#client--server-ids-601-603)

2.5 PLO message reference (server to client)

Field notation is in → §0.6, the message envelope in → §2.1.2, and the login reply sequences in → §2.3.6. The client-to-server messages that trigger many of these are in → §2.4, prop streams in → §2.6, and file downloads in → §2.7. Every id, with its wire byte, is indexed in → App. A.

> Status: draft. Layouts are given where a note measured or read them; ids with no layout in the notes say `unmeasured`. The staff tools' own server messages are in §7.

2.5.1 How a client reads a PLO

Rule	Builds	Source
Every handler checks a minimum length first and drops a short message silently	2.05, 2.17.1, 2.22, 2.31	[2.05] (protocol/packet-layouts.md#how-205-reads-a-message-205); [bin] (protocol/packet-layouts.md#server--client-ids-above-48-and-related-217-222-231-bin)
An id with no case in the receive switch falls to one shared default arm and is ignored without error	2.05 to 2.31	[2.05] (protocol/packet-ids.md#server--client-plo_); [bin][srv] (protocol/client-versions.md#server--client-ids-handled)
2.05 keeps two level pointers: the *named level*, set by every `PLO_LEVELNAME`, and the *object level*, which receives NPC messages (3, 150, 151, 166) and is set by `PLO_SETACTIVELEVEL`, or by each `PLO_LEVELNAME` when there is no NPC server	2.05	[2.05] (protocol/packet-layouts.md#how-205-reads-a-message-205)
Handlers for level objects (bombs, arrows, items, explosions, hits, trigger actions, horses, push-away) act only while the player stands in the named level	2.05	[2.05] (protocol/packet-layouts.md#how-205-reads-a-message-205)
A message the server re-emits for another player starts with `s player id`; with an unknown id, bombs, arrows, explosions and items are still shown without an owner, while fire spy, throw carried and player hits are dropped	2.05	[2.05] (protocol/packet-layouts.md#how-205-reads-a-message-205)
2.17.1, 2.22 and 2.31 read "comma text" through the client's comma-text parser, which allows double quotes	2.17.1, 2.22, 2.31	[bin] (protocol/packet-layouts.md#server--client-ids-above-48-and-related-217-222-231-bin)
The receive dispatcher takes the id as the first byte minus 32, checks it against a bound, and goes through an index table; the bound is id 185 in 2.17.1, 186 in 2.22 and 189 in 2.31	2.17.1, 2.22, 2.31	[bin] (client-internals/decomp-findings-2026-10-24-b/02-server-to-client-layouts.md#dispatcher-shape-all-builds)
4.20r6 and 4.20r8 take 100 and 108 before the main switch	4.20r6, 4.20r8	[bin] (protocol/v4x-clients.md#server--client-ids)

2.5.2 Ids handled by build

Builds before 2.00 are in → App. C.2. *Bound* is the highest id the receive switch tests, written as in the source.

Build	Count	Ids handled	Source
New Main 24, 26	51	0–45, 75, 100, 150, 151, 153 (bound `0x99`)	[bin] (version-profiles/plo-handled-ids-by-build.tsv)

Build	Count	Ids handled	Source
2.00 (rev 0–rev3)	53	0–45, 75, 100, 150, 151, 153–155 (bound 0x9b)	[bin] (version-profiles/plo-handled-ids-by-build.tsv) ; [bin] (client-internals/decomp-findings-2026-10-24-b/04-v2.00-client.md#server---client-ids-handled-200) ; [bin] (client-internals/decomp-findings-2026-10-24-c/plo_handled_ids_by_build.tsv)
2.01 (+rev1)	55	0–46, 75, 100, 150, 151, 153–156	[bin] (version-profiles/plo-handled-ids-by-build.tsv) ; [bin] (client-internals/decomp-findings-2026-10-24-c/plo_handled_ids_by_build.tsv)
2.02 (rev 0–rev2)	56	0–47, 75, 100, 150, 151, 153–156; disputed for rev0 and rev1 → App. E#7	[bin] (version-profiles/plo-handled-ids-by-build.tsv) ; [bin] (client-internals/decomp-findings-2026-10-24-c/plo_handled_ids_by_build.tsv)
2.03	58	0–48, 75, 100, 150, 151, 153–156, 165	[bin] (version-profiles/plo-handled-ids-by-build.tsv) ; [bin] (client-internals/decomp-findings-2026-10-24-c/plo_handled_ids_by_build.tsv)
2.04	58	0–25, 27–48, 75, 100, 150, 151, 153–156, 165, 166 (26 dropped)	[bin] (version-profiles/plo-handled-ids-by-build.tsv) ; [bin] (client-internals/decomp-findings-2026-10-24-c/plo_handled_ids_by_build.tsv)
2.05 (rev 0–rev2)	59	0–25, 27–48, 75, 100, 150, 151, 153–156, 165, 166, 168 (bound 0xa8); disputed for rev0 → App. E#7	[bin] (version-profiles/plo-handled-ids-by-build.tsv); [2.05] (protocol/packet-ids.md#server--client-plo_) ; [bin] (client-internals/decomp-findings-2026-10-24-c/plo_handled_ids_by_build.tsv)
2.1 test 2, test 3	61	the 2.05 set plus 49, 169	[bin] (version-profiles/plo-handled-ids-by-build.tsv) ; [bin] (client-internals/decomp-findings-2026-10-24-c/plo_handled_ids_by_build.tsv)
New World beta 1	53	the 2.05 set without 2, 4, 17, 18, 154, 155	[bin] (client-internals/decomp-findings-2026-10-24-c/08-test-builds-newworld-newmain.md#newworld-beta1--beta2)
New World beta 2	55	the beta 1 set plus 49, 169	[bin] (client-internals/decomp-findings-2026-10-24-c/08-test-builds-newworld-newmain.md#newworld-beta1--beta2)
2.17.1	79	0–25, 27–49, 68, 69, 75, 100, 150, 151, 153–156, 165, 166, 168–185	[bin] (version-profiles/plo-handled-ids-by-build.tsv)
2.22	80	the 2.17.1 set plus 186	[bin] (version-profiles/plo-handled-ids-by-build.tsv)
2.31	81	the 2.22 set without 44, plus 84 and 189	[bin] (version-profiles/plo-handled-ids-by-build.tsv); [bin] (protocol/client-versions.md#server--client-ids-handled)
3.04 (Linux)	79	0–49, 68, 69, 75, 84, 100–102, 150, 151, 153–156, 165, 166, 168, 169, 171–176, 178–181, 184–186	[bin] (protocol/v3x-clients.md#server--client-ids-handled)
3.153 (Mac OS X)	88	the 3.04 set without 44, plus 67, 74, 82, 104, 131–134, 187, 189	[bin] (protocol/v3x-clients.md#server--client-ids-handled)
4.20 r8	101	0–25, 27–43, 45–49, 67–69, 74, 75, 82, 84, 100–102, 104–108, 131–134, 150, 151, 153–156, 165, 166, 168–176, 178–187, 189–191, 193–196	[bin] (protocol/v4x-clients.md#server--client-ids)

Build	Count	Ids handled	Source
4.20r6 (Mac)	99	the 4.20r8 set without 41 and 196	[bin] (protocol/v4x-clients.md#server--client-ids)
5.00r7	100	0–25, 27–43, 45–49, 67–69, 74, 75, 82, 84, 101, 102, 104–107, 131–134, 140, 150, 151, 153–156, 165, 166, 168–176, 178–187, 189–191, 193–195, 197	[bin] (protocol/v5x-clients.md#server--client-ids-500r7)
6.01r1	91	the 6.03r7 set without 111	[bin] (protocol/v6x-clients.md#server--client-ids-601-603)
6.03r7	92	0–25, 27–40, 42, 43, 45–49, 67–69, 74, 82, 84, 101, 102, 104–107, 109, 111, 131, 134, 140, 150, 151, 153–156, 166, 168, 170, 172–174, 176, 178–180, 182, 183, 185–187, 189–191, 194, 195, 197, 198	[bin] (protocol/v6x-clients.md#server--client-ids-601-603)

Rule	Builds	Source
No build in the notes handles 26 (`PLO_NPCACTION`) after 2.03, or 152 and 167	2.04 to 2.31	[bin] (protocol/packet-ids.md#server--client-plo_); [bin] (protocol/client-versions.md#server--client-ids-handled)
Revisions of one version share every handler shape (2.05 rev1 differs by one constant in 8), so any revision stands for the others; on the wire only 2.02 rev2 differs, by its token and id 37	2.00 to 2.05	[bin] (client-internals/decomp-findings-2026-10-24-c/06-handler-changes-by-build.md#revisions-of-the-same-version)
2.00 has no active-level reference, since 156 arrives in 2.01; level-addressed messages act on the current level	2.00	[bin] (client-internals/decomp-findings-2026-10-24-c/04-plo-server-to-client.md#active-level-gating)
2.01 rewrites 0, 14 and 15 so the handler no longer waits for and then requests the level, the code side of map preloading; 154 and 155 go through a shared helper	2.01	[bin] (client-internals/decomp-findings-2026-10-24-c/06-handler-changes-by-build.md#200---201)
0, 6 and 14 gain checks that skip reloading a level that is already loaded	2.1 test 3	[bin] (client-internals/decomp-findings-2026-10-24-c/06-handler-changes-by-build.md#21-test2---test3); [bin] (client-internals/decomp-findings-2026-10-24-c/08-test-builds-newworld-newmain.md#21-test2-and-test3)
NewMain 24 and 26 have one handler set between them, but differ from 2.00 more than any 2.0x revision does, so NewMain is a separate branch	New Main	[bin] (client-internals/decomp-findings-2026-10-24-c/08-test-builds-newworld-newmain.md#newmain-24--26)
2.31 drops 44 (`HASNPCSERVER`); the 2.3 release notes say Classic switched to the NPC server	2.31	[bin][doc] (protocol/client-versions.md#server--client-ids-handled)
3.04 lacks 170, 177, 182 and 183, which 2.22 handles, and adds 84, 101 and 102	3.04	[bin] (version-profiles/3.04-3.153.md#server--client-ids)
4.20r8 drops nothing against 3.153 and adds 13 ids	4.20r8	[bin] (protocol/v4x-clients.md#server--client-ids)
5.00r7's list does not include 100	5.00r7	[bin] (protocol/v5x-clients.md#server--client-ids-500r7)
Against 5.00r7, 6.x drops 41, 75, 132, 133, 165, 169, 171, 175, 181, 184 and 193 and adds 109, 111 and 198	6.01r1, 6.03r7	[bin] (protocol/v6x-clients.md#server--client-ids-601-603)
A later client (5.07 and up, or 6.037, as the source note marks each id) ignores 26, 41, 44, 55, 56, 60, 78, 80, 81, 83, 132, 157, 159–164, 165, 167, 171, 175, 188 and 192	5.07, 6.037	[srv] (protocol/packet-ids.md#server--client-plo_)

2.5.3 Level data

Id	Name	Layout	Client action	Source
0	PLO_LEVELBOARD	a run of board changes, each `c x, c y, c w, c h + w × h s tile` (row-major), back to back; empty = "use your cached board"	parsing stops when fewer than 4 bytes are left or a run is shorter than `w*h*2+4`; a pending warp to this level completes first	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client)
1	PLO_LEVELLINK	`rest = dest x y w h newx newy`, split on spaces	see the rules below	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client); [bin] (version-profiles/plo-levellink-layouts.tsv)
2	PLO_BADDYPROPS	`c id` + BDPROP stream (→ §2.6)	an unknown id creates the baddy	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client)
3	PLO_NPCPROPS	`i3 id` + NPCPROP stream (→ §2.6); id > 0	ids below 1000 are level NPCs, created in the object level; 1000 and up are server-wide; a new NPC gets the `playerenters` event if the player is in that level	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client)
4	PLO_LEVELCHEST	`c opened, c x, c y` (whole tiles); if not opened, also `c item, c signindex`	a closed chest with fewer than 5 bytes is dropped; an opened chest draws nothing by itself, so on a board with no chest tiles the short form leaves no trace	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client); [cap] (client-internals/decomp-findings-2026-10-25-l/04-public-server-run.md#why-no-chests-were-drawn)
5	PLO_LEVELSIGN	`c x, c y` (whole tiles) + sign-alphabet text (→ §3)	needs at least 2 bytes	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client)
6	PLO_LEVELNAME	`rest` = level file name	sets the named level, and the object level when there is no NPC server	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client)
7	PLO_BOARDMODIFY	one run: `c x, c y, c w, c h + w × h s tile`, row-major	needs `len ≥ w*h*2+4`	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client)
10	PLO_ISLEADER	empty	this client runs the level's baddies and timeouts; ignored unless the player's level is the active level	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client)
17	PLO_HORSEADD	`c x×2, c y×2, c (dir & 3) + 4 × bushes` (bushes capped at 6), `rest` = image	`.gif` is added when the image has no extension	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client)
18	PLO_HORSEDEL	`c x×2, c y×2`; no player id	removes any horse within 2 tiles of that point	[2.05] (protocol/packet-layouts.md#level-data-server--client)
22	PLO_ITEMADD	`c x×2, c y×2, c item` (0–24); optional `s player` in bytes 3–4	unknown items ignored; with the player id, a dropped sword or shield copies that player's image	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client)
23	PLO_ITEMDEL	`c x×2, c y×2`; no player id	removes the item at exactly that point	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client)
39	PLO_LEVELMODTIME	2.05: `i5 modtime`, exactly 5 bytes; 2.17 to 2.31: `rest`, unparsed	2.05 writes the level to its web-level cache (→ §3); 2.17 to 2.31 hand the rest to the active gmap or level object, if one exists	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client); [bin] (version-profiles/plo-layouts-217-231.tsv)

Id	Name	Layout	Client action	Source
42	PLO_NEWWORLDTIME	i4 time (b0<<21 | b1<<14 | b2<<7 | b3)	needs at least 4 bytes	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client)
100	PLO_RAWDATA	i3 length (→ §2.1.3)	the next *length* bytes are one message	[srv] (protocol/codec-and-framing.md#raw-data-escape); [bin] (version-profiles/plo-layouts-217-231.tsv)
101	PLO_BOARDPACKET	behind 100: i3 (8194), then c 101 + 4096 × u16le raw tile ids + \n	replaces the 64 × 64 board	[srv] (protocol/packet-layouts.md#level-data-server--client)
107	PLO_BOARDLAYER	4.20r8: layer, x, y (each under 64), width and height (1–64), then two bytes per tile	a board change on a layer; layer 0 also clears the level's pending changes; 5.00r7 raises the script level to at least 5 and passes the data to a script object	[bin] (protocol/v4x-clients.md#server--client-ids); [bin] (protocol/v5x-clients.md#server--client-ids-500r7)
150	PLO_NPCDEL2	c n, n-byte level name, i3 npc id	removes that NPC from that level	[bin] (version-profiles/plo-layouts-217-231.tsv)
151	PLO_HIDENPCS	empty	removes NPCs with ids 1 to 999 from the active level (id 0 excluded)	[bin] (version-profiles/plo-layouts-217-231.tsv)
156	PLO_SETACTIVELEVEL	rest = level or .gmap name	sets the object level; a .gmap is loaded and made active; 2.31 adds a follow-up step whose purpose is unresolved	[srv][2.05] (protocol/packet-layouts.md#level-data-server--client); [bin] (version-profiles/plo-layouts-217-231.tsv)
185	PLO_BOARDHEIGHTS	c mapX, c mapY, c blockX, c blockY, c w−1, c h−1, then w × h cells of 2 raw bytes, row-major	height = (b0 − 0x20 − 0x32) + (b1 − 0x20) / 128; written into that gmap cell's height grid from blockX, blockY; mapX and mapY must lie in the gmap grid	[bin] (version-profiles/plo-layouts-217-231.tsv)
186	PLO_BOARDMODIFY2	c mapX, c mapY, c x, c y, c w, c h, then w × h s tile, row-major	tiles replace that gmap cell's board from x, y; needs len ≥ 6 + 2*w*h	[bin][cap] (version-profiles/plo-layouts-217-231.tsv)

Rule	Builds	Source
A link needs at least 5 space-separated fields, else the client raises "Wrong link data!"; x y w h are whole tiles read as floats; newx and newy are fields 5 and 6 or empty, and the literal -1 becomes playerx or playery	2.00, 2.04, 2.05 rev2, 2.17.1, 2.22, 2.31	[bin] (version-profiles/plo-levellink-layouts.tsv)
2.17.1 and 2.22 also have a helper that builds a default link with newx and newy preset to playerx and playery, used for editor and edge cases	2.17.1, 2.22	[bin] (version-profiles/plo-levellink-layouts.tsv)
dest is field 0, so a level name with a space cannot be linked	2.00 to 2.31	[bin] (version-profiles/plo-levellink-layouts.tsv)
The client takes a link when one point enters its rectangle: the player's centre (x+1.5, y+2), 1 to 1.5 tiles ahead in the walking direction; a 1×1 link off that centre line is walked past. Measured to half a tile in straight walks only	2.05 rev2 (lab, one fork server)	[cap] (protocol/link-test-2026-11-04.md#result)
Only PLO_BOARDPACKET carries raw little-endian tiles; every other board message uses GCHAR headers and GSHORT tiles	server	[srv] (protocol/packet-layouts.md#board-modify-correction)

Rule	Builds	Source
186: bytes after the tiles are not read, so a complete second record in the trailer is not applied; a payload one byte short is dropped silently with no disconnect; a block past column 63 is clipped	2.22, 2.31 (lab, injected frames; no server seen sending 186)	[cap] (version-profiles/lab-boardmodify2-186-trailer.tsv); [bin][cap] (protocol/packet-ids.md#server--client-plo_); [cap] (client-internals/decomp-findings-2026-10-24-b/07-lab-findings-46-186-200.md#plo_boardmodify2-186-222-and-231-client-reaction-observed-frames-synthetic)
In a two-client capture the server sent PLO_BOARDMODIFY with the reported bytes to both clients, the reporter included, and 15.0 s later (respawntime = 15) sent the original tiles back	2.05 with GServer-v2	[cap] (protocol/packet-layouts.md#board-change-on-the-wire-cap)
On level entry an empty PLO_LEVELBOARD followed the board packet and PLO_LEVELMODTIME	2.05 with GServer-v2	[cap] (protocol/packet-layouts.md#board-change-on-the-wire-cap)

2.5.3.1 Level send order (pre-2.1 server path)

Step	Messages	Source
1	if the client's cached modtime matches: the board changes since then, and nothing else	[srv] (protocol/packet-layouts.md#pre-21-level-sequence-sendlevel141-used-for-205-srv)
2	otherwise: RAWDATA + BOARDPACKET, then LEVELNAME (first level only), LEVELLINK × n, LEVELSIGN × n, LEVELMODTIME	[srv] (protocol/packet-layouts.md#pre-21-level-sequence-sendlevel141-used-for-205-srv)
3	then board changes, LEVELCHEST × n, HORSEADD × n, BADDYPROPS × n, ISLEADER (if leader), NEWWORLDTIME, NPCPROPS × n, and OTHERPLPROPS for each player in the level	[srv] (protocol/packet-layouts.md#pre-21-level-sequence-sendlevel141-used-for-205-srv)

2.5.4 Players and the local player

Id	Name	Layout	Client action	Source
8	PLO_OTHERPLPROPS	s id + PLPROP stream (→ §2.6); at least 2 bytes	an unknown id creates the player	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client)
9	PLO_PLAYERPROPS	PLPROP stream for the local player, no id (→ §2.6)	applies the props	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client)
14	PLO_PLAYERWARP	c x×2, c y×2, rest = level	the warp stays pending until the level data arrives	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client)
15	PLO_WARPFAILED	rest = level	2.05 ignores the text and shows a short failed-warp state	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client)
16	PLO_DISCMESSAGE	rest = text	stored as the disconnect reason; the server then closes the connection	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client)
25	PLO_SIGNATURE	c v	1: the client quits; 0: it quits unless the connected host is the loopback address; other values ignored	[bin] (version-profiles/plo-layouts-217-231.tsv)
28	PLO_FLAGSET	rest = name or name=value	adds the flag only if it is not already set	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client)
31	PLO_FLAGDEL	rest = flag name	removes the flag	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client)
38	PLO_PUSHAWAY	c dx, c dy, each v/16 − 4 (±2 tiles in 1/16 steps)	ignored unless the player's level is the active level	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client); [srv] (protocol/packet-ids.md#server--client-plo_)

Id	Name	Layout	Client action	Source
40	PLO_HURTPLAYER	`s attacker, c dx, c dy, c power`, optional `i3 npc`	see → §2.5.5	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client)
49	PLO_PLAYERWARP2	`c x×2, c y×2, c z` (unused), `c gmapX, c gmapY, rest` = level	gmapX and gmapY apply only when the level name ends in `.gmap`: they are clamped to the grid and the name is replaced by that cell's level	[bin] (version-profiles/plo-layouts-217-231.tsv)
154	PLO_FREEZEPLAYER2	empty	freezes the local player for 600 ticks unless unfrozen; skipped while some dialogs are open	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client); [bin] (version-profiles/plo-layouts-217-231.tsv)
155	PLO_UNFREEZEPLAYER	empty	unfreezes the local player	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client)
168	PLO_UNKNOWN168	payload ignored	resets the local player; sent by the login server	[bin] (protocol/packet-ids.md#server--client-plo_); [bin] (version-profiles/plo-layouts-217-231.tsv)
170	PLO_GHOSTMODE	`c flag`	2.17 to 2.31: ghost mode on when the flag is above 0; 5.00r7: stored as the connection mode, and a change resets a player field	[bin] (version-profiles/plo-layouts-217-231.tsv); [bin] (protocol/v5x-clients.md#server--client-ids-500r7)
173	PLO_GHOSTTEXT	`rest` = text	stored as the ghost-mode text	[bin] (version-profiles/plo-layouts-217-231.tsv)
174	PLO_GHOSTICON	`c flag`	shows the ghost icon when the flag is above 0	[bin] (version-profiles/plo-layouts-217-231.tsv)
181	PLO_UNKNOWN181	`s player id, c toggle`; at least 3 bytes	sets a boolean on that player	[bin] (version-profiles/plo-layouts-217-231.tsv)
55, 56	PLO_ADDPLAYER, PLO_DELPLAYER	unmeasured	ignored by 5.07 and later	[srv] (protocol/packet-ids.md#server--client-plo_)

2.5.5 Combat and level objects

These are the server's copies of the client actions in → §2.4.5; each carries the sender's `s player id` unless the row says otherwise.

Id	Name	Layout	Client action	Source
11	PLO_BOMBADD	`s player, c x×2, c y×2, c power, c fuse`	power = `v & 7`; the value 8 (bit 3) marks the bomb as that player's	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
12	PLO_BOMBDEL	`c x×2, c y×2`; no player id	removes the bomb at exactly that point	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
19	PLO_ARROWADD	`s player` + the 5 bytes of PLI_ARROWADD	needs at least 5 bytes; the last byte is decoded as `b − 33`	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
20	PLO_FIRESPY	`s player, c (a × 8 + b)`	only a is used	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
21	PLO_THROWCARRIED	`s player`	dropped when the player is unknown	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
27	PLO_BADDYHURT	`c id, c (dx + 4) × 16, c (dy + 4) × 16, c power`	applied only by the level leader	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)

Id	Name	Layout	Client action	Source
36	PLO_EXPLOSION	`s player, c radius, c x×2, c y×2, c power`	power & 7 clamped to 1–3; 8 (bit 3) = owned	[srv][2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
40	PLO_HURTPLAYER	`s attacker, c dx, c dy, c power`, optional `i3 npc`	see the rules below	[srv][2.05] (protocol/packet-layouts.md#hits-205)
46	PLO_HITOBJECTS	`s player, c power, c x×2, c y×2`, optional `i3 npc`	see the rules below	[srv][2.05] (protocol/packet-layouts.md#hits-205)
48	PLO_TRIGGERACTION	`s player, i3 npc, c x×2, c y×2`, `rest` = `action` or `action,params`; more than 5 bytes	fires `action` on the local player's scripts when the source is within 10 tiles; split at the first comma	[srv][2.05] (protocol/packet-layouts.md#hits-205)
175	PLO_SHOOT	`s player id`, `rest` = shot data (unmeasured)	only when the current level is the active level	[bin] (version-profiles/plo-layouts-217-231.tsv)
191	PLO_SHOOT2	4.20r8: x, y, z, three angle and speed values, a gani name with optional parameters, then a script string; 5.00r7: `s id` + data routed to an object	a projectile	[bin] (protocol/v4x-clients.md#server--client-ids); [bin] (protocol/v5x-clients.md#server--client-ids-500r7)

Rule	Builds	Source
PLO_HURTPLAYER: dx, dy decode as `(v − 0x60) / 16` (knockback −4 to 4) and power as `(v − 32) / 2` hearts, capped at 1.0	2.05	[2.05] (protocol/packet-layouts.md#hits-205)
With 6 or more bytes after the player id, the hit came from an NPC (i3), and distance is measured from that NPC	2.05	[2.05] (protocol/packet-layouts.md#hits-205)
A hit applies only when the attacker (player or NPC) is within 10 tiles and the local player can be hurt at that moment	2.05	[2.05] (protocol/packet-layouts.md#hits-205)
PLO_HITOBJECTS: power is `(v − 32) / 2`, capped at 1.0; it hurts the local player only when the player's centre is within 2 tiles of `(x, y)` on each axis	2.05	[2.05] (protocol/packet-layouts.md#hits-205)
40 and 46 share one handler, with one flag that picks the layout; the same handler takes UDP id 21 (→ §2.9)	2.05	[2.05] (protocol/packet-layouts.md#hits-205)
The trigger-action coordinates are half-tiles, not whole tiles as in the server source	2.05	[2.05] (protocol/packet-layouts.md#differences-from-gserver-worth-knowing-205)

How damage is applied after a hit is in → §4.3.

2.5.6 NPCs, weapons and scripts

Id	Name	Layout	Client action	Source
24	PLO_NPCMOVED	unmeasured	unmeasured	[srv] (protocol/packet-ids.md#server--client-plo_)
26	PLO_NPCACTION	`i3 npc id`, `rest`	handled by 2.00 to 2.03 only, and acts only while the player's level is the active level; purpose unknown	[srv] (protocol/packet-ids.md#server--client-plo_); [bin] (version-profiles/plo-handled-ids-by-build.tsv); [bin] (client-internals/decomp-findings-2026-10-24-c/04-plo-server-to-client.md#set-growth)
29	PLO_NPCDEL	unmeasured	unmeasured	[srv] (protocol/packet-ids.md#server--client-plo_)

Id	Name	Layout	Client action	Source
33	PLO_NPCWEAPONADD	`str name` + NPCPROP stream (normally `c 0` IMAGE + `str image`, `c 1` SCRIPT + `s len` + GS1 text)	a name the player already has is ignored; the weapon gets the `playerenters` event; a script line break is §, byte A7 from a latin-1 server and `C2 A7` from a UTF-8 server	[srv][2.05] (protocol/packet-layouts.md#weapons); [cap] (client-internals/decomp-findings-2026-10-25-l/04-public-server-run.md#weapon-scripts-and-line-breaks)
34	PLO_NPCWEAPONDEL	`rest` = weapon name	removes the weapon	[srv][2.05] (protocol/packet-layouts.md#weapons)
43	PLO_DEFAULTWEAPON	`c item index` (0–24: a built-in weapon such as bow, bomb, sword)	adds the built-in weapon	[srv][2.05] (protocol/packet-layouts.md#weapons)
44	PLO_HASNPCSERVER	empty	the client stops applying NPC props itself	[bin] (protocol/packet-ids.md#server--client-plo_); [bin] (version-profiles/plo-layouts-217-231.tsv)
131	PLO_NPCBYTECODE	`i3 code`, bytecode	script and class registration: 3.153 registers classes (`joinedclasses`) and `public.` variables from it, with 132 and 133 (→ §8)	[srv] (protocol/packet-ids.md#server--client-plo_); [bin] (protocol/v3x-clients.md#server--client-ids-handled)
133	PLO_UNKNOWN133	`rest` = text	maybe registers a class script	[srv] (protocol/packet-ids.md#server--client-plo_)
134	PLO_GANISCRIPT	gani text	3.153 parses the gani text in place	[srv] (protocol/packet-ids.md#server--client-plo_); [bin] (protocol/v3x-clients.md#server--client-ids-handled)
140	PLO_NPCWEAPONSCRIPT	`s info_length`, script	class and weapon script delivery (→ §8)	[srv] (protocol/packet-ids.md#server--client-plo_); [bin] (protocol/v5x-clients.md#server--client-ids-500r7)
165	PLO_MOVE	`i3 npc id` (> 0), `rest` = move data	passed to the NPC movement routine	[bin] (version-profiles/plo-layouts-217-231.tsv)
166	PLO_SHOWIMGNPC	`i3 npc id`, `rest` = props	applies the props to that NPC in the active level	[bin] (version-profiles/plo-layouts-217-231.tsv)
189	PLO_MOVE2	`i3 npc id`, `rest`	as 165, with an extra flag set	[bin] (version-profiles/plo-layouts-217-231.tsv)
193	PLO_UNKNOWN193	i5 time, file name	4.20r8 checks its cached copy (mod time, then CRC); 5.00r7 routes it to an NPC and raises the script level to at least 3	[bin] (protocol/v4x-clients.md#server--client-ids); [bin] (protocol/v5x-clients.md#server--client-ids-500r7)
194	PLO_CLEARWEAPONS	empty; precedes the weapon list	4.20r8 clears its weapon and class script lists; 5.00r7 clears every loaded level and the level caches	[srv] (protocol/packet-ids.md#server--client-plo_); [bin] (protocol/v4x-clients.md#server--client-ids); [bin] (protocol/v5x-clients.md#server--client-ids-500r7)
195	PLO_LOADGANI	`str gani`, then the gani text	loaded as a server-sent gani script	[srv] (protocol/packet-ids.md#server--client-plo_); [bin] (protocol/v4x-clients.md#server--client-ids)
196	(no gs2lib name)	as 193	4.20r8 reads it and does nothing else	[bin] (protocol/v4x-clients.md#server--client-ids)

Id	Name	Layout	Client action	Source
197	PLO_LOADSCRIPT	a list `type,name,saveToDisk,desKey,crc32`; type weapon, npc, class or gani; the key is two i5	script delivery (→ §8)	[srv] (protocol/packet-ids.md#server--client-plo_); [bin] (protocol/v5x-clients.md#server--client-ids-500r7)

Rule	Builds	Source
For v4 and later clients the server replaces the script prop with `NPCPROP_CLASS` and sends GS2 bytecode; 2.05 never sees that path	server	[srv] (protocol/packet-layouts.md#weapons)

2.5.7 Files and update packages

The download sequence, sizes and failure handling are in → §2.7.

Id	Name	Layout	Client action	Source
30	PLO_FILESENDFAILED	unmeasured	unmeasured	[srv] (protocol/packet-ids.md#server--client-plo_)
45	PLO_FILEUPTODATE	unmeasured	unmeasured	[srv] (protocol/packet-ids.md#server--client-plo_)
68	PLO_LARGEFILESTART	`rest` = file name	stores the name; 2.31 also clears the stored size	[bin] (version-profiles/plo-layouts-217-231.tsv)
69	PLO_LARGEFILEEND	`rest` = file name	if it equals the stored name, the accumulated stream is processed as one received file	[bin] (version-profiles/plo-layouts-217-231.tsv)
84	PLO_LARGEFILESIZE	`i5 size`	stored	[bin] (version-profiles/plo-layouts-217-231.tsv)
102	PLO_FILE	behind 100 (→ §2.7)	a received file	[srv] (protocol/codec-and-framing.md#raw-data-escape)
105	PLO_UPDATEPACKAGESIZE	package name and an i5 size	sets a size or modification time on a cached object	[bin] (protocol/v4x-clients.md#server--client-ids); [bin] (protocol/v5x-clients.md#server--client-ids-500r7)
106	PLO_UPDATEPACKAGEDONE	unmeasured	fires `onUpdatePackageDownloaded`, and `onPackagesDownloadComplete` after the last one; 4.20r8 restarts itself when its own executable was in a package	[bin] (protocol/v4x-clients.md#server--client-ids); [bin] (protocol/v5x-clients.md#server--client-ids-500r7)
109	PLO_UNKNOWN109	i5 time, then a c-length string A and a c-length string B	passes (B, A, time) to the file layer; most likely a file redirect	[bin][inf] (protocol/v6x-clients.md#server--client-ids-601-603)
111	PLO_SETNETCOOKIE	`rest`, no fields	written byte for byte to `creationtime.dat` in the client's folder, replacing it; nothing is sent back	[bin][inf] (protocol/v6x-clients.md#server--client-ids-601-603)
187	PLO_UPDATEPACKAGEISUPDATED	unmeasured	a package-update message	[srv] (protocol/packet-ids.md#server--client-plo_)
198	PLO_SERVEROPTIONS	i5 value	stored on the connection; nothing else in 6.03 reads it	[bin] (protocol/v6x-clients.md#server--client-ids-601-603)

2.5.8 Messages, windows and modes

Chat display and the private-message rules are in → §2.8.

Id	Name	Layout	Client action	Source
13	PLO_TOALL	`s player id, str message`	shown as a toall message	[srv][2.05] (protocol/chat-pm-rc.md#toall)
32	PLO_SHOWIMGPLAYER	`s sender id` + the sender's id 24 payload, relayed unchanged	no visible reaction to any tested payload (→ §2.4.9)	[cap] (protocol/packet-ids.md#server--client-plo_); [cap] (version-profiles/lab-showimg24-relay.tsv)
35	PLO_RC_ADMINMESSAGE	`rest` = `Admin <account>:` + 0xA7 + text	shown as an admin message	[srv][2.05] (protocol/chat-pm-rc.md#admin-and-staff)
37	PLO_PRIVATEMESSAGE	`s from id`, `rest` = `"type","message"` as a quoted comma list; type is a label such as `"Server Message:"` or empty	shown as a PM	[srv][2.05] (protocol/chat-pm-rc.md#private-messages)
41	PLO_STARTMESSAGE	`rest` = RTF or HTML text	trimmed and stored as `servermessage.rtf` or `servermessage.html`; if it differs from the stored copy it is rewritten and shown in the start-message viewer	[bin] (version-profiles/plo-layouts-217-231.tsv)
47	PLO_STAFFGUILDS	comma text, `"guild"`, repeated	stored as the staff-guild list; sent at login	[srv][2.05] (protocol/chat-pm-rc.md#admin-and-staff); [bin] (version-profiles/plo-layouts-217-231.tsv)
75	PLO_PROFILE	repeated `c n` + n bytes (trimmed); field 0 = account	delivered to the open profile window of that player, or the best match in the player list; 2.05 needs at least 10 fields (→ §2.8)	[bin] (version-profiles/plo-layouts-217-231.tsv); [2.05] (protocol/chat-pm-rc.md#profiles)
82	PLO_SERVERTEXT	unmeasured	the reply to PLI_REQUESTTEXT and PLI_SENDTEXT; 3.153 raises it to script as `receivetext`; 5.00r7 also feeds the buddy list with it	[srv] (protocol/packet-ids.md#server--client-plo_); [bin] (protocol/v3x-clients.md#server--client-ids-handled); [bin][inf] (protocol/v5x-clients.md#server--client-ids-500r7)
153	PLO_SAY2	`rest` = text; 0xA7 = newline	shown as an NPC sign message; signs use it too	[srv][2.05] (protocol/packet-layouts.md#player-state-server--client); [bin] (version-profiles/plo-layouts-217-231.tsv)
169	PLO_UPGRADEURL	`rest` = URL	stored; with a disconnect, the reconnect button becomes "Upgrade" and opens it	[bin] (protocol/packet-ids.md#server--client-plo_)
171	PLO_BIGMAP	comma text `txt,png,x,y`; at least 4 fields are used; x and y as floats	the big map	[bin] (version-profiles/plo-layouts-217-231.tsv)
172	PLO_MINIMAP	as 171	the minimap	[bin] (version-profiles/plo-layouts-217-231.tsv)
176	PLO_KINGDOMSMODE	empty	switches the client to Kingdoms mode	[bin] (version-profiles/plo-layouts-217-231.tsv)
177	PLO_NEWWORLDMODE	empty	switches the client to New World mode	[bin] (version-profiles/plo-layouts-217-231.tsv)

Id	Name	Layout	Client action	Source
178	PLO_SERVERWARP	comma text, at least 4 fields: 0 unused, 1 server name, 2 host, 3 port	see the rules below	[bin][cap] (version-profiles/plo-layouts-217-231.tsv); [cap] (version-profiles/lab-serverwarp178.tsv)
179	PLO_RPGWINDOW	rest = text	creates the RPG window on first use and gives it the text	[bin] (version-profiles/plo-layouts-217-231.tsv)
180	PLO_STATUSLIST	rest = text	passed to the status-list window if it exists	[bin] (version-profiles/plo-layouts-217-231.tsv)
182	PLO_LISTPROCESSES	empty	the client answers with PLI_PROCESSLIST (44); 4.20r8 lists the titles of all open windows	[bin] (version-profiles/plo-layouts-217-231.tsv); [bin] (protocol/v4x-clients.md#server--client-ids)
183	PLO_HASPROCESSRUNNING	rest = names	the client answers with PLI_PROCESSRUNNING (45): 1 if the string occurs in its own list, else 0	[bin] (version-profiles/plo-layouts-217-231.tsv)
184	PLO_TAKESCREENSHOT	rest = file name	accepted only with a .png extension; saves a screenshot under the screenshots folder; sends nothing back	[bin] (version-profiles/plo-layouts-217-231.tsv)
190	PLO_SERVERLISTCONNECTED	unmeasured	fires onServerListerConnect: the Login server has accepted the session (→ §2.3)	[bin] (protocol/v4x-clients.md#server--client-ids); [bin] (protocol/v5x-clients.md#server--client-ids-500r7)
108	PLO_UNKNOWN108	a subtype, then data	4.20r6 and 4.20r8: subtype 0 makes the client answer with id 162 (→ App. E#5); subtype 1 stores a value on the connection	[bin] (protocol/v4x-clients.md#server--client-ids)
104	PLO_UNKNOWN104	str data holding c, i5, i5	unmeasured	[srv] (protocol/packet-ids.md#server--client-plo_)

Rule	Builds	Source
On 178 the client closes the old connection and connects to the host in field 2 and the port in field 3 with a fresh login (new key, same stored credentials), with no listserver step	2.31 (lab)	[cap] (version-profiles/plo-layouts-217-231.tsv); [cap] (version-profiles/lab-serverwarp178.tsv); [cap] (client-internals/decomp-findings-2026-10-24-b/08-lab-findings-levelchange-24-178.md#c-plo_serverwarp-178)
Field 1 of 178 is not needed: a bogus name still connects; a decoy host in field 0 is never contacted	2.31 (lab)	[cap] (version-profiles/lab-serverwarp178.tsv)
With host and port in other positions the client makes no connection, closes the old one and shows "The connection couldn't be established!"; with three fields it does nothing	2.31 (lab)	[cap] (version-profiles/lab-serverwarp178.tsv)
The old connection closes 2 to 3 ms before the new connection's SYN	2.31 (lab)	[cap] (version-profiles/lab-serverwarp178.tsv)
178 is split by the client's comma-text reader: white space (bytes 1 to 0x20) before an item is skipped; a quoted item ends at its closing quote, two quotes in a row are one quote, and an unclosed quote runs to the end; an unquoted item ends at a comma or a byte of 0x20 or less; a comma at the very end adds no item, an empty item between commas is kept; a NUL ends the text	2.17.1, 2.22, 2.31	[bin] (client-internals/decomp-findings-2026-10-25-g/comma-text-rules.tsv)
So a server name with a space must be quoted: x,Main Server,h,1 reads as five items and connects to the host Server	2.17.1, 2.22, 2.31	[bin] (client-internals/decomp-findings-2026-10-25-g/03-id178-comma-text.md#the-splitter)

Rule	Builds	Source
Item 1 becomes the current server's name, without a two-character type prefix when its second character is a space; 2.31 also puts it in the window caption, alone if it contains "graal" in any case, else after `Graal: `; the name is not used to connect	2.17.1 to 2.31; caption 2.31	[bin] (client-internals/decomp-findings-2026-10-25-g/03-id178-comma-text.md#the-connect-routine)
A real service's 178 text, item 0 in particular, has never been captured	all	[doc] (client-internals/decomp-findings-2026-10-25-g/04-open-points.md)
182 and 183 are a process scan used as an anti-cheat check (→ §4.10)	5.00r7	[bin][inf] (protocol/v5x-clients.md#server--client-ids-500r7)

2.5.9 Staff and NPC-control messages

The RC ids (50 to 54, 57 to 59, 61 to 67, 70 to 74, 76, 77, 103) and the NC ids (78, 80, 157 to 164, 167, 188, 192) are documented with the staff tools in → §7. Game clients handle a few of them:

Id	Name	Builds	Use in a game client	Source
67	`PLO_RC_FILEBROWSER_MESSAGE`	3.153, 4.x, 5.00r7, 6.x	raised to script as a folder-log event	[bin] (protocol/v3x-clients.md#server--client-ids-handled)
74	`PLO_RC_CHAT`	3.153, 4.x, 5.00r7, 6.x	raised to script as an RC-chat event	[bin] (protocol/v3x-clients.md#server--client-ids-handled)
79	`PLO_NPCSERVERADDR`	none in the notes	two bytes (0, 2), then `host,port` text; 5.x ignores it	[srv] (protocol/packet-ids.md#server--client-plo_)

2.5.10 Other ids

Id	Name	Layout	Client action	Source
60	`PLO_UNKNOWN60`	unmeasured	ignored by 5.07 and later	[srv] (protocol/packet-ids.md#server--client-plo_)
81	`PLO_UNKNOWN81`	unmeasured	ignored by 5.07 and later	[srv] (protocol/packet-ids.md#server--client-plo_)
83	`PLO_UNKNOWN83`	unmeasured	ignored by 5.07 and later	[srv] (protocol/packet-ids.md#server--client-plo_)
110	`PLO_UNKNOWN110`	unmeasured	ignored by clients	[srv] (protocol/packet-ids.md#server--client-plo_)
124	`PLO_UNKNOWN124`	unmeasured	seen by RC3; possibly player flags (untested)	[srv] (protocol/packet-ids.md#server--client-plo_)
132	`PLO_UNKNOWN132`	unmeasured	5.07 only; 6.037 ignores it; 3.153 uses it for script and class registration	[srv] (protocol/packet-ids.md#server--client-plo_); [bin] (protocol/v3x-clients.md#server--client-ids-handled)
252	`PLO_SET_ENC_KEY`	sets the AES or RC4 key and IV; above 223, so not sendable as one GCHAR (→ §2.10)	unmeasured	[srv] (protocol/packet-ids.md#server--client-plo_)
253	`PLO_BUNDLE`	several messages in one (→ §2.10)	unmeasured	[srv] (protocol/packet-ids.md#server--client-plo_)

2.6 Props

Props are the typed fields of players, NPCs and baddies. The messages that carry them are in → §2.4.4 and → §2.5; field types are in → §0.6. Every prop id is indexed in → App. B.

> Status: draft.

2.6.1 Prop streams

Rule	Builds	Source
A prop stream is `(c id, value)*` running to the end of the message, with no length per prop	all binary generations	[srv] (protocol/player-props.md)
Streams appear in `PLI_PLAYERPROPS` (2), `PLO_PLAYERPROPS` (9, the local player, no prefix), `PLO_OTHERPLPROPS` (8, `s player id` first), `PLO_NPCPROPS` (3, `i3 npc id` first), `PLO_BADDYPROPS` (2, `c baddy id` first) and, for NPC props, `PLO_NPCWEAPONADD` (33)	server; 2.05	[srv] (protocol/player-props.md); [2.05] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
A reader must know the encoding of every id up to the highest it expects, since nothing marks where a prop ends	all	[srv] (protocol/player-props.md)
Each client parser is a loop over `(id, value)` with a switch on the id; an id the switch does not know ends the loop, the rest of the message is ignored, and props already applied stay applied	2.05, 2.22, 2.31	[2.05][2.22][2.31] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
An unknown id therefore cannot crash these clients, but one unknown prop placed early hides every prop after it	2.05, 2.22, 2.31	[2.05] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
The other-player stream checks the remaining length before multi-byte props and clamps string lengths; the self stream does not length-check the fixed-size props 3, 13, 14 and 24, so a truncated one reads a few bytes of garbage before the loop ends	2.05	[2.05] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
The self stream needs at least 2 bytes left to start a prop; a lone trailing id byte is dropped	2.00, 2.05	[bin] (version-profiles/plo-playerprops-layouts.tsv)
The other-player prefix is two 7-bit bytes, `(b0 − 0x20) × 0x80 + (b1 − 0x20)`; an unknown id creates the player record	2.00 to 2.31	[bin] (version-profiles/plo-otherplprops-layouts.tsv)
The NPC prefix is three 7-bit bytes; the handler needs more than 2 bytes; ids below 1000 are level NPCs, 1000 and up server-wide	2.00 to 2.31	[bin] (version-profiles/plo-npcprops-layouts.tsv)

2.6.2 PLPROP (player)

Encodings use the short names of → §0.6; `c(v + k)` means the value plus *k*, sent as a GCHAR.

Id	Name	Encoding	Notes	Source
0	NICKNAME	str	—	[srv] (protocol/player-props.md#plprop-player)
1	MAXPOWER	c	heart containers	[srv] (protocol/player-props.md#plprop-player)
2	CURPOWER	c	hearts × 2	[srv] (protocol/player-props.md#plprop-player)

Id	Name	Encoding	Notes	Source
3	RUPEESCOUNT	i3	gralats	[srv] (protocol/player-props.md#plprop-player)
4	ARROWSCOUNT	c	—	[srv] (protocol/player-props.md#plprop-player)
5	BOMBSCOUNT	c	—	[srv] (protocol/player-props.md#plprop-player)
6	GLOVEPOWER	c	—	[srv] (protocol/player-props.md#plprop-player)
7	BOMBPOWER	c	—	[srv] (protocol/player-props.md#plprop-player)
8	SWORDPOWER	c(power + 30) + str(image)	—	[srv] (protocol/player-props.md#plprop-player)
9	SHIELDPOWER	c(power + 10) + str(image)	—	[srv] (protocol/player-props.md#plprop-player)
10	GANI / BOWGIF	str (≥ 2.1) · pre-2.1: c(10 + len) + image, or c(bowpower) if no image	**2.05 uses the pre-2.1 form**	[srv][inf] (protocol/player-props.md#plprop-player)
11	HEADGIF	c(len + 100) + image	—	[srv] (protocol/player-props.md#plprop-player)
12	CURCHAT	str	chat bubble text	[srv] (protocol/player-props.md#plprop-player)
13	COLORS	5 × c	colour indexes in the order skin, coat, sleeves, shoes, belt (→ §2.6.3)	[srv] (protocol/player-props.md#plprop-player)
14	ID	s	—	[srv] (protocol/player-props.md#plprop-player)
15	X	c(x × 2)	half-tile	[srv] (protocol/player-props.md#plprop-player)
16	Y	c(y × 2)	—	[srv] (protocol/player-props.md#plprop-player)
17	SPRITE	c	% 4 = direction	[srv] (protocol/player-props.md#plprop-player)
18	STATUS	c	bit flags, see below	[srv] (protocol/player-props.md#plprop-player)
19	CARRYSPRITE	c	—	[srv] (protocol/player-props.md#plprop-player)
20	CURLEVEL	str	gmap name when on a gmap	[srv] (protocol/player-props.md#plprop-player)
21	HORSEGIF	str	—	[srv] (protocol/player-props.md#plprop-player)
22	HORSEBUSHES	c	—	[srv] (protocol/player-props.md#plprop-player)

Id	Name	Encoding	Notes	Source
23	EFFECTCOLORS	c	server always sends 0	[srv] (protocol/player-props.md#plprop-player)
24	CARRYNPC	i3	npc id	[srv] (protocol/player-props.md#plprop-player)
25	APCOUNTER	s(+1)	—	[srv] (protocol/player-props.md#plprop-player)
26	MAGICPOINTS	c	—	[srv] (protocol/player-props.md#plprop-player)
27	KILLSCOUNT	i3	—	[srv] (protocol/player-props.md#plprop-player)
28	DEATHSCOUNT	i3	—	[srv] (protocol/player-props.md#plprop-player)
29	ONLINESECS	i3	—	[srv] (protocol/player-props.md#plprop-player)
30	IPADDR	i5	—	[srv] (protocol/player-props.md#plprop-player)
31	UDPPORT	i3	—	[srv] (protocol/player-props.md#plprop-player)
32	ALIGNMENT	c	AP 0–100	[srv] (protocol/player-props.md#plprop-player)
33	ADDITFLAGS	c	—	[srv] (protocol/player-props.md#plprop-player)
34	ACCOUNTNAME	str	—	[srv] (protocol/player-props.md#plprop-player)
35	BODYIMG	str	—	[srv] (protocol/player-props.md#plprop-player)
36	RATING	i3	`(rating & 0xFFF) << 9 | (deviation & 0x1FF)`	[srv] (protocol/player-props.md#plprop-player)
37–41	GATTRIB1–5	str (≤ 223)	—	[srv] (protocol/player-props.md#plprop-player)
42	ATTACHNPC	c(type, 0 = npc) + i3(id)	—	[srv] (protocol/player-props.md#plprop-player)
43/44	GMAPLEVELX / Y	c	—	[srv] (protocol/player-props.md#plprop-player)
45	Z	c(z + 0.5 + 50)	—	[srv] (protocol/player-props.md#plprop-player)
46–49	GATTRIB6–9	str	—	[srv] (protocol/player-props.md#plprop-player)
50	JOINLEAVELVL	c	1 join, 0 leave	[srv] (protocol/player-props.md#plprop-player)

Id	Name	Encoding	Notes	Source
51	PCONNECTED	—	no payload; marks disconnect	[srv] (protocol/player-props.md#plprop-player)
52	PLANGUAGE	str	—	[srv] (protocol/player-props.md#plprop-player)
53	PSTATUSMSG	c	index into the server's status list	[srv] (protocol/player-props.md#plprop-player)
54–74	GATTRIB10–30	str	—	[srv] (protocol/player-props.md#plprop-player)
75	OSTYPE	str	2.19+ (`wind`)	[srv] (protocol/player-props.md#plprop-player)
76	TEXTCODEPAGE	i3	2.19+	[srv] (protocol/player-props.md#plprop-player)
77	UNKNOWN77	?	no client checked parses it (2.05, 2.22, 2.31); sending it aborts the rest of the stream	[srv][2.05][2.22][2.31] (protocol/player-props.md#plprop-player)
78–80	X2 / Y2 / Z2	s: `abs(v × 16) << 1`, low bit = sign	pixel precision, newer clients only	[srv] (protocol/player-props.md#plprop-player)
81	UNKNOWN81	c	player-list tab flag (unconfirmed). Above the highest id that 2.31 parses (80), so 2.31 and older stop reading at it	[srv][2.31] (protocol/player-props.md#plprop-player)
82	COMMUNITYNAME	str	—	[srv] (protocol/player-props.md#plprop-player)

2.6.2.1 STATUS bits (prop 18)

Bit	Meaning	Source
`0x01`	paused	[srv] (protocol/player-props.md#status-bits-prop-18-srv)
`0x02`	hidden	[srv] (protocol/player-props.md#status-bits-prop-18-srv)
`0x04`	male	[srv] (protocol/player-props.md#status-bits-prop-18-srv)
`0x08`	dead	[srv] (protocol/player-props.md#status-bits-prop-18-srv)
`0x10`	allow weapons	[srv] (protocol/player-props.md#status-bits-prop-18-srv)
`0x20`	hide sword	[srv] (protocol/player-props.md#status-bits-prop-18-srv)
`0x40`	has spin attack	[srv] (protocol/player-props.md#status-bits-prop-18-srv)

2.6.2.2 ADDITFLAGS bits (prop 33)

Bit	Meaning	Source
`0x01`	no mass message	[srv] (protocol/player-props.md#additflags-srv)
`0x02`	only show one level	[srv] (protocol/player-props.md#additflags-srv)
`0x04`	no toall	[srv] (protocol/player-props.md#additflags-srv)
`0x08`	voice disabled	[srv] (protocol/player-props.md#additflags-srv)

2.6.2.3 Encodings as 2.05 reads them

Prop	Rule	Source
8 SWORDPOWER	0–4 = the default `sword<n>.gif`; otherwise power + 30, then `str` image	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)

Prop	Rule	Source
9 SHIELDPOWER	0–3 = the defaults; otherwise power + 10, then `str` image	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)
10	the bow image or weapon (pre-2.1 BOWGIF), not GANI	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)
11 HEADGIF	`c n` below 100 → head<n>.gif; otherwise `c (len + 100)` + name, at most 100 bytes	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)
15, 16 X, Y	one GCHAR each, half-tiles	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)
17 SPRITE	direction + 4 × animation index; `0x80` in one special state	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)
19 CARRYSPRITE	−1 none, 0, 200, 201, or tile-based values	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)
30, 31 IPADDR, UDPPORT	feed the UDP peer channel (→ §2.9)	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)
36 RATING	rating = `b0 × 32 + ((b1 & 0x7F) >> 2)`, deviation = `(b1 & 3) × 128 + b2`	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)
51 PCONNECTED	no payload, but 2.05 skips one byte after it; harmless because the player is removed at once, but a server should put no props behind it	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)
7, 23	sent as constants: bomb power * (10) and effect colours 0	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)

2.6.3 Body colours and attributes

Rule	Builds	Source
A body picture is a palette image; before drawing, palette entry 5 takes the skin colour, 6 the coat, 4 the sleeves, 3 the shoes and 1 the belt; entry 0 always takes colour 18 (black, the outline)	2.05	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
The five colours of prop 13 are in the order skin, coat, sleeves, shoes, belt	2.05	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
The stock bodies follow that palette layout, and entry 2 is the clear colour	2.22 files	[file] (client-internals/body-colours-and-attributes.md#body-colours)
A new player is orange, darkred, black, darkred, black (2, 5, 18, 5, 18), with `body.png` and `head0.gif`; GServer-v2 gives a new account 2, 0, 10, 4, 18 instead	2.05; server	[2.05][srv] (client-internals/body-colours-and-attributes.md#body-colours)
A GANI sprite whose picture is ATTR1 to ATTR5 draws the player's attribute 1 to 5, which are props 37 to 41; 2.05 reads them for other players too	2.05	[2.05] (client-internals/body-colours-and-attributes.md#gani-attributes)
A player with no attribute of their own draws the GANI's DEFAULTATTR1 to DEFAULTATTR5; the stock `idle.gani` names `hat0.png` as attribute 1	2.22 files	[file] (client-internals/body-colours-and-attributes.md#gani-attributes)
GANI attribute props are strings that GANI PARAM/ATTR placeholders resolve against; 2.05 sends and receives GATTRIB1–5 but its expression table has no `gattrib` entry	2.05	[inf][2.05] (protocol/packet-layouts.md#gani)
On the wire, animation is only a prop: PLPROP_GANI (10) and NPCPROP_GANI (12) carry the GANI name, and the client fetches the file with PLI_WANTFILE	server	[srv] (protocol/packet-layouts.md#gani)

Index	Colour	RGB	Source
0	white	255, 255, 255	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
1	yellow	255, 255, 0	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
2	orange	255, 173, 107	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
3	pink	255, 132, 132	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
4	red	255, 0, 0	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
5	darkred	206, 24, 41	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
6	lightgreen	132, 255, 132	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
7	green	0, 255, 0	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
8	darkgreen	0, 198, 0	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
9	lightblue	132, 132, 255	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
10	blue	0, 0, 255	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
11	darkblue	0, 0, 198	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
12	brown	132, 0, 0	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
13	cynober	0, 255, 255	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
14	purple	255, 0, 255	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
15	darkpurple	132, 0, 132	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
16	lightgray	206, 206, 206	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
17	gray	132, 132, 132	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
18	black	0, 0, 8	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)
19	transparent	0, 0, 0	[2.05] (client-internals/body-colours-and-attributes.md#body-colours)

The looks commands that change these props are GS1 behaviour (→ §8).

2.6.4 NPCPROP

Id	Name	Encoding	Source
0	IMAGE	str	[srv] (protocol/player-props.md#npcprop)
1	SCRIPT	s(len ≤ 0x3FFF) + text	[srv] (protocol/player-props.md#npcprop)
2	X	c(px / 8) = half-tiles	[srv] (protocol/player-props.md#npcprop)
3	Y	c(px / 8)	[srv] (protocol/player-props.md#npcprop)
4	POWER	c	[srv] (protocol/player-props.md#npcprop)
5	RUPEES	i3	[srv] (protocol/player-props.md#npcprop)
6	ARROWS	c	[srv] (protocol/player-props.md#npcprop)
7	BOMBS	c	[srv] (protocol/player-props.md#npcprop)
8	GLOVEPOWER	c	[srv] (protocol/player-props.md#npcprop)
9	BOMBPOWER	c	[srv] (protocol/player-props.md#npcprop)
10	SWORDIMAGE	c(power + 30) + str, or c(0)	[srv] (protocol/player-props.md#npcprop)
11	SHIELDIMAGE	c(power + 10) + str, or c(0)	[srv] (protocol/player-props.md#npcprop)
12	GANI / BOWGIF	str	[srv] (protocol/player-props.md#npcprop)
13	VISFLAGS	c	[srv] (protocol/player-props.md#npcprop)

Id	Name	Encoding	Source
14	BLOCKFLAGS	c	[srv] (protocol/player-props.md#npcprop)
15	MESSAGE	str	[srv] (protocol/player-props.md#npcprop)
16	HURTDXDY	c(dx × 32 + 32), c(dy × 32 + 32)	[srv] (protocol/player-props.md#npcprop)
17	ID	i3	[srv] (protocol/player-props.md#npcprop)
18	SPRITE	c (pre-2.x clients get `% 4`)	[srv] (protocol/player-props.md#npcprop)
19	COLORS	5 × c	[srv] (protocol/player-props.md#npcprop)
20	NICKNAME	str	[srv] (protocol/player-props.md#npcprop)
21	HORSEIMAGE	str	[srv] (protocol/player-props.md#npcprop)
22	HEADIMAGE	c(len + 100) + image	[srv] (protocol/player-props.md#npcprop)
23–32	SAVE0–9	c each	[srv] (protocol/player-props.md#npcprop)
33	ALIGNMENT	c	[srv] (protocol/player-props.md#npcprop)
34	IMAGEPART	`s x, s y, c w, c h` (x, y clamped to 0–16000, w, h to 0–220)	[srv][2.05] (protocol/player-props.md#npcprop)
35	BODYIMAGE	str	[srv] (protocol/player-props.md#npcprop)
36–40	GATTRIB1–5	str	[srv] (protocol/player-props.md#npcprop)
41 / 42	GMAPLEVELX / Y	c	[srv] (protocol/player-props.md#npcprop)
43	Z	c(z + 50), z −50..170 (2.22, 2.31)	[srv][2.22] (protocol/player-props.md#npcprop)
44–47	GATTRIB6–9	str	[srv] (protocol/player-props.md#npcprop)
48	UNKNOWN48	not parsed by 2.05, 2.22 or 2.31; aborts the stream	[srv][2.22][2.31] (protocol/player-props.md#npcprop)
49–52	SCRIPTER, NAME, TYPE, CURLEVEL	str	[srv] (protocol/player-props.md#npcprop)
53–73	GATTRIB10–30	str	[srv] (protocol/player-props.md#npcprop)
74	CLASS	NPC-server class	[srv] (protocol/player-props.md#npcprop)
75 / 76	X2 / Y2	as PLPROP X2/Y2	[srv] (protocol/player-props.md#npcprop)
77	Z2	2 bytes, decoded like X2/Y2; sets the same Z (2.31)	[srv][2.31] (protocol/player-props.md#npcprop)

Rule	Builds	Source
The SCRIPT prop (1) carries GS1 source text, not bytecode, for pre-4.0 clients	server; 2.05	[srv][2.05] (protocol/player-props.md#npcprop)
SCRIPT is applied only while the NPC has no script yet; a second SCRIPT for the same NPC is ignored	2.05	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)
NPCPROP 43 is Z, `c (z + 50)` with z from −50 to 170, from 2.22 on; NPCPROP 77 is Z2 in 2.31	2.22, 2.31	[2.22][2.31] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)

2.6.5 BDPROP (baddy)

Id	Name	Encoding and notes	Source
0	ID	—	[srv] (protocol/player-props.md#bdprop-baddy)
1 / 2	X / Y	c(tiles × 2)	[srv] (protocol/player-props.md#bdprop-baddy)
3	TYPE	0 graysoldier … 9 dragon (see GS1 baddy names)	[srv] (protocol/player-props.md#bdprop-baddy)
4	POWERIMAGE	c(power) + str(image); an empty image means the default for the type	[srv][2.05] (protocol/player-props.md#bdprop-baddy)
5	MODE	0 walk, 1 look, 2 hunt, 3 hurt, 4 bumped, 5 die, 6 swamp-shot, 7 hare-jump, 9 dead; the 2.22 docs add 8 spider-shoot. On mode 5 the server switches to 9 after 2 s	[srv][doc] (protocol/player-props.md#bdprop-baddy)
6	ANI	c	[srv] (protocol/player-props.md#bdprop-baddy)
7	DIR	c(a × 4 + b): two directions in one byte, `a = v >> 2`, `b = v & 3` (which one is the head is not traced)	[srv][2.05] (protocol/player-props.md#bdprop-baddy)
8–10	VERSESIGHT / VERSEHURT / VERSEATTACK	str each	[srv] (protocol/player-props.md#bdprop-baddy)

The baddy verses come from the backslash-separated verse line of the level file (→ §3) [srv] (protocol/player-props.md#bdprop-baddy).

2.6.6 Ids each parser accepts

An id outside a parser's accepted set ends that message's stream (→ §2.6.1).

Stream	Build	Accepted ids	Stops at	Source
self (9)	2.00	0–35; disputed → App. E#8	36 and up	[bin] (version-profiles/plo-playerprops-layouts.tsv)
self (9)	2.04	between the 2.00 and 2.05 sets; not re-read	unmeasured	[bin] (version-profiles/plo-playerprops-layouts.tsv)
self (9)	2.05	0–42	43 and up, including 50 and 51	[2.05] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
self (9)	2.17.1, 2.22	through 74	50, 51, 53	[bin] (version-profiles/plo-playerprops-layouts.tsv); [2.22] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
self (9)	2.31	through 80, with X2, Y2, Z2 at 78–80	50, 51, 53, 75–77	[2.31] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
other players (8)	2.00	0–35	36 and up	[bin] (version-profiles/plo-otherplprops-layouts.tsv)
other players (8)	2.05	0, 2, 8–13, 15–21, 24, 30–42, 50, 51	1, 3–7, 14, 22, 23, 25–29, 43–49, 52 and up	[2.05] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)

Stream	Build	Accepted ids	Stops at	Source
other players (8)	2.22	highest 76	1, 3–7, 14, 22, 25–29	[2.22] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
other players (8)	2.31	0, 2, 8–13, 15–21, 23, 24, 30–76, 78–80	1, 3–7, 14, 22, 25–29, 77	[bin] (version-profiles/plo-otherplprops-layouts.tsv)
NPC (3, 33)	2.00	0–36; disputed → App. E#8	37 and up	[bin] (version-profiles/plo-npcprops-layouts.tsv)
NPC (3, 33)	2.05	0–40	41 and up	[2.05] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
NPC (3)	2.17.1	0–40, 44–47, 53–73	41–43, 48–52	[bin] (version-profiles/plo-npcprops-layouts.tsv)
NPC (3)	2.22	highest 73, including 43	48–52	[bin] (version-profiles/plo-npcprops-layouts.tsv); [2.22] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
NPC (3)	2.31	highest 77	48–52	[2.31] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
baddy (2)	2.05	0–10	11 and up	[2.05] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)

Rule	Builds	Source
A server that sends a 2.05 client another player's MAXPOWER (1), RUPEES (3), ID (14), HORSEBUSHES (22) or a counter (25–29) loses every prop after it in that message; send those to the player only, or last	2.05	[2.05] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
2.05 knows PLPROP 0–42, 50 and 51 only; it sends 0–24, 26, 31–33, 35 and 37–42, and only receives 25, 27–30, 34, 36, 50 and 51	2.05	[2.05] (protocol/player-props.md#205-coverage-update-2026-10-21-205)
2.05 sends props 0, 11, 13, 18, 31 and 33 together at session start, and 42, 15, 16 together on attach or a move	2.05	[2.05] (protocol/player-props.md#plprop-player)
No client checked parses PLPROP 77 or 81, or NPCPROP 48	2.05, 2.22, 2.31	[2.05][2.22][2.31] (protocol/player-props.md#unknown-prop-ids-205-checked-against-222-and-231)
The 2.22 send-side prop builder also knows 75 and 76	2.22	[bin] (version-profiles/plo-playerprops-layouts.tsv)
From 2.3 GServer-v2 puts X2 and Y2 before X and Y in other-player props	server	[srv] (version-profiles/2.22-2.31.md#wire-protocol)

Prop ranges for the 1.21 to 1.32 builds are in → App. C.6, and parser growth from 1.33 to the 2.1 tests in → App. C.3. Two notes disagree on the 2.00 ranges → App. E#8.

2.7 File transfer

The request messages are laid out in → §2.4.7 and the replies in → §2.5.7; the raw-data escape is in → §2.1.3. This chapter gives the sequence and the version rules.

> Status: draft.

2.7.1 Requests and replies

Request	Reply when the file is sent	Reply otherwise	Builds	Source
23 `PLI_WANTFILE` (name)	100 `RAWDATA` + 102 `FILE`	30 `FILESENDFAILED` (name)	1.21 to 6.x	[srv] (protocol/file-transfer.md#requests-client--server)
34 `PLI_UPDATEFILE` (i5 cached modtime, name)	100 + 102 when the modtime differs	below 2.1: 30; 2.1 and later: 45 `FILEUPTODATE`	1.42 to 6.x	[srv] (protocol/file-transfer.md#requests-client--server)
47 `PLI_VERIFYWANTSEND` (i5 `crc32`, name)	100 + 102 when the client's copy is stale	unmeasured	5.00r7, 5.24, 6.x	[srv] (protocol/file-transfer.md#requests-client--server); [cap] (protocol/file-transfer.md#what-5x-asks-for-at-login-cap)

Rule	Builds	Source
Below 2.1 the server adds `.gif` to a requested name with no extension	server	[srv] (protocol/file-transfer.md#requests-client--server)
Below 2.1, 30 also means "already up to date"; 2.05 handles 45, but the server does not send it to 2.05	server; 2.05	[srv] (protocol/file-transfer.md#replies-server--client)
Game clients cannot upload; RC upload uses the file-browser ids 89 to 98 and 65 to 67 (→ §7); disputed for 6.x → App. E#9	server	[srv] (protocol/file-transfer.md#upload-rc-only)

2.7.2 The `PLO_FILE` body

Client version (by the server's token table)	Layout behind 100	Source
below 2.1 (2.05)	`c 102`, `str filename`, then the file bytes; no modtime and no trailing `\n`; the RAWDATA length is 1 + 1 + name + data	[srv] (protocol/file-transfer.md#plo_file-body-srv)
2.1 to 2.13	`c 102`, `i5 modtime`, `str filename`, the bytes, then `\n`, in one piece	[srv] (protocol/file-transfer.md#plo_file-body-srv)
2.14 and later	as 2.1, split into 32000-byte chunks; a file over 32000 bytes is wrapped in 68 `LARGEFILESTART`, 84 `LARGEFILESIZE` and 69 `LARGEFILEEND`	[srv] (protocol/file-transfer.md#plo_file-body-srv)

Rule	Builds	Source
A client before 2.14 cannot receive a file over 64000 bytes; the server answers 30 instead	server	[srv] (protocol/file-transfer.md#plo_file-body-srv)
A 541,248-byte file reached 5.00r7 as 68, 84, then 102 chunks each with modtime and name, then 69, and was saved byte for byte	5.00r7 (lab, stock gs2lib)	[cap][bin] (protocol/file-transfer.md#what-5x-asks-for-at-login-cap)
On 69 the client compares the name with the stored one and processes the accumulated stream as one file (→ §2.5.7)	2.17.1 to 2.31	[bin] (version-profiles/plo-layouts-217-231.tsv)

2.7.3 The layout follows the server's token table

Rule	Builds	Source
Every version branch in the server's file code tests the version id, which comes only from the login token looked up in the server library's table; a token that is allowed but not in the table gets version 0, below every real version	server	[srv] (protocol/file-transfer.md#the-layout-follows-the-servers-token-table-not-the-client-2026-10-31)
Such a client gets every pre-2.1 path: `PL0_FILE` without modtime or `\n`, 30 for up to date, `.gif` added to names, the pre-2.1 level sequence with no 156, and no large-file messages	server	[srv] (protocol/file-transfer.md#the-layout-follows-the-servers-token-table-not-the-client-2026-10-31)
A version's rank is its position in the table; the server compares positions	server library	[src] (protocol/file-transfer.md#gs2libs-version-table-src)
The table holds, in order: `G3D22067` (5.00r7), `G3D14097` (5.12), `G3D26090` (5.30r3, 5.31r4, 6.00r2, 6.00r4), `G3D3007A` (6.01x), `G3D2505C` (6.034, 6.035), `G3D0311C` (6.03r7 Windows), `G3D0511C` (6.03r7 Linux); all sit above the 2.14 slot	server library	[src] (protocol/file-transfer.md#gs2libs-version-table-src)
`G3D15010` (5.24r3, 5.24r4, 6.00 beta2) and `G3D03048` (6.00 beta1) are not in the stock table	server library	[src] (protocol/file-transfer.md#gs2libs-version-table-src)
The client always reads `PL0_FILE` as `i5 modtime` + `str filename` + data; given the old layout it takes the name's length byte and first 4 characters as the modtime, and the 5th character as the length	5.24	[srv][inf] (protocol/file-transfer.md#the-524-png-as-file-name-bug)
For `sword1.png` that gives the error "Couldn't open ...1.png‰PNG... for writing"; with a table that knows `G3D15010` the same server's files were saved normally	5.24r4 (lab)	[cap] (protocol/file-transfer.md#the-524-png-as-file-name-bug)
A server for 5.x and 6.x clients must know every token it allows	server	[src] (protocol/file-transfer.md#gs2libs-version-table-src)
A client cannot tell the server which layout to use	all	[inf] (protocol/file-transfer.md#gs2libs-version-table-src)

2.7.4 Requests at login

Build	Requests seen	Source
2.05	for each file a level references (for example `head25.png`, `sword1.png`, `walk.gani`): 34 when cached, else 23; the server answers 100 + 102 or 30	[srv][inf][cap] (protocol/file-transfer.md#typical-flow-for-205-srvinf)
2.22	34 for the default images (sword, shield, body, bomb), then 23 (→ §2.4.11)	[cap] (version-profiles/2.22-2.31.md#wire-protocol)
5.24r4, 5.30r3, 6.03r7	`basepackage.gupd` first; with an empty cache as 23, twice	[cap] (protocol/file-transfer.md#what-5x-asks-for-at-login-cap)
5.00r7	`basepackage.gupd` as 47 when that server's copy is cached and as 23 when not; the cache is kept per server name	[cap] (protocol/file-transfer.md#what-5x-asks-for-at-login-cap)
5.x	then 34 for `idle.gani`, `sprites.png`, `body.png`, `head0.png`, `shield1.png` and the `guigames_*` images, and 23 for the fonts (`arial.ttf`, `arialbd.ttf`) and `tiledefs<server>.txt`	[cap] (protocol/file-transfer.md#what-5x-asks-for-at-login-cap)

Update packages and their messages (105, 106, 187) are in → §2.5.7.

2.7.5 The client's request lists

Rule	Builds	Source
The client keeps three name lists: wanted (asked with 23), update-asked (asked with 34) and settled (a reply came back)	2.00, 2.05 rev 2, 2.17.1, 2.22, 2.31	[bin] (client-internals/decomp-findings-2026-10-25-e/01-request-lists.md#lists)
No local copy: the name is appended to the wanted list unless already there, and sent while that list holds at most four names	same	[bin] (client-internals/decomp-findings-2026-10-25-e/01-request-lists.md#asking)
A local copy: the name is appended to the update-asked list unless it is there or settled, and sent while that list holds at most four; so each local file is checked at most once per connection	same	[bin] (client-internals/decomp-findings-2026-10-25-e/01-request-lists.md#asking)
The time in 34 is the local file's last-write time in Unix seconds; at login it matched the files' times for the sword, shield and body images	same; 2.22 (lab)	[bin][cap] (client-internals/decomp-findings-2026-10-25-e/01-request-lists.md#asking)
Builds with no id 34 never check a local copy	1.33 to 1.41	[bin] (client-internals/decomp-findings-2026-10-25-e/01-request-lists.md#asking)
Any reply (the file, up to date, send failed, or the end of a large file) removes the name from both lists and adds it to the settled list; if the name was among the first four, the name now fourth goes out; the two lists release independently	same	[bin] (client-internals/decomp-findings-2026-10-25-e/01-request-lists.md#replies)
Four of each kind were on the wire before the first reply	2.22 (lab)	[cap] (client-internals/decomp-findings-2026-10-25-e/README.md#top-facts)
The client never compares times; the server decides	same	[bin] (client-internals/decomp-findings-2026-10-25-e/README.md#top-facts)
A name answered with 30 is not asked for again on that connection: a missing step sound was asked for once however often the player walked	2.22 (lab)	[cap] (client-internals/decomp-findings-2026-10-25-e/01-request-lists.md#failed-names)
Images and ganis: a local file is used and checked, a missing one asked for; sounds: asked for only when missing, never checked; levels: no file transfer (board messages and the level modtime instead, → §3.8); a gmap: checked when present, asked for when missing	2.05 rev 2, 2.22; gmap 2.22 (lab)	[bin][cap] (client-internals/decomp-findings-2026-10-25-e/01-request-lists.md#who-asks)
In 2.05 rev 2 and 2.22 the request side was read from the matching reply-side code and the lab	2.05 rev 2, 2.22	[inf] (client-internals/decomp-findings-2026-10-25-e/request-rules.tsv)

2.7.6 Saving a download

Rule	Builds	Source
A received file is held in memory and written to disk only when saveimages is on (default on, → §3.9.3)	2.05 rev 2 to 2.31	[bin] (client-internals/decomp-findings-2026-10-25-e/02-save-rules.md#whether-a-file-is-written)
An empty or missing stream is never saved: nothing is written or deleted and the loaders are not told	1.39, 2.00, 2.05 rev 2, 2.17.1, 2.22, 2.31	[bin] (client-internals/decomp-findings-2026-10-25-k/02-save-folders-by-build.md#the-guard)
A file the client already has is replaced where it is; a new one is placed by name, compared without case	2.00 and later	[bin] (client-internals/decomp-findings-2026-10-25-e/02-save-rules.md#where)
From 2.1 the saved file gets the time from the reply, so the next check carries the server's time; before 2.1 it keeps the download time	2.1 and later; earlier	[bin] (client-internals/decomp-findings-2026-10-25-e/02-save-rules.md#time-of-the-saved-file)

Where a new download is saved, by build:

Builds	Folders	Source
1.33 to 1.41	everything in levels/webgifs; 1.39 saves only .gif, .png and .mng, deletes an old copy first and sets no time	[bin] (client-internals/decomp-findings-2026-10-25-k/02-save-folders-by-build.md#layouts)
1.42 NewMain	the earlier build adds ganis, heads, swords, shields; the later adds sounds for .wav and bomys	[bin] (client-internals/decomp-findings-2026-10-25-k/02-save-folders-by-build.md#layouts)
2.00, 2.01	sounds for .wav; under levels: bomys (names starting bomy or b_), ganis (.gani), emoticons (names starting emoticon), heads, bodies, swords, shields (names containing those words), else webgifs	[bin] (client-internals/decomp-findings-2026-10-25-k/02-save-folders-by-build.md#layouts)
2.02 to 2.05 rev 2, both 2.1 test builds	as 2.00, plus hats (hat then a digit as the fourth character)	[bin] (client-internals/decomp-findings-2026-10-25-k/02-save-folders-by-build.md#layouts)
2.17.1, 2.22, 2.31	as 2.05, plus maps for .gmap and for .txt, .gif, .png, .mng names containing map; a 2.22 gmap download was saved as maps/<name>	[bin][cap] (client-internals/decomp-findings-2026-10-25-k/save-folders-by-build.tsv); [cap] (client-internals/decomp-findings-2026-10-25-e/02-save-rules.md#where)
2.1 to 2.16	no binary; when maps arrived is unknown (→ App. E#10)	[doc] (client-internals/decomp-findings-2026-10-25-k/02-save-folders-by-build.md#layouts)

2.7.7 Finding a local file

Rule	Builds	Source
Names are folded A to Z only, then compared byte for byte; accented letters do not match across case	2.05 rev 2, 2.22	[bin] (client-internals/decomp-findings-2026-10-25-e/04-name-index-case-link.md#case-fold)
A file of that name in the current directory wins first: the game folder in 2.22, the levels folder in 2.05 rev 2 (or a world opened from elsewhere)	2.05 rev 2, 2.22	[bin] (client-internals/decomp-findings-2026-10-25-e/04-name-index-case-link.md#which-copy-is-used)
2.22 indexes the game folder at startup, files before subdirectories in listing order, skipping top-level pms, profiles and weblevels and .code files; for a duplicate name the newer last-write time wins, and an equal time keeps the first found	2.22	[bin] (client-internals/decomp-findings-2026-10-25-e/04-name-index-case-link.md#which-copy-is-used)
2.05 rev 2 looks up on demand: the current directory, then a remembered path, then a depth-first search of an opened world's folder and the game folder with the same skips; the first hit wins and is remembered	2.05 rev 2	[bin] (client-internals/decomp-findings-2026-10-25-e/04-name-index-case-link.md#which-copy-is-used)

Rule	Builds	Source
For two different files of one name, 2.22 draws the newer and 2.05 rev 2 the first found	2.05 rev 2, 2.22	[bin] (client-internals/decomp-findings-2026-10-25-e/name-index-rules.tsv)

2.8 Chat and private messages

The message layouts are in → §2.4.8 (client to server) and → §2.5.8 (server to client). How the client draws chat and nicknames is behaviour, in → §4.

> Status: draft.

2.8.1 Local chat

Rule	Builds	Source
Local chat is not a message: it is the CURCHAT prop (12) inside PLI_PLAYERPROPS; the server forwards it to the level as PLO_OTHERPLPROPS	server; 2.05	[srv] (protocol/chat-pm-rc.md#chat-bubble)
Server commands typed in chat (`setnick`, `/sethead` and the like) are parsed by the server from the same prop	server	[srv] (protocol/chat-pm-rc.md#chat-bubble)
When the chat-bar text, lower-cased and trimmed, starts with `toall:`, 2.05 sends the rest as PLI_TOALL (6) instead	2.05	[2.05] (protocol/chat-pm-rc.md#chat-bubble)
A line longer than 256 characters is cut to its first 253 characters plus `...` when sent; the receiver draws whatever arrives	2.05	[2.05] (client-internals/chat-text.md#length)
When the player's own line times out (3 s after the first move), the client sends an empty CURCHAT	2.05	[2.05][cap] (client-internals/chat-text.md#how-long-it-stays-live-205-rev-2-2026-10-30)
A typed `showname` is not sent to the server	2.05	[2.05][cap] (client-internals/chat-text.md#nickname-131-to-231)
Chat also goes peer to peer as UDP id 22 (→ §2.9)	2.00 to 2.05, 3.04	[2.05] (protocol/udp-p2p.md#udp-ids-own-dispatcher-not-plipl-numbering)

2.8.2 Private messages

Rule	Builds	Source
A PM names its recipients by player id (`s count`, then the ids), so one message can go to several players	server; 2.05	[srv][2.05] (protocol/chat-pm-rc.md#private-messages)
Mass PMs and guild PMs use the same message with different type labels, such as `"Server Message:"` or `"Guild message:"`	server	[srv] (protocol/chat-pm-rc.md#private-messages)
Players in jail can PM only RCs	server	[srv] (protocol/chat-pm-rc.md#private-messages)
ADDITFLAGS bit `0x01` blocks mass messages and `0x04` blocks toall (→ §2.6.2.2)	server	[srv] (protocol/player-props.md#additflags-srv)

2.8.3 Staff messages

Rule	Builds	Source
An admin message reaches a game client as PLO_RC_ADMINMESSAGE (35), `Admin <account>:` followed by `0xA7` and the text	server; 2.05	[srv][2.05] (protocol/chat-pm-rc.md#admin-and-staff)
The staff-guild list is sent at login as PLO_STAFFGUILDS (47)	server; 2.05	[srv][2.05] (protocol/chat-pm-rc.md#admin-and-staff)
The staff tools run their own logins: RC `GSERV023` to `GSERV025`, NC `NCL11012`, RC2 as type 6 on generation 5 (→ §7)	server	[srv] (protocol/chat-pm-rc.md#admin-and-staff)

2.8.4 Buddy and list text requests

Rule	Builds	Source
Buddy requests go in `PLI_SENDTEXT` as the tokens `GraalEngine`, `lister`, then `verifybuddies`, `addbuddy` or `deletebuddy`, then the account list	2.17+; server	[srv] (client-internals/feature-specs/14-friends-ignore.md#server-behaviour-srv)
The server forwards those three options to the listserver, never for guest accounts, and stores nothing itself	server	[srv] (client-internals/feature-specs/14-friends-ignore.md#server-behaviour-srv)
The listserver answers with `PLO_SERVERTEXT` (82) carrying `weapon`, `lister`, `buddylist`, then a comma list of accounts	listserver	[srv] (client-internals/feature-specs/14-friends-ignore.md#server-behaviour-srv)
`PLI_REQUESTTEXT` types `pmservers`, `pmserverplayers <server>`, `pmunmapserver` and `pmguilds` go to the listserver for the cross-server and guild tabs	server	[srv] (client-internals/feature-specs/14-friends-ignore.md#server-behaviour-srv)
A real 2.22 client sent no buddy request on add or delete; it only wrote `buddyaccounts.txt`, so the token layout on the wire is unmeasured (→ §4.10.5.2)	2.22	[cap] (client-internals/feature-specs/14-friends-ignore.md#lab-findings-from-a-real-222-client-2026-10-24-cap)
`PLI_MUTEPLAYER` (43) is `s playerId, c 1/0`; GServer-v2 registers no handler, so it counts as an invalid packet, and more than 5 invalid packets disconnect the client; two sends from a 2.22 client did not	2.17+; server	[srv][cap] (client-internals/feature-specs/14-friends-ignore.md#server-behaviour-srv)

2.8.5 Profiles

Rule	Builds	Source
`PLI_PROFILESET` and `PLO_PROFILE` are a run of `c length` + text fields, read until the message ends	2.05	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)
The profile is relayed from the listserver (→ §6)	2.05	[2.05] (protocol/chat-pm-rc.md#profiles)
2.05 sends PROFILESET from the profile window's Send button, which also works in another online player's window; Close on your own profile only keeps the edits locally	2.05	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)
On PROFILE, 2.05 needs at least 10 fields, drops field 0 and saves fields 1 to 9 to a local per-account profile file	2.05	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)
The in-game stats in the same window (online time, kills, deaths, rating and so on) are not in this message; they most likely come from the player's props	2.05	[inf] (protocol/chat-pm-rc.md#profile-field-list-205-205)

Field	Content	Source
0	account, the profile's owner	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)
1	real name	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)
2	age	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)
3	sex: `-` (unknown), `male` or `female`	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)
4	country	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)
5	ICQ number	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)
6	e-mail	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)

Field	Content	Source
7	homepage	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)
8	favourite Graal hangout	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)
9	favourite quote: the quote box's lines joined with spaces, trimmed, cut to 220 characters	[2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205)

Each text field is trimmed before sending [2.05] (protocol/chat-pm-rc.md#profile-field-list-205-205).

2.9 UDP peer-to-peer

A channel between clients that the game server never sees; the server only relays each player's address. The ids handled per build are in → App. C.4; the TCP twins of the level-object messages are in → §2.4.5.

> Status: draft.

2.9.1 Setup and addressing

Rule	Builds	Source
No open-source server implements this channel, because the server never sees it	server	[2.05] (protocol/udp-p2p.md)
The first build that links `bind`, `sendto` and `recvfrom`, and calls the feature "direct data packages", is 1.28 beta 2	1.28 beta 2	[bin] (protocol/gserv-1999.md#udp-128-beta-2-on)
Every 2.0x build, NewWorld and NewMain included, imports the same socket calls and has one send and one receive wrapper	2.00 to 2.1 tests	[bin] (client-internals/decomp-findings-2026-10-24-c/07-udp-channel.md)
At connect the client binds a UDP socket on `fixedudpport` when `fixedport` is set, otherwise on an ephemeral port; `noudp` turns the channel off	2.05	[2.05] (protocol/udp-p2p.md#setup)
1.31 adds `noudp=` and `fixedudpport=` to the client config; the catalogue gives 14899 as the fixed-port default	1.31	[bin] (protocol/gserv-1999.md#udp-128-beta-2-on)
`noudp` defaults to on	2.22, 2.31	[bin] (version-profiles/2.22-2.31.md#wire-protocol)
The client sends its own port as PLPROP UDPPORT (31, `i3`); a peer's address is PLPROP IPADDR (30, `i5`, a 32-bit address) plus UDPPORT, both arriving in `PL0_0THERPLPR0PS`	2.05	[2.05] (protocol/udp-p2p.md#setup)
A datagram is accepted only from the address and port of a known player	2.05	[2.05] (protocol/udp-p2p.md#datagram)

2.9.2 Datagram

Rule	Builds	Source
A datagram is `seq` (1 byte), then `c` UDP id and the fields, in the same G-type encodings as TCP; no zlib, no length prefix, no gen-3 insertion	2.05	[2.05] (protocol/udp-p2p.md#datagram)
The whole datagram is XORed with one key byte: `recipientId XOR 1`	2.05, 3.04	[2.05][cap] (protocol/udp-p2p.md#datagram)
In code the key is the low byte of the recipient's id field XOR the same field of the local player object, which reads 1 in practice	2.05	[2.05][cap] (protocol/udp-p2p.md#datagram)
In a three-player capture (ids 2, 3, 4) every datagram to player 2 used key 3, to player 3 key 2 and to player 4 key 5, whoever sent it	2.05	[cap] (protocol/udp-p2p.md#datagram)
`seq` counts from 0 per sender–recipient pair and wraps mod 256; a duplicate `seq` is dropped	2.05, 3.04	[2.05][cap] (protocol/udp-p2p.md#datagram)
1.28 beta 2 sends single, uncompressed, unframed messages straight to another player's address and port	1.28 beta 2	[cap] (protocol/gserv-1999.md#udp-128-beta-2-on)
The 2.05 receive wrapper drops a bit test on a length value and moves a field offset compared with 2.00–2.04; the datagram layout does not change	2.00 to 2.05	[bin] (client-internals/decomp-findings-2026-10-24-c/07-udp-channel.md)

2.9.3 Routing

Rule	Builds	Source
A level-local action goes over UDP only when every other player in the level has a UDP port; otherwise it goes over UDP to the peers that have one and over TCP to the server	2.05	[2.05] (protocol/udp-p2p.md#routing)
The TCP copy is also sent when the server has an NPC server (PLO_HASNPCSERVER)	2.05	[2.05] (protocol/packet-layouts.md#combat-and-objects-both-directions)
This rule covers the level-object messages only; player props stay on TCP	2.05	[cap] (protocol/packet-layouts.md#combat-and-objects-both-directions)
With UDP active, both clients still sent X, Y and sprite props over TCP; which condition picks the TCP copy when the server is in the level is not traced	2.05	[cap][inf] (protocol/udp-p2p.md#observed-on-the-wire-cap)
One player per level, the level leader picked with PLO_ISLEADER (10), sends the level NPCs' arrows to the others	1.28 beta 2	[cap] (protocol/gserv-1999.md#udp-128-beta-2-on)

2.9.4 UDP ids

The ids are their own space, not PLI or PLO numbers; the wire byte is again the id + `0x20` [2.05] (protocol/udp-p2p.md#udp-ids-own-dispatcher-not-plipl-numbering); [bin] (version-profiles/udp-ids-200-204.tsv).

Id	Meaning	Payload	TCP twin	Source
0	movement	six GCHARs: X × 2, Y × 2, SPRITE, STATUS, CARRYSPRITE (−1 none), and a letter that follows the animation	props 15–19	[2.05][cap] (protocol/udp-p2p.md#observed-on-the-wire-cap)
10, 11	bomb add, bomb delete	unmeasured beyond the TCP twins	PLI 4, 5	[2.05] (protocol/udp-p2p.md#udp-ids-own-dispatcher-not-plipl-numbering)
12	arrow add	the 5 bytes of relay A in 1.28 beta 2	PLI 9	[2.05] (protocol/udp-p2p.md#udp-ids-own-dispatcher-not-plipl-numbering); [cap] (protocol/gserv-1999.md#udp-128-beta-2-on)
13	fire spy	unmeasured	PLI 10	[2.05] (protocol/udp-p2p.md#udp-ids-own-dispatcher-not-plipl-numbering)
14	explosion	unmeasured	PLI 27	[2.05] (protocol/udp-p2p.md#udp-ids-own-dispatcher-not-plipl-numbering)
15	showimg	four GCHARs (index, x, y, name length), the image name, then a few trailing bytes; 3.04 sends one byte 9 about every 5.1 s per peer	PLI 24	[cap] (protocol/udp-p2p.md#observed-on-the-wire-cap); [cap] (protocol/udp-p2p.md#304-linux-lab-2026-10-30-cap)
16, 17	horse add, horse delete	unmeasured	PLI 7, 8	[2.05] (protocol/udp-p2p.md#udp-ids-own-dispatcher-not-plipl-numbering)
18	throw carried	unmeasured	PLI 11	[2.05] (protocol/udp-p2p.md#udp-ids-own-dispatcher-not-plipl-numbering)
19, 20	item add, item delete	unmeasured	PLI 12, 13	[2.05] (protocol/udp-p2p.md#udp-ids-own-dispatcher-not-plipl-numbering)
21	remote hit from GS1 `hitplayer`, sent only to the victim	`c (dx + 4) × 16, c (dy + 4) × 16, c halfhearts`, optional attacker-object bytes	PLO 40 shape	[2.05] (protocol/udp-p2p.md#payloads-of-ids-21-23-24)
22	chat	text, at most 253 bytes	CURCHAT prop	[2.05] (protocol/udp-p2p.md#udp-ids-own-dispatcher-not-plipl-numbering)

Id	Meaning	Payload	TCP twin	Source
23	level name	raw level name, no length byte; applied as CURLEVEL (20); no 2.05 sender	prop 20	[2.05][inf] (protocol/udp-p2p.md#payloads-of-ids-21-23-24)
24	entered level	`c x × 2`, `c y × 2`, raw level name; sent to peers in the 3 × 3 gmap neighbourhood	props 15, 16, 20	[2.05] (protocol/udp-p2p.md#payloads-of-ids-21-23-24)
25	hit objects	unmeasured	PLI 36	[2.05] (protocol/udp-p2p.md#udp-ids-own-dispatcher-not-plipl-numbering)
26	trigger action	for example NPC id 0, x × 2, y × 2, `leftmouse`	PLI 38	[cap] (protocol/udp-p2p.md#304-linux-lab-2026-10-30-cap)

Rule	Builds	Source
Movement goes out about every 100 ms while moving, then less often	2.05, 1.28 beta 2	[cap] (protocol/udp-p2p.md#observed-on-the-wire-cap); [cap] (protocol/gserv-1999.md#udp-128-beta-2-on)
The last movement byte was 55, 41 and 51 (`W`, `I`, `S` as characters) for walk, idle and sword, so it looks like the first letter of the gani name	2.05, 3.04	[cap][inf] (protocol/udp-p2p.md#observed-on-the-wire-cap)
Showimg is re-sent about every 3 s while shown	2.05	[cap] (protocol/udp-p2p.md#observed-on-the-wire-cap)
21's dx, dy is the direction from the script's `fromx`, `fromy` to the victim, clamped to −4..4; the receiver decodes `(b − 0x60) / 16` and damage `(b − 0x20) / 2` capped at 1.0; it is sent only when the victim is in the same level	2.05	[2.05] (protocol/udp-p2p.md#payloads-of-ids-21-23-24)
A newcomer sends one empty id 22 to each player already in the level	1.28 beta 2 to 1.32	[cap] (protocol/gserv-1999.md#udp-128-beta-2-on)
Ids 0, 15 and 22 go through a broadcast helper (prepend the id, send to every peer with a UDP port); the other ids through a peer-send wrapper	2.00, 2.05	[bin] (version-profiles/udp-ids-200-204.tsv)
UDP 25 and 26 first appear in the same builds as their TCP twins, PLI 36 (2.01) and PLI 38 (2.03)	2.01, 2.03	[bin] (version-profiles/udp-ids-200-204.tsv); [bin] (client-internals/decomp-findings-2026-10-24-c/07-udp-channel.md)
The Torque-era 3.04 client kept the channel byte for byte	3.04	[cap] (protocol/udp-p2p.md#304-linux-lab-2026-10-30-cap)
Whether 2.1 to 2.31 used the channel on the wire is unmeasured; 2.22 also used UDP for voice chat	2.1 to 2.31	[doc] (protocol/udp-p2p.md#open)

2.10 The 6.11 server-defined protocol

In 6.11 the client has no fixed protocol: a script fetched at login sets the framing, the cipher and both id tables. The classic generation-5 transport (→ §2.2) stays the default until that script switches it.

> Status: draft. The connector script itself was never captured (→ App. E#3), so the wire layout a 6.11 server uses is not in the notes.

2.10.1 The connector request

Rule	Builds	Source
On login the client sets the server name to `Login` and requests `con.png` and `con.gs` over HTTPS and `conf.gs` over plain HTTP from the official connector host (`<login-host>`)	6.11 Mac	[bin] (protocol/v6x-clients.md#connector-request)
Each step is retried up to twice, using a second host whose first label differs; after three failures the client shows a warning	6.11 Mac	[bin] (protocol/v6x-clients.md#connector-request)
The query is `?p=` + a comma-joined list `g=<client id>,p=mac,v=<version>,b=<build date and time>`, DES-ECB encrypted under a compiled-in key (whole 8-byte blocks only), base64-encoded with the client's own alphabet, then URL-encoded	6.11 Mac	[bin] (protocol/v6x-clients.md#connector-request)
The HTTPS connection checks the server against a certificate bundle compiled into the client, and sends `User-Agent: Graal/<version>`	6.11 Mac	[bin] (protocol/v6x-clients.md#connector-request)

2.10.2 The connector script

Rule	Builds	Source
The reply is loaded as the script object `StartScript_Connector`, and its `onCreated` runs; `scriptip`, `scriptsslcipher`, `scriptsslsubject` and `scriptsslissuer` are set first so the script can check where it came from	6.11 Mac	[bin] (protocol/v6x-clients.md#connector-reply)
If no connector script loads, a built-in `StartScript_Fail` is used	6.11 Mac	[bin] (protocol/v6x-clients.md#connector-reply)
The connection object gives the script `connecttogameserver`, `setprotocol`, `sendraw`, `senddata(id, data)`, `flush`, `setsslparameters`, `getsslcipher/subject/issuer`, `setencryptionparsekey`, `setencryptionout(type, key, iv)`, `getpassword`, `setloginaccount`, `requestbasefiles`, `setoutdatahandlers`, `clearindatahandlers`, `setindatahandlers`, `defaultinhandler(id, data)` and `tclient_setencryptionin(type, key, iv)`	6.11 Mac	[bin] (protocol/v6x-clients.md#what-the-script-controls)

Rule	Builds	Source
The script builds and sends the login itself, so the token, ids, header layout and cipher of 6.11 live in the server's script, not the client	6.11 Mac	[bin][inf] (protocol/v6x-clients.md#what-the-script-controls)

2.10.3 Framing and encryption

Rule	Builds	Source
`setprotocol(template)` replaces the generation-5 framing with a header described by the template; the header is as long as the template	6.11 Mac	[bin] (protocol/v6x-clients.md#what-the-script-controls)
Each template letter marks a field and repeating it sets the width, at most 4 bytes, big-endian: `I` packet counter (starting at 1), `L` length, `T` packet id, `E` another field	6.11 Mac	[bin] (protocol/v6x-clients.md#what-the-script-controls)
In that mode the stream can be wrapped in RC4 or AES-CBC (blocks padded with `0xFF`), set separately per direction; other names turn it off; TLS is also possible	6.11 Mac	[bin] (protocol/v6x-clients.md#what-the-script-controls)
None of this exists in 6.01 or 6.03: no connector, no RC4 or AES, no script-set id tables	6.01r1, 6.03r7	[bin] (protocol/v6x-clients.md#what-the-script-controls)
gs2lib names `PLI_SET_ENC_KEY` and `PLO_SET_ENC_KEY` (252, setting an AES or RC4 key and IV) and `PLI_BUNDLE` and `PLO_BUNDLE` (253, several messages in one); both are above 223, so they cannot be sent as one GCHAR	server library	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)

2.10.4 Id tables

Rule	Builds	Source
`setoutdatahandlers` loads a 256-entry table from internal message number to wire id; `0xFE` sends the message to a script event instead of the wire	6.11 Mac	[bin] (protocol/v6x-clients.md#what-the-script-controls)
`setindatahandlers` takes (wire id, handler) pairs; a handler below `0x5F` is one of 95 native handlers, `0xFE` sends the packet to the script, `0xFF` ignores it, and `0xFC`/`0xFD` set two special ids	6.11 Mac	[bin] (protocol/v6x-clients.md#what-the-script-controls)
The native handlers are also exposed to script by name, about 90 `tclient_*` functions, from `tclient_setplayerprops` and `tclient_setlevelboard` to `tclient_setghostsnear` and `tclient_setpackageend`	6.11 Mac	[bin] (protocol/v6x-clients.md#what-the-script-controls)
Slot 54 and slots above 56 have no event; the login and language messages are sent by the script itself; the slot order is not the wire order, so a 6.11 server must send the table	6.11 Mac	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)

The named send slots, each with the event 6.11 raises when its slot holds `0xFE`:

Slot	Event	Slot	Event	Slot	Event	Source
0	`onSendWantWeapScript`	19	`onSendTakeExtra`	38	`onSendLevelWarp`	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
1	`onSendWantClassScript`	20	`onSendRemoveExtra`	39	`onSendLevelWarpModtime`	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
2	`onSendWantGaniScript`	21	`onSendOpenChest`	40	`onSendDownloadFile`	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)

Slot	Event	Slot	Event	Slot	Event	Source
3	onSendToAllChat	22	onSendShot	41	onSendBoardModify	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
4	onSendIsPKer	23	onSendBaddyProperties	42	onSendBoardModify2	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
5	onSendCarryThrown	24	onSendBaddyHurt	43	onSendPreloadLevel	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
6	onSendAddHorse	25	onSendWeaponImage	44	onSendPlayerProperties	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
7	onSendDelHorse	26	onSendRCChat	45	onSendNPCProperties	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
8	onSendAddBomb	27	onSendRequestText	46	onSendFlag	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
9	onSendDelBomb	28	onSendSetText	47	onSendUnsetFlag	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
10	onSendServerWarp	29	onSendFileDeletion	48	onSendProjectile	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
11	onSendTriggerAction	30	onSendFolderDeletion	49	onSendUpdatePackage	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
12	onSendFireSpying	31	onSendFileRename	50	onSendPrivateMessage	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
13	onSendExplosion	32	onSendFilesMove	51	onSendUploadStart	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
14	onSendWeaponHit	33	onSendHaveWindow	52	onSendSaveFile	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
15	onSendPlayerHurt	34	onSendEnterLevel	53	onSendUploadEnd	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
16	onSendDeleteNPC	35	onSendWantImage	55	onSendPingAnswer	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
17	onSendDeleteWeapon	36	onSendImageUpdate	56	onSendWindowList	[bin] (protocol/v6x-clients.md#the-611-named-send-slots)
18	onSendExtra	37	onSendImageUpdateCRC			[bin] (protocol/v6x-clients.md#the-611-named-send-slots)

3 File formats

> Part index. Each chapter carries its own status line.

- → §3.1 Levels: Z3-V1, GR-V1, GLEVNW01 (`01-levels.md`)
- → §3.2 GRMAP001 and gmaps (`02-gmap.md`)
- → §3.3 GANI animations (`03-gani.md`)
- → §3.4 GOBJSET01 object sets (`04-gobjset.md`)
- → §3.5 Tile types and arrays.dat (`05-tiles-arrays.md`)
- → §3.6 Sign text alphabets (`06-sign-text.md`)
- → §3.7 Save games (graalsav) (`07-saves.md`)
- → §3.8 Web level cache (GWEBL001) (`08-caches.md`)
- → §3.9 Config and dat files (`09-config.md`)
- → §3.10 Image and sound asset conventions (`10-assets.md`)

3.1 Levels: Z3-V1, GR-V1, GLEVNW01

A Graal level is one 64×64 board of tiles plus its links, baddies, NPCs, chests and signs. Three file families carry it: the binary Z3-V1 (`.zelda`) and GR-V1 (`.graal`) formats, and the text GLEVNW01 (`.nw`) format. The wire form of a board is in → §2.

> Status: draft. The 2.05 GR-V reader and writer and the 2.31 reader are measured; the 2.05 loader's Z3-V1 path and every build after 2.31 are not.

3.1.0 Format selection

Item	Rule	Source
magics in the 2.05 string table	`Z3-V1.03`, `Z3-V1.04`, `GR-V1.00` to `GR-V1.03`, `GLEVNW01`; each set appears three times, one copy next to the GWEBL001 cache code (→ §3.8)	[2.05] (formats/level-binary.md)
server loader	picks the loader from the extension `.nw`, `.graal` or `.zelda`; for an unknown extension, from the 8-byte magic	[srv] (formats/glevnw01.txt)
server magic fallback	`GLEVNW01` → text loader; `GR-V1.01`, `GR-V1.02`, `GR-V1.03` → binary Graal loader; `Z3-V1.03`, `Z3-V1.04` → Zelda loader	[srv] (formats/glevnw01.txt)
real files	a reader must decide by the magic, not the extension: `.zelda` names hold GR-V1 files and `.graal` names hold `Z3-V1.04` files	[test] (formats/level-binary.md#z3-v1-differences) ; [test] (client-internals/decomp-findings-2026-10-25-j/03-extensions-and-magic.md)
`.zelda` names	694 of 801 distinct `.zelda`-named files in the old builds are `GR-V1.00` to `GR-V1.03`; 1st Graal's 16 levels are all `Z3-V1.04` under `.graal` names	[test] (client-internals/decomp-findings-2026-10-25-j/03-extensions-and-magic.md); [test] (client-internals/decomp-findings-2026-10-25-j/by-build.tsv)
client magics	every client tested, 1st Graal to 2.31 and the NewWorld betas, carries `Z3-V1.03` and `Z3-V1.04` and no third `Z3-V1` revision; the 1st Graal and 1.39 loaders compare the 8 magic bytes only	[bin] (client-internals/decomp-findings-2026-10-25-j/03-extensions-and-magic.md)
2.05 writer	always writes `GR-V1.03` for a level that is not saved as `.nw`	[2.05] (formats/level-binary.md#open)
2.31 writer	writes `GR-V1.04` only when the level has a height map, `GR-V1.03` otherwise	[2.31] (formats/level-binary-client.md#gr-v104-231)

Rule	Detail	Source
bare names	no client adds `.zelda` to a level name without an extension, and none swaps `.zelda` for `.graal` when a file is missing; a link or warp loads the name as written	[bin] (client-internals/bare-level-names.md#findings)
editor	the editor's open and save buttons (and from 1.3 the snapshot save) add `.graal` to a name typed without an extension	[bin] (client-internals/bare-level-names.md#findings)
built-in names	1st to 9th Graal compare the current level with fixed names such as `level1.zelda` and `cave2.zelda` or warp to them	[bin] (client-internals/bare-level-names.md#findings)
scope	read in 26 Windows builds, 1st Graal to 1.32 and 2.05; that the loader uses the name unchanged is inferred	[bin][inf] (client-internals/bare-level-names.md#confirmed-inferred-not-tested)

Which builds accept which magics is in the build catalogue (→ §1.1).

3.1.1 Z3-V1 (.zelda)

Z3-V1 is the Zelda Online level family. Its board coding is that of GR-V1 (→ §3.1.2.1); only the code width and the sections differ.

3.1.1.1 Layout

Order	Section	Shape	Source
1	magic	`char[8]`: `Z3-V1.02`, `Z3-V1.03` or `Z3-V1.04`	[srv] (formats/level-binary.md#layout)
2	board	bit stream until 4096 tiles are produced (→ §3.1.2.1)	[srv] (formats/level-binary.md#layout)
3	links	text lines until # or an empty line (→ §3.1.2.3)	[srv] (formats/level-binary.md#layout)
4	baddies	binary records until FF FF FF (→ §3.1.1.2)	[srv] (formats/level-binary.md#layout)
5	signs	text until an empty line or end of file (→ §3.1.2.6)	[srv] (formats/level-binary.md#layout)

Z3-V1 has **no NPC section and no chest section** [srv] (formats/level-binary.md#z3-v1-differences).

3.1.1.2 Revisions

Magic	Code width	Baddy record	Readers	Source
`Z3-V1.02`	11 bits	4 bytes, last byte always \n	only the 1998 Java Zelda Online games; no real file is in the notes	[srv][test] (formats/level-binary.md#z3-v1-differences); [src] (formats/grv1-decoder/README.md)
`Z3-V1.03`	12 bits	4 bytes: `x y type` then an empty line rest; no verse line	Zelda Online, Delphi and later clients, server	[srv][test] (formats/level-binary.md#z3-v1-differences)
`Z3-V1.04`	12 bits	`x y type` then the verse line (→ §3.1.2.4)	Delphi and later clients, server	[srv][test] (formats/level-binary.md#z3-v1-differences)

Rule	Detail	Source
link numbers	`Z3-V1.03` links are written with decimals (`31.0 42.0 2.0 1.0 43.5 41.0`), so the numbers are floating point	[test] (formats/level-binary.md#z3-v1-differences)
negative arrival	a negative newx or newy means "keep the player's own coordinate"	[test] (formats/level-binary.md#z3-v1-differences)
baddy bytes	the three raw bytes `x y type` are taken before the line end is looked for, because `0x0A` can be an x or a y	[test] (formats/level-binary.md#z3-v1-differences)
baddy types	values 0 to 6 occur; Zelda Online's names for them are green, blue and red soldier, shot soldier, swamp soldier, jumping hare, octopus	[test] (formats/level-binary.md#z3-v1-differences)
link destinations	no destination name contains a space; `Z3-V1.03` links name `.zelda` or `.editor` files, `Z3-V1.04` links `.graal`, `.zelda` or `.editor`	[test] (client-internals/decomp-findings-2026-10-25-j/02-links-and-arrivals.md#destination-extensions)
verses	every `Z3-V1.04` record carries the verse line, most with all three verses empty; no `Z3-V1.03` record has one	[test] (client-internals/decomp-findings-2026-10-25-j/01-record-shapes.md#computers-baddies)
a GR-V1 reader on Z3-V1	integer link parsing drops 124 decimal links, a computer line end left in the stream shifts every later record, and a negative arrival is taken as a literal coordinate	[test] (client-internals/decomp-findings-2026-10-25-j/05-old-reader-failures.md)
sample	the rules were checked against 139 distinct real files	[test] (formats/level-binary.md#z3-v1-differences) ; [test] (client-internals/decomp-findings-2026-10-25-j/format-stats.tsv); [test] (client-internals/decomp-findings-2026-10-25-j/04-loader-rules-compared.md)

3.1.2 GR-V1 (.graal)

3.1.2.0 Layout

Order	Section	Shape	Revisions	Source
1	magic	`char[8]`: `GR-V1.00` to `GR-V1.04`	all	[srv] (formats/level-binary.md#layout); [2.31] (formats/level-binary-client.md#gr-v104-231)
2	board	bit stream until 4096 tiles are produced (→ §3.1.2.1)	all	[srv][2.05] (formats/level-binary.md#board-bit-stream)
3	links	text lines until # or an empty line	all	[srv][2.05] (formats/level-binary-client.md#section-rules-in-the-client-reader-205)
4	baddies	3-byte records, each followed by a verse line, until `FF FF FF`	all	[srv][2.05] (formats/level-binary-client.md#section-rules-in-the-client-reader-205)
5	NPCs	text lines until # or an empty line	`GR-V1.00` and later	[srv][2.05] (formats/level-binary-client.md#section-rules-in-the-client-reader-205)
6	chests	text lines until #	`GR-V1.01` and later	[srv][2.05] (formats/level-binary-client.md#section-rules-in-the-client-reader-205)
7	signs	everything left (2.05); up to the first empty line (2.31)	all	[2.05][2.31] (formats/level-binary-client.md#agreement-with-level-binarymd)
8	heights	empty line, then a `HEIGHTS` block (→ §3.1.2.7)	`GR-V1.04`; accepted after any magic by 2.31	[2.31] (formats/level-binary-client.md#gr-v104-231)

The board is a packed bit stream with run-length codes, not a 64×64 array of 2-byte tiles; only `.nw` text and the wire board form store one 16-bit value per tile [srv] (formats/level-binary.md#correction).

3.1.2.1 Board bit stream

Rule	Detail	Source
bit order	bytes are appended to a little-endian bit buffer; each code is the bottom `width` bits	[srv] (formats/level-binary.md#board-bit-stream)
flag bit	the top bit of the code: `0x800` at 12 bits, `0x1000` at 13 bits	[srv] (formats/level-binary.md#board-bit-stream)
flag clear	the code is a tile id	[srv] (formats/level-binary.md#board-bit-stream)
flag set	a control code: `count = code & 0xFF`; if `code & 0x100` is set the next run is a double run	[srv] (formats/level-binary.md#board-bit-stream)
default count	`count = 1`: the next tile is written once	[srv] (formats/level-binary.md#board-bit-stream)
single run	the next tile is written `count` times, then `count` resets to 1	[srv] (formats/level-binary.md#board-bit-stream)
double run	two tiles A and B are read and `A B` is written `count` times (`{double 4} 5 6` → `5 6 5 6 5 6 5 6`), then the count resets	[srv] (formats/level-binary.md#board-bit-stream)
fill order	tiles fill `board[x + 64*y]` row by row, layer 0 only, until 4096 tiles are produced	[srv] (formats/level-binary.md#board-bit-stream)
double flag	the Java Zelda Online reader keeps the double flag until a double run completes, the GR-V1 reader clears it at the next control code; no real file shows a difference (0 of 663)	[test] (client-internals/decomp-findings-2026-10-25-j/04-loader-rules-compared.md)
tile range	a 12-bit file cannot address tiles ≥ 2048, because the flag takes the top bit; that is why 1.02 widened the code to 13 bits for the 4096-tile `pics1`	[inf] (formats/level-binary.md#board-bit-stream)

Code width per magic:

Magic	Width	Source
`Z3-V1.02`	11	[srv] (formats/level-binary.md#board-bit-stream)
`Z3-V1.03`, `Z3-V1.04`	12	[srv][2.05] (formats/level-binary.md#board-bit-stream)
`GR-V1.00`, `GR-V1.01`	12	[srv][2.05] (formats/level-binary.md#board-bit-stream)
`GR-V1.02`, `GR-V1.03`	13; the 2.05 loader compares the magic against 1.02 and 1.03 to pick it	[srv][2.05] (formats/level-binary.md#board-bit-stream)
`GR-V1.04`	13	[2.31] (formats/level-binary-client.md#agreement-with-level-binarymd)

3.1.2.2 Board writer (2.05)

The 2.05 writer walks the 4096 tiles in order and picks one case at each position [2.05] (formats/level-binary-client.md#board-writer-205):

Case	Codes emitted	Source
the next tile equals this one	`0x1000 | n`, then the tile; n counts 2 to 64 equal tiles	[2.05] (formats/level-binary-client.md#board-writer-205); [2.05] (formats/level-binary.md#open)
tiles `i, i+1` equal `i+2, i+3` (pattern AB AB)	`0x1100 | n`, then A, then B; n counts up to 64 pairs	[2.05] (formats/level-binary-client.md#board-writer-205)
otherwise	the tile, `& 0xFFF`	[2.05] (formats/level-binary-client.md#board-writer-205)
packing	13 bits per code, least significant bit first; the last partial byte is flushed	[2.05] (formats/level-binary-client.md#board-writer-205)
check	`write_board` in `tools/file-formats/gwebl.py` follows these rules and re-encodes 144 cached levels and 13 tutorial levels byte for byte	[test] (formats/level-binary-client.md)
full runs	single runs of exactly 64 occur in the embedded tutorial levels	[2.05] (formats/level-binary.md#open)

3.1.2.3 Links

Rule	Detail	Source
line	`dest x y w h newx newy`	[srv] (formats/level-binary.md#records)
spaces	the level name may contain spaces; the tokens before the last six are rejoined into `dest`	[srv] (formats/level-binary.md#records)
missing level	a link to a level that does not exist is dropped	[srv] (formats/level-binary.md#records)
arrival	`newx` and `newy` may be `playerx` or `playery`, optionally followed by a signed number (`playery+10`)	[srv] (formats/level-binary.md#records)
terminator	a line # or an empty line	[srv][2.05] (formats/level-binary.md#layout)
negative arrival	805 of 5381 distinct GR-V1 links in the old builds have a negative arrival; that 2.x reads it as "keep the player's coordinate", as Z3-V1 does, is inferred	[test][inf] (client-internals/decomp-findings-2026-10-25-j/02-links-and-arrivals.md#negative-arrivals)

How the client resolves an arrival expression is in → §4.

3.1.2.4 Baddies

Rule	Detail	Source
record	`i8 x, i8 y, i8 type`, then a `\n`-terminated verse line	[srv] (formats/level-binary.md#records)
verse line	three verses separated by `\`: sight, hurt, attack	[srv] (formats/level-binary.md#records)

Rule	Detail	Source
line end	the client reads up to a \n that comes after at least 3 bytes, so the binary bytes may contain 0x0A	[2.05] (formats/level-binary-client.md#section-rules-in-the-client-reader-205)
terminator	FF FF FF followed by an empty line; the client stops at a record whose first byte is ≥ 0x80	[srv] (formats/level-binary.md#records); [2.05] (formats/level-binary-client.md#section-rules-in-the-client-reader-205)
x of 0	a valid baddy; real levels have them; a reader that stops at x = 0 loses them and reads the rest of the baddy bytes as NPC lines	[test] (formats/grv1-decoder/README.md); [test] (client-internals/decomp-findings-2026-10-25-f/03-baddy-terminator.md)
corpus	902 baddies in 99 of 3113 levels; types 0 to 9 all occur	[test] (client-internals/decomp-findings-2026-10-25-f/03-baddy-terminator.md); [test] (client-internals/decomp-findings-2026-10-25-f/corpus-stats.tsv)
short record	the reference decoder also ends the section at a record shorter than 3 bytes	[test] (formats/grv1-decoder/README.md)

3.1.2.5 NPCs and chests

Record	Shape	Source
NPC	GCHAR x, GCHAR y, image#script; the script uses 0xA7 for a newline; an empty image is a script-only NPC	[srv] (formats/level-binary.md#records)
chest	GCHAR x, GCHAR y, GCHAR item, GCHAR signindex	[srv] (formats/level-binary.md#records)

Coordinates in these records are whole tiles (GCHAR), not half-tiles [srv] (formats/level-binary.md#records). NPC bytes are single bytes and can lie outside 0 to 63, as 165 NPCs in the corpus do. A level with no chests still ends the section with a bare #. Chest items 0 to 11, 16, 17, 19 and 24 occur, plus one 78, outside the 25-name table; 343 of 375 chests have sign index −1, which is "no sign" [test] (client-internals/decomp-findings-2026-10-25-f/01-section-shapes.md); [test] (client-internals/decomp-findings-2026-10-25-f/04-sign-and-chest-details.md). The chest item index table is in → §8.

3.1.2.6 Signs

Rule	Detail	Source
record	GCHAR x, GCHAR y, then text in the sign alphabet	[srv] (formats/level-binary.md#records)
alphabet	the same as signs on the wire: byte - 32 indexes the sign table (→ §3.6)	[2.05] (formats/level-binary-client.md#sign-text-in-files)
line break	the token v*eUWf + 0x80 decodes to #K(13) and a newline; it is the explicit line-break marker, not a space	[test] (formats/level-binary-client.md#sign-text-in-files)
end	2.05 reads signs to end of file; 2.31 stops at the first empty line, because a HEIGHTS block can follow	[2.05][2.31] (formats/level-binary-client.md#agreement-with-level-binarymd)
signs at (0,0)	common in real levels and hold dialogue text, so they are probably messages that NPC scripts show by index rather than signs on the map	[inf] (formats/level-binary-client.md#sign-text-in-files) ; [test] (client-internals/decomp-findings-2026-10-25-f/04-sign-and-chest-details.md)
no signs	a level with no signs ends after the chest #, with no empty line; in 3113 files nothing follows the signs	[test] (client-internals/decomp-findings-2026-10-25-f/01-section-shapes.md)

3.1.2.7 GR-V1.04 height block

GR-V1.04 is GR-V1.03 plus an optional height map [2.31] (formats/level-binary-client.md#gr-v104-231).

```
<empty line>
HEIGHTS
h00,h01,…,h08
…                          9 lines of 9 comma-separated decimals
h80,h81,…,h88
HEIGHTSEND
```

Rule	Detail	Source
position	after the signs and one empty line	[2.31] (formats/level-binary-client.md#gr-v104-231)
reader	accepts the block after the signs whatever the magic says; matches `HEIGHTSEND` case-insensitively; keeps at most 9 rows of 9 values	[2.31] (formats/level-binary-client.md#gr-v104-231)
board	unchanged: 13 bits, the same run codes	[2.31] (formats/level-binary-client.md#gr-v104-231)
same overlay	the 9×9 grid is the `HEIGHTS` overlay that GLEVNW01 carries and TerrainGenerator writes (→ §3.1.3.5)	[2.31] (formats/level-binary-client.md#gr-v104-231)
meaning	a 9×9 grid fits the corners of an 8×8 grid of 8×8-tile cells on a 64×64 level	[inf] (formats/level-binary-client.md#gr-v104-231)
where	in 2.31, GR-V parsing appears only in the web-level cache code, the one place that client both writes and reads GR-V1.04 (→ §3.8)	[2.31] (formats/level-binary-client.md#gr-v104-231)

3.1.2.8 Embedded tutorial levels (2.05)

The 2.05 executable carries the offline tutorial as 13 `RCDATA` resources, all plain `GR-V1.03` [2.05] (formats/level-binary-client.md#embedded-tutorial-levels-205):

Resource	Level name	Source
`TUTORIALA1` to `TUTORIALA4`	`tutorial_a-1.graal` to `tutorial_a-4.graal`	[2.05] (formats/level-binary-client.md#embedded-tutorial-levels-205)
`TUTORIALB1` to `TUTORIALB4`	`tutorial_b-1.graal` to `tutorial_b-4.graal`	[2.05] (formats/level-binary-client.md#embedded-tutorial-levels-205)
`TUTORIALB2H`	`tutorial_b-2_house.graal`	[2.05] (formats/level-binary-client.md#embedded-tutorial-levels-205)
`TUTORIALC1` to `TUTORIALC4`	`tutorial_c-1.graal` to `tutorial_c-4.graal`	[2.05] (formats/level-binary-client.md#embedded-tutorial-levels-205)

Rule	Detail	Source
selection	when the client is not connected and the requested level name starts with `tutorial`, it opens the matching resource instead of a file on disk	[2.05] (formats/level-binary-client.md#embedded-tutorial-levels-205)
grid	the levels form a 3×4 grid joined by edge links; the B2 house, B4 and B2 hold all the NPCs, chests and signs	[test] (formats/level-binary-client.md#embedded-tutorial-levels-205)
2.22	2.22 embeds 9 tutorial levels; all parse with the expected link grid, chest items and sign text	[test] (formats/grv1-decoder/README.md)
extraction	`tools/file-formats/extract_tutorials.py Graal.exe OUTDIR --json` writes the 13 `.graal` files and a parsed JSON for each	[src] (formats/level-binary-client.md#embedded-tutorial-levels-205)

3.1.2.9 Reference decoder

`formats/grv1-decoder/grv1.py` decodes `GR-V1.00` to `GR-V1.04` and `Z3-V1.02` to `Z3-V1.04` with the rules above; NPC scripts have `0xA7` turned into newlines, and `#K(13)` is dropped from sign text [src] (formats/grv1-decoder/README.md).

3.1.3 GLEVNW01 (.nw)

3.1.3.1 File

Rule	Detail	Source
form	line-oriented text; the first line is the signature `GLEVNW01`	[srv] (formats/glevnw01.txt)
editing	official notes say to edit it in an editor that does not wrap lines	[doc] (formats/glevnw01.txt)
transfer	servers send the same payload as for a binary `.graal`; only the on-disk form is larger	[srv] (formats/glevnw01.txt)

3.1.3.2 BOARD

```
BOARD x y width layerindex tilesdata
```

Rule	Detail	Source
meaning	one horizontal run of `width` tiles on the background, starting at `(x, y)`	[srv] (formats/glevnw01.txt)
tilesdata	`width`*2 characters; each tile is a 12-bit index into the tile list (`pics1.png`), two base64 characters, high 6 bits first	[srv] (formats/glevnw01.txt)
alphabet	`ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789+/`; `tile = index(left) << 6 | index(right)`	[srv] (formats/glevnw01.txt)
examples	`AA` is tile 0, `AB` tile 1, `//` tile 4095	[srv] (formats/glevnw01.txt)
trailing =	the client alphabet in the server's level code appends a `=`, which is not part of the 64-character alphabet	[srv] (formats/glevnw01.txt)
rejection	a line is ignored unless it has 6 tokens, the run stays inside x 0 to 63, and `tilesdata` has at least `width`*2 characters	[srv] (formats/glevnw01.txt)
storage	the tile goes to `levelTiles[x + y*64]`; `layerindex` is parsed and not used	[srv] (formats/glevnw01.txt)

3.1.3.3 Other records

Record	Shape	Rules	Source
link	`LINK dest x y width height newx newy`	`dest`, newx and newy must not contain spaces	[srv] (formats/glevnw01.txt)
sign	`SIGN x y`, text lines, `SIGNEND`	the body may be empty	[srv] (formats/glevnw01.txt)
NPC	`NPC image x y`, script lines, `NPCEND`	`-` if the NPC has no image; x and y may be floating point	[srv] (formats/glevnw01.txt)
baddy	`BADDY x y type`, verse lines, `BADDYEND`	type is a name or 0 to 9; verses may be shortened, but `BADDYEND` is required	[srv] (formats/glevnw01.txt)
chest	`CHEST x y item signindex`	item is a name or 0 to 24, resolved by the server's item table; an unknown item drops the chest	[srv] (formats/glevnw01.txt)

3.1.3.4 GSERVL01

Rule	Detail	Source
magic	GSERVL01, a second text-level signature known to the 3.x clients	[bin] (protocol/v3x-clients.md#file-formats)
parser	read by the same parser as GLEVNW01	[bin] (protocol/v3x-clients.md#file-formats)
keywords	BOARD, LINK, CHEST, SIGN, NPC…NPCEND, BADDY…BADDYEND, HEIGHTS… HEIGHTSEND, and REPLACENPCS…REPLACENPCSEND, which the client skips	[bin] (protocol/v3x-clients.md#file-formats)
later builds	4.x lists it among the magics it knows, as in 3.x	[bin] (protocol/v4x-clients.md)

3.1.3.5 Other writers of the same keywords

Three tools write level text, and only two of them produce what the loader accepts [GE] (formats/glevnw01.txt):

Writer	Output	Source
GraalEditor	opens *.graal and *.nw with one filter and treats both as text; does not require the signature; BOARD 0 is the tile layer as 64 lines of 64 decimal tile indices, not a base64 run; heights are a separate overlay, not a second BOARD; keywords LINK, SIGN, CHEST, NPC, BADDY, LEVELNAMES with fields in the order the dialogs collect them	[GE] (formats/glevnw01.txt)
GraalEditor compatibility	the decimal BOARD 0 text is not what the loader accepts; it must be rewritten as GLEVNW01 base64 runs before it loads, and a client .nw is not a decimal grid	[GE] (formats/glevnw01.txt)
TerrainGenerator	GLEVNW01 with 64 rows of tile 0 (each row 64 <y> and 128 A), a HEIGHTS block of nine lines of nine numbers closed by HEIGHTSEND, then a LINK on each edge that has a neighbour; no NPC section	[bin] (formats/glevnw01.txt)
DungeonGenerator	GLEVNW01 with base64 BOARD runs for 64 by 64 tiles, edge LINK lines and NPC blocks; stamp id 0 is replaced by a 4 by 4 ground fill before the board is stored, and stamp id 1 is a hole and is not written	[bin] (formats/glevnw01.txt)

The tools themselves are in → §11.

3.2 GRMAP001 and gmaps

A gmap is a grid of levels that the client walks across without a level warp. Its file is `GRMAP001` text. The 2.05-era text big map and minimap, which name a grid of levels for the map screen only, are a different format and are in → §3.2.4. How a client crosses a gmap edge on the wire is in → §4.10.

> Status: draft. The server reader and three tool writers are summarised; the 2.05 reader is read statically and may be dormant; how 2.17 and later clients lay out the gmap overview is unmeasured.

3.2.1 Three kinds of map

Kind	File	Server option	Effect	Source
level links	none; links inside each level (→ §3.1.2.3)	none	walking into a link rectangle makes the client ask for a level warp	[srv] (client-internals/feature-specs/06-gmap-overworld.md#three-different-maps-do-not-mix)
server big map	`bigmap.txt`-style text grid	`maps =`	tells the server which levels are adjacent, so players see and chat across edges; the client does nothing special	[srv] (client-internals/feature-specs/06-gmap-overworld.md#three-different-maps-do-not-mix)
gmap	`GRMAP001`	`gmaps =`	the client walks continuously across level borders	[srv] (client-internals/feature-specs/06-gmap-overworld.md#three-different-maps-do-not-mix)
map-screen overlays	text grid and image	`bigmap =`, `minimap =` (four comma fields `text,image,x,y`)	the server sends the overlay settings at login to 2.1 and later clients; the notes warn that setting `bigmap` breaks gmaps	[srv] (client-internals/feature-specs/06-gmap-overworld.md#three-different-maps-do-not-mix)

3.2.2 GRMAP001 as the server reads it

Key	Arguments	Meaning	Source
`GRMAP001`	none	signature, first line	[srv] (formats/gmap.md#keys-read-by-gserver-srv)
`WIDTH`, `HEIGHT`	n	grid size in levels	[srv] (formats/gmap.md#keys-read-by-gserver-srv)
`GENERATED`	one argument	read and stored; not used to rebuild names on the server side	[srv] (formats/gmap.md#keys-read-by-gserver-srv)
`LEVELNAMES` … `LEVELNAMESEND`	one grid row per line, quoted, comma-separated names	fills the grid row-major; names are lowercased for lookup	[srv] (formats/gmap.md#keys-read-by-gserver-srv)
`MAPIMG`	file	overview image	[srv] (formats/gmap.md#keys-read-by-gserver-srv)
`MINIMAPIMG`	file	minimap image	[srv] (formats/gmap.md#keys-read-by-gserver-srv)
`NOAUTOMAPPING`	none	flag	[srv] (formats/gmap.md#keys-read-by-gserver-srv)
`LOADFULLMAP`	none	flag: preload every level	[srv] (formats/gmap.md#keys-read-by-gserver-srv)

Key	Arguments	Meaning	Source
LOADATSTART … LOADATSTARTEND	name list	levels to preload	[srv] (formats/gmap.md#keys-read-by-gserver-srv)

Rule	Detail	Source
tokens	each line is split on whitespace after \r is stripped	[srv] (formats/gmap.md#keys-read-by-gserver-srv)
name rows	lines inside the LEVELNAMES block are parsed as quoted CSV	[srv] (formats/gmap.md#keys-read-by-gserver-srv)
coordinates	position in the world is cell*64 + local; local = global − 64 × cell	[cap] (client-internals/feature-specs/06-gmap-overworld.md#lab-capture-of-a-real-222-client-crossing-gmap-edges-2026-10-24-cap)
transfer	a client fetches the .gmap file itself as a file transfer when it does not have it	[cap] (client-internals/feature-specs/06-gmap-overworld.md#lab-capture-of-a-real-222-client-crossing-gmap-edges-2026-10-24-cap)

3.2.3 GRMAP001 as the tools write it

Three writers emit the signature, each with a different set of keys; GENERATED does not mean the same thing in all three [bin] (formats/grmap001.txt).

Writer	Keys in order	Level names	Source
Dungeon Generator (short form)	GRMAP001, WIDTH, HEIGHT, GENERATED <bottom-right level name>; no LEVELNAMES block	LEVELSTART, a column letter and a 1-based row; the grid is rebuilt from the last level's name in GENERATED	[bin] (formats/grmap001.txt)
TerrainGenerator (generator form)	GRMAP001, WIDTH, HEIGHT, GENERATED 1, GENSEED, GENBASE, GENEVENBORDERS (0 or 1), GENHEIGHT, GENCHAOS, LEVHEIGHT, LEVCHAOS, MAPIMG <basename>.png, one level name per row, HEIGHTMAP … HEIGHTMAPEND, RANDOMSEEDS … RANDOMSEEDSEND (one seed per level, when present)	basename, a column A then AA, a zero-padded row, and .nw; GENERATED 1 is a flag here	[bin] (formats/grmap001.txt)
GraalLevelGen	GRMAP001, WIDTH, HEIGHT, then a LEVELNAMES list of quoted, comma-separated file names	as listed	[bin] (formats/grmap001.txt)

Rule	Detail	Source
MAPIMG	the overview the client shows on the map list; the PNG TerrainGenerator saves from its preview dialog is a separate file	[bin] (formats/grmap001.txt)
HEIGHTMAP	the height field; a sibling _heights.txt holds the same numbers without the gmap wrapper	[bin] (formats/grmap001.txt)
per-level heights	each level file then carries a 9 by 9 HEIGHTS block (→ §3.1.2.7)	[bin] (formats/grmap001.txt)

3.2.4 Client readers by build

Build	Gmap reader	Map path in use	Source
2.05 rev 2	has the GRMAP001 string, a reader that knows WIDTH, HEIGHT, GENERATED, GENSEED, GENBASE, GENEVENBORDERS, GENHEIGHT, GENCHAOS, LEVHEIGHT, LEVCHAOS, MAPIMG, HEIGHTMAP … HEIGHTMAPEND, RANDOMSEEDS … RANDOMSEEDSEND and HEIGHTS, and a map class; no static caller of the reader was found, so it may be dormant	the text big map and minimap below; no gmap warp handler and no gmap cell prop name	[2.05] (formats/gmap.md#205)

Build	Gmap reader	Map path in use	Source
2.22	unmeasured	gmap crossing captured in the lab (→ §4.10)	[cap] (client-internals/feature-specs/06-gmap-overworld.md#lab-capture-of-a-real-222-client-crossing-gmap-edges-2026-10-24-cap)
2.31	unmeasured	gmap crossing captured in the lab (→ §4.10)	[cap] (client-internals/feature-specs/06-gmap-overworld.md#needs)

An earlier version of the gmap note said 2.05 had no `GRMAP001` string; that was wrong, and the build catalogue also lists it for 2.05 [2.05] (formats/gmap.md#205).

3.2.5 Text big map and minimap (2.05)

2.05 reads the big map and minimap as text grids; the GS1 commands `setmap` and `setminimap` load the same kind of file from a script [2.05] (formats/gmap.md#205-big-map-and-minimap-text-files-205).

Map	File	Image	Built-in integers	Source
big map	`bigmap.txt`	`map.png`	10, 16, 1688, 1024	[2.05] (formats/gmap.md#205-big-map-and-minimap-text-files-205)
minimap	`minimap.txt`	`map2.png`	664, 896	[2.05] (formats/gmap.md#205-big-map-and-minimap-text-files-205)
tutorial (offline)	`tutorialmap.txt`	`tutorialmap.png`	3, 4, 192, 486	[2.05] (formats/gmap.md#205-big-map-and-minimap-text-files-205)

Rule	Detail	Source
lines	plain text, one map row per line, each parsed as a comma-separated list with quoting allowed	[2.05] (formats/gmap.md#205-big-map-and-minimap-text-files-205)
cells	each cell is normalised and appended to one flat list; a cell is a level file name, and `x` marks an empty cell	[2.05] (formats/gmap.md#205-big-map-and-minimap-text-files-205)
shape	the column count is the number of cells on the first line; the row count is total cells ÷ columns	[2.05] (formats/gmap.md#205-big-map-and-minimap-text-files-205)
shipped file	the shipped `bigmap.txt` is 20 columns × 21 rows of quoted level names	[file] (formats/gmap.md#205-big-map-and-minimap-text-files-205)
server-list override	an entry with more than the usual 8 fields replaces these: field 8 big-map file, 9 its image, 10 and 11 two integers, 12 minimap file, 13 its image, 14 and 15 two integers	[2.05] (formats/gmap.md#205-big-map-and-minimap-text-files-205)
integers	stored as points in the player object; probably image offsets or scale, which fits the defaults, but their use is not traced	[inf] (formats/gmap.md#205-big-map-and-minimap-text-files-205)
later builds	the server-pushed forms `PLO_BIGMAP` (171) and `PLO_MINIMAP` (172, for example `zodiacminimap.txt,zodiacworldminimap3.png,10,10`) replace this; they carry file names plus a default x,y for heads of players off the map	[srv] (formats/gmap.md#205); [srv] (client-internals/feature-specs/06-gmap-overworld.md#three-different-maps-do-not-mix)

The server-list entry layout is in → §6.

3.3 GANI animations

A GANI file is the text animation format that GraalShop reads and writes and that clients load from `levels/ganis/`. This chapter gives its grammar as the editor reads it; how a client plays a GANI is in → §4.

> Status: draft. The grammar is from the GraalShop reader and writer and checked against client sample files; no client build's GANI loader is traced.

3.3.1 File

Rule	Detail	Source
form	line-oriented text; the first line is the discriminator	[bin] (formats/gani.txt)
case	the parser lowercases each line before comparing keywords; the writer emits uppercase keywords and space-padded columns	[bin] (formats/gani.txt)
samples	client samples under `levels/ganis/` and `levels/bomys/` match the sprite dialect, including files that omit the signature	[file] (formats/gani.txt)

First line	Mode value	Body	Source
`GANI0001`	0	sprite table, header, `ANI` … `ANIEND` (sprite mode)	[bin] (formats/gani.txt)
`GANI0002`	1	sprite table, header, `MOVIE` … `MOVIEEND` (movie mode)	[bin] (formats/gani.txt)

3.3.2 Grammar

The grammar as the editor accepts it [bin] (formats/gani.txt):

```
file        := signature? sprite* header* body
signature   := "GANI0001" | "GANI0002"
sprite      := "SPRITE" id sheet x y w h description
             | "ATTACHSPRITE"  id parent dx dy
             | "ATTACHSPRITE2" id parent dx dy
header      := "LOOP" | "CONTINUOUS" | "SINGLEDIRECTION"
             | "SETBACKTO" name
             | "BACKCOLOR" "#" hex6
             | "FRAMES" count
             | "DEFAULTHEAD" file | "DEFAULTBODY" file
             | "DEFAULTPARAM" n value | "DEFAULTATTR" n value
             | "COLOREFFECT" id args | "ZOOMEFFECT" id args
             | "ROTATEEFFECT" id args
             | "STRETCHXEFFECT" id args | "STRETCHYEFFECT" id args
             | "EFFECTMODE" id args
body        := ani | movie
ani         := "ANI" step* "ANIEND"
movie       := "MOVIE" actor* "MOVIEEND"
actor       := "ACTOR" name frame* "ACTOREND"
frame       := "FRAME" index token ("," token)*
step        := placement ("," placement)* | "WAIT" n | "PLAYSOUND" name x y
placement   := spriteId x y | "PARAM" n x y | "ATTR" n x y
script      := "SCRIPT" rawlines "SCRIPTEND"
```

3.3.3 Sprite rows

Rule	Detail	Source
SPRITE	id, sheet, x, y, w, h, description	[bin] (formats/gani.txt)
columns	the writer pads the id to 4, the sheet name to 14, and x, y, w and h to 4, then writes the description	[bin] (formats/gani.txt)
ATTACHSPRITE, ATTACHSPRITE2	hang a sprite on a parent sprite with a pixel offset; they are not a second sheet	[bin] (formats/gani.txt)

Client files therefore look like this [file] (formats/gani.txt):

```
SPRITE    0         SPRITES    0    0   24   12 shadow
SPRITE  200            HEAD    0    0   32   32 head up
SPRITE   20           SWORD   62    0   28   28 sword
```

3.3.4 Header

Keyword	Meaning	Source
LOOP, CONTINUOUS	playback flags	[bin] (formats/gani.txt)
SINGLEDIRECTION	each step holds one row instead of four; written only in sprite mode	[bin] (formats/gani.txt)
FRAMES	frame count; written only in movie mode	[bin] (formats/gani.txt)
SETBACKTO	names the GANI to return to	[bin] (formats/gani.txt)
BACKCOLOR	a #rrggbb fill behind the editor preview	[bin] (formats/gani.txt)
DEFAULTHEAD, DEFAULTBODY	reload the head and body images	[bin] (formats/gani.txt)
DEFAULTPARAM, DEFAULTATTR	seed the 30-row attribute grid	[bin] (formats/gani.txt)
COLOREFFECT, ZOOMEFFECT, ROTATEEFFECT, STRETCHXEFFECT, STRETCHYEFFECT, EFFECTMODE	stored as an id plus arguments and applied at blit time	[bin] (formats/gani.txt)
SCRIPT … SCRIPTEND	copied through as text; the editor does not execute it	[bin] (formats/gani.txt)

3.3.5 Sprite mode body

Rule	Detail	Source
bounds	ANI opens the step list and ANIEND closes it	[bin] (formats/gani.txt)
step	a comma-separated placement row, a WAIT hold, or a PLAYSOUND name x y cue	[bin] (formats/gani.txt)
placement	a sprite id plus x and y, or a PARAM or ATTR slot plus x and y	[bin] (formats/gani.txt)

Rule	Detail	Source
directions	a step holds four direction rows, or one row when SINGLEDIRECTION is set	[bin] (formats/gani.txt)
wait	the wait counter starts at 1; each WAIT adds to it	[bin] (formats/gani.txt)

3.3.6 Movie mode body

Rule	Detail	Source
bounds	MOVIE opens the actor list and MOVIEEND closes it	[bin] (formats/gani.txt)
actor	ACTOR name, a run of FRAME lines, ACTOREND	[bin] (formats/gani.txt)
actor kind	not a keyword in the frame line but an integer on the actor: 0 CHAR, 1 SPRITE, 2 SOUND, 3 TEXT	[bin] (formats/gani.txt)
frame	an index and a comma-separated token list; the tokens become key=value strings on the actor's frame record	[bin] (formats/gani.txt)

3.4 GOBJSET01 object sets

An object set is a list of named multi-tile stamps that an editor places on a board. It carries tile ids only; what a tile does is decided by the tile type table (→ §3.5).

3.4.1 Layout

```
GOBJSET01
OBJECT width height name
<height lines of width*2 base64 characters>
OBJECTEND
…
```

Rule	Detail	Source
signature	`GOBJSET01`	[file] (formats/gobjset01.md)
tile packing	the same 12-bit base64 packing as GLEVNW01 (→ §3.1.3.2)	[file] (formats/gobjset01.md)
name	may contain spaces	[file] (formats/gobjset01.md)
no other fields	there is no layer, link, NPC or type byte	[file] (formats/gobjset01.md)
placing	placing a stamp writes its tile ids onto the board; collision is decided later by the type table	[file] (formats/gobjset01.md)

3.4.2 Shipped sets

Item	Detail	Source
sets in the client tree	`objectsStandard`, `objectsHouse Inside`, `objectsCastle`, `objectsSnow`, `objectsVolcano`	[file] (formats/gobjset01.md)
objects	288 named multi-tile objects across the five sets	[file] (formats/gobjset01.md)
first stamp	`objectsStandard` begins with the 2×2 "Bush", base64 `ACAD` / `ASAT` → ids 2, 3, 18, 19	[file] (formats/gobjset01.md)
Bush collision	`disputed → App. E#100`	[file] (formats/gobjset01.md); [src] (formats/tile-walk.md#stamp-vocabulary)
generator stamps	in DungeonGenerator, stamp id 0 is a 4×4 ground fill and stamp id 1 is a hole that is not written (→ §3.1.3.5)	[bin] (formats/gobjset01.md)

3.5 Tile types and arrays.dat

A board stores tile ids; what a tile does is looked up in separate tables. This chapter gives the tile type values, the tile sheet geometry, the `arrays.dat` table file and the signed `arrays2.dat` with its `.sig` files. How movement uses these values is in → §4.2.

> Status: draft. Only part of the 25 `arrays.dat` tables is traced to a gameplay use; the signature key encoding is not pinned down (→ §3.5.5).

3.5.1 Tile sheet geometry

Rule	Detail	Source
id	a board stores a tile id 0 to 4095; the client does not inspect the tileset image for collision	[src] (formats/tiletypes.txt)
sheet	`pics1.png` is 2048×512, palette, 128 by 32 tiles of 16 px	[file] (formats/tiletypes.txt)
id to picture	`col = (id % 16) + (id // 512) * 16`, `row = (id // 16) % 32`, source rectangle `(col*16, row*16, 16, 16)`: the sheet is eight side-by-side blocks of 512 ids, each 16 columns by 32 rows, filled row by row; an earlier rule `col = id % 128`, `row = id / 128` is `disputed → App. E#102`	[file] (client-internals/client-assets.md#tile-sheets)

3.5.2 Tile type values

The official tile type names, as documented for `setshape2` in the 2001 feature notes [doc] (formats/tiletypes.txt):

Type	Official name	Source
0	nonblocking	[doc] (formats/tiletypes.txt)
2	hurting underground	[doc] (formats/tiletypes.txt)
3	chair	[doc] (formats/tiletypes.txt)
4	bed, upper side	[doc] (formats/tiletypes.txt)
5	bed, lower side	[doc] (formats/tiletypes.txt)
6	swamp	[doc] (formats/tiletypes.txt)
7	lava swamp	[doc] (formats/tiletypes.txt)
8	near water	[doc] (formats/tiletypes.txt)
9	near lava	[doc] (formats/tiletypes.txt)
11	water	[doc] (formats/tiletypes.txt)
12	lava	[doc] (formats/tiletypes.txt)
20	throw-through (fences)	[doc] (formats/tiletypes.txt)
21	jumping stone	[doc] (formats/tiletypes.txt)
22	blocking tile	[doc] (formats/tiletypes.txt)

3.5.2.1 A 4096-byte type table

A 4096-byte table copied from the standard tileset, one type byte per id, is kept in an outside client's source and written as `tiledefs.txt`: decimal, row-major, 128 values per line (the width of `pics1.png`) [src] (formats/tiletypes.txt). Its counts:

Value	Count	Official meaning	Source
0	823	nonblocking	[src] (formats/tiletypes.txt)
3	40	chair	[src] (formats/tiletypes.txt)

Value	Count	Official meaning	Source
4	4	bed, upper	[src] (formats/tiletypes.txt)
5	4	bed, lower	[src] (formats/tiletypes.txt)
6	12	swamp	[src] (formats/tiletypes.txt)
7	8	lava swamp	[src] (formats/tiletypes.txt)
8	30	near water	[src] (formats/tiletypes.txt)
11	47	water	[src] (formats/tiletypes.txt)
12	67	lava	[src] (formats/tiletypes.txt)
20	129	throw-through fence	[src] (formats/tiletypes.txt)
21	18	jumping stone	[src] (formats/tiletypes.txt)
22	2914	blocking	[src] (formats/tiletypes.txt)

Rule	Detail	Source
absent types	types 2 and 9 are in the official list but do not appear in this table	[src] (formats/tiletypes.txt)
first row	the first data line, ids 0 to 127, starts `0,0,20,20,…`: the top-left two tiles walk and the next two are walls	[src] (formats/tiletypes.txt)
that client's tests	wall if the type byte is ≥ 20 or the foot is outside the 64×64 board; water if it is 11; walkable otherwise	[src] (formats/tiletypes.txt)

3.5.2.2 Sheet types and the type-0 allow lists (official client)

Rule	Detail	Source
tile definition list	each entry is `image,prefix,type` with optional `x,y` offsets; the default is `pics1.png` of type 0, stock selection `pics1.png,,0`	[live] (formats/tile-walk.md#sheet-loading); [bin] (formats/tiletypes.txt)
sheet order	type 0 is the old Graal tile order (`pics1`), type 1 the new order; `addtiledef2` paints a rectangle on top of the base sheet; the offline tree holds `picso.png,offlinetutorial,1,0,0`	[file] (client-internals/client-assets.md#tiledef)
sheet choice	a level may reference several sheets; the active one is selected when the level is entered	[live] (formats/tile-walk.md#sheet-loading)
type-0 sheet	a tile walks only if its 12-bit id is in one of two internal allow lists (the "nowall" lists); an id in neither blocks	[live] (formats/tile-walk.md#collision-rules); [2.05] (formats/arrays-and-signatures.md#arraysdat)
allow lists	built when the sheet loads; not stored in `tiledefs.txt`, which names only the sheet, prefix and type; they are arrays 0 and 1 of `arrays.dat` (→ §3.5.3)	[live] (formats/tile-walk.md#type-0-allow-lists); [2.05] (formats/arrays-and-signatures.md#arraysdat)
type 1 and 2 sheets	consult a fixed 0/1 body-block grid; one extraction counted 1725 blocked and 2371 walkable cells	[live] (formats/tile-walk.md#collision-rules); [bin] (formats/tiletypes.txt)
type-0 mask	`disputed → App. E#101`	[live] (formats/tile-walk.md#type-0-allow-lists)
`setshape2`	blocks the body when the shape-tile type is greater than 19	[bin] (formats/tiletypes.txt)
throw	throw handling does not stop on type 0x14 (20)	[bin] (formats/tiletypes.txt); [live] (formats/tile-walk.md#collision-rules)
water	type 11 does not block; it changes the displayed tile and may set a swimming flag	[live] (formats/tile-walk.md#collision-rules)

The 2×2 Bush stamp ids 2, 3, 18 and 19 are walls in the 4096-byte table and walkable through the type-0 allow lists `disputed → App. E#100`.

3.5.3 arrays.dat

Rule	Detail	Source
integers	big-endian 32-bit	[2.05] (formats/arrays-and-signatures.md#arraysdat)
array header	`header = (kind << 16) | count`	[2.05] (formats/arrays-and-signatures.md#arraysdat)
kind 0	`count` int32 values follow	[2.05] (formats/arrays-and-signatures.md#arraysdat)
kind 1	`count` nested arrays follow, each with its own header	[2.05] (formats/arrays-and-signatures.md#arraysdat)
order	the client reads exactly 25 arrays in a fixed order into 25 globals at startup	[2.05] (formats/arrays-and-signatures.md#arraysdat)
missing file	the client quits with "File not found"; the file is not signed or checked	[2.05] (formats/arrays-and-signatures.md#summary)
2.22 file	11500 bytes; parses to exactly 25 arrays with nothing left over	[test] (formats/arrays-and-signatures.md#arraysdat)
tile ids	all tile ids in the tables index the 4096-tile `pics1` sheet	[2.05] (formats/arrays-and-signatures.md#arraysdat)

#	Shape	Content	Source
0	934 ints, max 4089	tile ids; with array 1, the legacy walkability test: in the old tile mode a tile blocks unless it is in array 0 or array 1	[2.05] (formats/arrays-and-signatures.md#arraysdat)
1	116 ints	tile ids, second list for that test	[2.05] (formats/arrays-and-signatures.md#arraysdat)
2	133 ints	tile ids; use not traced	[2.05] (formats/arrays-and-signatures.md#arraysdat)
3	76 ints	tile ids; use not traced	[2.05] (formats/arrays-and-signatures.md#arraysdat)
4	40 ints	tile ids; use not traced	[2.05] (formats/arrays-and-signatures.md#arraysdat)
5, 6	4 ints each	single 2×2 tile blocks (708–711, 724–727); use not traced	[2.05] (formats/arrays-and-signatures.md#arraysdat)
7	79 ints	tile ids; the legacy tile-type lookup maps a tile's position in this list to 4 classes (positions 0–11, 12–19, 20–62, 63+); the newer tile-type path returns types 6 to 9 for the same classes	[2.05] (formats/arrays-and-signatures.md#arraysdat)
8	4 lists, one per facing	ledge tiles (type 21): a ledge jump needs the tile to be in the list for the player's facing	[2.05] (formats/arrays-and-signatures.md#arraysdat)

#	Shape	Content	Source
9 to 12	small nested lists of 2×2 tile groups	level-object code: cut and lift	[2.05] (formats/arrays-and-signatures.md#arraysdat)
13	4 ints	one 2×2 tile block, used with array 14	[2.05] (formats/arrays-and-signatures.md#arraysdat)
14, 15	small nested lists of 2×2 tile groups	level-object code	[2.05] (formats/arrays-and-signatures.md#arraysdat)
16	2 × 2×2 blocks	stone blocks: entry 0 pushable, entry 1 pullable	[2.05] (formats/arrays-and-signatures.md#arraysdat)
17	124 × 4	packed records used by the horse object	[2.05] (formats/arrays-and-signatures.md#arraysdat)
18	nested	use not traced	[2.05] (formats/arrays-and-signatures.md#arraysdat)
19	nested	used by the leap object	[2.05] (formats/arrays-and-signatures.md#arraysdat)
20, 22, 23	nested	used by the baddy AI object	[2.05] (formats/arrays-and-signatures.md#arraysdat)
21	4 × 4	packed values; use not traced in detail	[2.05] (formats/arrays-and-signatures.md#arraysdat)
24	356 ints, packed	used by the item, bomb and extras objects	[2.05] (formats/arrays-and-signatures.md#arraysdat)

Meanings read from the uses in the 2.22 client [bin] (client-internals/decomp-findings-2026-10-25-l/02-arrays-dat.md#arrays-index--position-in-the-file):

#	Content (2.22)	Source
0	passable tile ids; any other tile is a wall (→ App. E#101, E#608)	[bin] (client-internals/decomp-findings-2026-10-25-l/02-arrays-dat.md#arrays-index--position-in-the-file)
1	liquid tile ids, water and lava	[bin] (client-internals/decomp-findings-2026-10-25-l/02-arrays-dat.md#arrays-index--position-in-the-file)
3	the liquid tiles that are lava	[bin] (client-internals/decomp-findings-2026-10-25-l/02-arrays-dat.md#arrays-index--position-in-the-file)

#	Content (2.22)	Source
9, 10	cuttable 2×2 objects and the tiles a cut leaves, indexed type, column, row: the bush `2 3 / 18 19` and `420 421 / 436 437`	[bin] (client-internals/decomp-findings-2026-10-25-l/02-arrays-dat.md#arrays-index--position-in-the-file)
11, 12	liftable 2×2 objects and the tiles a lift leaves: five types, among them the bush and the rock `34 35 / 50 51`	[bin] (client-internals/decomp-findings-2026-10-25-l/02-arrays-dat.md#arrays-index--position-in-the-file)
13	closed chest tiles 1784, 1785, 1800, 1801	[bin] (client-internals/decomp-findings-2026-10-25-l/02-arrays-dat.md#arrays-index--position-in-the-file)
14	open chest tiles 829, 845 / 830, 846	[bin] (client-internals/decomp-findings-2026-10-25-l/02-arrays-dat.md#arrays-index--position-in-the-file)

A loader that reads only the first four arrays as flat lists is right for 0 to 3 and misses the nested arrays 9 to 12 [bin] (client-internals/decomp-findings-2026-10-25-l/02-arrays-dat.md#arrays-index--position-in-the-file).

3.5.4 arrays2.dat

2.05 does not reference `arrays2.dat`, `graal.sig` or `arrays2.sig` and has no signature check [2.05] (formats/arrays-and-signatures.md#summary). In 2.17, 2.22 and 2.31 the file is plain text with CRLF lines, numbered from 0; 2.22 needs at least 5 lines [2.22] (formats/arrays-and-signatures.md#arrays2dat).

Line	Use	Source
0	main listserver host	[2.22] (formats/arrays-and-signatures.md#arrays2dat)
1	listserver port	[2.22] (formats/arrays-and-signatures.md#arrays2dat)
2	web host, joined into a URL template	[inf] (formats/arrays-and-signatures.md#arrays2dat)
3	registration page URL	[2.22] (formats/arrays-and-signatures.md#arrays2dat)
4	translation service URL	[2.22] (formats/arrays-and-signatures.md#arrays2dat)
5, 6	fallback listserver hosts, optional	[2.22] (formats/arrays-and-signatures.md#arrays2dat)

A fan-server copy of the file points line 0 at its own `<listserver-host>` with port 14922 and lists two other list hosts as fallbacks [test] (formats/arrays-and-signatures.md#arrays2dat). The 2.05 list host is hidden in the executable instead (→ §6).

3.5.5 graal.sig and arrays2.sig

Rule	Detail	Source
size	both files are 128 bytes, the size of an RSA-1024 signature	[test] (formats/arrays-and-signatures.md#the-signatures)
key	built at startup from an obfuscated hex string assembled one character at a time; each decoded byte is `not(b) - 10`	[2.22] (formats/arrays-and-signatures.md#the-signatures)
order	`graal.sig` is verified against the client's own executable file, then `arrays2.sig` against `arrays2.dat`	[2.22] (formats/arrays-and-signatures.md#the-signatures)
algorithm	an RSA component configured with a SHA-1 signing digest and a PKCS#1 scheme, judged from the component's class and property names; no signature has been verified	[inf] (formats/arrays-and-signatures.md#the-signatures)
failure	the client shows "The Graal execution file is corrupted. Please check for viruses or other programs that manipulate..." and exits	[2.22] (formats/arrays-and-signatures.md#the-signatures)
builds	2.17, 2.22 and 2.31 embed the same two key strings and the same error text	[2.17][2.22][2.31] (formats/arrays-and-signatures.md#the-signatures)
open	checking the shipped signatures against the decoded key with exponents 3, 17 and 65537 did not reproduce PKCS#1 padding; the key encoding is not pinned down	[test] (formats/arrays-and-signatures.md#the-signatures)

Rule	Detail	Source
fan-server 2.22	that executable has the three branches that raise the verify error rewritten so verification always passes; the 2.17 and 2.31 executables do not	[2.22][2.17][2.31] (formats/arrays-and-signatures.md#the-graal-reborn-222-exe-has-the-check-disabled)
consequence	an unpatched 2.17, 2.22 or 2.31 client refuses an edited `arrays2.dat` or executable; editing `arrays2.dat` alone fails the signature	[inf] (formats/arrays-and-signatures.md#the-graal-reborn-222-exe-has-the-check-disabled)

The check as a client integrity behaviour is in → §4.10.

3.6 Sign text alphabets

Binary sign text uses one packed alphabet. The same encoding is used in the sign sections of `.graal` and `.zelda` files (→ §3.1.2.6) and on the wire in `PLO_LEVELSIGN`; GLEVNW01 text signs are plain text [2.05][srv] (formats/sign-text.md).

3.6.1 Encoding

Rule	Detail	Source
record	`GCHAR x`, `GCHAR y`, then the text bytes	[2.05][srv] (formats/sign-text.md#encoding)
text byte	a GCHAR: `byte - 32` indexes a fixed 97-entry alphabet, the same string in the 2.05 client and in GServer-v2	[2.05][srv] (formats/sign-text.md#encoding)
codes 0 to 61	`A–Z`, `a–z`, `0–9`	[2.05][srv] (formats/sign-text.md#encoding)
codes 62 to 96	punctuation, space (code 89, byte `0x79`) and newline (code 96, byte `0x80`); several slots are # placeholders	[2.05][srv] (formats/sign-text.md#encoding)
glyph codes	one or two codes stand for # plus a symbol: #A #B #X #Y (button icons), #u #d #l #r (arrows), #x #y #z, #h and #4 (two codes each), and `#.`; code 86 is a literal #	[2.05][srv] (formats/sign-text.md#encoding)
other characters	written as the escape `#K(n)`, with n the decimal character code spelled in alphabet codes	[2.05][srv] (formats/sign-text.md#encoding)
controls	bytes under 32 are controls and are not shown	[2.05][srv] (formats/sign-text.md#encoding)
line break	the editor writes CRLF as `#K(13)` and the newline code; in a raw dump that is the bytes `v*eUWf` and a byte with the high bit set	[test] (formats/sign-text.md#encoding)

The alphabet, from code 0 [2.05] (formats/level-binary-client.md#sign-text-in-files):

```
ABCDEFGHIJKLMNOPQRSTUVWXYZabcdefghijklmnopqrstuvwxyz0123456789!?-.,#>()#####"####':/~&###
<####;\n
```

3.6.2 Corrections to earlier notes

Earlier claim	Correction	Source
two alphabets, one for files and one for the wire	one alphabet indexed by `byte - 32`	[test] (formats/sign-text.md#correction)
a `char + 89` file rule with "subtract 6 from A–Z"	it only looked right for some characters	[test] (formats/sign-text.md#correction)
`v*eUWf` + high byte = space	it is `#K(13)` + newline, a line break	[test] (formats/sign-text.md#correction)
garbled house-outside signs	viewers that drop non-ASCII bytes produce that appearance	[test] (formats/sign-text.md#correction)

NPC scripts in the same file are plain Latin-1 with `0xA7` as the newline, not sign-encoded [test] (formats/sign-text.md#correction).

3.6.3 GLEVNW01 signs

Rule	Detail	Source
record	`SIGN x y`, the text, `SIGNEND`; the body may be empty (→ §3.1.3.3)	[srv] (formats/sign-text.md#glevnw01-signs)
text	plain; no alphabet transform	[srv] (formats/sign-text.md#glevnw01-signs)

3.6.4 Reference decoder

Function	Use	Source
`decodeSignRecord(line)`	a raw sign line or a `PLO_LEVELSIGN` payload → `{ x, y, text }`	[src] (formats/sign-text.md#drop-in-decoder)
`decodeSignText(payload)`	the payload after the coordinates; by default the CR from `#K(13)` is dropped	[src] (formats/sign-text.md#drop-in-decoder)
`encodeSignText`, `encodeSignRecord`	the reverse	[src] (formats/sign-text.md#drop-in-decoder)
tests	`formats/sign-decoder/test.js` and `vectors.json`, from 11 real signs in GR-V1.03 levels	[test] (formats/sign-text.md#drop-in-decoder)

3.7 Save games (graalsav)

Clients from 1st Graal (September 1998) to the 1.4x builds write offline save games; from 2.00 the server owns the save [bin] (formats/graalsav.md); [2.05] (client-internals/feature-specs/05-save-load.md#what-it-is). This chapter gives the save container, its revisions, the DES-wrapped local files and the server's account file.

> Status: draft. Field names rest on 13th Graal's script variables; the Graalsav9 and Graalsv10 layouts come only from readers; the DES wrapping has not been round-tripped on a real save.

3.7.1 Container

Every revision is a Delphi `TWriter` value stream, not a hand-made binary layout. Each value starts with a one-byte type tag [bin] (formats/graalsav.md#container):

Tag	Type	Body	Source
2	Int8	1 byte	[bin] (formats/graalsav.md#container)
3	Int16	2 bytes, little-endian	[bin] (formats/graalsav.md#container)
4	Int32	4 bytes, little-endian	[bin] (formats/graalsav.md#container)
5	Extended	10-byte x87 float	[bin] (formats/graalsav.md#container)
6	String	1-byte length and the characters	[bin] (formats/graalsav.md#container)
12	LString	4-byte length and the characters; strings of 256 bytes or more	[bin] (formats/graalsav.md#container)
8, 9	False, True	none	[bin] (formats/graalsav.md#container)

Rule	Detail	Source
integer width	the writer picks the smallest of Int8, Int16 and Int32, so a reader must go by the tag	[bin] (formats/graalsav.md#container)
first value	always the magic string: `Graalsav1` to `Graalsav9`, `Graalsv10`, `Graalsv11`	[bin] (formats/graalsav.md#container)
field names	taken from 13th Graal's script variables (`playerrupees`, `playerdarts` ...), whose storage lines up with that build's save record; other builds are matched by position	[bin] (formats/graalsav.md)

3.7.2 Revisions by build

Magic	Written by	Read by	Extension	DES-wrapped	Source
Graalsav1	1st, 2nd, 3rd, 4th	same	`.save`	no	[bin] (formats/graalsav.md#versions-by-build)
Graalsav2	4th alpha, 5th, 5th alpha	same	`.save`	no	[bin] (formats/graalsav.md#versions-by-build)
Graalsav3	6th	same	`.save`	no	[bin] (formats/graalsav.md#versions-by-build)

Magic	Written by	Read by	Extension	DES-wrapped	Source
Graalsav4	7th, 8th	same	`.save`	no	[bin] (formats/graalsav.md#versions-by-build)
Graalsav5	9th alpha to 12th	same	`.save`	no	[bin] (formats/graalsav.md#versions-by-build)
Graalsav6	13th	same	`.save`	no	[bin] (formats/graalsav.md#versions-by-build)
Graalsav7	14th, 15th	same	`.save` (14th), `.gsave` (15th)	from 15th	[bin] (formats/graalsav.md#versions-by-build)
Graalsav8	1.1, 1.1.9	7 and 8	`.gsave`	yes	[bin] (formats/graalsav.md#versions-by-build)
Graalsav9, Graalsv10	no build in the set	1.21 to 1.32 readers	`.gsave`	yes	[bin] (formats/graalsav.md#versions-by-build)
Graalsv11	1.21 to 1.41 rev 1; writer traced in 1.21 and 1.32, later builds by string references	9, 10 and 11	`.gsave`	yes	[bin] (formats/graalsav.md#versions-by-build)

Rule	Detail	Source
4th alpha	built later than 4th, which is why it writes the newer revision	[bin] (formats/graalsav.md#versions-by-build)
1.33 to 1.41 rev 1	reference the three magics as 1.32 does (two references to `Graalsv11`, one each to `Graalsav9` and `Graalsv10`), so they write `Graalsv11`; the record itself was not re-read	[bin] (formats/graalsav.md#versions-by-build)
catalogue correction	earlier catalogue rows listed 1.21 to 1.39 as "Graalsav9"; a search for `Graalsav` misses `Graalsv10` and `Graalsv11`	[bin] (formats/graalsav.md#versions-by-build)
v9 and v10	presumably match saves from a 1.2x build that is not in the archive	[inf] (formats/graalsav.md#versions-by-build)
15th Graal	saving only works on a level from one named community server or on the online start level	[bin] (formats/graalsav.md#versions-by-build)
1.21 on	the client refuses to save online: "The server saves your game if you have an own account."	[bin] (formats/graalsav.md#versions-by-build)
2.00 on	the server owns the save; 2.05 contains "Save games are not supported anymore", and the offline F2/F3 save games ended with 1.41	[2.05] (client-internals/feature-specs/05-save-load.md#what-it-is)

3.7.3 File layout

```
magic                     String
mode                      Integer   0 one player, 1 two players, 2 online, 3/4 three/four
players (14th on)
[mode 2]      String, String        two online-session strings
player record × n         n = 1, 2, or the player count in 3/4-player modes
[Graalsav2+]  flag list   Integer count, then count × String
```

Rule	Detail	Source
layout	as above	[bin] (formats/graalsav.md#file-layout)
online strings	nickname and server	[inf] (formats/graalsav.md#file-layout)

Rule	Detail	Source
Graalsv11 header	`magic`, then `Integer players-1`, then the records and the flag list	[bin] (formats/graalsav.md#file-layout)
6 to 7	the change is in the header: the four-player mode	[bin] (formats/graalsav.md#graalsav7-and-graalsav8)

3.7.4 Player record

3.7.4.1 Graalsav1

Order	Type	Field	Source
1	String	level	[bin] (formats/graalsav.md#graalsav1)
2	String	nickname	[bin] (formats/graalsav.md#graalsav1)
3, 4	Float	x, y	[bin] (formats/graalsav.md#graalsav1)
5	Integer	dir	[inf] (formats/graalsav.md#graalsav1)
6	Integer	fullhearts	[inf] (formats/graalsav.md#graalsav1)
7	Float	hearts	[bin] (formats/graalsav.md#graalsav1)
8	Integer	rupees	[bin] (formats/graalsav.md#graalsav1)
9	Integer	darts, default 5	[bin] (formats/graalsav.md#graalsav1)
10	Integer	bombs, default 10	[bin] (formats/graalsav.md#graalsav1)
11	Integer	glovepower, default 2	[inf] (formats/graalsav.md#graalsav1)
12	4 × Boolean	flags; the first defaults to true	[bin] (formats/graalsav.md#graalsav1)
13	4 × Boolean	flags	[bin] (formats/graalsav.md#graalsav1)
14	4 × Integer	colours, the same four values the relay S message sends (→ §2.3.1)	[bin] (formats/graalsav.md#graalsav1)

3.7.4.2 Graalsav2 to Graalsav8

The record starts as Graalsav1 (level, nickname, x, y, dir, fullhearts, hearts, rupees, darts, bombs), then [bin] (formats/graalsav.md#graalsav2-to-graalsav6):

Field	2	3	4	5	6, 7, 8	Source
Integer glovepower	yes	yes	yes	yes	yes	[bin] (formats/graalsav.md#graalsav2-to-graalsav6)

Field	2	3	4	5	6, 7, 8	Source
Integer shieldpower, swordpower	yes	yes	yes	yes	yes	[bin] (formats/graalsav.md#graalsav2-to-graalsav6)
Integer shootpower	n/a	n/a	yes	yes	yes	[bin] (formats/graalsav.md#graalsav2-to-graalsav6)
one Boolean	yes	yes	n/a	n/a	n/a	[bin] (formats/graalsav.md#graalsav2-to-graalsav6)
5 × Integer colours	yes	yes	yes	yes	yes	[bin] (formats/graalsav.md#graalsav2-to-graalsav6)
Integer headset	yes	yes	yes	yes	yes	[bin] (formats/graalsav.md#graalsav2-to-graalsav6)
horse: String image, Integer bushes	n/a	n/a	n/a	yes	yes	[bin] (formats/graalsav.md#graalsav2-to-graalsav6)
carried NPCs: Integer n, then n × (String image, String script)	n/a	n/a	n/a	yes	yes	[bin] (formats/graalsav.md#graalsav2-to-graalsav6)
trailing Boolean	n/a	n/a	n/a	n/a	yes	[bin] (formats/graalsav.md#graalsav2-to-graalsav6)
trailing Integer	n/a	n/a	n/a	n/a	8 only; the 1.1.9 loader reads it only when the magic is 8 or above	[bin] (formats/graalsav.md#graalsav7-and-graalsav8)

The power order in the file is glove, shield, sword, then shoot; the colour count goes from 4 to 5 between Graalsav1 and Graalsav2 [bin] (formats/graalsav.md#graalsav2-to-graalsav6).

3.7.4.3 Graalsav9 to Graalsv11

The player is one string in the network player-props encoding [bin] (formats/graalsav.md#graalsav9-to-graalsv11-121-to-132):

```
String  props          concatenation over prop index i: chr(i+32) + the prop's encoded value
[v9]    String horse image, Integer bushes
Integer n, then n × (String image, String script)        carried and inventory NPCs
[v11]   Integer w, then w × String                        NPC weapon names
[v11]   Integer selected weapon index
```

Rule	Detail	Source
sources	`Graalsv11` is from the writer; the `[v9]` and `[v11]` conditions are from the reader	[bin] (formats/graalsav.md#graalsav9-to-graalsv11-121-to-132)
props codec	the same as the generation 2 protocol (→ App. B); a props decoder for generation 2 messages reads these saves	[bin] (formats/graalsav.md#graalsav9-to-graalsv11-121-to-132)

3.7.5 DES-wrapped local files

From 5th Graal the client carries a small DES class and two 8-byte keys, `AXBORG15` and `VERGRAAL`. They protect local files against editing; nothing on the network is encrypted in these builds [bin] (formats/graalsav.md#des-wrapped-files).

3.7.5.1 `.code` files (5th Graal on)

```
24 bytes  DES-ECB(AXBORG15, P)
24 bytes  DES-ECB(VERGRAAL, P)
```

Rule	Detail	Source
files	`guildname.code` (behind the `playeringuild` script check); in 15th Graal to 1.1.9 also `reg.code`, the registration key	[bin] (formats/graalsav.md#code-files-5th-graal-on)
P	a 24-byte buffer holding a Pascal short string: a length byte, then up to 23 characters	[bin] (formats/graalsav.md#code-files-5th-graal-on)
reader	decrypts both halves, requires them to be equal, and returns the string; a short file or differing halves read as ###	[bin] (formats/graalsav.md#code-files-5th-graal-on)
key broadcast	in 15th Graal to 1.1.9 the registration key is also sent as relay type 8, so two players with the same key are noticed (→ §2.3.1)	[bin] (formats/graalsav.md#code-files-5th-graal-on)

3.7.5.2 `.gsave` (15th Graal on)

```
u32le     n = length of the padded plaintext
n bytes   DES-ECB(AXBORG15, plaintext)
n bytes   DES-ECB(VERGRAAL, plaintext)
```

Rule	Detail	Source
plaintext	the `TWriter` stream of §3.7.1, padded to a multiple of 8 with random bytes	[bin] (formats/graalsav.md#gsave-15th-graal-on)
loader	decrypts both copies, compares them byte for byte, and parses only if they match	[bin] (formats/graalsav.md#gsave-15th-graal-on)
DES variant	the class takes a second constructor argument (7) whose role is not pinned down; the 2.05 DES is textbook DES-ECB with each key byte bit-reversed, and the same handling is probably at work here, so a round trip should try the keys bit-reversed first	[inf] (formats/graalsav.md#gsave-15th-graal-on)
open	no real save has been round-tripped	[bin] (formats/graalsav.md#gsave-15th-graal-on)

3.7.5.3 Saved passwords (1.3 test to 2.22)

The client saves account passwords in the registry under `HKEY_CURRENT_USER\Software\Graal\Client`, as `password` and as `<account>:pass` per account [2.05] (client-internals/saved-passwords.md#warning-for-anyone-holding-old-installs).

Rule	Detail	Source
cipher	DES-ECB with standard tables, each key byte bit-reversed before the key schedule; the same routine as the `.gsave` DES	[2.05][test] (client-internals/saved-passwords.md#cipher)
key	one fixed 8-byte key, the same in every copy of the client, so a saved value is as good as plain text to anyone who can read the registry or an export of it	[2.05] (client-internals/saved-passwords.md#warning-for-anyone-holding-old-installs)
builds	the same key and form from 1.3 test (Nov 1999) to 2.05; a value saved by 1.31 or 1.32 in the lab decrypts the same way	[test] (client-internals/saved-passwords.md#warning-for-anyone-holding-old-installs)
encrypt	cut the plaintext to 23 bytes; build 24 bytes: the length, the text, random padding; encrypt as three DES blocks; write each byte as two characters `(b >> 4) + 0x20`, `(b & 0x0F) + 0x20`	[2.05] (client-internals/saved-passwords.md#string-format-48-characters)
decrypt	input under 48 characters, or a first decrypted byte of 0 or 24 and above, gives an empty string	[2.05] (client-internals/saved-passwords.md#string-format-48-characters)

Rule	Detail	Source
same text, new value	the random padding makes every save of the same password differ	[2.05] (client-internals/saved-passwords.md#string-format-48-characters)
opting out	"Don't save the passwords" in More Options deletes the values when the client closes	[2.05] (client-internals/saved-passwords.md#warning-for-anyone-holding-old-installs)
2.22	a 49-character value that starts with a marker character is decrypted with a second key held by the options form; any other value with the first key, so old saves still load; new saves use the marker and the second key	[2.22] (client-internals/des-and-hidden-strings.md#des-saved-passwords-and-the-web-cache-pointers)
2.22 account name	stored with the byte transform of the login (→ §2.3.3)	[2.22] (client-internals/des-and-hidden-strings.md#des-saved-passwords-and-the-web-cache-pointers)
other uses	2.05 builds its hidden listserver host the same way, from a key and a 48-character text assembled one character at a time (→ §6.1.1); the web-level cache uses the same DES with per-block keys (→ §3.8)	[2.05][cap] (client-internals/des-and-hidden-strings.md#hidden-strings); [2.05] (client-internals/saved-passwords.md#other-uses-of-the-same-cipher)
what it protects	nothing; these schemes only keep the strings out of a casual string dump	[inf] (client-internals/des-and-hidden-strings.md#notes-for-tool-writers)

3.7.6 Server account file (GRACC001)

From 2.00 the save is the server's account file; the client receives it as its login burst (→ §2.3) [srv] (client-internals/feature-specs/05-save-load.md#what-it-is).

Rule	Detail	Source
file	`accounts/<name>.txt`; text, first line `GRACC001`, then `KEY value` lines, CRLF on write	[srv] (client-internals/feature-specs/05-save-load.md#server-file-gracc001-accountsnametxt-srv)
player keys	NAME, NICK (at most 223 characters), COMMUNITYNAME, LEVEL, X, Y, Z, MAXHP, HP, RUPEES, ANI, ARROWS, BOMBS, GLOVEP, SHIELDP, SWORDP, BOWP, BOW, HEAD, BODY, SWORD, SHIELD, COLORS (5 comma-separated), SPRITE, STATUS, MP, AP, APCOUNTER, ONSECS, IP, LANGUAGE, KILLS, DEATHS, RATING, DEVIATION, LASTSPARTIME	[srv] (client-internals/feature-specs/05-save-load.md#server-file-gracc001-accountsnametxt-srv)
repeated keys	FLAG (`name=value`), ATTR1 to ATTR30, WEAPON, CHEST (an `x:y:level` style string)	[srv] (client-internals/feature-specs/05-save-load.md#server-file-gracc001-accountsnametxt-srv)
admin keys	BANNED, BANREASON, BANLENGTH, COMMENTS, EMAIL, LOCALRIGHTS, IPRANGE, LOADONLY, FOLDERRIGHT (repeated), LASTFOLDER	[srv] (client-internals/feature-specs/05-save-load.md#server-file-gracc001-accountsnametxt-srv)
new account	the server clones `accounts/defaultaccount.txt`, applies `startlevel`, `startx` and `starty` from the server options if set, and saves the new file	[srv] (client-internals/feature-specs/05-save-load.md#server-file-gracc001-accountsnametxt-srv)

Rule	Detail	Source
guest	load-only, with a random `pc:` community name	[srv] (client-internals/feature-specs/05-save-load.md#server-file-gracc001-accountsnametxt-srv)
saving	on disconnect for accounts that are not load-only, and every 300 s while connected; not saved if the listserver login failed	[srv] (client-internals/feature-specs/05-save-load.md#server-file-gracc001-accountsnametxt-srv)

The client's other local files from 2.00 on are its config files (→ §3.9) and the web-level cache (→ §3.8) [2.05] (client-internals/feature-specs/05-save-load.md#what-it-is).

3.8 Web level cache (GWEBL001)

The web-level cache is a local copy of a server level, keyed by the level's modification time, so the client can skip downloading a level it already has. The format was traced in the 2.05 rev 2 reader and writer and confirmed on 144 real cache files from a 2.22 install: every one decrypts, passes the checksum, parses and re-encrypts byte for byte [2.05][test] (formats/gwebl001-cache.md).

3.8.1 When it is used

Event	Behaviour	Source
write	when the server sends the level's 5-byte modification time (`PLO_LEVELMODTIME`, id 39), the client saves the level it holds with that time	[2.05] (formats/gwebl001-cache.md#what-it-is)
read	when the client enters a level and a cache file exists, it builds the level from the file	[2.05] (formats/gwebl001-cache.md#what-it-is)
benefit	the server then presumably only needs to send levels that have changed	[inf] (formats/gwebl001-cache.md#what-it-is)
mismatch	a file whose header does not match the current server is deleted	[2.05] (formats/gwebl001-cache.md#what-it-is)

3.8.2 Path

```
<exe dir>\weblevels\<server host>\<level>-<server port>.code
```

Rule	Detail	Source
`<level>`	the level's file name, lowercased, with any path removed	[2.05] (formats/gwebl001-cache.md#path)
fallback	the reader also tries `<level>.code` without the port	[2.05] (formats/gwebl001-cache.md#path)
example	`weblevels\<public-host>\bank.nw-14803.code`	[test] (formats/gwebl001-cache.md#path)

3.8.3 Encrypted file

```
s32le      N            padded plaintext length + 8
N-8 bytes  ciphertext   DES-ECB, one fresh key per 8-byte block
8 bytes    checksum     bytewise sum (mod 256) of all plaintext blocks, per byte position
```

Rule	Detail	Source
layout	as above	[2.05] (formats/gwebl001-cache.md#file-layout-encrypted)
padding	the plaintext is padded with `0x0A` to a multiple of 8	[2.05] (formats/gwebl001-cache.md#file-layout-encrypted)
seed	`RandSeed = 0x04A80B38`; then for each byte c of `<level>`, `RandSeed = RandSeed * c` with 32-bit wrap	[2.05][test] (formats/gwebl001-cache.md#per-block-keys)
block key	8 calls to Delphi `Random(256)`, drawn fresh for every block in file order	[2.05][test] (formats/gwebl001-cache.md#per-block-keys)
`Random(n)`	`RandSeed = RandSeed * 0x08088405 + 1`, then `(RandSeed * n) >> 32`, unsigned	[2.05][test] (formats/gwebl001-cache.md#per-block-keys)
DES key bytes	bit-reversed, as in the 2.05 DES elsewhere (→ §3.7.5.2)	[2.05][test] (formats/gwebl001-cache.md#per-block-keys)
strength	the encryption needs only the level name; a wrong name shows up as a checksum mismatch	[2.05] (formats/gwebl001-cache.md#per-block-keys)

3.8.4 Plaintext

```
"GWEBL001"
GCHAR len + server host          must equal the connected host
GCHAR hi, GCHAR lo               server signature, (hi-32)*128 + (lo-32)
5 bytes                          level modification time, 5 × 7-bit GCHAR, big-endian
GCHAR len + level name           must equal <level> above
level body                       GR-V1.03 (2.05), or GR-V1.03 / GR-V1.04 (2.31)
```

Rule	Detail	Source
layout	as above	[2.05][2.31] (formats/gwebl001-cache.md#plaintext-layout)
signature check	passes when the stored value equals the value the server sent in PLO_SIGNATURE (id 25), or when the stored value is 1 and the server sent 73, GServer's usual value	[2.05] (formats/gwebl001-cache.md#plaintext-layout)
test files	all 144 hold 73	[test] (formats/gwebl001-cache.md#plaintext-layout)
modification time	decoded as a 35-bit integer it is Unix time; the test files decode to dates a little before their file timestamps (bank.nw decodes to 1247727798, 16 July 2009, and the file is dated 26 July 2009)	[test] (formats/gwebl001-cache.md#plaintext-layout)

3.8.5 Level body as cached

The body is an ordinary GR-V level (→ §3.1.2), but the writer fills only some sections [2.05] (formats/gwebl001-cache.md#level-body-as-cached):

Section	Cached	Source
board	yes, the 13-bit run-length stream	[2.05] (formats/gwebl001-cache.md#level-body-as-cached)
links	yes, `dest x y w h newx newy` lines, then #	[2.05] (formats/gwebl001-cache.md#level-body-as-cached)
baddies	no; terminator only, FF FF FF 0A	[2.05] (formats/gwebl001-cache.md#level-body-as-cached)
NPCs	no; empty, #	[2.05] (formats/gwebl001-cache.md#level-body-as-cached)
chests	no; empty, #	[2.05] (formats/gwebl001-cache.md#level-body-as-cached)
signs	yes, `x+32, y+32, sign text, \n`	[2.05] (formats/gwebl001-cache.md#level-body-as-cached)
heights	2.31 only: a HEIGHTS block when the level has a height map, which also switches the magic to GR-V1.04 (→ §3.1.2.7)	[2.31] (formats/gwebl001-cache.md#level-body-as-cached)

3.8.6 Tool

Command	Effect	Source
`python3 tools/file-formats/gwebl.py FILE.code`	summary	[src] (formats/gwebl001-cache.md#tool)
`… --json`	full parse	[src] (formats/gwebl001-cache.md#tool)
`… --raw plain.bin`	the decrypted stream	[src] (formats/gwebl001-cache.md#tool)

Command	Effect	Source
module	`encrypt_stream` re-creates files byte for byte; also `read_board`, `write_board` and `parse_grv` for plain `.graal` data	[src](formats/gwebl001-cache.md#tool)

3.9 Config and dat files

The client keeps its settings in `graalconfig.txt` next to the executable and its login details in the registry. This chapter gives both as 2.05 rev 2 reads and writes them, checked against a real `graalconfig.txt` written by 2.22, which uses the same keys in the same order [2.05][2.22] (client-internals/graalconfig-reference.md). The table files `arrays.dat` and `arrays2.dat` are in → §3.5.

> Status: draft. 2.05 is traced and 2.22 compared; the keys of 3.x and later clients are unmeasured.

3.9.1 Where settings live

Store	Location	Holds	Source
`graalconfig.txt`	next to the executable	every option on the More Options and Player List Options dialogs, the head, language, font, joystick and key bindings	[2.05] (client-internals/graalconfig-reference.md#where-settings-live)
registry	`HKEY_CURRENT_USER\Software\Graal\Client`	`nickname`, `accountname`, `password`, and one `<account>:pass` value per account	[2.05] (client-internals/graalconfig-reference.md#where-settings-live)

3.9.2 graalconfig.txt file rules

Rule	Detail	Source
form	a plain `key=value` list; key lookup is case-insensitive and order does not matter on read	[2.05] (client-internals/graalconfig-reference.md#where-settings-live)
read	once, when the options form is created at startup	[2.05] (client-internals/graalconfig-reference.md#when-it-is-read-and-written)
written	in full every time the More Options or Player List Options dialog closes, every key in the fixed order of §3.9.3; hand-added keys are lost on the next save	[2.05] (client-internals/graalconfig-reference.md#when-it-is-read-and-written)
missing file	the built-in defaults stay, and `language` is set to the English name of the Windows user locale (for example `English` or `German`)	[2.05] (client-internals/graalconfig-reference.md#when-it-is-read-and-written)
boolean	`true` when the lowercased value is exactly `true`, anything else `false`; written as `true` or `false`	[2.05] (client-internals/graalconfig-reference.md#value-parsing)
missing boolean	reads as `false`, not as the default; a partial file therefore switches off `musicon`, `soundfxon`, `saveimages`, `nohtmlimages` and `joystickon`	[2.05] (client-internals/graalconfig-reference.md#value-parsing)
integer	converted as decimal; a conversion error, an empty value or a missing key keeps the default	[2.05] (client-internals/graalconfig-reference.md#value-parsing)
string	used only when non-empty	[2.05] (client-internals/graalconfig-reference.md#value-parsing)

3.9.3 Keys in file order (2.05)

MO is the More Options dialog and PL the Player List Options dialog; labels are the dialog captions as the client spells them. [2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)

Key	Type	Default	Dialog label	Effect	Source
`waitvertical`	bool	false	MO "Wait for vertical blank (slower but smoother)"	vsync on flip	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`musicon`	bool	true	MO "Music support"	turning it off stops the music player	[2.05][inf] (client-internals/graalconfig-reference.md#keys-in-file-order)
`usewinamp`	bool	false	MO "Use WinAmp for music"	music goes through Winamp (→ §3.10.3)	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`soundfxon`	bool	true	MO "Sound FX's enabled"	sound effects	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`reversestereo`	bool	false	MO "Reverse stereo channels"	swaps channels	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`saveimages`	bool	true	MO "Save downloaded images (to \levels\webgifs)"	turning it on writes the images already in memory to disk	[2.05][inf] (client-internals/graalconfig-reference.md#keys-in-file-order)
`dontsavepms`	bool	false	PL "Don't save PMs/Profiles"	as labelled	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`compabilitydraw`	bool	false	MO "Draw in compatibility mode (very slow)"	spelling as in the client	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`nodecompressionerror`	bool	false	MO "Don't show decompression errors"	as labelled	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`head1`, then `head`	string	Fung Si Yun	none	read in that order, so head wins; only head= is written; the 2.22 file still holds the default	[2.05][2.22] (client-internals/graalconfig-reference.md#keys-in-file-order)
`joystickon`	bool	true	MO "Joystick support"	as labelled	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`nodirectinput`	bool	false	MO "Don't use DirectInput"	as labelled	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`joyselected` (also reads `joyselected1`)	string	`(none)`	Controls frame	`(none)`, `Joystick1` or `Joystick2`	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`keyset_<action>_joystickbutton`	string	per action	Controls frame	one of `Joy_Up`, `Joy_Down`, `Joy_Left`, `Joy_Right`, `Button1` to `Button12`	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)

Key	Type	Default	Dialog label	Effect	Source
keyset_<action>_key0 to _key2	string	per action	Controls frame	three key slots per action, by key name (Up, Num 8, Space, Tab …); empty means unbound	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
language	string	English	Connect frame language box	interface language	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
font	string	Tempus Sans ITC	MO "Font:"	chat and name font; the 2.22 folder ships this font file	[2.05][2.22] (client-internals/graalconfig-reference.md#keys-in-file-order)
fontsize	int	24	MO "Size:"	font size	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
fixedport	bool	false	MO "Use fixed port for data packages:"	fixes the UDP port	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
fixedudpport	int	14899	MO port box	used when fixedport is true	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
noudp	bool	false	MO "Don't use UDP for data packages"	changing the UDP options restarts the UDP socket (→ §2.9)	[2.05][inf] (client-internals/graalconfig-reference.md#keys-in-file-order)
nomassmessages	bool	false	PL "Ignore mass messages"	as labelled	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
nohtmlinpms	bool	false	PL "Don't use HTML for PMs"	as labelled	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
limitnicknames	bool	false	MO "Limit the lenght of nicknames to"	as labelled	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
nicknamelimit	int	50	MO length box	the limit	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
dontconnectlevels	bool	false	MO "Don't connect levels"	turning it on drops the cached neighbour levels	[2.05][inf] (client-internals/graalconfig-reference.md#keys-in-file-order)
notoalls	bool	false	MO "Ignore toalls"	turning it on also clears the toall history	[2.05][inf] (client-internals/graalconfig-reference.md#keys-in-file-order)
nohtmlimages	bool	true	PL "No images in PMs"	as labelled	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
sortplayersbyname	bool	false	PL "Playerlist: sort by nickname"	as labelled	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)

Key	Type	Default	Dialog label	Effect	Source
`dontloadlistheads`	bool	false	MO "Don't load heads for the playerlist / map"	as labelled	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`dontsavepasswords`	bool	false	MO "Don't save the passwords"	see §3.9.5	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`dontusedirectx7`	bool	false	MO "Don't use DirectX7"	as labelled	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
`fullscreenmode`	string	`1024x768`	MO "Fullscreen mode:"	one of `640x480`, `800x600`, `1024x768`, `1280x1024`, `1600x1200`; any other value keeps the default	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)

3.9.4 Key bindings

Rule	Detail	Source
position	the joystick and key block is written right after head=	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
actions	11, in file order: `up`, `left`, `down`, `right`, `item`, `attack`, `action`, `map`, `switchtochat`, `itemlist`, `pause`	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
duplicate key	a key name already bound earlier in the file is treated as `-` (unbound), so one key cannot drive two slots	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
rebinding	binding a key that another slot holds moves it, and the other slot gets the old key	[2.05] (client-internals/graalconfig-reference.md#keys-in-file-order)
default joystick buttons	in action order: `Joy_Up`, `Joy_Left`, `Joy_Down`, `Joy_Right`, `Button1` to `Button5`, `Button9`, `Button10`; the 2.22 file matches	[2.05][2.22] (client-internals/graalconfig-reference.md#keys-in-file-order)
2.22 default keys	arrows or IJKL, D, S or space, A, M, Tab, Q and P	[doc] (client-internals/config-and-input.md#update-2026-10-21-205-vs-222-graalconfig)
key names	2.05 holds only the bare prefix `keyset_` as a string, so it probably builds the binding key names at run time	[inf] (client-internals/config-and-input.md#update-2026-10-21-205-vs-222-graalconfig)

3.9.5 Registry values

Value	Read	Written	Source
`nickname`	at startup	when the dialogs close	[2.05] (client-internals/graalconfig-reference.md#registry-values)
`accountname`	at startup; a comma list whose first entry is the current account	when the dialogs close	[2.05] (client-internals/graalconfig-reference.md#registry-values)
`password`	at startup, decrypted	when the dialogs close, unless `dontsavepasswords` is true	[2.05] (client-internals/graalconfig-reference.md#registry-values)
`<account>:pass`	at startup for the current account, and again when the account box changes	as `password`	[2.05] (client-internals/graalconfig-reference.md#registry-values)

Rule	Detail	Source
clearing	when More Options closes with "Don't save the passwords" ticked, the client deletes `password` and every value whose name ends in `:pass` from the key	[2.05] (client-internals/graalconfig-reference.md#registry-values)

Rule	Detail	Source
encryption	password values are encrypted with a fixed DES key, the literal that older notes called a password mask	[2.05] (client-internals/graalconfig-reference.md#registry-values); [2.05] (client-internals/config-and-input.md#graalconfigtxt)

3.9.6 Keys by build

Key or group	2.05	2.22	Source
the keys of §3.9.3	read and written	present, same order	[2.05][2.22] (client-internals/graalconfig-reference.md)
`automapping`, the `micro*` and `voice*` voice-chat keys, `midivolume`, `mp3volume`, `radiovolume`, `sfxvolume`, `startinfullscreen`, `quickplayerworlds`	n/a	present	[2.05][doc] (client-internals/config-and-input.md#update-2026-10-21-205-vs-222-graalconfig)
`noudp`	default false	`noudp=true` in the compared file	[2.05][doc] (client-internals/config-and-input.md#update-2026-10-21-205-vs-222-graalconfig)
total	about 29 keys in the string table	86 keys in the compared file	[2.05][doc] (client-internals/config-and-input.md#update-2026-10-21-205-vs-222-graalconfig)
`serveraddr`, `loadcolorsfromserver`, `gamemode`	not read or written	unmeasured	[2.05] (client-internals/graalconfig-reference.md#keys-in-older-notes-that-205-does-not-read)

The three keys in the last row come from an older public key list [2.05] (client-internals/graalconfig-reference.md#keys-in-older-notes-that-205-does-not-read). The static survey places them in earlier clients, not later ones: `gamemode` from 1.1 to 1.41, `serveraddr` from 1.21 to 1.39, `loadcolorsfromserver` from 1.21 to NewMain [bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys).

3.9.6.1 Plain key strings added and dropped, 1998 to 2.31

From 1.38 some keys are built one character at a time, so a build can read a key that is missing here; three-letter fragments such as `ekv` are such keys [bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys).

Build	Added	Dropped	Source
6th Graal	`joystickon`, `musicon`, `reversestereo`, `soundfxon`, `waitvertical`		[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
8th Graal	`saveimages`		[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
1.1	`editnickname`, `gamemode`		[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
1.21	`accountname`, `loadcolorsfromserver`, `password`, `serveraddr`, `serverport`		[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
1.28 beta 2	`compabilitydraw`, `externalchat`, `language`, `nodecompressionerror`		[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)

Build	Added	Dropped	Source
1.3 test	font, fontsize	accountname, password	[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
1.31	fixedport, fixedudpport, noudp	editnickname	[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
1.32	nomassmessages		[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
1.33	nodirectinput		[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
1.35	nohtmlinpms		[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
1.36	limitnicknames, nicknamelimit	serverport	[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
1.38		externalchat	[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
1.39	dontconnectlevels, dontloadlistheads, nohtmlimages, notoalls, sortplayersbyname		[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
1.41 rev 1	dontsavepasswords, fullscreenmode, usewinamp	serveraddr	[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
NewMain 1.42	dontsavepms, dontusedirectx7, head, joyselected	gamemode	[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
2.00		loadcolorsfromserver	[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
2.1 test 2	startinfullscreen		[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
2.17.1	automapping, the micro* and voice* keys, midivolume, mp3volume, quickplayerworlds, radiovolume, sfxvolume	musicon, soundfxon, usewinamp	[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)
2.31	cachefilenames		[bin] (version-profiles/matrix/builds-deltas.md#graalconfigtxt-keys)

3.9.7 Dropped files

Rule	Detail	Source
gate	a file dragged onto the client window is handled only when an internal flag is set, the same flag that enables the editor drop target	[live] (client-internals/config-and-input.md#drop-files)

Rule	Detail	Source
`.txt`	spawns a script NPC	[live] (client-internals/config-and-input.md#drop-files)
`.gif`, `.png`, `.mng`	spawn an image NPC	[live] (client-internals/config-and-input.md#drop-files)
anything else	loaded as a level	[live] (client-internals/config-and-input.md#drop-files)
joystick	the device is activated when an internal bit is set; the WinMM joystick identifier is stored apart from the main window object	[live] (client-internals/config-and-input.md#device-path)

3.10 Image and sound asset conventions

This chapter lists the files a client install carries, the image sheets the engine expects, and which sound and music files each backend plays. How the client positions and mixes a sound in play is in → §4.7.

> Status: draft. The asset tree is from one official install; the 2.22 extension-to-backend dispatch and what 2.05 does after a missing sound arrives are untraced.

3.10.1 Image sheets

File	Size	Role	Source
`pics1.png`	2048×512	standard tiles, 4096 ids of 16 px (→ §3.5.1)	[file] (client-internals/client-assets.md#tile-sheets)
`picso.png`	1024×512	offline tileset, sheet type 1	[file] (client-internals/client-assets.md#tile-sheets)
`sprites.png`	128×452	sprite sheet	[file] (client-internals/client-assets.md#tile-sheets)
`letters.png`	—	bitmap font	[file] (client-internals/client-assets.md#tile-sheets)

The tile definition line that selects a sheet is in → §3.5.2.2.

3.10.2 The levels tree

Files on disk under `levels/` in the measured install [file] (client-internals/client-assets.md#levels-counts-files-on-disk):

Directory	Files	Contents	Source
`ganis/`	93	86 `.gani` (→ §3.3) plus images	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`midis/`	49	music	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`heads/`	35	33 png, 2 mng	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`weapons/`	33	weapon images, 2 txt	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`bomys/`	29	bomy images and ganis	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`emoticons/`	29	png and mng	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`webgifs/`	26	downloaded images, mostly png (→ §3.9.3 `saveimages`)	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`backpals/`	23	background palettes	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`bodies/`	20	body images	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`images/`	19	other images	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`smilies/`	15	—	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`baddies/`	10	one sheet per baddy	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
`shields/`	7	—	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)

Directory	Files	Contents	Source
swords/	6	—	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
npcs/	5	script text	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
horses/	4	—	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
hats/	4	—	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
tutorial/	4	—	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
letters/	2	blueletters, graal2002letters	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
snapshot/	2	ss_arrow, ss_bomb	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)
tiledefs/	1	the offline tile definition line	[file] (client-internals/client-assets.md#levels-counts-files-on-disk)

Top-level item	Contents	Source
maps/	bigmap.txt and minimap material (→ §3.2.5)	[file] (client-internals/client-assets.md#other-top-level)
arrays.dat, arrays2.dat, .sig files	→ §3.5.3 to §3.5.5	[file] (client-internals/client-assets.md#other-top-level)
bass.dll, basswma.dll, hvdi.dll	sound, WMA add-on and voice libraries (→ §3.10.4)	[file] (client-internals/client-assets.md#other-top-level)
readme.txt	documents HawkVoice (LGPL), joystick calibration and the classic controls: F1 options, F4 editor, F5 load, arrows or IJKL walk, S or space strike, A act, M map	[doc] (client-internals/client-assets.md#other-top-level)

3.10.3 Sound and music in 2.05

What	Backend	Source
sound effects (.wav)	DelphiX wave list on DirectSound	[2.05] (client-internals/sound-and-music.md#205-at-a-glance)
music, Windows MCI path	an invisible media player component (MCI)	[2.05] (client-internals/sound-and-music.md#205-at-a-glance)
music, Winamp path	a component that remote-controls a running Winamp 1.x or 2.x	[2.05] (client-internals/sound-and-music.md#205-at-a-glance)
music volume	midiOutSetVolume (MCI path) or Winamp IPC messages	[2.05] (client-internals/sound-and-music.md#205-at-a-glance)

Rule	Detail	Source
options	musicon, usewinamp, soundfxon, reversestereo (→ §3.9.3)	[2.05] (client-internals/sound-and-music.md#options-graalconfigtxt)
backend switch	chosen at game start and again whenever More Options closes; switching stops the current track and frees the old backend; turning music off stops the track	[2.05] (client-internals/sound-and-music.md#options-graalconfigtxt)
loop prefix	a leading + on a file name means looped; it is stripped	[2.05] (client-internals/sound-and-music.md#one-entry-point-for-every-sound)

Rule	Detail	Source
missing file	online, the client asks the server for the file by the same path that fetches images; the sound plays on a later trigger once the file has arrived	[2.05][inf] (client-internals/sound-and-music.md#one-entry-point-for-every-sound)
dispatch	`.wav` → effect path, only when `soundfxon`; any other extension → music path, only when `musicon`; `.mid` is the common case	[2.05] (client-internals/sound-and-music.md#one-entry-point-for-every-sound)
named effects	fifteen effects are known by name: `arrow`, `axe`, `bomb`, `crush`, `extra`, `fire`, `horse2`, `item`, `lift`, `lift2`, `put`, `steps`, `sword`, `thunder`, `water` (`.wav`)	[2.05] (client-internals/sound-and-music.md#effect-path-wav)
effect cache	each file is loaded once and cached by name; `stopsound` finds the cached wave by file name, ignoring the path	[2.05] (client-internals/sound-and-music.md#effect-path-wav)
MCI types	`.mid` goes to the MCI sequencer; other types depend on the MCI drivers installed	[2.05][inf] (client-internals/sound-and-music.md#music-path-mci)
MIDI loop	a `.mid` track replays when it ends, so MIDI music loops until `stopmidi` or another track	[2.05] (client-internals/sound-and-music.md#music-path-mci)
Winamp types	in Winamp mode music is whatever Winamp's input plugins accept, MP3 included	[2.05] (client-internals/sound-and-music.md#music-path-winamp)
Winamp start	the component finds the player by window class `Winamp v1.x`; if none is running it reads the Winamp path from `Software\CLASSES\Winamp.File\shell\Play\command` and starts it	[2.05] (client-internals/sound-and-music.md#music-path-winamp)

The voice cap, positional pan and volume rules and the GS1 sound commands are in → §4.7.

3.10.4 Sound libraries in 2.17, 2.22 and 2.31

Area	2.05	2.17, 2.22, 2.31	Source
library	Delphi X, MCI, Winamp	`bass.dll`, imported with the same functions in all three: samples, tracker music, file streams, URL streams, stream length and position, channel attributes and recording; `winmm` is still imported, with MIDI device queries added	[2.17][2.22][2.31] (client-internals/sound-and-music.md#217--222--231-bass-replaces-directsound-and-winamp)
effects	DirectSound wave list	BASS samples	[inf] (client-internals/sound-and-music.md#217--222--231-bass-replaces-directsound-and-winamp)
music files	MCI (MIDI) or Winamp	BASS streams and tracker modules; MIDI still through MCI	[inf] (client-internals/sound-and-music.md#217--222--231-bass-replaces-directsound-and-winamp)
internet radio	n/a	BASS URL streams, `radio` and `radiovolume`, judged from names	[inf] (client-internals/sound-and-music.md#217--222--231-bass-replaces-directsound-and-winamp)
Winamp	`usewinamp`	removed: no component, key or window class	[2.22] (client-internals/sound-and-music.md#217--222--231-bass-replaces-directsound-and-winamp)
volume options	none	`midivolume`, `mp3volume`, `radiovolume`, `sfxvolume`, `voicevolume`	[2.22] (client-internals/sound-and-music.md#217--222--231-bass-replaces-directsound-and-winamp)
voice chat	n/a	microphone capture through BASS recording, encoded with the `hvdi.dll` HawkVoice codec; an `emoticonmicro.png` icon shows over a talking player	[2.22] (client-internals/sound-and-music.md#voice-chat-222)

Area	2.05	2.17, 2.22, 2.31	Source
voice options	n/a	`microon`, `microinputdevice`, `microactivatebyvolume`, `microactivationlevel`, `microactivatekey`, `voicecodec`, `voicevolume`	[2.22] (client-internals/sound-and-music.md#voice-chat-222)
WMA	n/a	2.17.1 loads `basswma.dll` at run time by name and uses its stream functions; 2.22, 2.31 and GraalEditor 2.3.1.1 neither import nor name it, so the shipped DLL is a leftover and WMA music is not supported there	[2.17][2.22] (client-internals/sound-and-music.md#basswmadll)
extension dispatch	→ §3.10.3	unmeasured	[2.22] (client-internals/sound-and-music.md#open)

4 Client behaviour by version

> Part index. Each chapter carries its own status line.

- → §4.1 Frame loop and timing (`01-loop-timing.md`)
- → §4.2 Movement and collision (`02-movement-collision.md`)
- → §4.3 Combat, damage, hurt and knockback (`03-combat-damage.md`)
- → §4.4 Death clocks and respawn (`04-death-respawn.md`)
- → §4.5 Items, drops and board respawn (`05-items-board.md`)
- → §4.6 HUD, sign panel, chat text and options (`06-hud-ui.md`)
- → §4.7 Sound (`07-sound.md`)
- → §4.8 Offline and training mode (`08-offline.md`)
- → §4.9 The built-in editor (`09-editor.md`)
- → §4.10 Other client features (`10-feature-specs.md`)

4.1 Frame loop and timing

This chapter fixes the tick that every count in Part IV uses, how each build drives it, and the periodic work that runs off it. Durations of individual actions are in the chapter for each action (→ §4.1.5).

> Status: draft. The tick is measured for the Delphi builds (1st Graal to 2.31), Graal3D and 5.x/6.x on the wire; the 3.x and 4.x Torque-era loops are not in the notes (→ App. E#301).

4.1.1 The tick

Rule	Detail	Source
length	one game tick is 50 ms (20 ticks per second)	[bin] (version-profiles/matrix/README.md#acts-timing-movement-and-rules); [2.05] (client-internals/decomp-findings-2026-10-24/constants-205.tsv)
same in every Delphi build	the static survey of all 57 Delphi executables finds a 50 ms frame interval wherever a frame timer is present	[bin] (version-profiles/matrix/builds-static-survey.tsv)
tick counter	2.05 and 2.22 keep a per-player tick counter that rises by one every 50 ms whatever the player does	[live][2.22] (live-runs/2gta-222-2026-10-30/README.md)
5.x and 6.x	a moving 5.24r4, 5.30r3 or 6.03r7 client sends its position every 50 ms; their walk, push, lift and sword timings equal 2.05's in ticks	[wire] (protocol/v5v6-lab-play.md)
Graal3D	driven by a 50 ms DelphiX timer	[bin] (version-profiles/matrix/README.md#acts-timing-movement-and-rules)

4.1.2 What drives the tick, by build

Builds	Driver	Source
1st Graal to 4th Graal	a DelphiX timer on the main form, interval 50 ms	[bin] (version-profiles/matrix/builds-static-survey.tsv)
4th Graal alpha, 5th Graal, 5th Graal alpha	the survey finds no form timer (idle loop); the era summary says eras A to D all use the form timer: `disputed → App. E#300`	[bin] (version-profiles/matrix/builds-static-survey.tsv); [bin] (version-profiles/matrix/README.md#acts-timing-movement-and-rules)
6th Graal (beta5) to 1.41 rev1, and Graal3D	a DelphiX timer on the main form, interval 50 ms	[bin] (version-profiles/matrix/builds-static-survey.tsv)
NewMain 1.42 to 2.31 (every 2.0x, NewWorld betas 1 and 2, the 2.1 tests, 2.17.1, 2.22, 2.31)	no form timer; the frame runs from the application's idle handler behind a 50 ms gate	[bin] (version-profiles/matrix/builds-static-survey.tsv); [bin] (version-profiles/matrix/README.md#acts-timing-movement-and-rules)
2.31	in addition, the main form carries a 50 ms timer that runs the 64-tick integrity cycle (→ §4.1.3)	[2.31] (client-internals/decomp-findings-2026-10-24-b/03-periodic-ids-2.31.md)
3.x, 4.x	unmeasured	[doc] (version-profiles/matrix/README.md#acts-timing-movement-and-rules)

4.1.2.1 The frame function (idle-driven builds)

Rule	Detail	Source
order	after the 50 ms gate opens, one frame handles clock skew, network traffic, then drawing	[live] (client-internals/rendering-observations.md#frame-loop)

Rule	Detail	Source
no catch-up	a stall costs one frame; the client does not run extra ticks to catch up, and ignores any step count its pacer supplies	[live] (client-internals/rendering-observations.md#frame-loop)
input sampling	key state is read once per tick: a key released and pressed again inside one tick is not seen as a new press, and a tap shorter than a tick can be lost	[live][2.05] (client-internals/player-mannerisms-live.md#spin-attack-live-online)

4.1.3 Periodic work

Period	Work	Builds	Source
every tick	the player state machine steps (walk, push, hurt, death and other counters in §4.2 to §4.4)	all measured builds	[live][2.05] (client-internals/player-mannerisms-live.md)
every tick	the tile under the player is checked for water and lava, except while the sword swings, while riding, and when the player is not under local control	2.05	[2.05] (client-internals/water-and-ganis.md#water)
every tick	the probe point for links is taken and each link of the level is tested (→ §4.2.11)	2.05	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#following-a-link)
64 ticks (3.2 s)	a counter modulo 64 runs the integrity cycle: tick 1 a debugger probe and a 5000 ms timer armed, tick 10 the first tamper-check word, tick 20 a scan of top-level window titles and classes against a blacklist, tick 30 the second word, tick 31 `PLI_TAMPERCHECK` sent (→ §2.4, §4.10)	2.31 only; 2.17.1 and 2.22 have no such timer	[2.31] (client-internals/decomp-findings-2026-10-24-b/03-periodic-ids-2.31.md#pli-39-re-send-every-32-s)
about 5.1 s	a one-byte id 24 message (→ §2.4); tying it to the 5000 ms timer above is an inference	2.31; 2.22 only while another player is in the level	[2.31][inf] (client-internals/decomp-findings-2026-10-24-b/03-periodic-ids-2.31.md#pli-24-about-every-51-s-1-byte-lab-value-9)
5.0 s	the server's `PLO_NEWWORLDTIME`, seen alongside the client's id 24	2.22 against a public server	[cap] (live-runs/2gta-222-2026-10-30/README.md)

4.1.4 Movement updates on the wire

How often a moving client reports itself; the message layouts are in → §2.4.

Builds	What is sent while a key is held	Source
1.1.9 (relay)	a `D` line of type `P` on the level connection every 100 ms: x and y in half-tiles, then frame·4 + direction (→ §2.3)	[cap] (client-internals/live-trace-early.md#119-gserver-with-two-119-clients-relay)
1.21	one id 2 (player props) message every 100 ms with X, Y and SPRITE	[cap] (client-internals/live-trace-early.md#gserver-121-with-121-gserv007)
1.28 beta 2	UDP id 0 to each peer every 100 ms: X, Y, SPRITE, STATUS and CARRYSPRITE (→ §2.9)	[cap] (client-internals/live-trace-early.md#gserver-128-with-128-beta-2-gserv013)
1.3 test against GServer13	one id 2 message every 100 ms with X and SPRITE; speed 0.5 tile per tick	[cap] (client-internals/live-trace-early.md#gserver13-with-13-test-gserv014); [live] (client-internals/live-gserver13.md)
2.22	the X prop only on whole tiles (every other tick), plus the final half-tile when the player stops; the gani prop switches to the walk gani with the first step and back to idle 100 ms after the last	[cap] (live-runs/2gta-222-2026-10-30/README.md)

Builds	What is sent while a key is held	Source
2.22	a script that changes a client flag every tick makes the client send that flag every tick, one message each, uncoalesced	[cap] (live-runs/2gta-222-2026-10-30/README.md)
NewWorld beta	the idle gani is re-sent about every 100 ms, because the control NPC sets it each tick; 2.05 sends it only now and then	[cap] (client-internals/newworld-betas.md)
5.24r4, 5.30r3, 6.03r7	the position (x2/y2) every 50 ms	[wire] (protocol/v5v6-lab-play.md)
2.05	unmeasured as a rate; a hit sends one props message with CURPOWER, GANI and SPRITE in the hit's tick, then the knockback positions	[cap] (client-internals/live-trace-205.md)

4.1.5 Durations in ticks

The canonical place of each tick count:

- → §4.2.5 blocked steps before the push pose (18) and the push cycle (20)
- → §4.2.6 the ledge jump (8 ticks)
- → §4.2.8 the pull cycle (18 ticks)
- → §4.2.10 swimming and the lava burn (101 ticks)
- → §4.3 sword, spin, hurt and knockback
- → §4.4 the death timer

4.2 Movement and collision

This chapter gives the walking step, the wall test, what blocks a point, the blocked-step counter and the actions that hang off it (push pose, ledge jump, pushing blocks and NPCs), grab and pull, water and lava, link following, and the gates that stop movement. Tile type values and the `arrays.dat` lists are defined in → §3.5; this chapter only says how movement uses them. Units are tiles and 50 ms ticks (→ §4.1.1). Facings are 0 up, 1 left, 2 down, 3 right.

> Status: draft. The rule is read in full for 2.05, 2.17.1, 2.22 and 2.31; 1st Graal and 1.39 are read for the shape of the rule but not its offset tables; 3.x and 4.x are not in the notes. Open points are App. E#302 to E#309.

4.2.1 The step

Rule	Detail	Source
one key, one step	each arrow key held is one step per tick; with several held, the steps run in the order up, left, down, right, each from the position the previous one left; a diagonal is two steps	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#one-key-one-step)
facing	the step first sets the player's facing to its direction	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#one-key-one-step)
step vector	(0, -0.5), (-0.5, 0), (0, 0.5), (0.5, 0) by facing, times a speed factor	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#one-key-one-step)
positions	doubles, in tiles	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#one-key-one-step)
first tick	the first tick after the key goes down already moves the player	[live][2.05] (client-internals/player-mannerisms-live.md#walking-idle-and-running-in-place-live)
release	letting go switches to `idle` frame 0 in the next tick; there is no slow-down and no finishing step	[live][2.05] (client-internals/player-mannerisms-live.md#walking-idle-and-running-in-place-live)
walk animation	`walk` advances one frame per tick, 8 frames, 8-tick cycle	[live][2.05] (client-internals/player-mannerisms-live.md#summary-table-205-rev-2)

4.2.1.1 Speed factors

Factor	When	Builds	Source
1.0	on foot, carrying or not (carrying does not slow the walk)	2.05	[2.05] (client-internals/decomp-findings-2026-10-25-l/01-walking-body-box.md#step); [live][2.05] (client-internals/player-mannerisms-live.md#lifting-carrying-hiding-and-throwing-live)
1.2	a high speed state	2.05	[2.05] (client-internals/decomp-findings-2026-10-25-l/01-walking-body-box.md#step)
0.2	hidden under a carried object: 0.1 tile per tick	1.39, 1.41, 2.05, 2.22	[live][1.39][1.41][2.05] (client-internals/player-mannerisms-live.md#cross-version-table); [1.39] (version-profiles/matrix/rules-139.md#wall--walk-test-bin-lab)
about 0.2	hidden: 1.5 px per tick, steps of 2 and 1 px alternating	5.x, 6.x	[wire] (protocol/v5v6-lab-play.md)
0.4, 0.8	exist in the walking code; not tied to states (→ App. E#302)	2.05 to 2.31	[bin] (client-internals/decomp-findings-2026-10-25-m/open-questions.tsv)

Factor	When	Builds	Source
swimming	the same 0.5 tile per tick as walking	1.39, 1.41, 2.05, 2.22, 5.x, 6.x	[live] (client-internals/player-mannerisms-live.md#cross-version-table); [wire] (protocol/v5v6-lab-play.md)

4.2.1.2 Walking speed by build

Build	Step per tick	Speed	Source
1st Graal	0.5	10 tiles/s	[live] (version-profiles/matrix/live-runs-2026-10-30.md#walking-speed)
1.0 (10th)	0.5	10 tiles/s	[live] (version-profiles/matrix/live-runs-2026-10-30.md#walking-speed)
1.1 (15th)	0.5	10 tiles/s	[live] (version-profiles/matrix/live-runs-2026-10-30.md#walking-speed)
1.32	0.5	10 tiles/s	[live] (version-profiles/matrix/live-runs-2026-10-30.md#walking-speed)
1.39	0.5	10 tiles/s	[live][1.39] (client-internals/player-mannerisms-live.md#cross-version-table)
1.41 rev1	0.5; a second copy of y moves in 0.25 steps (the view or the drawn position)	10 tiles/s	[live] (version-profiles/matrix/live-runs-2026-10-30.md#walking-speed)
2.00	unmeasured	unmeasured	[live] (version-profiles/matrix/live-runs-2026-10-30.md#walking-speed)
2.05 rev2	0.5	10 tiles/s	[live] (version-profiles/matrix/live-runs-2026-10-30.md#walking-speed)
2.17.1, 2.31	0.5 (read from the walking routine)	10 tiles/s	[2.17.1][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md)
2.22	0.5 (live and from the walking routine)	10 tiles/s	[live][2.22] (live-runs/2gta-222-2026-10-30/README.md)
1.3 test (GServer13)	0.5	10 tiles/s	[live] (client-internals/live-gserver13.md)
5.24r4, 5.30r3, 6.03r7	8 px (0.5) per tick	10 tiles/s	[wire] (protocol/v5v6-lab-play.md)

4.2.2 The wall test

4.2.2.1 Probe points (2.05 to 2.31)

Six points per facing, offsets from the sprite's top-left, read at the position the step would reach. They are numbered 0 to 5 in the order given:

Facing	Fixed coordinate	The six points along the edge	Source
up	y + 1.0	x + 0.525, 1.475, 1.525, 2.475, 2.525, 0.475	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#the-probe-points)
left	x + 0.5	y + 1.025, 1.975, 2.025, 2.975, 3.025, 0.975	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#the-probe-points)
down	y + 3.0	x + 0.525, 1.475, 1.525, 2.475, 2.525, 0.475	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#the-probe-points)
right	x + 2.5	y + 1.025, 1.975, 2.025, 2.975, 3.025, 0.975	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#the-probe-points)

Rule	Detail	Source
body box	the points lie on the leading edge of a two-tile square, x + 0.5 to x + 2.5, y + 1 to y + 3	[2.05] (client-internals/decomp-findings-2026-10-25-l/01-walking-body-box.md#six-probe-points-per-facing)

Rule	Detail	Source
point to tile	`floor(v + 0.0001)`: a point on a tile line reads the tile that starts there; a value a hair under a whole number reads the tile above	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#the-probe-points)
off the board	a tile outside 0 to 63 is the neighbour level's on a gmap, or a fixed tile outside a gmap	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#the-probe-points)
off the board, default tileset	the wall test checks bounds first and calls off-board a wall; the type tests skip the bounds check and read id 632, which is in no list of the 2.22 `arrays.dat`	[2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#the-default-tileset)

4.2.2.2 The same test in earlier builds

Build	Rule	Source
1st Graal	six probes per facing, first four decide; a blocked step is undone on the stepped axis and the free axis snapped (facings 0 and 2 snap y down, 1 and 3 snap x onto a half-tile); a pattern of the six probes adds a nudge from a second table; the probe, nudge and step tables are unread (→ App. E#307)	[bin] (version-profiles/matrix/rules-1st-graal.md#wall-and-walk-test-bin)
1.1 (15th)	after bumping furniture x carried a 0.025 offset (26.925) while walking, which fits a nudge rule of the 2.05 kind; not read	[live][inf] (version-profiles/matrix/live-runs-2026-10-30.md#walking-speed)
1.39	six probe offsets per facing step; the numeric tables are unread (→ App. E#307)	[1.39] (version-profiles/matrix/rules-139.md#wall--walk-test-bin-lab)
1.39	walking into a character NPC's side nudged the player along it at 0.125 tile per tick, and `playertouchsme` did not fire on that contact	[live][1.39] (client-internals/player-mannerisms-live.md#notes-per-build)
2.05, 2.17.1, 2.22, 2.31	as §4.2.2.1 to §4.2.4; the four agree on every table and constant	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md)

4.2.3 What blocks a point

Tested in this order; the first hit blocks:

#	Test	Source
1	a list of objects with two flags and a player-sized body box, checked only when a client option is on (unidentified → App. E#303)	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#the-probe-points)
2	the level's NPCs that have a blocking shape or image	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#the-probe-points)
3	a list of 1.5 × 2 boxes with an enabled flag, x + 0.25 to x + 1.75, y to y + 2 (unidentified → App. E#303)	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#the-probe-points)
4	chests: each covers x to x + 2, y to y + 2, edges included, opened or not	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#the-probe-points)
5	the tile, which blocks unless it is in the passable list or the liquid list of `arrays.dat` (arrays 0 and 1, → §3.5.3)	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#the-probe-points)

4.2.3.1 How the tile is answered, by tileset

Tileset type	Wall	Source
0 (default, `pics1.png`)	not in array 0 and not in array 1; looked up by binary search of each sorted list, with no 4096-entry table built	[2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#the-default-tileset)
1, 2 (new kind)	a fixed rule on the tile id: id 32; rows 26 and 27 with column 2 or more; rows 30 to 47, 84 to 95, 116 to 127; row 128 or more with `row / 16` odd (row = id / 16, column = id % 16)	[2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#tilesets-of-the-new-kind)
5	no board rule; only NPC shapes have types	[2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#which-rule-applies)
any	a `setshape2` NPC shape the player stands on answers first	[2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#which-rule-applies)

Consequences on the default tileset:

Rule	Detail	Source
lava is liquid	the liquid list holds lava too, so lava is walkable and the swim test is true on it	[2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#the-default-tileset)
deep water	a tile in none of arrays 0 to 3 is a wall: the dark deep-water tile 3644 blocks, and `push` comes on the 18th refused step	[2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#the-default-tileset); [live] [2.05] (client-internals/live-trace-205.md#water-and-lava-live-new4nw)
shallow water	a tile only in the passable list is walked on, not swum in	[2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#the-default-tileset)
several lists	a tile can be in several lists; in the 2.22 file 6 of the 133 throw-through tiles are passable, and the 9 lava tiles that are not liquid are all passable, so they are walked on and still burn	[file][2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#the-default-tileset)
thrown objects	a thrown object stops on a tile that is not type 20 and that the wall test calls a wall	[2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#the-default-tileset)
scripts	`onwall(x, y)` is this wall test; `onwater(x, y)` is the type 11 test, true on lava; `tiletype(x, y)` returns -1 on the default tileset, 1 on type 5, and the fixed rule on types 1 and 2 (an NPC shape's type wins when above 1); 2.00 to 2.04 carry the same three names, their code not compared	[2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#script-functions)

4.2.4 Refused steps, nudges and snaps

Step	Rule	Source
decision	points 0 to 3 decide: all four free keeps the step, any blocked refuses it	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#decision)
1. corner nudge	if point 0 is blocked and points 2, 3 and 4 are free, add +0.1 across the edge (+x for up and down, +y for left and right); else if points 0 and 1 are free, point 3 is blocked and point 5 is free, add -0.1; the nudge is added to the stepped position	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#on-a-refused-step)

Step	Rule	Source
2. take the step back	subtract the step vector times the speed factor; the nudge stays	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#on-a-refused-step)
3. flush snap	along the facing only, when the coordinate has a fraction above 0: up `y = floor(y)`; left `x = floor(x + 0.5) - 0.5`; down `y = floor(y + 1)`; right `x = floor(x + 1.5) - 0.5`; there is no wall test on the snapped position	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#on-a-refused-step)
parent	on a platform object or with a parent, the snaps are relative to the parent's position	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#on-a-refused-step)
after every step	accepted or refused, y snaps to `floor(y + 0.0625)` if its fraction is under 0.0625 or over 0.9375, and x the same on `x + 0.5` (`x = floor(x + 0.5 + 0.0625) - 0.5`); then the position is stored, so a refused step still moves the player by the nudge and the snaps	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#after-every-step)
state	a refused step puts the player in the pushing mode (mode 1)	[2.05] (client-internals/decomp-findings-2026-10-25-l/01-walking-body-box.md#rule); [2.05] (client-internals/ledges-push-pull.md#player-modes-involved)

Consequences of the rule:

Rule	Detail	Source
left and up	the probe is inside the body box, so the body stops with its edge on the wall's tile line without a snap	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#what-this-means-for-stopping-positions)
right and down	the probe is the tile line one tile beyond the body; the step that would put the body's edge on a wall's line is refused, and the flush snap moves the body that half-tile	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#what-this-means-for-stopping-positions)
gaps	a two-wide body fits a two-tile gap and not a one-tile gap	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#what-this-means-for-stopping-positions)
corners	pressing against the end of a wall slides the body round it at 0.1 tile per tick, ten ticks per tile	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#what-this-means-for-stopping-positions)
sliding along a wall	holding Down into a wall and adding Right walks right at full speed with `walk`; the push pose ends in that tick	[live][2.05] (client-internals/player-mannerisms-live.md#walking-idle-and-running-in-place-live)
live confirmation	the stopping positions are read from the binary and not walked on a live 2.22 (→ App. E#304)	[bin] (client-internals/decomp-findings-2026-10-25-m/open-questions.tsv)

4.2.5 The blocked-step counter and the push pose

Rule	Detail	Source
counting	a step counts when it is refused and no corner nudge was applied; a free step or a nudge sets the count to 0, and so does the first refused step after a free one	[2.05] (client-internals/ledges-push-pull.md#the-ledge-test)
shared counter	the blocked-step count and the grab's pull cycle share one per-player counter	[2.05] (client-internals/ledges-push-pull.md#player-modes-involved)
running in place	a refused step still advances the walk frame; the gani stays `walk` until the push pose or a jump takes over	[live][2.05] (client-internals/live-trace-205.md#ledge-jump-and-running-in-place-live-new3nw)

Rule	Detail	Source
push pose	push replaces walk on the 18th blocked tick (0.9 s); at 38 the count goes back to 18 and the gani restarts at frame 0	[live][2.05] (client-internals/player-mannerisms-live.md#walking-idle-and-running-in-place-live)
push frames	5 frames × 4 ticks = the 20-tick wrap, so the pose loops seamlessly	[live][2.05] (client-internals/player-mannerisms-live.md#walking-idle-and-running-in-place-live)
sliding resets	after a slide the count goes on from 1, so the full 18 ticks pass again before push; pressing into the wall again after letting go starts at 1	[live][2.05] (client-internals/player-mannerisms-live.md#walking-idle-and-running-in-place-live)
on the wire	push is sprite steps 14 + frame (SPRITE 11, 12, 13, 11, 12)	[2.05] (client-internals/ledges-push-pull.md#sprites-on-the-wire)

By build:

Build	Push at	Wrap	Push frame length	Source
1st Graal	count 18 (the ledge test); push frames while the count is above 17 and not carrying	above 37 back to 18	unmeasured	[bin] (version-profiles/matrix/rules-1st-graal.md#wall-and-walk-test-bin)
1.39	18	38 to 18	about 2 ticks (video)	[live][1.39] (client-internals/player-mannerisms-live.md#cross-version-table); [1.39] (version-profiles/matrix/rules-139.md#wall--walk-test-bin-lab)
1.41	18	38 to 18	4 ticks (push.gani)	[live][1.41] (client-internals/player-mannerisms-live.md#cross-version-table)
2.05 rev 2	18	38 to 18	4 ticks	[live][2.05] (client-internals/player-mannerisms-live.md#cross-version-table)
2.22	the tick after 18 blocked ticks	37 back to 18	5 × 4 ticks, looping; a lab read of the frame field showed it cycling 0 to 3 each tick (→ App. E#305)	[live][2.22] (client-internals/player-mannerisms-live.md#cross-version-table); [live] (live-runs/2gta-222-2026-10-30/README.md)
5.24r4, 5.30r3, 6.03r7	0.85 to 0.90 s after the walk started	unmeasured	unmeasured	[wire] (protocol/v5v6-lab-play.md)

4.2.6 Ledge jumps

Rule	Detail	Source
when	the ledge test runs at blocked count 18, and again every 20 steps (the wrap)	[2.05] (client-internals/ledges-push-pull.md#the-ledge-test)
test point	from the player's position by facing up, left, down, right: (1.5, 0.5), (0, 2), (1.5, 3.5), (3, 2)	[2.05] (client-internals/ledges-push-pull.md#the-ledge-test)
ledge tile	the tile there must be a ledge tile for that facing: array 8 of arrays.dat on the default tileset (→ §3.5.3), or rows 30 and 31, columns 8 to 15, on a tileset of type 1 or 2	[2.05] (client-internals/ledges-push-pull.md#arraysdat)
landing	from (x + 1.5, y + 2): (0, -5), (-6.5, 3), (0, 7), (6.5, 3); the landing point must pass the walking test (not a wall, not a chest), or the player just keeps pushing	[2.05] (client-internals/ledges-push-pull.md#the-ledge-test)
sound	jump.wav at (x + 1.5, y + 2) when the jump starts	[2.05] (client-internals/ledges-push-pull.md#the-ledge-test)

Rule	Detail	Source
flight	8 ticks from the position after the refused step, the first in the same tick as the start; keys are not read until landing; the blocked-step counter counts down 7 to 0 as the jump timer	[2.05] (client-internals/ledges-push-pull.md#the-ledge-test); [live][2.05] (client-internals/live-trace-205.md#ledge-jump-and-running-in-place-live-new3nw)
gani	none of its own: the walk (or push) gani stays; no sprite of its own on the wire	[live][2.05] (client-internals/live-trace-205.md#ledge-jump-and-running-in-place-live-new3nw); [2.05] (client-internals/ledges-push-pull.md#sprites-on-the-wire)
mode	mode 2 for the 8 ticks	[2.05] (client-internals/ledges-push-pull.md#player-modes-involved)

The flight, per tick `i` = 0 to 7:

Facing	Position	Source
up	`y0 - [1.5, 2.7, 3.8, 4.5, 5.0, 5.2, 5.2, 5.0][i]`; live from y0 = 14: 12.5, 11.3, 10.2, 9.5, 9.0, 8.8, 8.8, 9.0	[2.05] (client-internals/ledges-push-pull.md#the-ledge-test); [live][2.05] (client-internals/live-trace-205.md#ledge-jump-and-running-in-place-live-new3nw)
down	`y0 + [-0.2, 0.2, 0.5, 1.5, 2.8, 4.0, 5.4, 7.0][i]`	[2.05] (client-internals/ledges-push-pull.md#the-ledge-test)
left, right	`x0 -/+ [1, 2, 3, 3.8, 4.5, 5.2, 5.8, 6.5][i]`, `y0 + [-0.2, -0.5, 0, 0.5, 1.1, 1.7, 2.3, 3.0][i]`	[2.05] (client-internals/ledges-push-pull.md#the-ledge-test)

Build	Difference	Source
1st Graal	at count 18 a passing ledge test sets the hop mode and the counter to 8; the hop follows a facing table for 8 ticks	[bin] (version-profiles/matrix/rules-1st-graal.md#wall-and-walk-test-bin)
2.22, 2.31	the same counts, probe points, flight curve and tables as 2.05; `jump.wav` plays only while the current gani is a built-in name (→ §4.2.12)	[2.22][2.31] (client-internals/ledges-push-pull.md#222-and-231)

4.2.7 Pushing blocks and NPCs

Rule	Detail	Source
when	at count 28 (checked before the step is counted, so on the 29th blocked step), and every 20 after	[2.05] (client-internals/ledges-push-pull.md#pushing)
trigger	the try first fires the trigger action pushed at the ledge test point, with the facing as its parameter (→ §4.2.9)	[2.05] (client-internals/ledges-push-pull.md#pushing)
block	the tile under that point is compared with the four tiles of the pushable block (array 16, entry 0, → §3.5.3); only that one tile is compared, and its place in the block gives the block's top-left	[2.05] (client-internals/ledges-push-pull.md#pushing)
target cells	from the block's top-left: up (0, -1) (1, -1); left (-1, 0) (-1, 1); down (0, 2) (1, 2); right (2, 0) (2, 1); none may be a wall, a chest, water, a chair or the lower half of a bed	[2.05] (client-internals/ledges-push-pull.md#pushing)
move	each old cell first takes the tile two cells ahead, then the block is written one cell ahead, so the row the block leaves gets the floor that was in front of it	[2.05] (client-internals/ledges-push-pull.md#pushing)
after	online, the 4 × 4 area round the old spot goes to the server as a board modify; the player moves one tile in the facing; `stonemove.wav` plays	[2.05] (client-internals/ledges-push-pull.md#pushing)

4.2.7.1 Pushed NPCs

Rule	Detail	Source
lookup	with no block found, the level's NPC hit test at the point: a character NPC in x + 0.5 to x + 2.5, y + 1 to y + 3; a `setshape2` NPC by its shape's tile types; a gani NPC by its animation; any other by its image box `[x, x + w/16) × [y, y + h/16)`, where a transparent pixel is a miss for plain image NPCs	[2.05] (client-internals/ledges-push-pull.md#pushed-and-pulled-npcs)
permission	the NPC needs `canbepushed` (GS1 command 44, cleared by 45, → §8.2)	[2.05] (client-internals/ledges-push-pull.md#pushed-and-pulled-npcs)
edge test	with the NPC left out of its own test, every point along the edge it moves into is checked, one per tile, `t` while `t < x + w` (or `y + h`): up (t, y - 1), left (x - 1, t), down (t, y + h), right (x + w, t); sizes in pixels / 16, character and gani NPCs count as 48 × 48	[2.05] (client-internals/ledges-push-pull.md#pushed-and-pulled-npcs)
blocked-point test for NPCs	other players (online only), blocking NPCs, two level object lists, and wall tiles; water does not stop an NPC	[2.05] (client-internals/ledges-push-pull.md#pushed-and-pulled-npcs)
move	the NPC moves one tile, the player one tile the same way, `waspushed` fires on the NPC (`actionpushed` has already fired), and the block sound plays	[2.05] (client-internals/ledges-push-pull.md#pushed-and-pulled-npcs)
water	a character NPC that crosses a water edge gets the splash and switches between `swim` and `idle`	[2.05] (client-internals/ledges-push-pull.md#pushed-and-pulled-npcs)
server	nothing is sent for the NPC move; the `pushed` trigger action is the only message	[2.05] (client-internals/ledges-push-pull.md#pushed-and-pulled-npcs)
2.22	the NPC lookup also searches the neighbouring map parts on a gmap	[2.22] (client-internals/ledges-push-pull.md#pushed-and-pulled-npcs)

4.2.8 Grab and pull

Rule	Detail	Source
A key order	throw what is carried; else lift (→ §4.5); else grab	[2.05] (client-internals/ledges-push-pull.md#grabbing-and-pulling)
grab test	by facing (1.5, 0.9), (0, 2), (1.5, 3.1), (3, 2): a wall tile, or an NPC that blocks; nothing is grabbed when another player stands there	[2.05] (client-internals/ledges-push-pull.md#grabbing-and-pulling)

Rule	Detail	Source
grab on NPCs	2.05 grabs a blocking NPC only when a per-NPC flag (most likely the character-shape flag) is clear; 2.22 and 2.31 grab any blocking NPC, and search the neighbouring map parts	[2.05][2.22] [2.31][inf] (client-internals/ledges-push-pull.md#222-and-231)
holding	the grab lasts while A is held; arrows do not move or turn the player; mode 5	[2.05] (client-internals/ledges-push-pull.md#grabbing-and-pulling)
pull cycle	the arrow away from the wall sets the counter to 2 and starts `pull`; the counter runs 2 to 19 and back to 2 (18 ticks); a pull is tried whenever the arrow is held at 19; letting go of any arrow sets the counter to 1	[2.05] (client-internals/ledges-push-pull.md#grabbing-and-pulling)
pull try	fires `pulled` one tile out from the body, (1.5, 0), (-0.5, 2), (1.5, 4), (3.5, 2) by facing, with the opposite facing as parameter, then reads the tile at the grab point	[2.05] (client-internals/ledges-push-pull.md#grabbing-and-pulling)
pullable block	array 16, entry 1 (→ §3.5.3); the two cells behind the player must be free, from the block's top-left: up (0, 4) (1, 4); left (4, 0) (4, 1); down (0, -3) (1, -3); right (-3, 0) (-3, 1); the block moves one tile towards the player (each old cell first takes the tile two cells behind), the player steps back one tile, and `lift.wav` plays, or `lift2.wav` by a player flag (→ App. E#306)	[2.05] (client-internals/ledges-push-pull.md#grabbing-and-pulling)
pulled NPCs	with no pullable block, an NPC with `canbepulled` (command 46, cleared by 47) is pulled: the two cells behind the player must be free from the whole-tile grab cell (up (0, 3) (1, 3); left (3, 0) (3, 1); down (0, -3) (1, -3); right (-3, 0) (-3, 1)), then the edge facing the player is checked (up (t, y + h), left (x + w, t), down (t, y - 1), right (x - 1, t)); NPC and player move one tile back and `waspulled` fires; nothing goes to the server	[2.05] (client-internals/ledges-push-pull.md#grabbing-and-pulling)
futile pull	on anything else (a wall, a rock too heavy for the glove, a push block) nothing moves; `pulled` still fires every 18 ticks and `pull` keeps playing	[2.05] (client-internals/ledges-push-pull.md#grabbing-and-pulling)
on the wire	`grab` is sprite step 19 (SPRITE 14); `pull` steps 20 + frame (SPRITE 15, 16, 17)	[2.05] (client-internals/ledges-push-pull.md#sprites-on-the-wire)

Pose timing by build:

Build	Pull frames	Arrow released during a pull	A released after a pull	Source
1.41	unmeasured	back to `grab`	`grab` stays until the next action	[live][1.41] (client-internals/player-mannerisms-live.md#cross-version-table)
2.05 rev 2	3 frames × 2 ticks	counter to 1; the pose stays `pull`, animating	`pull` loops until the next key moves the player	[live][2.05] (client-internals/player-mannerisms-live.md#grab-and-pull-live)
2.22	3 × 2 (`pull.gani`)	stays `pull`, looping	as 2.05	[live][2.22] (client-internals/player-mannerisms-live.md#cross-version-table)
5.x, 6.x	unmeasured	unmeasured	unmeasured; `grab` and `pull` start in the next tick, the stone stays put, dust is drawn at the feet	[wire][shot] (protocol/v5v6-lab-play.md)

4.2.9 The push and pull trigger actions

Rule	Detail	Source
local	NPCs at the point see `actionpushed` or `actionpulled`	[2.05] (client-internals/ledges-push-pull.md#summary)
server	`PLI_TRIGGERACTION` (38, → §2.4): GINT 0 (no NPC), `GCHAR x*2`, `GCHAR y*2`, the action name, then `,` and the parameter as a decimal string when there is one (`pushed,0`)	[2.05] (client-internals/ledges-push-pull.md#the-trigger-action-on-the-wire)
peers	2.05 also sends it to UDP peers as UDP id 26 (→ §2.9)	[2.05] (client-internals/ledges-push-pull.md#the-trigger-action-on-the-wire)
cadence, 2.05	one try on the 29th blocked step, then every 20 steps (push); every 18 ticks (pull)	[2.05] (client-internals/ledges-push-pull.md#pushing); [2.05] (client-internals/ledges-push-pull.md#grabbing-and-pulling)
cadence, 5.x and 6.x	`pushed,0` about 0.55 s after `push` starts, then every 1.0 s; `pulled,1` 0.85 s into a pull	[wire] (protocol/v5v6-lab-play.md)

4.2.10 Water and lava

Rule	Detail	Source
check	every tick, except while the sword swings, while riding, and when the player is not under local control	[2.05] (client-internals/water-and-ganis.md#water)
entry point	the switch to `swim` happens on the tick the point (x + 1.5, y + 2) is on a liquid tile; leaving happens when it is off, and `walk` comes back at frame 0	[live][2.05] (client-internals/player-mannerisms-live.md#swimming-live)
entering	when not already swimming: a splash (break effect 5) at the player's position; two more splashes when the player was not walking (a warp or a knock); then swim mode with `swim`, or `lava` when the tile type is 12	[2.05] (client-internals/water-and-ganis.md#water)
leaving	one splash, then back to standing	[2.05] (client-internals/water-and-ganis.md#water)
splash sound	`water.wav` at (x + 1, y + 1) from the effect's anchor; every other break effect plays `crush.wav`	[2.05] (client-internals/water-and-ganis.md#break-effect-sounds)
swim mode	mode 10: set on a type 11 tile; cannot be hurt; the sword needs no horse	[2.05] (client-internals/decomp-findings-2026-10-24/player-modes.tsv)
status bit	while crossing water 1.28 beta 2 sets STATUS bit 0x04 and sends prop 18	[cap] (client-internals/live-trace-early.md#gserver-128-with-128-beta-2-gserv013)

Swimming by build:

Build	Standing still in water	Swim frames	Source
1.39	only the head and a water ring are drawn; the ring keeps alternating, the head stays still	n/a (no ganis)	[live][1.39] (client-internals/player-mannerisms-live.md#notes-per-build)
1.41	`swim` frame resets to 0 and freezes	unmeasured	[live][1.41] (client-internals/player-mannerisms-live.md#cross-version-table)
2.05 rev 2	keeps animating; does not reset when the player turns	8 frames × 2 ticks (16-tick stroke)	[live][2.05] (client-internals/player-mannerisms-live.md#swimming-live)

Build	Standing still in water	Swim frames	Source
2.22	keeps animating; the frame resets to 0 at the stop, then runs on	8 × 2	[live][2.22] (client-internals/player-mannerisms-live.md#cross-version-table)
5.x, 6.x	unmeasured; `swim` from the first water step, only the head with a ripple, sideways strokes throw splashes	unmeasured	[wire][shot] (protocol/v5v6-lab-play.md)

4.2.10.1 Lava burn

Rule	Detail	Source
test	once per player tick, one point, the position plus (1.5, 2), against the lava list (array 3) only; the order of the two offsets is inferred	[2.05][inf] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#the-default-tileset)
counter	on lava a counter runs 0 to 100 and wraps to 0 on the next tick: one burn every 101 ticks; a tick off lava sets it to 0	[live][2.05] (client-internals/live-trace-205.md#water-and-lava-live-new4nw); [2.05] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#the-default-tileset)
burn	half a heart through the hurt routine, with a flag that skips the usual "can be hurt" checks (→ §4.3); reading that call as the hurt routine is inferred	[2.05][inf] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#the-default-tileset)
tiles	the bright-red lava tile 2882 is swum in (`lava` gani) and burns; the dark-red lava floor 2958 is walked on and burns	[live][2.05] (client-internals/live-trace-205.md#water-and-lava-live-new4nw)

4.2.11 Following links

Rule	Detail	Source
neighbours	on entering a level while a big map is loaded, the client fills a 3 × 3 table of level names round it from the big map's text grid (→ §3.2.5); cells off the grid stay empty; the step is skipped under an unidentified flag (→ App. E#308)	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#the-levels-neighbours)
probe	each tick, the player's position plus a per-facing offset (offsets unread → App. E#308)	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#following-a-link)
order	each link of the level in order; the first that fires is taken; offline the client loads the level, online it asks the server	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#following-a-link)

Link	Fires when	Arrival	Source
west edge: x 0, width 1, destination = west neighbour	probe x ≤ 0 facing left, or player x < -2.5, or (neighbour not loaded) probe in the rectangle	x = player x + 64, y unchanged	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#following-a-link)
east edge: x 63, width 1, destination = east neighbour	probe x ≥ 64 facing right, or player x > 63.5, or (not loaded) probe in the rectangle	x = player x - 64	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#following-a-link)

Link	Fires when	Arrival	Source
north edge: y 0, height 1, destination = north neighbour	probe y ≤ 0 facing up, or player y < -3.0, or (not loaded) probe in the rectangle	y = player y + 64	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#following-a-link)
south edge: y 63, height 1, destination = south neighbour	probe y ≥ 64 facing down, or player y > 63.0, or (not loaded) probe in the rectangle	y = player y - 64	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#following-a-link)
any other link	probe in the rectangle; a probe within 1 tile of an edge (x or y ≤ 1 or ≥ 63) counts only while the player faces out across that edge	the link's new-x and new-y, evaluated as expressions; a missing or `-1` value means `playerx` / `playery`	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#following-a-link)

Rule	Detail	Source
seam	an edge link between big-map neighbours keeps the player's exact offset across the seam (east off x 63.6 arrives at x -0.4), whatever its new-x says	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#following-a-link)
no big map	without a big map, or for a level that is not the neighbour, the link's own coordinates apply, and `playerx` on an east-edge link arrives on the east edge of the destination	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#following-a-link)
editor defaults	a new edge link gets new-x `0` on the east edge, new-y `0` on the south edge, new-x 61 on the west edge and new-y 61 on the north edge; the other coordinate stays `playerx` / `playery`	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#link-records-for-reference)
level-entry messages	what the client sends on each level entry is in → §2.4	[2.31] (client-internals/decomp-findings-2026-10-24-b/03-periodic-ids-2.31.md#pli-46-on-every-level-entry-gchar-1)

4.2.12 Gates that stop movement (2.22, 2.31)

Rule	Detail	Source
stop gates	walking, the ledge test, the A key and other gameplay code do nothing while either of two server-set flags is on; server messages 176 and 177 set them (→ §2.5.8); reading them as the full-stop messages is an inference	[2.22][2.31][inf] (client-internals/ledges-push-pull.md#222-and-231)
connection flag	a further flag on the server connection also stops gani changes, footsteps and the A key; its meaning was not found	[2.22] (client-internals/ledges-push-pull.md#222-and-231)
built-in gani gate	`jump.wav`, `stonemove.wav`, the pull sound and footsteps play only while the current gani is a built-in name (→ §4.10); the jump, push and pull still happen under a custom gani; sound names are bare (`jump.wav`) instead of under `sounds/`	[2.22][2.31] (client-internals/ledges-push-pull.md#222-and-231)
2.05	none of these gates	[2.05] (client-internals/ledges-push-pull.md#222-and-231)

4.3 Combat, damage, hurt and knockback

This chapter gives the sword swing and the spin, who judges a hit, the gates a hit must pass, the damage each source takes, the shared hurt routine, hurt mode, knockback and the hurt pose. The combat messages themselves are in → §2.4 and → §2.5.5; the STATUS bits are in → §2.6.2.1. Death is → §4.4.

> Status: draft. Read in full for 2.00 to 2.05 rev 2, 2.17.1, 2.22 and 2.31; 1st Graal and 1.39 for the main rules; 3.x to 6.x only as timed on the wire. Open points are App. E#309 to E#315.

4.3.1 Who judges a hit

Rule	Detail	Source
the victim decides	the victim's own client judges another player's swing from that player's props (position, direction, sprite) and applies the damage; the attacker only broadcasts its props	[2.05] (client-internals/decomp-findings-2026-10-24/04-sword-hitbox.md#who-decides-a-hit); [2.00] (client-internals/combat-200-204.md#the-same-in-every-build-200-to-205-rev-2)
bombs and arrows	the victim's client also judges a server-sent bomb (`PLO_BOMBADD`, 11), explosion (36) or shot (`PLO_ARROWADD`, 19), whoever owns it; no hit message is sent, only the victim's props message (CURPOWER and the `hurt` gani) and the knockback positions; a client's own bomb hurts its owner	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#b-bombs-arrows-baddies-who-owns-the-damage-part-2-fresh-runs); [2.05] (client-internals/damage-sources.md#other-players-bombs-blasts-and-shots)
hearts on the wire	the victim reports its own hearts: in the hit's tick one props message with CURPOWER, the `hurt` gani and one more prop; nothing from the server asks for it, and the server's own CURPOWER is not echoed	[cap][2.05] (client-internals/live-trace-205.md#hearts-are-reported-by-the-client); [lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#c-curpower-semantics-settled-for-222)
units	CURPOWER is in half-hearts, MAXPOWER in hearts; the client clamps CURPOWER to MAXPOWER × 2 before subtracting	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#c-curpower-semantics-settled-for-222)
killer report	when a hit by another player's sword, piece or shot, or a UDP hurt, leaves the hearts at 0 or below, the victim sends `PLI_CLAIMPKER` (14) with the attacker's id (→ §2.4)	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#damage-rules-that-feed-this); [2.05] (client-internals/damage-sources.md#who-can-hurt-whom)
integrity	hearts, AP and the heart cap are stored obfuscated twice and cross-checked; a mismatch raises a "memory problems" error (→ §4.10)	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#anti-cheat-note)

4.3.2 The sword swing

Rule	Detail	Source
preconditions	weapons allowed by the level or server, sword power not 0, not swimming without a horse, not already in the spin mode	[2.05] (client-internals/decomp-findings-2026-10-24/04-sword-hitbox.md#starting-a-swing)
mode and gani	mode 6; plays `sword` (`ridesword` on a horse) and the sword sound, then checks for a cuttable bush at the sword tip	[2.05] (client-internals/decomp-findings-2026-10-24/04-sword-hitbox.md#starting-a-swing); [2.05] (client-internals/water-and-ganis.md#built-in-animations)
tip probe	one tile beyond the middle of the leading edge: up (x + 1.5, y), left (x - 0.5, y + 2), down (x + 1.5, y + 4), right (x + 3.5, y + 2)	[2.05] (client-internals/decomp-findings-2026-10-25-l/01-walking-body-box.md#other-probes-same-routine-family)
timing	`sword` in the next tick, 5 frames of one tick, then `idle`; the bush is cut on the first swing frame; the player does not move while it plays, even with an arrow held; walking resumes the tick after, from walk frame 0	[live][2.05] (client-internals/player-mannerisms-live.md#sword-live)

Rule	Detail	Source
repeat	holding S does not swing again; each swing needs a new press; holding S gives no charge	[live][2.05] (client-internals/player-mannerisms-live.md#sword-live)
on the wire	the swing's sprite steps are 9 to 13 (SPRITE 6 to 10)	[2.05] (client-internals/decomp-findings-2026-10-24/04-sword-hitbox.md#starting-a-swing)
no sword	a player with sword power 0 started from the editor still plays `sword` and its sound, but no blade is drawn and nothing is cut	[live][2.05] (client-internals/player-mannerisms-live.md#sword-live)
carrying	the sword key with an object carried throws it instead (the same throw as the action key, → §4.5); no swing precondition is checked on that path	[2.05] (client-internals/decomp-findings-2026-10-28/01-sword-while-carrying.md); [live][2.22] (client-internals/player-mannerisms-live.md#notes-per-build)
`hidesword n`	sets a counter of n seconds (clamped 0 to 30, × 20 ticks) that counts down each tick; while it is 0 or more STATUS carries bit 0x20; it does not stop the local swing, only refuses the hit at the victim's end	[2.05] (client-internals/status-bit-0x20.md)
`hideplayer n`	the same clamp and scale on its own counter, which drives STATUS bit 0x02 (hidden)	[2.05] (client-internals/status-bit-0x20.md#how-the-bit-is-set)

By build:

Build	Swing	Cut	Source
1st Graal	hit frames 6 to 10	unmeasured	[bin] (version-profiles/matrix/rules-1st-graal.md#sword-reach-bin)
1.39	counter 1 to 6 (5 ticks), no gani	cuts the bush	[live][1.39] (client-internals/player-mannerisms-live.md#cross-version-table)
1.41	`sword` 5 ticks	on the first frame	[live][1.41] (client-internals/player-mannerisms-live.md#cross-version-table)
2.05 rev 2	`sword` 5 × 1 tick	on the first frame	[live][2.05] (client-internals/player-mannerisms-live.md#sword-live)
2.22	`sword`	the 2 × 2 leaves board change goes to the server in the same frame as the sword gani; idle follows 240 ms later	[cap] (live-runs/2gta-222-2026-10-30/README.md#what-the-client-sends-while-walking-and-cutting)
5.24r4, 5.30r3, 6.03r7	`sword` 5 ticks, `idle` 0.25 s later	the cut (board change) in the same tick as the swing starts	[wire] (protocol/v5v6-lab-play.md)

4.3.2.1 The spin attack

Rule	Detail	Source
start	a second S press while `sword` is still playing, with the has-spin status (STATUS bit 0x40); not a hold-to-charge	[live][2.05] (client-internals/player-mannerisms-live.md#spin-attack-live-online); [2.05] (client-internals/feature-specs/02-weapons-combat.md#205-sword-swing-and-hit-box-205dec)
new press	S must be up for at least one tick in between; a press on swing frame 2, 3 or 4 works; a press after the swing has ended starts a normal swing	[live][2.05] (client-internals/player-mannerisms-live.md#spin-attack-live-online)
timing	the swing is cut off: `spin` frame 0 shows in the tick after the second press; 13 frames of one tick (650 ms), then `idle`; the facing at the start is stored	[live][2.05] (client-internals/player-mannerisms-live.md#spin-attack-live-online)
mode	mode 13; a press during the spin is ignored and queues nothing	[live][2.05] (client-internals/player-mannerisms-live.md#spin-attack-live-online); [2.05] (client-internals/decomp-findings-2026-10-24/player-modes.tsv)

Rule	Detail	Source
wire	the client sends `GANI=spin` once at the start and `GANI=idle` at the end, and asks for `spin.gani` the first time	[live][2.05] (client-internals/player-mannerisms-live.md#spin-attack-live-online)
2.22	the same: a second press on frame 3 cut the swing off, `spin` ran 13 × 1 tick, a third press was ignored	[live][2.22] (client-internals/player-mannerisms-live.md#spin-attack-live-online)
1.39, 1.41	unmeasured (offline neither build could be given the has-spin status)	[live] (client-internals/player-mannerisms-live.md#open)
what it hits	which bushes the spin cuts and whether it hits players or NPCs: unmeasured	[live] (client-internals/player-mannerisms-live.md#spin-attack-live-online)

4.3.3 The sword hit test (victim's side)

A swing hits the local player only when every gate passes. In 2.05 rev 2 order:

#	Gate	Source
1	the attacker's player id is known	[2.05][2.22] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#gates-shared-by-the-sword-and-the-packets)
2	the attacker is on sprite steps 9 to 13 (SPRITE 6 to 10)	[2.05] (client-internals/decomp-findings-2026-10-24-d/hit-gates.tsv)
3	the victim can be hurt: not in hurt mode, not dead, not swimming, hearts above 0, and normal play or a sparring-zone level	[2.05] (client-internals/decomp-findings-2026-10-24-d/03-hit-gates-and-open-points.md#level-flags-seen)
4	the victim's AP is not exactly 100	[2.05] (client-internals/decomp-findings-2026-10-24-d/hit-gates.tsv)
5	the level has not run `noplayerkilling`; when the local player's own attack is refused by it, the client prints `nopkzone`	[2.05] (client-internals/decomp-findings-2026-10-24-d/03-hit-gates-and-open-points.md#gates)
6	the attacker has not already been judged in this second of the day (a per-attacker stamp, stored before the guild test); its place in the sword path is `disputed → App.E#310`	[2.05][2.22] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#gates-shared-by-the-sword-and-the-packets)
7	guild: the first bracketed text in the attacker's nickname (a `(`, then the first `)` after it with at least one character between) does not occur anywhere in the victim's nickname, case-sensitive	[2.05] (client-internals/decomp-findings-2026-10-24-d/03-hit-gates-and-open-points.md#gates)
8	the attacker's STATUS has neither 0x02 (hidden) nor 0x20 (sword hidden)	[2.05] (client-internals/decomp-findings-2026-10-24-d/03-hit-gates-and-open-points.md#gates)
9	the attacker is not kicking: 2.05 by the animation name `kick`, 2.17.1 and later by an object id of 3	[2.05][2.17.1] (client-internals/decomp-findings-2026-10-24-d/hit-gates.tsv)
10	reach box: the victim's centre 0 to 4 tiles ahead of the attacker's centre along its facing (both ends count) and strictly under 2 tiles to the side; centres are (x + 1.5, y + 2); skipped while the victim is in hurt mode	[2.05] (client-internals/decomp-findings-2026-10-24/04-sword-hitbox.md#who-decides-a-hit); [2.00] (client-internals/combat-200-204.md#the-same-in-every-build-200-to-205-rev-2)

By build:

Rule	1st Graal	1.39	2.00	2.01	2.02	2.03	2.04 = 2.05 rev 2; 2.17.1, 2.22, 2.31	Source
reach box	up to 4 ahead, sideways under 2; the 0 and 4 edges unread	0 to 4 ahead, sideways under 2	0 to 4, both ends; under 2	=	=	=	=	[bin] (version-profiles/matrix/rules-1st-graal.md#sword-reach-bin); [1.39] (version-profiles/matrix/rules-139.md#sword-reach-bin-lab); [2.00] (client-internals/combat-200-204.md#the-same-in-every-build-200-to-205-rev-2)
tested only in	the normal life-state	unmeasured	play state exactly 0 (its own victim test; no paused/editor test)	the shared can-be-hurt test	shared	shared	shared	[bin] (version-profiles/matrix/rules-1st-graal.md#sword-reach-bin); [2.00][2.01] (client-internals/combat-200-204.md#what-changes-by-build)
kick test	unmeasured	unmeasured	absent	by animation name	=	=	by name (2.05); object id 3 (2.17.1 on)	[2.00][2.01] (client-internals/combat-200-204.md#what-changes-by-build); [2.17.1] (client-internals/decomp-findings-2026-10-24-d/hit-gates.tsv)
no-kill flag	unmeasured	unmeasured	on the player, cleared on level load	on the player	on the player	on the player	on the level, read through a tamper-checked byte	[2.00][2.03][2.04] (client-internals/combat-200-204.md#what-changes-by-build)
`playerhurt` event on a hit	n/a	unmeasured	absent	present	present	present	present	[2.00][2.01] (client-internals/combat-200-204.md#what-changes-by-build)
props path: per-player link field equal	n/a	unmeasured	no	no	no	no	yes (→ §4.3.6)	[2.00][2.04] (client-internals/combat-200-204.md#what-changes-by-build)

4.3.4 Sword damage

Rule	Detail	Source
formula	half-hearts taken = the attacker's sword power as the victim last received it in SWORDPOWER (prop 8), limited to -20 to 20; a power-3 sword takes 1.5 hearts	[2.05][2.17.1][2.22][2.31] (client-internals/decomp-findings-2026-10-24-d/01-sword-damage.md#the-formula)
prop 8 on the wire	power 0 sends byte 0 (no image); power 1 to 4 with the matching default image (`sword1` to `sword4`) sends the power itself; anything else sends power + 30 (10 to 50) followed by the image name; the receiver reads a byte below 10 as the power and otherwise byte - 30 (→ §2.6)	[2.05] (client-internals/decomp-findings-2026-10-24-d/01-sword-damage.md#the-sword-power-byte-prop-8)
negative power	heals, but only at AP 40 or more; hearts stop at the maximum	[2.05] (client-internals/decomp-findings-2026-10-24-d/01-sword-damage.md#what-happens-to-the-hearts)

Rule	Detail	Source
not in the formula	the victim's shield power, item levels and the attacker's AP were not read into it	[2.05] (client-internals/decomp-findings-2026-10-24-d/damage-rules.tsv)

Build	Sword power store	Negative power	Source
1st Graal	fixed half a heart; the sword does not grow	n/a	[bin] (version-profiles/matrix/rules-1st-graal.md#sword-reach-bin)
2.00, 2.01, 2.02	the received power as is, in a 6-bit slot (0 to 63)	cannot occur: it wraps in the store (inferred from the mask)	[2.00][2.01][2.02][inf] (client-internals/combat-200-204.md#what-changes-by-build)
2.03 to 2.31	power + 20, limited to -20 to 20	heals at AP 40 or more	[2.03] (client-internals/combat-200-204.md#what-changes-by-build); [2.05] (client-internals/decomp-findings-2026-10-24-d/01-sword-damage.md)

4.3.5 Other damage sources (2.05)

Every source calls the same hurt routine (→ §4.3.7) with a power in hearts and a knockback direction.

Source	Hearts	Hit test	Source
explosion piece, type 1 (`putbomb 1`, `putexplosion`, the blast of a fireblast or nuke)	0.5	the piece's centre, offset by one tile, less than one tile from the player's centre (x + 1.5, y + 2) on both axes; the lab pattern is `disputed → App.E#311`	[2.05] (client-internals/damage-sources.md#bombs-and-explosions)
explosion piece, type 2 (`putbomb 2`, the super bomb)	0.5	as type 1	[2.05] (client-internals/damage-sources.md#powers)
explosion piece, type 3 (`putbomb 3`, the jolt bomb)	1	as type 1	[2.05] (client-internals/damage-sources.md#powers)
arrow (`shootarrow`)	0.5	the shot box against the player's box (→ §4.3.5.2)	[2.05] (client-internals/damage-sources.md#powers)
fireball (`shootfireball`)	1	as an arrow	[2.05] (client-internals/damage-sources.md#powers)
fireblast (`shootfireblast`)	0.5, then a blast of reach 2, type 1	as an arrow	[2.05] (client-internals/damage-sources.md#projectiles)
nuke (`shootnuke`)	1, then a blast of reach 2, type 1	as an arrow	[2.05] (client-internals/damage-sources.md#projectiles)
thrown object (`waspelt`)	1, or 0.5 when the player has exactly 1 heart	lands within 2 tiles of (x + 0.5, y + 1) on both axes	[2.05] (client-internals/damage-sources.md#thrown-objects)
baddy touch	0.5, or 1 for the golden warrior (type 7)	the player's centre within one tile of the baddy's point on both axes; the baddy may say its "hurts the player" verse	[2.05] (client-internals/damage-sources.md#baddies)
lava	0.5 every 101 ticks (→ §4.2.10.1)	the point (x + 1.5, y + 2) on lava	[live][2.05] (client-internals/live-trace-205.md#water-and-lava-live-new4nw)

Source	Hearts	Hit test	Source
PL0_HURTPLAYER (40)	its power byte - 32 in half-hearts, capped at one heart; 0 gives hurt mode and knockback with no hearts	→ §4.3.6	[2.05] (client-internals/decomp-findings-2026-10-24-d/01-sword-damage.md#the-hurt-packet-id-40-and-its-power); [lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#b-packet-40-hurt-player-real-client-as-victim-settled-for-the-cases-below)

Rule	Detail	Source
ownership	a bomb, blast, arrow or thrown object can belong to another player; online, a piece counts as a player's when it has an owner still in the player list; a player's source does not hurt in a level that ran `noplayerkilling`; a source with no owner (a baddy's, the level's, or one whose owner has left) always hurts	[2.05] (client-internals/damage-sources.md#who-can-hurt-whom); [2.05] (client-internals/damage-sources.md#when-a-piece-hurts-the-local-player)
one blast, one hit	overlapping pieces take hearts once, because the second finds the player in hurt mode	[2.05] (client-internals/damage-sources.md)
1.39 baddy touch	0.5 hearts (1 for type 7), within 1 tile on both axes of the baddy centre	[1.39] (version-profiles/matrix/rules-139.md#summary-table)
1st Graal	sword and baddy touch half a heart; baddy centre (x + 1, y + 1); two projectile types (sprites 65 and 261) half a heart, the others one heart; an arrow (sprites 107 to 110) flying opposite the player's facing, with the player's sprite index below 6, is blocked	[bin] (version-profiles/matrix/rules-1st-graal.md#hurt-and-knockback-bin)

4.3.5.1 Bombs and blasts

Rule	Detail	Source
`putbomb power,x,y`	powers 1 to 3 are the bomb, super bomb and jolt bomb of the built-in weapons (inferred from the weapon scripts); a local fuse is 60 ticks; water under (x + 1, y + 1) makes a splash and no bomb	[2.05][inf] (client-internals/damage-sources.md#bombs-and-explosions); [2.05] (client-internals/damage-sources.md#bomb-11)
going off	power 3 plays `thunder.wav`, any other `bomb.wav`; power 2 makes five type-2 blasts (four of reach 2 at the diagonal corners 2 tiles out, one of reach 4 in the middle); any other power one blast of reach 2 with the bomb's power as its type	[2.05] (client-internals/damage-sources.md#bombs-and-explosions)
blast shape	a cross: a centre piece and n pieces in each direction (n at most 15); the last piece of an arm uses the end picture; each piece lives 11 ticks and may hit on any of them	[2.05] (client-internals/damage-sources.md#bombs-and-explosions); [2.05] (client-internals/damage-sources.md#when-a-piece-hurts-the-local-player)
`putexplosion radius,x,y`	one blast of type 1 with that reach	[2.05] (client-internals/damage-sources.md#bombs-and-explosions)
knockback	along the piece's arm	[2.05] (client-internals/damage-sources.md#when-a-piece-hurts-the-local-player)
another player's bomb (11)	at least 4 bytes; power `v & 7`, bit 8 marks it as the sender's; the fuse byte is in ticks (the sender's copy is 5 ticks shorter than a local `putbomb`); put in the sender's level; nothing is sent back	[2.05] (client-internals/damage-sources.md#bomb-11)
another player's explosion (36)	at least 4 bytes; power `v & 7`, 0 read as 1 and above 3 read as 3; one cross, never the super-bomb split; one of two sounds by whether the power is 3 (that they are `thunder.wav` and `bomb.wav` is inferred)	[2.05][inf] (client-internals/damage-sources.md#explosion-36)

4.3.5.2 Shots

Rule	Detail	Source
local shots	the four shoot commands make a flying object of type 0 (arrow), 1 (fireball), 2 (fireblast) or 3 (nuke) at speed 1 and play `arrow.wav`	[2.05] (client-internals/damage-sources.md#projectiles)
another player's shot (19)	at least 5 bytes: x×2, y×2, flags, sprite, type (`byte - 33`); flags & 3 direction, 4 first removes a matching shot already held, 8 marks it as the sender's; arrows use sprites 107 + direction	[2.05] (client-internals/damage-sources.md#shot-19)
flight	1 tile per tick; past -2 or at 64 and beyond it moves to the neighbouring level of the loaded gmap block, or is gone	[2.05] (client-internals/damage-sources.md#when-a-shot-hurts-the-local-player)
hit box	size from the sprite table entry v (array 24): width `((v >> 6) & 63) / 16`, height `(v & 63) / 16`; hits when `x + w > px + 0.5`, `x < px + 2.5`, `y + h > py + 1`, `y < py + 3`	[2.05] (client-internals/damage-sources.md#when-a-shot-hurts-the-local-player)
skipped	while the shot bounces back, for a player's shot (one's own, or one received in a `noplayerkilling` level), or when the player cannot be hurt	[2.05] (client-internals/damage-sources.md#when-a-shot-hurts-the-local-player)
shield	the player faces the shot, shield power above 0, the shot is type 0 or 1, and the animation is one of two built-in names (inferred `idle` and `walk`); power 1: an arrow turns back for 3 ticks with a sound, any other sprite is gone; power 2 or more: the shot is sent back the other way and re-sent as a new shot	[2.05][inf] (client-internals/damage-sources.md#when-a-shot-hurts-the-local-player)
walls	whether arrows and fireballs pass walls is open (→ App. E#313)	[2.05] (client-internals/damage-sources.md#open)
lab	an injected shot hurt only when its last byte was `0x23` or more; meaning open (→ App. E#312)	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#b-bombs-arrows-baddies-who-owns-the-damage-part-2-fresh-runs)

4.3.6 Hits by message

Rule	`PLO_HURTPLAYER` (40)	`PLO_HITOBJECTS` (46)	Source
payload	attacker id, dx, dy, power; an optional trailing GINT3 NPC id	power, then the hit point's two bytes (half-tiles)	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#bc-addendum-fresh-runs-23102330-jst-packet-40-distance-gate); [2.05][2.22] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#packet-46-box)
gates 1 to 6 of §4.3.3	yes, without the attacker status test	yes	[2.05][2.22] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#gates-shared-by-the-sword-and-the-packets)
link field	the attacker's layer/parent link integer equals the victim's (new players start at 0; a horse does not change it)	as 40	[2.22] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#the-per-player-field-of-packet-40); [2.05][2.22] (client-internals/decomp-findings-2026-10-25-e/04-name-index-case-link.md#packet-40-link)
range	squared distance between the two X/Y positions at most 100 (lab: hits at 100.0, none at 110.25)	the player's centre (x + 1.5, y + 2) strictly under 2 tiles from the hit point on each axis	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#bc-addendum-fresh-runs-23102330-jst-packet-40-distance-gate); [2.05][2.22] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#packet-46-box)
power	`byte - 32` half-hearts, capped at 2	capped at one heart	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#b-packet-40-hurt-player-real-client-as-victim-settled-for-the-cases-below)

Rule	PLO_HURTPLAYER (40)	PLO_HITOBJECTS (46)	Source
knockback	(byte - 0x60) / 16 per axis, normalised to the 5-tile push; a zero vector goes 5 tiles east	the victim's centre minus the hit point	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#bc-addendum-fresh-runs-23102330-jst-packet-40-distance-gate); [2.05] [2.22] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#knockback-of-a-hit)
NPC id present	the hit is refused completely (no damage, no knockback); with only 2 trailing bytes it still hits	unmeasured	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#bc-addendum-fresh-runs-23102330-jst-packet-40-distance-gate)
unknown attacker	refused	refused	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#b-packet-40-hurt-player-real-client-as-victim-settled-for-the-cases-below)

Gate	2.00	2.01, 2.02, 2.03	2.04 = 2.05 rev 2	Source
id 46 handled	no	yes	yes	[2.00][2.01] (client-internals/combat-200-204.md#what-changes-by-build)
AP 100 immune	no	yes	yes	[2.00][2.01] (client-internals/combat-200-204.md#what-changes-by-build)
no-kill flag	no	yes	yes	[2.00][2.01] (client-internals/combat-200-204.md#what-changes-by-build)
once per attacker per second	no	yes	yes	[2.00][2.01] (client-internals/combat-200-204.md#what-changes-by-build)
guild test	yes	yes	yes	[2.00][2.01] (client-internals/combat-200-204.md#what-changes-by-build)
link field equal	no	no	yes	[2.00][2.04] (client-internals/combat-200-204.md#what-changes-by-build)
range and power cap	10 tiles, one heart	=	=	[2.00] (client-internals/combat-200-204.md#the-same-in-every-build-200-to-205-rev-2)

4.3.7 The hurt routine and hurt mode

Rule	Detail	Source
input	a power in hearts and a knockback vector	[2.05] (client-internals/decomp-findings-2026-10-24-d/01-sword-damage.md#what-happens-to-the-hearts)
refused	while the player cannot be hurt: in hurt mode (3), swimming (10), dead, or in the editor or paused outside a sparring zone; lava's call skips these checks	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#damage-rules-that-feed-this); [2.05] [inf] (client-internals/decomp-findings-2026-10-28-b/01-tile-type-tests.md#the-default-tileset)
hearts	the heart change is minus the power; a heal (negative power) applies only at AP 40 or more; hearts at 0 or below start the death (→ §4.4)	[2.05] (client-internals/decomp-findings-2026-10-24-d/01-sword-damage.md#what-happens-to-the-hearts)
mode	hurt mode (3) with the counter at 40, for every accepted hit, including power 0 and an ignored heal	[2.05][2.17.1][2.22][2.31] (client-internals/decomp-findings-2026-10-24-d/damage-rules.tsv)
event	the script event `playerhurt` fires	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#damage-rules-that-feed-this)
countdown	each tick the counter drops by one; at 0 the mode returns to normal; no new hit for the whole 40 ticks (2 s)	[2.05] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#knockback-of-a-hit); [live] [2.05] (client-internals/player-mannerisms-live.md#hurt-knockback-and-blinking-live)
arrows during hurt	held arrows are ignored while the counter is above 29; from the 12th tick after the hit (counter 28) a held arrow walks again and `walk` replaces `hurt`	[live][2.05] (client-internals/player-mannerisms-live.md#hurt-knockback-and-blinking-live)

Rule	Detail	Source
carrying	1.39: a positive-power hit may force a drop of the carried object; 1st Graal: the carry is dropped	[1.39] (version-profiles/matrix/rules-139.md#hurt-and-knockback-bin); [bin] (version-profiles/matrix/rules-1st-graal.md#hurt-and-knockback-bin)

4.3.7.1 Knockback

Rule	Detail	Source
vector	stored scaled to one tile; a zero vector becomes one tile east	[2.05][2.17.1][2.22][2.31] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#knockback-of-a-hit)
steps	while the counter reads 39 down to 35 after its decrement, the player moves one tile per tick along the vector: five tiles in 250 ms on open ground	[2.05][2.17.1][2.22][2.31] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#knockback-of-a-hit)
blocked	after each step twelve body-box points are tested with the walking rules (→ §4.2.3); if any is blocked the step is undone and tried again the next tick	[2.05][2.17.1][2.22][2.31] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#knockback-of-a-hit)
inside a blocking NPC	every step is refused and the player does not move	[live][2.05] (client-internals/player-mannerisms-live.md#hurt-knockback-and-blinking-live)
sword direction	the victim's centre minus the attacker's point (halved before normalising, where 2.00 and 2.02 keep it)	[2.05] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#knockback-of-a-hit); [2.00][2.02] (client-internals/combat-200-204.md#the-same-in-every-build-200-to-205-rev-2)
live	a touch NPC moved the player from (29.5, 48) to (28.79, 52.95), 5.0 tiles, along the line from the NPC image centre to the player's centre	[live][2.05] (client-internals/player-mannerisms-live.md#hurt-knockback-and-blinking-live)

The twelve points, offsets from (x, y), three per edge of a box 0.6 to 2.4 across and 1.1 to 2.9 down, centred on (x + 1.5, y + 2):

Point	x offset	y offset	Source
1	0.6	1.1	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)
2	1.5	1.1	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)
3	2.4	1.1	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)
4	0.6	1.1	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)
5	0.6	2.0	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)
6	0.6	2.9	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)
7	0.6	2.9	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)
8	1.5	2.9	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)
9	2.4	2.9	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)
10	2.4	1.1	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)
11	2.4	2.0	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)
12	2.4	2.9	[2.05] (client-internals/decomp-findings-2026-10-25-k/knockback-probes.tsv)

4.3.7.2 Hurt mode by build

Build	Lock-out	Knockback	Hurt pose	Blinking	Source
1st Graal	40 ticks; walking refused while the counter is 30 or more	5 steps; baddy and projectile hits an axis unit step from the attacker's facing, a sword hit the centre-to-centre delta; no zero-vector rule	the hurt body while the counter is above 30	odd counters draw a single frame	[bin] (version-profiles/matrix/rules-1st-graal.md#hurt-and-knockback-bin)

Build	Lock-out	Knockback	Hurt pose	Blinking	Source
1.39	40 ticks; a hit is also refused in mode 7	5 steps from the binary; 4 tiles live: `disputed → App. E#309`	about 10 ticks, the back with the arms out	after the pose, every other tick to 40	[1.39] (version-profiles/matrix/rules-139.md#hurt-and-knockback-bin); [live] [1.39] (client-internals/player-mannerisms-live.md#notes-per-build)
1.41	40	5 steps	`hurt` 31 ticks	from the hit, every other tick	[live][1.41] (client-internals/player-mannerisms-live.md#cross-version-table)
2.05 rev 2	40	5 steps	`hurt` 31 ticks (`hurt.gani` 1 frame, WAIT 30), keeping the facing; 12 ticks when an arrow is held	the body drawn on alternate ticks, only the shadow on the off ticks, for all 40 ticks, after `hurt` has given way	[live][2.05] (client-internals/player-mannerisms-live.md#hurt-knockback-and-blinking-live)
2.22	40	4.9 tiles on the same test	12 ticks with an arrow held	unmeasured	[live][2.22] (client-internals/player-mannerisms-live.md#cross-version-table)
5.24 r4, 5.30 r3, 6.03 r7	unmeasured	5 tiles in 5 ticks	`idle` 1.54 to 1.55 s after the hit	blinks	[wire] (protocol/v5v6-lab-play.md)

4.4 Death clocks and respawn

This chapter gives what the client does when its hearts reach zero, the death timer, the dead bit on the wire, the respawn key, the hearts and the destination after respawn, and the level-entry save that picks the destination. What the server does on the dead bit is in → §5. Damage is in → §4.3.

> Status: draft. Read for 1st Graal, 1.39, 2.00 to 2.05 rev 2, 2.17.1, 2.22 and 2.31; timed live for 2.05 and 2.22. Open points are App. E#314 and E#315.

4.4.1 Life-states

Value	Life-state	Source
0	playing	[2.05] (client-internals/decomp-findings-2026-10-24/life-states.tsv)
1	editor or pause	[2.05] (client-internals/decomp-findings-2026-10-24/life-states.tsv)
2	dead	[2.05] (client-internals/decomp-findings-2026-10-24/life-states.tsv); [1.39] (version-profiles/matrix/rules-139.md#death-and-respawn-bin)
4	a state in which any key restarts, with no timer test (1st Graal, 1.39); not named	[bin] (version-profiles/matrix/rules-1st-graal.md#death-and-the-restart-bin); [1.39] (version-profiles/matrix/rules-139.md#death-and-respawn-bin)

4.4.2 The death sequence (2.05)

Step	Rule	Source
trigger	any change that leaves hearts at 0 or below starts the sequence once; a per-frame check also forces the dead state whenever hearts are 0 or below	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#trigger)
1	the death timer starts at 0 and counts up one per tick; it stops at 30 (1.5 s)	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#sequence-on-the-dying-client)
2	the life-state becomes dead, the mode returns to 0 and the player faces down	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#sequence-on-the-dying-client)
3	the dead gani plays and the script event `playerdies` fires	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#sequence-on-the-dying-client)
4	controls are ignored until the timer passes 27 (1.35 s)	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#sequence-on-the-dying-client)
5	STATUS (prop 18) carries the dead bit 0x08 only while the life-state is dead and the timer is at least 28, so other players see the death about 1.4 s late	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#sequence-on-the-dying-client)
6	no dead sprite goes on the wire: SPRITE shows idle plus direction	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#sequence-on-the-dying-client)

On the wire:

Rule	Detail	Source
server CURPOWER 0	the client at once sends the dead gani and no CURPOWER; 27 ticks later STATUS 0x1C (dead bit set)	[live][cap][2.05] (client-internals/live-trace-205.md#server-curpower-0-death-and-respawn-live-online-2026-10-29)
2.22	dead gani and one more prop, no CURPOWER; then STATUS 28; the board goes black and an "End" scroll in the sign style appears under the player	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#c-curpower-semantics-settled-for-222); [cap] (live-runs/2gta-222-2026-10-30/README.md#baddies-death-and-respawn-2118-to-2121)
offline	burning to 0 hearts offline shows the End death scroll	[live][2.05] (client-internals/live-trace-205.md#water-and-lava-live-new4nw)
baddies	a leader client stops sending baddy props the moment it dies	[cap] (live-runs/2gta-222-2026-10-30/README.md#baddies-death-and-respawn-2118-to-2121)

4.4.3 Respawn

Rule	Detail	Source
key	once the timer has passed 27, any movement or action key respawns the player; there is no timer-only respawn, no respawn message and no server permission step	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#respawn); [cap] (live-runs/2gta-222-2026-10-30/README.md#baddies-death-and-respawn-2118-to-2121)
destination	the last saved level-entry point (level, x, y, direction; → §4.4.4)	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#respawn)
nothing saved, online	x 30.5, y 30.0, facing down, in the current level	[2.05] (client-internals/decomp-findings-2026-10-24/constants-205.tsv); [1.39] (version-profiles/matrix/rules-139.md#death-and-respawn-bin)
nothing saved, offline	the default start level `tutorial_b-4.graal` (axes of its start position unread)	[2.05] (client-internals/decomp-findings-2026-10-24/constants-205.tsv); [2.05] (client-internals/decomp-findings-2026-10-24-d/open-questions.tsv)
online warp	the client re-sends the level at the destination, so the server sees an ordinary level warp, not a "died" message; it does not wait for a reply	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#respawn); [live][cap][2.05] (client-internals/live-trace-205.md#server-curpower-0-death-and-respawn-live-online-2026-10-29)
2.05 frame	`PLI_LEVELWARP` (0) to the same level at the login point, and in the same frame CURPOWER, GANI `idle`, X, Y and STATUS with the dead bit cleared	[live][cap][2.05] (client-internals/live-trace-205.md#server-curpower-0-death-and-respawn-live-online-2026-10-29)
2.22 frame	the level warp, `PLI_TAMPERCHECK`, any script flags, then props with CURPOWER (full hearts), GANI `idle`, X, Y, SPRITE and STATUS; the server echoes the position and the hearts	[cap] (live-runs/2gta-222-2026-10-30/README.md#baddies-death-and-respawn-2118-to-2121)
dead bit	cleared when the life-state returns to playing	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#respawn)
repeat	a server that answers every respawn with CURPOWER 0 puts the client into a death loop	[live][2.05] (client-internals/live-trace-205.md#server-curpower-0-death-and-respawn-live-online-2026-10-29)

4.4.3.1 Hearts after respawn

AP	Hearts (2.05, 2.22)	Source
below 20	3	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#respawn); [lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#c-curpower-semantics-settled-for-222)
20 to 39	min(5, max hearts)	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#respawn); [lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#c-curpower-semantics-settled-for-222)

AP	Hearts (2.05, 2.22)	Source
40 or more	max hearts	[2.05] (client-internals/decomp-findings-2026-10-24/01-death-respawn.md#respawn); [lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#c-curpower-semantics-settled-for-222)
MAXPOWER 0	2.05 still reports CURPOWER 6 (3 hearts): the respawn hearts are not capped at a MAXPOWER of 0	[live][2.05] (client-internals/live-trace-205.md#server-curpower-0-death-and-respawn-live-online-2026-10-29)

4.4.3.2 By build

Rule	1st Graal	1.39	2.05 rev 2	2.22	Source
on death	hearts clamped to 0, counter and mode cleared, dead state; the relay object is notified; no inventory spill, no sound	hearts clamped to 0, counter 0, mode normal, dead state; `dead.wav`; the inventory partly spilled as floor items (rupees in 100/30/5/1 bands, arrows and bombs alike)	→ §4.4.2	→ §4.4.2	[bin] (version-profiles/matrix/rules-1st-graal.md#death-and-the-restart-bin); [1.39] (version-profiles/matrix/rules-139.md#death-and-respawn-bin)
timer	the up-count loop not found; the dead sprite and the key wait for a reading above 27	counts 0 to 30 and stops	0 to 30	unmeasured	[bin] (version-profiles/matrix/rules-1st-graal.md#death-and-the-restart-bin); [1.39] (version-profiles/matrix/rules-139.md#death-and-respawn-bin)
drawing	the board replaced by a filled rectangle and a text panel; the body uses sprite 157 once the counter is above 27	death sprites cycle between counters 4 and 27; a cleanup runs above 29	the dead gani	the "End" scroll	[bin] (version-profiles/matrix/rules-1st-graal.md#death-and-the-restart-bin); [1.39] (version-profiles/matrix/rules-139.md#death-and-respawn-bin)
dead bit 0x08	n/a (no STATUS prop)	only while dead and the counter is above 27	at 28 or more	sent	[bin] (version-profiles/matrix/rules-1st-graal.md#death-and-the-restart-bin); [1.39] (version-profiles/matrix/rules-139.md#summary-table)
key	above 27 (one key index does not restart)	above 27	above 27	a key, after 34 s in the run	[bin] (version-profiles/matrix/rules-1st-graal.md#death-and-the-restart-bin); [1.39] (version-profiles/matrix/rules-139.md#summary-table); [cap] (live-runs/2gta-222-2026-10-30/README.md#baddies-death-and-respawn-2118-to-2121)
destination	the new-game setup: `house1.graal` at (24.5, 30), facing 2; a second local player 12 tiles to the east	the saved entry, else (30.5, 30.0) online, else the built-in start-level table offline	→ §4.4.3	the level's entry point	[bin] (version-profiles/matrix/rules-1st-graal.md#death-and-the-restart-bin); [1.39] (version-profiles/matrix/rules-139.md#death-and-respawn-bin); [lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#d-end-of-respawn-save-settled-for-222-same-level-respawn)
hearts	3, and the heart cap set to 3	max hearts, no AP bands (1.39 has no AP display; the bands arrive with 1.41)	by AP	by AP	[bin] (version-profiles/matrix/rules-1st-graal.md#death-and-the-restart-bin); [1.39] (version-profiles/matrix/rules-139.md#summary-table)

4.4.4 The level-entry save

Run after every level entry, and at login; saving stores the new level's name, the player's x and y, and the direction as it is [2.05] (client-internals/decomp-findings-2026-10-24-d/02-respawn-entry-save.md#the-decision-run-after-every-level-entry).

Step	Condition	Result	Builds	Source
1	the level left is empty (login)	save	all read	[2.05] (client-internals/decomp-findings-2026-10-24-d/entry-save-rule.tsv)
2	index(new) × index(old) in the current gmap's level list is negative (-1 = absent; an index of 0 gives 0, so the first list entry behaves like "absent")	save	all read	[2.05] (client-internals/decomp-findings-2026-10-24-d/02-respawn-entry-save.md#the-decision-run-after-every-level-entry)
3	the name groups differ	save	2.00 (all revisions) to 2.05 rev 2	[2.05] (client-internals/decomp-findings-2026-10-24-d/02-respawn-entry-save.md#which-builds-have-the-group-test)
4	otherwise	do not save	all read	[2.05] (client-internals/decomp-findings-2026-10-24-d/entry-save-rule.tsv)

Rule	Detail	Source
name group	drop everything up to the last path delimiter, lower-case, then take the text up to and including the last § (byte 0xA7) or $; with neither, the group is empty, and two empty groups are equal; the extension stays	[2.05] (client-internals/decomp-findings-2026-10-24-d/02-respawn-entry-save.md#the-name-group-200-to-205)
effect	plain names such as `a.nw` and `b.nw` are always the same group; the test only separates names that use the delimiters	[2.05] (client-internals/decomp-findings-2026-10-24-d/02-respawn-entry-save.md#the-name-group-200-to-205)
no gmap	2.05 returns -1 for both indices when there is no current gmap; 2.00 to 2.04 take the list straight from the client's state, which is inferred to give the same result	[2.05][inf] (client-internals/decomp-findings-2026-10-24-d/02-respawn-entry-save.md#which-builds-have-the-group-test)
2.17.1, 2.22, 2.31	no group test (step 3 absent)	[2.17.1][2.22][2.31] (client-internals/decomp-findings-2026-10-24-d/02-respawn-entry-save.md#which-builds-have-the-group-test)
end-of-respawn save	the respawn routine calls the save once more, unconditionally; in the 2.22 lab, dying again before the server's warp reply still respawned at the entry point, not the death spot	[2.05] (client-internals/decomp-findings-2026-10-24-d/02-respawn-entry-save.md#after-the-respawn); [lab] [2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#d-end-of-respawn-save-settled-for-222-same-level-respawn)
gmap list order	the list's order and the timing of its update against the save are open (→ App. E#314)	[2.05] (client-internals/decomp-findings-2026-10-24-d/open-questions.tsv)

Builds that carry the group test's delimiter set, by a constant scan, with the rule itself read only where marked:

Builds	Delimiter constant	Source
1.1 to 1.36	absent	[bin] (client-internals/decomp-findings-2026-10-24-d/name-group-constant-by-build.tsv)
1.37, 1.38, 1.39, 1.41	present	[bin] (client-internals/decomp-findings-2026-10-24-d/name-group-constant-by-build.tsv)
2.00 to 2.05 rev 2, NewMain 24 and 26, NewWorld beta 1	present (rule read in all 2.00 to 2.05 builds)	[bin] (client-internals/decomp-findings-2026-10-24-d/name-group-constant-by-build.tsv)
2.1 test 2 and 3, NewWorld beta 2	absent; their entry code not read (→ App. E#315)	[bin] (client-internals/decomp-findings-2026-10-24-d/name-group-constant-by-build.tsv)
2.17.1, 2.22, 2.31	absent; entry code read, no group test	[bin] (client-internals/decomp-findings-2026-10-24-d/name-group-constant-by-build.tsv)

4.4.5 Server side (GServer-v2)

Death is client-driven; the server reacts only to the dead bit of STATUS (prop 18) changing [srv] (client-internals/feature-specs/04-death-respawn.md#what-it-is).

Event	Server action	Source
dies (bit 0 → 1)	outside a sparring zone, adds a death and drops random gralats between `mindeathgralats` and `maxdeathgralats` as a floor item, plus items when `dropitemsdead` is set; a dead level leader hands leadership to the next player, who gets `PL0_ISLEADER` (10)	[srv] (client-internals/feature-specs/04-death-respawn.md#exact-server-behaviour-tplayerpropscpp-status-case-srv)
revives (bit 1 → 0)	sets hearts by AP (under 20: 3; under 40: 5; else MAXPOWER; at least 0.5), pushes CURPOWER to the player and the level, and re-sends `ISLEADER` to a leader; it does not move the player	[srv] (client-internals/feature-specs/04-death-respawn.md#exact-server-behaviour-tplayerpropscpp-status-case-srv)
`PLI_CLAIMPKER` (14)	ignored for a self-kill or an unknown killer; in a sparring zone both players get a rating update (skipped when they share an IP); elsewhere the killer gets a kill and, with the AP system on, can lose AP	[srv] (client-internals/feature-specs/04-death-respawn.md#exact-server-behaviour-tplayerpropscpp-status-case-srv)
AP	with `apsystem` on, a per-second countdown raises AP by 1 (to 100) and reloads from `aptime0` to `aptime4` by AP band; it does not run while paused or in a sparring zone	[srv] (client-internals/feature-specs/04-death-respawn.md#exact-server-behaviour-tplayerpropscpp-status-case-srv)
hearts	no periodic regeneration; a client CURPOWER above the server's value is refused while AP is under 40	[srv] (client-internals/feature-specs/04-death-respawn.md#exact-server-behaviour-tplayerpropscpp-status-case-srv)
idle	`disconnectifnotmoved` with `maxnomovement` disconnects an idle player with a message	[srv] (client-internals/feature-specs/04-death-respawn.md#exact-server-behaviour-tplayerpropscpp-status-case-srv)

4.5 Items, drops and board respawn

This chapter gives the client's side of lifting, carrying, hiding and throwing, the sword cut, chests, items on the floor, who rolls item drops in each build, and what the client does (and does not do) about board respawn. The `arrays.dat` tables named here are defined in → §3.5.3; the server's respawn timers and drop tables are in → §5.

> Status: draft. Read for 1.3, 1.38 to 1.41 and 2.05 rev 2; lab-checked on 1.39, 2.05 rev 2 and 2.22. Open points are App. E#316 to E#318.

4.5.1 Matching a 2 × 2 object

Rule	Detail	Source
match	take the tile under the probe point; for each table entry in order, for each of its four tiles equal to that tile, the object's top-left is the point minus that tile's column and row; the four board tiles there must equal the entry's four, all on the board; the first full match wins	[2.05] (client-internals/decomp-findings-2026-10-25-l/03-cut-lift-chest.md#matching-a-2-x-2-object)
probe	one tile beyond the middle of the leading edge (→ §4.3.2)	[2.05] (client-internals/decomp-findings-2026-10-25-l/01-walking-body-box.md#other-probes-same-routine-family)
2.04 change	from 2.04 the whole 2 × 2 must match before anything happens; 1.3 matched only the single tile under the probe and wrote the leftover block at the derived corner	[bin][doc] (client-internals/board-respawn.md#sending-the-change)
writing	cut and lift share one routine: it writes the leftover tiles, then, only online, sends `PLI_BOARDMODIFY` (1) with the 2 × 2; nothing is sent if the tiles were already equal	[2.05] (client-internals/board-respawn.md#sending-the-change)

The objects (`arrays.dat`; the same in 1.21, 1.3 test and the 2.22 install, one index lower in 1.21):

Table	#	Object	Tiles (top row / bottom row)	Leaves	Carry sprite	Source
cut	0	bush	2 3 / 18 19	677 678 / 693 694	n/a	[bin] (client-internals/board-respawn.md#the-objects)
cut	1	tall grass	420 421 / 436 437	679 680 / 695 696	n/a	[bin] (client-internals/board-respawn.md#the-objects)
lift	0	bush	2 3 / 18 19	677 678 / 693 694	261	[bin] (client-internals/board-respawn.md#the-objects); [2.05] (client-internals/carry-hide-and-glove.md#glove-power-and-the-lift-table)
lift	1	sign	512 513 / 528 529	1802 1803 / 1818 1819	267	[bin] (client-internals/board-respawn.md#the-objects); [2.05] (client-internals/carry-hide-and-glove.md#glove-power-and-the-lift-table)
lift	2	pot (vase)	684 685 / 700 701	1770 1771 / 1786 1787	265	[bin] (client-internals/board-respawn.md#the-objects); [2.05] (client-internals/carry-hide-and-glove.md#glove-power-and-the-lift-table)
lift	3	light stone	34 35 / 50 51	1834 1835 / 1850 1851	263	[bin] (client-internals/board-respawn.md#the-objects); [2.05] (client-internals/carry-hide-and-glove.md#glove-power-and-the-lift-table)
lift	4	black stone	990 991 / 1006 1007	1834 1835 / 1850 1851	239	[bin] (client-internals/board-respawn.md#the-objects); [2.05] (client-internals/carry-hide-and-glove.md#glove-power-and-the-lift-table)

4.5.2 Lift, carry, hide and throw

Rule	Detail	Source
A key order	throw what is carried; else lift; else grab (→ §4.2.8)	[2.05] (client-internals/ledges-push-pull.md#grabbing-and-pulling)
glove	glove power g allows the first g + 2 lift entries; every player starts at glove 1 (bush, sign, pot); glove1 raises it to at least 2 (light stone), glove2 to at least 3 (black stone); online the power is player prop 6; the cut-and-lift note words the steps differently: disputed → App. E#316	[2.05] (client-internals/carry-hide-and-glove.md#glove-power-and-the-lift-table); [2.05] (client-internals/decomp-findings-2026-10-25-l/03-cut-lift-chest.md#lift-a-key)
NPCs	an NPC is lifted only when its script set canbecarried (command 41, cleared by 42), at the same point, with no glove check; it leaves the level's NPC list and nothing is sent	[2.05] (client-internals/carry-hide-and-glove.md#glove-power-and-the-lift-table)
lift	lift plays at once and the board tile becomes its leftover at once; a counter runs 4 down to 0 and the carried flag is set on the 5th tick; carrystill follows on the 7th tick (6 ticks of lift)	[live][2.05] (client-internals/player-mannerisms-live.md#lifting-carrying-hiding-and-throwing-live)
carry	carry (8 frames, one per tick) at the normal 0.5 tile per tick; the object is drawn over the head, the head and arms hidden behind it	[live][2.05] (client-internals/player-mannerisms-live.md#lifting-carrying-hiding-and-throwing-live)
carried image	the carried bush is the plain bush sprite, not the board's tiles (a bluish board bush is carried green)	[shot] (live-runs/2gta-222-2026-10-30/README.md#lift-hide-under-it-throw)
hide	D while carrying a bush, light stone, black stone or NPC hides the player under it (mode 14); D with a pot or sign does nothing; STATUS gets bit 0x02; the carry sprite prop does not change; only the object is drawn, where the player stands	[2.05] (client-internals/carry-hide-and-glove.md#hiding-mode-14); [live][2.05] (client-internals/player-mannerisms-live.md#lifting-carrying-hiding-and-throwing-live)
hidden walk	hiddenstill standing, hidden walking at 0.1 tile per tick (→ §4.2.1.1); hidden.gani is one frame, so the frame does not advance	[live][2.05] (client-internals/player-mannerisms-live.md#lifting-carrying-hiding-and-throwing-live)
hidden on the wire	hiddenstill is step 24, sprite 18 (as carrystill); hidden steps 25 to 28, sprites 19, 20, 19, 18	[2.05] (client-internals/carry-hide-and-glove.md#hiding-mode-14)
hits while hidden	the hit gate refuses hits on a hidden player unless a player byte that was not traced is set	[2.05] (client-internals/carry-hide-and-glove.md#hiding-mode-14)
leaving	D again, or A, goes back to carrystill in the same tick; out from under a bush (not an NPC) the client starts break effect 0 (the leaves) at (x + 0.5, y + 1)	[2.05] (client-internals/carry-hide-and-glove.md#hiding-mode-14)
throw	A, or S (also from hiding): idle in the same tick, the carried and hidden flags cleared, the carry sprite back to -1	[live][2.05] (client-internals/player-mannerisms-live.md#lifting-carrying-hiding-and-throwing-live); [2.05] (client-internals/decomp-findings-2026-10-28/01-sword-while-carrying.md)
flight	a thrown pot flies about 6 tiles in 9 ticks, its shadow along the ground; it breaks into four pieces that spread over 4 ticks; a landing does not touch the board, it only breaks into leaves (crush.wav, or water.wav for the water leap)	[live][2.05] (client-internals/player-mannerisms-live.md#lifting-carrying-hiding-and-throwing-live); [bin] (client-internals/board-respawn.md#the-objects)
D key while carrying	does not throw; with carry sprite 239, 261 or 263 it hides (above)	[2.05] (client-internals/decomp-findings-2026-10-28/01-sword-while-carrying.md)
stone without glove	A at a black stone with glove 1 does nothing: no lift and no grab	[live][2.05] (client-internals/player-mannerisms-live.md#lifting-carrying-hiding-and-throwing-live)

By build:

Build	Lift	Hide	Source
1.39	counter 4 to 0, carried on the 5th tick; no lift pose; carried bush id 261	hidden flag; walk 0.1 per tick	[live][1.39] (client-internals/player-mannerisms-live.md#cross-version-table)
1.41	`lift` 6 ticks, then `carrystill`	hidden flag, but the pose stays `carrystill` / `carry`, the legs cycling at 0.1 per tick; `hidden.gani` is shipped but not played	[live][1.41] (client-internals/player-mannerisms-live.md#notes-per-build)
2.05 rev 2, 2.22	as above	`hiddenstill` / `hidden`; D toggles	[live][2.05][2.22] (client-internals/player-mannerisms-live.md#cross-version-table); [shot] (live-runs/2gta-222-2026-10-30/README.md#lift-hide-under-it-throw)
5.24r4, 5.30r3, 6.03r7	`lift` and the board change in the key's tick, `carrystill` 0.30 s later; a throw sends id 11 and `idle` in the same tick	hidden at about 1.5 px per tick; D with a pot does nothing	[wire] (protocol/v5v6-lab-play.md)

4.5.3 Sword cut

Rule	Detail	Source
when	at the start of a swing, at the probe point, against the whole cut table with no glove	[2.05] (client-internals/decomp-findings-2026-10-25-l/03-cut-lift-chest.md#sword-cut)
effect	the four tiles become the entry's leftover tiles (array 10) through the shared write routine, then the leaves effect; the leaves fly up and out for about 8 ticks	[2.05] (client-internals/decomp-findings-2026-10-25-l/03-cut-lift-chest.md#sword-cut); [live][2.05] (client-internals/player-mannerisms-live.md#sword-live)
bombs	"bombs don't destroy half bushes" (2.05)	[doc] (client-internals/board-respawn.md#sending-the-change)
bomb blast, 1.x	the blast's four test points cut cut-entry 0 and destroy lift entries 1 to 3 (sign, pot, stone)	[bin] (client-internals/item-drops.md#the-1x-roll-bin)

4.5.4 Chests

Rule	Detail	Source
look	a chest record has no draw code in 2.05: a chest looks like its board tiles; the record adds the open action and a solid 2 × 2 footprint, opened or not	[2.05] (client-internals/decomp-findings-2026-10-25-l/03-cut-lift-chest.md#chests)
2.22 drawing	the full (closed) form of `PLO_LEVELCHEST` (4) is drawn as a closed chest and the short form as an open one, each a 2 × 2 block over the board with no board tile needed	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#part-1-fresh-runs-2026-10-25-22502310-jst-a-chests-real-222-client-labm-lab-plo_levelchest-id-4-injected)
solid	a closed chest stops the player when the body probes touch its 2 × 2 box; pushing against it plays `push`	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#part-1-fresh-runs-2026-10-25-22502310-jst-a-chests-real-222-client-labm-lab-plo_levelchest-id-4-injected)
opening	only facing up, from below, with the action key; facing down, left or right nothing is sent	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#part-1-fresh-runs-2026-10-25-22502310-jst-a-chests-real-222-client-labm-lab-plo_levelchest-id-4-injected)

Rule	Detail	Source
on opening	at once: asks for `chest.wav`, sends `PLI_OPENCHEST` (20) with the chest's top-left in whole tiles, draws the chest open locally without waiting; about 1.2 s later sends message 13 with the point one tile above the chest	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#part-1-fresh-runs-2026-10-25-22502310-jst-a-chests-real-222-client-labm-lab-plo_levelchest-id-4-injected)
sound	`chest.wav`, also when the chest has a sign; no item sound follows	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
record bytes	each + 32: opened flag, x, y, and only if not opened the item index and the sign index (→ §3.1)	[2.05] (client-internals/decomp-findings-2026-10-25-l/03-cut-lift-chest.md#chests)
the reward	what the server sends back for a known chest is unmeasured	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#part-1-fresh-runs-2026-10-25-22502310-jst-a-chests-real-222-client-labm-lab-plo_levelchest-id-4-injected)
2.05 opening sides	unmeasured	[lab] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#part-1-fresh-runs-2026-10-25-22502310-jst-a-chests-real-222-client-labm-lab-plo_levelchest-id-4-injected)

4.5.5 Item drops: who rolls

Builds	Who drops	What the client sends	Source
4th Graal alpha to 1.37	the client rolls	the roll as UDP 19 to the other players only; a pickup is applied locally, with UDP 20 and `PLI_ITEMDEL` (13); a timeout sends nothing	[bin] (client-internals/item-drops.md#summary)
1.38 to 1.41	the client rolls	the roll as UDP 19 and `PLI_ITEMADD` (12); online a pickup sends `PLI_ITEMTAKE` (32) and the server gives the item with props; a timeout sends `PLI_ITEMDEL` (13) (1.39 read and captured; 1.41 read for the roll only)	[bin][cap] (client-internals/item-drops.md#summary)
NewMain 1.42 to 2.31	the server (`PLO_ITEMADD`, 22); offline nothing drops	pickup `PLI_ITEMTAKE` (32); timeout `PLI_ITEMDEL` (13)	[bin][doc] (client-internals/item-drops.md#summary)

Rule	Detail	Source
the change	at 1.42: "bush rupees and pot hearts are server-side"; the roll is gone from NewMain 24 and 26 and still in 1.41	[doc][bin] (client-internals/item-drops.md#summary)
arrows and stones	no client ever rolls arrows, and no client drops anything for a stone	[bin] (client-internals/item-drops.md#summary)
2.05 offline	the cut routine makes only the leaves and the lift routine only the carried object, so an offline 2.05 bush or pot gives nothing	[bin] (client-internals/board-respawn.md#item-drops-from-142-on-server-side)
server drop timing	on a 2.22 server with pot drops on, the heart arrived in the same frame as the lift, at the pot's spot, and stayed when the pot was thrown	[cap] (live-runs/2gta-222-2026-10-30/README.md#pots)

4.5.5.1 The 1.x roll

One `Random(100)` per drop, with the Delphi generator `seed = seed × 134775813 + 1`, result `(seed × range) >> 32`:

Roll	Item	Source
0 to 24	green rupee (25 %)	[bin] (client-internals/item-drops.md#the-1x-roll-bin)
25, 26	blue rupee (2 %)	[bin] (client-internals/item-drops.md#the-1x-roll-bin)
27 to 31	heart (5 %)	[bin] (client-internals/item-drops.md#the-1x-roll-bin)

Roll	Item	Source
32 to 36	bombs (5 %)	[bin] (client-internals/item-drops.md#the-1x-roll-bin)
37 to 99	nothing (63 %)	[bin] (client-internals/item-drops.md#the-1x-roll-bin)

Rule	Detail	Source
same table	in every build from 4th Graal alpha to 1.41; from 1.38 the item is picked by its number in the item table (0, 1, 5, 3) instead of by name	[bin] (client-internals/item-drops.md#the-1x-roll-bin)
what rolls	every cut entry (tall grass too); a lifted bush (lift entry 0); a bush destroyed by a blast; a pot (lift entry 2) always leaves a heart, lifted or blasted; sign and stones nothing	[bin] (client-internals/item-drops.md#the-1x-roll-bin)
when	at the lift, not when the thrown pot breaks	[bin] (client-internals/board-respawn.md#item-drops-in-13-client-side)
where	at the object's top-left tile	[cap] (client-internals/item-drops.md#the-1x-roll-bin)
new item	life 164 and the announce flag	[bin] (client-internals/item-drops.md#the-1x-roll-bin)
1.3 to peers	UDP id 19, `chr(x·2+32) chr(y·2+32) chr(item+32)`, sent straight to each peer, only for item kinds marked in a per-item table; the board change still goes to the server over TCP	[bin] (client-internals/board-respawn.md#item-drops-in-13-client-side)

4.5.6 Items on the floor (2.05 rev 2)

Rule	Detail	Source
life	from 164, down 1 per tick; below 0 the item goes and, online, the client sends `PLI_ITEMDEL`; lab: 8.20 s from the drop	[bin][cap] (client-internals/item-drops.md#items-on-the-floor-205-rev-2-bin)
pickup	the item is not permanent (life below 100 at creation, such as a chest's display item with life 24), its life is below 160 (not in its first four ticks), and `	px + 1.5 - (ix + 1)
event	at life 140 `playerlaysitem` fires	[bin] (client-internals/item-drops.md#items-on-the-floor-205-rev-2-bin)
blink	not drawn while the life is below 37 and odd	[bin] (client-internals/item-drops.md#items-on-the-floor-205-rev-2-bin)
bounce	raised by `	sin(life / 3)
draw	the shadow, sprite 42, at (x + 0.5, y + 1 + h / 2 - 0.5) for an item h tiles high; the item centred on (x + 1, y + 1)	[bin] (client-internals/item-drops.md#items-on-the-floor-205-rev-2-bin)
sprites by item number	136, 137, 138, 67, 143, 139, 145, 351, 65, 350, 314, 315, 321, 323, 322, 324, 325, 328, 329, 330, 331, 332, 333, 334, 349	[bin] (client-internals/item-drops.md#items-on-the-floor-205-rev-2-bin)
floor items	`PLO_ITEMADD` makes an item only for the announce-flag items: 0 to 5, 7, 8, 19 and 24 (rupees, bombs, darts, heart, bow, bomb, gold rupee, spin attack)	[bin] (client-internals/item-drops.md#items-on-the-floor-205-rev-2-bin)
offline value	green 1, blue 5, red 30, gold 100 rupees (up to 9,999,999); bombs and darts 5 each (up to 99); a heart one whole heart; sound `extra.wav`	[bin] (client-internals/item-drops.md#items-on-the-floor-205-rev-2-bin)
online pickup	`PLI_ITEMTAKE` (32) with the item's three bytes; the server answers with props (for example RUPEESCOUNT)	[cap] (client-internals/item-drops.md#lab-runs-livecap)
1.3	the pickup test is an overlap of pixel boxes, and a timeout is silent	[bin] (client-internals/item-drops.md#items-on-the-floor-205-rev-2-bin)
1.x pickup box in tiles, draw order against players	unmeasured (→ App. E#317)	[bin] (client-internals/item-drops.md#open)

4.5.7 Board respawn on the client

Rule	Detail	Source
no client regrowth	no client regrows a cut or lifted object; the 1.3 and 2.05 board-modify record is built to send one change and has no timer; an offline 2.05 board polled for 633 s after a lift never changed back	[bin][live] (client-internals/board-respawn.md#no-client-side-respawn)
restore	a restore arrives as `PLO_BOARDMODIFY` (7) and is handled as any board modify; the timers and the echo rules are the server's (→ §5)	[bin] (client-internals/board-respawn.md#summary)
where	the restore puts the object back where it was lifted, since the board changed only at the lift	[bin] (client-internals/board-respawn.md#the-objects)
push and pull blocks	the 2.05 client sends the 4 × 4 area round the old spot (→ §4.2.7)	[2.05] (client-internals/ledges-push-pull.md#pushing)
re-entry offline	whether 2.05 reloads a level from disk on re-entry, which would bring a cut bush back, is unmeasured (→ App. E#318)	[bin] (client-internals/board-respawn.md#open)

4.6 HUD, sign panel, chat text and options

This chapter covers what the client draws over the game view: the status display, the sign panel, chat text and nicknames over players, and the item list window, plus what the in-game options change while playing. The file formats behind them are in → §3.6 (sign text), §3.9 (config) and §3.10 (sheets).

> Status: draft. The HUD, sign panel and item list are read from 2.05 rev 2; chat text is read from every Delphi build and checked live on 2.05 rev 2. The player list window is in → §4.10.

4.6.1 The status display

4.6.1.1 Sheet

Rule	Detail	Source
sheet	every build from 1.41 on ships a 224 × 145 `state.png`	[file] (client-internals/status-display.md#sheets)
2.22 against 2.05	the 2.22 sheet matches the 2.05 rev 2 and 2.1 test 3 sheets pixel for pixel except one 27 × 14 patch at (190, 60), which the status routine does not use	[file] (client-internals/status-display.md#sheets)
buttons	1.41's sheet has three blank buttons at x 158, 180 and 202; the 2.0-era NewMain sheets have one, at x 202, which the routine draws in all three slots	[file] (client-internals/status-display.md#sheets)

4.6.1.2 Layout (2.05 rev 2)

Positions are from the top-left of the game view; the routine's base point is (15, 15). Each part is drawn only when its `showstats` bit is set:

Part	`showstats` bit	Sheet rect (x, y, w, h)	Screen position	Source
rupee, bomb and arrow icons	0x02	(80, 33, 120, 16), one strip	(274, 30)	[2.05] (client-internals/status-display.md#layout)
rupee count	0x04	digit d at (14d, 81, 14, 16)	1 to 7 digits, no leading zeros, 12 px apart, centred on x 280, y 48	[2.05] (client-internals/status-display.md#layout)
bomb count	0x08	same digits	two digits at x 326 and 338, y 48	[2.05] (client-internals/status-display.md#layout)
arrow count	0x10	same digits	two digits at x 374 and 386, y 48	[2.05] (client-internals/status-display.md#layout)
hearts	0x20	big (40·half, 0, 40, 44); small (80 + 31·half, 0, 31, 33)	→ §4.6.1.3	[2.05] (client-internals/status-display.md#layout)
buttons	0x01	(202, 0, 22, 33)	x 15, 80 and 145, y 30	[2.05] (client-internals/status-display.md#layout)

Part	`showstats` bit	Sheet rect (x, y, w, h)	Screen position	Source
key names	0x01	text	on each button, 26 pt Helvetica, white with a black copy at +1, +1; smaller for longer key names; the item-list key drawn small at (147, 27)	[2.05] (client-internals/status-display.md#layout)
glove icon	0x01	sprite 145 for power 2, sprite 325 on top for power 3	(40, 29); the power printed at (60, 31) when 2 or more	[2.05] (client-internals/status-display.md#layout)
sword icon	0x01	sprite 17, from the player's sword image	(105, 31); the sword power printed at (125, 31)	[2.05] (client-internals/status-display.md#layout)
selected weapon	0x01	the weapon's whole image	x 170, centred on y 45	[2.05] (client-internals/status-display.md#layout)
alignment bar	0x40	frame (0, 97, 130, 22); fill (0, 119, AP, 10)	(15, 65); fill at +22, +6	[2.05] (client-internals/status-display.md#layout)
magic bar	0x80	frame (0, 49, 130, 22); fill (0, 71, MP, 10)	(15, 90); only with weapons on and MP above 0	[2.05] (client-internals/status-display.md#layout)

Rule	Detail	Source
other bits	0x100 and 0x200 belong to the player icons and the stacked weapon image list, drawn by the same routine but not part of the status display	[2.05] (client-internals/status-display.md#layout)
sprite rects	sprites 17, 145 and 325 come from `arrays.dat` array 24 (→ §3.5): 17 is (94, 36, 28, 28) of the sword image; 145 is (28, 212, 28, 32) and 325 is (0, 212, 28, 32) of `sprites.png`	[file] (client-internals/status-display.md#layout)
weapons flag	with weapons off only the first button, its key name and the alignment bar are drawn; the same flag gates the S and D keys	[2.05] (client-internals/status-display.md#layout)

4.6.1.3 Hearts

Rule	Detail	Source
which	one heart per container while power remains; empty containers are not drawn; heart i is full when power $- 1 \geq i$ and half when $i <$ power	[2.05] (client-internals/status-display.md#hearts)
big hearts	while power is 4 hearts or less; from (W − 198, 24), 45 px apart, W the view width	[2.05] (client-internals/status-display.md#hearts)
small hearts	above 4 hearts, 35 px apart; new rows start at hearts 5, 11, 16, 22 and 27; row $k + 1$ is $20(k + 1)$ px lower and shifted $\text{round}(17.5(k + 1)) - 35$ px sideways; the block moves 10 px down while power is below 6	[2.05] (client-internals/status-display.md#hearts)

4.6.1.4 The selected weapon

Rule	Detail	Source
one list	server weapons (`PLO_NPCWEAPONADD`) and built-in weapons (`PLO_DEFAULTWEAPON`) share one list in arrival order; a built-in weapon is skipped when the list already has its name (→ §2.5)	[2.05] (client-internals/status-display.md#the-selected-weapon-205-rev-2)
adding	every add selects the new last entry	[2.05] (client-internals/status-display.md#the-selected-weapon-205-rev-2)

Rule	Detail	Source
selecting index *i*	the first entry from *i* on, wrapping round, whose name is not empty and does not start with `-`; if every entry is hidden, *i* stays selected	[2.05] (client-internals/status-display.md#the-selected-weapon-205-rev-2)
deleting	the selection keeps its index if it still exists, otherwise it moves to the new last entry	[2.05] (client-internals/status-display.md#the-selected-weapon-205-rev-2)
built-in weapons	of the 25-entry item table, only ids 7 bow, 8 bomb, 12 superbomb, 20 fireball, 21 fireblast, 22 nukeshot and 23 joltbomb are weapons; the name is the item name with a capital first letter	[2.05] (client-internals/status-display.md#built-in-weapons)
images	`wbowi1` to `wbowi4` and `wbomb1` to `wbomb3` (`.gif` in 2.05); 2.22 asked for the same names as `.png`	[2.05][lab][2.22] (client-internals/status-display.md#built-in-weapons)

4.6.1.5 Not measured

Item	State	Source
2.22 and 2.31 status routines	unmeasured; their sheets and a 2.22 screenshot match 2.05	[cap] (client-internals/status-display.md#not-read)
builds before 2.05, including 1.41's three button cells	unmeasured	[doc] (client-internals/status-display.md#not-read)
font, colour and anchor of the glove and sword power text	unmeasured	[doc] (client-internals/status-display.md#not-read)

4.6.2 The sign panel (2.05)

4.6.2.1 Drawing

Rule	Detail	Source
sheet	`letters.png`, 256 × 224, 16 glyphs a row, each 16 × 32; glyph g at (g mod 16 × 16, g div 16 × 32); loaded once at start; the glyph order is the sign alphabet of → §3.6	[2.05] (client-internals/sign-panel.md#the-sheet)
advances	each glyph has its own advance, twice the client's half-width table (I 6 px, space 8, most letters 12)	[2.05] (client-internals/sign-panel.md#the-sheet)
frame glyphs	96 to 99 corners (top-left, top-right, bottom-left, bottom-right), 100 and 101 top and bottom edge, 102 and 103 left and right edge	[2.05] (client-internals/sign-panel.md#the-sheet)
button glyphs	91 to 94 the A, B, X and Y buttons; 77 to 80 up, down, left, right; 71 to 75, 87, 88 and 67 the other button pictures	[2.05] (client-internals/sign-panel.md#the-sheet)
fill colours	the pixels at (254, 223) and (255, 223) are the outer fill and, 8 px further in, the inner fill; a fill in the sheet's transparent colour is skipped; cream and cream on the 2.22 sheet, inner white on 1.39	[2.05][file] (client-internals/sign-panel.md#the-sheet)
size	24 cells wide and 9 tall (16 px cells), centred across the view	[2.05] (client-internals/sign-panel.md#the-panel)
vertical place	640-wide view: y 256, or y 128 when the player is more than 14 tiles down the view; 800: 18 tiles, y 320 or 160; 1024: 22 tiles, y 448 or 192; which two values are compared is inferred	[2.05][inf] (client-internals/sign-panel.md#the-panel)
frame order	top and bottom edges every 16 px, sides every 32 px, corners over them, then the two fills from (x + 16, y + 24)	[2.05] (client-internals/sign-panel.md#the-panel)
text	three lines a page from (x + 16, y + 16), 32 px apart; the text is split at newlines	[2.05] (client-internals/sign-panel.md#the-panel)

Rule	Detail	Source
# codes	`#A #B #X #Y #u #d #l #r #h #x #y #z #4 #.` draw button pictures; `#` and any other letter draws glyph 86 and moves 16 px; `#i` and a character shows an image from the level's list	[2.05] (client-internals/sign-panel.md#the-panel)
other characters	a character outside the glyph order is drawn in the system font, in the colour at (253, 223)	[2.05] (client-internals/sign-panel.md#the-panel)
check	not compared against a screenshot of a real client	[doc] (client-internals/sign-panel.md#confirmed-inferred-not-tested)

4.6.2.2 Keys while reading

Rule	Detail	Source
own state	reading is its own input state: the player does not walk, swing, lift or use an item; switch-to-chat still works, and while the chat bar has focus the sign keys are skipped	[2.05] (client-internals/sign-panel.md#keys-while-reading)
map key	ends reading at once	[2.05] (client-internals/sign-panel.md#keys-while-reading)
up	turns back three lines, unless on the first page	[2.05] (client-internals/sign-panel.md#keys-while-reading)
other game keys	left, down, right, D, S, A, item list and pause turn to the next three lines; when the page on show is not full, reading ends instead, so a sign of exactly three lines shows an empty second page before it closes	[2.05] (client-internals/sign-panel.md#keys-while-reading)
edge	a key counts only when no game key was down on the tick before; a held key does not repeat (→ §4.1.2.1)	[2.05] (client-internals/sign-panel.md#keys-while-reading)
side effects	`nextpage.wav` on each page turn (→ §4.7.3); `playerendsreading` runs when reading ends	[2.05][live] (client-internals/sign-panel.md#keys-while-reading); [live] (client-internals/live-checks-editor-offline-sound.md#sound-205)

4.6.2.3 NPCs with no image

Rule	Detail	Source
in game	an NPC with no image draws nothing of its own, only its text if any	[2.05] (client-internals/sign-panel.md#npcs-with-no-image)
gani and character NPCs	drawn from the gani, or the way players are drawn	[2.05] (client-internals/sign-panel.md#npcs-with-no-image)
placeholder	the placeholder picture appears only in the level editor's view (→ §4.9)	[2.05] (client-internals/sign-panel.md#npcs-with-no-image)

4.6.3 Chat text and nicknames

The chat line is player prop 12 (→ §2.6.2); the chat and PM messages are in → §2.8.

4.6.3.1 Font

Builds	Face	Height (px, cell)	Shadow	Source
1st Graal, 4th, 9th, 15th, 1.1.9, 1.21	`Dialog`	28	always blue	[bin] (client-internals/chat-text.md#font-per-build)
1.31, 1.38	`Dialog`	28	contrast rule	[bin] (client-internals/chat-text.md#font-per-build)
1.39, 1.41	`Helvetica`	24	contrast rule	[bin] (client-internals/chat-text.md#font-per-build)
NewMain 1.42, 2.00 to 2.05 rev 2, 2.17, 2.22, 2.31	`Tempus Sans ITC`, installed from the client folder	24	contrast rule	[bin] (client-internals/chat-text.md#font-per-build)

Rule	Detail	Source
drawing	straight onto the game view with the Windows canvas, bold, no bubble and no box	[bin][live][2.05] (client-internals/chat-text.md)
config	from 1.31 on, `font=` and `fontsize=` replace the defaults (→ §3.9.3)	[bin] (client-internals/chat-text.md#font-per-build)
height	the value is the cell height in pixels, not a point size; for Tempus Sans ITC a 24 px cell is an em of about 18.4 px, which the live capture matches	[bin][live][2.05] (client-internals/chat-text.md#font-per-build)
missing faces	`Dialog` does not exist on Windows and is mapped to a sans serif, normally Arial; `Helvetica` maps to Arial; neither was checked on real Windows	[inf] (client-internals/chat-text.md#font-per-build)
bold	Tempus Sans ITC has only a regular face, so Windows emboldens it: each glyph smeared one pixel right, each advance one pixel wider	[inf] (client-internals/chat-text.md#font-per-build)

4.6.3.2 Colour and position

Rule	Detail	Source
colour	chat is white	[bin] (client-internals/chat-text.md#colour-and-shadow)
passes	shadow at (+1, +1), then text at (0, 0)	[bin] (client-internals/chat-text.md#colour-and-shadow)
shadow, 1.21 and earlier	always RGB (0, 0, 255)	[bin] (client-internals/chat-text.md#colour-and-shadow)
shadow, 1.31 on	red and green 0 if the text's red or green is above 0xE0, otherwise 0xFF; blue equals the text's blue; the same routine draws nicknames and number pop-ups	[bin] (client-internals/chat-text.md#colour-and-shadow)
x	centred on the player's x + 1.5 tiles	[bin] (client-internals/chat-text.md#position)
y	the top of the text cell at the player's y − 1 tile, or y − 2 while carrying (the carry part read, not checked live)	[bin][live][2.05] (client-internals/chat-text.md#position)
clip	to the level view rectangle	[bin] (client-internals/chat-text.md#position)

4.6.3.3 Wrapping

Builds	Rule	Source
1.21 and earlier	one centred line, clipped, no wrapping	[bin] (client-internals/chat-text.md#wrapping-131-and-later)
1.31 and later	text wider than the view (640 px in 2.05) is split at spaces; a word that does not fit starts the next line, and a word too long for a line stands alone, unbroken	[bin] (client-internals/chat-text.md#wrapping-131-and-later)
1.31 and later	lines stack upward so the last line sits on the anchor; each line is centred on its own, then pushed left to fit the right edge and right to start at 0 or more	[bin] (client-internals/chat-text.md#wrapping-131-and-later)
6.x	wraps only text of 11 or more characters, breaking after any of `.,;+-?!`	[doc] (client-internals/chat-text.md#wrapping-131-and-later)

4.6.3.4 How long chat stays (2.05 rev 2)

Rule	Detail	Source
own line	stays while the player stands still (over 7 minutes seen); the first direction key starts a 60-tick (3 s) countdown, after which the line goes and the client sends an empty chat prop	[live][2.05] (client-internals/chat-text.md#how-long-it-stays-live-205-rev-2-2026-10-30)
another player's line	shows for at least 3 s from arrival and until that player's empty chat arrives; an empty line only waits out what is left of the 3 s, it does not start a fresh 3 s; with no empty line the text stays	[live][2.05] (client-internals/chat-text.md#how-long-it-stays-live-205-rev-2-2026-10-30)
length	a line longer than 256 characters is cut to 253 characters plus `...` when sent; the receiver draws what arrives	[bin] (client-internals/chat-text.md#length)
showap	a chat of exactly showap shows the player's AP instead of the text	[2.05] (client-internals/chat-text.md#special-texts-205)
other uses of the slot	while a per-player counter runs it shows two numbers joined by /; in some sprite states it shows an item image or a carried number; the chat is then not drawn; these branches were read but not named	[2.05] (client-internals/chat-text.md#special-texts-205)

4.6.3.5 Nicknames (1.31 to 2.31)

Rule	Detail	Source
position	drawn by the chat routine, centred on x + 1.5 tiles, the top of the text cell at y + 2.5 tiles	[bin][live][2.05] (client-internals/chat-text.md#nickname-131-to-231)
other players	always shown	[bin] (client-internals/chat-text.md#nickname-131-to-231)
own name	shown for 60 ticks (3 s) after setup (so at login), after an AP change, and after typing exactly showname, which is not sent	[bin][live][2.05] (client-internals/chat-text.md#nickname-131-to-231)
length	cut to `nicknamelimit` (default 50) only when `limitnicknames` is on (off by default)	[2.05] (client-internals/chat-text.md#nickname-131-to-231)
colour	follows AP (prop 32, → §2.6.2); 1.21 has no AP colours; the same ramp is in every build from 1.31 to 2.31	[bin] (client-internals/chat-text.md#nickname-131-to-231)

The AP colour ramp, each step `floor(n × k / d)` with a 0.0001 bias:

AP	Colour (R, G, B)	Source
0	(64, 0, 0)	[bin] (client-internals/chat-text.md#nickname-131-to-231)
1 to 26	(128 + (ap − 1)·127/25, 64 − (ap − 1)·64/25, same as G), dark red to red	[bin] (client-internals/chat-text.md#nickname-131-to-231)
27 to 49	(255, (ap − 26)·255/24, same as G), red to pale pink	[bin] (client-internals/chat-text.md#nickname-131-to-231)
50 to 55	white	[bin] (client-internals/chat-text.md#nickname-131-to-231)
56 to 60	(255 − (ap − 55)·127/5, 255, same as R), white to light green	[bin] (client-internals/chat-text.md#nickname-131-to-231)
61 to 70	(128 − (ap − 61)·128/10, 255, 128 + (ap − 61)·127/10), light green to cyan	[bin] (client-internals/chat-text.md#nickname-131-to-231)

AP	Colour (R, G, B)	Source
71 to 99	(0, 255 − (ap − 70)·255/29, 255), cyan to blue	[bin] (client-internals/chat-text.md#nickname-131-to-231)
100	(224, 196, 0), gold	[bin] (client-internals/chat-text.md#nickname-131-to-231)
live check	AP 0, 20, 40, 58, 65, 85 and 100 drew exactly (64,0,0), (224,16,16), (255,148,148), (179,255,179), (77,255,178), (0,124,255) and (224,196,0); AP 50, the value before any AP is sent, was white	[live][2.05] (client-internals/chat-text.md#nickname-131-to-231)

4.6.4 The item list (2.05)

Rule	Detail	Source
key	the `itemlist` action (index 9; default Q) opens and closes the list; it opens from the normal state only when a player flag allows it, flag not identified	[2.05] (client-internals/decomp-findings-2026-10-28/02-item-list.md)
content	the player's built-in and NPC weapons; a / in a weapon name makes a folder, and the current level's folders come first as 32 × 32 cells; names starting with - are skipped when selecting	[2.05] (client-internals/decomp-findings-2026-10-28/02-item-list.md)
window	framed, from (16, 80) to 16 px from the right and bottom edges; cells from (48, 112) to 48 px from the right and 64 px from the bottom	[2.05] (client-internals/decomp-findings-2026-10-28/02-item-list.md)
cells	max(image width, label width) + 10 wide, image height + label height + 15 tall, packed in the first free place left to right then down; image centred 5 px from the top, label centred below; a weapon with no image gets a 32 × 32 placeholder; the selected cell is filled RGB(255, 212, 128); font size 16; scroll arrows when cells are cut off	[2.05] (client-internals/decomp-findings-2026-10-28/02-item-list.md)
arrows	stretch the selected cell to the edge in that direction and take the overlapping cell with the closest near edge; moving selects the weapon at once and sends nothing	[2.05] (client-internals/decomp-findings-2026-10-28/02-item-list.md)
other keys	item key or Enter opens a folder, the attack key goes up a folder, other game keys and Esc close the list, a click selects a cell	[2.05] (client-internals/decomp-findings-2026-10-28/02-item-list.md)
Delete	removes the selected weapon; online it sends `PLI_NPCWEAPONDEL` (29) with the name, or `PLI_NPCDEL` (22) with the NPC id for an NPC that is not a weapon (→ §2.4)	[2.05] (client-internals/decomp-findings-2026-10-28/02-item-list.md)

4.6.5 Options at run time (2.05)

The file, its keys and its parse rules are in → §3.9; this table gives what each option does while playing.

Options	Effect	Source
`musicon`, `soundfxon`, `reversestereo`, `usewinamp`	sound and music (→ §4.7)	[2.05] (client-internals/feature-specs/09-options.md#what-the-options-change-at-run-time)

Options	Effect	Source
`noudp`, `fixedport`, `fixedudpport`	the UDP channel (→ §2.9)	[2.05] (client-internals/feature-specs/09-options.md#what-the-options-change-at-run-time)
`nomassmessages`, `notoalls`, `nohtmlinpms`, `nohtmlimages`, `sortplayersbyname`, `dontloadlistheads`	the player list and messages (→ §4.10)	[2.05] (client-internals/feature-specs/09-options.md#what-the-options-change-at-run-time)
`dontconnectlevels`	stops loading the neighbour levels of a gmap ahead of time	[2.05] (client-internals/feature-specs/09-options.md#what-the-options-change-at-run-time)
`limitnicknames`, `nicknamelimit`	cut drawn nicknames (→ §4.6.3.5)	[2.05] (client-internals/feature-specs/09-options.md#what-the-options-change-at-run-time)
network	none of the options is sent to the server	[2.05] (client-internals/feature-specs/09-options.md#what-it-is)
dialogs	More Options, Player List Options and the Controls frame; closing either dialog rewrites the whole file	[2.05] (client-internals/feature-specs/09-options.md#what-it-is)
later builds	2.17 and later add volume keys and an Audio/Microphone tab (→ §3.10.4)	[2.17][2.22] (client-internals/feature-specs/09-options.md#keys-file-order-default)

4.7 Sound

This chapter gives how the client places, limits and triggers sounds while playing. The backends, the options and the file types are in → §3.10.3 and §3.10.4.

> Status: draft. Read from 2.05 rev 2 and checked in a running 2.05 rev 2 client with an audio device; the 2.17 to 2.31 BASS clients are read only as far as → §3.10.4 goes.

4.7.1 Placing a sound (2.05)

Every sound goes through one entry point, called from about two dozen places: GS1 commands, gani `PLAYSOUND`, pickups, bombs, swords and baddies (→ §3.10.3 for the loop prefix, missing files and dispatch):

Rule	Detail	Source
arguments	the file, the level offset x and y, and the x and y of the sound in tiles	[2.05] (client-internals/sound-and-music.md#one-entry-point-for-every-sound)
listener	the player at (x + 1.5, y + 2)	[2.05] (client-internals/sound-and-music.md#one-entry-point-for-every-sound)
gmap	a sound from a neighbouring gmap level is first shifted by ±64 tiles (→ §3.2)	[2.05] (client-internals/sound-and-music.md#one-entry-point-for-every-sound)
pan	(sound x − listener x) × 150, clamped to ±10000 DirectSound pan units; `reversestereo` negates it	[2.05] (client-internals/sound-and-music.md#one-entry-point-for-every-sound)
volume	a negative attenuation growing with the distance on both axes, clamped at −10000 (hundredths of a dB)	[2.05] (client-internals/sound-and-music.md#one-entry-point-for-every-sound)
looped effects	kept in the player's looped-sound list; leaving a level or entering the editor stops every looped effect in it	[2.05] (client-internals/sound-and-music.md#effect-path-wav)
voices	at most 8 effects play at once; a ninth request is dropped	[2.05] (client-internals/sound-and-music.md#effect-path-wav)
reuse	a cached effect is replayed from its buffer with the new pan, volume and loop settings	[2.05] (client-internals/sound-and-music.md#effect-path-wav)
after a missing file arrives	what 2.05 does then is unmeasured	[doc] (client-internals/sound-and-music.md#open)

4.7.2 GS1 sound commands (2.05)

The commands' syntax is in → §8.2; their effect on the sound system:

Command	Effect	Source
`play file`	one-shot at the NPC's position, its x and y plus half its image size	[2.05] (client-internals/sound-and-music.md#gs1-sound-commands-in-205)
`playlooped file`	the same, with the loop prefix	[2.05] (client-internals/sound-and-music.md#gs1-sound-commands-in-205)
`stopsound file`	stops a cached effect found by file name; the path is ignored	[2.05] (client-internals/sound-and-music.md#gs1-sound-commands-in-205)
`stopmidi`	stops the current music, MCI or Winamp	[2.05] (client-internals/sound-and-music.md#gs1-sound-commands-in-205)

Command	Effect	Source
`setmusicvolume l,r`	MIDI-out volume (MCI) or Winamp volume and pan, each 0 to 1	[2.05] (client-internals/sound-and-music.md#gs1-sound-commands-in-205); [2.05] (client-internals/feature-specs/01-sound.md#exact-client-behaviour-from-client-internalssound-and-musicmd-205)
2.17 and later	the same commands plus `musicpos` and `musiclen`	[2.17][2.22] (client-internals/feature-specs/01-sound.md#exact-client-behaviour-from-client-internalssound-and-musicmd-205)

4.7.3 Which action plays which sound (2.05)

Matched by the client's lookup of each file under `sounds\` while the action ran:

Action	Sound	Source
walking on the ground	`steps.wav` when a move starts, then `steps2.wav` about every 0.2 s	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
swimming	`water.wav` about every 0.8 s (→ §4.2.10)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
riding a horse	`horse.wav` and `horse2.wav`, alternating about every 0.4 s	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
sword swing	`sword.wav`; with the battle axe, `axe.wav` (→ §4.3)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
lifting	`lift.wav` or `lift2.wav` by the gender byte (→ §4.7.4)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
getting hurt	the same `lift.wav` or `lift2.wav`, from `hurt`, an explosion, a thrown ball or a baddy hit	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
dying	`dead.wav` (→ §4.4)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
laying a bomb	`put.wav`	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
bomb or explosion	`bomb.wav`: `putbomb` of any power, `putexplosion`, `putexplosion2` power 2, the end of a fireblast or nuke	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
jolt explosion	`thunder.wav` (`putexplosion2` power 3)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
launching a projectile	`arrow.wav` for arrows, fireballs, fireblasts, nukes and `shootball`, and for arrows from shooting baddies and the dragon	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
fire spy	`fire.wav` (`spyfire`)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
a thrown object or a leap landing	`crush.wav`; `water.wav` for the water leap (type 5)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
picking up a laid item	`extra.wav` (→ §4.5)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
opening a chest	`chest.wav`, also when the chest has a sign; no item sound follows	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
turning a sign page	`nextpage.wav` (→ §4.6.2.2)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)

Action	Sound	Source
pushing a stone block	`stonemove.wav` about once a second while it moves (→ §4.2.7)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
ledge jump	`jump.wav` (→ §4.2.6)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
frog	`frog.wav` when it hops	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
baddy walking	`steps.wav`	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
baddy dies	`compudead.wav`	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
baddy draws its sword	`swordon.wav` at its call site; six baddy types tried, none played it	[2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
an arrow hitting the player	`arrowon.wav` at its call site; not reached in the lab	[2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
`item.wav`, `item2.wav`	used only in the sign-reading routine and a server-message handler; the chest and laid-item tests did not play them	[2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
another player's bomb or item, online	`bomb.wav` or `thunder.wav`, `item.wav`, `extra.wav`, from server messages	[2.05][inf] (client-internals/live-checks-editor-offline-sound.md#sound-205)
gani `PLAYSOUND`, GS1 `play` and `playlooped`	the named file	[2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)

4.7.4 lift.wav and lift2.wav

Rule	Detail	Source
choice	one byte on the player object picks the lift and hurt sound: set gives `lift.wav`, clear gives `lift2.wav`	[2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
default	new players start with the byte set	[2.05] (client-internals/live-checks-editor-offline-sound.md#sound-205)
`setgender female`	the sound changed to `lift2.wav`; that the command clears the byte is inferred from that	[live][inf] (client-internals/live-checks-editor-offline-sound.md#sound-205)
correction	an earlier note took `lift2.wav` for heavy objects and `fire.wav` for the fireball; both were wrong	[live] (client-internals/live-checks-editor-offline-sound.md#corrections-to-earlier-notes)

4.8 Offline and training mode

This chapter gives how the client plays without a game server: the ways in, where the player lands, where offline levels come from and what the embedded tutorial holds. The tutorial's level resources are in → §3.1.2.8; the level formats in → §3.1.

> Status: draft. Read from 2.05 rev 2 and checked live on 2.05 rev 2, 2.22 and 2.31; the special single-player level names are open (→ App. E#319).

4.8.1 What offline means

Rule	Detail	Source
condition	offline means the client has no game connection object; every check that refuses the editor, enables the tutorial resources or sends file requests tests that one condition	[2.05] (client-internals/offline-tutorial-205.md#how-the-client-goes-offline)
current level name	the global that holds the current level name is set only by the server's set-level message, so it stays empty in offline training; the level is still drawn and walked	[live] (client-internals/offline-training.md#difference-from-session-levels)
same rules	entering an offline level stores the level name and arrival point on the player; collision, signs, links and scripts run the same paths as for a disk level online	[live] (client-internals/offline-training.md#enter-path)

4.8.2 Ways in (2.05)

Way	Detail	Source
Training	a button on the start screen; starts the embedded tutorial at `tutorial_b-4.graal` and makes no connection	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#the-training-button)
Play offline	a button of the reconnect dialog, beside Reconnect and Open serverlist; choosing it frees the game socket	[2.05][live] (client-internals/offline-tutorial-205.md#how-the-client-goes-offline); [live] (client-internals/live-checks-editor-offline-sound.md#the-reconnect-dialog-and-play-offline)
command line	`Graal.exe level.graal` starts with a local player and opens the file in the editor; Play then runs it offline (→ §4.9)	[2.05] (client-internals/offline-tutorial-205.md#how-the-client-goes-offline)
F5 Load level	while online the client warns that loading a level disconnects from the server, then loads the file locally	[2.05] (client-internals/offline-tutorial-205.md#how-the-client-goes-offline)

4.8.2.1 The reconnect dialog

Rule	Detail	Source
texts	shown when the connection cannot be established, is dropped, or the server is full	[2.05] (client-internals/offline-tutorial-205.md#how-the-client-goes-offline)
no listserver	pressing Start with no listserver shows the connection error and stays on the server list, which has no Play offline button	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#the-reconnect-dialog-and-play-offline)
refused game port	the dialog appears; Play offline goes to the tutorial start, reading no level file from disk	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#the-reconnect-dialog-and-play-offline)
dropped after login	the dialog appears; Play offline reloads the current level from the game folder with the player where they stood	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#the-reconnect-dialog-and-play-offline)
half-closed socket	a normal close from the server side left the socket half-closed and showed no dialog; only a reset or socket error brought it up	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#the-reconnect-dialog-and-play-offline)

4.8.2.2 Where the player lands on Play offline

Case	Result	Source
the current level exists in the local folder	the player stays on it at the same coordinates	[2.05][live] (client-internals/offline-tutorial-205.md#where-the-player-lands)
it does not	the player goes to `tutorial_b-4.graal` at (32, 60); the big map switches to `tutorialmap.png` and the minimap to `tutorialmap2.png`, both with `tutorialmap.txt`; a first-offline-start flag is set	[2.05][live] (client-internals/offline-tutorial-205.md#where-the-player-lands)
the level still fails to load	a warning names the level and the client stays at the reconnect dialog	[2.05] (client-internals/offline-tutorial-205.md#where-the-player-lands)
dropped on a level not on disk	not repeated live; the code falls back to the tutorial	[2.05] (client-internals/live-checks-editor-offline-sound.md#the-reconnect-dialog-and-play-offline)

4.8.3 Where offline levels come from

Rule	Detail	Source
order	offline, a name starting with `tutorial` is read from the executable's resource (→ §3.1.2.8); any other name goes through the file-name cache to disk	[2.05] (client-internals/offline-tutorial-205.md#where-offline-levels-come-from)
embedded wins, 2.05	with a different `tutorial_b-4.graal` in the game folder, Training still loads the embedded level and does not look on disk	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#the-embedded-level-always-wins)
embedded wins, 2.22	with `offlinetutorial_b-2.graal` in the game folder, the client looks the name up once but still loads the embedded copy	[live][2.22] (client-internals/live-checks-editor-offline-sound.md#the-embedded-level-always-wins)
online	names starting with `tutorial` are not special and go to the server like any level, so a server can host its own	[2.05] (client-internals/offline-tutorial-205.md#where-offline-levels-come-from)
tile definitions	every client tested reads `levels/tiledefs/tiledefsOffline.txt` on Training, since the file is `tiledefs` + server name + `.txt` and the offline server name is `Offline`; the shipped file targets only `offlinetutorial*` levels, so it changes nothing in 2.05's tutorial	[live][2.05][2.22] (client-internals/live-checks-editor-offline-sound.md#tile-definitions-when-offline)
special names	`level1.graal`, `YoshiClanR2.zelda` and `YoshiClanFields.zelda`, the old single-player start levels, run an extra setup step on load; what it does is unmeasured (→ App. E#319)	[2.05] (client-internals/offline-tutorial-205.md#special-level-names)

4.8.4 The tutorial world

4.8.4.1 2.05

Rule	Detail	Source
grid	3 columns (a to c, west to east) by 4 rows (1 to 4, north to south), plus `tutorial_b-2_house.graal` linked from b-2 at (22, 26); edge links keep the other coordinate and land at 0 or 61	[test] (client-internals/offline-tutorial-205.md#the-tutorial-world)

Rule	Detail	Source
scenery levels	a-1 to a-4, c-1 to c-4 and b-1: 2 or 3 links, no NPCs, chests or signs	[test] (client-internals/offline-tutorial-205.md#the-tutorial-world)
b-4, the start	8 NPCs, 1 chest, 3 signs: a welcome sign, a gate NPC that opens when the player says the word the sign gives while standing in a band of rows, a `setmap` and `setminimap` NPC, two light NPCs and two character NPCs; the chest holds `glove1`	[test] (client-internals/offline-tutorial-205.md#the-tutorial-world)
b-3	a sign telling the player to cut the trees with the sword	[test] (client-internals/offline-tutorial-205.md#the-tutorial-world)
b-2	5 NPCs, 2 signs: an arrow box, a door that opens on the `opendoors` flag with `chest.wav`, character NPCs	[test] (client-internals/offline-tutorial-205.md#the-tutorial-world)
b-2 house	6 NPCs, 6 chests, 7 signs: a shopkeeper, a gate that slides open on `housegateopen` with `stonemove.wav`, lamps and lights; the chests hold battleaxe, lizardshield, spinattack, nukeshot, fullheart and superbomb	[test] (client-internals/offline-tutorial-205.md#the-tutorial-world)
signs at (0, 0)	not drawn on the board; chests and `say n` show them by index	[test] (client-internals/offline-tutorial-205.md#the-tutorial-world)
end	the last sign sends the player to F8, the server list, to go online	[test] (client-internals/offline-tutorial-205.md#the-tutorial-world)
map	`tutorialmap.txt`, set by the start level's `setmap` NPC	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#the-training-button)

4.8.4.2 Later builds

Builds	Detail	Source
2.17, 2.22, 2.31	9 embedded `GR-V1.03` levels, `offlinetutorial_a-1.graal` to `offlinetutorial_c-3.graal`	[2.17][2.22][2.31] (client-internals/offline-tutorial-205.md#later-clients)
2.22, 2.31	Training starts on `offlinetutorial_b-2.graal`, the centre of the grid; its NPCs play `cauldron_bubbles.wav` and `fire_torch0` to `fire_torch2` as soon as it loads; the maps are the client's own `bigmap.txt` and `minimap.txt` plus per-level map parts	[live] (client-internals/live-checks-editor-offline-sound.md#the-training-button)
start screens	2.05: Training, Options, Start; 2.22 adds Get an account!; 2.31 adds Editor, which launches the separate GraalEditor (→ §11.6)	[live] (client-internals/live-checks-editor-offline-sound.md#the-training-button)
GraalEditor 2.3.1.1	carries the same 9 names and `tiledefsOffline.txt`	[inf] (client-internals/offline-tutorial-205.md#later-clients)

4.9 The built-in editor

This chapter gives what is particular to the editor built into the 2.05 client: how play and editing switch, what entering the editor does to the game, and what it writes. The editor's window, dialogs and objects, and the later standalone GraalEditor, are in → §11.6; the level formats it reads and writes are in → §3.1.

> Status: draft. Read from 2.05 rev 2, with Generate map, Play and Edit flags checked live on 2.05 rev 2.

4.9.1 Where the editor lives

Builds	Editor	Source
2.05	a second mode of the main game window, not a separate program	[2.05] (client-internals/level-editor-205.md)
2.31	the client has no editor of its own; the start screen's Editor button starts `GraalEditor.exe` from the game folder as a separate process (→ §11.6)	[live] (client-internals/live-checks-editor-offline-sound.md#231-and-graaleditor)

4.9.2 Switching between play and editing (2.05)

Key	In game mode	Source
F4	opens the editor; refused while connected, with a message that the editor cannot be opened online (→ §4.8.1)	[2.05] (client-internals/level-editor-205.md#entering-and-leaving)
F5	the Load level dialog; online it first warns that loading disconnects from the server (→ §4.8.2)	[2.05] (client-internals/level-editor-205.md#entering-and-leaving)
P or the Play button	back to game mode on the current level (→ §4.9.4)	[2.05] (client-internals/level-editor-205.md#entering-and-leaving)
Close	closes the current level tab, asking to save a modified level	[2.05] (client-internals/level-editor-205.md#entering-and-leaving)

Rule	Detail	Source
teardown	entering the editor stops looped sounds (→ §4.7.1), frees the level's NPC and baddy instances, and opens the player's level as a tab re-read from disk, so board changes made while playing are not carried over; the view centres on the player	[2.05] (client-internals/level-editor-205.md#entering-and-leaving)
command line	`Graal.exe path\level.graal` skips the login screen, changes into the file's folder, creates a local player at (30, 30.5) and opens the file in the editor	[2.05] (client-internals/level-editor-205.md#entering-and-leaving)
editor keys	L, S, P and M (links, signs, play, generate map) are ignored while a combo box in the side panel has focus; the full key list is in → §11.6	[2.05] (client-internals/level-editor-205.md#editor-keys)
NPC with no image	shown with a placeholder picture only in the editor's view (→ §4.6.2.3)	[2.05] (client-internals/sign-panel.md#npcs-with-no-image)

4.9.3 Undo (2.05)

Rule	Detail	Source
scope	board tiles only, hence the button's name, Undo board copy; links, signs, NPCs, baddies and chests are not undoable	[2.05] (client-internals/level-editor-205.md#undo)

Rule	Detail	Source
record	before any board write: the rectangle (x, y, width, height) and the tiles it overwrites	[2.05] (client-internals/level-editor-205.md#undo)
stack	one per level tab, at most 128 records; the oldest is dropped when full	[2.05] (client-internals/level-editor-205.md#undo)
undo	pops the newest record, writes the old tiles back, re-selects the rectangle (or clears the selection if it was the whole board) and marks the level modified	[2.05] (client-internals/level-editor-205.md#undo)

4.9.4 Play, flags and the map image (2.05)

Rule	Detail	Source
Play saves first	Play on a named, modified level writes it to disk ("Level saved") before the Level options dialog asks for the start position; a level with no name must be saved under a name first; 2.22 does the same (→ §11.6)	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#play-saves-first); [2.05] (client-internals/level-editor-205.md#entering-and-leaving)
Edit flags	edits the client's in-memory flag list, not anything in the level; the list survives Play and a return to the editor, but is never written to disk or the registry and is gone after a restart	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#edit-flags)
Generate map (M)	after a prompt that it may take minutes, walks the level list of the loaded big map (`bigmap.txt` or the list a `setmap` NPC loaded) in list order, row by row	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#generate-map-m)
Generate map, each level	read from disk, its NPCs run into a 1024 × 1024 view, shrunk to 56 × 56 px; a missing or unreadable entry is a blank square; links are not followed, so a level not in the list (such as the tutorial house) is skipped	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#generate-map-m)
Generate map, output	offered in a save dialog as `graalmap.png` in the level's folder; the 3 × 4 tutorial list gave a 168 × 224 image; the level that was open is restored afterwards	[live][2.05] (client-internals/live-checks-editor-offline-sound.md#generate-map-m)

4.9.5 Saving and opening (2.05)

Rule	Detail	Source
format by extension	`.nw` is written as `GLEVNW01` text (→ §3.1); any other extension, normally `.graal`, always as `GR-V1.03`; the 2.05 editor cannot write older versions	[2.05] (client-internals/level-editor-205.md#saving)
`GR-V1.03` board	the writer emits a single run for two or more equal tiles and a double run when `A B A B` starts at the cursor, otherwise a plain tile (→ §3.1)	[2.05] (client-internals/level-editor-205.md#saving)
baddy section	ends with `FF FF FF 0A` on write; the reader does not need it (→ §3.1)	[2.05] (client-internals/level-editor-205.md#saving); [2.05] (client-internals/level-editor-205.md#loader-note-for-formatslevel-binarymd)
URL levels	saving a level opened from a URL falls through to Save as	[2.05] (client-internals/level-editor-205.md#saving)
opening	accepts `Z3-V1.03`, `Z3-V1.04`, `GR-V1.00` to `GR-V1.03` and `GLEVNW01`; anything else is refused as an unsupported file type	[2.05] (client-internals/level-editor-205.md#saving)
chest items	in index order: greenrupee, bluerupee, redrupee, bombs, darts, heart, glove1, bow, bomb, shield, sword, fullheart, superbomb, battleaxe, goldensword, mirrorshield, glove2, lizardshield, lizardsword, goldrupee, fireball, fireblast, nukeshot, joltbomb, spinattack (→ §4.5.4)	[2.05] (client-internals/level-editor-205.md#chests-tchestedit)
new chest	defaults to item 1 (bluerupee) and no sign (−1)	[2.05] (client-internals/level-editor-205.md#chests-tchestedit)

4.10 Other client features

This chapter collects the client features that sit outside the core game loop: the integrity checks, animation state as other players see it, key emotes, the player list and its social lists, and what the client does with the server's start message and disconnect texts. The message layouts are in → §2.4 and §2.5.

> Status: draft. The 2.05 parts are read from 2.05 rev 2; the 2.17 to 2.31 parts come from lab captures and reads of those clients. Open points are App. E#320 to E#322.

4.10.1 Integrity checks

4.10.1.1 The tamper check (2.17 to 2.31)

Rule	Detail	Source
builds	new in 2.17; absent from 2.05 and 1.x	[bin][cap] (client-internals/feature-specs/08-tampercheck.md#what-it-is-bincap)
message	PLI_TAMPERCHECK (39), 10 raw bytes, sent right after PLI_LANGUAGE (37) in the first frame after login; the per-build values are in → §2.4.9	[cap] (client-internals/feature-specs/08-tampercheck.md#what-it-is-bincap)
what the bytes are	two 32-bit words A then B, each written as 5 bytes in a form that is neither GINT5 nor masked: only the first and last byte are range-limited, so the middle three can fall below 0x20	[bin] (client-internals/decomp-findings-2026-10-24-b/01-tamper-check-id39.md#what-the-10-bytes-are)
the words	both are standard reflected CRC-32 values over the client's own image in memory, chained across ranges; A covers 8 or 9 ranges of the code, B one large code range and one block of static data	[bin] (client-internals/decomp-findings-2026-10-24-b/01-tamper-check-id39.md#what-the-10-bytes-are)
constant	nothing depends on the login key, account, level or time, so the value is a constant per build and changes only if the executable's bytes change; all six lab values were reproduced offline from the executables	[bin][cap] (client-internals/decomp-findings-2026-10-24-b/01-tamper-check-id39.md)
2.17	sent once at login; nothing more in about 90 s	[cap] (client-internals/decomp-findings-2026-10-24-b/01-tamper-check-id39.md#when-it-is-sent)
2.22	re-sent by the level-change routine, so on every gmap edge crossing and after every link request the client starts, but not after a server-side warp; no timer	[cap][bin] (client-internals/decomp-findings-2026-10-24-b/01-tamper-check-id39.md#when-it-is-sent); [cap] (client-internals/feature-specs/08-tampercheck.md#what-it-is-bincap)
2.31	re-sent every 3.2 s by the 64-tick integrity cycle of → §4.1.3; not re-sent on level entry	[bin][cap] (client-internals/decomp-findings-2026-10-24-b/03-periodic-ids-2.31.md#pli-39-re-send-every-32-s)
client check	the clients only send it; nothing on their receive side checks it	[bin] (client-internals/decomp-findings-2026-10-24-b/01-tamper-check-id39.md#does-the-server-verify-it)
servers	the open-source server reads and discards it; four public servers accepted a 2.22 login without it for at least 20 s, which shows only that they do not enforce it at login (→ App. E#320)	[srv] (client-internals/feature-specs/08-tampercheck.md#exact-server-behaviour-srv); [cap] (client-internals/feature-specs/08-tampercheck.md#public-servers)

4.10.1.2 Process list and window scan

Rule	Detail	Source
process list	the server sends PLO_LISTPROCESSES (182) once during login to 2.17 and later; the client can answer PLI_PROCESSLIST (44) with its running programs (→ §2.4.9)	[srv][inf] (client-internals/feature-specs/08-tampercheck.md#exact-server-behaviour-srv)

Rule	Detail	Source
2.31 window scan	tick 20 of the 64-tick cycle compares the titles and classes of top-level windows with an obfuscated blacklist; the list is not decoded and what the client does on a match is unknown (→ App. E#321)	[bin] (client-internals/feature-specs/08-tampercheck.md#what-it-is-bincap); [doc] (client-internals/feature-specs/README.md#settled-index-cross-cutting-items-status-as-of-the-lab-batch-c-2026-10-24)
2.31 debugger probe	tick 1 of the cycle probes for a debugger and arms a 5000 ms timer	[bin] (client-internals/decomp-findings-2026-10-24-b/03-periodic-ids-2.31.md#pli-39-re-send-every-32-s)

4.10.1.3 What outside input tools look like

The recovered trainers of the period are summarised here only for what the client's checks could see of them:

Rule	Detail	Source
how they act	every recovered sample types keys into a classic client window or writes text into it through window messages; none speaks the game protocol	[bin] (recovered-hacks/behaviour-models.md#what-a-server-can-see)
memory	no memory-writing or remote-thread imports were found in the unpacked samples, so none of them changes the client's image; the tamper check of → §4.10.1.1 cannot see them	[bin] (recovered-hacks/TRAINER_RE_NOTES.md); [inf] (recovered-hacks/THREAT_TAXONOMY.md)
what the server sees	only the resulting actions (dig, use, walk, chat), at the rates the tool's waits allow	[inf] (recovered-hacks/behaviour-models.md#what-a-server-can-see); [inf] (recovered-hacks/static-2026-11-01/README.md#what-a-checker-can-use)
window titles	some tools look for the client window by title, and some show their own named window; such a window is the kind of thing the 2.31 scan compares, but whether its list names any of them is unknown (→ App. E#321)	[bin] (recovered-hacks/behaviour-models.md); [inf] (recovered-hacks/TRAINER_RE_NOTES.md)

4.10.2 Animation state as others see it

4.10.2.1 2.05: the sprite prop

Rule	Detail	Source
form	prop 17 (→ §2.6.2) is direction + 4 × S	[2.05] (client-internals/feature-specs/11-emotes-ganis.md#205-sprite-prop-205dec); [2.05] (client-internals/decomp-findings-2026-10-24/02-sprites-poses-emotes.md#how-the-sprite-prop-17-is-built)
when S changes	only while the active gani is the default `def` gani and the step is 38 or lower; other ganis leave S alone, and a name the sender does not know gives 0; read from the sender, not confirmed on the wire	[2.05] (client-internals/feature-specs/11-emotes-ganis.md#205-sprite-prop-205dec)
hurt	S = 32, so the prop is 0x80 + direction	[2.05] (client-internals/feature-specs/11-emotes-ganis.md#205-sprite-prop-205dec)
step to S, steps 0 to 38	0, 4, 5, 4, 3, 2, 1, 2, 3, 6, 7, 8, 9, 10, 11, 12, 13, 11, 12, 14, 15, 16, 17, 23, 18, 19, 20, 19, 18, 21, 22, 21, 18, 24, 25, 26, 27, 28, 29; steps 9 to 13 are the sword swing (→ §4.3)	[2.05] (client-internals/decomp-findings-2026-10-24/sprite-step-table.tsv)

Rule	Detail	Source
gani to step	idle and def: frame; walk and walkslow: frame + 1; kick 9; sword, ridesword and spin: 9 + (frame mod 5); push 14 + frame; grab 19; pull 20 + frame; lift 23; carrystill and hiddenstill 24; carry and hidden 25 + frame; shoot 33; ridestill 34; ridefire 35; rideeat 36; ride 35 + (frame and 1); sit 37; sleep 38; hurt and ridehurt 39; dead 40, not sent	[2.05] (client-internals/decomp-findings-2026-10-24/gani-to-step.tsv)
built-in ganis	31 names, index 0 to 30: idle, walk, walkslow, kick, sword, ridesword, spin, push, grab, pull, lift, carrystill, hiddenstill, carry, hidden, shoot, ridestill, ridefire, rideeat, ride, sit, sleep, hurt, ridehurt, dead, maps1, maps2, maps3, swim, lava, carried	[2.05] (client-internals/decomp-findings-2026-10-24/gani-name-table.tsv)
not on the wire	swimming, lava and death are sent as S 0 plus direction; death travels only as the STATUS dead bit (→ §2.6.2.1, §4.4); there is no swim sprite, so another 2.05 client cannot tell a swimmer from an idle player	[2.05] (client-internals/decomp-findings-2026-10-24/02-sprites-poses-emotes.md#swim-lava-and-dead-on-the-wire)
STATUS as 2.05 sets it	0x01 editor or paused; 0x02 hidden (mode 14 or the hidden timer); 0x04 male; 0x08 dead, only once the death timer reaches 28 ticks; 0x10 weapons allowed; 0x20 sword hidden while the `hidesword` timer runs; 0x40 has the charged spin (→ §2.6.2.1)	[2.05] (client-internals/decomp-findings-2026-10-24/status-bits-prop18.tsv)
short codes	the client maps gani-name parts to one letter (attack A, carry C, dead D, hurt H, idle I, kick K, lift L, ride R, sit T, spin U, swim V, sword S, walk W, def X and others) to shorten cached gani file names	[2.05] (client-internals/decomp-findings-2026-10-24/gani-short-codes.tsv)

4.10.2.2 2.1 and later

Rule	Detail	Source
gani prop	prop 10 is the gani name as a string; a receiver that lacks `<name>.gani` asks the server for it	[srv][cap] (client-internals/feature-specs/11-emotes-ganis.md#a-animation-state)
2.22 cadence	→ §4.1.4	[cap] (live-runs/2gta-222-2026-10-30/README.md)

4.10.3 Key emotes (2.05 rev 2, same routine in 1.39)

Rule	Detail	Source
keys	Ctrl+A to Ctrl+Z, with no text field focused and the player neither dead nor locked	[2.05] (client-internals/feature-specs/11-emotes-ganis.md#b-emotes)
look up	the letter is lowercased and the client takes the first loaded file whose name starts with `emoticon_<letter>` and ends in png, mng or gif; there is no letter-to-name table, so the mapping is whatever files exist	[2.05] (client-internals/feature-specs/11-emotes-ganis.md#b-emotes); [2.05] (client-internals/decomp-findings-2026-10-24/02-sprites-poses-emotes.md#emotes-ctrlletter)
drawing	at the player's x + 2, y − 5 (a little lower when a second flag is set) for 50 ticks (2.5 s)	[2.05] (client-internals/feature-specs/11-emotes-ganis.md#b-emotes)
network	the key path sends nothing; `PLI_SHOWIMG` (24) is sent only by the script command `showimg` (→ §8.2)	[2.05] (client-internals/feature-specs/11-emotes-ganis.md#b-emotes)
others seeing it	unmeasured: captured SHOWIMG frames re-sending an AFK image about every 3 s are not explained by the key path (→ App. E#322)	[cap][inf] (client-internals/feature-specs/11-emotes-ganis.md#b-emotes)
earlier builds	1.38 introduced Ctrl+letter emotes, and 1.39 added letters	[doc] (client-internals/feature-specs/11-emotes-ganis.md#b-emotes)

4.10.4 Picking up and dropping items online

Rule	Detail	Source
2.05 pickup	sends `PLI_ITEMDEL` (13) with the item's half-tile position and changes its own props itself; the server only removes the item and tells the others	[cap][srv] (client-internals/feature-specs/03-inventory.md#item-pickup-and-chests-srvcap)
the other form	`PLI_ITEMTAKE` (32) also removes the item, and the server applies it to the player's props and echoes them	[srv] (client-internals/feature-specs/03-inventory.md#item-pickup-and-chests-srvcap)
drop	`PLI_ITEMADD` (12) with the position and item	[srv] (client-internals/feature-specs/03-inventory.md#item-pickup-and-chests-srvcap)
chest	`PLI_OPENCHEST` (20) with whole-tile x and y; the opened state is per account (→ §4.5.4)	[srv] (client-internals/feature-specs/03-inventory.md#item-pickup-and-chests-srvcap)

4.10.5 The player list and social lists

4.10.5.1 2.05 (F7)

Rule	Detail	Source
window	a stay-on-top form with a multi-select list, a caption "--- N online ---" and Profile and Options buttons; the Options window has 5 check boxes and the profile edit window 10 fields	[2.05] (client-internals/decomp-findings-2026-10-24/03-player-list-popup.md#player-list-window)
popup	right-click selects the row under the cursor and shows Message (default), History, Buddy, Ignore, Profile; Buddy and Ignore are checked when every selected player is in that list	[2.05] (client-internals/decomp-findings-2026-10-24/03-player-list-popup.md#player-list-window)
open	double-click or Enter opens the read window if a PM from that player is waiting, otherwise the compose window	[2.05] (client-internals/decomp-findings-2026-10-24/03-player-list-popup.md#player-list-window)
Message	one PM to all selected rows, reusing an open window for the same set; at most 50 recipients (→ §2.8)	[2.05] (client-internals/decomp-findings-2026-10-24/03-player-list-popup.md#actions)
Profile	the first selected player only; online it sends `PLI_PROFILEGET` (80), which the listserver answers (→ §6)	[2.05] (client-internals/decomp-findings-2026-10-24/03-player-list-popup.md#actions); [srv] (client-internals/feature-specs/10-right-click-menu.md#what-is-known-205inf)
Ignore	toggles each selected account in `ignoredaccounts.txt` and saves the buddy list too; local only; ignored players' chat and PMs are dropped	[2.05] (client-internals/decomp-findings-2026-10-24/03-player-list-popup.md#actions); [2.05] (client-internals/feature-specs/10-right-click-menu.md#what-is-known-205inf)
Buddy	toggles `buddyaccounts.txt`; Buddy and Ignore exclude each other	[2.05] (client-internals/decomp-findings-2026-10-24/03-player-list-popup.md#actions)
auto reply	the client holds the text "Your messages are ignored."; when it is sent was not traced	[2.05] (client-internals/decomp-findings-2026-10-24/03-player-list-popup.md#actions)
world click	there is no right-click handler on a player in the game view; the only player popup is the list's	[2.05] (client-internals/decomp-findings-2026-10-24/03-player-list-popup.md#not-found)
no player action form	the person-action form is the editor's NPC script editor (→ §4.9), not a trade or duel menu	[2.05] (client-internals/decomp-findings-2026-10-24/03-player-list-popup.md#person-action-is-not-a-player-menu)

4.10.5.2 By build

Builds	Change	Source
1.33	popup Message, Ignore, History; sections GPs, Guild, Normal, Ignore	[bin] (client-internals/feature-specs/10-right-click-menu.md#what-is-known-205inf)

Builds	Change	Source
1.34	ignore by account name; ignored players drawn red	[bin] (client-internals/feature-specs/14-friends-ignore.md#what-it-is)
1.35	Buddy, kept in `buddyaccounts.txt`	[bin] (client-internals/feature-specs/10-right-click-menu.md#what-is-known-205inf)
1.39	Profile and the Player List Options window	[bin] (client-internals/feature-specs/10-right-click-menu.md#what-is-known-205inf)
2.14 and later	a Staff section named from `PLO_STAFFGUILDS` (47); the status list `PLO_STATUSLIST` (180) with the player's own index in prop 53	[srv] (client-internals/feature-specs/14-friends-ignore.md#server-behaviour-srv)
2.22	sections Staff, Buddies, Guild, Normal, Ignore; popup Message, History, Buddy, Ignore, Mute, Profile	[cap] (client-internals/feature-specs/14-friends-ignore.md#lab-findings-from-a-real-222-client-2026-10-24-cap)
2.22 Buddy and Ignore	local only: they write `buddyaccounts.txt` or `ignoredaccounts.txt`, one account per line, delete an empty buddy file, move the entry in the list, and send nothing	[cap] (client-internals/feature-specs/14-friends-ignore.md#lab-findings-from-a-real-222-client-2026-10-24-cap)
2.22 Mute	sends `PLI_MUTEPLAYER` (43) with the player id and 1 or 0, one message per click; the entry does not move	[cap] (client-internals/feature-specs/14-friends-ignore.md#lab-findings-from-a-real-222-client-2026-10-24-cap)

4.10.6 Start message and disconnects

Rule	Detail	Source
start message	`PLO_STARTMESSAGE` (41) carries the server message as one line of HTML; 2.05 has a form for it; whether it shows at once or on request is unmeasured	[srv] (client-internals/feature-specs/12-motd-rules.md#packets-server-to-client-at-the-end-of-login-srv)
RPG window	`PLO_RPGWINDOW` (179), 2.1 and later: quoted, comma-separated lines naming the server and its software	[srv] (client-internals/feature-specs/12-motd-rules.md#packets-server-to-client-at-the-end-of-login-srv)
login order	level data, big map, minimap, RPG window, start message, an empty `PLO_SERVERTEXT` (82)	[srv] (client-internals/feature-specs/12-motd-rules.md#packets-server-to-client-at-the-end-of-login-srv)
disconnects	every refusal, kick and ban reaches the client as `PLO_DISCMESSAGE` (16) with the text, `\r` for a line break, then the socket closes; the client then shows the reconnect dialog (→ §4.8.2.1)	[srv] (client-internals/feature-specs/13-ban-kick.md#what-it-is)
upgrade	from 2.14, `PLO_UPGRADEURL` (169) with a disconnect turns Reconnect into an Upgrade button	[srv] (client-internals/feature-specs/13-ban-kick.md#exact-server-behaviour-messages-in-the-order-they-can-happen-srv)
server texts	the refusal texts and the order of the server's login checks are server behaviour (→ §5)	[srv] (client-internals/feature-specs/13-ban-kick.md#exact-server-behaviour-messages-in-the-order-they-can-happen-srv)

5 Game servers

This part describes the game servers that Graal clients talk to: the original 1999 GServer13, the 2006 39ster 2.1 remake, and GServer-v2 as the reference for later behaviour. Login frames are in → §2.3, ciphers in → §2.2, and the reply sequence after a good login in → §2.3.6. The listserver is in → §6, RC and NC in → §7.

> Status: draft. No official 2000–2002 GServer binary is in the archive, so the 2.x-era server rules here come from GServer-v2 and the 39ster remake, not from an original server.

5.1 GServer13 (1999)

GServer13 is the server shipped with the 1.3 test client in the `graal13b` archive; it was run with its own client under Wine on the loopback address [live][cap] (client-internals/live-gserver13.md).

5.1.1 Options and defaults

GServer13 opens a start dialog before it listens, and saves the values to `serverconfig.txt` [live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live):

Key	Dialog label	Default	Source
`serverport`	port	14804	[live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live)
`respawntime`	Respawn time (bushes etc.) in sec	10	[bin][live] (client-internals/board-respawn.md#options)
`horselifetime`	Horse lifetime in sec	10	[bin][live] (client-internals/board-respawn.md#options)
`apincrementtime`	Alignment increment time in sec	1200	[bin][live] (client-internals/board-respawn.md#options)
`baddyrespawntime`	Baddies respawn time in sec	60	[bin][live] (client-internals/board-respawn.md#options)

GServer 1.28 beta 2 (`GServerv.exe`) has the same four timer keys [bin] (client-internals/board-respawn.md#options). An RC of 2000 can set the same four timers over the wire (→ §7.1.4).

5.1.2 Accounts

Field of a new account (Edit accounts)	Default	Source
level and position	`onlinestartlocal.graal` at x 30.5, y 30	[live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live)
AP, MP	50, 0	[live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live)
hearts	3 of 3	[live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live)
rupees, bombs, arrows	0, 10, 5	[live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live)
glove, bomb, bow power	1, 1, 1	[live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live)
sword, shield	1 `sword1.gif`, 1 `shield1.gif`	[live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live)
head, colours	`head0.gif`; orange, white, blue, red, black	[live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live)
weapons, spin attack	weapons enabled, no spin attack	[live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live)

The rights levels are no rights, Graalian Police, Trusted Admin and Administrator [live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live). The 1.3 archive does not ship `onlinestartlocal.graal`; the lab copied it from the 15th Graal build, which uses the same `GR-V1.03` format [live] (client-internals/live-gserver13.md#server-settings-and-the-default-account-live). The account files of this server family are in → §5.4.1.

5.1.3 Login and play

Rule	Source
The client's first frame is a u16be length and zlib of the id-0 login line with `GSERV014`, account and password (→ §2.3.2)	[cap] (client-internals/live-gserver13.md#login-cap)
A wrong account is answered with id 16 and the text "An account with the name and password you typed in doesn't exist."; the client offers Reconnect or Play offline	[cap] (client-internals/live-gserver13.md#login-cap)
After a good login the board goes in a separate zlib frame (100, then a frame with 101 and 8192 bytes); the full reply order is in → §2.3.6.2	[cap] (client-internals/live-gserver13.md#login-cap)
The client then asks for each NPC image with id 23; the server answers id 30 (file not sent) for images it lacks	[cap] (client-internals/live-gserver13.md#login-cap)
The client sets flags from NPC scripts with id 18 and adds horses with id 7 (from `puthorse`); the server deletes each horse with id 18 about 9.5 to 9.8 s later, matching the 10 s horse lifetime on a 1 s tick	[cap] (client-internals/live-gserver13.md#login-cap)
The level leader (the player who sends level NPCs' arrows to the others over UDP) is picked by the server with id 10, sent at the end of the first player's login reply and to the next player when the leader leaves	[cap] (protocol/gserv-1999.md#udp-128-beta-2-on)
While a key is held the client sends id 2 with X (prop 15) and SPRITE (17) every 100 ms	[live][cap] (client-internals/live-gserver13.md#in-play-live-cap)
Chat is id 2 prop 12; the server relays it; 1.3 draws it with no bubble	[live][cap] (client-internals/live-gserver13.md#in-play-live-cap)
Horse deletes and bush regrowth land on the same fractional second, so the server checks its timers once a second	[cap] (client-internals/live-gserver13.md#login-cap)

5.1.4 Board changes and respawn

Step of `PLI_BOARDMODIFY` (1)	Rule	Source
1	needs at least 4 bytes, a current level, and `len ≥ 4 + 2·w·h`	[bin] (client-internals/board-respawn.md#pli_boardmodify-1-handling-bin)
2	writes the new tiles into the server's 64 × 64 board and keeps the old tiles	[bin] (client-internals/board-respawn.md#pli_boardmodify-1-handling-bin)
3	if a tile changed, relays `PLO_BOARDMODIFY` (7) with the client's bytes to every other player in the level; **the sender is not echoed**	[bin][live] (client-internals/board-respawn.md#pli_boardmodify-1-handling-bin)
4	a change of exactly 2 × 2 gets a countdown `Trunc(Random · R/2 + 3R/4)` with R = `respawntime`; any other size never respawns (1 × 1, 2 × 1, 3 × 3, the 4 × 4 push/pull block)	[bin][live] (client-internals/board-respawn.md#pli_boardmodify-1-handling-bin)
5	there is no tile check: any 2 × 2 change respawns, whatever the tiles	[bin] (client-internals/board-respawn.md#pli_boardmodify-1-handling-bin)

Step of PLI_BOARDMODIFY (1)	Rule	Source
6	the record joins the "changed since you left" list of every player who left the level since loading it; an earlier record with the same x, y, w, h is replaced	[bin] (client-internals/board-respawn.md#pli_boardmodify-1-handling-bin)

Respawn tick	Rule	Source
timer	a 1000 ms timer walks every loaded level, empty or not, and decrements each countdown	[bin] (client-internals/board-respawn.md#respawn-tick-bin)
restore	at < 1 the server removes the record, sends PLO_BOARDMODIFY (7) with the **old** tiles to everyone in the level, the original sender included, and writes them back into its board	[bin][live] (client-internals/board-respawn.md#respawn-tick-bin)
timing	for R = 10 the countdown starts at 7..12 and the first tick comes 0 to 1 s after the change, so the restore lands 6 to 12 s later; 7 and 12 are half as likely as 8 to 11	[bin] (client-internals/board-respawn.md#respawn-tick-bin)
measured	60 of 60 2 × 2 changes respawned after 6.21 to 11.98 s, mean 9.21 s; other sizes never respawned; restored tiles were the level's own tiles	[live] (client-internals/board-respawn.md#measured-live); [live] (version-profiles/lab-gs13-respawn.tsv)
re-entry	a player with no change list gets the level as on a first visit; otherwise only the queued changes go out and the list is dropped; the exact packet is hard to read	[bin][inf] (client-internals/board-respawn.md#level-re-entry-bin)
disconnect	the player's change lists are freed	[bin] (client-internals/board-respawn.md#level-re-entry-bin)

Item drops from cut bushes and pots are rolled by the 1.3 client, not by GServer13 [bin][cap] (client-internals/board-respawn.md#summary); the client side is in → §4.5.

5.1.5 Scripts

GServer13 has player and NPC props, including an NPC nickname field, but no GS1 command table and no `setcharprop` or `showcharacter` strings; NPC scripts of this era are carried to clients and run there [bin] (server-notes/setcharprop-hash-n.md#1-gserver13-1999-cannot-settle-the-value-question-by-lab). GServer 1.34's Windows binary lists `showcharacter` but not `setcharprop` [bin] (server-notes/setcharprop-hash-n.md#2-server-side-gs1-gserversource-archive). Server-side scripting is in → §5.5.

5.2 39ster 2.1 GServer

An open remake of the 1.39 server from December 2006, targeting 2.x clients; one Windows/POSIX C++ process with no script VM [src] (ash/ARCHITECTURE.md). The `ash/` notes describe this tree as written [src] (ash/INDEX.md).

5.2.1 Process and data

Item	Rule	Source
listen port	14822 by default	[src] (ash/INDEX.md)
threads	a main thread (accept loop with a 1 s `select`, `gameTime` + 1 each second, idle kick after 300 s) and one thread per connection; shared heaps, no locks	[src] (ash/ARCHITECTURE.md#2-process-and-thread-model)
player cap	`maxPlayers` 100, never enforced; player ids come from a free list	[src] (ash/ARCHITECTURE.md#2-process-and-thread-model)
data files	`data/` is walked once at boot; a file is found by basename, first match wins	[src] (ash/BLUEPRINT.md#source-tree)
accounts	`accounts/<name>`, plain text; loading is a stub that spawns every player on `level13.graal`	[src] (ash/ARCHITECTURE.md#4-data-stores)

Item	Rule	Source
weapons	`Weapons.txt`, INI style	[src] (ash/ARCHITECTURE.md#4-data-stores)
level formats	GR-V1.01 (12-bit codes), GR-V1.02 and 1.03 (13-bit codes) (→ §3.1); no `.nw` loader	[src] (ash/ALGORITHMS.md#a8--gr-v1-tile-bitstream); [src] (ash/ARCHITECTURE.md)

5.2.2 Connections and login

Item	Rule	Source
frame	u16be length (raw, not a G-integer) + zlib; the inflated body is `\n`-separated messages	[src] (ash/BLUEPRINT.md#wire-frame-blueprint)
player (type 0)	the login line carries a key; every later line drops one byte chosen by the LCG (→ §2.2.2)	[src] (ash/ARCHITECTURE.md#1-system-context); [src] (ash/ALGORITHMS.md#a4--player-descramble-drop-one-lcg)
RC (type 1)	same framing, no descramble; RC ids 51 to 87 are in the table but their bodies are empty stubs	[src] (ash/ARCHITECTURE.md#1-system-context)
login line	`c type`, `c key` (player only), version (7 bytes for a player, 8 for RC), 1 byte skipped, `str account`, `str password`; the login line itself is not scrambled	[src] (ash/BLUEPRINT.md#wire-frame-blueprint)
descramble	iterator starts at `0x04A80B38`; per line `iterator = iterator · 0x8088405 + key`, then the byte at `(iterator & 0xFFFF) mod count` is dropped	[src] (ash/ALGORITHMS.md#a4--player-descramble-drop-one-lcg)
G-integers	1 to 5 bytes, 7-bit groups high first, each + 32 (→ §2.1.1); widths used: 2 for player id and script length, 3 for rupees, kills, deaths and NPC id, 5 for file time	[src] (ash/ALGORITHMS.md#a1--gchar-numeric-codec)
head image	`c (100 + n)` then n bytes; an inbound length below 100 is a classic head index, `head{n}.gif`	[src] (ash/ALGORITHMS.md#a1--gchar-numeric-codec)

Server id	Role	Payload	Source
100	announced length	G-integer size of the next packet	[src] (ash/BLUEPRINT.md#server--client-special-ops-not-in-the-named-enum)
101	raw board	4096 native `short` tiles	[src] (ash/BLUEPRINT.md#server--client-special-ops-not-in-the-named-enum)
102	file	5-byte G-integer modification time, `c` name length, name, raw bytes	[src] (ash/BLUEPRINT.md#server--client-special-ops-not-in-the-named-enum)

The header comments of the 2006 tree give wrong opcode numbers after 83; the `msgFuncs[89]` table and the C enums are the reliable source [src] (ash/INDEX.md).

5.3 GServer-v2 reference behaviour

GServer-v2 is an open reimplementation, not an original server [src] (client-internals/board-respawn.md#gserver-v2-lab-server-src). Where this part says "GServer-v2", it means the behaviour read in its source or seen in the lab; it is the reference for what 2.x and later clients accept, not a record of the official servers.

5.3.1 Server source lineage

Tree	Era	Shape	Source
`linuxgserver131`	1.31	one `gserver.cpp` monolith, about 9k lines with zlib headers	[src] (server-notes/01-lineage.md#what-is-actually-source)
`sidrax-v1.4-srv-src-1.3.9`	1.3.9	`gserver.cpp`, `levels.cpp`, `npcs.cpp`, MySQL accounts	[src] (server-notes/01-lineage.md#what-is-actually-source)
`oldwin32-gserver1.39source`	1.39	same family, Win32	[src] (server-notes/01-lineage.md#what-is-actually-source)

Tree	Era	Shape	Source
gservernwsrc	NW, about 1.4	objects and Pascal strings	[src] (server-notes/01-lineage.md#what-is-actually-source)
gs2emu (d3ecab4)	Reborn 2.x	typed classes, about 18k lines	[src] (server-notes/01-lineage.md#what-is-actually-source)
Reborn binaries 2.1.0 to 2.4.0, GServer revisions 29, 31, 51	2.x, 2.17 line	binaries and default worlds only	[src] (server-notes/01-lineage.md#what-is-binary--world-only)

Era	What changed	Source
1.3.x monolith	one listen socket and one world; login, movement and chat in one file; Pascal strings (c length + bytes); MySQL or flat-file accounts; generation 3 at most	[src] (server-notes/01-lineage.md#what-changed-between-eras)
gs2emu / Reborn 2.x	`servers.txt` runs several worlds in one process; typed players (client, RC, NPC-server); generations 3, 4 and 5 chosen by the login type; `GRACC001` account files and `GLEVNW01` levels; RC file browser; listserver link (→ §6.5); optional NPC-server process	[src] (server-notes/01-lineage.md#what-changed-between-eras)
a Python rewrite of gs2emu, also called Ash	same on-disk layout and wire protocol; generations 3 to 5, client, RC and NPC-server logins, listserver link, accounts, levels with baddies, RC file browser, gmaps, baddies, NPC-server relay; no horses, no weapon bytecode; default port 14812	[src] (server-notes/01-lineage.md#what-changed-between-eras); [src] (server-notes/ash-overview.md#what-ash-implements)

Two different projects carry the name Ash (→ App. E#400).

5.3.2 Login and receive path

Rule	Source
Every multi-byte integer on the player socket is big-endian; frames are `u16be` length + payload	[src] (server-notes/03-protocol.md#protocol)
The login line is `c type`, `c key` (clients, and RC when the generation is above 3), 8 version bytes, `str account`, `str password`; an NPC-server adds a `s` listen port	[src] (server-notes/03-protocol.md#login-blob-first-line-after-zlib)
`type = 1 << value`; the type chooses the generation (client type 0 → 3, type 4 → 4, type 5 and RC2 → 5) and the full type table is in → §2.3.6.1	[src] (server-notes/04-algorithms.md#5-login)
The key resets both iterators; the start value per generation is `0, 0, 0x04A80B38, 0x04A80B38, 0x04A80B38, 0` and the limit is −1 (→ §2.2)	[src] (server-notes/04-algorithms.md#1-lcg-step)
Inbound generation 1 is plain, 2 and 3 are zlib, 4 and 5 are XOR then decompress; a generation-3 client line has its inserted byte removed before the id is read; the server's outbound generation 3 inserts nothing	[src] (server-notes/04-algorithms.md#4-inbound-frame); [src] (server-notes/04-algorithms.md#2-gen-3-dummy-byte-client--server-packets-only)
Login checks, in order: version in `allowedversions` (→ §5.4.2), IP ban, RC account on the staff list, NPC-server password; then the server asks the listserver with `SV0_VERIACC2` and continues when the answer comes (→ §6.5)	[src] (server-notes/04-algorithms.md#5-login)
An NPC-server login is answered with the NC maps, levels and weapons	[src] (server-notes/04-algorithms.md#5-login)
A trailing identity string (`win`, hardware ids, OS) is read and forwarded to the listserver	[src] (server-notes/lab-servers-2026-10-30.md#what-our-gserver-does-with-each-client-generation-src)

5.3.3 What real clients require of a server

These rules come from running real clients against GServer-v2 builds and fixing what they refused; each is a client requirement, not a server choice [lab] (tools/lab-servers/README.md#our-beta4-gserver-2026-10-30).

Requirement	Clients	Source
Send PLO_SETACTIVELEVEL (156) during the level load; without it the client waits on the loading screen	2.05 (and every Original-type client from 2.01 in the lab fix)	[src][lab] (server-notes/lab-servers-2026-10-30.md#2-gserver-v2-beta4-against-our-build); [lab] (tools/lab-servers/README.md#our-beta4-gserver-2026-10-30)
Send the first PLO_LEVELNAME before the board; after it, the client stays on the loading screen	2.05	[lab] (tools/lab-servers/README.md#our-beta4-gserver-2026-10-30)
Read the rest of the login frame: the client sends PLI_LANGUAGE in the same frame as the login line, and dropping it leaves the generation-3 iterator one step behind for the whole session ("Invalid PlayerProp" on 2.17.1)	2.05, 2.17.1	[lab] (tools/lab-servers/README.md#our-beta4-gserver-2026-10-30)
Treat a lone PLI_PACKETCOUNT (31) as an ordinary message: these clients send one every 10 s, and a server that reads the next frame as a login kicks them	1.33 to 1.39, and any 2.x client that sends 31 alone	[lab] (tools/lab-servers/README.md#133-to-139-on-beta4-2026-10-30)
Accept id 25 (the client's weapon upload)	1.33 to 1.37	[lab] (tools/lab-servers/README.md#133-to-139-on-beta4-2026-10-30)
Send prop 10 in the bow form, not as a gani	1.33 to 1.39	[lab] (tools/lab-servers/README.md#133-to-139-on-beta4-2026-10-30)
Do not save the hidden status bit (STATUS 0x02) into the account; a client that quit hidden logs in invisible	2.x (v4 lab)	[lab] (tools/lab-servers/README.md#our-beta4-gserver-2026-10-30)
Answer a 2.x client's zlib data error: 2.21 to 2.31 have a zlib bug on some chat; GServer-v2 either pads chat with a hidden clientside weapon (-gr_zlib_fix) or retries the frame as raw DEFLATE and rolls the generation-5 iterator back one step	2.21 to 2.31	[src] (server-notes/lab-servers-2026-10-30.md#2-gserver-v2-beta4-against-our-build)
A 5.24 client given the pre-2.1 file path saves PNG data as a file name and never gets files over 64000 bytes; with its own version slot it gets the modification time and the large-file messages	5.24r4	[cap] (tools/lab-servers/README.md#524-in-gs2libs-version-table-2026-10-31)

Real 2.05 rev 2, 2.17.1 and 2.22 reach the game and walk on a GServer-v2 build with these fixes, and real 1.33 to 1.39 play on its classic server [lab] (tools/lab-servers/README.md#tests-run-2026-10-30).

5.3.4 Version-dependent paths

Path	Rule	Source
version id	the token is looked up in a table; the position in the table is the version id, and comparisons are by id	[src] (tools/lab-servers/README.md#524-in-gs2libs-version-table-2026-10-31)
unknown token	an unknown token allowed by config gets id 0, sorts below everything, and is served every pre-2.1 path	[src] (server-notes/lab-servers-2026-10-30.md#what-our-gserver-does-with-each-client-generation-src)
level send below 2.1	the older GServer-v2 master sends a separate 1.41 level sequence with no 156; the newer branch has one level path for every version	[src] (server-notes/lab-servers-2026-10-30.md#2-gserver-v2-beta4-against-our-build)
default files	PLI_UPDATEFILE for a hard-coded list (idle.gani, walk.gani, sword?.png, body.png, the stock sounds, pics1.png and others) is answered "up to date" without checking times; PLI_WANTFILE has no such shortcut	[src] (server-notes/lab-servers-2026-10-30.md#what-our-gserver-does-with-each-client-generation-src)
default file to a pre-2.1 client	one build answers PLO_FILEUPTODATE (45), the other PLO_FILESENDFAILED (30); 2.05 accepts either	[lab] (server-notes/lab-servers-2026-10-30.md#beta4-built-and-run-against-real-clients-lab)

Path	Rule	Source
player class	chosen from the login type byte (type 0 is the Original class that 2.05 and 2.17 use), not from version config	[src] (server-notes/lab-servers-2026-10-30.md#beta4-built-and-run-against-real-clients-lab)
`generation` setting	server-wide; changes props, baddies, items and NPC details; 1.x clients need `classic`, and 2.x clients on a `classic` server get mangled packets	[src][lab] (server-notes/lab-servers-2026-10-30.md#beta4-built-and-run-against-real-clients-lab); [lab] (tools/lab-servers/README.md#133-to-139-on-beta4-2026-10-30)

5.3.5 Wire encodings the server writes

Item	Rule	Source
G-integers	`c` up to 223, `s` up to 12255, `i3` up to 1 568 735, `i4`, and `i5` for file size and time (→ §2.1.1)	[src] (server-notes/03-protocol.md#g-integers-cstring-writeg)
Pascal string	`c` length + raw bytes (the text is not offset)	[src] (server-notes/03-protocol.md#g-integers-cstring-writeg)
board	`PLO_RAWDATA` with `i3` 8194, then `PLO_BOARDPACKET` and 8192 raw tile bytes (64 × 64 u16)	[src] (server-notes/03-protocol.md#level-board)
file (2.1+)	`PLO_RAWDATA` with the total, then `PLO_FILE`: `i5` time, `c` name length, name, bytes (→ §2.7)	[src] (server-notes/03-protocol.md#file-send-21)
props	the prop encodings are in → §2.6	[src] (server-notes/03-protocol.md#player-props)

5.3.6 Word filter

Item	Rule	Source
file	`config/rules.txt` is the word filter, not a rules text players see	[srv] (client-internals/feature-specs/12-motd-rules.md#rules-in-gserver)
header	optional `WARNMESSAGE` and `SHOWWORDSTORC`, then `RULE` … `RULEEND` blocks	[srv] (client-internals/feature-specs/12-motd-rules.md#rules-in-gserver)
rule fields	`CHECK` (pm, chat, nick, toall), `MATCH` (? is a wildcard), `PRECISION` (number or percent), `WORDPOSITION` (full, start, part), `ACTION` (log, tellrc, replace, warn, jail, ban), optional per-rule `WARNMESSAGE`	[srv] (client-internals/feature-specs/12-motd-rules.md#rules-in-gserver)
applied to	chat (prop 12), PMs, toall and nickname changes	[srv] (client-internals/feature-specs/12-motd-rules.md#rules-in-gserver)
actions	`replace` swaps the matched letters for *; `warn` replaces the whole message with the warn text; `log` and `tellrc` go to the server log and RCs; `jail` and ban are stubs and do nothing	[srv] (client-internals/feature-specs/12-motd-rules.md#rules-in-gserver)
chat echo	when a chat line was changed, the server sends the new CURCHAT back to the sender so its own bubble shows the filtered text	[srv] (client-internals/feature-specs/12-motd-rules.md#rules-in-gserver)

5.3.7 Board respawn

Rule	Source
A change respawns only when its top-left tile before the change is one of 11 listed ids (bush, tall grass, pot, sign, stone, black stone and five others); otherwise it is kept with no timeout	[src] (client-internals/board-respawn.md#gserver-v2-lab-server-src)

Rule	Source
A change of any size can respawn, after exactly `respawntime` (default 15 s)	[src][cap] (client-internals/board-respawn.md#summary)
A new change deletes stored changes it fully covers	[src] (client-internals/board-respawn.md#gserver-v2-lab-server-src)
The change is relayed to the whole level, sender included	[cap] (client-internals/board-respawn.md#summary)
Restore swaps old and new tiles and keeps the record, so late joiners get the restored state	[src] (client-internals/board-respawn.md#gserver-v2-lab-server-src)
Drops, for clients at the 2.1 slot or above: bush, grass or swamp tiles with `bushitems` drop item `rand() % 6` `tiledroprate` % of the time (default 50); a pot with `vasesdrop` drops a heart	[src] (client-internals/board-respawn.md#gserver-v2-lab-server-src)
With `clientsidepushpull` on, a 4 × 4 change of a push-pull stone is refused and the board is sent back	[src] (client-internals/board-respawn.md#gserver-v2-lab-server-src)
The real 2.x GServer's timer, tile list and drop rates are unknown; no 2000–2002 official server binary has been found	[doc] (client-internals/board-respawn.md#open)

The 1.42 release moved bush and pot drops to the server [doc][bin] (client-internals/board-respawn.md#item-drops-from-142-on-server-side).

5.4 Accounts and allowed versions

5.4.1 Account stores

Server	Store	Layout	Source
1999 gserver tree	`accountslist.txt`, one account per line	`str` name, `str` password, a flag byte, a 3-byte file number	[src] (tools/classic-tools/gserverstat.md#what-it-reads)
1999 gserver tree	flag byte	bit 0 only-load, bits 1–3 admin level, bit 4 blocked	[src] (tools/classic-tools/gserverstat.md#what-it-reads)
1999 gserver tree	`accounts/accN.txt`	name, password, a prop list (`id + 32` and a value), the flag byte, two zero counts, weapons, action flags, opened chests	[src] (tools/classic-tools/gserverstat.md#what-it-reads)
1999 gserver tree	`accounts.txt`	read if present, imported, then removed	[src][bin] (tools/classic-tools/gserverstat.md#what-it-reads)
GServer 1.39 with MySQL	`accounts` table	`accname`, `encrpass` (traditional DES `crypt`), `email`, `blocked`, `onlyload`, `adminlevel`, `adminworlds`, `lastused`	[src][bin] (tools/server-tools-2001.md#gserver-139-web-scripts-2001)
gs2emu / GServer-v2	`GRACC001` files	text account files	[src] (server-notes/01-lineage.md#what-changed-between-eras)

Account rule	Source
The 1999 stat pages rank accounts by props 0 (nickname), 1 (max hearts), 3 (rupees, three 7-bit bytes), 27 (kills), 28 (deaths), 29 (online seconds) and 32 (alignment), and skip only-load accounts	[src][bin] (tools/classic-tools/gserverstat.md#what-it-reads)
GServer 1.39 servers wrote `playercount.txt`, `accountcount.txt`, `serveruptime.txt` and top lists as `nickname:value` lines for web pages	[src] (tools/server-tools-2001.md#gserver-139-web-scripts-2001)
The 1.39 chat commands listed for players: `sethead`, `setsword`, `setshield`, `setnick`, `showname`, `showap`, `showdeaths`, `showkills`, `showonlinetime`, `showguild`, `showadmins`, `unstuck me`, `toguild:`, `toall:`	[src] (tools/server-tools-2001.md#gserver-139-web-scripts-2001)
GServer-v2's stock default account has `MAXHP 3.00`, which its loader reads as 0 hearts	[lab] (tools/lab-servers/README.md#133-to-139-on-beta4-2026-10-30)

5.4.2 Allowed versions

The 1999 servers accept a fixed list of `GSERV` tokens with a minimum per connection type (→ §2.3.2). GServer-v2 reads the list from a file:

Rule	Source
Each line of `config/allowedversions.txt` is a token or a range `A:B`; a range allows every token whose position in the version table lies between A's and B's, both included	[src] (server-notes/public-allowed-versions.md#how-the-server-builds-the-list-src)
A token not in the table gets id 0 and lies in no range; a bare unknown token on its own line also resolves to 0, and so admits every unknown token	[src] (server-notes/public-allowed-versions.md#how-the-server-builds-the-list-src); [src] (tools/lab-servers/README.md#524-in-gs2libs-version-table-2026-10-31)
The refusal is "Your client version is not allowed on this server." followed by "Allowed:" and the list, each token printed by its table name and a range as `A - B`	[cap][src] (server-notes/public-allowed-versions.md#the-refusals-on-record)
A server prints its list only to a client it refuses	[inf] (server-notes/public-allowed-versions.md#open)
The stock file of the older master enables only `GNW03014` (2.22)	[src] (server-notes/public-allowed-versions.md#how-the-server-builds-the-list-src)
The newer branch groups ranges by `generation`: `classic` = 1.41r1 only, `newmain` = 2.22 to 2.31, `modern` = 5.12 to 6.037; stock, it refuses 2.05 and 2.17	[src] (server-notes/lab-servers-2026-10-30.md#beta4-built-and-run-against-real-clients-lab); [src] (tools/lab-servers/README.md#524-in-gs2libs-version-table-2026-10-31)
The listserver does not check the version; whether a build gets in is up to each game server's list	[cap] (server-notes/public-allowed-versions.md#what-that-means-for-5x-and-6x)

Token	Build	Name in the version table	Source
`G3D22067`	5.00r7	5.07	[src] (server-notes/public-allowed-versions.md#what-that-means-for-5x-and-6x)
`G3D14097`	5.12	5.12	[src] (server-notes/public-allowed-versions.md#what-that-means-for-5x-and-6x)
`G3D26090`	5.30r3, 5.31r4, 6.00r2, 6.00r4	5.31x	[src] (server-notes/public-allowed-versions.md#what-that-means-for-5x-and-6x)
`G3D3007A`	6.01x	6.015	[src] (server-notes/public-allowed-versions.md#what-that-means-for-5x-and-6x)
`G3D2505C`	6.034, 6.035	6.034/6.035	[src] (server-notes/public-allowed-versions.md#what-that-means-for-5x-and-6x)
`G3D0311C`, `G3D0511C`	6.03r7 Windows, Linux	6.037, 6.037 (Linux)	[src] (server-notes/public-allowed-versions.md#what-that-means-for-5x-and-6x)
`G3D15010`	5.24r3, 5.24r4, 6.00 beta2	not in the table	[src] (server-notes/public-allowed-versions.md#what-that-means-for-5x-and-6x)
`G3D03048`	6.00 beta1	not in the table	[src] (server-notes/public-allowed-versions.md#what-that-means-for-5x-and-6x)
`GNW03081`	2.05 rev 2	2.05r2 (newer branch only)	[src] (server-notes/lab-servers-2026-10-30.md#2-gserver-v2-beta4-against-our-build)

Refusal seen from a public server	Client token sent	Allowed list printed	Source
a GServer-v2 server reporting 4.0.0	`GNW14015` (2.3)	2.22, 5.07	[cap] (server-notes/public-allowed-versions.md#the-refusals-on-record)
a second server (signature 73)	`GNW03081` (2.05)	2.22, and the range 1.41r1 to iPhone Belote 1.5	[cap] (server-notes/public-allowed-versions.md#the-refusals-on-record)

The second server refused 2.05 although `GNW03081` lies inside its printed range in the current version table (→ App. E#404). No 5.x or 6.x client has logged in to a public server [cap] (server-notes/public-allowed-versions.md#which-client-versions-public-reborn-game-servers-allow).

5.5 NPC server

5.5.1 The 2001 NPC server

Rule	Source
The NPC server is a separate program that logs into the GServer like a player and owns the NPCs of a configured set of levels	[doc] (tools/npcserver-2001.md#1-what-the-npc-server-is)
Its main loop runs every 0.1 s: read from the GServer, run all NPC scripts, push changed NPC and player attributes, then sleep; 0.1 s is the smallest useful server-side `sleep`	[doc] (tools/npcserver-2001.md#1-what-the-npc-server-is)
Startup: load options, connect to the GServer, then (if connected) load the level list, then load NPCs from its database	[doc] (tools/npcserver-2001.md#1-what-the-npc-server-is)
From the GServer it receives player attributes, temporary level state (bombs, explosions, bush removal, dropped items), PMs, game time, levels to reload, and `server.` flag changes; the GServer decides what reaches clients	[doc] (tools/npcserver-2001.md#1-what-the-npc-server-is)
To the GServer it sends NPC props, player props changed by scripts, PM replies, weapon grants and warps	[inf][doc] (tools/npcserver-2001.md#implied-gserver-link-inference-needs-a-capture-to-confirm)
The packet ids and port of the GServer–NPC-server link are unmeasured	[doc] (tools/npcserver-2001.md#open-questions)
Scripts run on the server by default; `//#CLIENTSIDE` splits a script, and the part after it is sent to clients	[doc] (tools/npcserver-2001.md#2-server-side-scripting-model)
Server-only events: `created`, `initialized`, `playerenters`, `playerleaves` (not on disconnect), `playertouchsme` (needs `setshape`), `playerchats`, `npcwarped`, `exploded`	[doc] (tools/npcserver-2001.md#2-server-side-scripting-model)
The server never loads images, so `setshape 1,w,h` gives an NPC its touch rectangle	[doc] (tools/npcserver-2001.md#3-commands)

Variable	Scope	Persisted	Source
`name`	current script or class	no	[doc] (tools/npcserver-2001.md#4-variables-and-flags)
`this.name`	current NPC	yes, for database NPCs	[doc] (tools/npcserver-2001.md#4-variables-and-flags)
`level.name`	the NPC's level	no	[doc] (tools/npcserver-2001.md#4-variables-and-flags)
`server.name` (number)	this NPC server only	no	[doc] (tools/npcserver-2001.md#4-variables-and-flags)
flags	current player	yes, in the account	[doc] (tools/npcserver-2001.md#4-variables-and-flags)
`client.` flags	current player, also sent to the client, which can change them	yes	[doc] (tools/npcserver-2001.md#4-variables-and-flags)
`this.` flags	current NPC	yes, for database NPCs	[doc] (tools/npcserver-2001.md#4-variables-and-flags)
`server.` flags	world-wide, shared with other NPC servers of the world, never sent to players	yes	[doc] (tools/npcserver-2001.md#4-variables-and-flags)

Command group	Commands	Source
flow and core	`destroy`, `if`, `while`, `for`, `break`, `continue`, `sleep`, `set`, `unset`, `setstring`, `message`	[doc] (tools/npcserver-2001.md#3-commands)

Command group	Commands	Source
NPC appearance and state	setgif/setimg, setgifpart/setimgpart, setshape, drawoverplayer, drawunderplayer, dontblock, blockagain, show, hide, noplayeronwall, showcharacter, setcharprop, setcharani, setchargender, putbomb, putexplosion, putexplosion2, setarray	[doc] (tools/npcserver-2001.md#3-commands)
server-side additions	canwarp, canwarp2, cannotwarp, callnpc, with (players[i]), with (getplayer(acct)), with (npcs[i]), with (getnpcs(name)), join, warpto, tokenize, tokenize2	[doc] (tools/npcserver-2001.md#3-commands)
acting on the current player	setplayerprop, setani, setgender, say, enableweapons, disableweapons, toweapons, addweapon, removeweapon, setlevel2, freezeplayer2, unfreezeplayer	[doc] (tools/npcserver-2001.md#3-commands)
string helpers	strlen, startswith(part,str), indexof(part,str) (−1 if absent), #e(start,len,str), #T(str) (trim), tokenscount, #t(i)	[doc] (tools/npcserver-2001.md#3-commands)

Command	Rule	Source
noplayeronwall	onwall and onwall2 stop counting players on this NPC	[doc] (tools/npcserver-2001.md#3-commands)
canwarp	database NPCs only; the NPC warps through links exactly as a player would	[doc] (tools/npcserver-2001.md#3-commands)
canwarp2	database NPCs only; for wandering overworld NPCs: normal links, checks onwall at the destination, refuses small links such as house doors	[doc] (tools/npcserver-2001.md#3-commands)
with	swaps the current player or NPC for the block; commands are skipped if the target does not exist; local variables still resolve to the caller; sleep, break and continue inside the block leave the swapped context in place	[doc] (tools/npcserver-2001.md#3-commands)
join class	loads class.txt from the scripts folder; an NPC holds its own script then each joined class; a class loaded by another NPC is shared, so updating it updates every member; later scripts can override functions	[doc] (tools/npcserver-2001.md#3-commands)
warpto	moves the NPC to another level; database NPCs only	[doc] (tools/npcserver-2001.md#3-commands)
addweapon	like toweapons, but the weapon script comes from the database and the weapon must already exist	[doc] (tools/npcserver-2001.md#3-commands)
tokenize, tokenize2	tokenize splits on spaces and commas; tokenize2 adds extra delimiters	[doc] (tools/npcserver-2001.md#3-commands)

Client support for these commands per build is in → §8.1 and → §8.2; the PM admin interface and the NC socket that replaced it are in → §7.3.

5.5.2 Message codes on a script-running server

Rule	Source
In a setcharprop value, #n expands to the nickname of the player who triggered the script; #n(id) picks another player by id	[srv] (server-notes/setcharprop-hash-n.md#answer)

Rule	Source
`setcharprop` always expands its second argument before applying it	[srv] (server-notes/setcharprop-hash-n.md#2-server-side-gs1-gserversource-archive)
When the first argument is #n, the expanded text becomes the NPC's nickname and its `showcharacter` nickname, and the NPC nickname prop is marked for sending	[srv] (server-notes/setcharprop-hash-n.md#2-server-side-gs1-gserversource-archive)
#F is the current player's level name, which can differ from #L (the NPC's level)	[doc] (tools/npcserver-2001.md#6-message-codes)
A capture that proves the server-side #n expansion on a script-running server is still open	[inf] (server-notes/setcharprop-hash-n.md#still-open)

5.5.3 Later NPC servers

Server	Rule	Source
GServer-v2, newer branch	adds a GS1 NPC server, tested on the 2.x generation; the V8 GS2 NPC server stays optional	[src] (server-notes/lab-servers-2026-10-30.md#2-gserver-v2-beta4-against-our-build)
GServer-v2, newer branch, own notes	1.38 to 1.411 and the 2.x/3.x generation are supported client-authoritative; clientside GS1 works up to 5.007; server-side GS2 is not supported	[src] (server-notes/lab-servers-2026-10-30.md#2-gserver-v2-beta4-against-our-build)
the Python gs2emu rewrite	announces the NPC server to clients and relays the address query (→ §7.2)	[src] (server-notes/ash-overview.md#what-ash-implements)

6 Listserver

The listserver gives a client the list of game servers and, from 2.22, checks its account. Two links exist: client ↔ listserver (§6.1 to §6.4) and game server ↔ listserver (§6.5) [srv] (protocol/listserver.md#listserver-protocol). Framing and ciphers are in → §2.1 and → §2.2. From 4.x the first connection goes to a game server named `Login` instead (→ §2.3.5).

> Status: draft. Real-client captures exist for 2.00, 2.05, 2.17.1, 2.22 and 2.31 against a lab listserver only; the public-service runs used a reimplementation's frames. No 2000–2002 listserver binary or capture exists.

6.1 Requests

6.1.1 Where each build looks for the list

Builds	Host and port	List name	Source
1.33 to 1.39	no listserver; the player types a game-server address	n/a	[bin] (protocol/login-frames-by-version-2026-10-25.md#real-versus-seihai-per-profile-seihai-main-21a37a5); [lab] (tools/lab-servers/README.md#133-to-139-on-beta4-2026-10-30)
1.41	unmeasured: only a patched copy survives	unmeasured	[bin] (protocol/login-frames-by-version-2026-10-25.md#the-three-real-shapes)
1.42	`<listserver-host>` built in, port 14921	newmain	[bin] (protocol/login-frames-by-version-2026-10-25.md#the-three-real-shapes)
2.00 to 2.05	built in as a DES-hidden string assembled one character at a time, port 14921; `arrays2.dat` is not read	newmain	[2.05] (protocol/listserver.md#205-specifics-205); [bin] (protocol/login-frames-by-version-2026-10-25.md#the-three-real-shapes)
2.1 test builds	port 14922 built in	unmeasured	[bin] (protocol/login-frames-by-version-2026-10-25.md#the-three-real-shapes)
2.17 to 2.31	`arrays2.dat`, checked against `arrays2.sig`: line 0 host, line 1 port (14922), lines 5 and 6 fallback hosts, tried in order (→ §3.5)	newmain	[bin] (protocol/listserver.md#217-list-host-bin); [2.22] [cap] (protocol/listserver.md#222-specifics-222cap)
RC 1.41 (2002)	a DES-hidden host to which the code appends the top-level domain, port 14922	rc	[bin] (tools/npc-control.md#listserver-login-new-in-141-bin)

The 2.22 client keeps the hosts from `arrays2.dat` in memory under a one-byte transform, so a plain memory search misses them [2.22] (protocol/listserver.md#222-specifics-222cap).

6.1.2 Client → listserver ids

Id	Name	Layout	Source
0	V1VER	`rest` = list name, such as `newmain` (RC 1.41: `rc`)	[srv] (protocol/listserver.md#client--listserver-1); [bin] (tools/npc-control.md#listserver-login-new-in-141-bin)
1	SERVERLIST	`str account`, `str password`	[srv] (protocol/listserver.md#client--listserver-1)
4	V2VER	`c key` + `newmain`; selects the 2.2 bz2 mode	[srv] (protocol/listserver.md#client--listserver-1)
5	V2SERVERLISTRC	`c key` (RC)	[srv] (protocol/listserver.md#client--listserver-1)

Id	Name	Layout	Source
7	V2ENCRYPTKEYCL	c key + GNW30123 + newmain	[bin] (protocol/listserver.md#client--listserver-1)
223	GRSECURELOGIN	a Graal Reborn extension	[srv] (protocol/listserver.md#client--listserver-1)

6.1.3 Login shapes

The listserver login differs by client generation, not by build; inside a generation every observed build sends the same bytes [cap] (protocol/login-frames-by-version-2026-10-25.md#short-answer).

Shape	Builds seen with a real client	Client → listserver	Source
A: list name only	2.00 (client side), 2.05 (port 14921)	one plain zlib frame: id 0 + newmain + \n (9 bytes decoded, 17 on the wire); no account, password, token or key; the client then closes	[cap] (protocol/login-frames-by-version-2026-10-25.md#the-three-real-shapes)
B: id 0 then id 1	2.17.1 (port 14922 from its signed arrays2.dat)	one plain zlib frame holding id 0 newmain and id 1 (str account, str password); no token, key or cipher	[cap] (protocol/login-frames-by-version-2026-10-25.md#the-three-real-shapes); [cap] (protocol/listserver.md#217-list-host-bin)
C: id 7, then generation 5	2.22, 2.31 (port 14922)	frame 1, plain zlib, 18 bytes decoded: id 7, one key byte, GNW30123, newmain, \n; frame 2, generation 5 with compression byte 0x02: id 1, XORed with the key	[cap] (protocol/login-frames-by-version-2026-10-25.md#the-three-real-shapes)
generation 4	2.19 to 2.21	id 4 c key + token, then bz2 frames with partial XOR (limit 4)	[srv] (protocol/listserver.md#client--listserver); [lab] (tools/lab-servers/README.md#our-listserver-version-2)
RC 1.41	RC 1.41 (2002)	id 0 rc, then id 1 str account, str password	[bin] (tools/npc-control.md#listserver-login-new-in-141-bin)

Rule	Source
GNW30123 is the listserver token of 2.22 and 2.31, shared by both; it is not a game build token (2.22 logs in to a game server as GNW03014, 2.31 as GNW28015)	[bin][cap] (protocol/listserver.md#222-specifics-222cap); [cap] (protocol/login-frames-by-version-2026-10-25.md#the-three-real-shapes)
Per-build tokens such as GNW24011 (2.00), GNW03081 (2.05) and GNW22122 (2.17.1) go to the game server, never to the listserver	[cap] (protocol/login-frames-by-version-2026-10-25.md#the-three-real-shapes)
The classic frames (A and B) carry no field that could name a build	[cap] (protocol/login-frames-by-version-2026-10-25.md#does-the-evidence-support-classic-logins-look-fake)
The real clients compress at zlib's best level (stream starts 78 DA); a 78 9C stream with the same deflate body is answered by the public service for shape C	[cap] (protocol/login-frames-by-version-2026-10-25.md#one-byte-level-difference-that-applies-to-every-seihai-frame)
The 2.05 list connection derives from the game connection's base class and keeps its base send path, which has no generation-3 insertion	[2.05] (protocol/listserver.md#205-specifics-205)
A real 2.00 sends one frame (id 0 + newmain), makes no second listserver connection, and accepts a 2.17-style two-frame reply from a lab forwarder	[lab] (protocol/listserver.md#205-specifics-205); [cap] (client-internals/decomp-findings-2026-10-24-b/07-lab-findings-46-186-200.md#200-listserver-handshake-client-side-observed-list-reply-synthetic); [cap] (version-profiles/lab-200-listserver-handshake.tsv)
After the list, 2.17 logs in to the game server as type 0 with a key, GNW22122, account and password, in generation 3	[cap] (protocol/listserver.md#217-list-host-bin)

Rule	Source
What 1.41, 1.42, 2.01 to 2.04 and 2.1 to 2.16 send is unmeasured; 1.42 and 2.01 to 2.04 share 2.00's built-in list data and are expected to send shape A	[inf] (protocol/login-frames-by-version-2026-10-25.md#real-versus-seihai-per-profile-seihai-main-21a37a5)

6.2 Replies

6.2.1 Listserver → client ids

Id	Name	Layout	Source
0	SVRLIST	`c` count, then one record per server (→ §6.4)	[srv][cap] (protocol/listserver.md#listserver--client)
1	NULL	disputed → App. E#401	[srv] (protocol/listserver.md#listserver--client); [bin] (tools/npc-control.md#listserver-login-new-in-141-bin)
2	STATUS	disputed → App. E#401	[srv] (protocol/listserver.md#listserver--client); [bin] (tools/npc-control.md#listserver-login-new-in-141-bin)
3	SITEURL	URL	[srv] (protocol/listserver.md#listserver--client)
4	ERROR	text; the connection closes	[srv] (protocol/listserver.md#listserver--client); [bin] (tools/npc-control.md#listserver-login-new-in-141-bin)
5	UPGURL	donate or upgrade URL	[srv] (protocol/listserver.md#listserver--client)

Rule	Builds	Source
The receive handler switches on ids 0 and 1 only	2.05	[2.05] (protocol/listserver.md#205-specifics-205); [2.05] (client-internals/client-architecture-205.md#networking-classes)
A client fills the list as soon as id 0 arrives after its id 0 `newmain`; it does not wait for id 1, and closes the socket about 2 s after it gets the list	2.05	[cap] (protocol/listserver.md#205-specifics-205)
The window shows the fields in record order: name, language, description, URL, version, players, address, port	2.05	[cap] (protocol/listserver.md#205-specifics-205)
The list client acts on ids 0 and 1 only; any other id, including 2, 3 and 5, is ignored	2.00	[bin] (client-internals/v2.00-connect-path.md#list-reply-bin)
Id 1 carries one server record in the id 0 record layout and appends it to the list; when it is then the only row, it is selected	2.00	[bin] (client-internals/v2.00-connect-path.md#list-reply-bin)
After id 0 the client fills the list and closes the list socket; a row is drawn only with at least 6 fields, and the detail view needs at least 8	2.00	[bin] (client-internals/v2.00-connect-path.md#list-reply-bin)
RC 1.41 reads 0 (server list), 1 (one more server entry), 2 (end, disconnect) and 4 (error text, disconnect), and shows name and player count	RC 1.41	[bin] (tools/npc-control.md#listserver-login-new-in-141-bin)
The 2004 RC logs in with client id 5, an empty token and an empty list name, then `str account`, `str password`; it shows name and player count from an id 0 list, and id 4 text in a message box	RC 2004	[cap] (protocol/rc-capture-2026-11-04.md#rc-list-from-the-listserver)
2.22 shows the status lines and the Show More and Donations buttons, and files tagged servers under tabs (→ §6.4)	2.22	[lab] (tools/lab-servers/README.md#tests-run-2026-10-30)

6.2.2 Reply frames seen

Login	Reply	Source
shape B from a real 2.17, lab listserver	two plain zlib frames: id 0 (the list); then id 2 status, id 3 site URL, id 5 donation URL	[cap] (protocol/listserver.md#217-list-host-bin)

Login	Reply	Source
shape C, port 14922 of the public service	two generation-5 frames with compression byte `0x04` (zlib): id 0 list; then id 2 status, id 3 site URL, id 5 upgrade URL	[cap] (protocol/public-listserver-2026-10-25.md#what-the-runs-settle)
shape C, port 14923 of the public service	three generation-5 frames: the legacy list first, sent straight after the id 7 frame before the client's id 1, then the two frames above	[cap] (protocol/public-listserver-2026-10-25.md#what-the-runs-settle)
classic (shape B frame), port 14923 of the public service	one plain zlib frame, id 0 only, no status, site or upgrade lines, no trailing NUL bytes; the server then waits	[cap] (protocol/public-listserver-2026-10-25.md#what-the-runs-settle)

Detail	Value	Source
trailing bytes	generation-5 frames end with two `0x00` bytes after the last \n; the plain legacy frame does not	[cap] (protocol/public-listserver-2026-10-25.md#frame-and-field-layout-derived)
status frame	id 2 text of two lines joined by CR (a welcome line and "N server(s) online"), id 3 site URL, id 5 upgrade URL, \n after each	[cap] (protocol/public-listserver-2026-10-25.md#frame-and-field-layout-derived)
status count	the number in the status text is the server's own text and need not match the row count	[cap] (protocol/public-listserver-2026-10-25.md#what-the-runs-settle)
timing, port 14922	both frames about 0.4 s after the client's second frame, on a link with a 0.2 s round trip	[cap] (protocol/public-listserver-2026-10-25.md#timing-users-link-round-trip-about-02-s)
timing, port 14923, shape C	the legacy list within a few ms of the handshake; the two current frames about 0.35 s after the client's id 1	[cap] (protocol/public-listserver-2026-10-25.md#timing-users-link-round-trip-about-02-s)
timing, port 14923, classic	about 0.2 s after the login	[cap] (protocol/public-listserver-2026-10-25.md#timing-users-link-round-trip-about-02-s)

6.2.3 The classic `serverlist3` listserver

Rule	Source
One process with two listen ports, one for clients and one for game servers, set by `clientsport` and `serversport` in a `key=value` config; also `playerlimit`, `databaseserver` and `showmore`	[bin] (tools/classic-tools/serverlist3.md#start-and-config)
Frame: `u16be` length + zlib; lines start with `id + 32`; strings are `count + 32` and the bytes; an inflated buffer past 16383 bytes or an outgoing buffer past 4095 bytes counts as a broken link	[bin] (tools/classic-tools/serverlist3.md#frame)
Client id 0 is the edition text, which filters the directory to rows whose edition contains it and that were written in the last 10 minutes; id 1 is account and password; other ids are ignored; there is no key, token or cipher	[bin] (tools/classic-tools/serverlist3.md#what-a-client-sends)
Passwords are checked locally against a `crypt` hash of the first 8 characters with a two-character salt	[bin] (tools/classic-tools/serverlist3.md#what-a-client-sends)
A hard failure (unknown account, wrong password, ban) is id 4 with the text, then the link closes	[bin] (tools/classic-tools/serverlist3.md#what-the-client-is-shown)
A success is id 0 (the directory) and id 2 (status text), then the link closes; the trial and free-server texts also send id 3 first with the `showmore` URL	[bin] (tools/classic-tools/serverlist3.md#what-the-client-is-shown)
Status texts: a welcome line, a staff sentence, a remaining-time sentence, the trial sentence or the free-server sentence	[bin] (tools/classic-tools/serverlist3.md#what-the-client-is-shown)

6.3 Login-type matrix

On 2026-10-25 the public listserver was asked for its list 38 times, once per port for 19 version profiles of a reimplementation, with no game-server login [cap] (protocol/listserver-login-type-matrix-2026-10-25.md). The service chooses its reply by **login type**, not by version or token [cap] (protocol/listserver-login-type-matrix-2026-10-25.md#result).

Login sent	Profiles	Port 14922	Port 14923	Source
classic: one plain zlib frame with id 0 `newmain` and id 1	1.41, 1.42, 2.00, 2.01, 2.02, 2.02 rev 2, 2.03, 2.04, 2.1, 2.12 to 2.17	connection accepted, no reply and no close for the 5 s timeout, in all 15	legacy list in all 15: one plain zlib id 0 frame, 2 or 4 rows	[cap] (protocol/listserver-login-type-matrix-2026-10-25.tsv); [cap] (protocol/listserver-login-type-matrix-2026-10-25.md#result)
generation 5: id 7 + `GNW30123` + `newmain`, then id 1	2.22, 2.3, 2.31, and the 2.05 profile of the run	modern list (8 rows) with status, site and upgrade lines	the legacy list first, then the modern list (12 or 17 rows in all) with the status lines	[cap] (protocol/listserver-login-type-matrix-2026-10-25.tsv); [cap] (protocol/listserver-login-type-matrix-2026-10-25.md#result)
any	2.05 profile	port 14921: connection refused at once	n/a	[cap] (protocol/public-listserver-2026-10-25.tsv); [cap] (protocol/public-listserver-2026-10-25.md#what-the-runs-settle)

Rule	Source
A port that gets a list: a classic login on 14923; a generation-5 login on 14922 (14923 also answers it)	[cap] (protocol/listserver-login-type-matrix-2026-10-25.md#result)
The legacy list's row count varied between runs (2 or 4) with no link to the version; the 2-row form carried `P` and `H` prefixes and the 4-row form none	[cap] (protocol/listserver-login-type-matrix-2026-10-25.md#result); [cap] (protocol/login-frames-by-version-2026-10-25.md#what-the-matrix-shows-public-service-seihai-only)
The 2.05 rows of the matrix are a generation-5 login, not a classic 2.05 login; a classic 2.05 login was not run	[cap] (protocol/login-frames-by-version-2026-10-25.md#problem-in-the-matrix-data-the-205-rows-are-not-a-classic-205-login)
No real client of these versions was captured against the public service; the runs say which login type the service answers on which port, not what a real 1.41 to 2.17 client sends	[cap] (protocol/listserver-login-type-matrix-2026-10-25.md#what-this-does-not-show)
The 2001-era server tools list clients on 14922 and older clients on 14923 (`clientPort0ld`)	[src] (tools/server-tools-2001.md#gs2emu-serverlist-thedgtl-snapshot-de66a3f)

Why the public 14922 ignores a classic login, and what it does with a real 2.17.1, is open (→ App. E#403). Whether the 14923 legacy list is a real 2000-era reply is disputed (→ App. E#402).

6.4 Server-list records

Field	Content	Source
—	each record starts with `c` field count, 8 in every capture	[cap] (protocol/public-listserver-2026-10-25.md#frame-and-field-layout-derived)
1	type prefix and name	[srv][cap] (protocol/listserver.md#listserver--client)
2	language	[srv][cap] (protocol/listserver.md#listserver--client)
3	description	[srv][cap] (protocol/listserver.md#listserver--client)
4	URL	[srv][cap] (protocol/listserver.md#listserver--client)
5	version text	[srv][cap] (protocol/listserver.md#listserver--client)
6	player count, as text	[srv][cap] (protocol/listserver.md#listserver--client)
7	address, as text	[srv][cap] (protocol/listserver.md#listserver--client)
8	port, as text	[srv][cap] (protocol/listserver.md#listserver--client)

Every field is a `str` [cap] (protocol/public-listserver-2026-10-25.md#frame-and-field-layout-derived). The list body ends with `\n` after the last record [cap] (protocol/public-listserver-2026-10-25.md#frame-and-field-layout-derived).

Type prefix	Tier or tab	HQ level in the lab listserver	Source
`U`	hidden	0	[srv] (protocol/listserver.md#listserver--client); [lab] (tools/lab-servers/README.md#our-listserver-version-2)

Type prefix	Tier or tab	HQ level in the lab listserver	Source
H	bronze (hosted)	1	[srv] (protocol/listserver.md#listserver--client); [lab] (tools/lab-servers/README.md#our-listserver-version-2)
none	silver, shown as "Classic" by 2.17.1	2	[srv] (protocol/listserver.md#listserver--client); [lab] (tools/lab-servers/README.md#tests-run-2026-10-30)
P	gold	3	[srv] (protocol/listserver.md#listserver--client); [lab] (tools/lab-servers/README.md#our-listserver-version-2)
3	3D	4	[srv] (protocol/listserver.md#listserver--client); [lab] (tools/lab-servers/README.md#our-listserver-version-2)

Rule	Source
Pre-2.2 clients get no prefix for bronze	[srv] (protocol/listserver.md#listserver--client)
Connect needs a selected record with at least 8 fields and takes the address and port from fields 7 and 8; with either empty it shows "The connection could not be established!" without dialling (2.00)	[bin] (client-internals/v2.00-connect-path.md#connect-bin)
A missing core asset (`objects.dat`, `arrays.dat` or one of the main images) sets a fatal flag at Start that is never cleared; the server list still works, but Connect then returns before creating the player, leaving a grey game window with no socket (2.00)	[bin] (client-internals/v2.00-connect-path.md#the-fatal-flag); [cap] (client-internals/v2.00-connect-path.md#lab-rule-for-200)
The public 14923 legacy list sent untagged rows in its 4-row form; 3 of 4 rows also appear in the current list with the same address, port and version text	[cap] (protocol/public-listserver-2026-10-25.md#frame-and-field-layout-derived)
A 2003 fan listserver sets the `P` prefix for a server marked `Favorite=Yes`, so the client files it under Gold	[bin] (tools/server-tools-2001.md#win32-listserver-by-rick-2003--contains-a-password-stealer-bin)
Field bytes are passed through as sent: a captured address field had lost its leading characters, and a name carried three literal `0x3F` bytes	[cap] (protocol/public-listserver-2026-10-25.md#what-the-runs-settle)

6.5 Game server link

Rule	Source
Listserver packets on this link have no length prefix: `id + payload + \n`	[src] (server-notes/03-protocol.md#protocol)
Game servers register on port 14900 in the Reborn listserver	[src] (tools/server-tools-2001.md#gs2emu-serverlist-thedgtl-snapshot-de66a3f)
A GServer-v2 opens with a plain `REGISTERV3` line, then zlib frames: `NEWSERVER`, `SERVERHQPASS` / `SERVERHQLEVEL`, `SETPLYR` and `PLYRADD` per player, `VERIACC2` per login, and `SENDTEXT` with its allowed versions; the older master never pings	[src] (server-notes/lab-servers-2026-10-30.md#what-our-gserver-does-with-each-client-generation-src)
The newer GServer-v2 sends `SETIP` every 60 s as a keep-alive, and its `VERIACC2` carries a `s` identity length	[src] (server-notes/lab-servers-2026-10-30.md#2-gserver-v2-beta4-against-our-build)
Id 16 from a GServer-v2 is a pong and 99 is the listserver's ping	[src] (tools/lab-servers/README.md#what-changed-from-the-stub-of-2026-10-22)

Direction	Id	Name	Source
server → list	0	SETNAME	[src] (server-notes/03-protocol.md#listserver-ids)
server → list	7	SETPLYR	[src] (server-notes/03-protocol.md#listserver-ids)
server → list	14	PLYRADD	[src] (server-notes/03-protocol.md#listserver-ids)
server → list	15	PLYRREM	[src] (server-notes/03-protocol.md#listserver-ids)

Direction	Id	Name	Source
server → list	16	PING	[src] (server-notes/03-protocol.md#listserver-ids)
server → list	17	VERIACC2	[src] (server-notes/03-protocol.md#listserver-ids)
server → list	22	NEWSERVER	[src] (server-notes/03-protocol.md#listserver-ids)
server → list	23	SERVERHQPASS	[src] (server-notes/03-protocol.md#listserver-ids)
server → list	24	SERVERHQLEVEL	[src] (server-notes/03-protocol.md#listserver-ids)
list → server	8	ERRMSG	[src] (server-notes/03-protocol.md#listserver-ids)
list → server	11	VERIACC2	[src] (server-notes/03-protocol.md#listserver-ids)
list → server	15 to 17	FILESTART3, FILEDATA3, FILEEND3	[src] (server-notes/03-protocol.md#listserver-ids)
list → server	19, 20	request and send text	[srv] (protocol/listserver.md#game-server--listserver-summary-srv)
list → server	29	PM player	[srv] (protocol/listserver.md#game-server--listserver-summary-srv)
list → server	30	assign PC id	[srv] (protocol/listserver.md#game-server--listserver-summary-srv)
list → server	99	PING	[src] (server-notes/03-protocol.md#listserver-ids)
list → server	100	RAWDATA	[src] (server-notes/03-protocol.md#listserver-ids)

The other server-to-list ids (0 to 31) set description, language, version, URL, address, port and player count, verify guilds, get and set profiles, request files, the list, server info and buddies, and PM a player [srv] (protocol/listserver.md#game-server--listserver-summary-srv). No open listserver serves files to a game server [src] (server-notes/lab-servers-2026-10-30.md#3-listservers).

Classic `serverlist3` game-server link	Source
The first line is the login: kind 0 or 1, then `GLSER001` or `GLSER002`, then the world name; the socket address must match the address stored for that world	[bin] (tools/classic-tools/serverlist3.md#what-a-game-server-sends)
Id 4: player add or update (2-byte id, a flag byte, c length, name); with flag bit 1 set and kind 0 the listserver answers id 7 with the player id and, when queued, "N ghost(s) before you"	[bin] (tools/classic-tools/serverlist3.md#what-a-game-server-sends)
Id 5: player leave (2-byte id)	[bin] (tools/classic-tools/serverlist3.md#what-a-game-server-sends)
The public directory is a database table, not built from these messages; the link carries presence, the ghost queue and the address check	[bin] (tools/classic-tools/serverlist3.md#what-a-game-server-sends)

Reborn listserver data	Source
Tables for users (account, password, salt, activation, ban, profile fields), servers (name, address, port, players, description, URL, language, a two-character type, version, uptime), per-server HQ password and level, IP bans, guilds and guild members	[src] (tools/server-tools-2001.md#gs2emu-serverlist-thedgtl-snapshot-de66a3f)
Account passwords are `md5(md5(password) + salt)` with a three-character salt	[src] (tools/server-tools-2001.md#gs2emu-serverlist-thedgtl-snapshot-de66a3f)

7 RC and NC

RemoteControl (RC) is the staff client for a game server; NPC Control (NC) is its connection to the NPC server (→ §5.5). Both are Delphi programs that reuse the game client's connection class [bin] (tools/npc-control.md#program-shape-bin). The game-server login types are in → §2.3.6.1, framing in → §2.1.4 and ciphers in → §2.2. Staff ids that game clients also handle are in → §2.4.10 and → §2.5.9.

> Status: draft. Every layout here is a static read of the RC binaries, except those in → §7.1.8, which one capture of the 2004 RC against a GServer-v2 fork shows. The checks that would settle the rest are listed in the protocol note's loopback list.

7.1 RC login and commands

7.1.1 Builds

Archive	Title date	Login token	Forms	Adds	Source
rc133	none	`GSERV014`	7	accounts, players, send, update levels	[bin] (tools/remotecontrol-lineage.md#builds)
rc139rev2	2000/8/19	`GSERV019`	15	chat window, flag lists, toalls, SQL account list, options, warp	[bin] (tools/remotecontrol-lineage.md#builds)
rc1025	2000/10/25	`GSERV020`	15	same forms	[bin] (tools/remotecontrol-lineage.md#builds)
rc62101	2001/01/27	`GSERV021`	15	same forms	[bin] (tools/remotecontrol-lineage.md#builds)
rc8212001	2001/08/20	`GSERV022`	18	rights, comments, local ban; server options leave the start dialog	[bin] (tools/remotecontrol-lineage.md#builds)
rc201	2002/01/04	`GSERV022`	18	new main window layout	[bin] (tools/remotecontrol-lineage.md#builds)
rc141	2002/05/02	`GSERV023`	26	NPC Control, server list, file browser, short message popup	[bin] (tools/remotecontrol-lineage.md#builds); [bin] (tools/npc-control.md#program-shape-bin)
RemoteControl in the Graal Reborn folder (2004)	n/a	`GSERV024`	25	RC and NC in one program, file browser, server list	[bin] (tools/remotecontrol.md)

The archive names do not sort by date: rc141 is the newest of the 2000–2002 set [bin] (tools/remotecontrol-lineage.md). RemoteControl builds of 1999 log in with `GSERV007` (shipped with 1.21) and `GSERV011` (shipped with 1.28 and 1.3) [bin] (protocol/gserv-1999.md#login). A GServer-v2 also knows `GSERV025` (→ App. E#405).

7.1.2 Login

Builds	First message	Mode during login	Mode after	Source
1999 to rc141 (2002)	type 1, token, `str account`, `str password`; no key byte	u16be length + zlib, no cipher, no type byte	unchanged	[bin] (tools/remotecontrol-lineage.md#what-stays-the-same-bin); [bin] (tools/npc-control.md#logins-bin)

Builds	First message	Mode during login	Mode after	Source
2004	type 6, `c key` (`rand & 0x7F`), `GSERV024`, `str account`, `str password`	u16be length + zlib, no cipher, no type byte	generation 5 (type byte + partial XOR)	[bin] (tools/remotecontrol.md#logins-bin)

Rule	Source
Type 6 is what the server calls RC2; the 2004 Delphi RC already uses it, so "RC2" in server tables means the generation-5 RC login, not the later GTK program	[bin][srv] (tools/remotecontrol.md#logins-bin)
The 2004 tokens are plain string constants, unlike the game clients' tokens built one character at a time	[bin] (tools/remotecontrol.md#logins-bin)
The connect dialog of every build has a built-in default account; the start dialog up to 2001/01/27 defaults to the loopback address and port 14804	[bin] (tools/remotecontrol-lineage.md#start-dialog-and-the-old-option-packets-bin)
From 2002/05/02 the address fields are gone and the server is picked from the listserver (→ §6.1.1)	[bin] (tools/remotecontrol-lineage.md#start-dialog-and-the-old-option-packets-bin)
In the 39ster 2.1 remake an RC connection (type 1) skips the per-line descramble (→ §5.2.2)	[src] (ash/ARCHITECTURE.md#1-system-context)
A GServer-v2 refuses an RC whose account is not on the staff list	[src] (server-notes/04-algorithms.md#5-login)

Right after login the RC sends one `PLI_PLAYERPROPS` (2):

Builds	Props	Source
2000/8/19 on	0 (NICKNAME) = the nickname from the options; closing the player list with "Go in away mode" sends it again with `-away` appended	[bin] (tools/remotecontrol-lineage.md#start-dialog-and-the-old-option-packets-bin)
rc141 (2002)	33 (ADDITFLAGS) = 1 only when "Ignore mass messages" is ticked; 0 (NICKNAME); 53 (PSTATUSMSG) = the selected status index	[bin] (tools/npc-control.md#logins-bin)
2004	a short message whose prop list is not traced	[bin] (tools/remotecontrol.md#logins-bin)

7.1.3 Framing

Builds	Rule	Source
all	messages are `c id` + payload + `\n`, G-integers + `0x20`, batched and flushed at 0x1000 bytes	[bin] (tools/remotecontrol.md#codec); [bin] (tools/npc-control.md#framing-both-sockets-bin)
2000 to 2002	u16be length + zlib (zlib 1.1.2); no cipher, type byte, key byte or per-message insertion	[bin] (tools/remotecontrol-lineage.md#what-stays-the-same-bin)
2002	large uploads: id 50 (`PLI_RAWDATA`) with an `i3` length, a flush, then one frame of `c id` + header + raw bytes with no `\n`; receiving id 100 switches the reader to raw mode for that many bytes	[bin] (tools/npc-control.md#framing-both-sockets-bin)
2002	more than 256 KB queued unsent drops the socket	[bin] (tools/npc-control.md#framing-both-sockets-bin)
2004, RC socket	client-side generation 5: each flush picks raw, zlib or bz2 by size, writes a type byte, and XORs the first 12 or 4 dwords with the `0x04A80B38`-seeded LCG plus the key, one step per 4-byte group present; send and receive keep separate iterators with the same key (→ §2.2.3)	[bin] (tools/remotecontrol.md#codec)

7.1.4 Start dialog and server timers

Up to 2001/01/27 the start dialog asks for four server timers, each with a "don't change" box; each value goes out after login as an `i4`, only when its box is unticked [bin] (tools/remotecontrol-lineage.md#start-dialog-and-the-old-option-packets-bin):

Field	Default	Id	Name	Source
Respawn time (bushes etc.) in sec	10	55	RC_RESPAWNSET	[bin] (tools/remotecontrol-lineage.md#start-dialog-and-the-old-option-packets-bin)
Horse lifetime in sec	10	56	RC_HORSELIFESET	[bin] (tools/remotecontrol-lineage.md#start-dialog-and-the-old-option-packets-bin)
Alignment increment time in sec	1200	57	RC_APINCREMENTSET	[bin] (tools/remotecontrol-lineage.md#start-dialog-and-the-old-option-packets-bin)
Baddies respawn time in sec	60	58	RC_BADDYRESPAWNSET	[bin] (tools/remotecontrol-lineage.md#start-dialog-and-the-old-option-packets-bin)

These are GServer13's four timer options (→ §5.1.1). From 2001/08/20 the timers are gone from the dialog and 55 to 58 are no longer sent [bin] (tools/remotecontrol-lineage.md#start-dialog-and-the-old-option-packets-bin).

7.1.5 Client → server ids by build

Constant ids at the RC socket's send call sites; ids sent through a computed value are not included, so each row is a floor [bin] (tools/remotecontrol-lineage.md#client--server-ids-per-build-bin); [bin] (tools/remotecontrol.md#ids).

Build	Ids	Source
rc133	1, 51–64	[bin] (tools/remotecontrol-lineage.md#client--server-ids-per-build-bin)
2000/8/19, 2000/10/25	1, 2, 6, 28, 55–58, 61–64, 67–79, 82	[bin] (tools/remotecontrol-lineage.md#client--server-ids-per-build-bin)
2001/01/27	1, 2, 6, 28, 55–58, 61–64, 68, 70–74, 76–79, 82	[bin] (tools/remotecontrol-lineage.md#client--server-ids-per-build-bin)
2001/08/20	1, 2, 6, 28, 61–64, 68, 70–79, 82–88	[bin] (tools/remotecontrol-lineage.md#client--server-ids-per-build-bin)
2002/01/04	1, 2, 6, 28, 61–64, 68–79, 82–88	[bin] (tools/remotecontrol-lineage.md#client--server-ids-per-build-bin)
2002/05/02	1, 2, 6, 28, 51–54, 61–64, 68–79, 82–92, 94, 96–98, plus the NC ids (→ §7.3.2)	[bin] (tools/remotecontrol-lineage.md#client--server-ids-per-build-bin)
2004	2, 6, 28, 51–54, 61–64, 68–76, 78, 79, 82, 84, 86, 88–92, 94, 96–98	[bin] (tools/remotecontrol.md#ids)
2004, on the wire	also 83, 85 and 87 (rights, comments and ban get), with 73 and 76; the static row is a floor	[cap] (protocol/rc-capture-2026-11-04.md#ids-against-the-static-read)

The later names cover server options, folder config, player props get and set, accounts, bans, rights, comments and the file browser [srv] (protocol/chat-pm-rc.md#admin-and-staff). Ids 51 to 54 changed meaning:

Id	rc133 use	Later name	Source
51	account list: Edit (get account)	RC_SERVEROPTIONSGET	[bin] (tools/remotecontrol-lineage.md#ids-5154-changed-meaning-bin)
52, 53	edit account: Apply	RC_SERVEROPTIONSSET, RC_FOLDERCONFIGGET	[bin] (tools/remotecontrol-lineage.md#ids-5154-changed-meaning-bin)
54	account list: Delete	RC_FOLDERCONFIGSET	[bin] (tools/remotecontrol-lineage.md#ids-5154-changed-meaning-bin)

Id	rc133 use	Later name	Source
59	player list: Edit (get player props)	RC_PLAYERPROPSGET	[bin] (tools/remotecontrol-lineage.md#ids-5154-changed-meaning-bin)
60	edit player: Apply	RC_PLAYERPROPSSET	[bin] (tools/remotecontrol-lineage.md#ids-5154-changed-meaning-bin)

From 2000/8/19 accounts move to 70 to 78, and 51 to 54 are unused until 2002/05/02 brings them back as server options and folder config; a server that talks to rc133 cannot use the later names for 51 to 54 [bin][inf] (tools/remotecontrol-lineage.md#ids-5154-changed-meaning-bin).

2004 file operation	Rule	Source
folder change	id 0x5a	[bin] (tools/remotecontrol.md#file-and-account-operations)
upload	framed by ids 0x9b, 0x5d, 0x9c	[bin] (tools/remotecontrol.md#file-and-account-operations)
delete	needs the rights string to contain `w`	[bin] (tools/remotecontrol.md#file-and-account-operations)

7.1.6 Server → RC ids

Builds	Ids handled on the RC socket	Source
2004	13, 16, 30, 35, 37, 55–59, 61–70, 72–74, 76, 77, 79, 100, 103, 180	[bin] (tools/remotecontrol.md#ids)

Id	Name	Layout	Source
13	TOALL	`s player id`, `str message` (→ §2.8)	[srv][2.05] (protocol/chat-pm-rc.md#toall)
16	DISCMESSAGE	`rest` = text; a disconnect box	[bin] (tools/npc-control.md#nc--client-ids-handled)
35	RC_ADMINMESSAGE	`rest` = `Admin <account>:\xa7<text>`	[srv][2.05] (protocol/chat-pm-rc.md#admin-and-staff)
37	PRIVATEMESSAGE	`s from id`, then a quoted, comma-separated type label and message (→ §2.8)	[srv][2.05] (protocol/chat-pm-rc.md#private-messages)
47	STAFFGUILDS	`"guild"`, repeated, sent at login	[srv][2.05] (protocol/chat-pm-rc.md#admin-and-staff)
0x37	(player list)	creates or updates a player row: a 14-bit id, then a prop stream (the note gives the id in hex only); defaults are 3 hearts, `sword1.gif`, `body.png`, `onlinestartlocal.graal`	[bin] (tools/remotecontrol.md#player-list)
74	RC_CHAT	an RC console line	[srv] (protocol/chat-pm-rc.md#admin-and-staff)
79	NPCSERVERADDR	`s player id`, then `ip,port` (→ §7.2.2)	[bin] (tools/npc-control.md#finding-the-npc-server-bin)

Players in jail can PM only RCs [srv] (protocol/chat-pm-rc.md#private-messages). The layouts of the other RC ids are unmeasured in the notes [bin] (tools/remotecontrol.md#ids), except those seen on the wire in → §7.1.8 [cap] (protocol/rc-capture-2026-11-04.md#player-props).

7.1.7 Features and local settings

Build	Feature	Source
rc133	player list (nickname, account, level, head, id), disconnect, edit player, message, message to all, account list (add, delete, edit), update levels; the account form already carries rights (Load only, Graalian Police, Trusted Admin, Administrator, No rights, Blocked), look, position, items, weapons, opened chests, NPC flags, MP, AP, spin attack	[bin] (tools/remotecontrol-lineage.md#feature-timeline-from-forms-bin)
2000/8/19	chat window (RTF), server flags, toalls, SQL account list with a filter, message editor, options (ignore mass messages, no HTML in PMs, no images in PMs, away on close), warp player	[bin] (tools/remotecontrol-lineage.md#feature-timeline-from-forms-bin)
2001/08/20	edit rights, edit comments, local ban	[bin] (tools/remotecontrol-lineage.md#feature-timeline-from-forms-bin)
2002/05/02	NPC Control (NPC list, add NPC, local NPCs, classes, weapons, level list), listserver login and server list, file browser with drag and drop, short message popup	[bin] (tools/remotecontrol-lineage.md#feature-timeline-from-forms-bin)

Setting	Where	Source
options: `nickname=`, `nomassmessages=`, `nohtmlinpms=`, `nohtmlimages=`, `attachaway=`	`controlconfig.txt` beside the exe; the later GTK RC2 uses `control2config.txt`	[bin] (tools/npc-control.md#local-settings-bin); [bin] (tools/remotecontrol.md#file-and-account-operations)
account (plain) and password (DES-obfuscated, 48 printable characters)	registry `HKCU\Software\Graal\Remote Control`	[bin] (tools/npc-control.md#local-settings-bin); [bin] (tools/remotecontrol-lineage.md#what-stays-the-same-bin)

The password scheme is a 24-byte block (length byte, up to 23 bytes of text, random padding), encrypted as three DES ECB blocks with an LSB-first key, each output byte written as two nibbles + `0x20` [bin] (tools/npc-control.md#password-obfuscation). The same scheme, with other keys, hides the RC 1.41 and 2.05 list hosts and the 1.1 guild files [bin] (tools/npc-control.md#password-obfuscation); [bin] (tools/server-tools-2001.md#graal-11-guildmaker-1999-bin).

7.1.8 Player admin on the wire (2004)

One session of the 2004 RC against a GServer-v2 fork, with one 2.22 player online; each row is a single observation [cap] (protocol/rc-capture-2026-11-04.md#setup). Notation as in → §2.1.

Step	Rule	Source
after login	server → RC in this order: 25, 194, 74 three times, 190 (empty), 47, 180, 103, one 55 per player online, then 74 `Currently online: ...`	[cap] (protocol/rc-capture-2026-11-04.md#after-the-rc-login)
after login	the RC then sends 2 PLAYERPROPS twice and the server answers with 9	[cap] (protocol/rc-capture-2026-11-04.md#after-the-rc-login)

Dir	Id	Layout	Source
S→RC	103 RC_MAXUPLOADFILESIZE	i5 bytes; 20971520 seen	[cap] (protocol/rc-capture-2026-11-04.md#after-the-rc-login)
S→RC	55 ADDPLAYER	`s player id`, `str account`, then props as id + value	[cap] (protocol/rc-capture-2026-11-04.md#after-the-rc-login)

Dir	Id	Layout	Source
RC →S	73 RC_PLAYERPROPSGET2	`s player id`	[cap] (protocol/rc-capture-2026-11-04.md#player-props)
S→ RC	72 RC_PLAYERPROPSGET	`s player id`, `str account`, `str world`, `c n`, n bytes of props, tail	[cap] (protocol/rc-capture-2026-11-04.md#player-props)
RC →S	76 RC_PLAYERPROPSSET2	`str account`, `str world`, `c n`, n bytes of props, tail, then one more byte (0 seen; → App. E#421)	[cap] (protocol/rc-capture-2026-11-04.md#player-props)
both	props tail	`s`, `s` (flag and chest counts, order unseen → App. E#420), `c weapon count`, `str` per weapon	[cap] (protocol/rc-capture-2026-11-04.md#player-props)
RC →S	83 RC_PLAYERRIGHTSGET	account as `rest`	[cap] (protocol/rc-capture-2026-11-04.md#rights)
S→ RC	62 RC_PLAYERRIGHTSGET	`str account`, `i5 rights`, `str IP range`, `s n` + n bytes of folder rights	[cap] (protocol/rc-capture-2026-11-04.md#rights)
RC →S	84 RC_PLAYERRIGHTSSET	same layout as 62; folder lines as CSV (→ App. E#422)	[cap] (protocol/rc-capture-2026-11-04.md#rights)
RC →S	85 RC_PLAYERCOMMENTSGET	account as `rest`	[cap] (protocol/rc-capture-2026-11-04.md#comments)
S→ RC	63 RC_PLAYERCOMMENTSGET	`str account`, comments as CSV (`rest`), one item per text box line	[cap] (protocol/rc-capture-2026-11-04.md#comments)
RC →S	86 RC_PLAYERCOMMENTSSET	`str account`, comments as CSV (`rest`)	[cap] (protocol/rc-capture-2026-11-04.md#comments)
RC →S	87 RC_PLAYERBANGET	account as `rest`	[cap] (protocol/rc-capture-2026-11-04.md#ban)
S→ RC	64 RC_PLAYERBANGET	`str account`, `c banned`, reason (`rest`)	[cap] (protocol/rc-capture-2026-11-04.md#ban)
RC →S	88 RC_PLAYERBANSET	`str account`, `c banned`, reason as CSV (`rest`)	[cap] (protocol/rc-capture-2026-11-04.md#ban)

Observation	Rule	Source
props sent back	the RC's 76 leaves out props 0, 30, 34 and 36, adds 21, 22 and 23 as one zero byte each, and sends 29 as 0	[cap] (protocol/rc-capture-2026-11-04.md#player-props)
rights bits	the dialog's check boxes map to bits from bit 0; "Warpto x y" and "Admin message" gave 513 (bits 0 and 9)	[cap] (protocol/rc-capture-2026-11-04.md#rights)
ban menu	the player menu has Edit Ban only: the account ban, with no IP or computer id ban	[cap] (protocol/rc-capture-2026-11-04.md#ban)

7.2 NC login

7.2.1 Login and framing

Rule	Source
The NC socket's first message is type 3, `NCL11012`, `str account`, `str password`, with no key byte, sent right after the socket connects	[bin] (tools/npc-control.md#logins-bin); [bin] (protocol/npc-control.md#login)
The NC socket stays in fixed zlib framing (u16be length + zlib) for the whole session: no type byte, no cipher	[bin] (tools/remotecontrol.md#logins-bin); [bin] (protocol/npc-control.md#login)
The NC login did not change between rc141 (2002) and the 2004 RC	[bin] (protocol/npc-control.md#login)

Rule	Source
A GServer-v2 also knows the NC token `NCL21075` (→ §2.3.6.1)	[srv] (protocol/login-and-walk.md#how-the-server-parses-it-srv)
Status bar text after login: "Connected to NPC-Server"	[bin] (tools/npc-control.md#finding-the-npc-server-bin)

7.2.2 Finding the NPC server

The RC never asks for an NC address [bin] (tools/npc-control.md#finding-the-npc-server-bin):

Step	Rule	Source
1	any player whose nickname contains `(Server)` is treated as an NPC-server player	[bin] (tools/npc-control.md#finding-the-npc-server-bin)
2	for each, the RC sends id 94 (`PLI_NPCSERVERQUERY`): `s player id` + the text `location`, when the player appears and again whenever the NC socket drops	[bin] (tools/npc-control.md#finding-the-npc-server-bin)
3	the server answers id 79 (`PLO_NPCSERVERADDR`): `s player id`, then `ip,port`	[bin] (tools/npc-control.md#finding-the-npc-server-bin)
4	the RC opens the NC socket only if it is not already open and that player's nickname still contains `(Server)`, then sends the NC login	[bin] (tools/npc-control.md#finding-the-npc-server-bin)

Rule	Source
The binary's default NC fields (the loopback address and 14804) are leftovers of the old start dialog and unused once discovery works	[bin][inf] (tools/npc-control.md#finding-the-npc-server-bin); [bin] (protocol/npc-control.md#finding-the-npc-server-bin)
The 2001 NPC Control text says the NewWorld game server exposed NC on port 14950; NewWorld accounts could connect, main-server accounts needed admin level 4 or higher	[doc] (tools/npc-control.md#what-the-npccontrol-13-bundle-is-doc)
The 2001 bundle says adding, viewing and changing classes did not work yet	[doc] (tools/npc-control.md#what-the-npccontrol-13-bundle-is-doc)

7.3 NPC, class and weapon messages

On the NC socket, received ids 33, 80, 157 to 164 and 167 go to the NC handler; everything else, 16 included, goes to the RC handler [bin] (tools/npc-control.md#routing-on-the-nc-socket-bin).

7.3.1 NC → client

Id	Name	Layout	Effect	Source
16	DISCMESSAGE	`rest` = text	disconnect box	[bin] (tools/npc-control.md#nc--client-ids-handled)
33	NPCWEAPONADD	`str name`, `str image`, `s len`, `len` bytes of script	weapon script editor	[bin] (tools/npc-control.md#nc--client-ids-handled)
80	NC_LEVELLIST	`rest` = level list, one per line	level list editor	[bin] (tools/npc-control.md#nc--client-ids-handled)
157	NC_NPCATTRIBUTES	`rest` = attribute text, one per line	attribute viewer (view only)	[bin] (tools/npc-control.md#nc--client-ids-handled)
158	NC_NPCADD	`i3 id`, then props in any order and any subset: `c 50` + `str name`, `c 51` + `str type`, `c 52` + `str level`	adds or updates an NPC list row	[bin] (tools/npc-control.md#nc--client-ids-handled)
159	NC_NPCDELETE	`i3 id`	removes the row	[bin] (tools/npc-control.md#nc--client-ids-handled)

Id	Name	Layout	Effect	Source
160	NC_NPCSCRIPT	`i3 id`, `rest` = script	script editor	[bin] (tools/npc-control.md#nc--client-ids-handled)
161	NC_NPCFLAGS	`i3 id`, `rest` = flags, one per line	flag editor	[bin] (tools/npc-control.md#nc--client-ids-handled)
162	NC_CLASSGET	`str name`, `rest` = script	class editor	[bin] (tools/npc-control.md#nc--client-ids-handled)
163	NC_CLASSADD	`rest` = class name	adds to the class list	[bin] (tools/npc-control.md#nc--client-ids-handled)
164	NC_LEVELDUMP	`rest` = text	dump viewer (view only)	[bin] (tools/npc-control.md#nc--client-ids-handled)
167	NC_WEAPONLISTGET	repeated `str name`	weapon list	[bin] (tools/npc-control.md#nc--client-ids-handled)
188	NC_CLASSDELETE	class name, form unmeasured	handled from 2004, not in 2002	[bin][inf] (protocol/npc-control.md#nc--client-server-ids)

7.3.2 Client → NC

Id	Name	From	Payload	Source
3	(login)	connect	→ §7.2.1	[bin] (tools/npc-control.md#client--nc-ids-sent)
79	RC_CHAT	a console line typed in NC mode	text	[bin] (tools/npc-control.md#client--nc-ids-sent)
103	NC_NPCGET	NPC list: open, Enter, Attributes	`i3 id`	[bin] (tools/npc-control.md#client--nc-ids-sent)
104	NC_NPCDELETE	NPC list: Delete, after a confirm	`i3 id`	[bin] (tools/npc-control.md#client--nc-ids-sent)
105	NC_NPCRESET	NPC list: Reset	`i3 id`	[bin] (tools/npc-control.md#client--nc-ids-sent)
106	NC_NPCSCRIPTGET	NPC list: Script	`i3 id`	[bin] (tools/npc-control.md#client--nc-ids-sent)
107	NC_NPCWARP	Quick warp or the warp dialog	`i3 id`, `c x`×2, `c y`×2, level	[bin] (tools/npc-control.md#client--nc-ids-sent)
108	NC_NPCFLAGSGET	NPC list: Edit flags	`i3 id`	[bin] (tools/npc-control.md#client--nc-ids-sent)
109	NC_NPCSCRIPTSET	script editor: Save	`i3 id`, script	[bin] (tools/npc-control.md#client--nc-ids-sent)
110	NC_NPCFLAGSSET	flag editor: Save	`i3 id`, flags	[bin] (tools/npc-control.md#client--nc-ids-sent)
111	NC_NPCADD	Add NPC dialog	comma text: name, id, type, scripter, start level, x, y; type is one of `OBJECT`, `ANIMAL`, `POLICE`, `HORSE`, `BOAT`, `HOUSEC`	[bin] (tools/npc-control.md#client--nc-ids-sent)
112	NC_CLASSEDIT	class list: open	class name	[bin] (tools/npc-control.md#client--nc-ids-sent)

Id	Name	From	Payload	Source
113	NC_CLASSADD	class editor: Save	`str name`, `rest` = script	[bin] (tools/npc-control.md#client--nc-ids-sent)
114	NC_LOCALNPCSGET	Local NPCs dialog	level name	[bin] (tools/npc-control.md#client--nc-ids-sent)
115	NC_WEAPONLISTGET	console button	empty	[bin] (tools/npc-control.md#client--nc-ids-sent)
116	NC_WEAPONGET	weapon list: open	weapon name	[bin] (tools/npc-control.md#client--nc-ids-sent)
117	NC_WEAPONADD	weapon editor: Save	`str name`, `str image`, `rest` = script	[bin] (tools/npc-control.md#client--nc-ids-sent)
118	NC_WEAPONDELETE	weapon list: Delete	weapon name	[bin] (tools/npc-control.md#client--nc-ids-sent)
119	NC_CLASSDELETE	class delete; sent from 2004, no button in 2002	class name	[bin] (protocol/npc-control.md#client--nc-client-ids)
150	NC_LEVELLISTGET	console button	empty	[bin] (tools/npc-control.md#client--nc-ids-sent)
151	NC_LEVELLISTSET	level list editor: Save	level list	[bin] (tools/npc-control.md#client--nc-ids-sent)

Rule	Source
All NC texts open in one shared editor window with a mode; Save sends 109, 110, 113, 117 or 151 by mode; attributes (157) and dumps (164) can only be saved locally as `.rtf`	[bin] (tools/npc-control.md#client--nc-ids-sent)
The reply id and layout after id 114 (local NPCs) are unmeasured	[inf] (protocol/npc-control.md#check-on-loopback-inf)
Whether 33 carries the script length as `s` or `i3` for large scripts, and whether big set messages go through the raw-data path, is unmeasured	[inf] (protocol/npc-control.md#check-on-loopback-inf)

7.3.3 The 2001 PM admin interface

Before the NC socket, admins controlled the 2001 NPC server by PMs to its player login; only senders whose nickname carries `(NPC Admin)` were answered, and an unknown command such as `help` returned the command list [doc] (tools/npcserver-2001.md#5-admin-interface-pm-based). Multi-line commands put their payload on the lines after the first [doc] (tools/npcserver-2001.md#5-admin-interface-pm-based).

Command	Payload and effect	Source
`add npc <name>`	next lines: id, type (up to 6 characters), scripter, level, x, y; database NPC ids are unique and at least 1000; level and x, y become the reset position	[doc] (tools/npcserver-2001.md#5-admin-interface-pm-based)
`reset npc <name>`	resets every attribute but the script and returns the NPC to its start; the name is case-sensitive	[doc] (tools/npcserver-2001.md#5-admin-interface-pm-based)
`update script <name>`	the following lines replace the script; a running `sleep` is cut	[doc] (tools/npcserver-2001.md#5-admin-interface-pm-based)
`get script <name>`	returns an NPC's or a class's script	[doc] (tools/npcserver-2001.md#5-admin-interface-pm-based)

Command	Payload and effect	Source
`update class <name>`	replaces a class script, saved to the scripts folder; members pick it up at once	[doc] (tools/npcserver-2001.md#5-admin-interface-pm-based)
`get npc list`	lists database NPCs (id ≥ 1000) with positions	[doc] (tools/npcserver-2001.md#5-admin-interface-pm-based)
`dump vars`, `dump vars <name>`	dumps the level's or an NPC's variables	[doc] (tools/npcserver-2001.md#5-admin-interface-pm-based)
`reload level list`	rereads the text file of levels this server handles	[doc] (tools/npcserver-2001.md#5-admin-interface-pm-based)

The NC Add NPC dialog has the same seven fields as `add npc`, so id 111 is most likely that command moved onto its own message [inf][doc] (protocol/npc-control.md#client--nc-client-ids). The RC finds the NPC server by `(Server)` in the nickname rather than `(NPC Admin)` [bin] (tools/npcserver-2001.md#open-questions).

8 NPC scripting

> Part index. Each chapter carries its own status line.

- → §8.1 GS1 language (`01-gs1-language.md`)
- → §8.2 GS1 reference by version (`02-gs1-reference.md`)
- → §8.3 GS2 language (`03-gs2-language.md`)
- → §8.4 GS2 bytecode (`04-gs2-bytecode.md`)

8.1 GS1 language

This chapter describes GS1, the NPC and weapon script language of the 1.x and 2.x clients, as the 2.05 rev 2 client parses and runs it. The command-by-command reference, the expression tables and the per-build changes are in → §8.2. The later language is in → §8.3.

> Status: draft. The full # message-code set, the member-name mapping of the object arrays and the `move` option bits are still open (→ App. E#203, App. E#204, App. E#206).

8.1.1 Where scripts run

Item	Rule	Source
host	Classic NPCs are pure client-side; the archived script corpus shows no separate server VM	[doc] (npc-library/archival-scripting-notes.md#runtime-model-highlights-classic-client)
corpus	the archived level scripts sort into genres; baddies (452 scripts) and hazards (192) lead, and most baddy NPCs are event skeletons that react to the player entering or touching them, an item drop, or all baddies dying	[doc] (npc-library/behaviours.md#combat--baddies-452-scripts); [doc] (npc-library/behaviours.md#combat--hazards-192-scripts)
kinds of script	a level NPC, a weapon NPC (made by `toweapons -Name`), and a child NPC made by `putnpc`	[doc] (npc-library/archival-scripting-notes.md#runtime-model-highlights-classic-client)
delivery to 2.05	NPC and weapon scripts arrive as source text in the NPC script prop (`NPCPROP_SCRIPT`)	[2.05] (npc-library/gs1-reference-205.md#execution-model-205)
weapon delivery	PLO 33 NPCWEAPONADD carries `str name` and an NPC prop stream with image and script; a name the player already has is ignored; 2.05 fires `playerenters` on the new weapon	[srv][2.05] (client-internals/feature-specs/02-weapons-combat.md#weapon-lists-server-to-client-srv205)
pre-v4 clients	the server sends the weapon script as plain GS1 text	[srv] (client-internals/feature-specs/02-weapons-combat.md#weapon-lists-server-to-client-srv205)
level editor	each level NPC has an image (or none) and a GS1 script; the 2.05 editor's script window has *Test* (a syntax check) and *Style*	[2.05] (client-internals/level-editor-205.md)
NPC server present	when the server sends PLO 44 (`PLO_HASNPCSERVER`), several visual commands are allowed online (→ §8.1.9) and the client stops applying NPC props itself	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works); [bin] (protocol/packet-ids.md)
2.3 and later	the classic client-side NPC mode is removed; Classic moves to the NPC server	[doc] (client-internals/feature-timeline.md)
4.04 rev 2	still runs GS1 weapon scripts, but sends only `client.` string flags to the server; `set`, `setstring gr.*` and `setplayerprop #c` from a weapon send nothing	[live] (client-internals/live-behaviour-404.md)

8.1.2 Compilation

2.05 does not compile GS1 to bytecode. It parses the script text once, on level load, and interprets the result [2.05] (npc-library/gs1-reference-205.md#execution-model-205; npc-library/gs1-command-reference-205.md#how-the-interpreter-works).

Item	Rule	Source
compiled form	a flat list of statement nodes	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)

Item	Rule	Source
statement node	a command number, up to two string arguments, up to eight numeric expression arguments, a jump target and a pre-resolved name index (item, baddy, colour and the like)	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
command number	600 + the command's index in the command table; `if` is 605 (→ §8.2.4)	[2.05] (npc-library/gs1-reference-205.md#execution-model-205)
expressions	resolved through a second table of functions and variables with numeric ids 100–825 (→ §8.2.5) and a small operator table	[2.05] (npc-library/gs1-reference-205.md#execution-model-205)
argument parsing	each command has an argument signature (→ §8.2.1); the compiler uses it to parse the arguments and to build the usage text of its error messages	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
compiled-away keywords	`else`, `function` and closing braces never reach the executor; they become four internal node kinds: assignment or expression statement, unconditional jump, function call, and end of function body	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
name matching	command names that end in a space in the table (`say `, `set `, `setlevel `, `setimg `, `move ` and others) require an argument; the others are matched as whole words	[inf] (npc-library/gs1-reference-205.md#execution-model-205)
missing files	offline, the compiler rejects a filename argument naming a missing file unless it starts with `http://`	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
OpenGS1	OpenGS1's stack bytecode is an independent reimplementation; 2.05 does not share its opcode set (→ §8.4.3)	[2.05] (npc-library/gs1-reference-205.md#execution-model-205)

8.1.3 Execution

Item	Rule	Source
run	every event runs the whole node list from the top	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
event test	events are flags tested with `if (name)`; the engine sets the flag before it runs the script (→ §8.1.5)	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
`if`	evaluates its condition and jumps past its block when the condition is false; the threshold for "true" is disputed → App. E#200	[2.05] (npc-library/gs1-reference-205.md#execution-model-205; npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
loops	`for` and `while` are the same loop node; the condition is checked on each pass and the loop stops after 10,000 passes, as the 2000 documentation says	[2.05][doc] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
function calls	a call pushes a return point; nesting stops at 10,000 deep	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
`return`	pops the call stack and resumes after the call; with an empty stack (or at the end of a function body) it ends the script	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
string arguments	trimmed, and message codes (`#c`, `#v(...)` and others) are expanded before use, for almost every command	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)

These commands end the current script run after they execute:

Command	Condition	Source
`setlevel`, `setlevel2`	always	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
`destroy`	always	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
`sleep`	non-zero argument	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
`toinventory`	offline	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
`hide`	on an NPC that is following the player	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
`break`, `continue`	outside a loop	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)

8.1.4 Timeouts and sleep

Item	Rule	Source
timer	a script sets `timeout = N` (seconds); when it expires the script runs with the `timeout` flag set	[doc] (npc-library/archival-scripting-notes.md#runtime-model-highlights-classic-client)
common rates	0.05 s for AI and HUDs, 0.1–0.2 s for menus, 0.5–1.0 s for idle NPCs, in the archived corpus	[doc] (npc-library/archival-scripting-notes.md#runtime-model-highlights-classic-client)
`sleep n`	sets the NPC's timeout to *n*, remembers the next statement, and ends the run; the next `timeout` event resumes there	[2.05] (npc-library/gs1-command-reference-205.md#control-flow)
`sleep 0`	only clears the timer display	[2.05] (npc-library/gs1-command-reference-205.md#control-flow)
`timershow`	draws the NPC's timeout counter over it; cleared by `message` and by `sleep 0`	[2.05] (npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour)
who gets timeouts	only the level leader receives an NPC's timeout events, unless the NPC runs `timereverywhere`	[2.05] (npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour)

8.1.5 Events

The engine sets one of these flags and runs the script [2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205). The scrape count is the number of the 3,172 scraped level scripts that test the event, where the notes counted it.

Event	Scrape count	Source
`created`	unmeasured	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
`playerenters`	2,268	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205); [file] (npc-library/gs1-commands.md#event-frequency-reminder)
`timeout`	1,456	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205); [file] (npc-library/gs1-commands.md#event-frequency-reminder)
`playertouchsme`	1,986	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205); [file] (npc-library/gs1-commands.md#event-frequency-reminder)
`playertouchsother`	1,074	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205); [file] (npc-library/gs1-commands.md#event-frequency-reminder)
`playerendsreading`	1,123	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205); [file] (npc-library/gs1-commands.md#event-frequency-reminder)

Event	Scrape count	Source
`playerchats`	unmeasured	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
`playersays(…)`, `playersays2(…)`	unmeasured; these are functions, not flags	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
`wasshot`	728	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205); [file] (npc-library/gs1-commands.md#event-frequency-reminder)
`wasshooted`, `washit`, `wasthrown`, `waspelt`, `peltwith…`	unmeasured	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
`exploded`	unmeasured	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
`compusdied`	1,122	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205); [file] (npc-library/gs1-commands.md#event-frequency-reminder)
`compsdead`	unmeasured	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
`playerdies`, `playerhurt`, `playerhurted`	unmeasured	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
`playerlaysitem`, `weaponfired`	unmeasured	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
`action…`	unmeasured; any action name sent by `triggeraction`	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
`shotbyplayer`, `shotbybaddy`	unmeasured	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
`main`	unmeasured; sits in the same string cluster, probably a weapon's main event or entry	[inf] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)

Events documented for 2.22 but absent in 2.05: `actionleftmouse`, `actionrightmouse`, `actiondoublemouse`, `actionprojectile`, `keypressed`, `mousedown`, `playeronline`, `playerpaused` [2.05][doc] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205).

How an event reaches another script:

Mechanism	Rule	Source
`callnpc i, event`	runs NPC *i*'s script with the given event flag set; the documentation says parameters after the event arrive as `#p(n)`	[2.05][doc] (npc-library/gs1-command-reference-205.md#control-flow)
`callweapon i, event`	moves weapon *i* to the player's centre, then runs its script with the event flag set	[2.05] (npc-library/gs1-command-reference-205.md#control-flow)
`triggeraction x, y, action, params`	fires `action` with its parameters at the object on (x, y) locally, and sends it online (→ §8.2.3)	[2.05] (npc-library/gs1-command-reference-205.md#control-flow)
`triggeraction 0,0,clientside,…`	from 2.21, reaches a weapon's client-side part	[doc] (client-internals/feature-timeline.md)

8.1.6 State flags

The engine keeps these flags current; a script tests them with `if` [2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205).

Group	Flags	Source
NPC	`visible`, `isweapon`, `followsplayer`, `isleader`, `isonmap`	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)

Group	Flags	Source
carrying	`carrying`, `carries…`, `carriesnpc`	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
player	`weaponsenabled`, `playeronhorse`, `canspin`, `playerswimming`, `playerismale`, `playerisfemale`, `playerreading`, `playerattached`, `firedonhorse`	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)
level	`issparringzone`, `sparringzone`, `nopkzone`, `lighteffectsenabled`	[2.05] (npc-library/gs1-reference-205.md#built-in-flags-and-events-205)

8.1.7 Flags, variables and persistence

Item	Rule	Source
flags	`set name` sets a flag, `unset name` clears it; the documentation says a `server.` prefix makes it server-wide	[2.05][doc] (npc-library/gs1-command-reference-205.md#flags-and-variables)
string variables	`setstring name,value` stores the flag `name=value`; an empty value deletes it	[2.05] (npc-library/gs1-command-reference-205.md#flags-and-variables)
arrays	`setarray var,n` resizes an array variable to *n* elements; ignored unless the variable is plain or `this.`	[2.05] (npc-library/gs1-command-reference-205.md#flags-and-variables)
NPC-local variables	`this.` prefix	[doc] (npc-library/gs1-reference-205.md#operators-and-syntax-doc)
NPC save slots	`save[n]`, reached as `this.save[i]` or `npcs[i].save[i]` (→ §8.2.5)	[2.05] (npc-library/gs1-reference-205.md#functions-and-variables-205); [doc] (npc-library/archival-scripting-notes.md#runtime-model-highlights-classic-client)
persistence	scripts persist state through `toweapons`, `save[n]` and `client.*` flags	[doc] (npc-library/archival-scripting-notes.md#runtime-model-highlights-classic-client)
tokens	`tokenize` splits a string into the global token list on spaces and commas (quotes group); `tokenize2` adds delimiter characters; `tokenscount` gives the count	[2.05] (npc-library/gs1-command-reference-205.md#flags-and-variables; npc-library/gs1-reference-205.md#functions-and-variables-205)
2.19	`this.` strings on the client; on NPC-server servers the client stops sending strings that are not `client.`	[doc] (client-internals/feature-timeline.md)
4.04 rev 2	only `client.` string flags are sent to the server	[live] (client-internals/live-behaviour-404.md)
image ids	HUD and shop scripts in the archived corpus collide often in the `showimg` index band around 300 and above	[doc] (npc-library/archival-scripting-notes.md#runtime-model-highlights-classic-client)

8.1.8 Syntax and operators

Item	Forms	Source
assignment	`=`, `+=`, `-=`, `*=`, `/=`, `++`, `--`	[doc] (npc-library/gs1-reference-205.md#operators-and-syntax-doc)
arithmetic	`+ - * / %`, `^` for power	[doc] (npc-library/gs1-reference-205.md#operators-and-syntax-doc)
comparison	`== != < > <= >=`	[doc] (npc-library/gs1-reference-205.md#operators-and-syntax-doc)
logic	`&&`, `||`, `!`	[doc] (npc-library/gs1-reference-205.md#operators-and-syntax-doc)
membership	`in` for ranges and sets: `x in |0,64|`, `2 in {1,2,3}`	[doc] (npc-library/gs1-reference-205.md#operators-and-syntax-doc)

Item	Forms	Source
control flow	`if`/`else`, `for`, `while`, `break`, `continue`, `function name() {}` with `name();`, `return`, `sleep`, `timeout = n`	[doc] (npc-library/gs1-reference-205.md#operators-and-syntax-doc)
string codes	# codes such as `#s(var)` and `#v(expr)`; the full 2.05 set is open → App. E#203	[doc] (npc-library/gs1-reference-205.md#operators-and-syntax-doc)
character props	`setcharprop` and `setplayerprop` take a property code (#1, #2, #3, #5, #8, `#C0`..`#C5`, `#P1`..`#P5`, `#c`, `#n`) as their first word (→ §8.2.4.5)	[2.05] (npc-library/gs1-command-reference-205.md#character-npcs-showcharacter)
not commands	`warp` and `jump` are not GS1 commands in 2.05; warping is `setlevel` / `setlevel2` or an assignment to `playerx` / `playery`	[2.05] (npc-library/gs1-command-reference-205.md#corrections-to-gs1-commandsmd)

8.1.9 Execution contexts

Context	Rule	Source
weapon NPCs act as the player	when the running NPC is one of the player's weapons, commands that spawn things (bombs, explosions, horses, projectiles, `showimg` layers, `hitobjects`, `triggeraction`, `reflectarrow`) use the player's level and count the player as the source	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
online gate	`seteffect`, `setcoloreffect`, `setzoomeffect`, `changeimgcolors`, `changeimgzoom`, `addtiledef`, `addtiledef2`, `removetiledefs`, `replaceani` and `attachplayertoobj` run only offline, or online when the server announced an NPC server (PLO 44); online without an NPC server they do nothing	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
editor preview	when the in-game editor builds a preview copy of a `showcharacter` NPC, it runs that NPC's `playerenters` in a restricted mode	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
preview whitelist	only `if`, `else`, `for`, `while`, `setgif`, `setgifpart`, `setimg`, `setimgpart`, `setmap`, `showcharacter`, `setcharprop`, `setcharani`, `setbackpal`, `setbacktile`, `setbacktile2`, `addtiledef`, `addtiledef2`, `setz`, `loadmap`, function calls and `++`/`--` style assignments run; everything else is skipped	[2.05][inf] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
online side effects	some commands also send a TCP or UDP message when connected (→ §8.2.3)	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)

8.1.10 The script corpus

The notes hold two collections of real GS1 scripts, the evidence for the frequency counts in this part:

Collection	Content	Source
level scrape	3,172 unique scripts from GR-V1 `.graal` levels; the scrape reported 3,187, and 15 records were lost to truncated source pages	[file] (npc-library/README.md)
level scrape by genre	combat 1,400; interaction 602; dialogue 373; spawn 328; misc 257; object 168; transport 59	[file] (npc-library/README.md#counts)
archived catalogue	1,520 unique script bodies from 6,275 placements, with a body-hash catalogue; held off-repository	[doc] (npc-library/archival-scripting-notes.md#source-archive)

The archived catalogue classifies about 1,519 documented scripts [doc] (npc-library/archival-scripting-notes.md#classification-summary-from-ash):

Class	Approx. count	Source
quest / dialogue	516	[doc] (npc-library/archival-scripting-notes.md#classification-summary-from-ash)
level prop / trigger	390	[doc] (npc-library/archival-scripting-notes.md#classification-summary-from-ash)
farm / planting	136	[doc] (npc-library/archival-scripting-notes.md#classification-summary-from-ash)
player weapon / projectile	135	[doc] (npc-library/archival-scripting-notes.md#classification-summary-from-ash)
shop / merchant	111	[doc] (npc-library/archival-scripting-notes.md#classification-summary-from-ash)
minigame	59	[doc] (npc-library/archival-scripting-notes.md#classification-summary-from-ash)
enemy / baddy AI	44	[doc] (npc-library/archival-scripting-notes.md#classification-summary-from-ash)
inventory / equipment UI	37	[doc] (npc-library/archival-scripting-notes.md#classification-summary-from-ash)
HUD / stats overlay	20	[doc] (npc-library/archival-scripting-notes.md#classification-summary-from-ash)

The commands most used in the level scrape are `hide` and `show` (in most `playertouchsme` scripts), then the level-change commands, `hurt`, `say`, and `set` / `unset` [file] (npc-library/gs1-commands.md#offline-wiring-priority).

8.2 GS1 reference by version

This chapter is the GS1 reference: argument types, name tables, every command of the 2.05 rev 2 table with what 2.05 does, the expression tables, constants, and how the command table grew from 1.33 to the NewWorld betas. How scripts are compiled and run is in → §8.1.

> Status: draft. Command behaviour was read from 2.05 rev 2 only; other builds are covered by table membership, not by executor reads (→ §8.2.7). Four argument signatures are disputed (→ App. E#201).

8.2.1 Argument signature letters

Each command in the table carries one letter per argument. The compiler uses the letters to parse and to label its usage messages [2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works).

Letter	Compiler's label	Accepts	Source
`R`	number	any numeric expression	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works; npc-library/gs1-reference-205.md#argument-signature-letters)
`S`	string	a string expression: #s codes, quoted text, string variables; message codes are expanded at run time	[2.05][inf] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works; npc-library/gs1-reference-205.md#argument-signature-letters)
`M`	string	raw text up to `;`, such as flag and level names; message codes are expanded at run time	[2.05][inf] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works; npc-library/gs1-reference-205.md#argument-signature-letters)
`F`	filename	a file name	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
`Q`	filename	a file name that names a script file	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
`I`	itemname	one of the 25 item names (→ §8.2.2)	[2.05][doc] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works; npc-library/gs1-reference-205.md#argument-signature-letters)
`B`	baddyname	one of the 10 baddy names	[2.05][doc] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works; npc-library/gs1-reference-205.md#argument-signature-letters)
`C`	colorname	one of the 20 colour names, or a number	[2.05][doc] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works; npc-library/gs1-reference-205.md#argument-signature-letters)
`D`	direction	`up` `left` `down` `right` (0..3), or a numeric expression	[2.05][doc] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works; npc-library/gs1-reference-205.md#argument-signature-letters)
`L`	carry objectname	one of the 11 carry objects	[2.05][doc] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works; npc-library/gs1-reference-205.md#argument-signature-letters)
`O`	objectname	an entry from the editor's predefined tile-objects list (`tileobjects/`)	[2.05][inf] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works; npc-library/gs1-reference-205.md#argument-signature-letters)

8.2.2 Name tables

Table	Index: name	Source
items (I)	0 greenrupee, 1 bluerupee, 2 redrupee, 3 bombs, 4 darts, 5 heart, 6 glove1, 7 bow, 8 bomb, 9 shield, 10 sword, 11 fullheart, 12 superbomb, 13 battleaxe, 14 goldensword, 15 mirrorshield, 16 glove2, 17 lizardshield, 18 lizardsword, 19 goldrupee, 20 fireball, 21 fireblast, 22 nukeshot, 23 joltbomb, 24 spinattack	[2.05][doc][srv] (npc-library/gs1-command-reference-205.md#name-tables; npc-library/gs1-reference-205.md#constants-doc)
baddies (B)	0 graysoldier, 1 bluesoldier, 2 redsoldier, 3 shootingsoldier, 4 swampsoldier, 5 frog, 6 octopus, 7 goldenwarrior, 8 lizardon, 9 dragon	[2.05][doc] (npc-library/gs1-command-reference-205.md#name-tables; npc-library/gs1-reference-205.md#constants-doc)
harebaddy	2.05 ships `baddyhare.gif`, so a hare type exists; its index is open → App. E#205	[file] (npc-library/gs1-reference-205.md#constants-doc)
colours (C)	0 white, 1 yellow, 2 orange, 3 pink, 4 red, 5 darkred, 6 lightgreen, 7 green, 8 darkgreen, 9 lightblue, 10 blue, 11 darkblue, 12 brown, 13 cynober, 14 purple, 15 darkpurple, 16 lightgray, 17 gray, 18 black, 19 transparent; the same list indexes `PLPROP_COLORS`	[2.05][srv] (npc-library/gs1-command-reference-205.md#name-tables; npc-library/gs1-reference-205.md#constants-doc)
carry objects (L)	0 bush, 1 sign, 2 vase, 3 stone, 4 blackstone, 5 bomb, 6 hotbomb, 7 superbomb, 8 joltbomb, 9 hotjoltbomb, 10 none	[2.05] (npc-library/gs1-command-reference-205.md#name-tables)
directions (D)	0 up, 1 left, 2 down, 3 right; numeric values are reduced mod 4, negative values too	[2.05] (npc-library/gs1-command-reference-205.md#name-tables)
direction table	the four names follow the command table directly; a dump that reads two records past `hide` finds up and `right` and wrongly counts 163 commands	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works; client-internals/newworld-betas.md#corrections-to-other-notes)

8.2.3 Online side effects

When the client is connected, some commands also tell the server (TCP, PLI ids → App. A) or nearby peers (UDP ids → §2.9) [2.05] (npc-library/gs1-command-reference-205.md#online-side-effects). A level-local action goes over UDP only when every other player in the level has a UDP port, otherwise over both [2.05] (npc-library/gs1-command-reference-205.md#online-side-effects).

Command	TCP (PLI)	UDP	Source
`take`, `take2`, `removeitem`	13 ITEMDEL	20 item del	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`putcomp`, `putnewcomp`	17 BADDYADD, instead of spawning locally	n/a	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`putnpc`	21 PUTNPC, instead of creating locally	n/a	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`destroy`	29 NPCWEAPONDEL if the NPC is also a weapon, then 22 NPCDEL	n/a	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`toweapons`	33 WEAPONADD	n/a	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`hitcompu`	16 BADDYHURT	n/a	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)

Command	TCP (PLI)	UDP	Source
`hitplayer` on a remote player	n/a	21 remote hit, to the victim only	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`putexplosion`, `putexplosion2`	27 EXPLOSION	14 explosion	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`spyfire`	10 FIRESPY	13 fire spy	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`hitobjects`	36 HITOBJECTS	25 hit objects	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`triggeraction`	38 TRIGGERACTION	26 trigger action	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`removebomb`	n/a	11 bomb del	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`removehorse`, `takehorse`	n/a	17 horse del	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`reflectarrow`	n/a	12 arrow, re-sent with the new direction	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)
`setplayerprop #c,...`	chat prop	22 chat	[2.05] (npc-library/gs1-command-reference-205.md#online-side-effects)

8.2.4 Commands of 2.05 rev 2

The 2.05 table holds 161 commands, ops 600 to 760 (op = 600 + table index); `show` (759) and `hide` (760), the two most common commands in the scraped library, are its last two entries [2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-commands.md#update-2026-10-21). The table is the same in 2.05 rev 1 and rev 2 [bin] (client-internals/build-diff-204-205.md#205-rev1--rev2).

How to read the tables below:

- *Op* is the 2.05 command number. Commands are inserted in the middle of the table in later builds, so the index of a command changes between builds and an op is valid only for 2.05 [bin] (client-internals/cross-version-timeline.md#gs1-commands).
- *In the table from* is the first build whose table has the command; 1.33 is the earliest build read, so "1.33" means "1.33 or earlier" [bin] (client-internals/cross-version-timeline.md#gs1-commands).
- A † marks a command that is not in the 2.22 `commands.rtf`: removed, renamed (`setgif` → `setimg`, `setgifpart` → `setimgpart`) or undocumented [2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205).
- Positions are in tiles unless the row says otherwise; "doc" means the 2000 `npcprogramming.doc` or the `newfeatures` logs, and where the client and the documentation disagree the row gives the client's behaviour [2.05][doc] (npc-library/gs1-command-reference-205.md#commands).

8.2.4.1 Messages, signs, sound, files

Op	Command	Args	In the table from	What 2.05 does	Source
600	`say`	R	1.33	Shows sign *n* of the current level (index into the level's sign list). Skipped if the index is out of range or the player is already in the sign-reading state. Puts the player into the reading state.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
606	`play`	F	1.33	Plays a sound positioned at the NPC's centre. Online, a file that is not found locally is requested by its raw name. Doc: midis loop.	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
635	`openurl`	M	1.33; takes arguments from 2.01	Opens the URL in the system browser (doc: with a confirm dialog, no `http://` needed).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
637	`stopmidi`	—	1.33	Stops the current midi.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
648	`message`	S	1.33	Sets the text drawn above the NPC. An empty string clears it. Also turns off `timershow`.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
676	`showfile`	F	1.33	Opens the file with its associated program, like a double-click.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
678	`setbackpal`	F	1.33	Copies the 256-colour palette of the given image into the tileset and into every loaded level's tile image, then repaints. Remembers the file so the same palette is not reloaded.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
679	`setletters`	F	1.33	Replaces the sign font image (default `letters.png`) if the file exists.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
702	`playlooped`	F	1.37	Same, but the file is queued as a looped sound (internally the name gets a + prefix). Stops on level exit or `stopsound`.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
703	`stopsound`	F	1.37	Stops that (looped) sound.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
711	say2	S	1.41 rev1	Same as say, but shows the given text directly instead of a level sign.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
739	setmusicvolume	RR	2.0	Sets music volume, left and right (doc: 0..1).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
740	openurl2	MRR	2.01	Opens the URL in the client's own HTML window of the given width and height.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#messages-signs-sound-files) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

8.2.4.2 Flags and variables

Op	Command	Args	In the table from	What 2.05 does	Source
604	set	M	1.33	Sets a flag (doc: `server.` prefix makes it server-wide).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#flags-and-variables) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
607	unset	M	1.33	Clears a flag.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#flags-and-variables) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
666	setarray	RR	1.33	Resizes the array variable named by the first argument to *n* elements. Ignored unless the first argument is a plain or `this.` variable.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#flags-and-variables) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
677	setstring	SS	1.33	Sets string variable *name* = *value* (stored as the flag `name=value`). Empty value deletes it.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#flags-and-variables) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
742	tokenize	S	2.02 rev2	Splits the expanded string into the global token list on spaces and commas (quotes group).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#flags-and-variables) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
743	tokenize2	SS	2.02 rev2	Same, with extra delimiter characters added to the space/comma set.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#flags-and-variables) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

8.2.4.3 Control flow

Op	Command	Args	In the table from	What 2.05 does	Source
605	`if`	—	1.33	Evaluates the condition; jumps past the block when it is 0.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
634	`else`	—	1.33	Compiled away into a jump.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
660	`for`	—	1.33	Loop; condition checked each pass; stops after 10,000 passes.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
667	`sleep`	R	1.33	Sets the NPC's timeout to the value and remembers the next statement; the next `timeout` event resumes there. `sleep 0` just clears the timer display. Ends the current run.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
668	`while`	—	1.33	Same loop node as `for`.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
710	`callnpc`	RS	1.41 rev1	Runs NPC *index*'s script with the given event flag set. Doc: params after the event are passed as `#p(n)`.	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
712	`break`	—	1.41 rev1	Jumps to the end of the enclosing loop. Outside a loop it ends the script.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
713	`continue`	—	1.41 rev1	Jumps to the loop's next pass. Outside a loop it ends the script.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
714	`callweapon`	RS	1.41 rev1	Moves weapon *index* to the player's centre, then runs its script with the given event flag set.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
716	`function`	—	1.41 rev1	Compiled away. Calls to a named function become a call node (max depth 10,000).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
717	`return`	—	1.41 rev1	Pops the call stack and resumes after the call. With an empty stack it ends the script.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
744	`trigeraction`	RRSS	2.03	Fires *action* with *params* at the object on (x, y) locally, and sends it online. From a weapon, the sender is the player.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#control-flow) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

8.2.4.4 NPC appearance and behaviour

Op	Command	Args	In the table from	What 2.05 does	Source
608	`setgif` †	F	1.33	Sets the NPC image, shows the NPC, and resets the visible part to the whole image.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
619	`dontblocklocal`	—	1.34	Same, local only.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
620	`drawoverplayer`	—	1.33	Draw layer +1.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
631	`timershow`	—	1.33	Draws the NPC's timeout counter over it. Cleared by `message` and `sleep 0`.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
638	`blockagainlocal`	—	1.34	Same, local only.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
639	`drawunderplayer`	—	1.33	Draw layer −1.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
641	`canbecarried`	—	1.33	Player can / cannot lift the NPC.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
642	cannotbecarried	—	1.33	Player can / cannot lift the NPC.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
644	canbepushed	—	1.33	Player can / cannot push it.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
645	cannotbepushed	—	1.33	Player can / cannot push it.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
646	canbepulled	—	1.33	Player can / cannot pull it.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
647	cannotbepulled	—	1.33	Player can / cannot pull it.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
652	putnpc	QQRR	1.33	Offline: creates a new NPC with that image and script file at (x, y) and runs its `playerenters`. Online: asks the server instead.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
653	destroy	—	1.33	Removes the NPC from the level (and from the weapon list), sends the deletes online, ends the run.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
670	timereverywhere	—	1.33	This client gets the NPC's timeout events even when it is not the level leader.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
680	setgifpart †	FRRRR	1.33	Same, then shows only (x, y, w, h) of the image in pixels. x, y are clamped to 0..16000 and w, h to 0..220, then trimmed to the image size.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
686	showlocal	—	1.34	Same, but marked local (not synced).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
687	`hidelocal`	—	1.34	Same, marked local.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
688	`dontblock`	—	1.33	Player can walk through the NPC.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
689	`blockagain`	—	1.33	Blocks again and also resets the draw layer to normal (undoes `drawoverplayer`/`drawunderplayer`).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
719	`setimg`	F	1.41 rev1	Sets the NPC image, shows the NPC, and resets the visible part to the whole image.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
720	`setimgpart`	FRRRR	1.41 rev1	Same, then shows only (x, y, w, h) of the image in pixels. x, y are clamped to 0..16000 and w, h to 0..220, then trimmed to the image size.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
725	`setshape`	RRR	NewMain 1.42; takes arguments from 2.04	Type 0 or 1 (1 = fixed rectangle), width and height in pixels. Other types are ignored.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
726	`seteffect`	RRRR	1.33; out 1.34–1.41; back NewMain 1.42	Global screen tint r, g, b, a (day/night). Online gate applies.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
727	`setcoloreffect`	RRRR	NewMain 1.42	Per-NPC colour multiply r, g, b, a. Online gate applies.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
728	`setzoomeffect`	R	NewMain 1.42	Per-NPC zoom factor. Online gate applies.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
729	`drawaslight`	—	NewMain 1.42	Draw layer 9: above the day/night effect, below the status bar.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
745	`move`	RRRR	2.03	Smoothly moves the NPC by dx, dy over *time* seconds with option bits. (option bits not decoded)	[2.05][inf] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
751	`setshape2`	RRR	2.04	Makes the NPC block like a small level: width and height in tiles, plus a tile-type array (literal or array variable) of w×h entries. A short or missing array clears the shape.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
759	`show`	—	1.33	Shows the NPC (re-applies its image).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
760	`hide`	—	1.33	Hides the NPC (only if it is in the current level). If it was following the player, it stops following and the run ends.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

8.2.4.5 Character NPCs (`showcharacter`)

Op	Command	Args	In the table from	What 2.05 does	Source
633	`followplayer`	—	1.33	Offline only: NPC follows the player and stops blocking. Ignored online (matches the doc).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#character-npcs-showcharacter) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
673	`showcharacter`	—	1.33	Draws the NPC as a player character (once; later calls do nothing).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#character-npcs-showcharacter) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
675	setcharprop	MM	1.33	Sets a character property from a message code: #1 sword image, #2 shield image, #3 head image, #5 horse image (creates the horse if needed), #8 body image, #C0..#C5 colours (name or number), #P1..#P5 attributes, #c chat text (not in preview mode), #n nickname. Other codes are ignored. The first word is the property code (not expanded as a message). Codes inside the value are expanded first; on script-running servers #n in that value is the triggering player's nickname, while property code #n writes the NPC's nickname — see `server-notes/setcharprop-hash-n.md`.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#character-npcs-showcharacter) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
684	carryobject	L	1.34	Character carries the object (0..10; none = 10). Only for `showcharacter` NPCs; switches to the carry gani.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#character-npcs-showcharacter) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
685	throwcarry	—	1.34	Throws the carried object if the character is in a carry sprite; bombs become live bombs.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#character-npcs-showcharacter) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
701	takehorse	R	1.36	Character mounts horse *index* (removes it from the level). Only for `showcharacter` NPCs.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#character-npcs-showcharacter) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
709	setcharani	SS	1.39	Sets the character's gani and its parameter string.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#character-npcs-showcharacter) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
733	setchargender	S	2.0	`male` sets male; anything else sets female.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#character-npcs-showcharacter) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

8.2.4.6 The player

Op	Command	Args	In the table from	What 2.05 does	Source
603	hurt	R	1.33	Hurts the player by half-hearts and knocks them away from the NPC's centre. Skipped when the player cannot be hurt right now (dead, in a protected state, or flagged unhittable on an NPC-server). 0 or negative heals (doc).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
610	setlevel	M	1.33; takes arguments from 1.38	Warps the player to that level at the current x, y. Ends the run.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
611	setskincolor	C	1.33	Player colour slots 0..4.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
612	setcoatcolor	C	1.33	Player colour slots 0..4.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
613	setsleevecolor	C	1.33	Player colour slots 0..4.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
614	setshoecolor	C	1.33	Player colour slots 0..4.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
615	setbeltcolor	C	1.33	Player colour slots 0..4.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
616	`setplayerdir`	`D`	1.33	Player direction, reduced mod 4. Accepts a name or an expression (doc says names only).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
618	`takeplayercarry`	—	1.33	Removes what the player carries and resets the gani.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
621	`enableweapons`	—	1.33	Turns player weapons on / off.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
622	`disableweapons`	—	1.33	Turns player weapons on / off.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
632	`toinventory`	disputed → App. E#201	1.33	Sets the flag and moves the NPC into the inventory. Offline it leaves the level and the run ends; online a copy goes to the inventory and the NPC stays. Only for NPCs of the current level.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
649	`setsword`	`MR`	1.33	Sword image and power.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
650	`setshield`	`MR`	1.33	Shield image and power.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
654	`freezeplayer`	`R`	1.33	Freezes the player for 0..30 seconds (clamped) and drops what they carry.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
655	`sethead`	`F`	1.33	Player head image.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
656	`toweapons`	`M`	1.33	Adds a copy of the NPC as weapon *name* (replacing one with the same name, in the same slot), runs its `playerenters`, selects it, and sends it online. A `/` in the name keeps only the part after it.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
659	`hidesword`	R	1.33	Hides the sword for 0..30 seconds.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
663	`setbow` †	S	1.33	Sets the bow image on the running NPC's character, not on the player. (differs from doc, which says the player's bow)	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
669	`hideplayer`	R	1.33	Hides the player for 0..30 seconds.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
674	`setplayerprop`	MM	1.33	Same codes as `setcharprop` but on the player: `#1 #2 #3 #5 #8 #C0..#C5 #P1..#P5 #c`. #c sets the chat text with a 60-tick timeout and sends it online. #7 is not handled (doc lists it).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
683	`setbody`	F	1.33	Player body image (skipped if it is the same file).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
691	`takeplayerhorse`	—	1.36	Removes the player's horse.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
705	`setlevel2`	MRR	1.38	Warps to that level at (x, y). Ends the run.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
708	`setani`	SS	1.39	Player gani and parameter string.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
723	`showstats`	R	1.41 rev1	Bitmask of status-bar parts to show (doc: 1 ASD, 2 icons, 4 rupees, 8 bombs, ...).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
732	`setgender`	S	2.0	`male` sets male; anything else sets female.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
737	enabledefmovement	—	2.0	Turns the built-in movement and controls on / off.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
738	disabledefmovement	—	2.0	Turns the built-in movement and controls on / off.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
749	replaceani	disputed → App. E#201	2.04	Replaces one of the 31 default gani names (walk, idle, sword, ...) with another gani file. Online gate applies.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
750	attachplayertoobj	RR	2.04	Type 0 only (NPC): attaches the player to the NPC with that id. Online gate applies.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
752	detachplayer	—	2.04	Detaches the player if attached.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
753	enableselectweapons	—	2.04	Allows / blocks opening the weapon selection.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
754	disableselectweapons	—	2.04	Allows / blocks opening the weapon selection.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#the-player) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

8.2.4.7 Combat, items, level objects

Op	Command	Args	In the table from	What 2.05 does	Source
601	lay	I	1.33; takes arguments from 1.36	Drops the item next to the NPC, to the right or below depending on where the player stands (doc).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
602	take	I	1.33	Picks up the first item of that type lying within the NPC's radius (half its diagonal) and gives it to the NPC.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
609	removecompus	—	1.33	Deletes every baddy (doc: causes `compusdied`).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
617	putcomp	BRR	1.33	Offline: spawns that baddy at (x, y). Online: asks the server.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
623	shootarrow	D	1.33	Fires that projectile from the NPC's edge in the given direction (reduced mod 4). Drawn on the NPC's layer, or the player's for weapons.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
624	shootfireball	D	1.33	Fires that projectile from the NPC's edge in the given direction (reduced mod 4). Drawn on the NPC's layer, or the player's for weapons.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
625	shootfireblast	D	1.33	Fires that projectile from the NPC's edge in the given direction (reduced mod 4). Drawn on the NPC's layer, or the player's for weapons.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
626	shootnuke	D	1.33	Fires that projectile from the NPC's edge in the given direction (reduced mod 4). Drawn on the NPC's layer, or the player's for weapons.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
627	putbomb	RRR	1.33	Bomb of power 0..2 at (x, y).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
628	putexplosion	RRR	1.33; takes arguments from 1.41 rev1	Explosion of radius *r* at (x, y), with sound.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
629	putleaps	RRR	1.33	Leaps of type 0..5 at (x, y) (bush, swamp, stone, sign, ball, water).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
630	shootball	—	1.33	Fires a ball from the NPC's centre toward the player.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
640	`puthorse`	disputed → App.E#201	1.33	Horse with that image at (x, y).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
643	`noplayerkilling`	—	1.33	Turns off player-vs-player damage in this level.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
651	`putnewcomp`	BRRMR	1.33	Same, plus image and power.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
661	`hitcompu`	RRRR	1.33	index, power, fromx, fromy: damages and knocks back the baddy, and sends it online.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
662	`hitplayer`	RRRR	1.33	index, half-hearts, fromx, fromy. Local player: hurt with knockback. Another player: UDP 21 to that player only (skipped if they are dead or protected). An NPC: treated like `hitnpc`.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
692	`removebomb`	R	1.36	Deletes bomb *index*.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
693	`removearrow`	R	1.36	Deletes arrow *index* (local only, doc).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
694	`removeitem`	R	1.36	Deletes item *index*.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
695	`removeexplo`	R	1.36	Deletes explosion *index*.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
696	`removehorse`	R	1.36	Deletes horse *index*.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
697	explodebomb	R	1.36	Detonates bomb *index* now.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
698	reflectarrow	R	1.36	Reverses arrow *index*. Against a character NPC it bounces off the side it hits; otherwise both directions flip. The arrow's sprite turns to its new direction and the shooter becomes the player when called from a weapon.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
699	lay2	IRR	1.36	Drops the item at (x, y).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
700	take2	R	1.36	Picks up item *index*.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
706	hitnpc	RRRR	1.38	NPC *index* gets the damage; its script gets `washit` style handling (no automatic push, doc).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
715	spyfire	RR	1.41 rev1	Spawns a fire-spy effect just in front of the player (player position plus their facing direction) and sends it online. (role of the two arguments)	[2.05][inf] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
722	putexplosion2	RRRR	1.41 rev1	Power, radius, x, y; power 3 is a jolt-bomb explosion.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
741	hitobjects	RRR	2.01	Hits everything at (x, y) with *power*, like a sword, and sends it online.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#combat-items-level-objects) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

8.2.4.8 Board, map, images

Op	Command	Args	In the table from	What 2.05 does	Source
657	showimg	disputed → App. E#201	1.33	Image *index* at (x, y), on the NPC's layer +1 (player's for weapons). Re-using an index replaces it; the same file only moves. A name starting with `@` is a text image.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
658	hideimg	R	1.33	Deletes image *index*.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
664	updateboard	RRRR	1.33	Redraws the board rectangle (x, y, w, h) after `board[]` writes (doc: also syncs online).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
665	putobject	0RR	1.33	Offline only: stamps a predefined tile object on the 64×64 board at (x, y), clipped to the board, then redraws. Ignored online (doc: security).	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
671	setminimap	FFRR	1.33	Same for the minimap.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
672	setmap	FFRR	1.33	Loads the big map image and its level-names file, sets the player's head-icon position, and recomputes "is on map" for every player. Both files must exist.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
704	changeimgpart	RRRRR	1.38	Visible part of image *index*; x, y clamped to 0..16000, w, h to 0..220.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
718	changeimgvis	RR	1.41 rev1	Layer of image *index*: 0, 1, 2 = under / level with / over the NPC; 3 = light layer (like `drawaslight`); 4 = screen (over the status bar).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
730	changeimgcolors	RRRRR	New Main 1.42	Colour multiply for image *index*. Online gate applies.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
731	changeimgzoom	RR	New Main 1.42	Zoom for image *index*. Online gate applies.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
746	addtiledef	SSR	2.04	Use image *file* instead of `pics1.png` for levels starting with *prefix*. Online gate applies.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
747	addtiledef2	SSRR	2.04	Same, but pastes the image at (x, y) of the tileset. Online gate applies.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
748	removetiledefs	S	2.04	Removes tile definitions for *prefix*. Online gate applies.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
758	loadmap	F	2.05 rev1	Loads a gmap file; if the current level belongs to it (and the editor is closed), rebuilds the level view.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#board-map-images) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

8.2.4.9 Parsed but ignored by the 2.05 client

Op	Command	Args	In the table from	What 2.05 does	Source
636	seturllevel	M	1.33	parsed, no executor case; doc describes it as an offline URL warp; not implemented here.	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
681	setfocus	RR	1.33	parsed, no executor case; later logs mention a camera focus at (x, y).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
682	resetfocus	—	1.33	parsed, no executor case	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
690	setbacktile †	R	1.34; takes arguments from 2.05 rev1	parsed, no executor case; allowed in preview mode but still a no-op.	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
707	canwarp2 †	—	NewMain 1.42	parsed, no executor case; warp permission is handled by the server in this era (inf).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
721	drawovertrees †	—	1.41 rev1	parsed, no executor case	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
724	canwarp †	—	1.41 rev1	parsed, no executor case	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

Op	Command	Args	In the table from	What 2.05 does	Source
734	`join` †	S	2.0	parsed, no executor case; server-side class join in later versions (inf).	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
735	`freezeplayer2` †	—	2.0	parsed, no executor case	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
736	`unfreezeplayer`	—	2.0	parsed, no executor case	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
755	`setbacktile2` †	RRRRR	2.05 rev1	parsed, no executor case	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
756	`setz` †	RRRRRRR	2.05 rev1	parsed, no executor case	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)
757	`updateterrain`	—	2.05 rev1	parsed, no executor case	[2.05] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#parsed-but-ignored-by-the-205-client) [bin] (client-internals/cross-version-timeline.md#gs1-commands)

8.2.5 Functions and variables

Expressions resolve names through a table of functions and variables with numeric ids [2.05] (npc-library/gs1-reference-205.md#functions-and-variables-205).

Ids	Group	Names	Source
100–102	board and conversion	`board[R]`, `strtofloat(S)`, `timevar`	[2.05] (npc-library/gs1-reference-205.md#functions-and-variables-205)
200–218	object counts and globals	`compuscount`, `playerscount`, `bombscount`, `arrowscount`, `itemscount`, `exploscount`, `horsescount`, `npcscount`, `signscount`, `selectedweapon`, `screenwidth`, `screenheight`, `tokenscount`, `paramscount`, `actionplayer`, `weaponscount`, `levelorgx`, `levelorgy`	[2.05] (npc-library/gs1-reference-205.md#functions-and-variables-205)
400–455	maths, tests, string tests	`random(R,R)`, `sin(R)`, `cos(R)`, `arctan(R)`, `int(R)`, `abs(R)`, `arraylen(R)`, `testcompu(R,R)`, `testplayer(R,R)`, `testbomb(R,R)`, `testitem(R,R)`, `testexplo(R,R)`, `testhorse(R,R)`, `testnpc(R,R)`, `testsign(R,R)`, `tiletype(R,R)`, `onwall(R,R)`, `playersays(S)`, `playersays(R,S)`, `playersays2(S)`, `playersays2(R,S)`, `strequals(S,S)`, `strcontains(S,S)`, `hasweapon(S)`, `onwater(R,R)`, `keydown(R)`, `startswith(S,S)`, `indexof(S,S)`, `strlen(S)`, `onmapx(S)`, `onmapy(S)`, `vecx(R)`, `vecy(R)`, `getz(R,R)`	[2.05] (npc-library/gs1-reference-205.md#functions-and-variables-205)

Ids	Group	Names	Source
600–606	compus[i].*	7 members: x, y, type, dir, headdir, power, mode	[2.05][doc] [inf] (npc-library/gs1-reference-205.md#functions-and-variables-205)
640–643	bombs[i].*	4 members: x, y, power, time	[2.05][doc] [inf] (npc-library/gs1-reference-205.md#functions-and-variables-205)
660–666	arrows[i].*	7 members: x, y, dx, dy, dir, type, from	[2.05][doc] [inf] (npc-library/gs1-reference-205.md#functions-and-variables-205)
680–683	items[i].*	4 members: x, y, type, time	[2.05][doc] [inf] (npc-library/gs1-reference-205.md#functions-and-variables-205)
700–704	explos[i].*	5 members: x, y, power, time, dir	[2.05][doc] [inf] (npc-library/gs1-reference-205.md#functions-and-variables-205)
720–726	horses[i].*	7 members: x, y, dir, bushes, bombs, bombpower, type	[2.05][doc] [inf] (npc-library/gs1-reference-205.md#functions-and-variables-205)
740–758	this.* and npcs[i].*	x, y, width, height, rupees, darts, bombs, hearts, timeout, glovepower, swordpower, shieldpower, hurtdx, hurtdy, id, save[2], dir, sprite, ap; save takes an index	[2.05] (npc-library/gs1-reference-205.md#functions-and-variables-205)
780–781	signs[i].*	2 members: x, y	[2.05][doc] [inf] (npc-library/gs1-reference-205.md#functions-and-variables-205)
800–819	player* and players[i].*	playerx, playery, playerrupees, playerdarts, playerbombs, playerhearts, playerfullhearts, playerdir, playerglovepower, playerbombpower, playershootpower, playerswordpower, playershieldpower, playerheadset, playersprite, playerid, playersaysnumber, playermp, playerap, playerhurtdx	[2.05] (npc-library/gs1-reference-205.md#functions-and-variables-205)

Ids	Group	Names	Source
820–825	player* and players[i].*	playerhurtdy, playerfreezetime, playerhurtpower, playerattachid, playerattachtype, playerz	[2.05] (npc-library/gs1-reference-205.md#functions-and-variables-205)

The member names of compus[i], bombs[i], arrows[i], items[i], explos[i], horses[i] and signs[i] are not in the expression table. Each group's member count matches the 2.22 documentation exactly, and the names above are the documentation's names in its order; the name-to-id mapping stays an inference until the member parser is traced → App. E#204 [2.05][doc][inf] (npc-library/gs1-reference-205.md#functions-and-variables-205).

The NewWorld betas add eight time variables after the direction list. They are variables (also usable as #v(name) codes), not commands; the server sends the time in packet 42 [bin][doc] (client-internals/newworld-betas.md#gs1-changes):

Variable	Range	Source
nwday	1…28	[doc] (client-internals/newworld-betas.md#gs1-changes)
nwweek	1…40	[doc] (client-internals/newworld-betas.md#gs1-changes)
nwyear	1000…	[doc] (client-internals/newworld-betas.md#gs1-changes)
nwweekday	1…7	[doc] (client-internals/newworld-betas.md#gs1-changes)
nwmonth	1…10	[doc] (client-internals/newworld-betas.md#gs1-changes)
nwtime	0…1439 minutes	[doc] (client-internals/newworld-betas.md#gs1-changes)
nwhour	0…23	[doc] (client-internals/newworld-betas.md#gs1-changes)
nwmin	0…59	[doc] (client-internals/newworld-betas.md#gs1-changes)

8.2.6 Constants

Constant set	Values	Source
baddy modes	0 walking, 1 looking, 2 hunting, 3 hurt, 4 bumped, 5 dying, 6 shooting (swamp), 7 jumping (frog), 8 shooting (spider), 9 dead	[doc] (npc-library/gs1-reference-205.md#constants-doc)
player sprites	0 idle; 1–8 walk; 9–13 sword; 14–18 push; 19–22 pull; 23 lift; 24 carry idle; 25–32 carry walk; 33 shoot; 34–36 ride; 37 sit; 38 sleep; 39 hurt; 40 dead; 39 and 40 only on showcharacter NPCs	[doc] (npc-library/gs1-reference-205.md#constants-doc)
tile types (tiletype(x,y))	0 nonblocking, 2 hurting underground, 3 chair, 4 bed top, 5 bed bottom, 6 swamp, 7 lava swamp, 8 near water, 9 near lava, 11 water, 12 lava, 20 throw-through, 21 jumping stone, 22 blocking	[doc] (npc-library/gs1-reference-205.md#constants-doc)
move options	cachetype 0–2; +4 block check; +8 event when done; +16 apply direction; 2.05's handling of the bits is not decoded → App. E#206	[doc] (npc-library/gs1-reference-205.md#constants-doc); [2.05][inf] (npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour)
showstats mask	1 ASD, 2 icons, 4 rupees, 8 bombs, and further bits the notes do not list	[doc] (npc-library/gs1-command-reference-205.md#the-player)
changeimgvis layers	0, 1, 2 under, level with, over the NPC; 3 light layer, as drawaslight; 4 screen, over the status bar	[2.05] (npc-library/gs1-command-reference-205.md#board-map-images)
putleaps types	0 bush, 1 swamp, 2 stone, 3 sign, 4 ball, 5 water	[2.05] (npc-library/gs1-command-reference-205.md#combat-items-level-objects)
seteffectmode (2.22)	0 light, 1 transparency, 2 holes	[doc] (npc-library/gs1-reference-205.md#constants-doc)

8.2.7 The command table by build

The executor matches commands against one table of fixed-size records [bin] (client-internals/cross-version-timeline.md#gs1-commands). In 2.05 and the NewWorld betas a record is 12 bytes: name, a flags word and an argument-signature string [bin] (client-internals/newworld-betas.md#gs1-changes). Other keyword lists that follow the table in some builds (*=, :=, the 1.33 soldier names, the NewWorld time variables) are not counted [bin] (client-internals/cross-version-timeline.md#gs1-commands).

Build	Commands	Added over the previous row	Source
1.33	88	n/a; earliest build read	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.34	93	`dontblocklocal`, `blockagainlocal`, `showlocal`, `hidelocal`, `carryobject`, `throwcarry`, `setbacktile`; `seteffect` and `unseteffect` leave the table	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.35	93	none; same table as 1.34	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.36	104	`lay2`, `take2`, `takehorse`, `takeplayerhorse`, `removebomb`, `removearrow`, `removeitem`, `removeexplo`, `removehorse`, `explodebomb`, `reflectarrow`; `lay` now takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.37	106	`playlooped`, `stopsound`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.38	109	`setlevel2`, `changeimgpart`, `hitnpc`; `setlevel` now takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.39	111	`setani`, `setcharani`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.41 rev 1	126	`canwarp`, `callnpc`, `say2`, `break`, `continue`, `callweapon`, `spyfire`, `function`, `return`, `changeimgvis`, `setimg`, `setimgpart`, `drawovertrees`, `putexplosion2`, `showstats`; `putexplosion` now takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
NewMain 1.42	134	`canwarp2`, `setshape`, `seteffect` (back), `setcoloreffect`, `setzoomeffect`, `drawaslight`, `changeimgcolors`, `changeimgzoom`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.0	142	`setgender`, `setchargender`, `join`, `freezeplayer2`, `unfreezeplayer`, `enabledefmovement`, `disabledefmovement`, `setmusicvolume`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.01	144	`openurl2`, `hitobjects`; `openurl` now takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.02 rev 2	146	`tokenize`, `tokenize2`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.03	148	`triggeraction`, `move`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.04	157	`addtiledef`, `addtiledef2`, `removetiledefs`, `replaceani`, `attachplayertoobj`, `detachplayer`, `setshape2`, `enableselectweapons`, `disableselectweapons`; `setshape` now takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.05 rev 1, rev 2	161	`setbacktile2`, `setz`, `updateterrain`, `loadmap`; `setbacktile` now takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands; client-internals/build-diff-204-205.md#204--205)
2.1 tests	161	none; same table as 2.05	[bin] (client-internals/cross-version-timeline.md#gs1-commands)

Build	Commands	Added over the previous row	Source
New World beta 1	161	none; same table as 2.05	[bin] (client-internals/cross-version-timeline.md#gs1-commands; client-internals/newworld-betas.md#gs1-changes)
New World beta 2	165	`addstring`, `insertstring`, `removestring`, `deletestring`, inserted after `loadmap`	[bin] (client-internals/cross-version-timeline.md#gs1-commands; client-internals/newworld-betas.md#gs1-changes)

Every command or variable the 2.01 to 2.05 change logs name appears in that build's string table; the only undocumented additions are `actionplayer`, `lighteffectsenabled` (2.03) and the `bomy_*` animation names [bin][doc] (client-internals/decomp-findings-2026-10-24-c/09-features-classes-vocabulary.md#vocabulary-scriptidentifier-strings-resources-excluded); [bin] (client-internals/decomp-findings-2026-10-24-c/script_vocab_deltas.tsv). The 2.1 tests add the strings `ascii`, `defnpc`, `playeronline`, `startinfullscreen` and `tiles` (test 2) and `groundheights`, `waterheight`, `playerhp` and `playermaxhp` (test 3) outside the command table [bin] (client-internals/decomp-findings-2026-10-24-c/08-test-builds-newworld-newmain.md#21-test2-and-test3).

Being in the table does not mean a command does anything. Executor cases were read for 2.05 (→ §8.2.4.9) and for the NewWorld betas against their bases; other builds have not been checked [bin] (client-internals/cross-version-timeline.md#gs1-commands).

8.2.7.1 NewWorld betas

Item	Rule	Source
string lists (beta 2 only)	`addstring list,text` (S,S), `insertstring list,index,text` (S,R,S), `removestring list,text` (S,S), `deletestring list,index` (S,R), with `sarraylen(list)`, `lindexof(list,text)` and `#I(list,i)`	[bin] (client-internals/newworld-betas.md#gs1-changes)
list storage	a list is a normal string flag in comma form: `numbers="12 34",567,"8,9"`	[doc] (client-internals/newworld-betas.md#gs1-changes)
live in the betas, ignored in 2.05	`setfocus`, `resetfocus`, `setbacktile`, `drawovertrees`, `setbacktile2`, `setz`, `updateterrain`	[bin] (client-internals/newworld-betas.md#gs1-changes)
parsed but ignored in the betas, live in 2.05	`removecompus`, `putcomp`, `putnewcomp`, `hitcompu`; `puthorse`, `takehorse`, `takeplayerhorse`, `removehorse`; `carryobject`, `throwcarry`, `takeplayercarry`; `setsword`, `hidesword`, `setbow`; `freezeplayer`; `enabledefmovement`, `disabledefmovement`; `toinventory`; `followplayer`	[bin] (client-internals/newworld-betas.md#gs1-changes)
why	the control NPC script (`controlnpc.txt`) takes over movement, swords, carrying and horses	[bin][doc] (client-internals/newworld-betas.md#gs1-changes)

8.2.7.2 Scripting changes from 2.1 to 2.31

From the release notes; the binaries have not been read for the command tables of these builds [doc] (client-internals/feature-timeline.md).

Version	Scripting change	Source
2.02	string parsing	[doc] (client-internals/feature-timeline.md)
2.04	gani `PLAYSOUND PARAM1`, `replacegani`	[doc] (client-internals/feature-timeline.md)
2.1	gani scripts (`SCRIPT … SCRIPTEND`), string arrays, `play2`, `graalversion`	[doc] (client-internals/feature-timeline.md)

Version	Scripting change	Source
2.12	`wraptext`, `wraptext2`	[doc] (client-internals/feature-timeline.md)
2.15	`showimg2`	[doc] (client-internals/feature-timeline.md)
2.16	`seteffectmode` (real transparency); gani attributes `#P10`–`#P30`	[doc] (client-internals/feature-timeline.md)
2.17	sound through BASS, with the same GS1 sound commands plus `musicpos` and `musiclen`	[2.17][2.22] (client-internals/feature-specs/01-sound.md)
2.19	`this.` strings on the client; on NPC-server servers only `client.` strings are sent	[doc] (client-internals/feature-timeline.md)
2.2	`updateboard2` for server-side tile changes on gmaps	[doc] (client-internals/feature-timeline.md)
2.21	`triggeraction 0,0,clientside,…` reaches a weapon's client-side part	[doc] (client-internals/feature-timeline.md)
2.22	`gravity`, `downloadsize`	[doc] (client-internals/feature-timeline.md)
2.3	classic client-side NPC mode removed; `timevar2`	[doc] (client-internals/feature-timeline.md)
2.22 (symbol table, against 2.17)	added `gravity`, `textwidth(zoom,font,style,text)`, `textheight(zoom,font,style)`, `isthisflagset`, `isplayerflagset`, and a #R message code with a string argument; removed `isflagset`, which the two new tests replace	[bin] (version-profiles/2.22-2.31.md#gs1-script-differences-from-the-clients-symbol-table-bin)
2.31 (symbol table, against 2.22)	added `timevar2`, `downloadsize`, `onwall2(x,y,w,h)`, `onwater2(x,y,w,h)`; nothing removed	[bin] (version-profiles/2.22-2.31.md#gs1-script-differences-from-the-clients-symbol-table-bin)
2.22 `downloadsize`	announced in the 2.22 notes but absent from the 2.22 symbol table; a 2.22 script sees it as undefined (0)	[bin][doc][inf] (version-profiles/2.22-2.31.md#gs1-script-differences-from-the-clients-symbol-table-bin)

8.2.7.3 2.05 against the 2.22 documentation

Item	Names	Source
documented for 2.22, absent in 2.05	`addstring`, `insertstring`, `replacestring`, `removestring`, `deletestring`, `putnpc2`, `showtext`, `showtext2`, `showpoly`, `showani`, `play2`, `setshootparams`, `shoot`, `enablefeatures`, `seteffectmode`, `changeimgmode`, and the mouse and `keypressed` family	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205)
in 2.05, not in the 2.22 `commands.rtf`	the † rows of §8.2.4	[2.05][doc] (npc-library/gs1-reference-205.md#commands-161-205)

8.2.8 Superseded readings

An earlier catalogue of GS1 commands, built from public tables and the scrape before 2.05 was read, is corrected by the 2.05 reading [2.05] (npc-library/gs1-command-reference-205.md#corrections-to-gs1-commandsmd). The corrected values are the ones in this chapter.

Earlier reading	Corrected	Source
R is the literal string "real"	R is a numeric expression	[2.05] (npc-library/gs1-command-reference-205.md#corrections-to-gs1-commandsmd); [file] (npc-library/gs1-commands.md#pattern-letters)

Earlier reading	Corrected	Source
`D` is 0 south, 1 west, 2 east, 3 north	`D` is 0 up, 1 left, 2 down, 3 right, or a number	[2.05] (npc-library/gs1-command-reference-205.md#corrections-to-gs1-commandsmd); [file] (npc-library/gs1-commands.md#pattern-letters)
`warp` and `jump` are commands	neither is a GS1 command in 2.05 (→ §8.1.8)	[2.05] (npc-library/gs1-command-reference-205.md#corrections-to-gs1-commandsmd)
`canwarp`, `canwarp2`, `join`, `setfocus`, `resetfocus`, `unfreezeplayer` act	they compile but do nothing in 2.05	[2.05] (npc-library/gs1-command-reference-205.md#corrections-to-gs1-commandsmd)
`spyfire` is a debug hook	`spyfire` is a game effect with a network message (PLI 10)	[2.05] (npc-library/gs1-command-reference-205.md#corrections-to-gs1-commandsmd)
the 2.05 table has 163 commands, or ends at 758	161 commands, 600–760, ending with `show` and `hide`	[2.05] (npc-library/gs1-commands.md#update-2026-10-21; npc-library/gs1-reference-205.md#commands-161-205)
2.05 runs OpenGS1-style bytecode	2.05 interprets a parse tree built from script text (→ §8.1.2)	[2.05] (npc-library/gs1-commands.md#update-2026-10-21)

8.3 GS2 language

GS2 is the script language of the 4.x and later clients. The notes record where GS2 appears, how it is delivered and the names of the events and functions seen in clients and labs. They do not document its syntax, standard library or event model; that gap is → App. E#207. GS1 is in → §8.1.

> Status: draft. Complete for what the notes hold; the language itself is open (→ App. E#207).

8.3.1 Where GS2 appears

Builds	Scripting	Source
2.3	classic client-side NPC mode removed; Classic moves to the NPC server	[doc] (client-internals/feature-timeline.md)
2.22, 2.31	GS1 only	[bin][doc] (version-profiles/matrix/README.md#thinks-wire-and-scripting)
3.x (`G3D03014`, `G3D28015`)	GS1 natively, with TorqueScript `.dso` files for menus; 3.153 handles the NPC bytecode ids	[bin] (version-profiles/matrix/README.md#thinks-wire-and-scripting)
4.x (`G3D17026`, `G3D27076`, `G3D20126`)	GS2 bytecode, server-pushed ganis (PLO 195), update packages `GRPKG001`	[bin][cap][doc] (version-profiles/matrix/README.md#thinks-wire-and-scripting)
4.04 rev 2	still runs GS1 weapon scripts and sends only `client.` string flags (→ §8.1.1)	[live] (client-internals/live-behaviour-404.md)
4.x 3D engine	the Torque console and TorqueScript are gone; every engine object is exposed to Graal's GS2 script objects; there is no TorqueScript compiler and `.dso` files are never requested	[bin] (graal3d/torque-engine-4x.md#overview)
5.x (`G3D22067`, `G3D14097`, `G3D15010`, `G3D26090`)	GS2, with the `GRESCR01` script cache	[bin][cap][doc] (version-profiles/matrix/README.md#thinks-wire-and-scripting)
6.x	GS2; in 6.11 even the packet id tables come from the server's script	[bin][cap][doc] (version-profiles/matrix/README.md#thinks-wire-and-scripting)

8.3.2 Delivery

How the server hands GS2 code to a client. Which message id carries the compiled code is disputed → App. E#202.

Item	Rule	Source
NPC script	for v4 and later clients the NPC script prop is replaced by `NPCPROP_CLASS` plus a bytecode message; 2.05 never sees that path	[srv] (protocol/packet-layouts.md); [srv] [2.05] (client-internals/feature-specs/02-weapons-combat.md#weapon-lists-server-to-client-srv205)
NPC bytecode	`PLO_NPCBYTECODE` (131), `i3` code then the bytecode; layout and version gate → §8.4.1	[srv] (protocol/packet-ids.md; protocol/graphics-movement-bytecode.md#on-wire)
weapon script	`PLO_NPCWEAPONSCRIPT` (140): `s` `info_length`, then the script	[srv] (protocol/packet-ids.md)
script load	`PLO_LOADSCRIPT` (197): a list `type,name,saveToDisk,desKey,crc32`; type is weapon, npc, class or gani; the key is two `i5`	[srv] (protocol/packet-ids.md)

Item	Rule	Source
3D mode	the 3D server, not the TCP game server, delivers the client's GS2 weapons and flags in 3D mode, as text commands `addweapon`, `weaponscript`, `deleteweapon` and `addnpc` (gives a named client-side object its script)	[bin][inf] (graal3d/torque-engine-4x.md); [bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)

8.3.3 Names seen in clients and labs

These GS2 names appear in the notes; each row gives only what the source says about it:

Name	Kind	What the source says	Source
`sendtext`	function	a message of four strings `(weapon, type, option, data)`; the 3D plug-in's text messages have this shape	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
`receivetext`	event	a weapon named in such a message receives it	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
`onRCChat`	event	receives messages of type `rc`	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll; graal3d/torque-engine-4x.md)
`onRPGMessage(data)`	event	receives the `rpg` command	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll; graal3d/torque-engine-4x.md)
`action<name>`	event	a `t` line in an `events` message triggers it on a weapon or GUI (`weapon`, `gui`) or an NPC (`npc`)	[bin] (graal3d/torque-engine-4x.md)
`trigger`, `getweapon`	client events	the client batches them per connection as `t …` and `w …` lines (others as `s …`) in one `events` message	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
`onConnectionAccepted`, `onConnectRequestTimedOut`, `onConnectRequestRejected`, `onConnectionDropped`, `onConnectionError`	events on the universe	fired for each 3D connection outcome	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
`onGraal3DLoginStep`	event	reports the three Torque mission-load phases, with the datablock and ghost events	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
`getDatablockProperties(obj, list)`	function	lists datablock fields (name, type and attributes) for GS2 scripts	[bin] (graal3d/torque-engine-4x.md; protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
`addtiledef("pics1.png", "", 0)`	function	called from a GS2 weapon - `System` on a 5.x/6.x lab server; the full level drew after the warp with or without that weapon	[shot] (protocol/v5v6-lab-play.md#the-login-server-mode)

8.3.4 Tools around GS1 and GS2

Tool	Relevance	Source
LevelScriptEditor	lists every NPC of a folder of `.nw` levels and replaces identical scripts in one go; it reads like a tool for porting GS1 scripts to GS2 on a big world	[src][inf] (client-internals/reference-reimplementations/levelscripteditor.md#what-it-is)

8.4 GS2 bytecode

This chapter covers compiled scripts on the wire. 2.05 and earlier clients receive GS1 as text and never see bytecode (→ §8.1.2). The format of the bytecode that 4.x and later clients receive is not in the notes (→ App. E#207). The only bytecode format the notes describe is OpenGS1's, a reimplementation, given in → §8.4.3 for reference.

> Status: draft. The GS2 bytecode format itself is open (→ App. E#207); the id that carries it is disputed (→ App. E#202).

8.4.1 Transport

Item	Rule	Source
message	`PLO_NPCBYTECODE` (131, wire byte `0xa3`): `i3` code, then the bytecode	[srv] (protocol/packet-ids.md)
version gate	sent only to clients at or above the GServer-v2 version constant `CLVER_4_0211`, and only when the bytecode is non-empty	[srv] (protocol/graphics-movement-bytecode.md#on-wire)
framing	wrapped in `PLO_RAWDATA` with a length prefix (→ §2.1.2)	[srv] (protocol/graphics-movement-bytecode.md#on-wire)
with NPC props	an NPC is sent as `PLO_NPCPROPS` plus the optional `PLO_NPCBYTECODE`	[srv] (protocol/graphics-movement-bytecode.md#other-level-objects)
2.05	the 2.05 receive switch does not handle it	[srv][2.05] (protocol/packet-ids.md)
other ids	a second reading puts GS2 bytecode on id 197 → App. E#202	[srv] (protocol/packet-layouts.md; client-internals/feature-specs/02-weapons-combat.md#weapon-lists-server-to-client-srv205)
how the client stores it	not traced	[inf] (protocol/graphics-movement-bytecode.md#what-still-needs-capture-or-confirmation)

8.4.2 Format

The notes do not decode the bytecode a 4.x or later client receives, and do not say whether it shares anything with OpenGS1's format → App. E#207 [inf] (protocol/graphics-movement-bytecode.md#what-still-needs-capture-or-confirmation).

8.4.3 OpenGS1 bytecode (reference reimplementation)

OpenGS1 defines a stack-based bytecode VM for GS1 [src] (protocol/graphics-movement-bytecode.md#scripting--gs1-bytecode). It is an independent reimplementation: 2.05 does not share its opcode set and does not run bytecode at all [2.05] (npc-library/gs1-reference-205.md#execution-model-205). How the notes first read this format is → App. E#208.

File layout, in order:

Part	Content	Source
1	u32 body offset	[src] (protocol/graphics-movement-bytecode.md#bytecode-file-layout-bytecodecpp)
2	u32 string-constant count, then length-prefixed strings	[src] (protocol/graphics-movement-bytecode.md#bytecode-file-layout-bytecodecpp)
3	u32 number-constant count, then `f32` values	[src] (protocol/graphics-movement-bytecode.md#bytecode-file-layout-bytecodecpp)
4	the bytecode body, starting at the body offset	[src] (protocol/graphics-movement-bytecode.md#bytecode-file-layout-bytecodecpp)

A packed value is 4 bytes:

Field	Bits	Content	Source
valueType	4	constant number, constant string, constant array, or named	[src] (protocol/graphics-movement-bytecode.md#packedvalue-4-bytes)
value	16	an index into the constant table or the name table	[src] (protocol/graphics-movement-bytecode.md#packedvalue-4-bytes)
unused	12	n/a	[src] (protocol/graphics-movement-bytecode.md#packedvalue-4-bytes)

Opcodes, by name; the notes give no numeric values:

Opcode	Meaning	Source
OP_PUSH	push a packed value	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)
OP_ASSIGN	S(1) = S(0)	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)
OP_ARR_SET, OP_ARR_GET	array access	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)
OP_ADD … OP_POW	arithmetic	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)
OP_INC, OP_INCPUSH, OP_DEC, OP_DECPUSH	increment and decrement	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)
OP_CALL, OP_CMD_CALL	function call, command call	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)
OP_JMP, OP_JAL, OP_RET	control flow	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)
OP_EQ, OP_LT, OP_GT, OP_LTE, OP_GTE	comparisons	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)
OP_NOT, OP_AND, OP_OR	logic	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)
OP_JEZ, OP_JNZ	conditional jumps	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)
OP_STOP	halt	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)
OP_DBG_OUT	debug output	[src] (protocol/graphics-movement-bytecode.md#opcodes-from-operationhpp)

9 Graal3D (2000)

Graal3D is a separate 3D side project from February 2000, with its own Delphi client (`Graal3D.exe`) and its own server (`GServer.exe`). It does not talk to the 2D GServer or the list server. The Torque-based 3D of the 4.x clients is a different program (→ §10).

> Status: draft.

9.0 What ships and how it relates to 2D Graal

Fact	Source
The archive `graal3d.7z` holds the client, the server, `graal3dconfig.txt`, `serverconfig.txt`, two levels (`level1.x`, `startlevel.x`), eight player-model meshes and their BMP textures	[file] (graal3d/README.md#what-ships)
Both programs carry resource stamps from 2000-02-05, between 2D releases 1.32 (`GSERV018`) and 1.33 (the first `GNW` token)	[bin] (graal3d/README.md#graal3d-february-2000)
The client is Delphi with DelphiX `TDXDraw` in Direct3D retained mode (z-buffer, textures), a 50 ms `TDXTimer`, and zlib 1.1.2 linked	[bin] (graal3d/README.md#what-ships)
Neither program has `.graal`/`.nw` loaders, GANI, tiles or NPC scripting	[bin] (graal3d/README.md#what-ships)
The server's default TCP port is 14901, its levels live in `levels\`, and accounts in `accounts.txt`	[file] (graal3d/README.md#what-ships)
The server holds at most 200 players; a player id is the lowest free slot from 1 to 200	[bin] (graal3d/README.md#server-windows)
Every Delphi class in the Graal3D server also exists in the 2D `GServer13.exe` of 1999-11-29; the 2D server has 23 more (NPC, baddy, chest, horse, level-link and board-modify classes, and their deltas)	[bin] (graal3d/README.md#how-it-relates-to-the-2d-graal-of-the-same-time)
The account record is 44 bytes in both servers; the level object shrinks from 8,244 bytes (a 64×64 board) to 24 bytes (name, file text, players in it, players already sent the file)	[bin] (graal3d/README.md#how-it-relates-to-the-2d-graal-of-the-same-time)
The per-observer "unsent props" tracker, the admin forms and the guild-tag check are carried over from the 2D server	[bin] (graal3d/README.md#how-it-relates-to-the-2d-graal-of-the-same-time)
The client keeps the 2D client's class names for the main form, the socket object, the connect window, the options window, the self-check and the DES helper; the level, NPC, weapon and script classes are gone	[bin] (graal3d/README.md#how-it-relates-to-the-2d-graal-of-the-same-time)
Each frame tick the client reads TCP, reads UDP, diffs its own player props and queues the changes, then flushes the TCP send buffer	[bin] (graal3d/README.md#client-windows)
The client windows are the main 3D view with a chat line, the connect window (nickname, host, port 14901, the built-in guest account), "Hardware options" (fixed UDP port, "Don't use UDP") and "Player look"	[bin] (graal3d/README.md#client-windows)
The server windows are a start dialog, the main window (player count, "Edit accounts", "Update levels", a tray menu), an account list and editor, an online player list, a message window and a level-reload window	[bin] (graal3d/README.md#server-windows)

9.1 Protocol

9.1.1 Encoding and frames

Rule	Source
The text codec is the 2D one: `c` is one byte + 32, `sN` is an N-byte integer with 7 bits per byte, most significant first, each byte + 32, and `str` is `c(len)` then the bytes (→ §2.1)	[bin] (graal3d/protocol.md#encoding)
A TCP frame is a 2-byte big-endian length, then one zlib block holding one or more messages, each `c(id)` + payload + `\n`	[bin] (graal3d/protocol.md#tcp-frames)
Both sides compress when the queue passes 4096 bytes or on the frame-tick flush; the client sends in 256-byte slices and inflates into a buffer of up to 256,000 bytes	[bin] (graal3d/protocol.md#tcp-frames)
A client drops the link when more than 256 KiB of compressed output is pending	[bin] (graal3d/protocol.md#tcp-frames)
There is no key byte and no per-packet scramble	[bin][cap] (graal3d/protocol.md#tcp-frames)
Text with line breaks (level files, admin messages) writes each `\n` as § (0xA7)	[cap] (graal3d/protocol.md#tcp-frames)
The client sends its login line and its first props line in one frame; the server answers each event with its own frame, so a server may flush per event	[cap] (graal3d/protocol.md#tcp-frames)
Packet ids and the prop table are numbered from scratch, not shared with 2D; positions are 3D, the angle is one byte and colours are 24-bit	[bin] (graal3d/README.md#how-it-relates-to-the-2d-graal-of-the-same-time)

9.1.2 Login

Rule	Source
On TCP accept, before login, the server sends id 6 with `c(0)`	[cap] (graal3d/protocol.md#login)
The client's first line is `c(0)`, the version `GR3DV002`, `str(account)`, `str(password)`; the server reads the version at bytes 2 to 9	[bin][cap] (graal3d/protocol.md#login)
`GR3DV001` is recognised and rejected with id 5 and the text "You need at least Graal3D beta2 to play on this server."	[bin] (graal3d/protocol.md#login)
A missing, in-use or disabled account gets id 5 with the matching text, then the server closes	[cap] (graal3d/protocol.md#login)
The server still parses the rest of a rejected frame and applies the props line to a throwaway player; a server need not copy this	[cap] (graal3d/protocol.md#login)
Id 1 with the same layout is an admin login; it needs admin level 1 or more and gets the account list (50) and the player list (55)	[bin][cap] (graal3d/protocol.md#login)
A normal account already in use is rejected; a load-only account may log in many times at once	[cap] (graal3d/protocol.md#login)

Login order on the server [cap] (graal3d/protocol.md#login):

Step	Server action	Source
1	Gives the lowest free id, links the account and applies its saved props; a non-empty saved nickname is relayed at once to the others as 2 + prop 12	[cap] (graal3d/protocol.md#login)

Step	Server action	Source
2	Sends 3 with props 0–5	[cap] (graal3d/protocol.md#login)
3	Exchanges 2 with all of props 0–17 between the new player and every other player	[cap] (graal3d/protocol.md#login)
4	Sends the player to their level (→ §9.3.2)	[cap] (graal3d/protocol.md#login)
5	Tells admin sessions with 55	[cap] (graal3d/protocol.md#login)
6	Handles the client's first props line like any other, so the others see props 12 and 15 again	[cap] (graal3d/protocol.md#login)

9.1.3 Server → client ids

Id	Name in the notes	Payload	Client action	Source
0	LEVEL	level text with § for \n, or empty	saves and loads it as a .x mesh; empty means "you already have it" (→ §9.3)	[bin][cap] (graal3d/protocol.md#server--client)
1	LEVELNAME	rest = file name	stored as the current level name	[bin][cap] (graal3d/protocol.md#server--client)
2	OTHERPLPROPS	c(player id) + props	creates the player on first sight, then applies the props	[bin][cap] (graal3d/protocol.md#server--client)
3	PLAYERPROPS	props	applies to the local player	[bin][cap] (graal3d/protocol.md#server--client)
4	leader	empty	sent to the first player in a level and to the next when the first leaves; sets a flag nothing reads	[bin][cap] (graal3d/protocol.md#server--client)
5	DISCMESSAGE	rest = text	shown when the socket closes; empty gives "You have been disconnected from the server."	[bin][cap] (graal3d/protocol.md#server--client)
6	session	c(v)	stores a value nothing reads; the server sends 0 on accept	[bin][cap] (graal3d/protocol.md#server--client)
7	FILESENDFAILED	rest = file name	not handled	[bin] (graal3d/protocol.md#server--client)
8	ADMINMESSAGE	rest, § = line break	shown in an "Administrator message" box	[bin][cap] (graal3d/protocol.md#server--client)
16	2D disconnect message	text	not handled; sent only by the server's player-list kick button, so that kick arrives silently	[bin] (graal3d/README.md#how-it-relates-to-the-2d-graal-of-the-same-time)
50–59	admin replies	→ §9.2.3	admin sessions only	[bin][cap] (graal3d/protocol.md#admin-commands)
100	RAWDATA	s3(len)	flags the next line as raw, then discards it	[bin] (graal3d/protocol.md#server--client)
101	FILE	str(name) + bytes	the server's answer to request 2; unused by this client	[bin] (graal3d/protocol.md#server--client)

9.1.4 Client → server ids

Id	Payload	Server action	Source
0	login (→ §9.1.2)		[bin][cap] (graal3d/protocol.md#client--server)
1	props	applies, then relays (→ §9.2.1)	[bin][cap] (graal3d/protocol.md#client--server)
2	file name	looks in `levels\`; sends 100 + 101 if the file is under 64,500 bytes, else 7; the shipped client never sends it	[bin] (graal3d/protocol.md#client--server)
51–64	admin commands	→ §9.2.3	[bin][cap] (graal3d/protocol.md#admin-commands)

9.1.5 UDP between clients

Rule	Source
Unless `noudp` is set, the client binds one UDP socket on port 0 or on the fixed port from its config, and announces the bound port in prop 15	[bin] (graal3d/protocol.md#udp-client-to-client)
The server adds the peer's address as prop 16; a client keeps as peers the players flagged `50 = 1` that have both a port and an address	[bin] (graal3d/protocol.md#udp-client-to-client)
Datagrams have no length prefix and no compression	[bin] (graal3d/protocol.md#udp-client-to-client)
First byte `c(0)`: the values of props 1–5 back to back without prop ids (9 bytes), a position update; 8 bytes or fewer is ignored	[bin] (graal3d/protocol.md#udp-client-to-client)
First byte `c(1)`: chat text, at most 200 bytes; longer text is cut and ends in `...`	[bin] (graal3d/protocol.md#udp-client-to-client)
A datagram is accepted only from the address and port of a known player	[bin] (graal3d/protocol.md#udp-client-to-client)
Position goes to each peer only when props 1–5 changed since the last tick; chat goes right after Enter, and also to the server as prop 13 in the next props diff	[bin] (graal3d/protocol.md#udp-client-to-client)
In 2D clients of the same time UDP went to the server; in Graal3D it goes peer to peer (the 2D peer channel is → §2.9)	[bin] (graal3d/README.md#how-it-relates-to-the-2d-graal-of-the-same-time)
UDP between two real clients was not seen in the lab, because Wine has no retained-mode `d3drm.dll`	[lab] (graal3d/lab-2026-10-30.md#not-confirmed)

9.2 Props and accounts

9.2.1 Player props

The same ids are used in both directions and inside UDP; a props payload is a run of `c(prop id)` + value with no separators [bin] (graal3d/protocol.md#player-properties).

Id	Value	Field	Default	Client sends	Source
0	c	player id (1–200)	—	no	[bin] (graal3d/protocol.md#player-properties)
1	s2	x, `(v − 8000) / 2`	0	yes	[bin] (graal3d/protocol.md#player-properties)
2	s2	y, same scale	0	yes	[bin] (graal3d/protocol.md#player-properties)

Id	Value	Field	Default	Client sends	Source
3	s2	z, same scale	0	yes	[bin] (graal3d/protocol.md#player-properties)
4	c	angle, v × 2 degrees	0	yes	[bin] (graal3d/protocol.md#player-properties)
5	s2	animation tick counter (→ §9.6.3)	0	yes	[bin] (graal3d/client-behaviour.md#animation-counter-prop-5)
6	str	body model	`body.x`	yes	[bin] (graal3d/protocol.md#player-properties)
7	str	head model	`head.x`	yes	[bin] (graal3d/protocol.md#player-properties)
8	str	body texture, reloaded on change	`body.bmp`	yes	[bin] (graal3d/protocol.md#player-properties)
9	str	head texture	`man.bmp`	yes	[bin] (graal3d/protocol.md#player-properties)
10	4 × 6 bits	arm colour, 24-bit RGB	`FF0000`	yes	[bin] (graal3d/protocol.md#player-properties)
11	4 × 6 bits	shoe colour	`800000`	yes	[bin] (graal3d/protocol.md#player-properties)
12	str, at most 200	nickname, optional (`guild`)	—	yes	[bin] (graal3d/protocol.md#player-properties)
13	str, at most 200	chat text	—	yes	[bin] (graal3d/protocol.md#player-properties)
14	str	level file	`level1.x` in the client, `startlevel.x` in the server	no	[bin][cap] (graal3d/protocol.md#player-properties)
15	s3	UDP port, 0 = none	0	yes, its bound port	[bin][cap] (graal3d/protocol.md#player-properties)
16	5 bytes: 4 bits, then 4 × 7 bits	IPv4 address: the in-memory address word read as a little-endian integer	—	no; the server fills in the TCP peer address	[bin][cap] (graal3d/protocol.md#player-properties)
17	4 × 6 bits	hand colour	`FDCC47`	yes	[bin] (graal3d/protocol.md#player-properties)
50	c	1 = this player is in your level (add as UDP peer), 0 = not	—	server only	[bin][cap] (graal3d/protocol.md#player-properties)
51	none	player left: remove	—	server only	[bin][cap] (graal3d/protocol.md#player-properties)

Rule	Source
Prop 10 colours the two arms, 17 the two hands and 11 the two feet	[bin] (graal3d/protocol.md#player-properties)
Each tick the client re-encodes every prop it sends, compares with the last tick and sends the changed ones as one id-1 message; the first send carries the full look (6–12 and 15)	[bin] (graal3d/protocol.md#player-properties)

Rule	Source
The wire range of props 1–3 is −4000 to 4095.5 in half-unit steps, which covers the ±4000 world clamp (→ §9.5)	[bin] (graal3d/client-behaviour.md#walking-and-running)

9.2.2 Server relay rules

Rule	Source
On prop 12 the server splits `nick (guild)` and keeps the tag only if `guilds\guild<guild>.txt` lists the account; if the text had a tag, the result is echoed back with 3; a plain nickname gets no echo	[cap] (graal3d/protocol.md#server-relay-rules)
The nickname then goes to every player (2) and every admin session (59)	[cap] (graal3d/protocol.md#server-relay-rules)
Prop 15 goes to every player as 2 + prop 15	[cap] (graal3d/protocol.md#server-relay-rules)
Prop 14 from a client is stored and relayed to the level, but does not move the player; only the server moves players, at login and when an admin edits an online player (60)	[cap][bin] (graal3d/protocol.md#server-relay-rules)
Prop 16 from a client is ignored while the player has a live socket	[bin] (graal3d/protocol.md#server-relay-rules)
All changed props in the line go as one 2 line to the players in the same level, leaving out props 1–5 and 13 when both players have a UDP port; when nothing is left, nothing is sent	[cap] (graal3d/protocol.md#server-relay-rules)
Each observer has its own dirty set, so a player who was skipped gets the missing props later	[bin] (graal3d/protocol.md#server-relay-rules)
On disconnect every player gets `c(id)` + prop 51 and admin sessions get 56 + `c(id)`	[cap] (graal3d/protocol.md#level-change)

9.2.3 Admin protocol

The admin protocol is read from the server alone; no Graal3D admin client is in the archives [bin] (graal3d/README.md#open-points). Each command needs a minimum admin level (0 none, 1 and 2 staff, 3 full), and an admin can act only on accounts at or below their own level unless they are level 3 [bin] (graal3d/protocol.md#admin-commands). An account id is a 28-bit number sent as 4 bytes of 7 bits [bin] (graal3d/protocol.md#admin-commands).

Id in	Payload	Effect or reply	Source
51	account id	reply 54, the account record	[bin] (graal3d/protocol.md#admin-commands)
52	account id, `str` name, `str` password, `str` props, `c` flags	edit account; 51 (changed) and 52 (renamed) to admins	[bin] (graal3d/protocol.md#admin-commands)
53	account record	add account; 50 to admins	[bin] (graal3d/protocol.md#admin-commands)
54	account id	delete account; 53 to admins	[bin] (graal3d/protocol.md#admin-commands)
59	`c(player id)`	reply 57, that player's account	[bin] (graal3d/protocol.md#admin-commands)
60	`c(player id)` + account record	edit an online player's account	[bin] (graal3d/protocol.md#admin-commands)

Id in	Payload	Effect or reply	Source
61	`c(player id)`	kick: the player gets 5 "The server administrator has disconnected you." and the others get prop 51	[cap] (graal3d/protocol.md#admin-commands)
62	`s2(count)` + `str` names	reload these levels from disk	[bin] (graal3d/protocol.md#admin-commands)
63	text	message to all players (8) as `Administrator <account>:§<text>`	[cap] (graal3d/protocol.md#admin-commands)
64	`c(player id)` + text	message to one player (8)	[bin] (graal3d/protocol.md#admin-commands)

Id out	Meaning	Layout	Source
50	account added	account id, `c(folder)`, name to end of line; one per account after an admin login	[cap] (graal3d/protocol.md#admin-commands)
51	account changed	account id, `c(flags)`; sent when a bound player leaves and the record is saved	[cap] (graal3d/protocol.md#admin-commands)
52	account renamed	unmeasured	[bin] (graal3d/protocol.md#admin-commands)
53	account deleted	unmeasured	[bin] (graal3d/protocol.md#admin-commands)
54	account record	account id, `str(password field)` (came back empty), `str(props)`, `c(flags)`	[cap] (graal3d/protocol.md#admin-commands)
55	player joined	`c(player id)`, `str(account)`, prop 14	[cap] (graal3d/protocol.md#admin-commands)
56	player left	`c(player id)`	[cap] (graal3d/protocol.md#admin-commands)
57	player's account	unmeasured	[bin] (graal3d/protocol.md#admin-commands)
58	player's account renamed	unmeasured	[bin] (graal3d/protocol.md#admin-commands)
59	player props	`c(player id)` + props: the nickname, the full props on login, prop 14 on level entry	[cap] (graal3d/protocol.md#admin-commands)

Rule	Source
The folder byte in 50 fits 1 for a normal account plus 4 × the admin level, and 0 for a load-only account; the formula is fitted to three accounts	[cap][inf] (graal3d/protocol.md#admin-commands)
55 and 56 match the 2D add- and remove-player ids; the names of the other admin ids are descriptive, not recovered	[bin] (graal3d/protocol.md#admin-commands)
The admin kick path (61) sends 5, which the client shows; the player-list kick button sends the 2D id 16, which it ignores	[bin] (graal3d/README.md#how-it-relates-to-the-2d-graal-of-the-same-time)
Commands 51–54, 59, 60 and 62 were not exercised live	[lab] (graal3d/lab-2026-10-30.md#not-confirmed)

9.2.4 Account and config files

Rule	Source
`accounts.txt` holds one account per `\n`-terminated line in the +32 codec: `str(name) str(password) str(saved props) c(flags)` and seven `c(0)` bytes; a hand-written file is accepted	[cap] (graal3d/formats.md#accountstxt-server)
The saved props are the player's prop run, props 1–14, 16 and 17 (no 0 or 15); empty until the player first leaves, written back when a bound player leaves or is edited, and applied at the next login, nickname and chat text included	[cap] (graal3d/formats.md#accountstxt-server)
Flags: bit 0 = load only (the session is not tied to the record, props are not saved back, many logins may share it), bits 1–3 = admin level, bit 4 = blocked	[cap] (graal3d/protocol.md#admin-commands)
The server UI names the levels No rights (0), Graalian Police (1), Trusted Admin (2) and Administrator (3)	[bin] (graal3d/formats.md#accountstxt-server)
The seven trailing bytes are the empty count-prefixed lists of the 2D account record; nothing in Graal3D fills them	[bin] (graal3d/formats.md#accountstxt-server)
The server rewrites `accounts.txt` whenever an account changes	[bin] (graal3d/formats.md#accountstxt-server)
`guilds\guild<name>.txt` lists one account per line; a nickname keeps its `(name)` tag only if the file exists and lists the account, as in the 2D server	[bin] (graal3d/formats.md#guildsguildnametxt-server)
`serverconfig.txt` holds `serverport=14901`	[file] (graal3d/formats.md#serverconfigtxt-server)
The client stores nickname, account and password in the registry under `HKCU\Software\Graal3D`, section `Client`; the password is three DES blocks under a key built into the exe, written as 48 printable characters (obfuscation, not protection; the key is not given here)	[bin] (graal3d/formats.md#registry-client)

Client config `graal3dconfig.txt`, `key=value` lines read at start and written back on change [file] (graal3d/formats.md#graal3dconfigtxt-client):

Key	Shipped value	Meaning	Source
`serveraddr`	`<public-host>`	server host or address; the connect window defaults to the loopback address	[file] (graal3d/formats.md#graal3dconfigtxt-client)
`serverport`	`14901`	TCP port	[file] (graal3d/formats.md#graal3dconfigtxt-client)
`fixedport`	`false`	bind UDP to `fixedudpport` instead of a random port	[file] (graal3d/formats.md#graal3dconfigtxt-client)
`fixedudpport`	`14898`	fixed UDP port	[file] (graal3d/formats.md#graal3dconfigtxt-client)
`noudp`	`false`	no UDP socket; position and chat go only through the server	[file] (graal3d/formats.md#graal3dconfigtxt-client)
`bodyx`, `headx`, `bodytex`	`body.x`, `head.x`, `body.bmp`	props 6, 7 and 8	[file] (graal3d/formats.md#graal3dconfigtxt-client)

Key	Shipped value	Meaning	Source
`armcolor`, `shoecolor`, `handcolor`	`FF0000`, `800000`, `FDCC47`	props 10, 11 and 17	[file] (graal3d/formats.md#graal3dconfigtxt-client)
`accountname`, `password`	—	also accepted by the reader	[bin] (graal3d/formats.md#graal3dconfigtxt-client)

9.3 Levels (.x)

9.3.1 Level files

Rule	Source
A level is a DirectX retained-mode `.x` text file (`xof 0302txt 0064`); there is no tile layer, NPC or link data	[file] (graal3d/formats.md#levels-levelsx-on-the-server)
`startlevel.x` is the server's start level and `level1.x` the client's default; the two hold the same scene and differ only in line ends and trailing spaces	[file] (graal3d/formats.md#levels-levelsx-on-the-server)
The shipped scene is one 139-vertex mesh `obj` with nine materials and seven textures (`grass`, `way`, `stones`, `wall`, `tower`, `platform`, `clouds`)	[file] (graal3d/formats.md#levels-levelsx-on-the-server)
The server caches each level's text and sends it inline in id 0; the client saves it under the level name in its own folder and loads it with Direct3D RM	[bin][cap] (graal3d/formats.md#levels-levelsx-on-the-server)
Textures are not transferred: the BMPs a level names must already be in the client folder	[bin] (graal3d/formats.md#levels-levelsx-on-the-server)
The client's saved copy was byte-identical to the server's file in the lab	[cap] (graal3d/protocol.md#level-change)

9.3.2 Level change

Step	Server action	Source
1	Finds the level in its cache or loads `levels\<name>`	[bin] (graal3d/protocol.md#level-change)
2	Sends 1 (name), then 0 with the text, or 0 empty if this player was already sent that level; the "already sent" list is per player, so a second login gets the full text again	[cap] (graal3d/protocol.md#level-change)
3	For each player already in the level, sends both sides 2 with `50 = 1` and the props not yet seen	[cap] (graal3d/protocol.md#level-change)
4	Sends 4 if the player is alone in the level	[cap] (graal3d/protocol.md#level-change)
5	Tells admin sessions with 59	[cap] (graal3d/protocol.md#level-change)
—	On leaving a level the others get `50 = 0`	[cap] (graal3d/protocol.md#level-change)

9.3.3 Player model and textures

Rule	Source
The player is eight `.x` meshes (`xof 0303txt 0032`): body, head, two arms, two hands, two feet, each with normals, texture coordinates and a material	[file] (graal3d/formats.md#player-model-client)
`body.x` is textured with `body.bmp` (128×128, 24-bit; `body1.bmp` is an alternative) and `head.x` with `man.bmp` (64×64)	[file] (graal3d/formats.md#player-model-client)
Mesh and texture names travel as props 6–9, so they may point at any file present on every client; colours are set as material colours at run time	[bin] (graal3d/formats.md#player-model-client)
Textures are plain Windows BMPs, 24-bit or 8-bit RLE, mostly 128×128 (`clouds.bmp` is 256×256)	[file] (graal3d/formats.md#textures)

9.4 Camera

Quantity or rule	Value	Source
Follow distance	220 behind the player	[bin] (graal3d/camera.md#status-for-g3d-2)
Look height	80 above the player on Y	[bin] (graal3d/camera.md#status-for-g3d-2)
Gap closing per tick	factor `((d − 220) / 220)² / 10` with the sign of `d − 220`, `d` the horizontal distance; skipped if the factor is 0 or 100 or more	[bin] (graal3d/camera.md#follow-rule-numpad-0-toggles-follow-onoff)
Y follow per tick	`(player_y + 80 − camera_y) / 10`	[bin] (graal3d/camera.md#follow-rule-numpad-0-toggles-follow-onoff)
Turn	up to ±5° per tick toward the player, more when the player is behind the camera	[bin] (graal3d/camera.md#follow-rule-numpad-0-toggles-follow-onoff)
Aim	`LookAt` the player, then pitch −0.15 rad about the camera's local X axis	[bin] (graal3d/camera.md#follow-rule-numpad-0-toggles-follow-onoff)
Follow toggle	Numpad 0; with follow off the camera stays where it is	[bin] (graal3d/camera.md#follow-rule-numpad-0-toggles-follow-onoff)
Facing sent	prop 4 is the angle of the camera-to-player direction, sent as degrees / 2	[bin] (graal3d/camera.md#follow-rule-numpad-0-toggles-follow-onoff)
Surface	640×456 pixels	[bin] (graal3d/camera.md#display-size-aspect-context-only)
Field	the Direct3D RM initial field 0.5, a square window; the exe sets no field	[doc][bin] (graal3d/camera.md#field-of-view)
Opening	`2 × atan(0.5)` ≈ 53.13° on either axis of that window	[inf] (graal3d/camera.md#field-of-view)
Front plane	1 (library initial value)	[doc] (graal3d/camera.md#field-of-view)
Back plane	disputed → App. E#500	[doc] (graal3d/camera.md#still-needed)
Lab	none of these figures is lab-confirmed; Wine's retained-mode stubs break camera frames	[lab] (graal3d/camera.md#still-needed)

9.5 Collision

Quantity	Value	Source
Body radius (horizontal half-extent)	5	[bin] (graal3d/collision.md#what-the-collider-needs)
Upper probe height	+35 above the feet	[bin] (graal3d/collision.md#what-the-collider-needs)
Upper probe lateral offsets	0, +10 and −10 on the strafe axis only	[bin] (graal3d/collision.md#what-the-collider-needs)
Max step height	30	[bin] (graal3d/collision.md#what-the-collider-needs)
Walkable face	normal Y > 0	[bin] (graal3d/collision.md#what-the-collider-needs)
Feet-mesh attach	normal Y > 0.2; steeper walkable faces snap the point but skip the attach	[bin] (graal3d/collision.md#what-the-collider-needs)
Wall already clear	new signed distance ≥ 3	[bin] (graal3d/collision.md#what-the-collider-needs)
Wall started inside	previous signed distance < −5	[bin] (graal3d/collision.md#what-the-collider-needs)
World clamp, x and z	±4000; a step that would leave is dropped	[bin] (graal3d/collision.md#what-the-collider-needs)
Sliding	subtract `distance × normal` from the test point	[bin] (graal3d/collision.md#what-the-collider-needs)

Order of tests for one attempted step:

Step	Action	Source
1	Build the candidate position from the walk, run or jump delta (→ §9.6)	[bin] (graal3d/collision.md#order-of-tests-one-attempted-step)
2	Load every face of the level mesh, filled at level load, and transform its vertices to world space	[bin] (graal3d/collision.md#order-of-tests-one-attempted-step)
3	Run the body probes against every face; a blocked probe stops the rest	[bin] (graal3d/collision.md#body-probes)
4	Reject the step if a probe is blocked, otherwise keep the (possibly slid) position	[bin] (graal3d/collision.md#order-of-tests-one-attempted-step)
5	Drop the step if x or z leaves ±4000	[bin] (graal3d/collision.md#order-of-tests-one-attempted-step)

Rule	Source
Ground probes: four points in the feet plane at ±5 on the facing axis and ±5 on the strafe axis, a plus shape, not a filled disc	[bin] (graal3d/collision.md#body-probes)
Upper probes: Y + 35 at lateral 0, +10, −10; they use only the wall and ceiling path	[bin] (graal3d/collision.md#body-probes)
A face is used only if the probe's (x, z) lies inside its horizontal projection, by a winding test that tries both edge orders	[bin] (graal3d/collision.md#over-face-test)
Walkable face, ground probes only: signed distance > 30 lands on the face; < −30 fails (step too high); otherwise the probe slides onto the plane, with the mesh attach when normal Y > 0.2	[bin] (graal3d/collision.md#walkable-faces-normal-y--0)
Wall or ceiling (normal Y ≤ 0): the face is dropped if the new distance is ≥ 3, the previous distance < −5, or the move has zero length; otherwise the new point slides onto the plane	[bin] (graal3d/collision.md#walls-and-ceilings-normal-y--0)
No floor friction, extra ground stickiness or ceiling thickness shows in the routine's constants	[bin] (graal3d/collision.md#what-this-does-not-settle)
A disc of radius 5 is an approximation of the plus-shaped probe set, not what the binary runs	[inf] (graal3d/collision.md#what-this-does-not-settle)

Rule	Source
Collision is not lab-confirmed; Wine's face query is a stub	[lab] (graal3d/collision.md#what-this-does-not-settle)

9.6 Client behaviour

9.6.1 Controls

Key	Action	Source
Numpad 8 / 2	forward / back, relative to the camera's view direction	[bin] (graal3d/client-behaviour.md#controls)
Numpad 4 / 6	strafe left / right	[bin] (graal3d/client-behaviour.md#controls)
Numpad 5	jump	[bin] (graal3d/client-behaviour.md#controls)
Numpad 0	camera follow on or off	[bin] (graal3d/client-behaviour.md#controls)
Enter	send the chat line	[bin] (graal3d/client-behaviour.md#controls)
Ctrl+Enter	fullscreen on or off	[bin] (graal3d/client-behaviour.md#controls)
F1 / F2	Hardware options / Player look	[bin] (graal3d/client-behaviour.md#controls)
Esc	quit	[bin] (graal3d/client-behaviour.md#controls)
—	movement keys are polled only while the chat line lacks focus and the window is active	[bin] (graal3d/client-behaviour.md#controls)

9.6.2 Walking, running and jumping

Rule	Source
Walking speed is 8 units per 50 ms tick (160 per second)	[bin] (graal3d/client-behaviour.md#walking-and-running)
After 25 ticks of walking without stopping the speed becomes ×1.5 (12 per tick); stopping resets the count	[bin] (graal3d/client-behaviour.md#walking-and-running)
Holding only "back" walks backwards along the facing at key-down, not following the camera	[bin] (graal3d/client-behaviour.md#walking-and-running)
While walking, y gets `abs(sin(3.14 × 2 × (anim − 1) / 40)) × 3` added (the program uses 3.14, not π)	[bin] (graal3d/client-behaviour.md#walking-and-running)
The jump climbs by a fixed table of 10, 10, 9, 8, 6, 3, 1 units per tick (47 in 7 ticks), walks the table back down with negative speeds, then falls at −10 per tick until it lands	[bin] (graal3d/client-behaviour.md#jumping)
Releasing the key early starts the fall at once from the top of the table, so a tap gives a short hop; a new jump needs the key released and pressed again	[bin] (graal3d/client-behaviour.md#jumping)

9.6.3 Animation counter

Rule	Source
Prop 5 is a tick counter for the walk and jump poses, not a pose id	[bin] (graal3d/client-behaviour.md#animation-counter-prop-5)
Starting to walk sets it to 0 or 20 at random, so players do not step in time; a jump sets it to 40	[bin] (graal3d/client-behaviour.md#animation-counter-prop-5)

Rule	Source
Where it advances each tick was not traced; that standing still freezes it is an inference	[inf] (graal3d/client-behaviour.md#animation-counter-prop-5)

9.6.4 Chat

Rule	Source
There are no chat bubbles; a chat line (prop 13 or a UDP chat datagram) is added to a chat log as `nickname: text`	[bin] (graal3d/client-behaviour.md#chat)
The log is drawn at the bottom left of the view, x = 5, newest line at y ≈ 451, older lines above, in white 14 px "Dialog"	[bin] (graal3d/client-behaviour.md#chat)
Each line stays 10 seconds (200 ticks)	[bin] (graal3d/client-behaviour.md#chat)
A line over 200 characters is cut and ends in `...`	[bin] (graal3d/client-behaviour.md#chat)
The "Player look" window edits props 6–11 and 17 and saves them in `graal3dconfig.txt`	[bin] (graal3d/client-behaviour.md#look)

9.6.5 Lab coverage

Fact	Source
Lab-confirmed on 2026-10-30: the accept and reject flow, the login order, the full props exchange, level name, text and leader, prop 50, the UDP relay rule, the guild-tag echo, prop 14 not moving the player, saved props, load-only logins, the disconnect messages, the admin login replies and `accounts.txt`	[lab] (graal3d/lab-2026-10-30.md#what-was-confirmed)
Not confirmed: UDP between two real clients, admin commands 51–54, 59, 60 and 62, and the two kick buttons	[lab] (graal3d/lab-2026-10-30.md#not-confirmed)
Ids 4 and 6 set client values that nothing reads	[bin][cap] (graal3d/README.md#open-points)

10 Torque-era 3D (3.x to 5.12)

From 2004 to 2007 the Graal client was built on GarageGames' Torque Game Engine (TGE). Torque gave the console, the script VM, the GUI, the resource manager, the renderer and, for 3D servers, the UDP netcode; Graal's 2D game, GS1/GS2 and the Graal TCP protocol stayed native code on top. The February 2000 DirectX Graal3D is unrelated (→ §9).

> Status: draft.

10.1 Engine lineage

10.1.1 Builds

Build	Date	Shape	Torque part	3D world	Game string / protocol	Menu scripts	Source
1998 to 2.31	—	Delphi	none	none	n/a	n/a	[bin] (torque/wire-by-build.md#which-builds-are-torque)
3.04 Linux	Jan 2004	one C++ binary (SDL, OpenAL)	full TGE, version number 1100	TGE classes, no 3D scripts shipped	`Torque Game Engine Demo` / 9	TorqueScript `.dso`	[bin] (torque/README.md#lineage-at-a-glance)
3.153 Mac	Feb 2005	one C++ binary (Carbon, PPC)	full TGE with Graal's net-layer changes	stock TGE world and `main3D.cs` client scripts	`Graal3D` / 19	TorqueScript `.dso`	[bin][file] (torque/README.md#lineage-at-a-glance)
4.04r2 Win	Mar 2006	Delphi launcher, `GraalEngine.dll` (2D) and `Graal3DEngine.dll` (3D plug-in)	2D engine on Torque base classes; the 3D DLL near-stock TGE	stock TGE with datablocks	`Graal3D` / 19	GS2 and native	[bin] (torque/README.md#lineage-at-a-glance)
4.20r6 Mac	Sep 2006	one C++ binary (PPC)	Torque core and netcode, Graal game layer	`GameObject` model, no datablocks	`Graal3D` / 21	GS2 and native	[bin] (torque/wire-by-build.md#which-builds-are-torque)
4.20r8 Win	Jan 2007	2D engine on Torque base classes	—	unmeasured: the upload has no `Graal3DEngine.dll`	unmeasured	GS2 and native	[bin] (torque/wire-by-build.md#which-builds-are-torque)
5.12 Win	Oct 2007	Delphi launcher and one engine DLL, still named `Graal3DEngine.dll`	Torque core and netcode; the game layer is Graal's own	`GameObject` and `GameShape*`, no datablocks, no `Player`/`ShapeBase`	`Graal3D` / 24	GS2 and native	[bin] (torque/README.md#lineage-at-a-glance)
5.24r3 and later	2010+	engine linked into `Graal.exe`	no Torque 3D; only the name `GuiGraal3DCtrl` is left	none	n/a	GS2	[bin] (torque/engine-and-formats.md#which-torque)

Build	Date	Shape	Torque part	3D world	Game string / protocol	Menu scripts	Source
6.x	2010–2014	native C++ per platform	none	none	n/a	GS2	[bin] (torque/README.md#lineage-at-a-glance)

10.1.2 Engine version and changes

Fact	Source
3.04 reports version 1100 (formatted "1.100"), uses TGE's default game string, the home folder `~/.garagegames/` (3.151 changed it to `~/.graal`) and TGE source paths, so 3.x is built on a TGE 1.1–1.2 era code base	[bin] (torque/engine-and-formats.md#which-torque)
3.153 is the same engine for PowerPC and Carbon, plus the community resources `AdvancedCamera`, `fxFoliageReplicator`, `twSurfaceReference` and `volumeLight`, a 3D Studio loader and a Quake 3 `.md3` loader	[bin] (torque/engine-and-formats.md#which-torque)
The 4.04r2 `Graal3DEngine.dll` is near-stock TGE (`Player`, `AIPlayer`, vehicles, projectiles, `InteriorInstance`, `TSStatic`, the `fx*` replicators, every matching datablock) and imports 623 symbols from `GraalEngine.dll`	[bin] (torque/engine-and-formats.md#which-torque)
The 4.04r2 DLL matches open TGE 1.x in its out-of-band packets, packet header, Huffman table, rate limits, DTS/DIF/TER versions and class list, and lacks TGE 1.4's lighting, so it is of the TGE 1.2/1.3 line	[bin][inf] (graal3d/torque-engine-4x.md#torque-version)
5.12 keeps Torque's core (console, sim objects, `NetConnection`/`GameConnection`, ghosting, `SceneObject`, terrain, interiors, DTS, sky, sun, water, the editors) but drops the TGE game layer: no datablocks, no `GameBase`/`ShapeBase`/`Player`, vehicles, projectiles, items, `ScriptObject` or audio emitters	[bin] (torque/engine-and-formats.md#which-torque)
4.20r6 already has the 5.12 model, so the TGE game layer was dropped between 4.04r2 (Feb 2006 token) and 4.20r6 (Jul 2006 token)	[bin] (torque/wire-by-build.md#the-handshake-per-build)
3.04 has no Graal changes in Torque's net layer and none of the Graal net classes	[bin] (torque/wire-by-build.md#the-handshake-per-build)
Graal's net-layer changes (class table in the accept, 12-bit class ids, 14-bit length prefixes, BigData and terrain events) are already in 3.153	[bin] (torque/wire-by-build.md#the-handshake-per-build)

The Graal changes a compatible peer adds to open TGE [bin] (graal3d/torque-engine-4x.md#torque-version):

Change	Source
a class table in the connect accept, with 12-bit class ids in the server's numbering (→ §10.3.3)	[bin] (graal3d/torque-engine-4x.md#torque-version)
a 14-bit length prefix on every event, ghost update and datablock	[bin] (graal3d/torque-engine-4x.md#length-prefixes-graal-change)
a class id on every ghost update and a raw index width	[bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes)
12-bit datablock ids (TGE uses 10)	[bin] (graal3d/torque-engine-4x.md#events)
a server time and client-reported state in the game packets (→ §10.3.5)	[bin] (graal3d/torque-engine-4x.md#server--client-data-packet)

Change	Source
mission acks without a sequence number (4.04r2)	[bin] (graal3d/torque-engine-4x.md#mission-start)
the BigData and terrain events (→ §10.5.3)	[bin] (graal3d/torque-engine-4x.md#graal-events-and-text-big-data-channel)
GS2 in place of TorqueScript, from 4.x on	[bin] (graal3d/torque-engine-4x.md#overview)

10.1.3 Rendering

Fact	Source
Every build has Torque's display devices: OpenGL, a Direct3D wrapper (`opengl2d3d.dll` / `glu2d3d.dll`) and Voodoo2 (full screen only); the minimum mode is 640×480	[bin] (torque/engine-and-formats.md#rendering)
3.x uses SDL and OpenGL on Linux, AGL and DrawSprocket on Mac; 4.x and 5.12 on Windows draw 2D with DirectDraw/Direct3D 7 in the Delphi launcher and host the Torque canvas for 3D (`GuiGraal3DCtrl`)	[bin] (torque/engine-and-formats.md#rendering)
In 4.04r2 the loaded plug-in registers its own handler for every drawing callback of the 2D engine, so in 3D mode the Torque DLL owns the window and the 2D game draws through Torque's canvas	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
Scene lighting is Torque's, cached per mission and run as the last mission-load phase	[bin] (torque/engine-and-formats.md#rendering)
Under Wine with software OpenGL the 5.12 Torque side starts on switching to 3D, but the launcher faults re-initialising DirectDraw and stops driving the engine's frame export; the 3D view was not seen	[run] (torque/engine-and-formats.md#rendering)

10.2 Join sequence

Step	What happens	Source
1	The server list marks a 3D server with the name prefix 3 (the "Graal 3D" tab)	[bin] (protocol/v3x-clients.md#listserver)
2	Choosing it runs one routine in 3.153, 4.04r2 and 5.12: strip the two-character prefix, remember UDP port = listed port + 1, set the 3D-mode flag and start the Torque side	[bin] (torque/network.md#how-a-client-gets-to-a-3d-server)
3a	3.153 creates a `SimGroup` named `MissionGroup` and runs `main3D.cs`, which creates `GameConnection(ServerConnection)` and calls `connect()` with the stored address	[bin][file] (torque/network.md#how-a-client-gets-to-a-3d-server)
3b	4.04r2 and 5.12 first ask the launcher (a non-zero answer cancels), then have it connect to host, port + 1, with the UDP option; 4.04r2 loads `Graal3DEngine.dll` on demand at this point	[bin] (torque/network.md#how-a-client-gets-to-a-3d-server)
4	In both 2D and 3D the client opens the normal Graal TCP connection to the listed port and logs in as usual (→ §2.3)	[bin] (torque/network.md#how-a-client-gets-to-a-3d-server)
5	The Torque handshake runs on UDP (→ §10.3), then the mission download (→ §10.3.6)	[bin][run] (torque/network.md#the-full-handshake-live)

Rule	Source
A 3D server is two servers on one host: a Graal gserver on TCP *P* and a Torque server on UDP *P* + 1	[bin][run] (torque/network.md#how-a-client-gets-to-a-3d-server)
The client's own UDP port is random unless the fixed-port option is on, then `fixedudpport` (default 14899)	[bin][file] (torque/network.md#how-a-client-gets-to-a-3d-server)
28000 is only the Torque address parser's default for an address without a port	[bin] (torque/wire-by-build.md#the-handshake-per-build)

Rule	Source
Seen live: the 5.12 client sent the Torque challenge to UDP *P* + 1 and opened TCP *P* in the same millisecond; the TCP side carried the ordinary 5.x login frame with the built-in guest account, then ids 37 and 23	[run] (torque/network.md#how-a-client-gets-to-a-3d-server)
The 3D connection lives only as long as the TCP one: when the lab TCP listener closed, the client reported "You have been disconnected from the game server." and sent a Torque disconnect	[run] (torque/network.md#the-full-handshake-live)
In 4.04r2 a Torque time-out ("The 3D server connection has timed out") also drops the TCP session	[bin] (graal3d/torque-engine-4x.md#callbacks-the-dll-gives-the-engine)
4.04r2 gives the 3D side the account name, nickname and password through the engine; no session key or cookie is passed	[bin] (graal3d/torque-engine-4x.md#state-the-dll-receives)
If the 4.04r2 plug-in is missing, the launcher has the text "You need to download Graal3D at … for playing this server!"; when it shows it was not traced	[bin][inf] (graal3d/torque-engine-4x.md#connecting)
The 4.04r2 launcher still registers `graal3://` against `Graal3.exe`, the 3.x name	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
Each outcome fires a GS2 event: `onConnectionAccepted`, `onConnectRequestTimedOut`, `onConnectRequestRejected`, `onConnectionDropped`, `onConnectionError`	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)

TCP traffic in 3D mode, 4.04r2 [bin] (graal3d/torque-engine-4x.md#the-tcp-gserver-connection-in-3d-mode):

Traffic	In 3D mode	Source
S→C 0 `LEVELBOARD`, 14 `PLAYERWARP`, 28 `FLAGSET`, 31 `FLAGDEL`, 49 `PLAYERWARP2`, 101 `BOARDPACKET`	ignored; level, warps and flags come from the 3D server	[bin] (graal3d/torque-engine-4x.md#the-tcp-gserver-connection-in-3d-mode)
level requests at start-up, getweapon, client-variable and flag sets, server-side triggers	rerouted to the 3D server (→ §10.5.3)	[bin] (graal3d/torque-engine-4x.md#the-tcp-gserver-connection-in-3d-mode)
chat, PMs, the player list, file transfer, RC	not touched by the switch, so presumably still TCP	[inf] (graal3d/torque-engine-4x.md#the-tcp-gserver-connection-in-3d-mode)
missing 3D resources	fetched with the normal TCP file download by lower-case name, except `.dso`, `.cs` and `.ter` (→ §2.7)	[bin] (graal3d/torque-engine-4x.md#the-tcp-gserver-connection-in-3d-mode)

10.3 Handshake by build

10.3.1 Info packets

The first bit of a datagram picks the kind: 1 is a data packet of an established connection, 0 an info packet whose first byte is the type; all integers are little-endian [bin] (torque/network.md#torque-packet-framing).

Type	Name	Direction	Body	Source
26 (0x1A)	ConnectChallengeRequest	C→S	U32 connect sequence	[bin][run] (torque/network.md#torque-packet-framing)
28 (0x1C)	ConnectChallengeReject	S→C	not handled by the Graal clients	[bin] (torque/network.md#torque-packet-framing)
30 (0x1E)	ConnectChallengeResponse	S→C	U32 sequence, 4 × U32 address digest	[bin][run] (torque/network.md#torque-packet-framing)
32 (0x20)	ConnectRequest	C→S	U32 sequence, 4 × U32 digest, string `GameConnection`, then the request fields (→ §10.3.2)	[bin][run] (torque/network.md#torque-packet-framing)

Type	Name	Direction	Body	Source
34 (0x22)	ConnectReject	S→C	U32 sequence, string reason	[bin] (torque/network.md#torque-packet-framing)
36 (0x24)	ConnectAccept	S→C	U32 sequence, the class table (from 3.153), U32 protocol (→ §10.3.3)	[bin][run] (torque/network.md#torque-packet-framing)
38 (0x26)	Disconnect	both	U32 sequence, string reason	[bin][run] (torque/network.md#torque-packet-framing)

Rule	Source
Types below 23 go to the master-server and LAN query handler	[bin] (torque/network.md#torque-packet-framing)
A string is Torque `writeString`: one flag bit (1 = Huffman with Torque's fixed tree, 0 = raw), a U8 length, then the codes or bytes; fields after the first string are not byte-aligned	[bin][run] (torque/network.md#torque-packet-framing)
The address digest is TGE's MD5 over the client address, the sequence and a random server key	[bin][inf] (graal3d/torque-engine-4x.md#out-of-band-handshake)
There is no encryption beyond Torque's framing; the join password travels in clear	[bin] (torque/network.md#the-connect-request-by-version)

10.3.2 Connect request by build

The request writes the net class group (U32, 0 = game) and the class CRC (U32), then the `GameConnection` fields [bin] (torque/network.md#the-connect-request-by-version).

Field	3.04	3.153	4.04r2	4.20r6	5.12	Source
game string	`Torque Game Engine Demo`	`Graal3D`	`Graal3D`	`Graal3D`	`Graal3D`	[bin] (torque/wire-by-build.md#the-handshake-per-build)
current = minimum protocol	9	19	19	21	24	[bin] (torque/wire-by-build.md#the-handshake-per-build)
class CRC	stock table, value unmeasured	unmeasured	disputed → App. E#501	unmeasured	`0xFFFFFFFF` (static table, never written)	[bin][run] (torque/wire-by-build.md#the-handshake-per-build)
join password	the server-password pref	set by the engine, not traced	the account password, plain text	not traced	the account password, plain text	[bin] (torque/network.md#the-connect-request-by-version)
argc (at most 16), then strings	script args	engine-set, not traced	2: account, nickname	not traced	2: account, nickname	[bin] (torque/network.md#the-connect-request-by-version)
server side of the request	stock script `onConnectRequest`	stock script `onConnectRequest`	native	native	native	[bin] (torque/wire-by-build.md#the-handshake-per-build)

Rule	Source
The live 5.12 request was 71 bytes and decoded completely: group 0, CRC `0xFFFFFFFF`, `Graal3D`, 24/24, the built-in guest password as join password, argc 2 (`guest`, empty nickname), no bits left over	[run] (torque/network.md#the-full-handshake-live)
A 5.12 server rejects a game string other than `Graal3D` with `CHR_GAME` ("Only official Graal clients are allowed to connect."), current < 24 with `CHR_PROTOCOL_LESS`, minimum > 24 with `CHR_PROTOCOL_GREATER`, and argc > 16 with `CR_INVALID_ARGS`	[bin] (torque/network.md#the-connect-request-by-version)

Rule	Source
4.04r2 knows the same reasons plus `CHR_PROTOCOL` and `CHR_INVALID`; it rejects < 19 and a minimum ≥ 20	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
Stock Torque rejects a CRC mismatch with `CHR_INVALID`; in 5.12 the class table in the accept takes the CRC's place	[bin][run] (torque/network.md#the-connect-request-by-version)

10.3.3 Connect accept and class table

Field	Width	Meaning	Source
flag	1 bit	1 = another entry follows, 0 = end of table	[bin][run] (torque/network.md#the-connect-request-by-version)
class type	12 bits	0 object, 1 datablock, 2 event (3.153, 4.04r2); 0 object, 1 event (4.20r6, 5.12; the event meaning is inferred)	[bin][inf] (torque/network.md#the-connect-request-by-version)
server class id	12 bits	0 to 255	[bin] (torque/network.md#the-connect-request-by-version)
class name	string	Huffman string, such as `NetStringEvent`	[bin] (torque/network.md#the-connect-request-by-version)
protocol	U32	after the end flag; must equal the build's protocol	[bin][run] (torque/network.md#the-connect-request-by-version)

Rule	Source
For each entry the client finds its own class of that name, type and group and records server id ↔ local id in two maps of 256 slots per group and type; class ids on the wire are then the server's	[bin] (torque/network.md#the-connect-request-by-version)
A name the client does not know maps to "unknown" and its data is skipped	[bin] (graal3d/torque-engine-4x.md#out-of-band-handshake)
Until a table is read the map is the identity (3.153)	[bin] (torque/wire-by-build.md#the-handshake-per-build)
An accept with an empty table (one 0 bit) and protocol 24 is enough for 5.12 to reach "Connection established"	[run] (torque/network.md#the-connect-request-by-version)
A stock accept without the table bit is read one bit off, gives protocol 12 and is silently rejected by 5.12	[run] (torque/network.md#the-connect-request-by-version)
The class table is in 3.153 and 4.04r2 with three types, in 4.20r6 and 5.12 with two types; 3.04 has none	[bin] (torque/wire-by-build.md#the-handshake-per-build)
Each build accepts exactly its own protocol (9, 19, 19, 21, 24); anything else gives `CHR_PROTOCOL`	[bin] (torque/network.md#the-connect-request-by-version)

10.3.4 Live handshake (5.12)

Time	Direction	Packet	Source
+0.000 s	C→S	challenge request, sequence 1024	[run] (torque/network.md#the-full-handshake-live)

Time	Direction	Packet	Source
+0.000 s	S→C	challenge response with the lab server's digest	[run] (torque/network.md#the-full-handshake-live)
+3.009 s	C→S	connect request, 71 bytes (the 3 s gap was the lab injector's own wait)	[run] (torque/network.md#the-full-handshake-live)
+3.009 s	S→C	connect accept: sequence, empty class table, protocol 24	[run] (torque/network.md#the-full-handshake-live)
+3.043 s	C→S	first data packet, sequence 1, with the "max rate" flag and two 12-bit values (2080 and 368 in the stock field order)	[run] (torque/network.md#after-the-connection-is-accepted)
then	C→S	one 17-byte data packet a second with no events, and a ping after about 4.6 s with nothing acknowledged	[run] (torque/network.md#after-the-connection-is-accepted)
+18.02 s	C→S	disconnect, sequence 1024, empty reason	[run] (torque/network.md#the-full-handshake-live)

The client's console logged the same steps: "Sending Connect challenge Request", "Got Connect challenge Response", "Sending Connect Request", "Connection established", and at the end "Issuing Disconnect packet." [run] (torque/network.md#the-full-handshake-live).

10.3.5 Data packets

Rule	Source
Header (stock TGE): flag 1, a sequence-parity bit, 9-bit sequence, 9-bit highest ack, 2-bit type (0 data, 1 ping, 2 ack), 3-bit ack byte count, then the ack mask	[bin][run] (torque/network.md#after-the-connection-is-accepted)
Then a flag → 10-bit update delay and 10-bit packet size, and a flag → two 10-bit max-rate values	[bin] (graal3d/torque-engine-4x.md#packet-header-and-rates)
4.04r2 rate prefs: to server 32 (8–32), to client 32 (1–32), packet size 800 with an allowed range of 100–450; the update delay is 1024 / rate ms, so in practice 32 packets/s of 450 bytes	[bin] (graal3d/torque-engine-4x.md#packet-header-and-rates)

Game-packet bodies after the header [bin] (graal3d/torque-engine-4x.md#server--client-data-packet); [bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era):

Part	4.04r2	4.20r6, 5.12	Source
S→C move ack	int32 last move ack	none	[bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)
S→C damage flash and white-out	flag → { flag → float7; flag → float7 }	none	[bin] (graal3d/torque-engine-4x.md#server--client-data-packet)
S→C control object	flag → flag (dirty) → u32 server time, int12 ghost index, the object's data; else 3 × F32 compression point	flag → flag → u32 server time (ms), int12 ghost index of a `GameObject`, its data; the first control object starts login step 5	[bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)
S→C camera	flag → int12 camera ghost, flag → camera data	the same; no flag clears the camera	[bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)
S→C field of view	flag → int8	flag → int8	[bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)
C→S first flag	first person [inf]	set when a connection float is 0; meaning not traced	[inf] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)

Part	4.04r2	4.20r6, 5.12	Source
C→S position	3 × F32 position of the control object, flag → int12 its ghost index	the same, or the ghost of the object it is mounted on	[bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)
C→S state	flag → the control object's data; else int32 first move index, int5 move count (up to 30), the moves	flag → the control object's data; no move branch	[bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)
C→S field of view	flag → int8	flag → int8, clamped to 1–179	[bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)
Both, then	events, then ghosts	events, then ghosts	[bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)

Rule	Source
The u32 server time is a Graal addition; the client keeps it as its synchronised clock	[bin] (graal3d/torque-engine-4x.md#server--client-data-packet)
In 5.12 the client always reports its own position and never sends a Torque `Move`; position is client-reported as in 2D Graal	[bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)
Unguaranteed events: flag 1, int12 class id, sized payload, …, flag 0; guaranteed-ordered events add a flag (next in sequence) else int7 sequence	[bin] (graal3d/torque-engine-4x.md#events)
Ghosts: flag, int4 index width (sent as is; TGE sends width − 3), then per ghost flag 1, index, flag (delete), and if not deleted int12 class id on every update and the sized data; flag 0 ends	[bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes)
New ghosts go into the client's `MissionGroup`; in 4.04r2 they wait while TCP file downloads are running	[bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes)
In 5.12 the class id reader takes 12 bits and applies the server-to-local map; events and ghosts each have their own length wrappers	[bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)

10.3.6 Mission download

Build	Phases	Source
3.153, 4.04r2 (stock TGE)	1 "Download Datablocks & Targets", 2 "Download Ghost Objects", 3 "Mission Lighting"	[bin][file] (torque/network.md#after-the-connection-is-accepted)
5.12	1 "Download Ghost Objects", 2 "Mission Lighting"	[bin] (torque/network.md#after-the-connection-is-accepted)

Rule	Source
The server drives the phases with `MissionStartPhaseN` remote commands and the client answers `MissionStartPhaseNAck`	[bin] (torque/network.md#after-the-connection-is-accepted)
In 3.153 these are script functions; phase 1 acks with the sequence number, and phase 3 lights the scene and acks when lighting is done	[file] (torque/network.md#after-the-connection-is-accepted)
In 4.04r2 they are native; the acks carry only the command tag, no sequence number; every other `commandToClient` name is ignored	[bin] (graal3d/torque-engine-4x.md#mission-start)
4.04r2 raises `onGraal3DLoginStep` 1 to 5: phases 1, 2 and 3, lighting done (`$missionRunning = 1`), and the first control object (switch to the 3D GUI)	[bin] (graal3d/torque-engine-4x.md#mission-start)
5.12 starts the value at −1 and raises 2 (phase 1), 3 (phase 2), 4 (lighting done), 5 (first control object) and 6 (TCP file downloads finished, the `GraalControl3D` GUI attached)	[bin] (torque/wire-by-build.md#512-game-packets-the-gameobject-era)
The 3.153 loading screen is a snowball rolling across `LoadingGui`, with the Christmas 2004 3D images	[file] (torque/scripting.md#what-the-3153-scripts-do)

10.4 DSO v33

Graal 3.x keeps menus, options, key binds, message boxes and the 3D mission-loading flow in compiled TorqueScript; from 4.x on no `.dso` files ship [file] (torque/scripting.md#torquescript-in-graal-3x-the-dso-format-and-the-shipped-scripts).

10.4.1 Shipped scripts

Build	Scripts	Source
3.04 Linux	39 compiled scripts in the TGE starter layout; a PM window and a player list in script; no 3D scripts	[file] (torque/scripting.md#torquescript-in-graal-3x-the-dso-format-and-the-shipped-scripts)
3.153 Mac	44 compiled scripts (`scripts/`, `scripts2D/`, `scriptsgui/`) and 13 plain `.cs` files (prefs, key configs, nine language tables); adds the 3D client scripts	[file] (torque/scripting.md#torquescript-in-graal-3x-the-dso-format-and-the-shipped-scripts)
4.x and later	none; menus are native or GS2	[file] (torque/scripting.md#torquescript-in-graal-3x-the-dso-format-and-the-shipped-scripts)

10.4.2 File layout

All integers little-endian [file] (torque/scripting.md#file-format-version-33).

Field	Notes	Source
U32 version	33	[file] (torque/scripting.md#file-format-version-33)
U32 size, bytes	global string table, NUL-separated	[file] (torque/scripting.md#file-format-version-33)
U32 count, count × F64	global float table	[file] (torque/scripting.md#file-format-version-33)
U32 size, bytes	function string table	[file] (torque/scripting.md#file-format-version-33)
U32 count, count × F64	function float table	[file] (torque/scripting.md#file-format-version-33)
U32 codeLength, U32 lineBreakPairCount		[file] (torque/scripting.md#file-format-version-33)
code	`codeLength` words; each one byte, or `0xFF` then a U32	[file] (torque/scripting.md#file-format-version-33)
lineBreakPairCount × 2 × U32	line and ip pairs	[file] (torque/scripting.md#file-format-version-33)
U32 identCount, then per ident U32 string offset, U32 count, count × U32 ip	patches identifier words in the code with global-string entries	[file] (torque/scripting.md#file-format-version-33)

Rule	Source
Code inside a function body uses the function tables; code outside uses the global tables	[file] (torque/scripting.md#file-format-version-33)
The Mac PPC build writes the F64 tables big-endian while the rest stays little-endian; the 3.04 Linux files are little-endian throughout	[file] (torque/scripting.md#file-format-version-33)

Rule	Source
An argument the body never uses gets an ident offset just past the end of the global string table, so its name is lost	[file] (torque/scripting.md#file-format-version-33)
All 83 shipped files decode with the opcode table below and no other	[file] (torque/scripting.md#graals-opcode-table)

10.4.3 Opcodes

The numbering differs from stock TGE 1.x: two unused opcodes follow `CREATE_OBJECT` and one follows `CALLFUNC` [file] (torque/scripting.md#graals-opcode-table).

#	Opcode	#	Opcode	#	Opcode	Source
0	FUNC_DECL	28	SHL	56	STR_TO_UINT	[file] (torque/scripting.md#graals-opcode-table)
1	CREATE_OBJECT	29	AND	57	STR_TO_FLT	[file] (torque/scripting.md#graals-opcode-table)
2	unused	30	OR	58	STR_TO_NONE	[file] (torque/scripting.md#graals-opcode-table)
3	unused	31	ADD	59	FLT_TO_UINT	[file] (torque/scripting.md#graals-opcode-table)
4	ADD_OBJECT	32	SUB	60	FLT_TO_STR	[file] (torque/scripting.md#graals-opcode-table)
5	END_OBJECT	33	MUL	61	FLT_TO_NONE	[file] (torque/scripting.md#graals-opcode-table)
6	JMPIFFNOT	34	DIV	62	UINT_TO_FLT	[file] (torque/scripting.md#graals-opcode-table)
7	JMPIFNOT	35	NEG	63	UINT_TO_STR	[file] (torque/scripting.md#graals-opcode-table)
8	JMPIFF	36	SETCURVAR	64	UINT_TO_NONE	[file] (torque/scripting.md#graals-opcode-table)
9	JMPIF	37	SETCURVAR_CREATE	65	LOADIMMED_UINT	[file] (torque/scripting.md#graals-opcode-table)
10	JMPIFNOT_NP	38	SETCURVAR_ARRAY	66	LOADIMMED_FLT	[file] (torque/scripting.md#graals-opcode-table)
11	JMPIF_NP	39	SETCURVAR_ARRAY_CREATE	67	TAG_TO_STR	[file] (torque/scripting.md#graals-opcode-table)
12	JMP	40	LOADVAR_UINT	68	LOADIMMED_STR	[file] (torque/scripting.md#graals-opcode-table)
13	RETURN	41	LOADVAR_FLT	69	LOADIMMED_IDENT	[file] (torque/scripting.md#graals-opcode-table)
14	CMPEQ	42	LOADVAR_STR	70	CALLFUNC_RESOLVE	[file] (torque/scripting.md#graals-opcode-table)
15	CMPGR	43	SAVEVAR_UINT	71	CALLFUNC	[file] (torque/scripting.md#graals-opcode-table)
16	CMPGE	44	SAVEVAR_FLT	72	unused	[file] (torque/scripting.md#graals-opcode-table)
17	CMPLT	45	SAVEVAR_STR	73	ADVANCE_STR	[file] (torque/scripting.md#graals-opcode-table)
18	CMPLE	46	SETCUROBJECT	74	ADVANCE_STR_APPENDCHAR	[file] (torque/scripting.md#graals-opcode-table)
19	CMPNE	47	SETCUROBJECT_NEW	75	ADVANCE_STR_COMMA	[file] (torque/scripting.md#graals-opcode-table)
20	XOR	48	SETCURFIELD	76	ADVANCE_STR_NUL	[file] (torque/scripting.md#graals-opcode-table)
21	MOD	49	SETCURFIELD_ARRAY	77	REWIND_STR	[file] (torque/scripting.md#graals-opcode-table)
22	BITAND	50	LOADFIELD_UINT	78	TERMINATE_REWIND_STR	[file] (torque/scripting.md#graals-opcode-table)

#	Opcode	#	Opcode	#	Opcode	Source
23	BITOR	51	LOADFIELD_FLT	79	COMPARE_STR	[file] (torque/scripting.md#graals-opcode-table)
24	NOT	52	LOADFIELD_STR	80	PUSH	[file] (torque/scripting.md#graals-opcode-table)
25	NOTF	53	SAVEFIELD_UINT	81	PUSH_FRAME	[file] (torque/scripting.md#graals-opcode-table)
26	ONESCOMPLEMENT	54	SAVEFIELD_FLT	82	BREAK	[file] (torque/scripting.md#graals-opcode-table)
27	SHR	55	SAVEFIELD_STR	—	—	[file] (torque/scripting.md#graals-opcode-table)

Rule	Source
Operands: `FUNC_DECL` name, namespace, package, hasBody, end ip, argc, then argc arg idents; `CREATE_OBJECT` parent ident, isDatablock, fail ip; `ADD_OBJECT` and `END_OBJECT` one flag; jumps one target	[file] (torque/scripting.md#graals-opcode-table)
`SETCURVAR`, `SETCURVAR_CREATE`, `SETCURFIELD`, the `LOADIMMED_*` opcodes, `TAG_TO_STR` and `ADVANCE_STR_APPENDCHAR` take one operand; `CALLFUNC` and `CALLFUNC_RESOLVE` take name, namespace and call type (0 function, 1 method, 2 parent); the rest take none	[file] (torque/scripting.md#graals-opcode-table)
The operand shape is the same as stock TGE 1.x; opcodes 2, 3 and 72 never occur, so what they were is unknown	[file] (torque/scripting.md#graals-opcode-table)
The repo has a disassembler and a decompiler for this format under `torque/tools/`	[doc] (torque/scripting.md#the-decoder)

10.4.4 What the 3.153 scripts do

Script area	Behaviour	Source
start-up	`main.cs` opens the console, runs `main2D.cs` and writes default prefs (fixed UDP port 14899, font size 22, OpenAL); `main2D.cs` loads the GUI and binds and shows the 2D main menu; the 2D game is the native control `Adventure` (`GuiGraalCtrl`)	[file] (torque/scripting.md#what-the-3153-scripts-do)
menus	call native `Adventure.*` methods (nickname, account, password, language, server list, offline mode, direct connect, connect, account fields); with `$serverstartconnect` set and an account filled in, the menu goes straight to the server list, which is how `-connect` and `graal://` links join	[file] (torque/scripting.md#what-the-3153-scripts-do)
2D keys	arrows move, A/S/D act (Space = S), Q, P, M, T microphone, F5 load level, F7 player list, F8 server list; global F1 help, F2 console, F3 options, Alt+Enter full screen, Esc	[file] (torque/scripting.md#what-the-3153-scripts-do)
3D keys	WASD and arrows move and also drive the Graal player; mouse look with sensitivity clamped to 0.05–2.0 and scaled by FOV/90; Space jump, left mouse fire, right mouse first person, E free look	[file] (torque/scripting.md#what-the-3153-scripts-do)
3D start	`main3D.cs` maps nine terrain textures to detail textures and footstep sounds, loads the play, loading and download GUIs, sets net port 0, creates `GameConnection(ServerConnection)` and connects	[file] (torque/scripting.md#what-the-3153-scripts-do)
connection errors	every Torque reject code maps to a message box, including the Graal protocol-less and protocol-greater texts	[file] (torque/scripting.md#what-the-3153-scripts-do)
chat over Torque	stock `clientCmdChatMessage` / `clientCmdServerMessage`, ending in native `onChatMessage` / `onServerMessage`	[file] (torque/scripting.md#what-the-3153-scripts-do)

Script area	Behaviour	Source
3.04	the key map equals 3.153's 2D map, named `moveMap`, with F3 bound in the map	[file] (torque/scripting.md#what-the-3153-scripts-do)

10.5 Object model

10.5.1 Networked classes

Build	Networked classes	Source
3.04	stock TGE; no Graal net classes	[bin] (torque/wire-by-build.md#the-handshake-per-build)
3.153	the TGE game, datablock and Graal event classes of 4.04r2, without `AudioProfile`	[bin] (torque/wire-by-build.md#the-networked-class-sets)
4.04r2	117 class registrations, 98 networked: 49 objects, 34 datablocks, 15 events	[bin] (torque/wire-by-build.md#the-networked-class-sets)
4.20r6 objects	`Camera`, `DynamicGrass`, `DynamicGrassReplicator`, `DynamicShapeReplicator`, `GameObject`, `SimpleNetObject`, `Sky`, `Sun`, `TerrainBlock`, `WaterBlock`, `fxSunLight`, `twSurfaceReference`	[bin] (torque/wire-by-build.md#the-networked-class-sets)
4.20r6 events	`BigDataChunkEvent`, `BigDataStartEvent`, `ConnectionMessageEvent`, `GhostAlwaysObjectEvent`, `NetStringEvent`, `RemoteCommandEvent`, `RequestReloadEvent` (C→S), `SimpleMessageEvent`, `TerrainChunkEvent`, `TerrainRequestUpdatesEvent` (C→S)	[bin] (torque/wire-by-build.md#the-networked-class-sets)
5.12	10 objects (4.20r6's without `SimpleNetObject` and `twSurfaceReference`) and 9 events (without `SimpleMessageEvent`)	[bin] (torque/wire-by-build.md#the-networked-class-sets)

4.04r2 networked classes, all stock TGE unless marked [bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes):

Group	Classes	Source
shapes and players	`AIPlayer`, `Player`, `ShapeBase`, `StaticShape`, `Item`, `TSStatic`, `Shape3ds` (Graal), `Projectile`, `Debris`, `Splash`	[bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes)
vehicles	`FlyingVehicle`, `HoverVehicle`, `WheeledVehicle`, `VehicleBlocker`	[bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes)
cameras	`Camera`, `PathCamera`, `AdvancedCamera`	[bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes)
world	`InteriorInstance`, `PathedInterior`, `TerrainBlock`, `WaterBlock`, `Sky`, `Sun`, `Precipitation`, `Lightning`	[bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes)
mission and markers	`GameBase`, `Marker`, `MissionArea`, `MissionMarker`, `SpawnSphere`, `WayPoint`, `Trigger`, `PhysicalZone`, `ScopeAlwaysShape`, `SimpleNetObject`	[bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes)

Group	Classes	Source
effects	`AudioEmitter`, `ParticleEmitterNode`, `FractalTree`, `fxFoliageReplicator`, `fxGrassReplicator`, `fxShapeReplicator`, `fxShapeReplicatedStatic`, `fxLight`, `fxSunLight`, `volumeLight`, `twSurfaceReference`	[bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes)
Graal replicators	`DynamicGrass`, `DynamicGrassReplicator`, `DynamicShapeReplicator`	[bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes)
datablocks	the stock `…Data` classes for shapes, images, items, players, cameras, vehicles, projectiles, effects, particles, audio, markers, interiors and triggers	[bin] (graal3d/torque-engine-4x.md#ghosts-graal-changes)

Rule	Source
In 4.04r2 `Item`, Sky, Sun and Path register by string address, so a plain string search misses them	[bin] (torque/wire-by-build.md#the-networked-class-sets)
In 4.04r2 ShapeBase carries a Graal chat string drawn above the head, with emoticon handling; how the client's own chat reaches the 3D server was not found and probably goes over TCP	[bin][inf] (graal3d/torque-engine-4x.md#ghosts-graal-changes)
Every other class registers with group mask 0 (not networked); the `GameShape*` classes ride inside a `GameObject` update	[bin] (torque/wire-by-build.md#the-networked-class-sets)

10.5.2 The 5.12 `GameObject` model

All classes are script-visible through GS2 property and method tables [bin] (torque/engine-and-formats.md#the-512-3d-object-model).

Class	What it is	Notable members	Source
`GameObject`	every 3D thing, a ghosted `SceneObject`	mounting, node position and transform, interior checks, drop height, box collision, the controlling client, show-images on 3D objects, `onMount`, `onUnmount`, `onAdd`, `onRemove`; script variables and attributes synced to clients	[bin] (torque/engine-and-formats.md#the-512-3d-object-model)
`GameShapeDTS`	Torque DTS model	`playAnimation(slot, name, flags)` with flags 1 hold, 2 at end, 4 attack; `playAnimation2` with start, end, time scale and transition; pause, stop, node and material lists, `onEndAnimation`	[bin] (torque/engine-and-formats.md#the-512-3d-object-model)
`GameShapeDIF`	Torque interior	`activateLight`, `deactivateLight`	[bin] (torque/engine-and-formats.md#the-512-3d-object-model)
`GameShape3DS`	3D Studio model	collision hulls are meshes named `Col-…`, closed and convex	[bin] (torque/engine-and-formats.md#the-512-3d-object-model)
`GameShapeFT`	3D text from a TrueType font		[bin] (torque/engine-and-formats.md#the-512-3d-object-model)

Class	What it is	Notable members	Source
`GameShape`	base class	pixel and vertex shaders (GLSL), fractal-tree fields	[bin] (torque/engine-and-formats.md#the-512-3d-object-model)
`GameMovementInterpolate`, `…List`, `…Rigid`	how an object moves	`move3d`, `finishMoves`, `addRigidMove`, `applyImpulse`, mass, `onImpact`, `onCollision`, `onEnterLiquid`, `onLeaveLiquid`	[bin] (torque/engine-and-formats.md#the-512-3d-object-model)
`Camera`	the view	modes `free` and `lookat`, follow an attached object, look position and vector, Euler rotation	[bin] (torque/engine-and-formats.md#the-512-3d-object-model)
`DynamicGrass`, `DynamicGrassReplicator`, `DynamicShapeReplicator`	ground cover	terrain alignment, slope limits, allowed surfaces, per-grid counts	[bin] (torque/engine-and-formats.md#the-512-3d-object-model)
`Sky`, `Sun`, `WaterBlock`, `fxSunLight`, `TerrainBlock`	stock Torque	three cloud layers, fog volumes, fog storms with server-synchronised time	[bin] (torque/engine-and-formats.md#the-512-3d-object-model)

Rule	Source
The 2D Graal player is still the player; 3D objects can carry ganis and show-images	[bin] (torque/engine-and-formats.md#the-512-3d-object-model)
The GS2 properties on `GameObject` are `alpha`, `attachedtoobject`, `attr`, `attrobjects`, `chat`, `collision`, `iseditorobject`, `liquidheight`, `liquidtype`, `movetype`, `nickcolor`, `nickname`, `scaledshapebox`, `shadow`, `shape`, `shapebox`, `showterrain`, `skin`, `solid`, `velocity`, `watercoverage`	[bin] (torque/wire-by-build.md#the-512-gameobject-update)

`GameObject` update, 5.12: thirteen flag-guarded sections; the mask bits are the server's dirty bits, and on the wire each section is one flag [bin] (torque/wire-by-build.md#the-512-gameobject-update).

#	Dirty bits	Payload after the flag	Source
1	`0x1`	string, the object's name	[bin] (torque/wire-by-build.md#the-512-gameobject-update)
2	`0x2` or `0x200`	3 × 32 bits (velocity, inferred)	[bin][inf] (torque/wire-by-build.md#the-512-gameobject-update)
3	`0x8`	string, a setting of the movement object	[bin] (torque/wire-by-build.md#the-512-gameobject-update)
4	`0x10`	string, also passed to the movement object	[bin] (torque/wire-by-build.md#the-512-gameobject-update)
5	`0x20`	transform: position, rotation, then 32 bits; the packing is unmeasured	[bin] (torque/wire-by-build.md#the-512-gameobject-update)

#	Dirty bits	Payload after the flag	Source
6	`0x80`	string that creates or replaces the shape object (shape, inferred)	[bin][inf] (torque/wire-by-build.md#the-512-gameobject-update)
7	`0x40`, with a shape object	the shape object's own data	[bin] (torque/wire-by-build.md#the-512-gameobject-update)
8	any of `0xF00`, with a movement object	int14 bit length, then the movement object's data	[bin] (torque/wire-by-build.md#the-512-gameobject-update)
9	`0x1000`	string, the chat text (with emoticons)	[bin] (torque/wire-by-build.md#the-512-gameobject-update)
10	always 0 from this DLL	the reader takes a string and loads it as a script or image; what a real server sent is unknown	[bin] (torque/wire-by-build.md#the-512-gameobject-update)
11	any of `0x3FC000`	the attribute list	[bin] (torque/wire-by-build.md#the-512-gameobject-update)
12	`0x400000`	string, then 4 colour bytes RGBA 0–255 (nickname and `nickcolor`, inferred)	[bin][inf] (torque/wire-by-build.md#the-512-gameobject-update)
13	`0x800000`	int16 of booleans; bits `0x01`, `0x04`, `0x08`, `0x10`, `0x20`, `0x40` are used	[bin] (torque/wire-by-build.md#the-512-gameobject-update)

10.5.3 Graal events over Torque

Event	Direction	Layout	Source
`BigDataStartEvent`	both	int16 transfer id, three strings (s1, s2, s3, each up to 255 characters), int24 total length, int3 coding (0 none, 1 zlib, 2 bzip2)	[bin] (graal3d/torque-engine-4x.md#graal-events-and-text-big-data-channel)
`BigDataChunkEvent`	both	int16 transfer id, int8 length (up to 191), the bytes	[bin] (graal3d/torque-engine-4x.md#graal-events-and-text-big-data-channel)
`TerrainChunkEvent`	S→C	int4 chunk x, int4 chunk y, u32 (a stamp, inferred), int10 length, then a zlib 16 × 16 piece of heights and materials; 16 × 16 chunks make a 256 × 256 terrain	[bin][inf] (graal3d/torque-engine-4x.md#terrain-streaming-graal)
`TerrainRequestUpdatesEvent`	C→S	256 × u32, one per chunk, telling the server what the client already has	[bin] (graal3d/torque-engine-4x.md#events)
`RequestReloadEvent`	C→S	asks the server to resend an object	[bin] (graal3d/torque-engine-4x.md#events)
`SimDataBlockEvent` (4.04r2)	S→C	int12 (id − 3), int12 class id, int12 index, int13 total, sized datablock data	[bin] (graal3d/torque-engine-4x.md#events)
`RemoteCommandEvent`	both	int5 argc, tagged-string arguments; the 4.04r2 client acts only on the mission phases	[bin] (graal3d/torque-engine-4x.md#events)
others	—	`GhostAlwaysObjectEvent`, `NetStringEvent`, `ConnectionMessageEvent`, `PathManagerEvent`, `SimpleMessageEvent`, the audio and lightning events, stock TGE	[bin] (graal3d/torque-engine-4x.md#events)

Rule	Source
A BigData message has the shape of GS2 `sendtext`: weapon, type, option, data	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)

Rule	Source
The sender uses zlib from 50 bytes and bzip2 from 4096 bytes, and sends the start event then up to 16 chunks per call; the TCP framing uses nearly the same limits (48 and 4096, → §2.2)	[bin] (graal3d/torque-engine-4x.md#graal-events-and-text-big-data-channel); [bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
S→C with s1 set: the weapon of that name gets `onReceiveText(s2, s3, params)`	[bin] (graal3d/torque-engine-4x.md#graal-events-and-text-big-data-channel)
S→C with s1 empty, s2 is a command: `set` / `unset` a player flag; `events` (lines `s <flag>`, or `t <target>,<kind>,<action>,<params>` on a weapon, GUI or NPC; kind `rc` goes to `onRCChat` and `torque` is ignored); `addweapon` (with a 5-byte Graal integer, a time stamp inferred); `weaponscript`; `deleteweapon`; `addnpc` (a script for a named client-side object); `setmaterials` (terrain materials); `rpg` (→ `onRPGMessage`)	[bin][inf] (graal3d/torque-engine-4x.md#graal-events-and-text-big-data-channel)
In 3D mode the 3D server, not the TCP gserver, delivers the client's GS2 weapons and flags	[bin][inf] (graal3d/torque-engine-4x.md#graal-events-and-text-big-data-channel)
C→S the engine batches its game events as one message with `s2 = "events"`: `w <weapon>` from getweapon, `s <name>=<value>` from client-variable and flag sets, `t weapon,<name>,<params>` from a server-side trigger	[bin] (graal3d/torque-engine-4x.md#graal-events-and-text-big-data-channel)
The 4.04r2 DLL's server-side BigData handler is empty	[bin] (graal3d/torque-engine-4x.md#graal-events-and-text-big-data-channel)
The 5.12 client streams terrain from the server instead of reading `.ter` files ("Loading terrain from terrain chunks."); terrain textures must be 256×256 at 24 bits	[bin] (torque/engine-and-formats.md#3d-resource-formats)
The 4.04r2 client caches terrain under `levels3d/terrain/` and never requests `.ter` over TCP	[bin][inf] (graal3d/torque-engine-4x.md#terrain-streaming-graal)

10.6 3D formats

Ext	Format	Builds	Source
`.dts`, `.dsq`	Torque shape and sequence; 4.04r2 reads up to version 24; 5.12 rejects shapes that are too old	3.x, 4.04r2, 5.12	[bin] (torque/engine-and-formats.md#3d-resource-formats); [bin] (graal3d/torque-engine-4x.md#files)
`.dif`	Torque interior; 4.04r2 reads resource version 44, interior version 0	all	[bin] (graal3d/torque-engine-4x.md#files)
`.ter`	Torque terrain, version 3 or older, from `levels3d/terrain/`	3.x, 4.04r2	[bin] (graal3d/torque-engine-4x.md#files)
terrain chunks	streamed with `TerrainChunkEvent` (→ §10.5.3)	5.12	[bin] (torque/engine-and-formats.md#3d-resource-formats)
`.3ds`	3D Studio (`Shape3ds`, then `GameShape3DS`)	3.153, 4.04r2, 5.12	[bin] (torque/engine-and-formats.md#3d-resource-formats)
`.md3`	Quake 3 model in head, upper and lower parts, version 15	3.153, 4.04r2	[bin] (torque/engine-and-formats.md#3d-resource-formats); [bin] (graal3d/torque-engine-4x.md#files)
`.mis`	Torque mission file; 5.12 has no `.mis` loader and builds its `MissionGroup` from ghosts (inferred)	3.x; 4.04r2 disputed → App. E#502	[bin][inf] (torque/engine-and-formats.md#3d-resource-formats); [bin][inf] (graal3d/torque-engine-4x.md#files)
`.ml`, `.ifl`, `.dml`	material lists and image-file lists for animated textures	stock Torque	[bin] (torque/engine-and-formats.md#3d-resource-formats); [bin] (graal3d/torque-engine-4x.md#files)
`.gft`	Torque cached GUI font	3.x, 4.04r2, 5.12	[bin] (torque/engine-and-formats.md#3d-resource-formats); [bin] (graal3d/torque-engine-4x.md#files)

Ext	Format	Builds	Source
`.dso`	compiled TorqueScript (→ §10.4)	3.x; 4.04r2 disputed → App. E#503	[file] (torque/scripting.md#torquescript-in-graal-3x-the-dso-format-and-the-shipped-scripts); [bin] (graal3d/torque-engine-4x.md#files)
images	`.png`, `.jpg`, `.jp2`, `.jng`, `.mng`, `.gif`, `.bmp`, `.bm8`, `.tga`, `.dds`	4.04r2	[bin] (graal3d/torque-engine-4x.md#files)
`.zip`, `.gpake`	archives the resource manager opens	4.04r2	[bin] (graal3d/torque-engine-4x.md#files)

Rule	Source
No `.dts`, `.dif`, `.ter` or `.mis` file ships with any Graal client: the 3D content always came from the server	[file] (torque/engine-and-formats.md#3d-resource-formats)
3.153 maps terrain textures to detail textures and footstep sound types in script (→ §10.4.4)	[file] (torque/engine-and-formats.md#3d-resource-formats)
4.04r2 ships 3D GUI skins in `levels3d/gui/`, help pages in `levels3d/help/*.hfl` and nine `languages3D/language*.cs` tables of `$language::Name = "text";` lines, probably loaded by server GUI scripts	[file][inf] (graal3d/torque-engine-4x.md#files)

11 Tools and the formats they define

The official and community tools around Graal, covered for the files they read and write and the behaviour that bears on those files. The byte-level formats themselves are in Part III; the staff wire protocols are in → §7.

> Status: draft.

11.0 Tool map

Tool	Role	Files it defines or writes	Section	Source
Built-in level editor	editor inside each client up to 2.22	GR-V1.03, GLEVNW01, GOBJSET01 sets, map images	→ §11.6	[2.22] (tools/level-editor/README.md#classic-graal-level-editor-notes-by-version)
GraalEditor	standalone level editor from 2.3	GLEVNW01 with `HEIGHTS`, gmaps	→ §11.1	[GE] (tools/graaleditor-vs-205.md#lineage)
GraalShop	GANI editor	`GANI0001`, `GANI0002`	→ §11.2	[bin] (tools/graaleditor-vs-205.md#graalshop-the-gani-editor)
TerrainGenerator, DungeonGenerator, GraalLevelGen	world generators	GLEVNW01, GR-V1.03, GRMAP001, `bigmap.txt`, `GDUNG001` options	→ §11.3	[bin][doc] (tools/generators-222/README.md#graal-222-generators)
gif2mng	image converter	PNG, MNG	→ §11.4	[bin] (tools/gif2mng.md#gif2mng)
RemoteControl, NPC Control	staff clients	— (wire only)	→ §11.5, → §7	[bin] (recovered-drive-tools/ghidra-analysis/RemoteControl_classic/SUMMARY.md)
TextConvert, Character Designer, GChar, gserverstat and others	converters, offline editors, server-side helpers	GLEVNW01, `CHR:1.00`, HTML leaderboards	→ §11.7	[bin] (tools/classic-tools/README.md#classic-server-side-tools)

11.1 GraalEditor

11.1.1 Lineage and package

Fact	Source
Up to 2.22 the editor is built into the client (F4, refused online); 2.31 has no editor forms, and its start screen has an Editor button that launches `GraalEditor.exe` from the game folder	[GE][2.05] (tools/graaleditor-vs-205.md#lineage)
GraalEditor 2.3.1.1 (2005) is the same `TAdventure` program with the online client removed: the 2.05 main form, tool forms, toolbar hints and even game-only strings remain	[GE] (tools/graaleditor-vs-205.md#lineage)
It keeps the whole game model and GS1 interpreter, so Play runs a level locally; it has no network client, list-server host or login token	[GE] (tools/graaleditor-vs-205.md#lineage)
The shipped copy is a Thinstall package (magic `0x82D0B188`); the Delphi image is recovered by running the stub decompressor and its block codec	[bin] (tools/graaleditor.md#packing)
Version resource: company `Linux Cyberjoueur`, file version 2.3.1.1, internal name `Graal editor`, product `GraalEditor`	[bin] (tools/graaleditor.md#packing)
The main form is `TAdventure`, the document `TAdventureLevel`, the unit `UAdventureEditor`	[GE] (tools/graaleditor.md#document-model)

Fact	Source
`Greaditor.exe` (2001, in the New World beta folders) is a separate UPX-packed editor that was not unpacked	[bin] (tools/level-editor/version-deltas.md#1-the-standalone-editors-one-line-each)

11.1.2 Files

Rule	Source
Open: GR-V1.00–1.03, Z3-V1.03/1.04 and GLEVNW01, with the filter `Graal levels (*.graal, *.nw)`	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)
Save: GLEVNW01 with a new `HEIGHTS` section; the GR-V1 writer strings are still present; "Level(s) saved" means several levels can be saved at once	[GE][inf] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)
The `BOARD 0` layer as GraalEditor writes it	disputed → App. E#600
Heights are a `TLevelHeights` overlay per level, not a second board	[GE] (tools/graaleditor.md#document-model)
Sections read and written: `BOARD`, `LINK`, `SIGN`, `CHEST`, `NPC`, `BADDY`, `LEVELNAMES`, in the field order of the dialogs (→ §3.1.3)	[GE] (tools/graaleditor.md#document-model)
The editor stores no tile types; collision comes from the 4096-byte type table (→ §3.5.2)	[GE] (tools/graaleditor.md#collision)
gmaps are wired in: the `GRMAP001` reader plus `LEVELNAMES`, `MAPIMG`, `MINIMAPIMG`, `NOAUTOMAPPING` and the generator keys (→ §3.2); a `TEditorLevelCache` holds neighbours	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)
2.05 already has a dormant `GRMAP001` reader with no caller	[2.05][inf] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)
Offline data: `offlinetutorial_*` (3×3) levels and `tiledefsOffline.txt`	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)

11.1.3 Additions over the 2.05 built-in editor

Area	GraalEditor	Source
terrain	per-level height overlay, "Terrain Height Spot", "Show terrain grid"	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)
palette	a fourth mode, Predefined NPCs, with a parameter grid	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)
NPCs	one `TEditNPC` window with image, script, Test, Style and Find; scripts may carry `//#EDIT` and `//#COMMENTS` markers	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)
character NPC	the same fields on four tabs (Look, Movement, Attack, General)	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)
tiles	Export Tiles Array: the selection as one line of tile indices and as a script array	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)
view and log	Hide npcs, Show terrain grid, a log window	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)

Area	GraalEditor	Source
sound	BASS, with the function set the 2.22 client imports	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)
integrity	the client self-check plus RSA/SHA-1 checks of `GraalEditor.sig` and `arrays2.sig`; live, the editor starts with no `GraalEditor.sig` and with a corrupted `arrays2.sig`, so the check does not gate start-up (→ §3.5.5)	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)
unchanged	the tile undo model, the link, sign, chest and baddy dialogs, GOBJSET01 sets, right-click copy, Generate map → `graalmap.png`	[GE] (tools/graaleditor-vs-205.md#what-graaleditor-adds-over-205)

11.2 GraalShop

Fact	Source
`GraalShop.exe` 1.1.1.0 ("Gani Editor", 2006) and `GraalShop_old.exe` (2002) are Delphi programs with one main form, "Graal Shop - The Graal Animation Editor", and an Add sprite dialog; they edit `.gani` files only	[bin] (tools/graaleditor-vs-205.md#graalshop-the-gani-editor)
The name is a pun on Photoshop, not a store	[doc] (tools/graalshop.md#graalshop)
The model holds sprites with attachments (`ATTACHSPRITE`, `ATTACHSPRITE2`), frames of placements and sounds, and in movie mode actors and actor modifiers; undo and redo are a list of edit operations	[bin] (tools/graaleditor-vs-205.md#model)
Sprite mode (`GANI0001`): direction selector, sprite-definition grid, frame strip and canvas; play, add, delete, copy and insert frames, add sound, add, import or prune sprite definitions, reverse the animation, undo, redo, file actions; selected sprites can be moved, raised, lowered and keyed by x/y	[bin] (tools/graaleditor-vs-205.md#workflow)
Movie mode (`GANI0002`): a timeline of CHAR, SPRITE, SOUND and TEXT actors; TEXT has font, zoom, size, alignment, wordwrap and shading; CHAR has ani, dir, head, body, sword, shield, horse, colours, attr and param; options Automatic Gani Setting and Graal2002 ganis	[bin] (tools/graaleditor-vs-205.md#workflow)
Header fields are edited in a grid and written as `LOOP`, `BACKCOLOR`, `SETBACKTO`, `DEFAULTHEAD`, `DEFAULTBODY`, `DEFAULTPARAM`, `DEFAULTATTR`, the effect keys and `SCRIPT` … `SCRIPTEND` (→ §3.3.4)	[bin] (tools/graaleditor-vs-205.md#workflow)
The script block is copied, not run	[bin] (tools/graalshop.md#contract)
Preview uses DirectDraw 7 and plays `PLAYSOUND` steps through DirectSound; default art is `sprites.png`, `body.png`, `head0.png`, `shield1.png`, `ride.png`, `soundicon.png`; the Graal folder is found through `Graal.exe` and `FILENAMECACHE.txt`, and `pms`, `profiles` and `weblevels` are searched	[bin] (tools/graaleditor-vs-205.md#workflow)
`GraalShop_old` has sprite mode only: no `GANI0002`, `MOVIE`, `ACTOR`, `ATTACHSPRITE2` or `ROTATEEFFECT`	[bin] (tools/graaleditor-vs-205.md#workflow)
2.05 plays only the sprite-mode subset (`SPRITE`, `ANI` … `ANIEND`, `WAIT`, `PLAYSOUND`, `LOOP`, `CONTINUOUS`, `SETBACKTO`, `SINGLEDIRECTION`, `COLOREFFECT`, `ZOOMEFFECT`), so movie-mode ganis and attachments from the 2006 GraalShop do not play there	[2.05] (tools/graaleditor-vs-205.md#how-this-compares-with-what-205-can-play)
The offline tile definition in the client tree is `picso.png,offlinetutorial,1,0,0`	[file] (tools/graalshop.md#contract)

11.3 Generators

Three writers emit `GLEVNW01` and `GRMAP001`, each with a different key set; one writer's output is not a valid input for another's loader (the key sets are → §3.2.3) [bin] (tools/generators.md#generators).

11.3.1 Builds

Program	Date	Writes	Source
TerrainGenerator.exe	2003-12-16	GRMAP001 generator form and GLEVNW01 levels of tile 0 with a 9×9 HEIGHTS block and edge links	[bin] (tools/generators-222/README.md#graal-222-generators)
dungeongen.exe and the Linux dungeongen	2003-05-23	GLEVNW01 levels from GDUNG001 options, short GRMAP001	[bin] (tools/generators-222/README.md#graal-222-generators)
GraalLevelGen_nw-gmap.exe	2003-03-07	GLEVNW01 and a LEVELNAMES gmap; GR-V1.03 when the output name does not end in .nw	[bin] (tools/generators-222/README.md#not-reimplemented)
GraalLevelGen_graal-txt.exe	2004-01-18	GR-V1.03 levels and bigmap.txt	[bin] (tools/level-tools-gaps/README.md#graal-txt-and-nw-txt)
GraalLevelGen_nw-txt.exe	2001-06-30	GLEVNW01 levels and bigmap.txt	[bin] (tools/level-tools-gaps/README.md#graal-txt-and-nw-txt)
GraalLevelGen.exe	December 2000	levels named .graal and bigmap.txt; the readme says New World format and not for 1.39 or older	[doc][bin] (tools/classic-tools/graal-level-gen.md#graallevelgen-december-2000)
GraalLevelGenNW.exe	2001-10-08	a New World levels tree from a map image, with calendar fields and per-level options such as whether warping is allowed	[bin] (recovered-drive-tools/ghidra-analysis/GraalLevelGenNW/SUMMARY.md)

No generator imports a socket library; output stays in the chosen folder [bin] (tools/classic-tools/graal-level-gen.md#what-it-never-sends).

11.3.2 TerrainGenerator

Rule	Source
Dialog fields: grid width and height, base, height, chaos, even borders, an islands preset (even borders, the rim kept at the base) and a seed	[doc][bin] (tools/generators-222/README.md#terrain)
The height field is plasma midpoint displacement	[bin] (tools/generators.md#terraingenerator)
gmap keys in order: WIDTH, HEIGHT, GENERATED 1, GENSEED, GENBASE, GENEVENBORDERS, GENHEIGHT, GENCHAOS, LEVHEIGHT, LEVCHAOS, MAPIMG, one level name per line, HEIGHTMAP, RANDOMSEEDS	[bin] (tools/generators-222/README.md#terrain)
Level names: world name, an A…Z/AA column, a zero-padded row, .nw	[bin] (tools/generators-222/README.md#terrain)
A level is 64 BOARD rows of tile 0, a HEIGHTS block of nine lines of nine numbers, and a LINK on each edge with a neighbour; no NPC section	[bin] (tools/generators-222/README.md#terrain)
Edge arrivals: west 61, east 0, north 61, south 0; the other axis is playerx or playery	[bin] (tools/generators-222/README.md#terrain)

11.3.3 DungeonGenerator

Rule	Source
dungeonoptions.txt starts with GDUNG001; a LAYOUTFILE of X (wall) and . (path) wins over LAYOUTSTYLE	[file] (tools/generators-222/README.md#dungeon)

Rule	Source
Each layout cell is 8 tiles wide and 11 tall, cut into 64×64 levels named `LEVELSTART`, a column letter and a 1-based row (`cave_A1.nw`)	[doc][bin] (tools/generators-222/README.md#dungeon)
The gmap is three lines after the signature: `WIDTH`, `HEIGHT`, `GENERATED` with the bottom-right name	[bin] (tools/generators-222/README.md#dungeon)
Each level is one base64 `BOARD`, edge links and `NPC` blocks; stamp 0 becomes a 4×4 ground fill, stamp 1 is a hole and is not written	[bin] (tools/generators.md#dungeongenerator)
Place names for NPCs and 88-id `TILES` blocks: `path`; `walln`/`walls`/`wallw`/`walle` (one open side, wall opposite); `wallnw`/`wallne`/`wallsw`/`wallse` (one open diagonal); the …2 corner forms (two adjacent open sides); `walld1` (open north and south) and `walld2` (open east and west); `wall` otherwise	[doc][file][bin] (tools/generators-222/README.md#dungeon)
`NPC` lines take `image`, `script`, `blocking`, `underplayer`, `dominant`, `chance` (1–100, `%` optional), `place`, `x`, `y`; false `blocking` adds `dontblock`, `underplayer` adds `drawunderplayer`	[bin] (tools/generators-222/README.md#dungeon)
`ADDTESTNPCS` adds one test NPC per level running `addtiledef`, `addtiledef2` and `loadmap`	[doc][bin] (tools/generators-222/README.md#dungeon)
`LAYOUTSYMMETRY` 1 none, 2 vertical mirror, 3 horizontal mirror, 4 both, 0 picks 1–4 from the seed; `DOUBLE 1` draws at half size and scales up; `LAYOUTORIENTATION` is stored	[doc][bin] (tools/generators-222/README.md#dungeon)
The random styles (maze, onion, spiral, rogue, snake, square spiral) come from the linked DungeonMaker 0.9 library, which reads `layouts/rooms*`, `layouts/stats*` and `layouts/constants`; its crawler steps are unmeasured	[bin][doc] (tools/level-tools-gaps/README.md#dungeonmaker-09)

11.3.4 GraalLevelGen

Rule	Source
The input picture is 64 pixels per level across and down, saved as gif or 8-bit png	[doc] (tools/classic-tools/graal-level-gen.md#what-the-operator-does)
The pixel grid is `(width + 5) // 6` by `(height + 5) // 6` cells, each read at `(x × 6, y × 6)`; a colour matches when the sum of absolute channel differences is under 32	[bin] (tools/generators-222/README.md#level-generator); [bin] (tools/level-tools-gaps/README.md#water-mountain-and-sand-edges)
Forest blits a 6×5 stamp of tiles 25–30, 41–46, 57–62, 73–78, 89–94 at `(x × 6, y × 6 + 1)`, shifted 3 tiles left on odd rows	[bin] (tools/generators-222/README.md#level-generator)
Other colours write one fill tile: water 12, mountains 80, flowers 4, sand 5, puddle 6, puddle stone 7, small stones 8, big stone 9, swamp 10, bushes 11; grass and unknown colours write nothing	[bin] (tools/generators-222/README.md#level-generator)
Water, mountain and sand edges are blitted from neighbour flags; the stamp-to-shore table is unmeasured	[bin] (tools/level-tools-gaps/README.md#water-mountain-and-sand-edges)
nw-gmap names are `world_a-01.nw` to `world_z-99.nw`, and the gmap is `WIDTH`, `HEIGHT` and quoted `LEVELNAMES` rows; edge links use the 0/61 arrivals	[doc][bin] (tools/generators-222/README.md#level-generator)
graal-txt names are prefix + column + `-` + row + `.graal`; nw-txt names drop the dash and end in `.nw`; column letters are a + index mod 26 with a leading letter from 26 levels across; the row is 1-based with leading zeros from 10 and 100 levels down	[bin] (tools/level-tools-gaps/README.md#graal-txt-and-nw-txt)
`bigmap.txt` is one row per line of `"name","name"` (the editor's M-key map)	[bin][doc] (tools/level-tools-gaps/README.md#graal-txt-and-nw-txt)
A second run on existing levels keeps their links and NPCs and paints only the new non-grass tiles	[doc] (tools/generators-222/README.md#level-generator)

Colour table, the same in the December 2000 build and the nw-gmap build [doc][bin] (tools/classic-tools/graal-level-gen.md#what-the-operator-does); [doc][bin] (tools/generators-222/README.md#level-generator):

Index	Meaning	RGB	Source
0	grass, not drawn	24, 140, 24	[doc] (tools/generators-222/README.md#level-generator)
1	forest	181, 239, 189	[doc] (tools/generators-222/README.md#level-generator)
2	water	0, 0, 255	[doc] (tools/generators-222/README.md#level-generator)
3	mountains	128, 0, 0	[doc] (tools/generators-222/README.md#level-generator)
4	flowers	255, 0, 0	[doc] (tools/generators-222/README.md#level-generator)
5	sand	150, 110, 0	[doc] (tools/generators-222/README.md#level-generator)
6	puddle	0, 255, 255	[doc] (tools/generators-222/README.md#level-generator)
7	puddle stone	0, 179, 179	[doc] (tools/generators-222/README.md#level-generator)
8	small sand stones	113, 82, 0	[doc] (tools/generators-222/README.md#level-generator)
9	big sand stone	83, 61, 0	[doc] (tools/generators-222/README.md#level-generator)
10	swamp	0, 106, 0	[doc] (tools/generators-222/README.md#level-generator)
11	bushes	0, 66, 0	[doc] (tools/generators-222/README.md#level-generator)

11.4 gif2mng

Rule	Source
A GIF converter with banner date 2004-06-03, units `UGif` and `UMNG`, zlib linked	[bin] (tools/gif2mng.md#gif2mng)
One frame becomes PNG, two or more become MNG	[bin] (tools/gif2mng.md#gif2mng)
`-b` rewrites the existing palette to the seven Graal body-colour slots and forces transparent index 2; it does not add a second palette (the slots' draw roles are → §2.6.3)	[bin] (tools/gif2mng.md#gif2mng)
The seven slot RGB values and the MNG `TERM` delay fields are unmeasured	[doc] (tools/level-tools-gaps/README.md#gif2mng)

11.5 RemoteControl and NPC Control

The RC and NC wire protocols, logins and commands are → §7; the RC lineage from `GSERV014` to `GSERV023` is → §7.1. This section lists the recovered staff-client binaries and what each shows.

Binary	Tool kit	Behaviour	Source
`RemoteControl.exe` (classic `rc.rar` line)	Delphi	server-list connect screen; registry prefs, a local startup script and password fields; NPC attributes, script and flags, NPC level list, local NPCs; player count and NPC-server status; account and player editors; deflate	[bin] (recovered-drive-tools/ghidra-analysis/RemoteControl_classic/SUMMARY.md)
custom `remotecontrol.exe` and a UPX `Agret_RC.exe` variant	Delphi	the classic RC role with a blue theme and a text config with nickname and UI-colour keys; the unpacked `Agret_RC.exe` is a different build	[bin] (recovered-drive-tools/ghidra-analysis/remote control_custom/SUMMARY.md)
`RemoteControl-NW.exe` (New World packs, NC V2 pack)	Delphi	account and player editors and a database account search by admin level, admin worlds and block state; admin message and broadcast, edit account, update level, disconnect, ban, away-on-close, HTML in PMs	[bin] (recovered-drive-tools/ghidra-analysis/RemoteControl-NW/SUMMARY.md)

Binary	Toolkit	Behaviour	Source
`RemoteControl3.exe` (RC3 beta, also in a 5.x pack)	GTK on Windows	server list, version and password fields; list-server and separate NPC-server connections; player list with mass PM and admin message, NPC lists, level-NPC dump, folder and config pages, profile, rights, account and access editors; ban with time and reason, warps, disconnect, update level, separate NC chat; bzip2 and zlib; highlighters for Graal script, ini, Pascal and C++	[bin] (recovered-drive-tools/ghidra-analysis/RemoteControl3/SUMMARY.md)
`NPCControl.exe` (`nc7132001`, July 2001)	Delphi	a text config of server name, port and password mirrored in the registry; NPC and weapon management, take a script from a class, edit weapon scripts, local NPCs; deflate	[bin] (recovered-drive-tools/ghidra-analysis/NPCControl/SUMMARY.md)

11.6 Level editor

The built-in editor, with 2.22 as the reference and every other build as a delta. The 2.05-specific in-client behaviour is → §4.9. Formats it reads and writes are → §3.1 [2.22] (tools/level-editor/README.md#classic-graal-level-editor-notes-by-version).

11.6.1 Entering, window and modes (2.22)

Rule	Source
The editor is the `TAdventure` main form in a second mode; the same exe plays and edits	[2.22] (tools/level-editor/222-reference.md#1-entering-and-leaving)
F4 opens it and is refused online ("You cannot open the editor in the online mode."); from a tutorial level it opens a new empty level	[2.22] (tools/level-editor/222-reference.md#1-entering-and-leaving)
F5 opens Load level; P or Play returns to the game on the current level, saving every modified level first since 2.13	[2.22] (tools/level-editor/222-reference.md#1-entering-and-leaving)
A level path as the first command-line argument skips the login screen and opens it in the editor	[2.22] (tools/level-editor/222-reference.md#1-entering-and-leaving)
F10 toggles the window border; Esc warns before leaving; dropping a text file creates an NPC from it	[2.22] (tools/level-editor/222-reference.md#1-entering-and-leaving)
Board view: 64 × 64 tiles at 16 px with scroll bars, wheel and arrow-key scrolling	[2.22] (tools/level-editor/222-reference.md#2-window-and-tool-modes)
Toolbar order: New, Open, Save, Save as, Undo board copy, Cut, Copy, Paste, Delete, Turn selection into a warp field, Edit links, Edit signs, selection-as-text (2.16), Close, Play	[2.22] (tools/level-editor/222-reference.md#2-window-and-tool-modes)
One tab per open level; Close asks about unsaved changes	[2.22] (tools/level-editor/222-reference.md#2-window-and-tool-modes)

Rule	Source
Palette modes: predefined object (stamp), tile palette, baddies/NPCs/chests, predefined NPCs (`defnpc*.txt` with an #EDIT grid)	[2.22] (tools/level-editor/222-reference.md#2-window-and-tool-modes)
Blocking, foreground and terrain overlays are computed from the type table, not stored	[2.22] (tools/level-editor/222-reference.md#2-window-and-tool-modes)
Editor control-NPCs (2.13): `editnpc<Name>.txt` scripts editor actions with `mousex`, `mousey`, `mousescreenx`, `mousescreeny` and `mousebuttons` (left 1, middle 2, right 4); only tile changes are supported	[2.22] (tools/level-editor/222-reference.md#2-window-and-tool-modes)
M builds a PNG map image of the level, and of a gmap from 2.16 rev 7	[2.22] (tools/level-editor/222-reference.md#2-window-and-tool-modes)
The mouse state machine (select, drag, drop, double-click by kind) is known only in outline: left drag selects or paints, right button on an object clones it, right button elsewhere flood-fills, double-click opens an object's dialog or makes a tile the background tile	[2.22] (tools/level-editor/222-reference.md#2-window-and-tool-modes)

11.6.2 Level model (2.22)

List	Element	Source
board	64 × 64 16-bit tile ids, index y × 64 + x	[2.22] (tools/level-editor/222-reference.md#3-level-model)
links	destination, x, y, w, h, new-x text, new-y text	[2.22] (tools/level-editor/222-reference.md#3-level-model)
signs	x, y in tiles, text	[2.22] (tools/level-editor/222-reference.md#3-level-model)
chests	x, y, item, sign index	[2.22] (tools/level-editor/222-reference.md#3-level-model)
NPCs	image or none, x, y in fractional tiles, script	[2.22] (tools/level-editor/222-reference.md#3-level-model)
baddies	x, y, type, three verse strings	[2.22] (tools/level-editor/222-reference.md#3-level-model)
heights	optional 9 × 9 floats	[2.22] (tools/level-editor/222-reference.md#3-level-model)
cache	recently loaded levels, for the link walker and gmaps (2.17 on)	[2.22] (tools/level-editor/222-reference.md#3-level-model)

11.6.3 Open and save (2.22)

Rule	Source
The open filter is `Graal levels (*.graal, *.nw)`; several files open in their own tabs	[2.22] (tools/level-editor/222-reference.md#open)
The loader branches on the first eight bytes: `GLEVNW01` text; `GR-V1.00–1.03` binary (13-bit codes for 1.02/1.03, 12-bit for 1.00/1.01); `Z3-V1.03/1.04`; anything else is rejected ("Unsupported file type!", no caller found)	[2.22] (tools/level-editor/222-reference.md#open)
`GR-V1.04` is read only by the weblevels cache path, not by the editor	[2.22] (tools/level-editor/222-reference.md#open)

Rule	Source
The text reader rules (layer field ignored, row skipping, 12-bit tile ids, `LINK` defaults, `BADDY` clamp, `HEIGHTS` only when an overlay exists) are → §3.1.3	[2.22] (tools/level-editor/222-reference.md#the-glevnw01-text-reader-222)
Save (Ctrl+S) is save-all: it commits a floating paste, then saves every modified level with a file name plus the current one, deleting the old file and writing in place; a modified map is saved too; the status bar shows "N Level(s) saved"	[2.22] (tools/level-editor/222-reference.md#save-222)
The extension picks the format: `.nw` writes GLEVNW01, anything else GR-V1.03; the editor never writes Z3-V1, GR-V1.00–1.02 or GR-V1.04	[2.22] (tools/level-editor/222-reference.md#save-222)
The text writer emits `HEIGHTS` only when the overlay is marked modified (2.05 never does)	[2.22] (tools/level-editor/222-reference.md#save-222)
Chests and baddies are written only when two obfuscated session-mode flags are both clear; otherwise only the terminators; the same flags gate the loaders; that they are a New World mode switch is an inference	[2.22][inf] (tools/level-editor/222-reference.md#save-222)
2.22 deletes before writing, so a crash loses the level; the 2.3 change log says the editor then writes a copy and renames it	[2.22] (tools/level-editor/222-reference.md#safety)

11.6.4 Editing tools (2.22)

Rule	Source
Tile ids are 0–4095 on the 2048 × 512 sheet; the id → sheet mapping is → §3.5.1	[2.22] (tools/level-editor/222-reference.md#5-tiles-and-the-sheet)
A tile definition line is `image,prefix,type,x,y`; type 0 is the old order and type 1 the new; type 4 and its x offsets are unmeasured	[2.22][inf] (tools/level-editor/222-reference.md#5-tiles-and-the-sheet)
Palette: click picks one tile, drag picks a block that pastes as a floating selection; double-click sets the background tile, which fills new levels	[2.22] (tools/level-editor/222-reference.md#5-tiles-and-the-sheet)
Undo covers board tiles only: a rectangle snapshot before each write, at most 128 records per tab	[2.22] (tools/level-editor/222-reference.md#6-selection-paste-undo-fill)
Draw mode: with one tile selected, left drag paints (1.41); flood fill with the right button (2.00), Shift for complete objects only, and the background tile when nothing is selected	[2.22] (tools/level-editor/222-reference.md#6-selection-paste-undo-fill)
Turn selection into a warp field: a link over the selection with new x `playerx` and new y `playery`, then the first matching edge rule: starts in column 63 → new x 0; row 63 → new y 0; column 0 and one wide → new x 61; row 0 and one high → new y 61	[2.22] (tools/level-editor/222-reference.md#6-selection-paste-undo-fill)
Stamps come from `objects<set>.txt` in GOBJSET01 (→ §3.4); add-selection inserts after the chosen stamp and Save objects rewrites the set; placing writes tile ids only; sets arrived in 2.04	[2.22] (tools/level-editor/222-reference.md#7-stamps-predefined-objects)
Edit links (L): Linkfile, X, Y, W, H, New X, New Y; Load opens the destination in a tab; the dialog refuses invalid fields with "Wrong values! Type in a filename…"; arrival text is stored as typed	[2.22] (tools/level-editor/222-reference.md#8-links)
Whether the engine evaluates an arrival such as `playery+10` at warp time	disputed → App. E#601

Rule	Source
Edit signs (S): X, Y, New, Delete; a (0, 0) sign is not drawn and is used by index from chests and `say n`; text is in the sign alphabet (→ §3.6); trailing empty lines are removed	[2.22] (tools/level-editor/222-reference.md#9-signs)
Chest dialog: item, sign index, Sign button; items in index order `greenrupee, bluerupee, redrupee, bombs, darts, heart, glove1, bow, bomb, shield, sword, fullheart, superbomb, battleaxe, goldensword, mirrorshield, glove2, lizardshield, lizardsword, goldrupee, fireball, fireblast, nukeshot, joltbomb, spinattack`; a new chest is item 1, sign −1	[2.22] (tools/level-editor/222-reference.md#10-chests)
Baddy dialog: ten types (Gray, Blue, Red, Shooting and Swamp Soldier, Frog, Spider, Golden Warrior, Lizardon, Dragon) and three verses; types 4–6 disable and save empty verses; a new baddy says "Stop!" and "Ouch!"	[2.22] (tools/level-editor/222-reference.md#11-baddies)
NPC dialog: image, script, Test, Style, Find; `//#EDIT name value` and `//#COMMENTS text` markers; predefined NPCs replace each named variable on drop; NPC positions survive a binary save only as whole and half tiles	[2.22][inf] (tools/level-editor/222-reference.md#12-npcs)
Multi-level editing (2.12): when an NPC calls `setmap` the neighbours show together; scroll by dragging the player or with arrows; a big cursor on maps (2.13)	[2.22] (tools/level-editor/222-reference.md#14-tabs-maps-and-multi-level-editing)
Other dialogs: load by name, log window, `TShowImg` (role unmeasured), variable editor, tile-array window	[2.22] (tools/level-editor/222-reference.md#15-other-dialogs)

Keys, 2.22 [2.22] (tools/level-editor/222-reference.md#13-keys):

Key	Action	Source
Ctrl+N, Ctrl+O, Ctrl+S	New, Open, Save (save-all)	[2.22] (tools/level-editor/222-reference.md#13-keys)
Ctrl+Z, Alt+Backspace	Undo board copy	[2.22] (tools/level-editor/222-reference.md#13-keys)
Ctrl+X / Shift+Del, Ctrl+C / Ctrl+Ins, Ctrl+V / Shift+Ins, Del	Cut, Copy, Paste, Delete	[2.22] (tools/level-editor/222-reference.md#13-keys)
L, S, P, M	Edit links, Edit signs, Play, map image; ignored while a side-panel combo has focus	[2.22] (tools/level-editor/222-reference.md#13-keys)
Ctrl+Q	batch re-save of every level named in the loaded map, by extension (2.17.1, 2.22)	[2.22] (tools/level-editor/222-reference.md#13-keys)
F10, Esc	border toggle, warning before leaving	[2.22] (tools/level-editor/222-reference.md#13-keys)

11.6.5 Builds as deltas from 2.22

Builds	Delta	Source
2.17.1	none found; same class list	[2.17] (tools/level-editor/version-deltas.md#5-deltas-as-a-checklist-per-build-relative-to-222)
2.05, rev 1, rev 2	no heights, cache, tiles-array or log windows, no predefined NPCs; separate person-image and script dialogs; a sign-codes window; open filter `*.graal`; the map parser present from rev 1 but dormant; GLEVNW01 first appears; the chest clamp is gone	[2.05] (tools/level-editor/version-deltas.md#3-formats-per-build)
2.0 rev 3 to 2.04	no GLEVNW01 reader or writer; `.zelda` dropped from the open filter at 2.0 rev 3 (the reader still accepts it); object sets and tile definitions from 2.04; chests clamped to 0–62 and NPCs to 0–63 on load	[2.04][inf] (tools/level-editor/version-deltas.md#3-formats-per-build)

Builds	Delta	Source
2nd Graal to 2.0 rev 2	read GR-V1.00–1.03 and Z3-V1.03/1.04, save GR-V1.03; filter `*.graal`, `*.zelda`; same clamps	[1.39][2.00] (tools/level-editor/version-deltas.md#3-formats-per-build)
1.28b2 to 1.33	baddy class still named opponent; character NPC dialog present	[bin] (tools/level-editor/version-deltas.md#2-class-timeline-of-the-built-in-editor)
before 1.28b2	no character dialog or level class; 2nd Graal adds chest and person dialogs, 3rd Graal the flags dialog; 1st Graal reads only Z3-V1 and has no chest or NPC dialog	[bin] (tools/level-editor/version-deltas.md#2-class-timeline-of-the-built-in-editor)
New World betas	no chest and no baddy editor or class; heights and map in beta 1; text format in the beta's own `levelformat.txt`	[bin] (tools/level-editor/version-deltas.md#2-class-timeline-of-the-built-in-editor)
2.1 test builds	combined NPC dialog, predefined NPCs, no sign-codes window	[bin][doc] (tools/level-editor/version-deltas.md#2-class-timeline-of-the-built-in-editor)
2.31	no editor dialogs; the level model and GR-V1.04 reader remain for the game; no GLEVNW01	[2.31] (tools/level-editor/version-deltas.md#3-formats-per-build)

Rule	Source
The load clamps are compares against 62 for chests and 63 for NPCs in the 1.39, 2.00 and 2.04 loaders, absent from 2.05 on; a chest at x = 63 loads at 62 up to 2.04	[bin] (tools/level-editor/version-deltas.md#3-formats-per-build)
The baddy section ends at a record shorter than three bytes or whose first byte is 0x80 or more, with no x = 0 test, in 1.39, 2.00, 2.05, 2.17.1 and 2.22	[bin] (tools/level-editor/formats-crossref.md#batch-f-decomp-findings-2026-10-25-f-pr-39)
The editor writes double runs (ABAB…, up to 64 pairs) in GR-V1 boards, and a newline after each chest record	[2.22] (tools/level-editor/formats-crossref.md#formatslevel-binarymd-and-level-binary-clientmd)

Change-log dated editor features, from the shipped `newfeatures` files [doc] (tools/level-editor/version-deltas.md#4-change-log-dated-features):

Version	Features	Source
1.41	drag painting with one tile, M map PNG, status-bar position and tile, double-click background tile, no selection outside the level, drop a text file to make an NPC, trailing sign lines trimmed, new levels use the background tile	[doc] (tools/level-editor/version-deltas.md#4-change-log-dated-features)
2.00	flood fill with the right button	[doc] (tools/level-editor/version-deltas.md#4-change-log-dated-features)
2.04	object sets; `addtiledef`, `addtiledef2`, `removetiledefs`	[doc] (tools/level-editor/version-deltas.md#4-change-log-dated-features)
2.1	predefined NPCs, merged sign-code and add-NPC windows, multi-tile palette selection, F4 on a tutorial opens an empty level	[doc] (tools/level-editor/version-deltas.md#4-change-log-dated-features)

Version	Features	Source
2.12	multi-level editing on `setmap`, scrolling by player drag or arrows	[doc] (tools/level-editor/version-deltas.md#4-change-log-dated-features)
2.13	scrollbars on single levels, big cursor on maps, Play saves all, editor control-NPCs	[doc] (tools/level-editor/version-deltas.md#4-change-log-dated-features)
2.13 rev 3	dropped objects can be released and moved; caching and Play position fixes	[doc] (tools/level-editor/version-deltas.md#4-change-log-dated-features)
2.15	wheel scrolling, arrow scrolling on single levels	[doc] (tools/level-editor/version-deltas.md#4-change-log-dated-features)
2.16	selection-as-text button, `showimg` and `putnpc` in editor NPCs, map-level fixes; M on gmaps from rev 7	[doc] (tools/level-editor/version-deltas.md#4-change-log-dated-features)
2.3	editor and client become separate exes; write-copy-and-rename saving; filename cache	[doc] (tools/level-editor/version-deltas.md#4-change-log-dated-features)

11.7 Other tools

Tool	What it does	Files	Source
TextConvert	on start, walks the current folder and subfolders and converts every GR-V1 level to GLEVNW01, links included; no socket	GR-V1.00–1.03 in, GLEVNW01 out; errors `Wrong link data!`, `Wrong chest data!`, `Decompression error!`	[bin][doc] (tools/classic-tools/text-convert.md#textconvert)
Graal Character Designer 1.2.7 (Nov 1999) and GChar 1.0	offline character-look editors; need `headsconfig.txt`, `levels\heads\`, `levels\gifs\`, `levels\webgifs\`; import `*.chi` files and rename them `*.chr`; no socket	`CHR:1.00<name>=<head gif>,<male|female>,<five colour names>`, one line; GChar keeps `GChar.ini` (`drawmethod`, `headmode`, `graaldir`)	[bin][doc][file] (tools/classic-tools/character-designer.md#graal-character-designer-and-gchar)
`gserverstat`, `gserverstat2`, `statmaker` (1999 gserver tree)	HTML leaderboards on standard output, 14 modes (kills default), top 50, load-only accounts and the account named `super` skipped; `statmaker` loops with 10 s and 600 s sleeps; no socket	reads `weapons.txt`, `accountslist.txt`, `accounts/accN.txt` (→ §5.1)	[bin][src][file] (tools/classic-tools/gserverstat.md#gserverstat-and-gserverstat2)
Classic listserver `serverlist3`	list server ELF with MySQL; protocol → §6	config keys `clientsport`, `serversport`, `playerlimit`, `databaseserver`, `showmore`; status files beside the process	[bin] (tools/classic-tools/serverlist3.md#start-and-config)
Win32 `ListServer.exe`	Delphi directory helper reading an ini server list (→ §7)	ini, registry	[bin] (recovered-drive-tools/ghidra-analysis/ListServer/SUMMARY.md)
Script Editor 2.53	GS1 NPC and weapon script editor with templates and a find-NPC helper	GS1 scripts	[bin] (recovered-drive-tools/ghidra-analysis/scripteditor/SUMMARY.md)

Tool	What it does	Files	Source
BytecodeViewer	console dump of a GS2 bytecode capture: skips the first byte, then the information section, function names, strings and bytecode, with opcode names from `BytecodeViewer_defs.txt` (GS2 bytecode → §8)	GS2 bytecode	[src] (recovered-drive-tools/ghidra-analysis/BytecodeViewer/SUMMARY.md)
Assimilator ("GIF2PNG")	offline batch checker and converter for scripts, levels, text and animations	asset files	[bin] (recovered-drive-tools/ghidra-analysis/Assimilator/SUMMARY.md)
PNGView	offline image viewer	PNG	[bin] (recovered-drive-tools/ghidra-analysis/PNGView/SUMMARY.md)
TilesEditor (39ster, Reborn line)	offline 64-bit Qt editor for levels, tiles, links, chests and NPCs (anonymous and external classes), layers, grid snap, map export	levels	[bin] (recovered-drive-tools/ghidra-analysis/TilesEditor/SUMMARY.md)
DClient setup	a VB6-era installer bootstrap; its cabinet is truncated, so the client cannot be rebuilt	—	[bin] (recovered-drive-tools/ghidra-analysis/DClient_setup/SUMMARY.md)

Recovered server binaries, described in Part V [bin] (recovered-drive-tools/ghidra-analysis/gserver13/SUMMARY.md):

Binary	Notes	Source
`gserver13` (Linux, Dec 1999)	unstripped; account parse, level enter and update, guild verify, IP bans (→ §5.1)	[bin] (recovered-drive-tools/ghidra-analysis/gserver13/SUMMARY.md)
`gserver` (Linux) and `gserver.exe` (Win32), about June 2002	the same classic server with source beside it: NW level load, RC player props, MySQL accounts	[bin] (recovered-drive-tools/ghidra-analysis/gserver-linux/SUMMARY.md); [bin] (recovered-drive-tools/ghidra-analysis/gserver-win/SUMMARY.md); [bin] (recovered-drive-tools/ghidra-analysis/gserver_pe_GServersource/SUMMARY.md)
Matriark `gserver` (pack labelled 1.39)	stripped static build with database accounts, guilds, RC and NC sessions and leaderboard queries; the pack's bug note says head, shield and sword path checks disagree with the paths used when sending level files	[bin][doc] (recovered-drive-tools/ghidra-analysis/Matriark_gserver/SUMMARY.md)

Appendix A. Message index

Every client-to-server (PLI) and server-to-client (PLO) id with its wire byte and the section that documents it. Names follow → §0.7; the wire byte is the id + `0x20` (→ §2.1.2). The *2.05* column says whether 2.05 rev 2 sends (PLI) or handles (PLO) the id. UDP peer ids are a separate id space (→ §2.9).

A.1 Client to server

Id	Wire	Name	2.05	Section	Source
0	`0x20`	`PLI_LEVELWARP`	sent	→ §2.4.3	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
1	`0x21`	`PLI_BOARDMODIFY`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
2	`0x22`	`PLI_PLAYERPROPS`	sent	→ §2.4.4	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
3	`0x23`	`PLI_NPCPROPS`	sent	→ §2.4.4	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
4	`0x24`	`PLI_BOMBADD`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
5	`0x25`	`PLI_BOMBDEL`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
6	`0x26`	`PLI_TOALL`	sent	→ §2.4.8	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
7	`0x27`	`PLI_HORSEADD`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
8	`0x28`	`PLI_HORSEDEL`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
9	`0x29`	`PLI_ARROWADD`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
10	`0x2a`	`PLI_FIRESPY`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
11	`0x2b`	`PLI_THROWCARRIED`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
12	`0x2c`	`PLI_ITEMADD`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
13	`0x2d`	`PLI_ITEMDEL`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
14	`0x2e`	`PLI_CLAIMPKER`	sent	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
15	`0x2f`	`PLI_BADDYPROPS`	sent	→ §2.4.4	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
16	`0x30`	`PLI_BADDYHURT`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
17	`0x31`	`PLI_BADDYADD`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
18	`0x32`	`PLI_FLAGSET`	sent	→ §2.4.6	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
19	`0x33`	`PLI_FLAGDEL`	sent	→ §2.4.6	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
20	`0x34`	`PLI_OPENCHEST`	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
21	`0x35`	`PLI_PUTNPC`	sent	→ §2.4.6	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)

Id	Wire	Name	2.05	Section	Source
22	0x36	PLI_NPCDEL	sent	→ §2.4.6	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
23	0x37	PLI_WANTFILE	sent	→ §2.4.7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
24	0x38	PLI_SHOWIMGPLAYER	sent	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
25	0x39	PLI_UNKNOWN25	no	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
26	0x3a	PLI_HURTPLAYER	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
27	0x3b	PLI_EXPLOSION	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
28	0x3c	PLI_PRIVATEMESSAGE	sent	→ §2.4.8	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
29	0x3d	PLI_NPCWEAPONDEL	sent	→ §2.4.6	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
30	0x3e	PLI_LEVELWARPMOD	sent	→ §2.4.3	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
31	0x3f	PLI_PACKETCOUNT	no	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
32	0x40	PLI_ITEMTAKE	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
33	0x41	PLI_WEAPONADD	sent	→ §2.4.6	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
34	0x42	PLI_UPDATEFILE	sent	→ §2.4.7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
35	0x43	PLI_ADJACENTLEVEL	sent	→ §2.4.3	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
36	0x44	PLI_HITOBJECTS	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
37	0x45	PLI_LANGUAGE	sent	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
38	0x46	PLI_TRIGGERACTION	sent	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
39	0x47	PLI_TAMPERCHECK	no	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
40	0x48	PLI_SHOOT	no	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
41	0x49	PLI_SERVERWARP	no	→ §2.4.3	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
43	0x4b	PLI_MUTEPLAYER	no	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
44	0x4c	PLI_PROCESSLIST	no	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
45	0x4d	PLI_PROCESSRUNNING	no	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
46	0x4e	PLI_ENTERLEVEL	no	→ §2.4.3	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
47	0x4f	PLI_VERIFYWANTSEND	no	→ §2.4.7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
48	0x50	PLI_SHOOT2	no	→ §2.4.5	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
50	0x52	PLI_RAWDATA	no	→ §2.4.7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
51	0x53	PLI_RC_SERVEROPTIONSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
52	0x54	PLI_RC_SERVEROPTIONSSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)

Id	Wire	Name	2.05	Section	Source
53	0x55	PLI_RC_FOLDERCONFIGGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
54	0x56	PLI_RC_FOLDERCONFIGSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
55	0x57	PLI_RC_RESPAWNSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
56	0x58	PLI_RC_HORSELIFESET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
57	0x59	PLI_RC_APINCREMENTSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
58	0x5a	PLI_RC_BADDYRESPAWNSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
59	0x5b	PLI_RC_PLAYERPROPSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
60	0x5c	PLI_RC_PLAYERPROPSSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
61	0x5d	PLI_RC_DISCONNECTPLAYER	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
62	0x5e	PLI_RC_UPDATELEVELS	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
63	0x5f	PLI_RC_ADMINMESSAGE	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
64	0x60	PLI_RC_PRIVADMINMESSAGE	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
65	0x61	PLI_RC_LISTRCS	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
66	0x62	PLI_RC_DISCONNECTRC	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
67	0x63	PLI_RC_APPLYREASON	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
68	0x64	PLI_RC_SERVERFLAGSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
69	0x65	PLI_RC_SERVERFLAGSSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
70	0x66	PLI_RC_ACCOUNTADD	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
71	0x67	PLI_RC_ACCOUNTDEL	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
72	0x68	PLI_RC_ACCOUNTLISTGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
73	0x69	PLI_RC_PLAYERPROPSGET2	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
74	0x6a	PLI_RC_PLAYERPROPSGET3	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
75	0x6b	PLI_RC_PLAYERPROPSRESET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
76	0x6c	PLI_RC_PLAYERPROPSSET2	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
77	0x6d	PLI_RC_ACCOUNTGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
78	0x6e	PLI_RC_ACCOUNTSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
79	0x6f	PLI_RC_CHAT	no	→ §2.4.10	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
80	0x70	PLI_PROFILEGET	sent	→ §2.4.8	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
81	0x71	PLI_PROFILESET	sent	→ §2.4.8	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)

Id	Wire	Name	2.05	Section	Source
82	0x72	PLI_RC_WARPPLAYER	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
83	0x73	PLI_RC_PLAYERRIGHTSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
84	0x74	PLI_RC_PLAYERRIGHTSSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
85	0x75	PLI_RC_PLAYERCOMMENTSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
86	0x76	PLI_RC_PLAYERCOMMENTSSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
87	0x77	PLI_RC_PLAYERBANGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
88	0x78	PLI_RC_PLAYERBANSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
89	0x79	PLI_RC_FILEBROWSER_START	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
90	0x7a	PLI_RC_FILEBROWSER_CD	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
91	0x7b	PLI_RC_FILEBROWSER_END	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
92	0x7c	PLI_RC_FILEBROWSER_DOWN	no	→ §2.4.10	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
93	0x7d	PLI_RC_FILEBROWSER_UP	no	→ §2.4.10	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
94	0x7e	PLI_NPCSERVERQUERY	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
96	0x80	PLI_RC_FILEBROWSER_MOVE	no	→ §2.4.10	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
97	0x81	PLI_RC_FILEBROWSER_DELETE	no	→ §2.4.10	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
98	0x82	PLI_RC_FILEBROWSER_RENAME	no	→ §2.4.10	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
103	0x87	PLI_NC_NPCGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
104	0x88	PLI_NC_NPCDELETE	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
105	0x89	PLI_NC_NPCRESET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
106	0x8a	PLI_NC_NPCSCRIPTGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
107	0x8b	PLI_NC_NPCWARP	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
108	0x8c	PLI_NC_NPCFLAGSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
109	0x8d	PLI_NC_NPCSCRIPTSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
110	0x8e	PLI_NC_NPCFLAGSSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
111	0x8f	PLI_NC_NPCADD	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
112	0x90	PLI_NC_CLASSEDIT	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
113	0x91	PLI_NC_CLASSADD	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
114	0x92	PLI_NC_LOCALNPCSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
115	0x93	PLI_NC_WEAPONLISTGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)

Id	Wire	Name	2.05	Section	Source
116	0x94	PLI_NC_WEAPONGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
117	0x95	PLI_NC_WEAPONADD	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
118	0x96	PLI_NC_WEAPONDELETE	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
119	0x97	PLI_NC_CLASSDELETE	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
130	0xa2	PLI_REQUESTUPDATEBOARD	no	→ §2.4.7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
150	0xb6	PLI_NC_LEVELLISTGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
151	0xb7	PLI_NC_LEVELLISTSET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
152	0xb8	PLI_REQUESTTEXT	no	→ §2.4.7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
154	0xba	PLI_SENDTEXT	no	→ §2.4.7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
155	0xbb	PLI_RC_LARGEFILESTART	no	→ §2.4.10	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
156	0xbc	PLI_RC_LARGEFILEEND	no	→ §2.4.10	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
157	0xbd	PLI_UPDATEGANI	no	→ §2.4.7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
158	0xbe	PLI_UPDATESCRIPT	no	→ §2.4.7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
159	0xbf	PLI_UPDATEPACKAGEREQUESTFILE	no	→ §2.4.7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
160	0xc0	PLI_RC_FOLDERDELETE	no	→ §2.4.10	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
161	0xc1	PLI_UPDATECLASS	no	→ §2.4.7	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
162	0xc2	PLI_RC_UNKNOWN162	no	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
252	n/a (above 223)	PLI_SET_ENC_KEY	no	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)
253	n/a (above 223)	PLI_BUNDLE	no	→ §2.4.9	[srv][2.05] (protocol/packet-ids.md#client--server-pli_)

A.2 Server to client

Id	Wire	Name	2.05	Section	Source
0	0x20	PLO_LEVELBOARD	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
1	0x21	PLO_LEVELLINK	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
2	0x22	PLO_BADDYPROPS	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
3	0x23	PLO_NPCPROPS	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
4	0x24	PLO_LEVELCHEST	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
5	0x25	PLO_LEVELSIGN	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
6	0x26	PLO_LEVELNAME	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)

Id	Wire	Name	2.05	Section	Source
7	0x27	PLO_BOARDMODIFY	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
8	0x28	PLO_OTHERPLPROPS	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
9	0x29	PLO_PLAYERPROPS	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
10	0x2a	PLO_ISLEADER	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
11	0x2b	PLO_BOMBADD	handled	→ §2.5.5	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
12	0x2c	PLO_BOMBDEL	handled	→ §2.5.5	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
13	0x2d	PLO_TOALL	handled	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
14	0x2e	PLO_PLAYERWARP	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
15	0x2f	PLO_WARPFAILED	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
16	0x30	PLO_DISCMESSAGE	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
17	0x31	PLO_HORSEADD	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
18	0x32	PLO_HORSEDEL	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
19	0x33	PLO_ARROWADD	handled	→ §2.5.5	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
20	0x34	PLO_FIRESPY	handled	→ §2.5.5	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
21	0x35	PLO_THROWCARRIED	handled	→ §2.5.5	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
22	0x36	PLO_ITEMADD	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
23	0x37	PLO_ITEMDEL	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
24	0x38	PLO_NPCMOVED	handled	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
25	0x39	PLO_SIGNATURE	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
26	0x3a	PLO_NPCACTION	no	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
27	0x3b	PLO_BADDYHURT	handled	→ §2.5.5	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
28	0x3c	PLO_FLAGSET	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
29	0x3d	PLO_NPCDEL	handled	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
30	0x3e	PLO_FILESENDFAILED	handled	→ §2.5.7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
31	0x3f	PLO_FLAGDEL	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
32	0x40	PLO_SHOWIMGPLAYER	handled	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
33	0x41	PLO_NPCWEAPONADD	handled	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
34	0x42	PLO_NPCWEAPONDEL	handled	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
35	0x43	PLO_RC_ADMINMESSAGE	handled	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)

Id	Wire	Name	2.05	Section	Source
36	0x44	PLO_EXPLOSION	handled	→ §2.5.5	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
37	0x45	PLO_PRIVATEMESSAGE	handled	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
38	0x46	PLO_PUSHAWAY	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
39	0x47	PLO_LEVELMODTIME	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
40	0x48	PLO_HURTPLAYER	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
41	0x49	PLO_STARTMESSAGE	handled	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
42	0x4a	PLO_NEWWORLDTIME	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
43	0x4b	PLO_DEFAULTWEAPON	handled	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
44	0x4c	PLO_HASNPCSERVER	handled	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
45	0x4d	PLO_FILEUPTODATE	handled	→ §2.5.7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
46	0x4e	PLO_HITOBJECTS	handled	→ §2.5.5	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
47	0x4f	PLO_STAFFGUILDS	handled	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
48	0x50	PLO_TRIGGERACTION	handled	→ §2.5.5	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
49	0x51	PLO_PLAYERWARP2	no	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
50	0x52	PLO_RC_ACCOUNTADD	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
51	0x53	PLO_RC_ACCOUNTSTATUS	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
52	0x54	PLO_RC_ACCOUNTNAME	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
53	0x55	PLO_RC_ACCOUNTDEL	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
54	0x56	PLO_RC_ACCOUNTPROPS	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
55	0x57	PLO_ADDPLAYER	no	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
56	0x58	PLO_DELPLAYER	no	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
57	0x59	PLO_RC_ACCOUNTPROPSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
58	0x5a	PLO_RC_ACCOUNTCHANGE	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
59	0x5b	PLO_RC_PLAYERPROPSCHANGE	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
60	0x5c	PLO_UNKNOWN60	no	→ §2.5.10	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
61	0x5d	PLO_RC_SERVERFLAGSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
62	0x5e	PLO_RC_PLAYERRIGHTSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
63	0x5f	PLO_RC_PLAYERCOMMENTSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
64	0x60	PLO_RC_PLAYERBANGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)

Id	Wire	Name	2.05	Section	Source
65	0x61	PLO_RC_FILEBROWSER_DIRLIST	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
66	0x62	PLO_RC_FILEBROWSER_DIR	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
67	0x63	PLO_RC_FILEBROWSER_MESSAGE	no	→ §2.5.9	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
68	0x64	PLO_LARGEFILESTART	no	→ §2.5.7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
69	0x65	PLO_LARGEFILEEND	no	→ §2.5.7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
70	0x66	PLO_RC_ACCOUNTLISTGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
71	0x67	PLO_RC_PLAYERPROPS	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
72	0x68	PLO_RC_PLAYERPROPSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
73	0x69	PLO_RC_ACCOUNTGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
74	0x6a	PLO_RC_CHAT	no	→ §2.5.9	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
75	0x6b	PLO_PROFILE	handled	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
76	0x6c	PLO_RC_SERVEROPTIONSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
77	0x6d	PLO_RC_FOLDERCONFIGGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
78	0x6e	PLO_NC_CONTROL	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
79	0x6f	PLO_NPCSERVERADDR	no	→ §2.5.9	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
80	0x70	PLO_NC_LEVELLIST	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
81	0x71	PLO_UNKNOWN81	no	→ §2.5.10	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
82	0x72	PLO_SERVERTEXT	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
83	0x73	PLO_UNKNOWN83	no	→ §2.5.10	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
84	0x74	PLO_LARGEFILESIZE	no	→ §2.5.7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
100	0x84	PLO_RAWDATA	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
103	0x87	PLO_RC_MAXUPLOADFILESIZE	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
104	0x88	PLO_UNKNOWN104	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
105	0x89	PLO_UPDATEPACKAGESIZE	no	→ §2.5.7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
106	0x8a	PLO_UPDATEPACKAGEDONE	no	→ §2.5.7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
107	0x8b	PLO_BOARDLAYER	no	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
108	0x8c	PLO_UNKNOWN108	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
109	0x8d	PLO_UNKNOWN109	no	→ §2.5.7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
110	0x8e	PLO_UNKNOWN110	no	→ §2.5.10	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)

Id	Wire	Name	2.05	Section	Source
111	0x8f	PLO_SETNETCOOKIE	no	→ §2.5.7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
124	0x9c	PLO_UNKNOWN124	no	→ §2.5.10	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
131	0xa3	PLO_NPCBYTECODE	no	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
132	0xa4	PLO_UNKNOWN132	no	→ §2.5.10	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
133	0xa5	PLO_UNKNOWN133	no	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
134	0xa6	PLO_GANISCRIPT	no	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
140	0xac	PLO_NPCWEAPONSCRIPT	no	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
150	0xb6	PLO_NPCDEL2	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
151	0xb7	PLO_HIDENPCS	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
153	0xb9	PLO_SAY2	handled	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
154	0xba	PLO_FREEZEPLAYER2	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
155	0xbb	PLO_UNFREEZEPLAYER	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
156	0xbc	PLO_SETACTIVELEVEL	handled	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
157	0xbd	PLO_NC_NPCATTRIBUTES	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
158	0xbe	PLO_NC_NPCADD	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
159	0xbf	PLO_NC_NPCDELETE	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
160	0xc0	PLO_NC_NPCSCRIPT	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
161	0xc1	PLO_NC_NPCFLAGS	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
162	0xc2	PLO_NC_CLASSGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
163	0xc3	PLO_NC_CLASSADD	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
164	0xc4	PLO_NC_LEVELDUMP	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
165	0xc5	PLO_MOVE	handled	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
166	0xc6	PLO_SHOWIMGNPC	handled	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
167	0xc7	PLO_NC_WEAPONLISTGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
168	0xc8	PLO_UNKNOWN168	handled	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
169	0xc9	PLO_UPGRADEURL	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
170	0xca	PLO_GHOSTMODE	no	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
171	0xcb	PLO_BIGMAP	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
172	0xcc	PLO_MINIMAP	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)

Id	Wire	Name	2.05	Section	Source
173	0xcd	PLO_GHOSTTEXT	no	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
174	0xce	PLO_GHOSTICON	no	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
175	0xcf	PLO_SHOOT	no	→ §2.5.5	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
176	0xd0	PLO_KINGDOMSMODE	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
177	0xd1	PLO_NEWWORLDMODE	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
178	0xd2	PLO_SERVERWARP	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
179	0xd3	PLO_RPGWINDOW	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
180	0xd4	PLO_STATUSLIST	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
181	0xd5	PLO_UNKNOWN181	no	→ §2.5.4	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
182	0xd6	PLO_LISTPROCESSES	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
183	0xd7	PLO_HASPROCESSRUNNING	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
184	0xd8	PLO_TAKESCREENSHOT	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
185	0xd9	PLO_BOARDHEIGHTS	no	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
186	0xda	PLO_BOARDMODIFY2	no	→ §2.5.3	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
187	0xdb	PLO_UPDATEPACKAGEISUPDATED	no	→ §2.5.7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
188	0xdc	PLO_NC_CLASSDELETE	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
189	0xdd	PLO_MOVE2	no	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
190	0xde	PLO_SERVERLISTCONNECTED	no	→ §2.5.8	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
191	0xdf	PLO_SHOOT2	no	→ §2.5.5	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
192	0xe0	PLO_NC_WEAPONGET	no	→ §7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
193	0xe1	PLO_UNKNOWN193	no	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
194	0xe2	PLO_CLEARWEAPONS	no	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
195	0xe3	PLO_LOADGANI	no	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
197	0xe5	PLO_LOADSCRIPT	no	→ §2.5.6	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
198	0xe6	PLO_SERVEROPTIONS	no	→ §2.5.7	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
252	n/a (above 223)	PLO_SET_ENC_KEY	no	→ §2.5.10	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)
253	n/a (above 223)	PLO_BUNDLE	no	→ §2.5.10	[srv][2.05] (protocol/packet-ids.md#server--client-plo_)

A.3 Former names

An older gs2lib copy (the GServer-v2 submodule of 2025-01) uses these names for the same ids:

Former name	Id	Current name	Source
PLI_SHOWIMG	24	PLI_SHOWIMGPLAYER	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)
PLI_MAPINFO	39	PLI_TAMPERCHECK	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)
PLI_UNKNOWN46	46	PLI_ENTERLEVEL	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)
PLO_SHOWIMG	32	PLO_SHOWIMGPLAYER	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)
PLO_FULLSTOP	176	PLO_KINGDOMSMODE	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)
PLO_FULLSTOP2	177	PLO_NEWWORLDMODE	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)
PLO_UNKNOWN183	183	PLO_HASPROCESSRUNNING	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)
PLO_UNKNOWN184	184	PLO_TAKESCREENSHOT	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)
PLO_UNKNOWN185	185	PLO_BOARDHEIGHTS	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)
PLO_UNKNOWN186	186	PLO_BOARDMODIFY2	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)
PLO_UNKNOWN190	190	PLO_SERVERLISTCONNECTED	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)

Rule	Source
The newer gs2lib copy adds PLI_PROCESSRUNNING (45), PLI_SET_ENC_KEY (252), PLI_BUNDLE (253), PLO_SETNETCOOKIE (111) and PLO_UPGRADEURL (169)	[srv] (protocol/packet-ids.md#renames-between-gs2lib-versions)
An older note names 100, 101 and 102 FILELEN, TILES and FILEBLOB; the roles are those of PLO_RAWDATA, PLO_BOARDPACKET and PLO_FILE	[doc] (protocol/packet-ids.md#conflicts-with-older-notes)
Provisional ids from early notes (login 0x01, level request 0x10, player update 0x20) are retired: the login is type 0, PLI_LEVELWARP is 0 and PLI_PLAYERPROPS is 2	[doc] (protocol/packet-ids.md#conflicts-with-older-notes)

Appendix B. Prop index

Every player (PLPROP), NPC (NPCPROP) and baddy (BDPROP) prop id with its name and the section that gives its encoding. Prop streams are defined in → §2.6.1.

B.1 Player, NPC and baddy props

Kind	Id	Name	Section	Source
PLPROP	0	NICKNAME	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	1	MAXPOWER	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	2	CURPOWER	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	3	RUPEESCOUNT	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	4	ARROWSCOUNT	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	5	BOMBSCOUNT	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	6	GLOVEPOWER	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	7	BOMBPOWER	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	8	SWORDPOWER	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	9	SHIELDPOWER	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	10	GANI / BOWGIF	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	11	HEADGIF	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	12	CURCHAT	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	13	COLORS	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	14	ID	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	15	X	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	16	Y	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	17	SPRITE	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	18	STATUS	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	19	CARRYSPRITE	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	20	CURLEVEL	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	21	HORSEGIF	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	22	HORSEBUSHES	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	23	EFFECTCOLORS	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	24	CARRYNPC	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	25	APCOUNTER	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	26	MAGICPOINTS	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	27	KILLSCOUNT	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	28	DEATHSCOUNT	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	29	ONLINESECS	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	30	IPADDR	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	31	UDPPORT	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	32	ALIGNMENT	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	33	ADDITFLAGS	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	34	ACCOUNTNAME	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	35	BODYIMG	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	36	RATING	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	37	GATTRIB1	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	38	GATTRIB2	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	39	GATTRIB3	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	40	GATTRIB4	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	41	GATTRIB5	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)

Kind	Id	Name	Section	Source
PLPROP	42	ATTACHNPC	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	43	GMAPLEVELX	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	44	GMAPLEVELY	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	45	Z	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	46	GATTRIB6	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	47	GATTRIB7	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	48	GATTRIB8	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	49	GATTRIB9	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	50	JOINLEAVELVL	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	51	PCONNECTED	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	52	PLANGUAGE	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	53	PSTATUSMSG	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	54	GATTRIB10	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	55	GATTRIB11	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	56	GATTRIB12	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	57	GATTRIB13	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	58	GATTRIB14	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	59	GATTRIB15	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	60	GATTRIB16	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	61	GATTRIB17	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	62	GATTRIB18	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	63	GATTRIB19	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	64	GATTRIB20	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	65	GATTRIB21	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	66	GATTRIB22	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	67	GATTRIB23	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	68	GATTRIB24	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	69	GATTRIB25	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	70	GATTRIB26	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	71	GATTRIB27	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	72	GATTRIB28	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	73	GATTRIB29	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	74	GATTRIB30	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	75	OSTYPE	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	76	TEXTCODEPAGE	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	77	UNKNOWN77	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	78	X2	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	79	Y2	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	80	Z2	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	81	UNKNOWN81	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
PLPROP	82	COMMUNITYNAME	→ §2.6.2	[srv] (protocol/player-props.md#plprop-player)
NPCPROP	0	IMAGE	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	1	SCRIPT	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	2	X	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	3	Y	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	4	POWER	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	5	RUPEES	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	6	ARROWS	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	7	BOMBS	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	8	GLOVEPOWER	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	9	BOMBPOWER	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)

Kind	Id	Name	Section	Source
NPCPROP	10	SWORDIMAGE	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	11	SHIELDIMAGE	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	12	GANI / BOWGIF	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	13	VISFLAGS	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	14	BLOCKFLAGS	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	15	MESSAGE	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	16	HURTDXDY	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	17	ID	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	18	SPRITE	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	19	COLORS	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	20	NICKNAME	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	21	HORSEIMAGE	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	22	HEADIMAGE	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	23	SAVE0	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	24	SAVE1	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	25	SAVE2	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	26	SAVE3	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	27	SAVE4	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	28	SAVE5	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	29	SAVE6	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	30	SAVE7	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	31	SAVE8	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	32	SAVE9	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	33	ALIGNMENT	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	34	IMAGEPART	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	35	BODYIMAGE	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	36	GATTRIB1	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	37	GATTRIB2	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	38	GATTRIB3	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	39	GATTRIB4	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	40	GATTRIB5	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	41	GMAPLEVELX	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	42	GMAPLEVELY	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	43	Z	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	44	GATTRIB6	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	45	GATTRIB7	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	46	GATTRIB8	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	47	GATTRIB9	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	48	UNKNOWN48	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	49	SCRIPTER	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	50	NAME	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	51	TYPE	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	52	CURLEVEL	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	53	GATTRIB10	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	54	GATTRIB11	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	55	GATTRIB12	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	56	GATTRIB13	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	57	GATTRIB14	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	58	GATTRIB15	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	59	GATTRIB16	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	60	GATTRIB17	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)

Kind	Id	Name	Section	Source
NPCPROP	61	GATTRIB18	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	62	GATTRIB19	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	63	GATTRIB20	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	64	GATTRIB21	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	65	GATTRIB22	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	66	GATTRIB23	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	67	GATTRIB24	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	68	GATTRIB25	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	69	GATTRIB26	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	70	GATTRIB27	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	71	GATTRIB28	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	72	GATTRIB29	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	73	GATTRIB30	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	74	CLASS	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	75	X2	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	76	Y2	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
NPCPROP	77	Z2	→ §2.6.4	[srv] (protocol/player-props.md#npcprop)
BDPROP	0	ID	→ §2.6.5	[srv] (protocol/player-props.md#bdprop-baddy)
BDPROP	1	X	→ §2.6.5	[srv] (protocol/player-props.md#bdprop-baddy)
BDPROP	2	Y	→ §2.6.5	[srv] (protocol/player-props.md#bdprop-baddy)
BDPROP	3	TYPE	→ §2.6.5	[srv] (protocol/player-props.md#bdprop-baddy)
BDPROP	4	POWERIMAGE	→ §2.6.5	[srv] (protocol/player-props.md#bdprop-baddy)
BDPROP	5	MODE	→ §2.6.5	[srv] (protocol/player-props.md#bdprop-baddy)
BDPROP	6	ANI	→ §2.6.5	[srv] (protocol/player-props.md#bdprop-baddy)
BDPROP	7	DIR	→ §2.6.5	[srv] (protocol/player-props.md#bdprop-baddy)
BDPROP	8	VERSESIGHT	→ §2.6.5	[srv] (protocol/player-props.md#bdprop-baddy)
BDPROP	9	VERSEHURT	→ §2.6.5	[srv] (protocol/player-props.md#bdprop-baddy)
BDPROP	10	VERSEATTACK	→ §2.6.5	[srv] (protocol/player-props.md#bdprop-baddy)

Appendix C. Version matrix

Build and era names are defined in → §1.1 and → §1.6. Each cell points back to the canonical section; behaviour rules by build (timing, movement, combat, death) are tabulated in → §4.1 to → §4.4 and are not repeated here.

C.1 Wire and scripting by era

Era	Login token	Cipher gen	Message ids handled or sent	Props parsed	Scripting	Source
Z	none	text lines (→ §2.3.1)	text lobby lines	n/a	none	[bin][cap][doc] (version-profiles/matrix/README.md#things-wire-and-scripting)
A	none	1 (no zlib)	relay lobby `N`, `P`, `C`, `D` lines; `D` payload types per build	relay `D` types	GS1-like; `this.` variables from 1.1 beta1	[bin][cap][doc] (version-profiles/matrix/README.md#things-wire-and-scripting); [bin] (client-internals/version-profiles/v1998-v11.toml)
B	`GSERV007` to `GSERV018`	2 (zlib, no cipher)	GSERV id ranges (→ §2.3.2)	unmeasured	GS1; NPC weapons from 1.21	[bin][cap][doc] (version-profiles/matrix/README.md#things-wire-and-scripting)
3D0	`GR3DV002`	own (2D codec, no key byte)	own renumbered ids (→ §9)	3D floats, 24-bit colours	none	[bin][cap][doc] (version-profiles/matrix/README.md#things-wire-and-scripting)
C	`GNW20020` to `GNW07080`	2	1.33: S→C 0–41, 100; C→S 0–31; 45 and 35 ids by 1.39	own 0–33/34, others 21–23 ids	GS1, 88 to 111 commands	[bin][cap][doc] (version-profiles/matrix/README.md#things-wire-and-scripting); [bin][doc] (client-internals/versions-133-139.md#protocol-changes)
D	`GNW13110`	3	1.41 additions (→ C.2)	props to 36	GS1, 126 commands	[bin][cap][doc] (version-profiles/matrix/README.md#things-wire-and-scripting)
E	`GNW08011` to `GNW03081`; NewWorld `G2N15041`	3	2.05: 59 S→C ids	to 42 (2.05)	GS1, 134 to 161 commands; ganis from 1.42	[bin][cap][doc] (version-profiles/matrix/README.md#things-wire-and-scripting)
F	`GNW31101` (2.1) to `GNW22122` (2.17)	3	2.17: 79 S→C, 61 C→S; TAMPERCHECK (39) from 2.17	to 74 (gani attributes from 2.16)	GS1 and gani scripts (2.1); `this.` client strings (2.19)	[bin][cap][doc] (version-profiles/matrix/README.md#things-wire-and-scripting)
G	`GNW03014`, `GNW14015`, `GNW28015`; listserver `GNW30123`	5 (compression byte, partial XOR, bz2)	2.22: 80; 2.31: 81	to 74 (2.22); 80 with X2/Y2/Z2 (2.31)	GS1; classic client-side NPC mode removed in 2.3	[bin][cap][doc] (version-profiles/matrix/README.md#things-wire-and-scripting)
H	`G3D03014`, `G3D28015`; listserver `G3D30123`	5	79 (3.04); 88 (3.153)	as G	GS1 native, TorqueScript `.dso` for menus; NPC bytecode ids (3.153)	[bin][cap][doc] (version-profiles/matrix/README.md#things-wire-and-scripting)

Era	Login token	Cipher gen	Message ids handled or sent	Props parsed	Scripting	Source
I	G3D17026, G3D27076, G3D20126	5	98 to 101	unmeasured	GS2 bytecode; server-pushed ganis (195); update packages GRPKG001	[bin][cap][doc] (version-profiles/matrix/README.md#thinks-wire-and-scripting)
J	G3D22067, G3D14097, G3D15010, G3D26090	5	95 to 100	unmeasured	GS2; GRESCR01 script cache; anti-cheat process scan (182, 183)	[bin][cap][doc] (version-profiles/matrix/README.md#thinks-wire-and-scripting)
K	G3D0108A, G3D0511C (Linux), G3D0311C (Windows 6.03); 6.11 set by script	5; 6.11 can switch to a script-defined header with RC4, AES or TLS	91 to 97	unmeasured	GS2; in 6.11 the packet id tables come from the server script (→ §2.10)	[bin][cap][doc] (version-profiles/matrix/README.md#thinks-wire-and-scripting)

Login-time extras a client must send: PLI_LANGUAGE (37) from 2.02 rev2 [bin] (version-profiles/matrix/README.md#thinks-wire-and-scripting); PLI_TAMPERCHECK (39) once per session in 2.17, on gmap crossings and level links in 2.22, and every 3.2 s in 2.31 [cap] (version-profiles/matrix/README.md#thinks-wire-and-scripting); machine-id MD5s after the password from 4.x on [bin] (version-profiles/matrix/README.md#thinks-wire-and-scripting). Server discovery by era is in → §6.

C.2 Server-to-client ids handled, 1.33 to 2.1 tests

Ids are logical (→ §0.7) and count only ids that reach a real handler.

Build	Change	Source
1.33 to 1.37	43 ids: 0–41, 100	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
1.38	+ 43	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
1.39	+ 75 (45 ids: 0–41, 43, 75, 100)	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
1.41 rev1	same set as 1.39	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
NewMain 1.42	+ 42, 44, 45, 150, 151, 153	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
2.0	+ 154, 155	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
2.01	+ 46, 156	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
2.02 rev2	+ 47	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
2.03	+ 48, 165	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
2.04	+ 166; − 26	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
2.05 rev1, rev2	+ 168	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
NewWorld betas	2.05 minus 2, 4, 17, 18, 154, 155	[bin] (client-internals/cross-version-timeline.md#server--client-packets)
2.1 tests	2.05 plus 49, 169	[bin] (client-internals/cross-version-timeline.md#server--client-packets)

Later builds: → §2.5 and the per-build tables in `version-profiles/plo-handled-ids-by-build.tsv`, which App. A indexes.

C.3 Prop parsers

Parser	1.33	Added later	Source
player props	0–33 [inf]	34 (1.39); 35, 36 (1.41); 37–41 (2.02 rev2); 42 (2.04); 43–45 (2.1 tests)	[bin][inf] (client-internals/cross-version-timeline.md#prop-parsers)
NPC props	0–34 [inf]	35 (1.41); 36–40 (2.02 rev2); 41–43 (2.1 tests)	[bin][inf] (client-internals/cross-version-timeline.md#prop-parsers)
baddy props	0–10	none; the parser is gone in both NewWorld betas	[bin] (client-internals/cross-version-timeline.md#prop-parsers)
sparse player-prop sub-parser	21 ids up to 51	23 ids (1.39); 31 ids (2.05); role unconfirmed, likely props of other players	[bin][inf] (client-internals/cross-version-timeline.md#prop-parsers)

The 1.33 player/NPC split is assumed from 1.34, where the player parser is the one that calls the sparse sub-parser [inf] (client-internals/cross-version-timeline.md#prop-parsers). Prop encodings: → §2.6.

C.4 UDP ids handled

Build	Ids	Source
1.33 to 1.37	0, 10–23	[bin] (client-internals/cross-version-timeline.md#udp-packets)
1.38	+ 24	[bin] (client-internals/cross-version-timeline.md#udp-packets)
2.01	+ 25	[bin] (client-internals/cross-version-timeline.md#udp-packets)
2.03	+ 26	[bin] (client-internals/cross-version-timeline.md#udp-packets)
2.1 tests	+ 27	[bin] (client-internals/cross-version-timeline.md#udp-packets)
NewWorld betas	drop 16, 17	[bin] (client-internals/cross-version-timeline.md#udp-packets)

UDP messages: → §2.9.

C.5 GS1 command table size

Build	Commands	Added	Source
1.33	88	—	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.34	93	`dontblocklocal`, `blockagainlocal`, `showlocal`, `hidelocal`, `carryobject`, `throwcarry`, `setbacktile`; `seteffect` and `unseteffect` leave the table	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.35	93	same table as 1.34	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.36	104	`lay2`, `take2`, `takehorse`, `takeplayerhorse`, `removebomb`, `removearrow`, `removeitem`, `removeexplo`, `removehorse`, `explodebomb`, `reflectarrow`; `lay` takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.37	106	`playlooped`, `stopsound`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)

Build	Commands	Added	Source
1.38	109	`setlevel2`, `changeimgpart`, `hitnpc`; `setlevel` takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.39	111	`setani`, `setcharani`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
1.41 rev1	126	`canwarp`, `callnpc`, `say2`, `break`, `continue`, `callweapon`, `spyfire`, `function`, `return`, `changeimgvis`, `setimg`, `setimgpart`, `drawovertrees`, `putexplosion2`, `showstats`; `putexplosion` takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
NewMain 1.42	134	`canwarp2`, `setshape`, `seteffect`, `setcoloreffect`, `setzoomeffect`, `drawaslight`, `changeimgcolors`, `changeimgzoom`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.0	142	`setgender`, `setchargender`, `join`, `freezeplayer2`, `unfreezeplayer`, `enabledefmovement`, `disabledefmovement`, `setmusicvolume`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.01	144	`openurl2`, `hitobjects`; `openurl` takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.02 rev2	146	`tokenize`, `tokenize2`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.03	148	`triggeraction`, `move`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.04	157	`addtiledef`, `addtiledef2`, `removetiledefs`, `replaceani`, `attachplayertoobj`, `detachplayer`, `setshape2`, `enableselectweapons`, `disableselectweapons`; `setshape` takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.05 rev1, rev2	161	`setbacktile2`, `setz`, `updateterrain`, `loadmap`; `setbacktile` takes arguments	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
2.1 tests	161	same table as 2.05	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
NewWorld beta 1	161	same table as 2.05	[bin] (client-internals/cross-version-timeline.md#gs1-commands)
NewWorld beta 2	165	`addstring`, `insertstring`, `removestring`, `deletestring`	[bin] (client-internals/cross-version-timeline.md#gs1-commands)

A command in the table may have no executor case and be parsed and ignored; new commands are inserted mid-table, so command indexes differ between builds [bin] (client-internals/cross-version-timeline.md#gs1-commands). The commands themselves: → §8.2.

C.6 Generation 2 builds, 1.21 to 1.32

Build	S→C ids handled	Player props parsed	Player props sent	NPC props	Baddy props	Source
1.21	0–34 except 30, plus 100	0–24	0–24	0–17	0–10	[bin] (protocol/gserv-1999.md#message-ids); [bin] (protocol/gserv-1999.md#props-by-build)
1.28 beta 2	0–38, plus 100	0–31	0–24, 26, 31	0–32	0–10	[bin] (protocol/gserv-1999.md#message-ids); [bin] (protocol/gserv-1999.md#props-by-build)
1.3 test	0–39, plus 100	0–32	0–24, 26, 31, 32	0–34	0–10	[bin] (protocol/gserv-1999.md#message-ids); [bin] (protocol/gserv-1999.md#props-by-build)
1.31	0–40, plus 100	0–32	0–24, 26, 31, 32	0–34	0–10	[bin] (protocol/gserv-1999.md#message-ids); [bin] (protocol/gserv-1999.md#props-by-build)
1.32	0–41, plus 100	0–33	0–24, 26, 31–33	0–34	0–10	[bin] (protocol/gserv-1999.md#message-ids); [bin] (protocol/gserv-1999.md#props-by-build)

Prop numbers grow at the end, so a decoder can use the later tables and stop at the build's last id [bin] (protocol/gserv-1999.md#props-by-build). The meanings of ids 35 to 41 in this generation are unmeasured [bin] (protocol/gserv-1999.md#open-questions).

Client-to-server ids sent by these builds are in → §2.4.2.

C.7 Lab runs of later real clients

Each run used an unmodified client against a GServer-v2 lab on a loopback-only network, with hosts redirected by name [cap] (version-profiles/lab-304-linux-2026-10-30.md#setup-for-repeating-it); [cap] (version-profiles/lab-v5-2026-10-30.md#setup-for-repeating-it).

Builds	How it ran	What it settled	Source
3.04 (Linux)	the original packed executable on Xvfb with software GL	the listserver login is one plain frame holding ids 7 and 1, answered in gen 5; the game login frame is plain zlib with 5, 37 and two 34 requests, and every later frame is gen 5, including a bz2 reply	[cap] (version-profiles/lab-304-linux-2026-10-30.md#startup-and-listserver); [cap] (version-profiles/lab-304-linux-2026-10-30.md#game-login)
4.20r6, 3.153 (Mac)	Mac OS X 10.4 in an emulated PowerPC machine	4.20r6 sends the login with a four-field comma list `mac`, an empty MAC hash, a 32-hex hash, and an empty field; 3.153 walks with X2 and Y2	[cap] (version-profiles/lab-mac-tiger-2026-10-31.md#login-line-captured-run2pcap-proxylog); [cap] (version-profiles/lab-mac-tiger-2026-10-31.md#walking-x2y2-confirmed)
5.00r7, 5.12, 5.24r4	Wine on Xvfb behind a gen-5 logging proxy	the server reached through the licence file is treated as the Login server, where no characters and no HUD are drawn; stock gs2lib does not know the 5.24 token and serves it the pre-2.1 file layout	[cap] (version-profiles/lab-v5-2026-10-30.md#login-mode-why-nothing-is-drawn-at-first); [cap] (version-profiles/lab-v5-2026-10-30.md#setup-for-repeating-it)

Appendix D. Glossary

Short definitions of the terms the manual uses, each with the section that defines it. A glossary entry adds no fact of its own. Where a term has several meanings in the notes, the entry lists the meanings and points to where each one applies.

> Status: draft.

Term	Meaning	Defined in
1st Graal ... 15th Graal	early 1998–1999 Delphi builds named by archive order	→ §1.1.1
`ADDITFLAGS`	a bit field in the player props	→ §2.6.2
admin level	the 0–3 rights value in an account record (Graal3D: No rights, Graalian Police, Trusted Admin, Administrator)	→ §5.4, → §9.2.4
allowed versions	the server's list of version tokens it lets log in	→ §5.4
archive name	the name of the archive a build was found in, such as `graal121`	→ §0.9
arrays.dat, arrays2.dat	client data files holding the tile allow lists and other tables	→ §3.5.3, → §3.5.4
arrival	the newx/newy text of a link; a number, `playerx`/`playery`, or one of those with an offset	→ §3.1.3, → App. E#601
baddy	a level opponent with a type and three verse lines; `BDPROP` props	→ §3.1, → §2.6.5
BDPROP	a baddy prop id	→ §2.6.5, → App. B
BigData events	the two Torque events that carry Graal text and scripts over the 3D connection	→ §10.5.3
`bigmap.txt`	the text world map of older builds and generators: one row per line of quoted level names	→ §3.2.5, → §11.3.4
board	one level's tile grid	→ §0.8, → §3.1
board modify	a message that changes a rectangle of board tiles	→ §2.4.5, → §2.5.3
build	one client executable by release name and revision, such as `2.05 rev 2`	→ §0.9, → §1.1
build tag	an evidence tag naming the build a claim was read from, such as `[2.05]`	→ App. G.2
`CHR:1.00`	the one-line character file of the Character Designer and GChar	→ §11.7
class table	the list of class names and ids in a Torque connect accept, a Graal change from 3.153 on	→ §10.3.3
client type	the login type value that selects a server code path (0, 4, 5, 6)	→ §0.9, → §2.3
control-NPC	an `editnpc*.txt` script that drives the built-in editor	→ §11.6.1
DSO	compiled TorqueScript, file version 33 in the 3.x clients	→ §10.4
encryption generation	the transport scheme 1–6, in the GServer-v2 numbering	→ §0.9, → §2.2.1
era	a heading aid grouping builds; never a scope for a fact	→ §0.9
frame	one transport unit carrying one or more messages	→ §0.7, → §2.1.4
GANI	the text animation format, `GANI0001` sprite mode and `GANI0002` movie mode	→ §3.3
`GameObject`	the 3D object class of the 4.20r6 and 5.12 clients	→ §10.5.2
GCHAR, GSHORT, GINT3–5, GSTRING	Graal's printable integer and string encodings, short names `c`, `s`, `i3`–`i5`, `str`	→ §0.6, → §2.1.1
`GDUNG001`	the DungeonGenerator options file	→ §11.3.3
gen 3 insertion	the per-message byte the generation-3 client inserts	→ §2.2.2
`GENERATED`	a gmap key whose meaning depends on the writer: a level name, a flag, or absent	→ §3.2.3
GLEVNW01	the text level format (`.nw`)	→ §3.1.3
gmap, GRMAP001	a world map naming a grid of level files	→ §3.2
GOBJSET01	the stamp-set format the editors use	→ §3.4

Term	Meaning	Defined in
GR-V1	the binary `.graal` level family, 1.00 to 1.04	→ §3.1.2
`GR3DV002`	the version string of the February 2000 Graal3D client	→ §9.1.2
Graal3D	either the February 2000 DirectX program or the Torque 3D side of 3.x–5.12; the manual says which	→ §9, → §10
GraalEditor	the standalone level editor from 2.3	→ §11.1
GraalShop	the GANI editor	→ §11.2
GS1	the original NPC scripting language, interpreted from text	→ §8.1
GS2	the later scripting language, compiled to bytecode	→ §8.3, → §8.4
GServer13	the 1999 game server	→ §5.1
GServer-v2	the open-source server used as `[srv]` reference behaviour	→ §5.3
guest login	the clients' built-in guest account	→ §0.10
GWEBL001	the encrypted web-level cache format	→ §3.8
half-tile	8 px; the unit of most 2D coordinate fields	→ §0.8
`HEIGHTS`	a 9 × 9 height block in a level file; distinct from a gmap's `HEIGHTMAP`	→ §3.1.2.7, → App. E#606
leader	the player picked to run level-local actions for a level	→ §2.9.3, → §9.1.3
level-format generation	the level file family and revision	→ §0.9, → §1.5
listserver	the directory server a client asks for the server list	→ §6
load only	an account flag: the session is not tied to the record and may be shared	→ §5.4, → §9.2.4
logical id	the decimal message id used in this manual; the wire byte is the id + `0x20`	→ §0.7
message	one id, its payload and, in most generations, a `\n` terminator	→ §0.7, → §2.1.2
mission phases	the Torque `MissionStartPhaseN` steps after a 3D connect	→ §10.3.6
NC, NPC Control	the staff client for an NPC server	→ §7.2, → §11.5
NewMain, NewWorld	the 2000–2001 builds of the New World line	→ §1.1
NPC server	the server-side host for NPC scripts	→ §7
NPCPROP	an NPC prop id	→ §2.6.4, → App. B
offline mode	single-player play without a server	→ §4.8
PLI	a client-to-server message, names `PLI_*`	→ §0.7, → §2.4, → App. A
PLO	a server-to-client message, names `PLO_*`	→ §0.7, → §2.5, → App. A
PLPROP	a player prop id	→ §2.6.2, → App. B
predefined NPC	a `defnpc*.txt` script with `#EDIT` parameters, placed from the editor palette	→ §11.6.1
prop stream	a run of prop id and value pairs in one message	→ §2.6.1
raw data	the escape that carries a binary payload behind `PLO_RAWDATA`	→ §2.1.3
RC, RemoteControl	the staff client for a game server	→ §7.1, → §11.5
server-defined protocol	the 6.11 scheme in which a connector script defines the transport	→ §2.10
sign alphabet	the code table used for sign text in binary levels	→ §3.6
stamp	a predefined tile object from a GOBJSET01 set	→ §3.4, → §11.6.4
tick	one step of a client's game loop; its length is per build	→ §0.8, → §4.1
tile	one board cell, 16 px on a side	→ §0.8
tile definition	a line `image,prefix,type,x,y` swapping the tile sheet for levels whose name starts with a prefix	→ §11.6.4
tile type	the per-tile value from the type table that decides walls and water	→ §3.5.2
TGE, Torque	GarageGames' Torque Game Engine, the base of the 3.x–5.12 clients	→ §10.1
UDP id	a message on the peer-to-peer UDP channel	→ §0.7, → §2.9.4

Term	Meaning	Defined in
unmeasured	a table cell value meaning no build was measured	→ §0.4
version token	the login version string a build sends, such as `GNW03081`	→ §0.9, → §1.2
warp field	a link created from an editor selection	→ §11.6.4
wire byte	the byte on the wire for a GCHAR id, the logical id + `0x20`	→ §0.7
Zelda Online	the 1997–1998 Java precursor and its `.zelda` (Z3-V1) levels	→ §1.1.1, → §3.1.1

Appendix E. Open questions and disagreements

Each entry quotes the claims, names their sources, and says what would settle it. No entry is resolved by guess (→ §0.4). Entries are numbered in the order they were added; the number never changes.

> Status: draft. Seeded from each part as it was written, and from the 0.0.2 disagreements chapter and CONSOLIDATED.md (E#600–608). Id ranges: E#1–99 front matter, Part I and Part II; E#100s Part III; E#200s Part VIII; E#300s Part IV; E#400s Parts V–VII; E#500s Parts IX–X; E#600s Part XI and the seed.

E#1 Compiler of the early Delphi builds

Item	Content	Source
Claim A	1st Graal, 3rd Graal to 1.1.9 and 1.21 use the Delphi 2 class-metadata layout (self pointer 64, class name 32 bytes before the VMT)	[bin] (client-internals/build-catalogue.md#patterns-worth-knowing)
Claim B	the same builds carry the RTL copyright "1983,97 Borland", a year late for Delphi 2	[bin] (client-internals/build-catalogue.md#patterns-worth-knowing)
Open	which compiler built them; a Delphi-2-level compiler shipped in 1997, such as C++Builder 1's, would fit both, but it has not been checked	[inf] (client-internals/build-catalogue.md#patterns-worth-knowing)
Settles it	matching the RTL of these builds against C++Builder 1 and Delphi 2 runtime libraries	[inf] (client-internals/build-catalogue.md#patterns-worth-knowing)

E#2 Generation-5 step count of 5.12

Item	Content	Source
Claim A	5.12's frame decoder steps the iterator `len / 4` times, rounded down, for zlib and bz2 frames (capped at 4), and rounds up for raw frames; it encodes the same way; a proxy that re-encodes with round-down stepping lets 5.12 play against GServer-v2	[bin][wire] (protocol/v5v6-lab-play.md#512s-cipher-step-count)
Claim B	every later frame of ten 5.x and 6.x Windows captures, 5.12 among them, decodes with one step per group present, rounded up	[cap] (protocol/v5v6-windows-unpacked.md)
Open	whether the 5.12 captures held a compressed frame short enough (under 16 bytes, length not a multiple of 4) to tell the rules apart	[inf] (protocol/v5v6-lab-play.md#512s-cipher-step-count)
Settles it	decoding a 5.12 capture that contains such a frame under both rules	[inf] (protocol/v5v6-lab-play.md#512s-cipher-step-count)

E#3 Public `Login` server session (5.x, 6.x)

Item	Content	Source
Unobserved	the bytes a public 5.x or 6.x `Login` server sends, including the payload of S→C 190	[doc] (protocol/login-session-known.md#still-unobserved)
Unobserved	the GUI script text for the server list, the account screens and the `ChatBar` control	[doc] (protocol/login-session-known.md#still-unobserved)
Unobserved	the file list inside that server's `basepackage.gupd`, and any later update package	[doc] (protocol/login-session-known.md#still-unobserved)
Unobserved	the backdrop level that server chooses	[doc] (protocol/login-session-known.md#still-unobserved)

Item	Content	Source
Unobserved	the 6.11 connector script body, and with it the framing, cipher and login token a 6.11 client uses	[doc] (protocol/login-session-known.md#still-unobserved)
Settles it	a capture of a real `Login` session, which needs the owner's written permission because it touches a public server	[doc] (AGENTS.md#labs)

E#4 Whether 5.00r7 sends id 11

Item	Content	Source
Claim A	the 5.00r7 send wrappers send 1, 2, 4–10, 12–19, …: no 11 (`PLI_THROWCARRIED`)	[bin] (protocol/v5x-clients.md#client--server-ids-500r7)
Claim B	the 5.00r7 profile lists 4–19, which includes 11; 5.12, 5.24r4 and 5.30r3 were seen sending 11 in captures	[bin][cap] (version-profiles/5.00-6.03.toml)
Open	which list is right for 5.00r7; both are floors, since sends through a variable are not captured	[bin] (protocol/v5x-clients.md#client--server-ids-500r7)
Settles it	a 5.00r7 capture of a throw, or a re-read of the send wrappers for id 11	[inf] (protocol/v5x-clients.md#client--server-ids-500r7)

E#5 Meaning of id 162 from the client

Item	Content	Source
Claim A	gs2lib names 162 `PLI_RC_UNKNOWN162`: an empty message from the RC3 beta	[srv] (protocol/packet-ids.md#client--server-pli_)
Claim B	4.20r8 sends 162 as the answer to S→C 108	[bin] (protocol/v4x-clients.md#client--server-ids)
Claim C	6.01r1 and 6.03r7 send 162 as a reply to a server request: `!` followed by a 2-byte value (two 7-bit groups, capped at `0x6FFF`) the server sent; probably the ping answer that 6.11 calls `onSendPingAnswer`	[bin][inf] (protocol/v6x-clients.md#client--server-ids-601-603)
Open	whether B and C are the same message, and how the RC3 beta's empty 162 relates to them	[inf] (protocol/v6x-clients.md#client--server-ids-601-603)
Settles it	a capture of a 4.x or 6.x client answering S→C 108	[inf] (protocol/v4x-clients.md#client--server-ids)

E#6 Client-to-server constants of 1.21 and 1.32

Item	Content	Source
Claim A	the ids visible as constants are 0 and 10 to 26 in 1.21, and 0 and 10 to 30 in 1.32	[bin] (protocol/gserv-1999.md#message-ids)
Claim B	the per-build profile lists 1.21 as 0, 10–13, 15–26 (no 14) and 1.32 as 0, 10–21, 23–30 (no 22)	[bin] (client-internals/version-profiles/v121-v132.toml)
Open	whether A is a range summary of B or B missed a send site; both are floors, since some sends pass the id in a register	[bin] (client-internals/version-profiles/v121-v132.toml)
Settles it	a re-read of the 1.21 and 1.32 send sites for ids 14 and 22	[inf] (protocol/gserv-1999.md#open-questions)

E#7 First builds that handle S→C 47 and 168

Item	Content	Source
Claim A	the cross-version timeline adds 47 at 2.02 rev2 and 168 at 2.05 rev1	[bin] (client-internals/cross-version-timeline.md#server--client-packets)

Item	Content	Source
Claim B	the per-build handled-id table lists 47 already in 2.02 and 2.02 rev1, and 168 already in 2.05 rev0	[bin] (version-profiles/plo-handled-ids-by-build.tsv) ; [bin] (client-internals/decomp-findings-2026-10-24-c/04-plo-server-to-client.md#set-growth)
Open	whether the timeline grouped the revisions or the table read the wrong revision	[inf] (version-profiles/plo-handled-ids-by-build.tsv)
Settles it	a re-read of the receive switch of 2.02 rev0 and 2.05 rev0 for cases 47 and 168	[inf] (version-profiles/plo-handled-ids-by-build.tsv)

E#8 Prop parser ranges of 2.00

Item	Content	Source
Claim A	the cross-version timeline has the player-prop parser gain 35 and 36 in 1.41, and the NPC-prop parser gain 36 to 40 only in 2.02 rev2	[bin][inf] (client-internals/cross-version-timeline.md#prop-parsers)
Claim B	the 2.00 layout tables have the self and other-player streams accept 0–35 (no 36) and the NPC stream accept 0–36	[bin] (version-profiles/plo-playerprops-layouts.tsv); [bin] (version-profiles/plo-npcprops-layouts.tsv)
Open	whether 2.00 parses player prop 36 and NPC prop 36	[inf] (version-profiles/plo-npcprops-layouts.tsv)
Settles it	a re-read of the 2.00 prop switches at ids 36	[inf] (version-profiles/plo-playerprops-layouts.tsv)

E#9 Uploads from game clients

Item	Content	Source
Claim A	game clients cannot upload; the RC file-browser ids are for RemoteControl only	[srv] (protocol/file-transfer.md#upload-rc-only)
Claim B	6.01r1 and 6.03r7 send the file-browser and upload ids (93, 96–98, 155, 156) with an upload queue, and 3.153 and 5.00r7 send several of them	[bin] (protocol/v6x-clients.md#client--server-ids-601-603); [bin] (version-profiles/3.04-3.153.md#client--server-ids)
Open	whether a server accepts these from a game connection, and which game feature sends them	[inf] (protocol/v6x-clients.md#client--server-ids-601-603)
Settles it	a capture of a 6.x game client using the upload path against a server	[inf] (protocol/v6x-clients.md#client--server-ids-601-603)

E#10 When the `maps` download folder arrived

Item	Content	Source
Open	the 2.1 test builds save downloads with no `maps` rule and 2.17.1 has one; no client between them has been found, so the build that added it is unknown	[bin] (client-internals/decomp-findings-2026-10-25-k/03-open-points.md)
Settles it	the save routine of any client from 2.1 to 2.16, if one turns up	[inf] (client-internals/decomp-findings-2026-10-25-e/03-open-points.md)

E#11 Client-to-server ids the NewWorld betas lack

Item	Content	Source
Claim A	the betas send the 2.1 test set without 7, 8, 11, 15, 16, 17 and 20	[bin] (client-internals/version-profiles/141-217.toml)
Claim B	the betas send 34 ids, the 2.1 test set without 7, 11, 15, 16 and 17 only	[bin] (client-internals/decomp-findings-2026-10-24-c/08-test-builds-newworld-newmain.md#newworld-beta1--beta2)

Item	Content	Source
Open	the two reads disagree on 8 (`H0RSEDEL`) and 20 (`0PENCHEST`); Claim B's count of 34 is 39 less five	[inf] (client-internals/decomp-findings-2026-10-24-c/05-pli-client-to-server.md)
Settles it	a re-read of the beta send sites for constants 8 and 20	[inf] (client-internals/decomp-findings-2026-10-24-c/08-test-builds-newworld-newmain.md#newworld-beta1--beta2)

E#12 Whether the token digits are a build date

Item	Content	Source
Claim A	the `GNW` digits are DDMMY, on or a few days before the link date	[bin] (client-internals/build-catalogue.md#patterns-worth-knowing)
Claim B	the dates fit the change-log order of 2.01 to 2.05, but nothing in the binary names the format	[inf] (client-internals/decomp-findings-2026-10-24-c/10-open-questions-and-corrections.md#open)
Open	the reading is a strong fit, not a stated format	[inf] (client-internals/decomp-findings-2026-10-24-c/02-login-token-and-version.md#reading-the-digits-inferred)
Settles it	a vendor note or server source that decodes the token	[inf] (client-internals/decomp-findings-2026-10-24-c/10-open-questions-and-corrections.md#open)

E#100 Collision of the Bush stamp tiles

Item	Content	Source
Claim A	the 2×2 Bush stamp writes ids 2, 3, 18 and 19, whose type bytes are 20 and 22, so the bush is wall	[file] (formats/gobjset01.md); [src] (formats/tiletypes.txt)
Claim B	ids 2, 3, 18 and 19 appear in the type-0 walk tables and are therefore walkable when the sheet is type 0	[live] (formats/tile-walk.md#stamp-vocabulary)
Open	Claim A reads a 4096-byte type table kept by an outside client; Claim B reads the official client's type-0 allow lists; which one the official client applies to these ids on a type-0 sheet is not settled	[inf] (formats/tiletypes.txt)
Settles it	looking up ids 2, 3, 18 and 19 in arrays 0 and 1 of a shipped `arrays.dat` (→ §3.5.3), or walking into a placed bush in a lab client	[inf] (formats/tile-walk.md#type-0-allow-lists)

E#101 Size of the type-0 allow lists

Item	Content	Source
Claim A	the type-0 allow lists are arrays 0 and 1 of `arrays.dat`, of 934 and 116 tile ids	[2.05] (formats/arrays-and-signatures.md#arraysdat)
Claim B	one exported type-0 mask counted 3195 non-blocking and 682 blocking tiles	[live] (formats/tile-walk.md#type-0-allow-lists)
Open	934 + 116 ids do not add up to 3195 walkable tiles, and 3195 + 682 is not 4096; the notes do not say how the exported mask was made	[inf] (formats/tile-walk.md#type-0-allow-lists)
Settles it	rebuilding the mask from arrays 0 and 1 of a shipped `arrays.dat` and comparing it with the exported one	[inf] (formats/arrays-and-signatures.md#arraysdat)

E#102 Tile id to pixel position on pics1

Item	Content	Source
Claim A	`col = id % 128`, `row = id / 128`, pixel `(col * 16, row * 16)`	[file] (formats/tiletypes.txt)

Item	Content	Source
Claim B	`col = (id % 16) + (id // 512) * 16`, `row = (id // 16) % 32`: eight side-by-side blocks of 512 ids, each 16 columns by 32 rows; Claim A is called wrong in a correction dated 2026-10-25	[file] (client-internals/client-assets.md#tile-sheets)
Open	the manual follows Claim B, the later correction; Claim A's note has not been updated	[inf] (client-internals/client-assets.md#tile-sheets)
Settles it	correcting `formats/tiletypes.txt`, or rendering a known board with both rules against a client screenshot	[inf] (client-internals/client-assets.md#tile-sheets)

E#200 When a GS1 condition counts as true

Item	Content	Source
Claim A	in 2.05 a condition counts as true when it evaluates to 1.0 or more	[2.05] (npc-library/gs1-reference-205.md#execution-model-205)
Claim B	`if` evaluates its expression and jumps past its block when it is 0	[2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
Open	how 2.05 treats a condition between 0 and 1, or a negative one; the claims agree only for 0 and for values of 1 or more	[inf] (npc-library/gs1-reference-205.md#execution-model-205; npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
Settles it	reading the 2.05 `if` executor's comparison, or running `if (0.5)` and `if (-1)` in a lab	[inf] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)

E#201 Four GS1 argument signatures in 2.05

Item	Content	Source
Claim A	`toinventory` is `S`; `puthorse` is `MRR`; `showimg` is `RMRR`; `replaceani` is `MM`	[2.05] (npc-library/gs1-reference-205.md#commands-161-205)
Claim B	`toinventory` is `M`; `puthorse` is `F,R,R`; `showimg` is `R,F,R,R`; `replaceani` is `S,S`	[2.05] (npc-library/gs1-command-reference-205.md#commands)
Open	which letters the 2.05 table holds for these four commands; both notes say they read the same table	[inf] (npc-library/gs1-reference-205.md#commands-161-205; npc-library/gs1-command-reference-205.md#commands)
Settles it	re-reading the signature strings of ops 632, 640, 657 and 749 in the 2.05 rev 2 table	[inf] (npc-library/gs1-reference-205.md#commands-161-205)

E#202 Which message carries GS2 bytecode

Item	Content	Source
Claim A	NPC bytecode is `PLO_NPCBYTECODE`, id 131, sent to clients at or above `CLVER_4_0211` and wrapped in `PLO_RAWDATA`	[srv] (protocol/packet-ids.md; protocol/graphics-movement-bytecode.md#on-wire)
Claim B	for v4 and later clients the script prop is replaced by `NPCPROP_CLASS` plus id 197 (`PLO_UNKNOWN197`, "GS2 bytecode"); the weapon-list note also lists 197 as the bytecode that replaces the script for GS2	[srv] (protocol/packet-layouts.md; client-internals/feature-specs/02-weapons-combat.md#weapon-lists-server-to-client-srv205)
Claim C	id 197 is `PLO_LOADSCRIPT`, a list `type,name,saveToDisk,desKey,crc32`	[srv] (protocol/packet-ids.md)
Open	whether 197 carries bytecode, a script-load descriptor, or both in different server versions, and how it relates to 131	[inf] (protocol/packet-ids.md; protocol/packet-layouts.md)
Settles it	a capture of a 4.x or 5.x client receiving an NPC with a GS2 script from a lab server	[inf] (protocol/graphics-movement-bytecode.md#what-still-needs-capture-or-confirmation)

E#203 The 2.05 `#` message-code set

Item	Content	Source
Known	codes such as `#s(var)`, `#v(expr)`, `#c`, `#p(n)` and the `setcharprop` property codes are in use	[doc] (npc-library/gs1-reference-205.md#operators-and-syntax-doc); [2.05] (npc-library/gs1-command-reference-205.md#how-the-interpreter-works)
Open	the full set of # codes 2.05 expands in string arguments	[doc] (npc-library/gs1-reference-205.md#operators-and-syntax-doc)
Settles it	reading the 2.05 message-code expander	[inf] (npc-library/gs1-reference-205.md#operators-and-syntax-doc)

E#204 Member names of the GS1 object arrays

Item	Content	Source
Known	each of `compus[i]`, `bombs[i]`, `arrows[i]`, `items[i]`, `explos[i]`, `horses[i]` and `signs[i]` has as many member ids in 2.05 as the 2.22 documentation has names	[2.05][doc] (npc-library/gs1-reference-205.md#functions-and-variables-205)
Open	which name maps to which id; the manual gives the documentation's names in its order, which is an inference	[inf] (npc-library/gs1-reference-205.md#functions-and-variables-205)
Settles it	tracing the 2.05 member parser	[inf] (npc-library/gs1-reference-205.md#functions-and-variables-205)

E#205 Index of the hare baddy

Item	Content	Source
Known	2.05 ships `baddyhare.gif`, so a hare baddy type exists; the baddy name table has 10 names, 0–9	[file][2.05] (npc-library/gs1-reference-205.md#constants-doc; npc-library/gs1-command-reference-205.md#name-tables)
Open	the hare's type index	[doc] (npc-library/gs1-reference-205.md#constants-doc)
Settles it	reading the 2.05 baddy image selection by type	[inf] (npc-library/gs1-reference-205.md#constants-doc)

E#206 `move` option bits in 2.05

Item	Content	Source
Known	the documentation gives cachetype 0–2, +4 block check, +8 event when done, +16 apply direction	[doc] (npc-library/gs1-reference-205.md#constants-doc)
Open	how 2.05 handles each bit; the 2.05 reading did not decode them	[inf] (npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour)
Settles it	reading the 2.05 `move` executor	[inf] (npc-library/gs1-command-reference-205.md#npc-appearance-and-behaviour)

E#207 GS2 language and bytecode format

Item	Content	Source
Known	4.x and later clients run GS2 and receive it compiled; 3.153 already handles the NPC bytecode ids	[bin][cap][doc] (version-profiles/matrix/README.md#thinks-wire-and-scripting)
Open	GS2 syntax, standard library and event model; the bytecode format; how the client stores received bytecode; whether later clients use another format	[inf] (protocol/graphics-movement-bytecode.md#what-still-needs-capture-or-confirmation)

Item	Content	Source
Settles it	a lab note decoding a captured bytecode message from a 4.x client session, and a note on the GS2 documentation in the archived websites	[inf] (protocol/graphics-movement-bytecode.md#what-still-needs-capture-or-confirmation)

E#208 What the OpenGS1 opcode set describes

Item	Content	Source
Claim A	the graphics-and-bytecode note calls the OpenGS1 VM "GS1 bytecode" and lists the full GS1 opcode set and constant-table layout as known	[src] (protocol/graphics-movement-bytecode.md#scripting--gs1-bytecode)
Claim B	2.05 does not compile GS1 to bytecode; OpenGS1's opcode set is an independent reimplementation that 2.05 does not share	[2.05] (npc-library/gs1-reference-205.md#execution-model-205; npc-library/gs1-commands.md#update-2026-10-21)
Open	whether OpenGS1's format matches any official client's bytecode (3.153 or 4.x), or only OpenGS1 itself	[inf] (protocol/graphics-movement-bytecode.md#what-still-needs-capture-or-confirmation)
Settles it	comparing a captured official bytecode message (→ App. E#207) with the OpenGS1 layout	[inf] (protocol/graphics-movement-bytecode.md#what-still-needs-capture-or-confirmation)

E#300 Frame driver of 4th Graal alpha, 5th Graal and 5th Graal alpha

Item	Content	Source
Claim A	the static survey records no form timer for 4th Graal alpha, 5th Graal and 5th Graal alpha (idle loop), while 4th Graal and 6th Graal on either side have the 50 ms form timer	[bin] (version-profiles/matrix/builds-static-survey.tsv)
Claim B	eras A to D all drive the tick from a DelphiX timer with interval 50 on the main form	[bin] (version-profiles/matrix/README.md#acts-timing-movement-and-rules)
Open	whether these three builds run from the idle handler, or keep a timer the survey did not find; the tick length is 50 ms either way only if the idle gate is 50 ms there too	[inf] (version-profiles/matrix/builds-static-survey.tsv)
Settles it	reading the main form and the frame routine of the three builds	[inf] (version-profiles/matrix/README.md#what-is-still-unknown)

E#301 Frame loop of the 3.x and 4.x clients

Item	Content	Source
Unobserved	the 3.x and 4.x Torque-era clients' tick length and loop driver; the 5.x and 6.x clients are timed only from their 50 ms position messages	[wire] (protocol/v5v6-lab-play.md); [doc] (version-profiles/matrix/README.md#acts-timing-movement-and-rules)
Settles it	a timed walk of a 3.x and a 4.x client on the lab server	[inf] (version-profiles/matrix/README.md#what-is-still-unknown)

E#302 Walking speed factors other than 1.0, 1.2 and the hidden 0.2

Item	Content	Source
Known	the walking code multiplies the half-tile step by a speed factor: 1.0 on foot, 1.2 in a high speed state; hiding gives 0.1 tile per tick	[2.05] (client-internals/decomp-findings-2026-10-25-l/01-walking-body-box.md#step); [live] (client-internals/player-mannerisms-live.md#lifting-carrying-hiding-and-throwing-live)

Item	Content	Source
Open	factors 0.4 and 0.8 also exist (0.8 depends on a carried object); which states use them is not traced; carrying measured live does not slow the walk	[bin] (client-internals/decomp-findings-2026-10-25-m/open-questions.tsv)
Settles it	a lab walk in each candidate state (riding, shield, carrying each object kind) on 2.05	[inf] (client-internals/decomp-findings-2026-10-25-m/open-questions.tsv)

E#303 Two unidentified lists in the blocked-point test

Item	Content	Source
Open	the blocked-point test reads a list of objects with two flags and a player-sized body box (only with a client option on), and a list of 1.5 × 2 boxes with an enabled flag; neither is identified (signs, baddies, dropped objects and other players are candidates)	[bin] (client-internals/decomp-findings-2026-10-25-m/open-questions.tsv)
Settles it	tracing what fills each list, or a lab test of walking into each kind of level object	[inf] (client-internals/decomp-findings-2026-10-25-m/open-questions.tsv)

E#304 Stopping positions and the wire rounding of nudged positions

Item	Content	Source
Known	the flush snap and the 0.1 nudges are read from the 2.05, 2.17.1, 2.22 and 2.31 walking routines	[2.05][2.22][2.31] (client-internals/decomp-findings-2026-10-25-m/01-walking-rule.md#on-a-refused-step)
Open	the stopping positions are not confirmed on a live client; how the client rounds a position that is not on a half-tile when it sends it is not captured	[bin] (client-internals/decomp-findings-2026-10-25-m/open-questions.tsv)
Settles it	a lab walk towards a rock from both half-tile alignments on 2.22, with the props messages captured	[inf] (client-internals/decomp-findings-2026-10-25-m/open-questions.tsv)

E#305 The 2.22 push frame field

Item	Content	Source
Claim A	2.22 plays push as 5 frames × 4 ticks, looping, as 2.05 does	[live][2.22] (client-internals/player-mannerisms-live.md#cross-version-table)
Claim B	a memory trace of 2.22 against a public server showed a push frame field cycling 0 to 3 every tick from the moment the blocked count passed 18	[live] (live-runs/2gta-222-2026-10-30/README.md)
Open	whether the traced field is the gani frame or another per-tick counter; the second run used the server's own push gani	[inf] (live-runs/2gta-222-2026-10-30/README.md)
Settles it	reading the gani frame field of 2.22 offline with the stock `push.gani`	[inf] (client-internals/player-mannerisms-live.md#open)

E#306 The flag that picks `lift2.wav` for a pull

Item	Content	Source
Open	a successful pull plays `lift.wav`, or `lift2.wav` by a player flag that was not traced	[2.05] (client-internals/ledges-push-pull.md#grabbing-and-pulling)
Settles it	tracing the flag in 2.05's pull routine	[inf] (client-internals/ledges-push-pull.md#open)

E#307 Probe, nudge and step tables of 1st Graal and 1.39

Item	Content	Source
Known	both builds test six probes per facing step and use the 18 / 38 blocked-step counter; 1st Graal has a nudge table	[bin] (version-profiles/matrix/rules-1st-graal.md#wall-and-walk-test-bin); [1.39] (version-profiles/matrix/rules-139.md#wall--walk-test-bin-lab)
Open	the numeric probe offsets, nudge offsets and step table of both builds; whether they equal 2.05's	[bin] (version-profiles/matrix/rules-139.md#still-open)
Settles it	reading the tables from both exports	[inf] (version-profiles/matrix/rules-1st-graal.md)

E#308 Link probe offsets and the neighbour-table flag

Item	Content	Source
Open	the per-facing probe offsets for link following; the flag that skips the 3 × 3 neighbour table; which big map 2.05 uses offline and whether 2.22 does the same with `PLO_BIGMAP` (171); how walking behaves past the edge before the link fires when the neighbour is loaded	[2.05] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#open)
Settles it	reading the probe table and the flag in 2.05, and a lab crossing on 2.22 with a big map	[inf] (client-internals/decomp-findings-2026-10-28/03-edge-links.md#open)

E#309 Knockback distance of 1.39

Item	Content	Source
Claim A	the 1.39 binary moves the player one tile per tick while the hurt counter reads 39 down to 35: five steps	[1.39] (version-profiles/matrix/rules-139.md#hurt-and-knockback-bin)
Claim B	the offline 1.39 run measured four tiles of knockback on the touch-NPC test	[live][1.39] (client-internals/player-mannerisms-live.md#cross-version-table)
Open	whether the live run lost a step to collision or to the ±1 tick of the video timing	[inf] (version-profiles/matrix/rules-139.md#hurt-and-knockback-bin)
Settles it	a fresh 1.39 knockback re-measure on open ground	[inf] (version-profiles/matrix/rules-139.md#still-open)

E#310 Where the once-per-second hit stamp sits

Item	Content	Source
Claim A	the per-attacker second-of-day stamp is shared by the sword and both hit messages, and is stored before the guild test	[2.05][2.22] (client-internals/decomp-findings-2026-10-25-k/01-knockback-and-packet-hits.md#gates-shared-by-the-sword-and-the-packets); [2.05] (client-internals/decomp-findings-2026-10-24-d/03-hit-gates-and-open-points.md#gates)
Claim B	in 2.00 to 2.05 rev 2 the stamp is in the hurt-message path only; it was not found in the sword judge of any build	[2.00][2.05] (client-internals/combat-200-204.md#what-changes-by-build)
Settles it	a lab test of two sword hits from one attacker inside one second on 2.05	[inf] (client-internals/combat-200-204.md#confirmed-inferred-not-tested)

E#311 Blast hit zone and the occasional full-heart hit

Item	Content	Source
Claim A	a blast piece hurts when its centre, offset by one tile, is under one tile from the player's centre on both axes	[2.05] (client-internals/damage-sources.md#bombs-and-explosions)
Claim B	in the 2.22 lab, with a bomb 0.5 or 1.5 tiles off the victim's row, hits came at y offsets +1, +3 and -3 but not at 0 or ±2, and at x offsets ±4 and +5 but not 0 or ±1: a pattern keyed on the bomb's half-tile y rather than a plain radius; a few trials took a full heart instead of a half without a clear rule	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#b-bombs-arrows-baddies-who-owns-the-damage-part-2-fresh-runs)
Settles it	reading the 2.22 explosion routine and repeating the sweep with one bomb per trial	[inf] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#b-bombs-arrows-baddies-who-owns-the-damage-part-2-fresh-runs)

E#312 The last byte of a shot

Item	Content	Source
Claim A	the last byte of `PLO_ARROWADD` (19) is the shot type, `byte - 33`	[2.05] (client-internals/damage-sources.md#shot-19)
Claim B	in the 2.22 lab an injected shot hurt only when that byte was `0x23` or more; `0x20` to `0x22` did nothing, and a real arrow from another 2.22 client carried `0x21` and never hurt	[lab][2.22] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#b-bombs-arrows-baddies-who-owns-the-damage-part-2-fresh-runs)
Settles it	reading the 2.22 shot receiver, or a capture of a real arrow that hurts	[inf] (client-internals/lab-captures-2026-10-25/01-damage-chest-respawn-weapons.md#b-bombs-arrows-baddies-who-owns-the-damage-part-2-fresh-runs)

E#313 Do arrows and fireballs pass walls

Item	Content	Source
Open	in the 2.05 shot flow types 0 and 1 skip the blocking test that stops types 2 and 3, which would let arrows and fireballs fly through walls; the test's arguments were not recovered	[2.05] (client-internals/damage-sources.md#open)
Settles it	a lab shot at a wall on 2.05	[inf] (client-internals/damage-sources.md#open)

E#314 Gmap list order and the level-entry save

Item	Content	Source
Open	the order of the gmap's level list and when it is updated against the entry save decide whether entering a gmap from outside saves the respawn point	[2.05] (client-internals/decomp-findings-2026-10-24-d/open-questions.tsv)
Settles it	a lab walk from a plain level into a gmap and back on 2.05, dying on each side	[inf] (client-internals/decomp-findings-2026-10-24-d/open-questions.tsv)

E#315 The level-entry save in the 2.1 test builds

Item	Content	Source
Open	2.1 test 2 and 3 lack the delimiter constant of the name-group test, so they are expected to have no group test; their entry code was not read	[bin][inf] (client-internals/decomp-findings-2026-10-24-d/02-respawn-entry-save.md#which-builds-have-the-group-test)
Settles it	reading the entry routine of the two builds	[inf] (client-internals/decomp-findings-2026-10-24-d/open-questions.tsv)

E#316 Glove power steps

Item	Content	Source
Claim A	every player starts at glove 1, which lifts the bush, sign and pot; `glove1` adds the light stone and `glove2` the black stone	[2.05][srv] (client-internals/carry-hide-and-glove.md#glove-power-and-the-lift-table)
Claim B	a bare hand lifts the first two entries, power 1 adds the pot, power 2 the light and black stones together	[2.05] (client-internals/decomp-findings-2026-10-25-l/03-cut-lift-chest.md#lift-a-key)
Settles it	a lab lift of each object on 2.05 at glove 0, 1, 2 and 3	[inf] (client-internals/carry-hide-and-glove.md#glove-power-and-the-lift-table)

E#317 The 1.x pickup box and item draw order

Item	Content	Source
Open	only 1.3's pixel overlap for picking up a floor item was read; the 1.x box in tiles, and whether items are drawn under or over players, are not known	[bin] (client-internals/item-drops.md#open)
Settles it	reading the pickup test of 1.39, and a screenshot of a player standing on an item	[inf] (client-internals/item-drops.md#open)

E#318 Level reload on re-entry offline

Item	Content	Source
Open	whether 2.05 offline reloads a level from disk when the player re-enters it, which would bring a cut bush back	[bin] (client-internals/board-respawn.md#open)
Settles it	a lab run on 2.05 offline: cut a bush, leave the level, come back and read the board	[inf] (client-internals/board-respawn.md#open)

E#319 The single-player start levels

Item	Content	Source
Open	loading `level1.graal`, `YoshiClanR2.zelda` or `YoshiClanFields.zelda` runs an extra setup step in 2.05; what the step does was not traced	[2.05] (client-internals/offline-tutorial-205.md#special-level-names)
Settles it	loading each name offline on 2.05 in the lab and comparing the player and level state with another level	[inf] (client-internals/offline-tutorial-205.md#special-level-names)

E#320 Does any server check the tamper check

Item	Content	Source
Open	the clients only send `PLI_TAMPERCHECK` (39); the open-source server discards it, and four public servers accepted a 2.22 login without it for 20 s; whether any server checks the value, or its periodic re-send from 2.31, is unknown	[srv][cap] (client-internals/feature-specs/08-tampercheck.md#public-servers)

Item	Content	Source
Settles it	a 2.31 session on a public server with the re-send withheld or altered, which needs the user's approval for a public login	[inf] (client-internals/feature-specs/08-tampercheck.md#needs)

E#321 The 2.31 window blacklist

Item	Content	Source
Open	2.31 compares top-level window titles and classes with an obfuscated list every 3.2 s; the list and the client's reaction to a match are not decoded	[bin] (client-internals/feature-specs/README.md#settled-index-cross-cutting-items-status-as-of-the-lab-batch-c-2026-10-24)
Settles it	decoding the list in 2.31 and a lab run with a window of a listed title open	[inf] (client-internals/decomp-findings-2026-10-24-b/05-open-items.md)

E#322 How others see a key emote

Item	Content	Source
Open	the 2.05 Ctrl+letter path draws the emote locally and sends nothing, yet captures show `PLI_SHOWIMG` frames re-sending an AFK image about every 3 s; they may come from a server script calling `showimg` or from an untraced path	[2.05][cap] (client-internals/feature-specs/11-emotes-ganis.md#b-emotes)
Settles it	a lab capture of a real 2.05 client pressing Ctrl+A next to a second client	[inf] (client-internals/feature-specs/README.md#still-open-after-the-2026-10-24-decompile-pass-new-or-remaining)

E#323 Level-object routines not read (2.22)

Item	Content	Source
Open	the throw of a lifted object, what a cut or lift drops, the carrying pose, the chest and sign probe points, the glove property value the server sends, snapping after a step, and the swim and carry speed factors were not read	[bin] (client-internals/decomp-findings-2026-10-25-l/open-questions.tsv)
Settles it	reading those routines in the 2.22 client, or a lab run per item	[inf] (client-internals/decomp-findings-2026-10-25-l/open-questions.tsv)

E#400 Two projects called Ash

Item	Content	Source
Claim A	Ash is a Python rewrite of the gs2emu protocol, with its own world and default port 14812; it is not the Reborn 2.4 binary	[src] (server-notes/01-lineage.md#what-is-binary--world-only); [src] (server-notes/ash-overview.md)
Claim B	Ash is the blueprint pack of the 2006 39ster 2.1 GServer, which describes that server as written, not a rewrite; default port 14822	[src] (ash/INDEX.md); [src] (ash/ARCHIVAL-NOTE.md)
Open	whether the two are one project at two stages or two projects that share a name; the manual keeps them apart (→ §5.2, → §5.3.1)	[inf] (ash/ARCHIVAL-NOTE.md)
Settles it	the Drive archives of both, compared by date and content	[doc] (ash/ARCHIVAL-NOTE.md#full-archive)

E#401 Listserver reply ids 1 and 2

Item	Content	Source
Claim A	id 1 is NULL (empty) and id 2 is STATUS text with `\r` as a line break; 2.17, 2.22 and the public service send status text as id 2	[srv] (protocol/listserver.md#listserver--client); [cap] (protocol/public-listserver-2026-10-25.md#frame-and-field-layout-derived)
Claim B	RC 1.41 (2002) reads id 1 as one more server entry and id 2 as the end of the list, then disconnects	[bin] (tools/npc-control.md#listserver-login-new-in-141-bin)
Open	whether the 2002 listserver used 1 and 2 differently from the later Reborn listserver, or the RC reads them its own way	[inf] (tools/npc-control.md#listserver-login-new-in-141-bin)
Settles it	a 2002-era listserver binary or capture, or a re-read of the 2.05 handler for id 1	[inf] (protocol/listserver.md#205-specifics-205)

E#402 The 2000-era listserver reply

Item	Content	Source
Claim A	the real 2.00-era listserver reply is observed: one plain zlib frame, id 0 only, 4 rows, no status, site or upgrade lines (the public service's 14923 reply to a classic login)	[cap] (protocol/public-listserver-2026-10-25.md#what-the-runs-settle)
Claim B	what a real 2.00-era listserver sends is still unobserved; the two-frame list that 2.00 accepted in the lab came from a lab forwarder	[lab] (protocol/listserver.md#205-specifics-205)
Open	whether today's 14923 legacy reply is the same as the service of 2000 to 2002	[inf] (protocol/listserver-login-type-matrix-2026-10-25.md#result)
Settles it	a capture of a real 2.00 or 2.05 against the 14923 service, or a 2000-era listserver binary	[inf] (protocol/login-frames-by-version-2026-10-25.md#captures-that-would-settle-it-all-read-only-list-requests)

E#403 Classic logins on port 14922

Item	Content	Source
Claim A	a real 2.17.1 sends a classic login (shape B) to port 14922, the port in its signed `arrays2.dat`	[cap] (protocol/login-frames-by-version-2026-10-25.md#the-three-real-shapes)
Claim B	the public 14922 accepts a classic login and never answers it; the 2001-era tools put pre-2.2 clients on 14923	[cap] (protocol/listserver-login-type-matrix-2026-10-25.md#result); [src] (tools/server-tools-2001.md#gs2emu-serverlist-thedgtl-snapshot-de66a3f)
Open	whether the 2001-era service answered classic logins on 14922, whether 2.17.1 was built for another listserver, and whether the zlib header (`78 DA` against `78 9C`) matters for a classic frame	[inf] (server-notes/lab-servers-2026-10-30.md#real-clients-against-it-today-lab); [inf] (protocol/login-frames-by-version-2026-10-25.md#does-the-evidence-support-classic-logins-look-fake)
Settles it	a list-only capture of a pristine 2.17.1 against the public service; this touches a public server and needs the owner's written permission	[doc] (AGENTS.md#labs)

E#404 A public server refusing 2.05 inside its own range

Item	Content	Source
Claim A	the server's printed range runs from 1.41r1 to iPhone Belote 1.5, which in the current version table covers every 2.x to 6.x entry, 2.05 included	[src][inf] (server-notes/public-allowed-versions.md#what-that-means-for-5x-and-6x)
Claim B	the same server refused 2.05 (`GNW03081`)	[cap] (server-notes/public-allowed-versions.md#the-refusals-on-record)

Item	Content	Source
Open	which version table the server runs; an older table with no 2.05 entry would explain both	[inf] (server-notes/public-allowed-versions.md#what-that-means-for-5x-and-6x)
Settles it	one 5.00r7 or 6.03r7 login to that server, with the owner's written permission	[inf] (server-notes/public-allowed-versions.md#open)

E#405 RC token `GSERV025`

Item	Content	Source
Claim A	the server knows the RC tokens `GSERV023` to `GSERV025`	[srv] (protocol/chat-pm-rc.md#admin-and-staff)
Claim B	the RC builds read send `GSERV007`, `GSERV011`, `GSERV014` and `GSERV019` to `GSERV024`; none sends `GSERV025`	[bin] (protocol/gserv-1999.md#login); [bin] (tools/remotecontrol-lineage.md#builds); [bin] (tools/remotecontrol.md#logins-bin)
Open	which RC build sends `GSERV025`, and with which login type	[inf] (tools/remotecontrol-lineage.md)
Settles it	a static read of the later RC builds in the Drive RC folder (rc2, the custom RemoteControl, the RC3 beta)	[inf] (protocol/npc-control.md#drive-material-inventory-only-not-in-git)

E#420 Flag and chest counts in the RC props tail

Item	Content	Source
Claim A	the tail after the props block in 72 and 76 holds two s counts, then the weapons; one counts flags and one opened chests	[cap] (protocol/rc-capture-2026-11-04.md#player-props)
Open	which s is which; both were 0 in the only capture	[cap] (protocol/rc-capture-2026-11-04.md#player-props)
Settles it	a capture of 72 for a player with some flags and no opened chests	[inf] (protocol/rc-capture-2026-11-04.md#player-props)

E#421 The last byte of RC_PLAYERPROPSSET2

Item	Content	Source
Claim A	the 2004 RC's 76 ends with one byte after the weapons, 0 in the only capture	[cap] (protocol/rc-capture-2026-11-04.md#player-props)
Open	what the byte means; the server's 72 has no such byte	[cap] (protocol/rc-capture-2026-11-04.md#player-props)
Settles it	a static read of the RC's 76 builder, or a capture after changing other dialog fields	[inf] (protocol/rc-capture-2026-11-04.md#player-props)

E#422 Quoting of folder rights

Item	Content	Source
Claim A	the 2004 RC sends folder rights lines as CSV, so `rw levels/*` goes out quoted, and splits what it reads back the same way	[cap] (protocol/rc-capture-2026-11-04.md#rights)
Claim B	a GServer-v2 fork returned the line unquoted, and the RC showed it as two lines	[cap] (protocol/rc-capture-2026-11-04.md#rights)
Open	whether the original server stored and returned the CSV form, which would make the fork's reply a fork bug	[inf] (protocol/rc-capture-2026-11-04.md#rights)
Settles it	a capture of 62 from an original 2.x GServer, or a static read of its rights reply	[inf] (protocol/rc-capture-2026-11-04.md#rights)

E#500 Back clipping plane of the Graal3D client

Item	Content	Source
Claim A	the Direct3D retained-mode library's initial back plane is 100, and the February 2000 client sets no field of its own	[doc] (graal3d/camera.md#field-of-view)
Claim B	a later public DelphiX fork calls `SetBack(5000)` during retained-mode init; that fork starts after 2004, so it is not the unit linked in 2000	[doc] (graal3d/camera.md#still-needed)
Open	which back plane the February 2000 exe uses; the static read that found no field constant did not look for `SetBack`	[inf] (graal3d/camera.md#still-needed)
Settles it	a static read of the exe for a `SetBack` call, or a lab run with the native `d3drm.dll`	[inf] (graal3d/camera.md#still-needed)

E#501 Class CRC of the 4.04r2 3D plug-in

Item	Content	Source
Claim A	the 4.04r2 connect request carries the stock Torque class CRC, so a server needs exactly the same list of networked classes or the client is rejected with `CHR_INVALID`	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
Claim B	the 4.04r2 CRC table is static data holding `0xFFFFFFFF` for group 0 with no writer, so nothing computes a CRC, as in 5.12	[bin] (torque/wire-by-build.md#the-handshake-per-build)
Open	which value a real 4.04r2 client sends; Claim B is the later, more specific read, and the 4.x plug-in note now defers to it	[inf] (graal3d/torque-engine-4x.md#not-done)
Settles it	a capture of a 4.04r2 connect request	[inf] (graal3d/torque-engine-4x.md#not-done)

E#502 Whether 4.04r2 loads `.mis` mission files

Item	Content	Source
Claim A	`.mis` mission files are used by 3.x and 4.04r2	[bin] (torque/engine-and-formats.md#3d-resource-formats)
Claim B	the 4.04r2 plug-in does not use `.mis`; the mission arrives over the network	[bin][inf] (graal3d/torque-engine-4x.md#files)
Open	whether the 4.04r2 loader is present but unused, or absent	[inf] (graal3d/torque-engine-4x.md#files)
Settles it	a static read of the 4.04r2 plug-in for the mission loader and its callers, or a capture from a 4.04r2 3D session	[inf] (graal3d/torque-engine-4x.md#not-done)

E#503 Whether 4.04r2 reads `.dso` compiled TorqueScript

Item	Content	Source
Claim A	the 4.04r2 plug-in's formats include `.dso` compiled TorqueScript	[bin] (protocol/v4x-clients.md#the-3d-plug-in-graal3denginedll)
Claim B	the 4.04r2 plug-in has no TorqueScript compiler or console, and `.dso` and `.cs` files are never requested or used	[bin] (graal3d/torque-engine-4x.md#overview)
Open	whether a `.dso` reader remains in the plug-in without being reachable	[inf] (graal3d/torque-engine-4x.md#files)
Settles it	a static read of the 4.04r2 plug-in for the code-block loader and its callers	[inf] (graal3d/torque-engine-4x.md#not-done)

E#600 Tile layer as GraalEditor writes it

Item	Content	Source
Claim A	GraalEditor 2.3.1.1 writes `BOARD 0` as 64 lines of 64 decimal tile indices, which the client GLEVNW01 loader does not accept until rewritten as base64 runs	[GE] (tools/graaleditor.md#document-model); [GE] (formats/glevnw01.txt)
Claim B	the 2.05 and 2.22 built-in editors, from which GraalEditor is derived, write base64 `BOARD` rows	[2.22] (tools/level-editor/version-deltas.md#3-formats-per-build)
Open	which form a file saved by GraalEditor itself holds; the claim was never checked against such a file	[inf] (tools/level-editor/version-deltas.md#3-formats-per-build)
Settles it	saving a level from GraalEditor 2.3.1.1 and reading its `BOARD` rows	[inf] (tools/level-editor/version-deltas.md#3-formats-per-build)

E#601 Whether link arrival offsets are evaluated

Item	Content	Source
Claim A	link arrival fields may hold `playerx` or `playery` plus a signed offset such as `playery+10`; 21 arrivals in a 3113-level archive have that form, and the link dialog says math and variables may be used	[2.22] (tools/level-editor/222-reference.md#8-links)
Claim B	the editors store the arrival text verbatim and do not parse it	[2.22] (tools/level-editor/formats-crossref.md#batch-f-decomp-findings-2026-10-25-f-pr-39)
Open	whether the client evaluates the offset when the link fires, or treats it as the bare variable	[inf] (tools/level-editor/222-reference.md#16-open)
Settles it	a lab run through a link with an offset arrival, or a trace of the warp code	[inf] (tools/level-editor/222-reference.md#16-open)

E#602 Code width of `Z3-V1` boards

Item	Content	Source
Claim A	`Z3-V1.02` uses 11-bit codes and `Z3-V1.03`/`.04` use 12; only the 1998 Java Zelda Online games read `.02`	[srv] (formats/level-binary.md#z3-v1-differences)
Claim B	the 2.05 and 2.31 readers use 12 bits for both `Z3-V1` versions they handle; the 2.05 string table lists `.03` and `.04` only	[2.05][2.31] (formats/level-binary-client.md)
Open	the two agree on `.03` and `.04`; read as a width for every Z3 file they do not, and whether 2.05's reader is ever called on a `.02` magic is unmeasured	[doc] (docs/book/04-where-the-notes-disagree.md#how-many-bits-a-z3-code-uses)
Settles it	a `Z3-V1.02` file decoded both ways, and a trace of whether 2.05 reads that magic	[doc] (docs/book/04-where-the-notes-disagree.md#how-many-bits-a-z3-code-uses)

E#603 Whole tiles and fractions in level records

Item	Content	Source
Claim A	coordinates in the binary level records are whole tiles	[srv] (formats/level-binary.md#records)
Claim B	`Z3-V1.03` link numbers are floating point and include halves such as `43.5`; GLEVNW01 allows fractional NPC positions; of 18,710 arrival fields in an archive, 12,035 were plain numbers, halves included	[srv] (formats/level-binary.md#z3-v1-differences); [GE] (formats/glevnw01.txt); [test] (client-internals/decomp-findings-2026-10-25-f/02-link-arrival-expressions.md)
Open	which fields the whole-tile sentence covers	[doc] (docs/book/04-where-the-notes-disagree.md#whole-tiles-and-fractions)

Item	Content	Source
Settles it	a field list for link box, link arrival, baddy, NPC, chest and sign, each with its stated type and one real file of each magic	[doc] (docs/book/04-where-the-notes-disagree.md#whole-tiles-and-fractions)

E#604 Where a binary sign list ends

Item	Content	Source
Claim A	2.05 reads signs to the end of the file; 2.31 stops at the first empty line, because a `HEIGHTS` block can follow	[2.05][2.31] (formats/level-binary-client.md#agreement-with-level-binarymd)
Claim B	the sign section runs until an empty line or the end of the file	[srv] (formats/level-binary.md#layout)
Open	both client behaviours fit Claim B but not each other on a file with signs, a blank line and a `HEIGHTS` block; the server layout does not say which bytes are signs there	[doc] (docs/book/04-where-the-notes-disagree.md#where-a-sign-list-ends)
Settles it	one such file read by the 2.05, 2.31 and server readers, with the sign count each keeps	[doc] (docs/book/04-where-the-notes-disagree.md#where-a-sign-list-ends)

E#605 Client listserver frames in two vocabularies

Item	Content	Source
Claim A	2.00 and 2.05 send id 0 and `newmain` in one plain zlib frame; 2.17 adds id 1 with account and password; 2.22 and 2.31 send id 7, a key byte, `GNW30123` and `newmain`, then a generation-5 frame with id 1	[cap] (protocol/login-frames-by-version-2026-10-25.md#the-three-real-shapes)
Claim B	the classic Server Lister ELF acts on two client lines, an edition line (id 0) and an account line (id 1), and has no version token and no later cipher	[bin] (tools/classic-tools/serverlist3.md#what-a-client-sends)
Open	whether "edition" and "list id 0 and `newmain`" are one observation, and whether "no version token" and "the client sends `GNW30123`" describe two programs; no capture has been laid beside both	[doc] (docs/book/04-where-the-notes-disagree.md#listserver-shapes)
Settles it	loopback captures of 2.05, 2.17 and 2.22 written out frame by frame beside both descriptions	[doc] (docs/book/04-where-the-notes-disagree.md#listserver-shapes)

E#606 Whether a gmap carries `HEIGHTS` blocks

Item	Content	Source
Claim A	a gmap is "a level list that carries HEIGHTS blocks and is not itself a board"	[doc] (protocol/format-notes-from-salvage.md)
Claim B	`HEIGHTS` belongs to each level file, `HEIGHTMAP` belongs to the TerrainGenerator gmap only, and a server reads a gmap that has neither; `GENERATED` means a file name, a flag, or is absent, depending on the writer	[bin] (formats/grmap001.txt); [srv] (formats/gmap.md)
Open	whether the salvage sentence means the level's `HEIGHTS`, the gmap's `HEIGHTMAP`, or both	[doc] (docs/book/04-where-the-notes-disagree.md#what-generated-names-and-where-heights-lives)
Settles it	a correction in the salvage note naming which block it means	[doc] (docs/book/04-where-the-notes-disagree.md#what-generated-names-and-where-heights-lives)

E#607 Leftover lines in the protocol cheat sheet

Item	Content	Source
Claim A	the cheat sheet still shows a raw byte after the version, says RC clients do not scramble, names 100, 101 and 102 as file length, tiles and file blob, and cuts frames into 1,024-byte slices; the recovery note shows guessed ids 1 (login), 16 (level request) and 32 (player update)	[doc] (protocol/PROTOCOL-CHEATSHEET.md); [doc] (protocol/login-recovery-start.md)
Claim B	the login has no mystery byte (the extra byte is the generation-3 insertion, and RC 1.x is generation 3 too); 100, 101 and 102 are raw-data, board and file; no 1,024-byte slice is given; the login is type 0, the level request client id 0 and a walk client id 2; directions are 0 up, 1 left, 2 down, 3 right	[doc] (protocol/login-and-walk.md); [doc] (protocol/codec-and-framing.md)
Open	these are withdrawals written in the later notes; the manual follows the later notes (→ §2.1–2.3), but no capture has been set beside each leftover line	[doc] (docs/book/04-where-the-notes-disagree.md#a-sketch-that-still-prints-another-frame)
Settles it	a 2.05 login, a generation-3 RC session, a generation-5 flush and four labelled facings, each set beside the line it tests	[doc] (docs/book/04-where-the-notes-disagree.md#a-sketch-that-still-prints-another-frame)

E#608 How many `arrays.dat` lists make a tile passable

Item	Content	Source
Claim A	the first array of `arrays.dat` is the passable list; tiles in it get type 0 and everything else type 22	[doc] (CONSOLIDATED.md#tiles)
Claim B	in the old tile mode a tile blocks unless it is in array 0 (934 ids) or array 1 (116 ids)	[2.05] (formats/arrays-and-signatures.md)
Open	whether the one-list summary predates the 2.05 read or describes another reader; the list sizes are → App. E#101	[inf] (formats/arrays-and-signatures.md)
Settles it	a 2.05 session recording, for ids in array 1 only, whether the player can walk on them	[inf] (formats/arrays-and-signatures.md)

Appendix F. Source map

> Status: stub. Filled by PR 15 of the work plan in issue #191. Main sources: manual/sources.tsv.

Appendix G. Evidence-tag legend

How tags are written and cited is fixed in → §0.3. This appendix lists every tag the manual uses.

G.1 Kind-of-evidence tags

Tag	The claim comes from
`[bin]`	a static read of a program: headers, resources, strings, constants, code paths, described as behaviour
`[file]`	a file shipped with a build or found in an archive
`[doc]`	a readme, manual, release note, or a repository rule document
`[srv]`	server source, summarised through the notes
`[src]`	other source code, summarised through the notes
`[lab]`	a client this project ran in a loopback lab
`[live]`	a client this project ran and watched in play
`[cap]`	a packet capture
`[wire]`	bytes read on the wire during a loopback session
`[run]`	a loopback session log
`[shot]`	a screenshot or still from a run
`[test]`	a check of a reader or rule against real files
`[inf]`	an inference that the source note marked as one (→ §0.3)
`[GE]`	the GraalEditor notes
`[checked]`	a search of named builds for a script word or string

Source of this legend: the reading guide [doc] (docs/book/00-how-to-read.md#what-a-source-tag-means) and the agent rules [doc] (AGENTS.md#no-real-servers-no-binary-addresses).

G.2 Build tags

A build tag names the client build the claim was read from, written with the release number as the notes write it: `[2.05]`, `[2.22]`, `[2.31]`, `[1.39]`, `[5.12]` and so on. A build tag implies a static read or trace of that build unless another tag on the same claim says otherwise. The full list of builds and their names is → §1.1.

G.3 Combining tags

Form	Meaning
`[srv][2.05]`	both kinds of evidence agree on the claim
`[2.05][inf]`	read from 2.05, with an inference step the source marked
`[cap][inf]`	seen in a capture, with an inference step the source marked